BIF 101- Introduction to Bioinformatics Lec 1- 150

Effort by Amina Zee

Amina Zee

CELL BIOLOGY

Lec 01

Unit of Life

Amina Zee

Unit of Life

Cell Structure

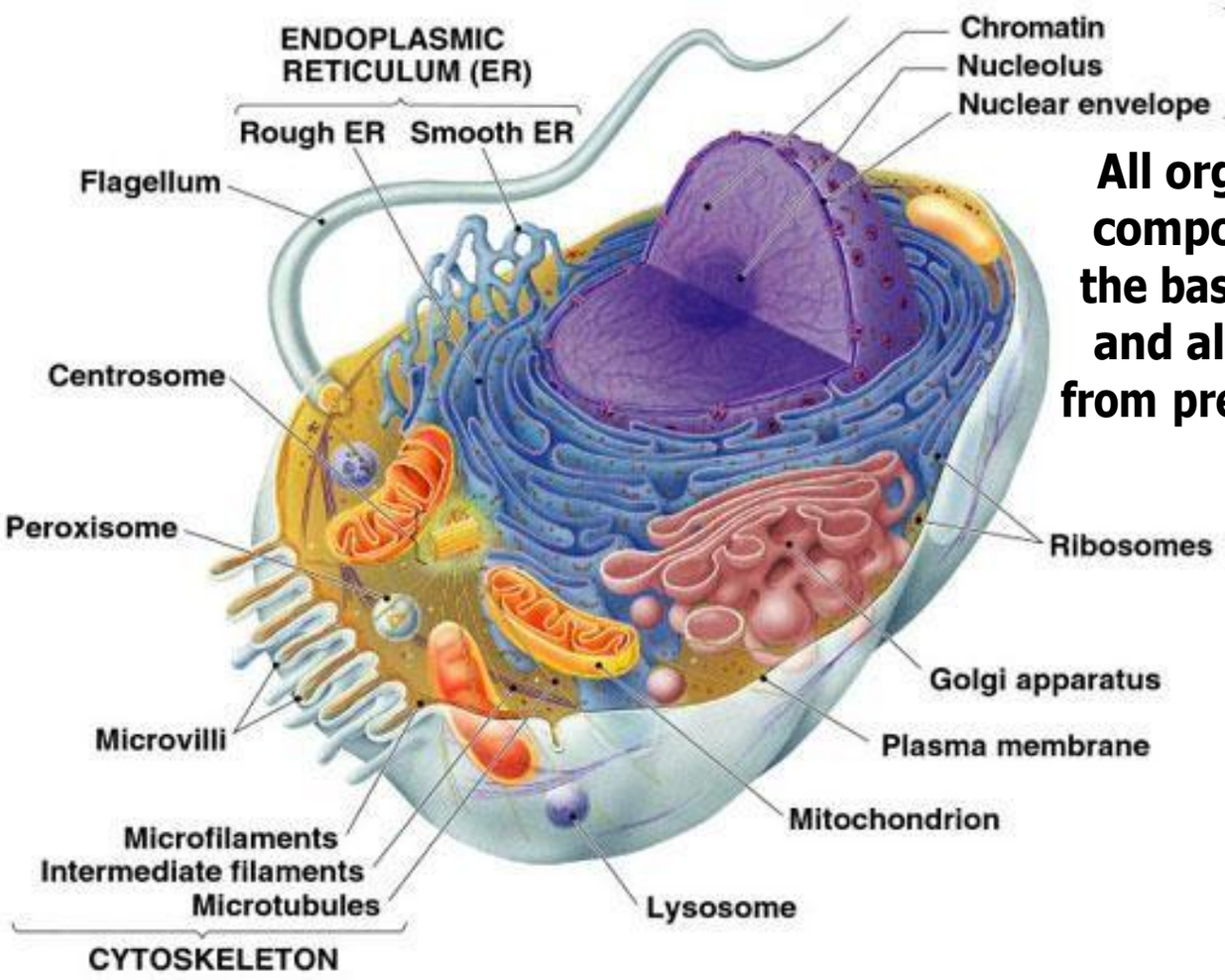
Cellular compartments

Organelles

Structure & Function

Amina Zee

Unit of Life



All organisms are composed of cells the basic unit of life and all cells come from preexisting cells

Amina Zee

Not a random collection of molecules

Life

collection of macromolecules that can perform unique functions because the are enclosed in structural compartment that provides consistency (homeostasis).

Amina Zee

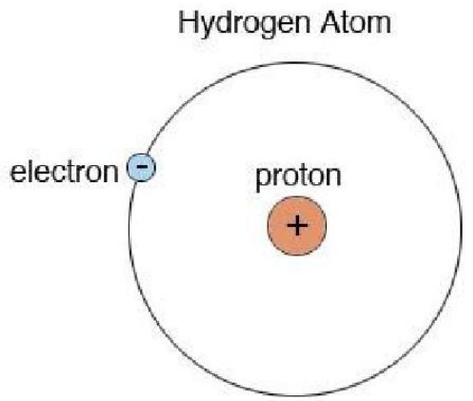
CELL BIOLOGY

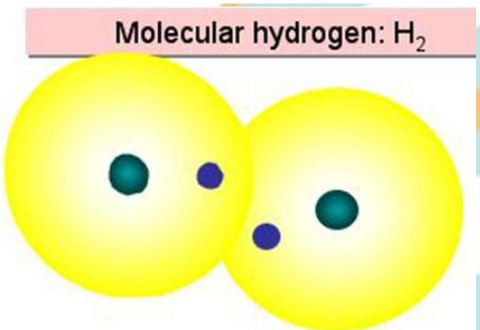
Lec 02
Composition of Matter

Amina Zee

Composition of Matter

Atomic Structure

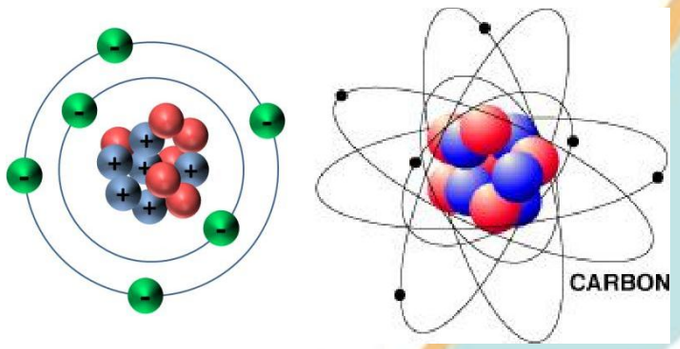


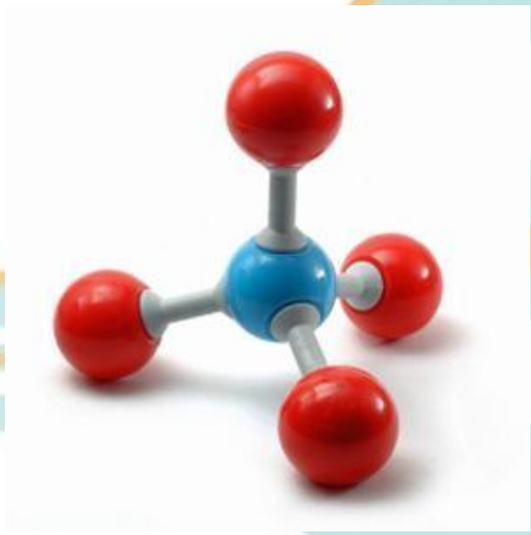


Amina Zee

Composition of Matter

Carbon Atom



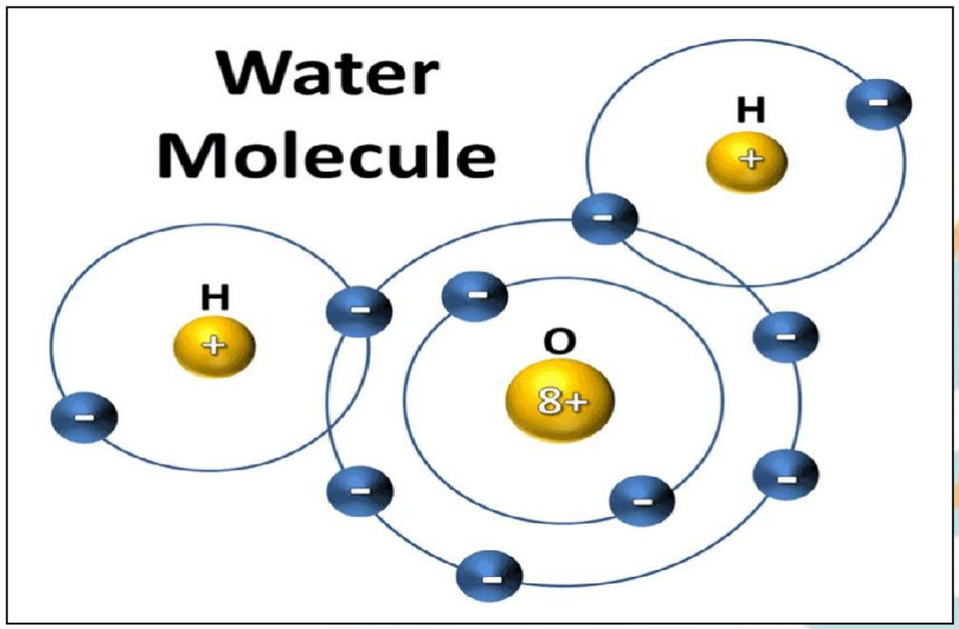


Amina Zee

Composition of Matter

Electronegativity

The tendency of an atom to attract electrons towards itself.

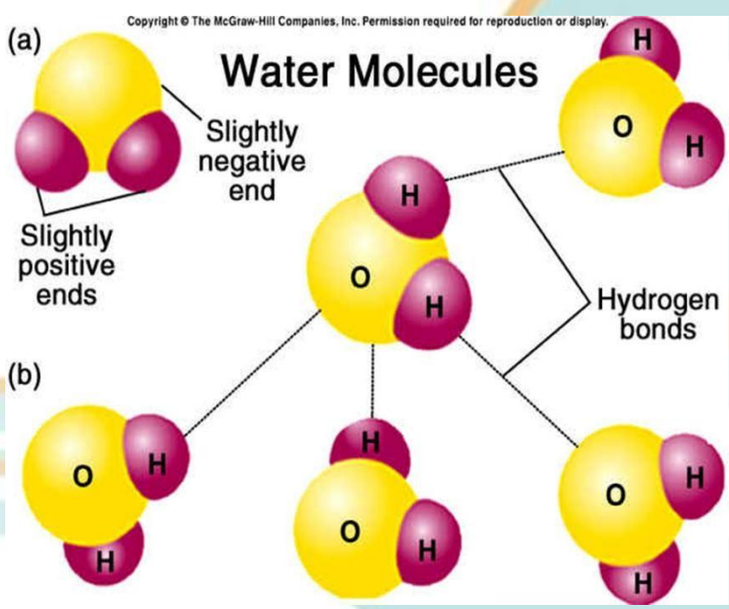


Amina Zee

Composition of Matter

Polarity of H2O

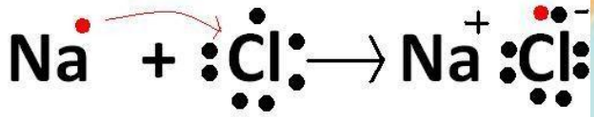
Results in Hydrogen Bonding

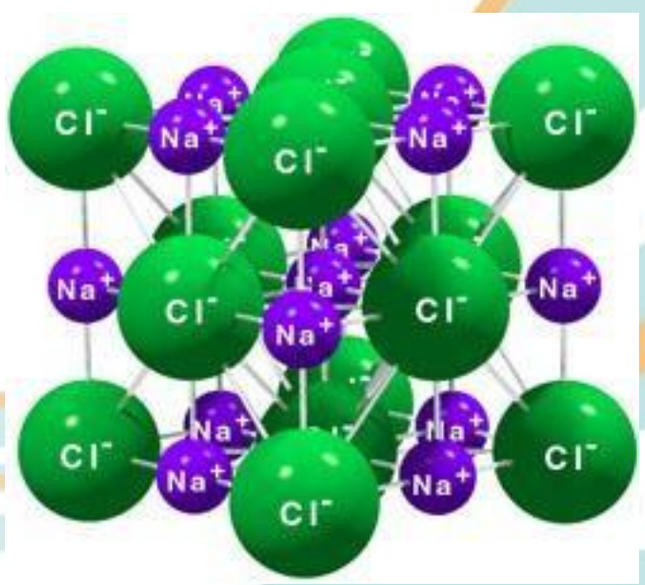


Amina Zee

Composition of Matter

Ionic Bond





Amina Zee

CELL BIOLOGY

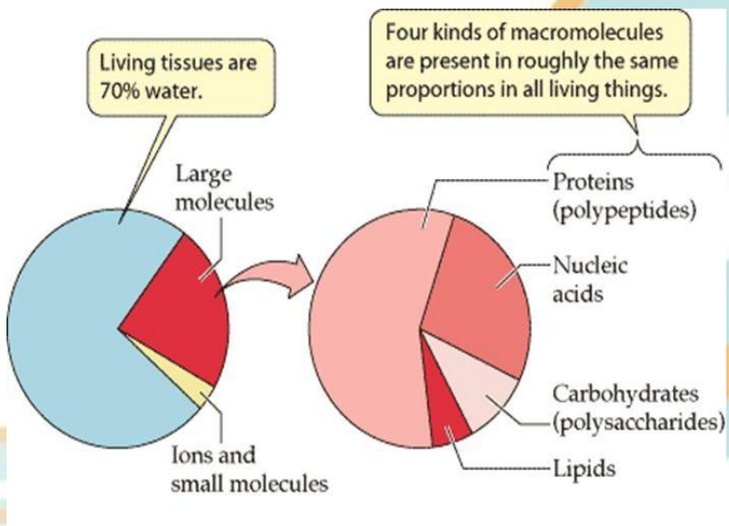
Lec 03

Molecules of Life

Amina Zee

Molecules of Life

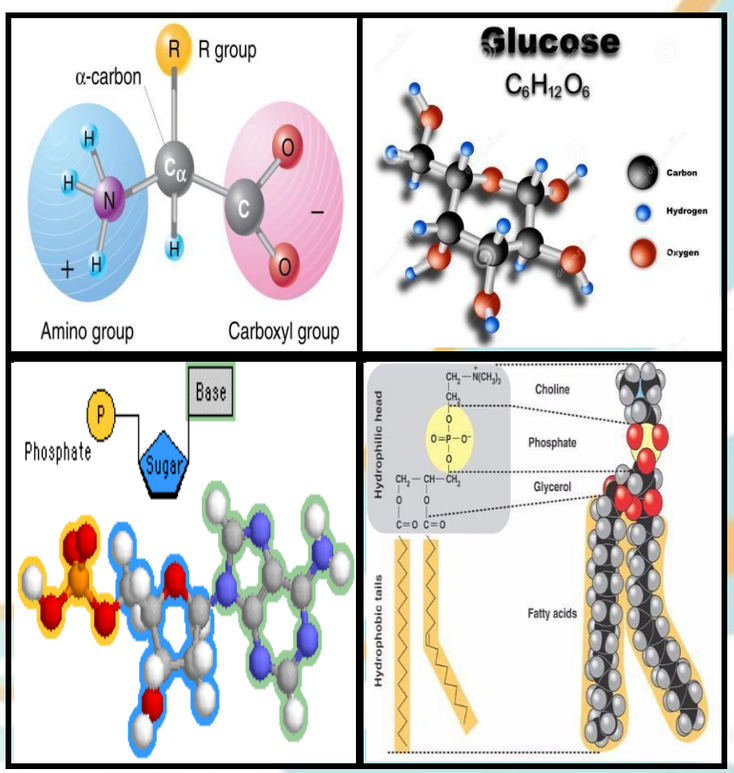
Composition of Life



Amina Zee

Molecules of Life

Building Blocks of Life



Amina Zee

Molecules of Life

Polymers & Monomers

Macromolecules are polymers of smaller monomers molecules

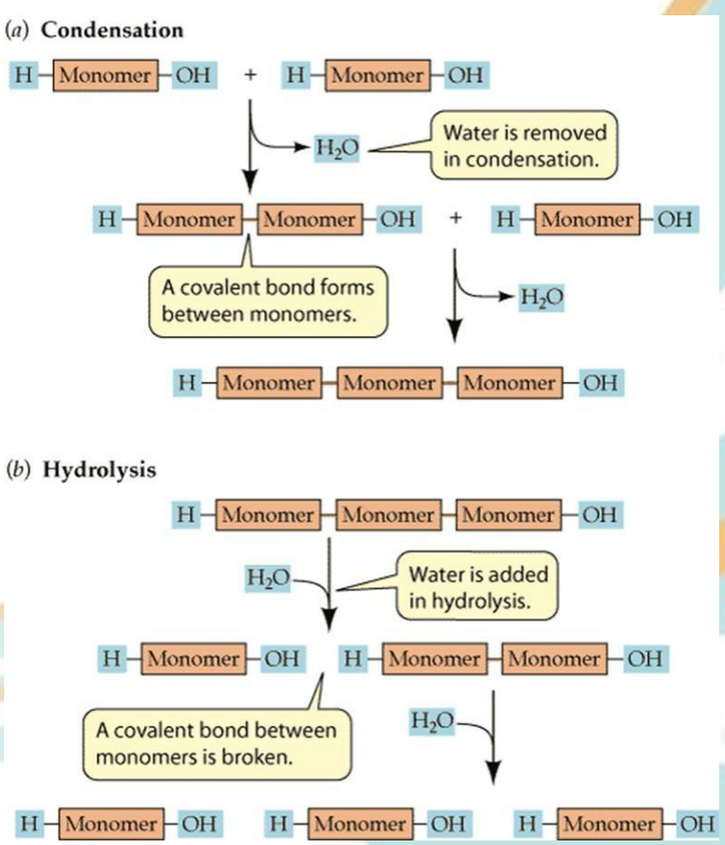
Condensation reactions: Monomers join H_2O Removal

Hydrolysis reactions: Use water to break polymers into monomers.

Amina Zee

Molecules of Life

Condensation & Hydrolysis



Amina Zee

CELL BIOLOGY

Journey into the Cell

Amina Zee

Unit of Life

View animation

END

Amina Zee

CELL BIOLOGY

Lec 05

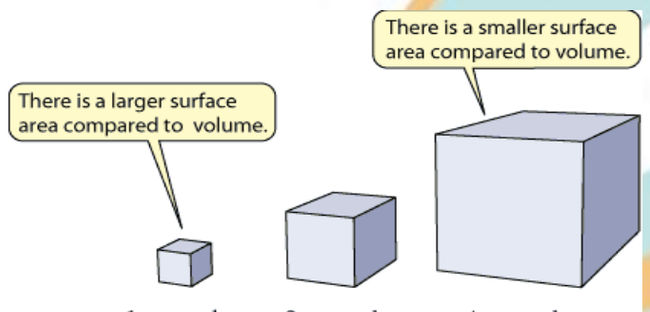
Size Matters

Amina Zee

Size Matters

Cell Structure

Cells are small to maintain large surface area to volume ratio



	1-mm cube	2-mm cube	4-mm cube
Surface area	6 sides × 1^2 = 6 mm^2	6 sides × 2^2 = 24 mm^2	6 sides × 4^2 = 96 mm^2
Volume	1^3 = 1 mm^3	2^3 = 8 mm^3	4^3 = 64 mm^3
Surface area-to-volume ratio	6/1	3/1	1.5/1

Amina Zee

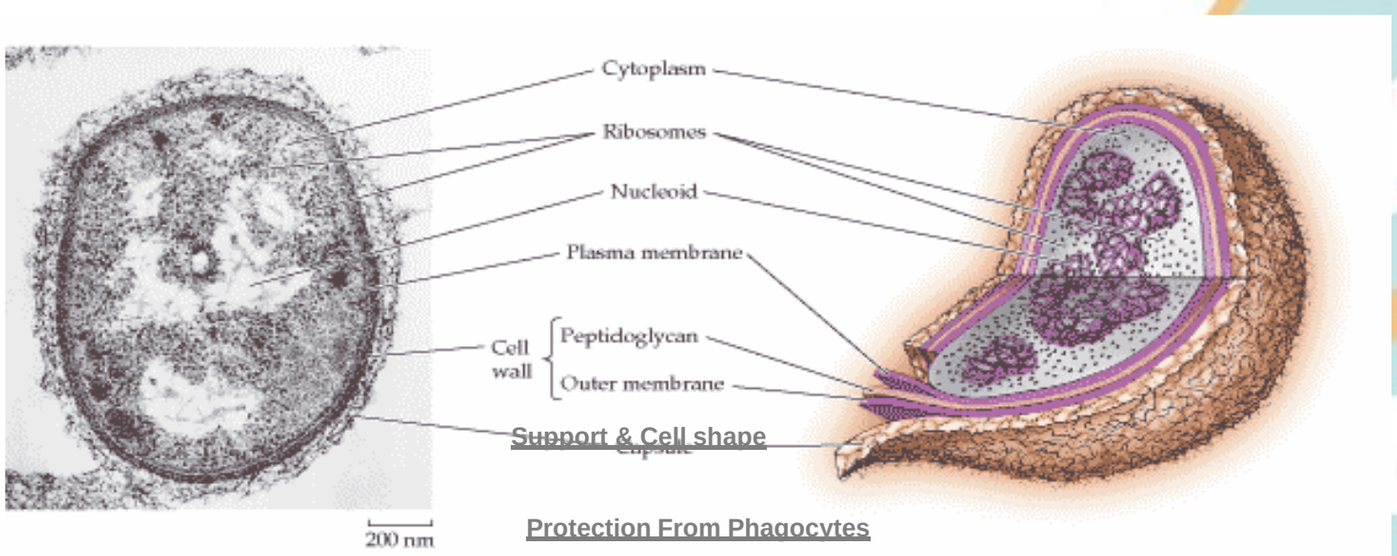
Prokaryotic Cells

No membrane enclosed internal compartments.

Plasma membrane regulates traffic (barrier).

Nucleoid region contains DNA.

Most have cell wall.



Support & Cell shape

Protection From Phagocytes

Amina Zee

Prokaryotic Cells

Special Features

Cyanobacteria
Chlorophyll containing have folds of plasma membrane, other have mesosomes (energy).

Some have actin like filaments and other have Flagella made-up of Flagellin.

END

Amina Zee

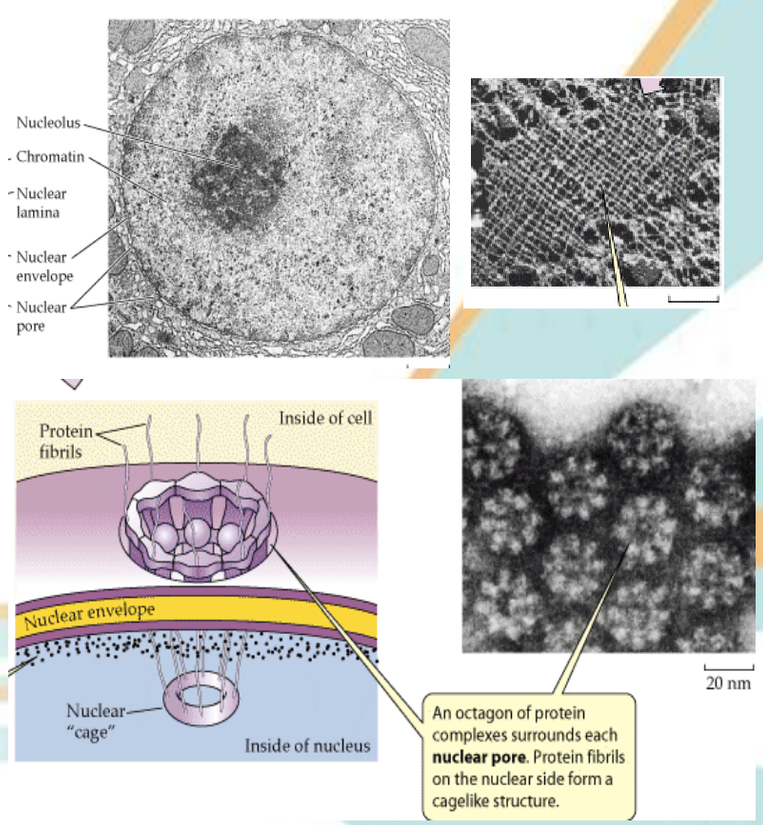
Lec 06

The nucleus

Amina Zee

The Nucleus

Nuclear EMs



Amina Zee

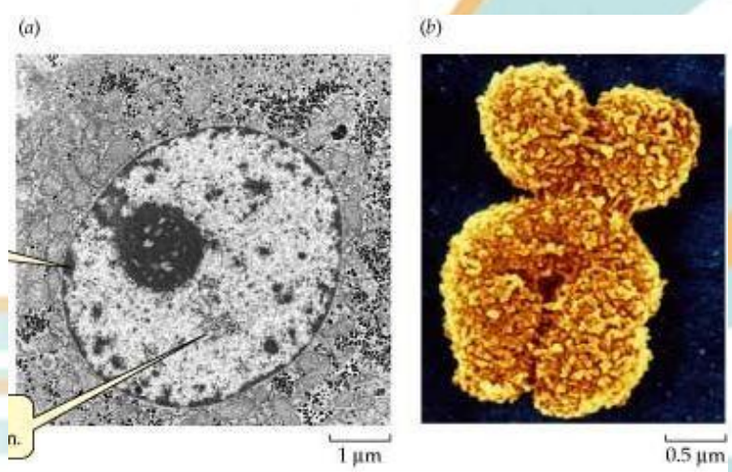
The Nucleus

Chromatin

Euchromatin

Hetrochromatin

Chromosome



END

Amina Zee

Lec 07

Introduction to molecular biology

Amina Zee

Introduction to Molecular Biology

- Molecular Biology
- (BIO-302)
- In the broader terms, definition of molecular biology includes all aspects of the study of life from a molecular perspective.

Amina Zee

Introduction to Molecular Biology

- More precisely, the term “Molecular Biology” refers to the biology of the molecules related to genes, gene products and heredity.
- In other words, the term molecular biology is often substituted for a more appropriate term, M**olecular Genetics.**

Amina Zee

Introduction to Molecular Biology

- So molecular biology grew out of the disciplines of genetics and biochemistry.
- In the present age, world is in the midst of two scientific revolutions. One is information technology and the other is Molecular Biology.

Introduction to Molecular Biology

- Both deal with the handling of large amounts of information.
- Molecular Biology has revolutionized the biological sciences as well especially in the fields of Health Sciences and Agricultural Sciences.

Amina Zee

Lec 08

History of molecular biology

Amina Zee

History of Molecular Biology

- Molecular Biology takes its roots mainly from the the disciplines of Genetics and Biochemistry.
- Some important discoveries which led to the emergence of Molecular Biology as a discipline include:-

Amina Zee

History of Molecular Biology

Biochemistry

- Synthesis of Urea by Friedrich Wohler (1828).
- The first enzyme diastase was discovered by Anselme Payen in 1833.

Genetics

- Mendel's Experiments on garden pea by Gregor Mendel in 1865.
- Chromosome theory of inheritance by Thomas Hunt Morgan in 1910.

Amina Zee

History of Molecular Biology

Biochemistry

- The Discovery of DNA by Friedrich Miescher in 1869.
- Eduard Buchner discovered cell-free fermentation in1897.
- James B. Sumner crystallized enzyme Urease in 1926.

Genetics

- Barbara McClintock and Harriet Creighton provided physical evidence of recombination in 1931.
- One-gene/one-enzyme hypothesis proposed by George Beadle and Edward Tatum in 1941.

Amina Zee

History of Molecular Biology

Biochemistry

- James Watson and Francis Crick discovered the double helical structure of DNA in 1953.

Genetics

- Oswald Avery and his colleagues demonstrated in 1944 that DNA is the hereditary material.
- It was further confirmed by Alfred Hershey and Martha Chase in 1952.

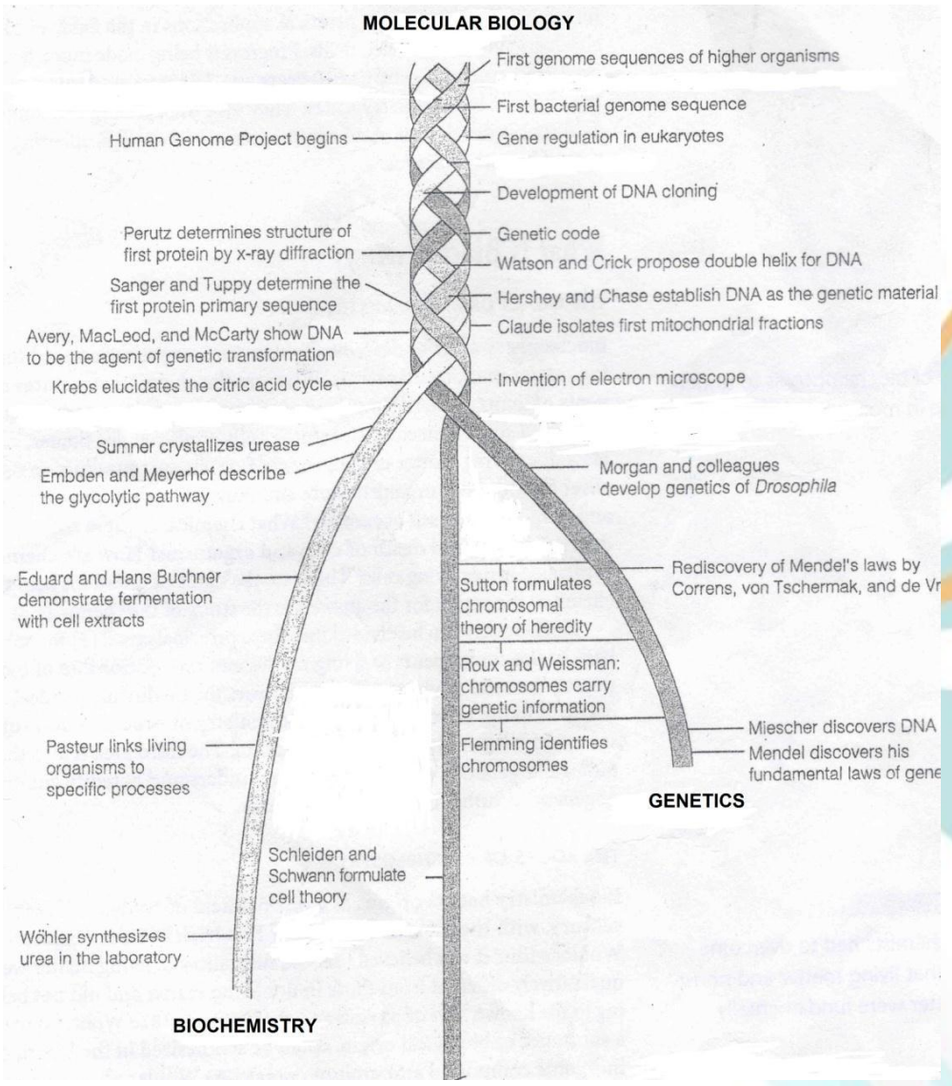
Amina Zee

History of Molecular Biology

- At this point, the discoveries in the two fields led to the foundation of a new discipline called Molecular Biology.

Amina Zee

History of Molecular Biology



Amina Zee

Achievements of Molecular Biology

- In 1957, Francis Crick laid out the "Central dogma of moleular biology" which foretold the relationship between DNA, RNA, and proteins.

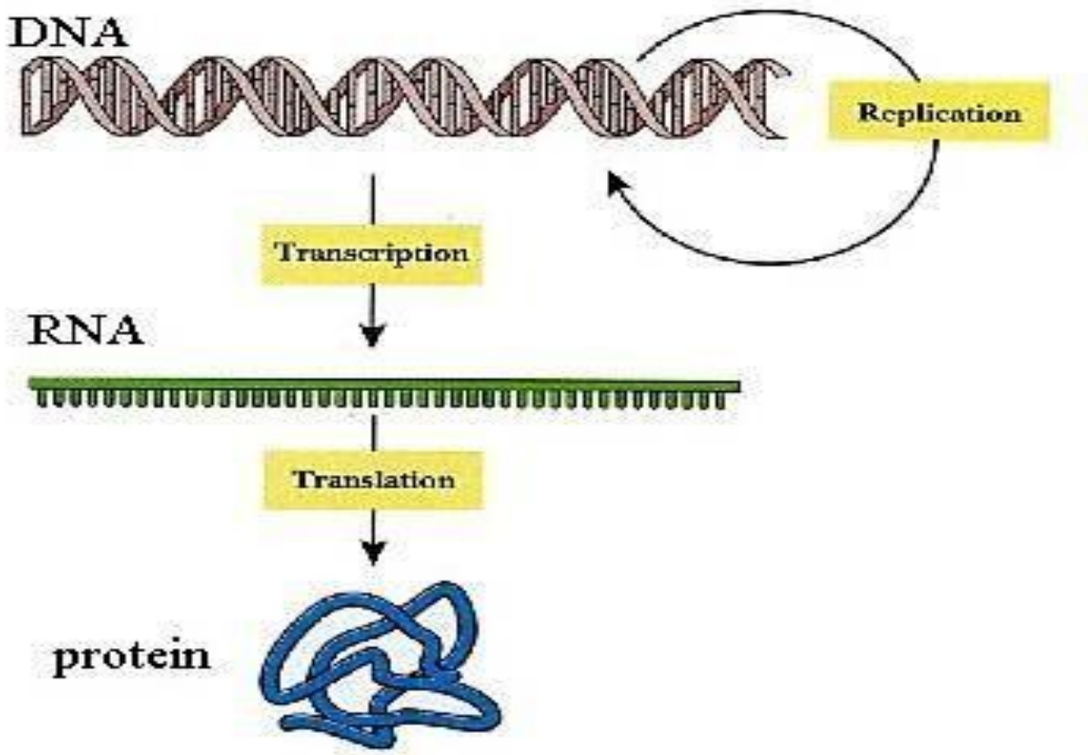
Amina Zee

Lec 09

Achievements of molecular biology

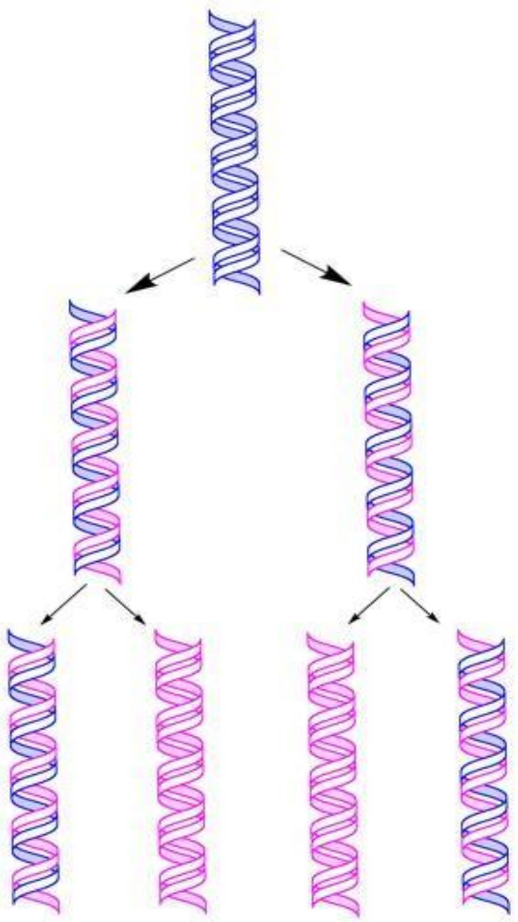
Amina Zee

Achievements of Molecular Biology



Amina Zee

Achievements of Molecular Biology



- In 1958, Mathew Meselson & Franklin Stahl proved that DNA replication was semi-conservative.

Amina Zee

Achievements of Molecular Biology

- Marshall Nirenberg and Gobind Khorana working independently cracked the code in the early 1960s.
- They found that 3 bases constitute a code word, called a **codon**, that stands for one amino acid.

Amina Zee

Achievements of Molecular Biology

Gene Cloning

- Since 1970s, scientists have learned to isolate genes, place them in new organisms and reproduce them by a set of techniques collectively known as **gene cloning**.

Amina Zee

Achievements of Molecular Biology

Genetically Modified Organisms (GMOs)

- The technique of Gene cloning led to the creation of a large number of genetically modified organisms with desirable characters.

Amina Zee

Achievements of Molecular Biology

Human Genome Project

- The Human Genome Project (HGP) was launched in 1990 and completed in 2003.

Amina Zee

Lec 10

Nucleic acid

Amina Zee

Nucleic Acids

Amina Zee

Nucleic Acids

- Nucleic acids are important group of biomolecules which are responsible for storage & transmission of hereditary information.
- Like proteins and polysaccharides, nucleic acids are also polymeric compounds.

Nucleic Acids

The repeating units in the nucleic acids are.

Nucleotides.

Nucleic acids

- Ribonucleic acids (RNA)
- Deoxyribonucleic acids (DNA)

Amina Zee

Lec 11

chemical composition of DNA

Amina Zee

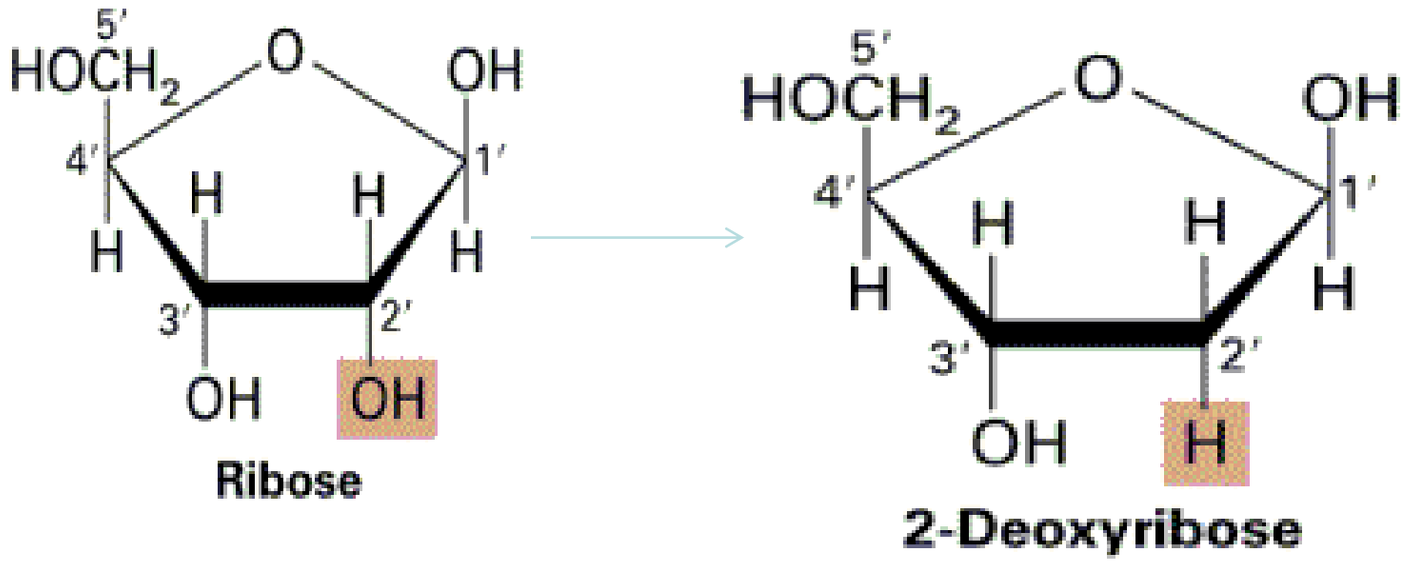
Chemical composition of DNA

- DNA is a polymer of Deoxyribonucleotides.
- Deoxyribonucleotide is composed of three components:
  - Deoxyribose
  - Nitrogenous Base
  - Phosphoric acid

Amina Zee

Chemical composition of DNA

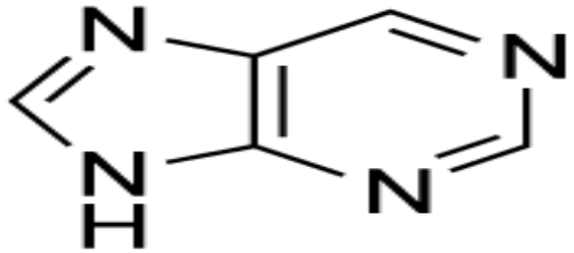
Deoxyribose (a pentose sugar derivative)



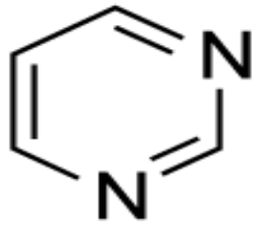
Amina Zee

Chemical composition of DNA

Nitrogenous Bases



purine

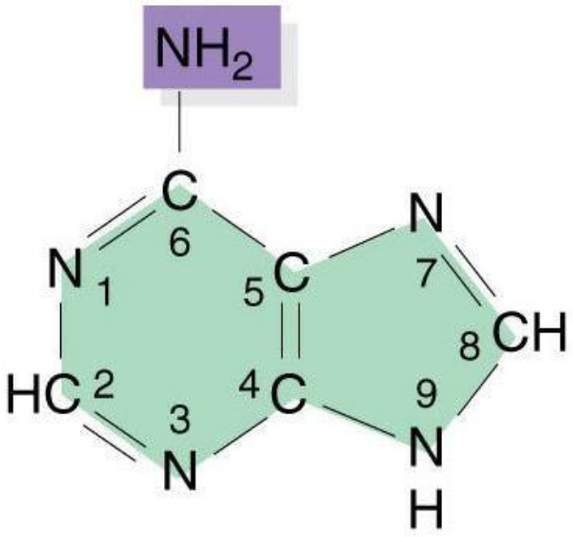


pyrimidine

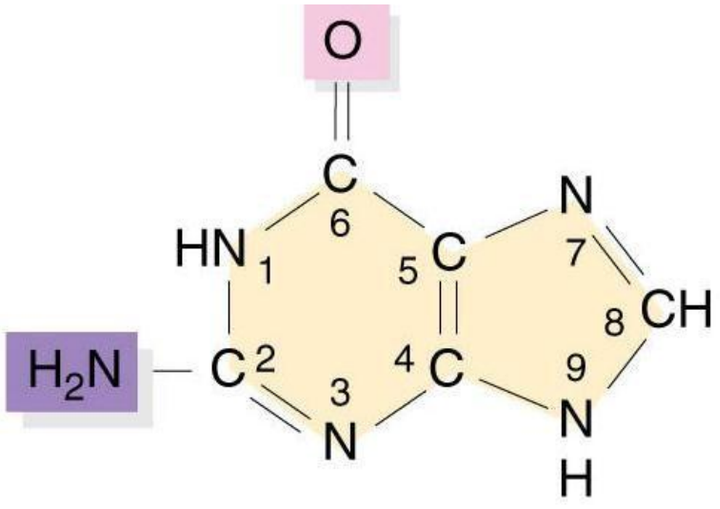
Amina Zee

Chemical composition of DNA

Purines



Adenine (A)

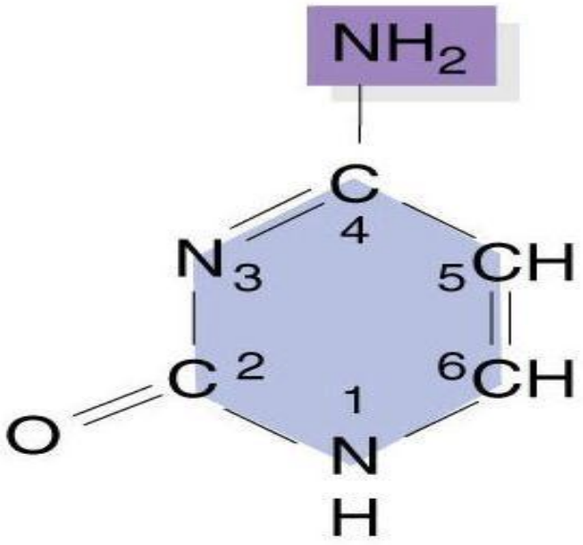


Guanine (G)

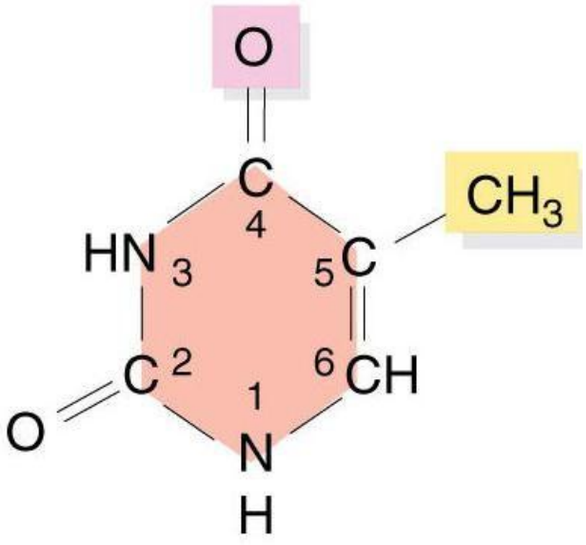
Amina Zee

Chemical composition of DNA

Pyrimidines



Cytosine (C)

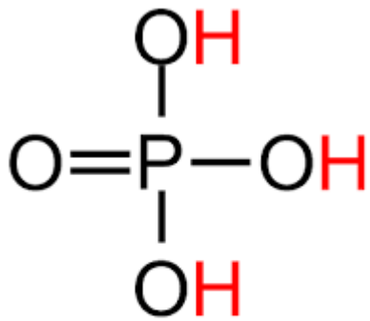


Thymine (T)

Amina Zee

Chemical composition of DNA

Phosphoric acid



H_3PO_4

end

Amina Zee

Lec 12

Nucleoside & Nucleotide

Amina Zee

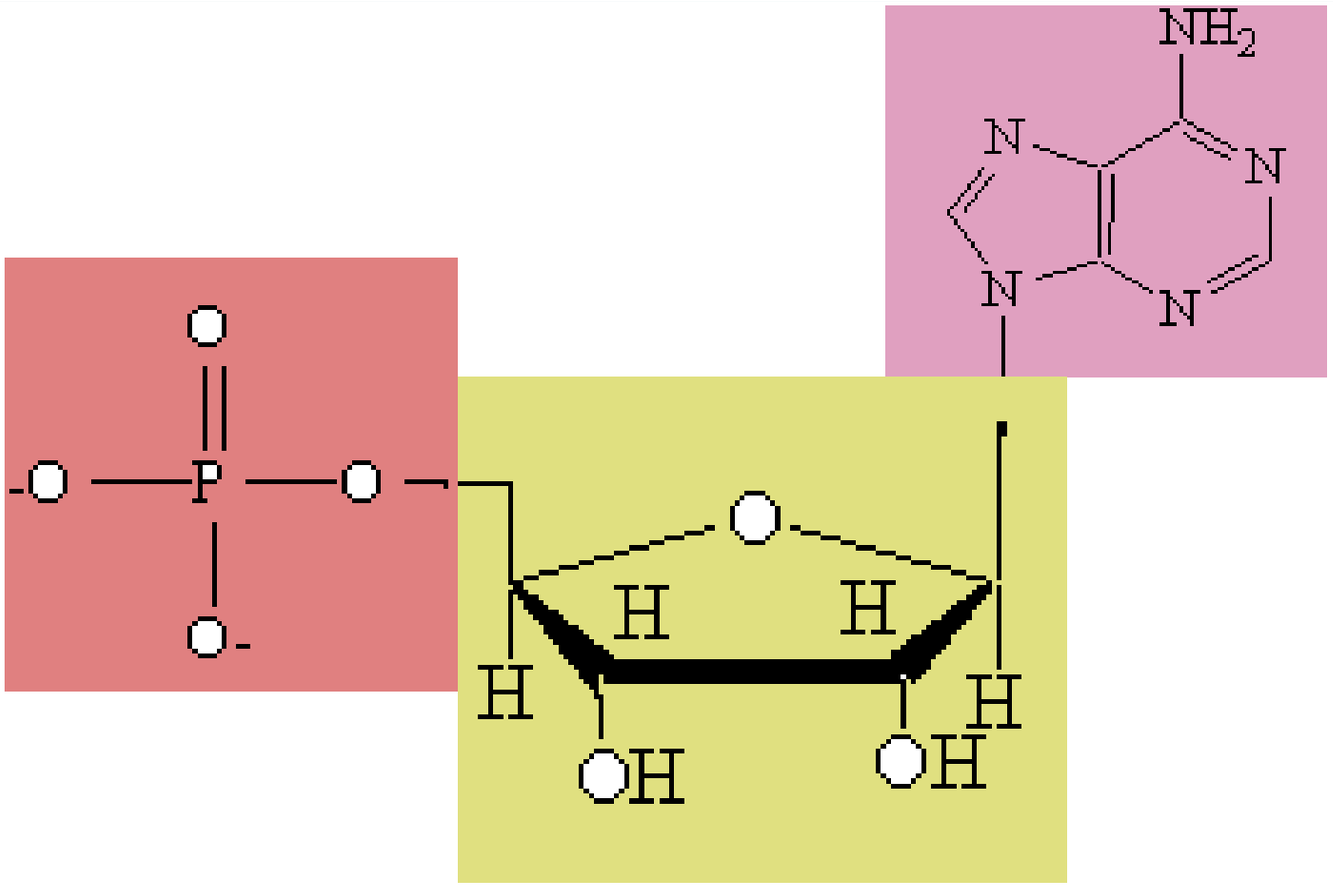
Nucleoside & Nucleotide

- A molecule containing all these three components is called a nucleotide.

- While a molecule without the phosphate group is called a nucleoside.

Amina Zee

Nucleoside & Nucleotide



Amina Zee

Nucleoside & Nucleotide

Nucleotide =
Nucleoside +
Phophoric acid
&
Nucleoside =
Nucleotide –
Phoshoric acid

end

Amina Zee

Lec 13

Types of Deoxyribonucleotides

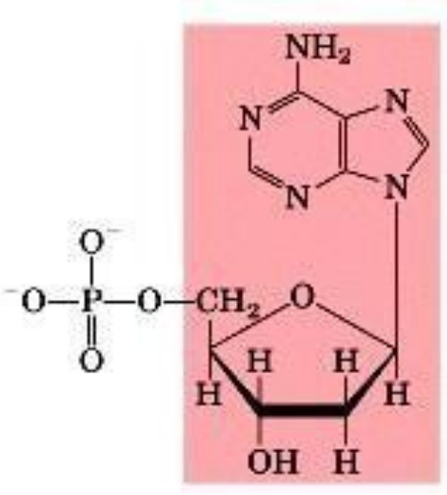
Amina Zee

Types of Deoxyribonucleotides

- There are four types of Deoxy-ribonucleotides.

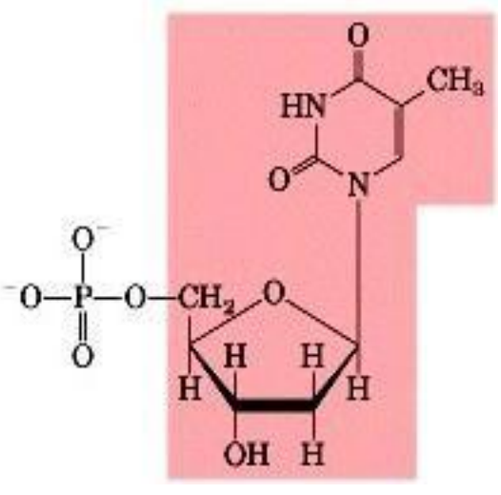
Amina Zee

Types of Deoxyribonucleotides



Deoxyadenosine

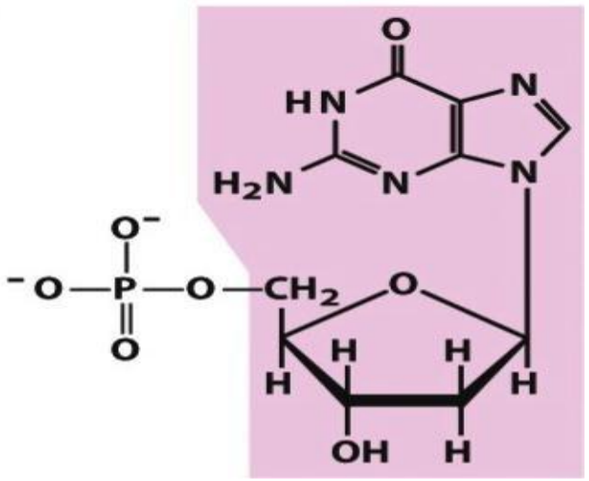
Deoxyadenylate



Deoxythymidine

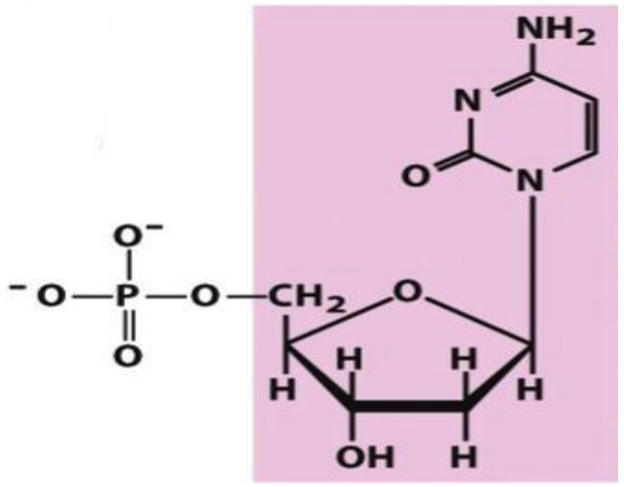
Deoxythymidylate

Types of Deoxyribonucleotides



Deoxyguanosine

Deoxyguanylate



Deoxycytidine

Deoxycytidylate

Amina Zee

Types of Deoxyribonucleotides

- These four deoxy-ribonucleotides make the structural units of DNA.

end

Amina Zee

Lec 14

How do Deoxyribonucleotides Join?

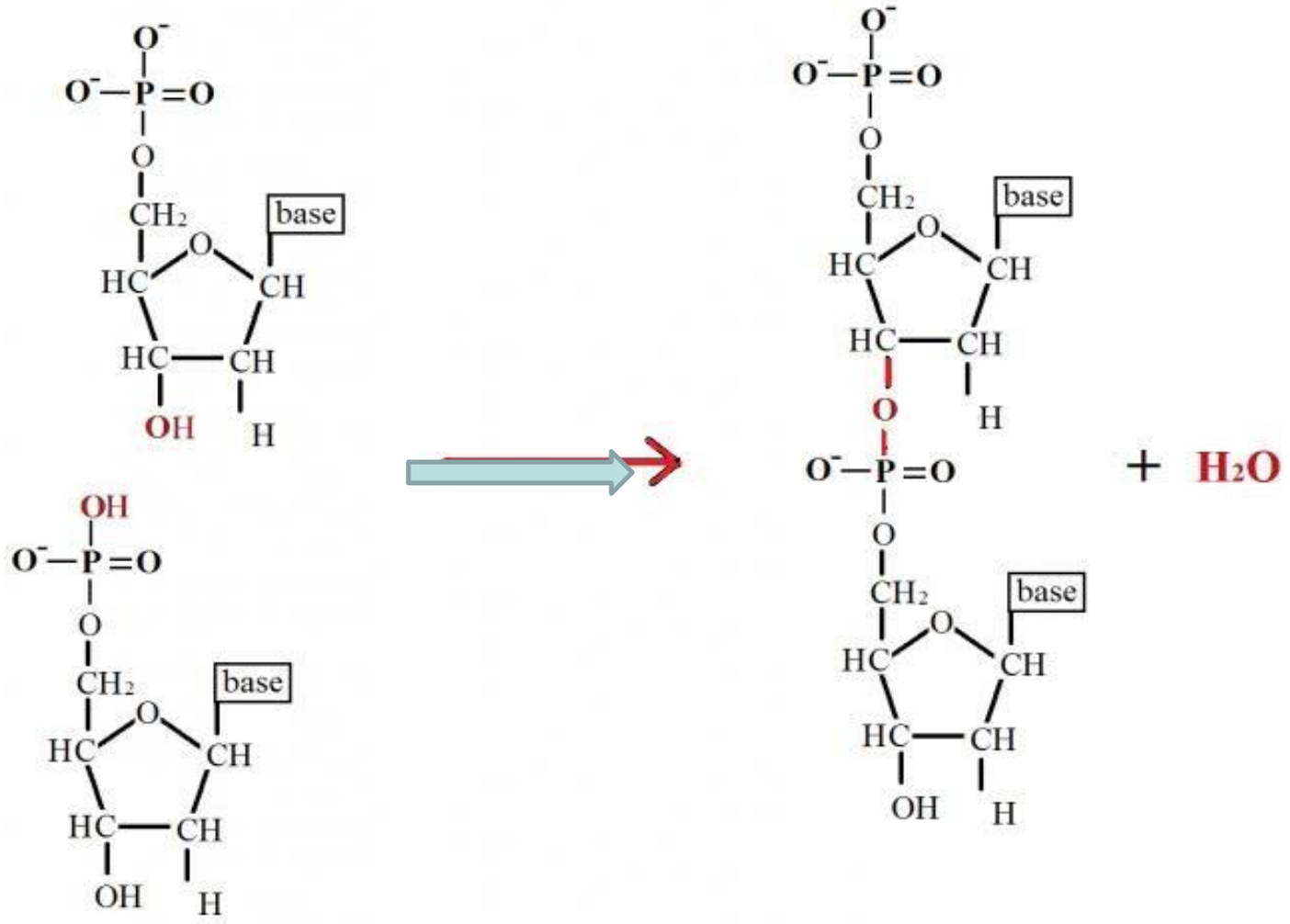
Amina Zee

How do Deoxyribonucleotides Join?

- The successive nucleotides of DNA are joined together through phospho-diester linkages.

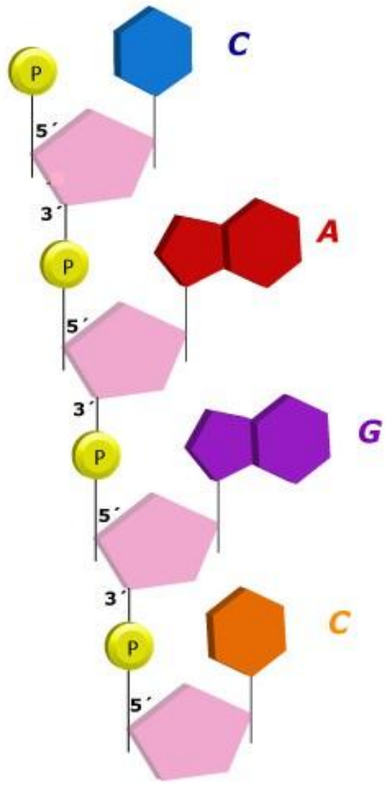
Amina Zee

How do Deoxyribonucleotides Join?



Amina Zee

How do Deoxyribonucleotides Join?



Amina Zee

Lec 15

Structure of DNA

Amina Zee

Structure of DNA

Work of Chargaff (Late 1940s)

- The discovery of the structure of DNA is one of the greatest events in the history of science.

Amina Zee

Structure of DNA

Work of Chargaff (Late 1940s)

- Erwin Chargaff and his colleagues provided a most important clue to the structure of DNA.
- The work of Chargaff led him to following conclusions, also called “Chargaff Rules”:-

Structure of DNA

Work of Chargaff (Late 1940s)

1. Base composition of DNA varies from one species to another.

Amina Zee

Structure of DNA

Work of Chargaff (Late 1940s)

2. The DNA isolated from different tissues of the same species have the same base composition.

Amina Zee

Structure of DNA

Work of Chargaff (Late 1940s)

3. The base composition of DNA in a given species does not change with an organism's age, nutritional state, or changing environment.

Structure of DNA

Work of Chargaff (Late 1940s)

4. In DNA, the number of adenosine residues is equal to the number of thymidine (A=T) and the number of guanosine residues is equal to the number of cytidine (G=C).

Amina Zee

Structure of DNA

Work of Chargaff (Late 1940s)

- It means that the sum of the purine residues equals the sum of the pyrimidine residues (AG=TC).

Introduction to Genetics

LEC 16

WHAT IS GENETICS

Amina Zee

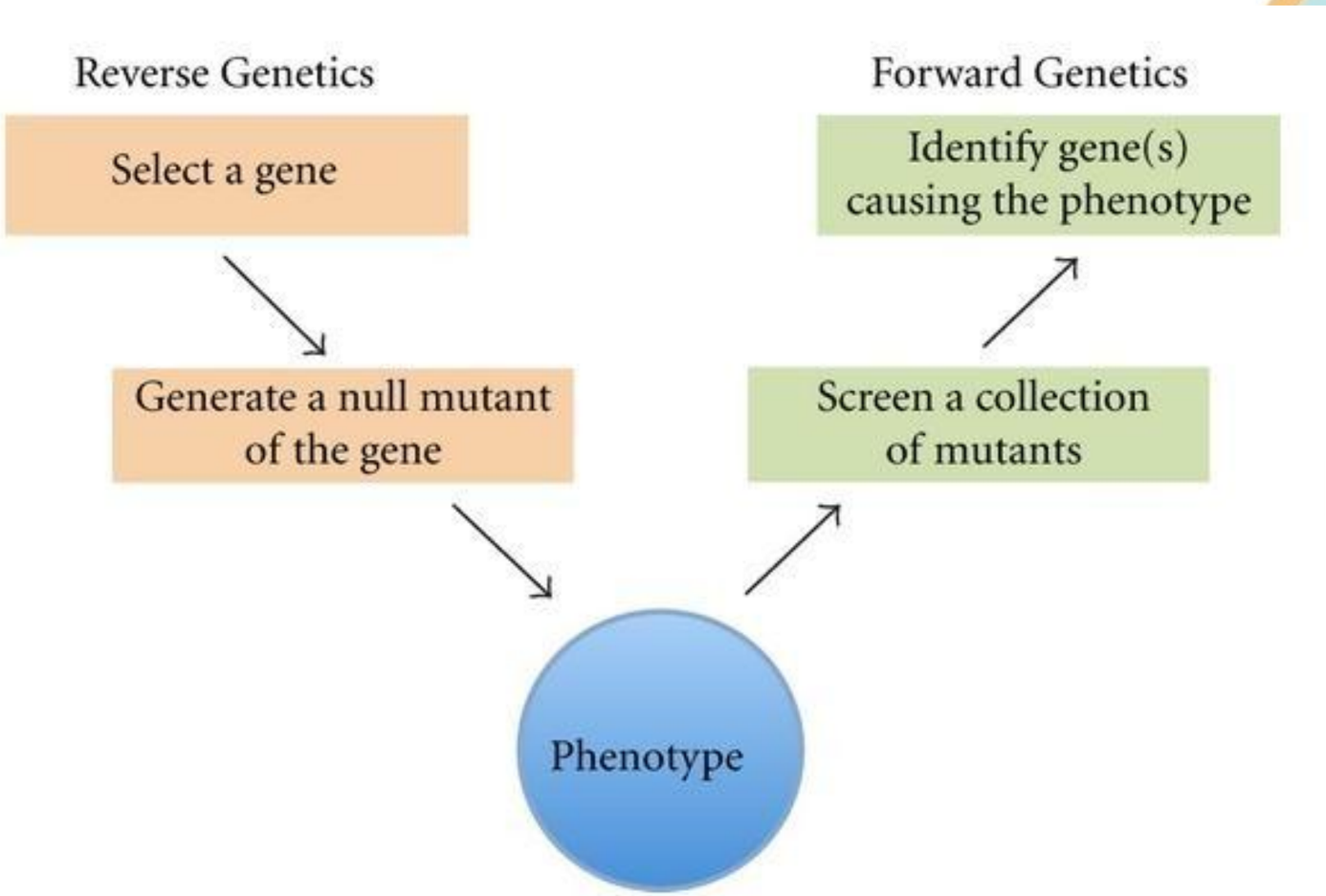
What is Genetics

DEFINITION

- Study of genes, heredity, variation.
- A field of Biology.
- The principles of heredity.
- Mendel unaware chromosomes, gene.

Amina Zee

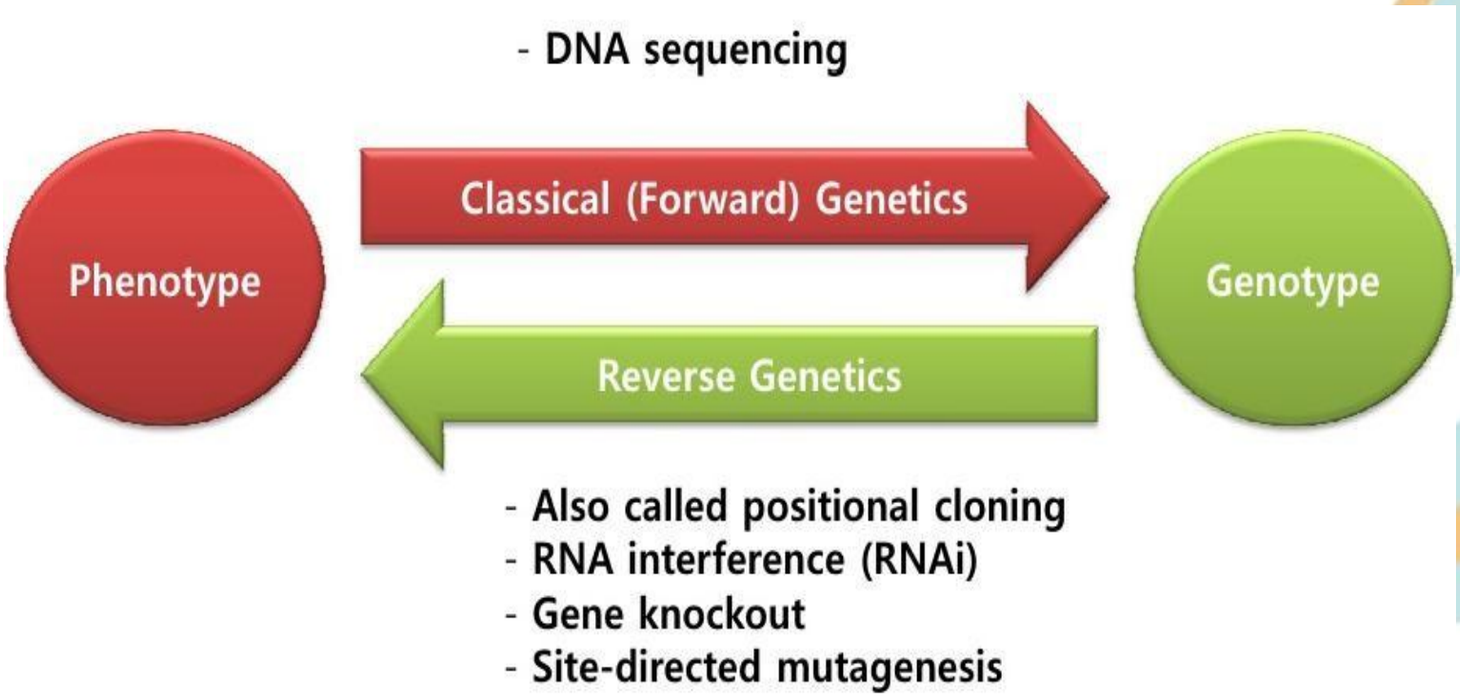
What is Genetics



Amina Zee

What is Genetics

GENETICS



Amina Zee

What is Genetics

GARDEN PEA

- Seed in a variety of shapes and colors.
- Self, cross pollinate.
- Takes up little space.
- Short generation time
- Produces many offspring.

Amina Zee

What is Genetics

MENDEL WAS FORTUNATE

- Peas - in many varieties.
- Strict control over which plants mated.
- The pea traits are distinct, contrasting.

Amina Zee

What is Genetics

MENDEL STUDIED THE TRAITS

	Flower color	Flower position	Seed color	Seed shape	Pod shape	Pod color	Stem length
P	Purple × White	Axial × Terminal	Yellow × Green	Round × Wrinkled	Inflated × Constricted	Green × Yellow	Tall × Dwarf
F_1	Purple	Axial	Yellow	Round	Inflated	Green	Tall

Amina Zee

What is Genetics

GENETICS

- Study of genes, chromosomes and heredity is called as Genetics

Amina Zee

Introduction to Genetics

LEC 17

SUB DISCIPLINES OF GENETICS

Amina Zee

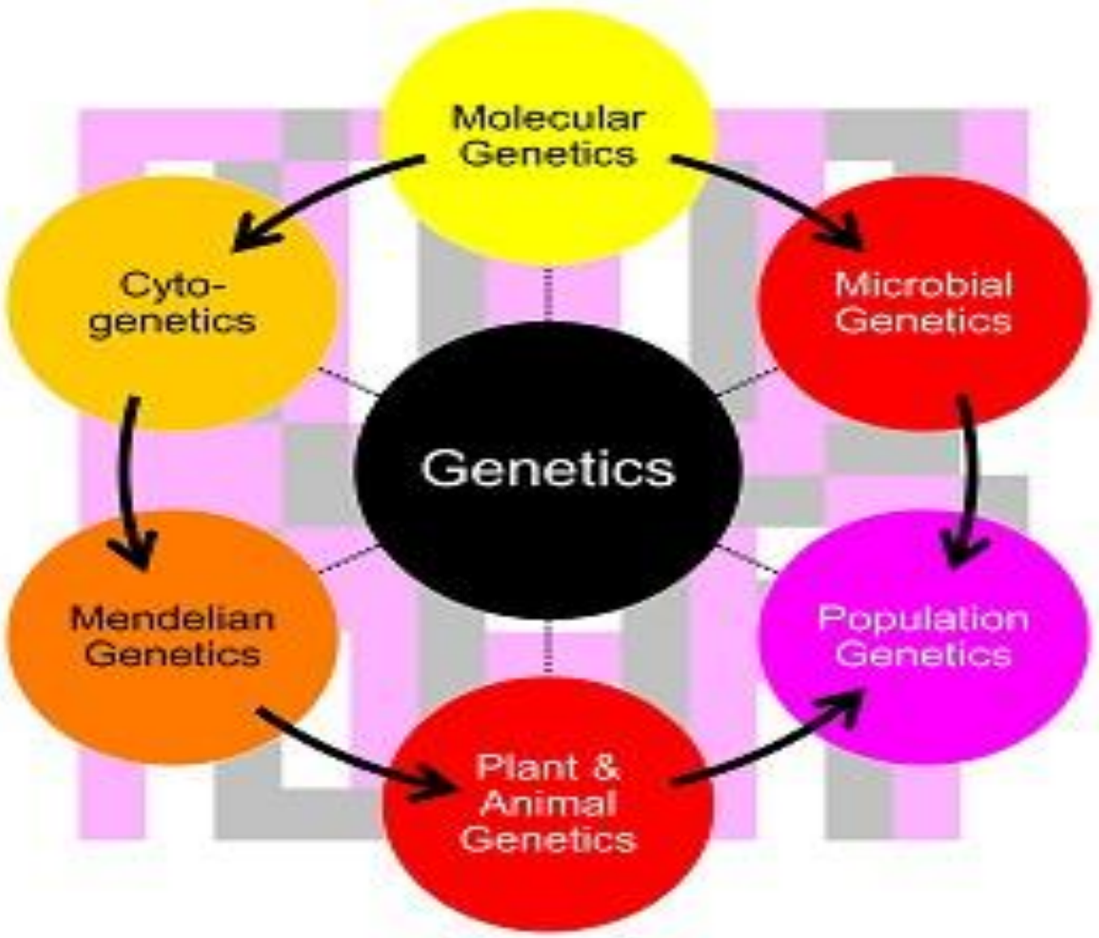
Sub Disciplines of Genetics

FOUR SUB DISCIPLINES

- Transmission (Classical) Genetics
- Population Genetics
- Quantitative Genetics
- Molecular Genetics

Amina Zee

Sub Disciplines of Genetics



Amina Zee

Sub Disciplines of Genetics

DEVELOPMENTS IN GENETICS

- Historically, transmission genetics developed first, followed by population genetics, quantitative genetics and finally molecular genetics.

Amina Zee

Sub Disciplines of Genetics

TRANSMISSION OR CLASSICAL GENETICS

- Deals with movement of genes and genetic traits from parents to offspring. Mendel s' Laws
- Deals with genetic recombinations.

Amina Zee

Sub Disciplines of Genetics

POPULATION GENETICS

- Study traits in a group of population.

- Studies heredity in groups for traits determined by one or a few genes.

Amina Zee

Sub Disciplines of Genetics

QUANTITATIVE GENETICS

- Studies group hereditary for traits determined by many genes simultaneously.

- Skin color, height, eye color

Amina Zee

Sub Disciplines of Genetics

MOLECULAR GENETICS

- Deals with the molecular structure and function of genes.

Amina Zee

Introduction to Genetics

LEC 18

GENETICS TERMINOLOGIES

Amina Zee

Genetics Terminologies

GENETICS TERMINOLOGIES

- Character: heritable feature (skin color, height etc).
- Trait: variant for a character (i.e. brown, black, white etc).
- True-breed: all offspring of same variety.

Amina Zee

Genetics Terminologies

DIFFERENT GENERATIONS OF A CROSS

- P generation (parents).
- F1 generation (1st filial generation).
- F2 generation (2nd filial generation).

Amina Zee

Genetics Terminologies

PURE CROSS AND HYBRID CROSS

- Pure Cross

True breeding X True breeding

WW X ww

- Hybrid Cross

F1 generation X F1 generation

Ww X Ww

Amina Zee

Genetics Terminologies

GENOTYPE AND PHENOTYPE

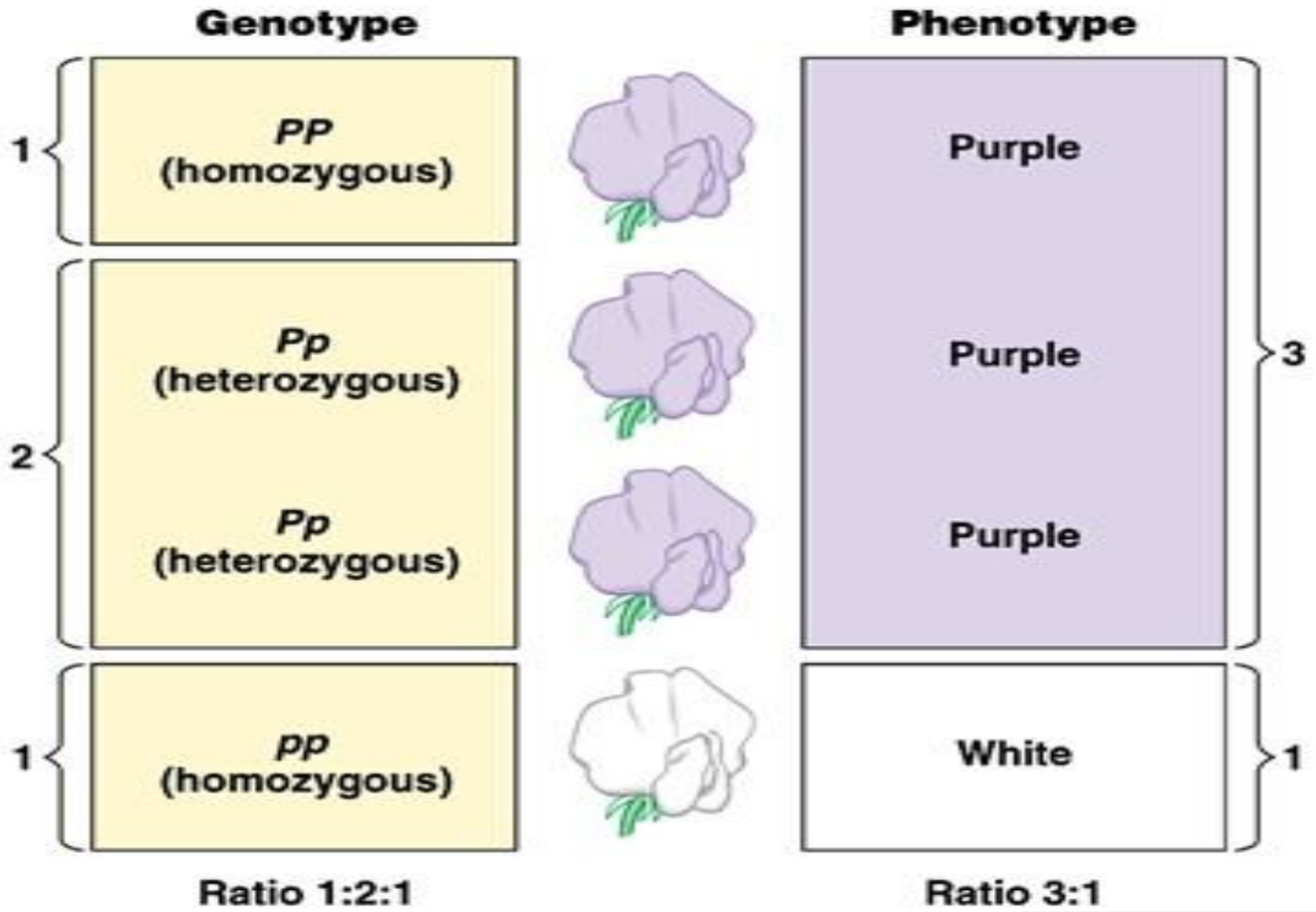
- Genetic make-up of an organism is called Genotype.

- Physical appearance of an organism is called Phenotype.

Amina Zee

Genetics Terminologies

GENOTYPE AND PHENOTYPE



Amina Zee

Genetics Terminologies

DOMINANT AND RECESSIVE

- Dominant: when one characteristic expresses itself over the other i.e. round over wrinkled.
- Recessive: The trait that does not show through in the first generation i.e. wrinkled.

Amina Zee

Lec 19

Genome Informatics

Amina Zee

Genome Informatics

Introduction

- Genome sequencing provides the sequences of all the genes of an organism
- A major application of Bioinformatics is analysis of full genomes that have been sequenced

Amina Zee

Genome Informatics

Introduction

- Challenge is to identify those genes that are predicted to have a particular biological function

Amina Zee

Genome Informatics

Genomics

Study of all of a person's genes (the **Genome**), including interactions of those genes with each other and with the person's environment **(NHGRI)**

Amina Zee

Genome Informatics

Genome Informatics

Genome informatics is the field in which computational and statistical techniques are applied to derive biological information from genome sequences

Amina Zee

Genome Informatics

Genome informatics includes methods to analyze DNA sequence information and to predict protein sequenc e and structure

(Iossifov, et al. 2014)

Amina Zee

Genome Informatics

Genome Sequences

Availability of genome sequences facilitates;

- The discovery and utilization of sequence polymorphisms
- Opportunity to explore genetic variability both between and within the organisms

Amina Zee

Genome Informatics

Genome Analysis

Following tasks;

- Sequencing
- Assembly
- Repeat identification and masking out
- Gene prediction
- Looking for EST and cDNA sequences
- Genome annotation

Amina Zee

Genome Informatics

Genome Analysis

- Expression analysis
- Metabolic pathways and regulation studies
- Functional genomics
- Gene location/gene map identification

Amina Zee

Genome Informatics

Genome Analysis

- Comparative genomics
- Identify clusters of functionally related genes
- Evolutionary modeling
- Self-comparison of proteome

Amina Zee

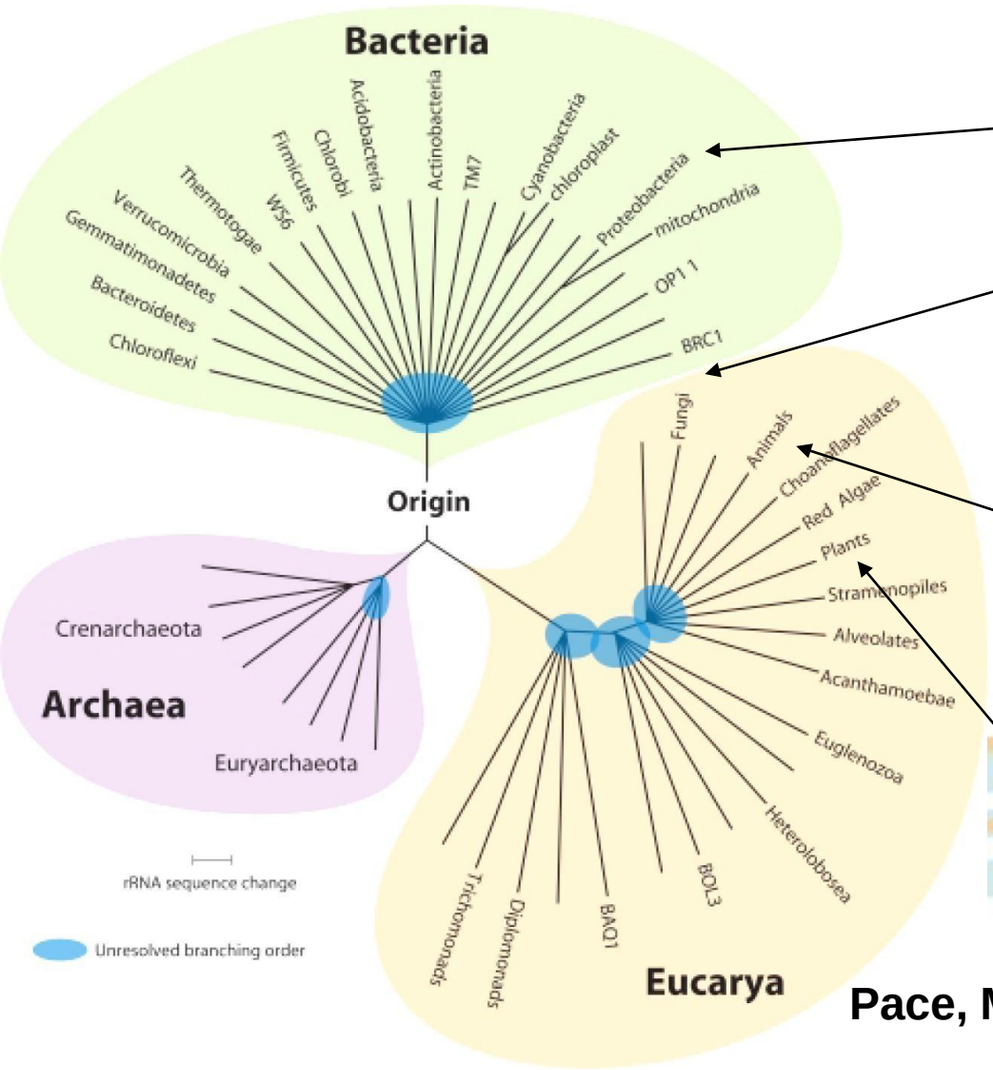
Genome Informatics

Model organisms

- E. coli – bacteria
- S. cerevisiae – yeast
- C. elegans – worm
- D.melanogaster – fly
- Danio rerio – zebrafish
- Mus musculus - mouse
- Homo sapiens – you and me
- Arabidopsis - plant

Amina Zee

Genome Informatics



E. coli is here

Yeast (*Saccharomyces*)

Worms, flies, fish, mice

Arabidosis, rice, soybean

Pace, Microbiol. Mol. Biol. Rev. 73:565 (2009)

Amina Zee

Genome Informatics

Conclusions

- Sequencing and analysis of full genomes paves the way for future discoveries
- Different model organisms can help explore our Genome and what maters most for us

Amina Zee

Lec 20

Prokaryotic Genome

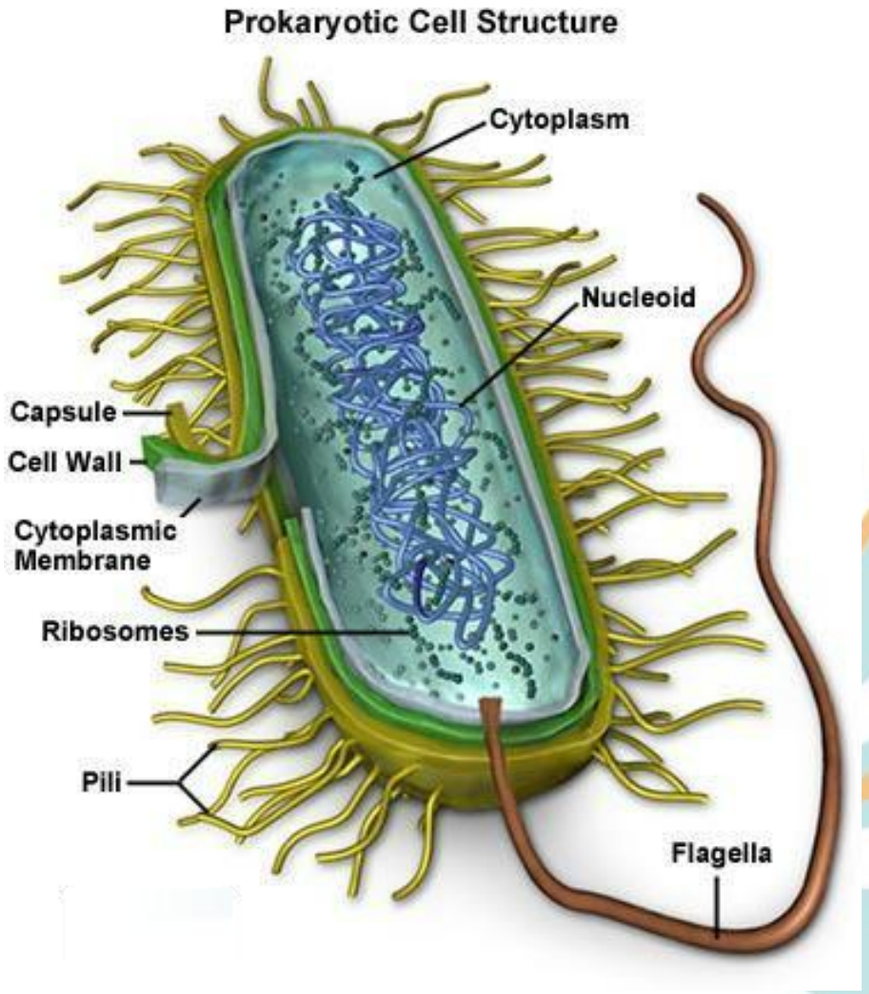
Amina Zee

Prokaryotic Genome

- Prokaryotes are the organisms whose Genetic material (DNA) is not enclosed in a nuclear membrane
- No membrane bound organelles

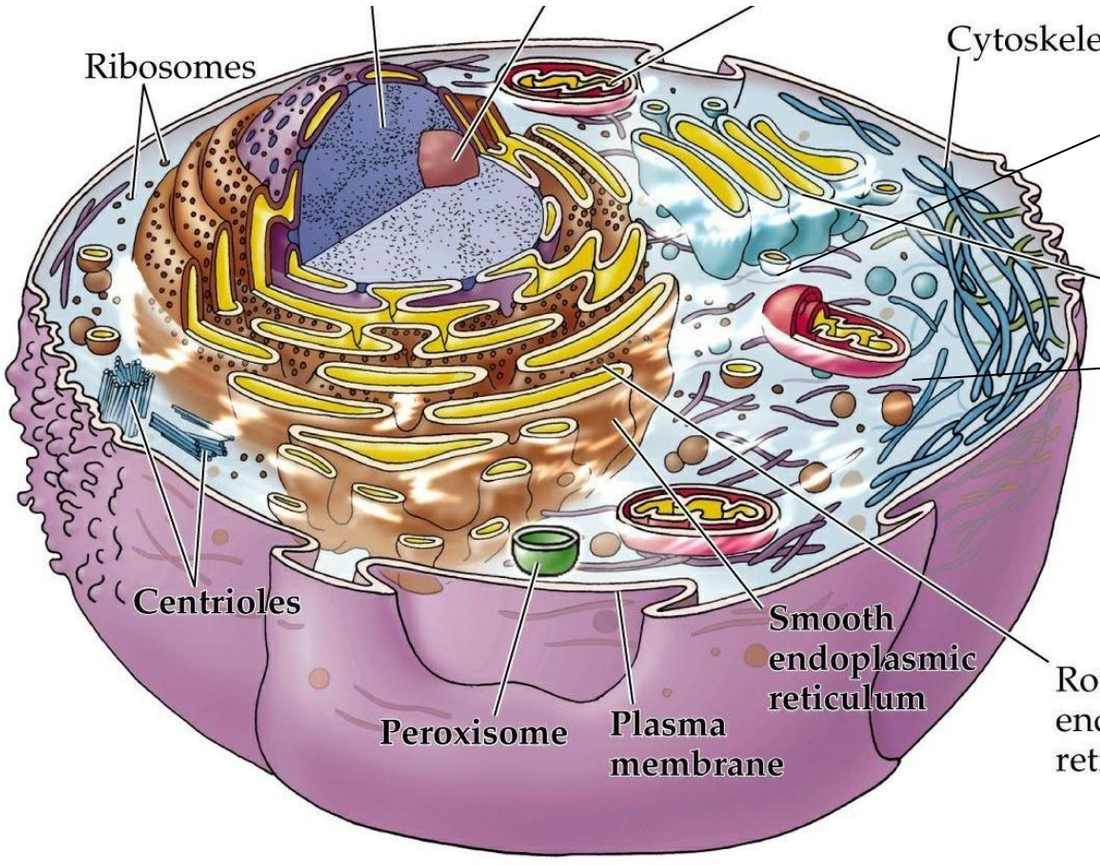
Amina Zee

Prokaryotic Genome

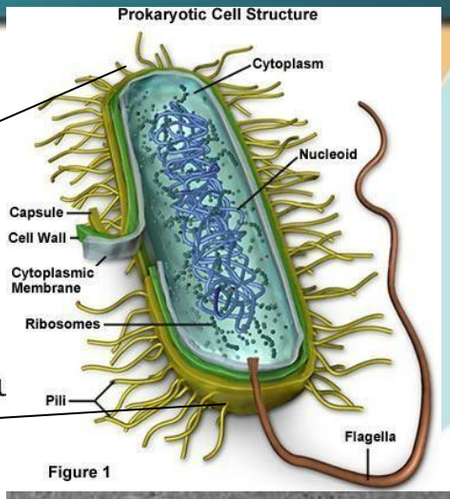


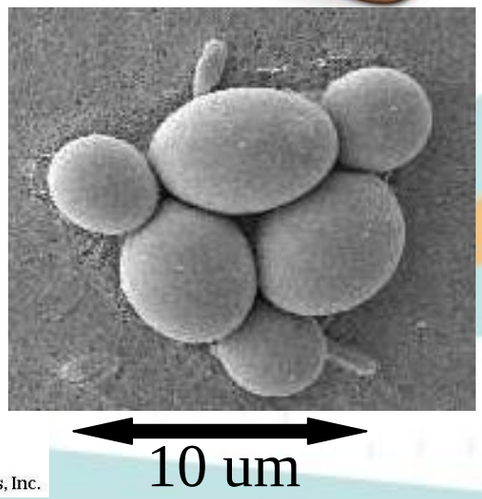
Amina Zee

Prokaryotic Genome



© 2001 Sinauer Associates, Inc.





Mitochondria evolved from a bacterial endosymbiont

Amina Zee

Prokaryotic Genome

- First prokaryotic Genome sequenced was that of *Hemophilus influenzae*
- Paved the way for sequencing of many other organisms

Amina Zee

Prokaryotic Genome

Selection criteria

Following are the criteria for organisms selected for sequencing;

- They had been subjected to a detailed biological analysis and thus were model organisms

Amina Zee

Prokaryotic Genome

Selection criteria

- important human pathogens
- They were of phylogenetic interest
- Sequences were annotated as they were sequenced

Amina Zee

Prokaryotic Genome

Table 10.2. *Features of representative prokaryotic genomes*

Organism (reference)	Phylogenetic group	Genome size (Mbp) (no. protein-encoding genes)	Novel functions
Escherichia coli (Blattner et al. 1997)	Bacteria	4.6 (4288)	model organism
Methanococcus jannaschii (Bult et al. 1996)	Archaea	1.66 (1682)[a]	grows at high temperature and pressure and produces methane
Hemophilus influenzae (Fleischmann et al. 1995)	Bacteria	1.83 (1743)	human pathogen
Mycoplasma pneumoniae (Himmelreich et al. 1996)	Bacteria	0.82 (676)	human pathogen that grows inside cells; metabolically weak
Bacillus subtilis (Kunst et al. 1997)	Bacteria	4.2 (4098)	model organism
Aquifex aeolicus (Deckert et al. 1998)	Bacteria	1.55 (1512)[b]	ancient species, grows at high temperature and can grow in a hydrogen, oxygen, carbon dioxide atmosphere in the presence of only mineral salts
Synechocystis sp. (Kaneko et al. 1996a,b)	Bacteria	3.57 (3168)	ancient organism that produces oxygen by light-harvesting; may have oxygenated atmosphere

Mount

Amina Zee

Prokaryotic Genome

Conclusions

- Prokaryotes are simple Genomes
- Easy models to study Biochemistry and Molecular biology of life processes
- Sequencing is done on economically important organisms

Amina Zee

Lec 21
Eukaryotic Genome

Amina Zee

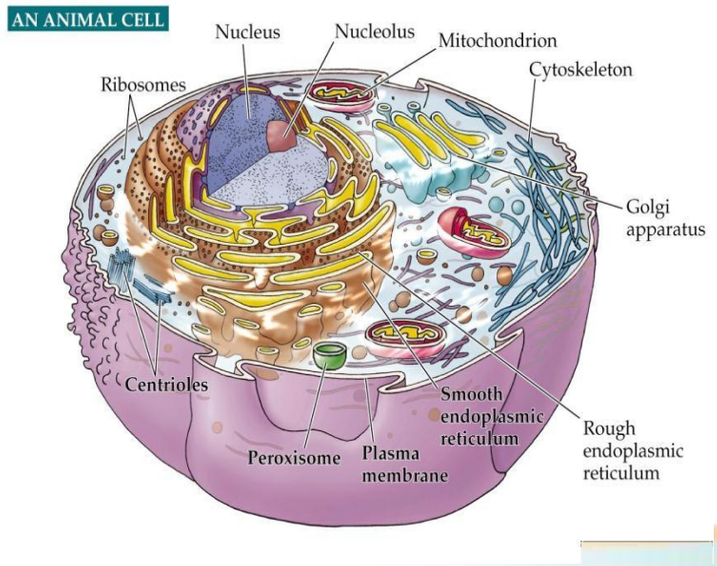
Introduction:

- We have seen that prokaryotes are simple genomes in comparison to eukaryotes which are relatively complicated. So distinct properties of eukaryotes are mentioned below:
- Eukaryotes have larger genomes
- Have tandem repeats

Amina Zee

Introduction:

- Have introns in their protein-coding genes (introns are between the exons which are the protein coding regions within the genes).
- Heterochromatin and euchromatin region (eukaryotes have complicated genome, so the chromosome is grouped as heterochromatin; densely packed region and euchromatin; lightly packed region).



- Here, in this diagram we can see a typical eukaryotic cell which is pretty stuffed as compare to prokaryotic one. We have nucleus in the middle, channels coming out of the nucleus known as *endoplasmic reticulum* (helps in transportation), ribosomes (for protein synthesis), mitochondria (energy synthesis), we can also see the cytoskeleton that makes the structure of this cell intact and Golgi apparatus (are concerned with the secretion).
  So complicated membrane bounded organelles are present in the eukaryotes

Amina Zee

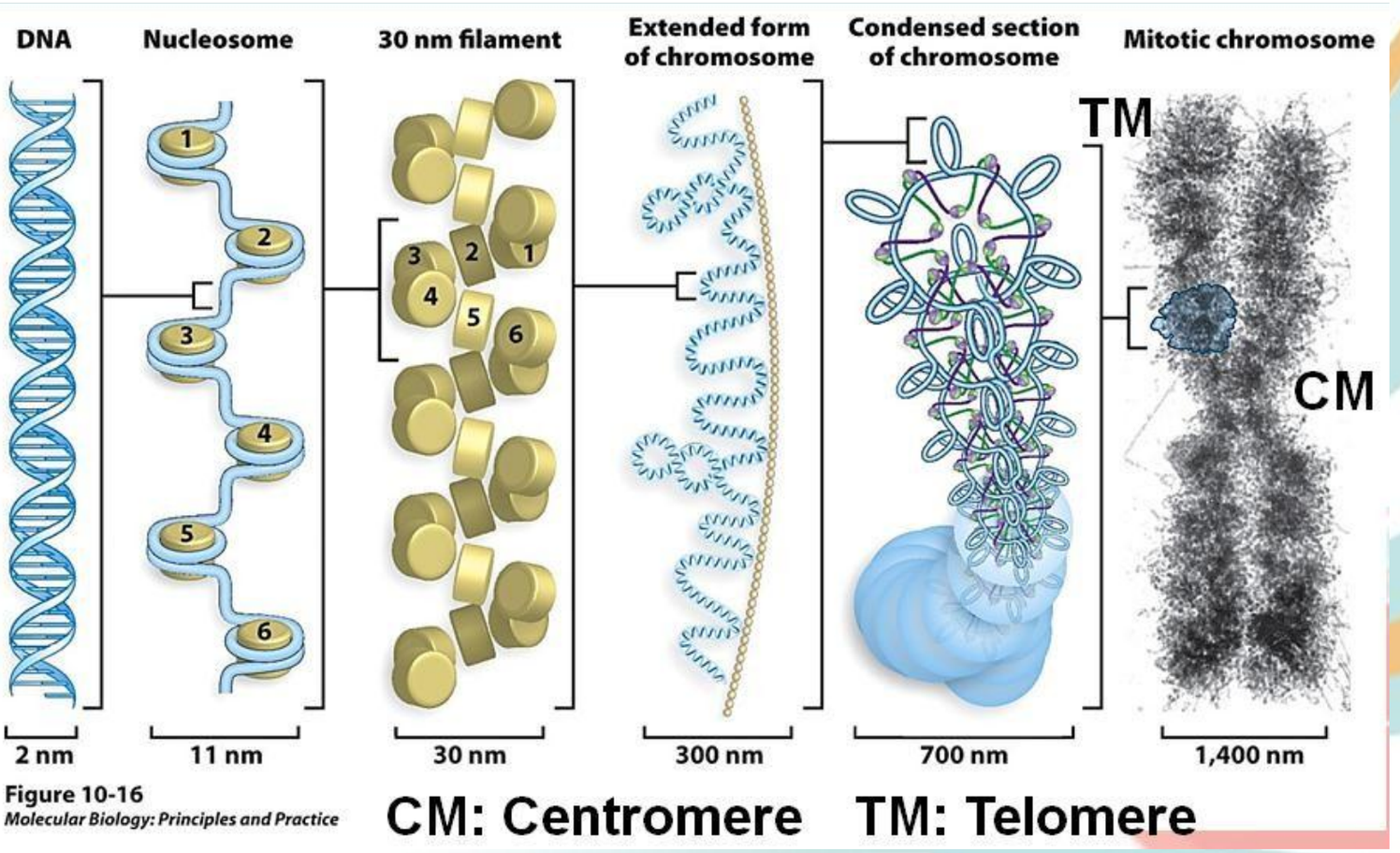


Figure 10-16
Molecular Biology: Principles and Practice

Here, in this diagram we see the connection between the DNA and the chromosomes

Amina Zee

- On the left-hand side, we see a DNA strand that is a 2nm wide strand. So, the DNA wraps over the protein complex molecules (histones - labelled as 1, 2, 3...), and this structure is known as nucleosome.
- Then these histones, turn around and makes a wider structure and it makes a 3nm filament (in third section). Then these nucleosome structures supercoil on their selves to make those further bigger fibres and until they reach the chromosome the width is 1,400 nm.

Amina Zee

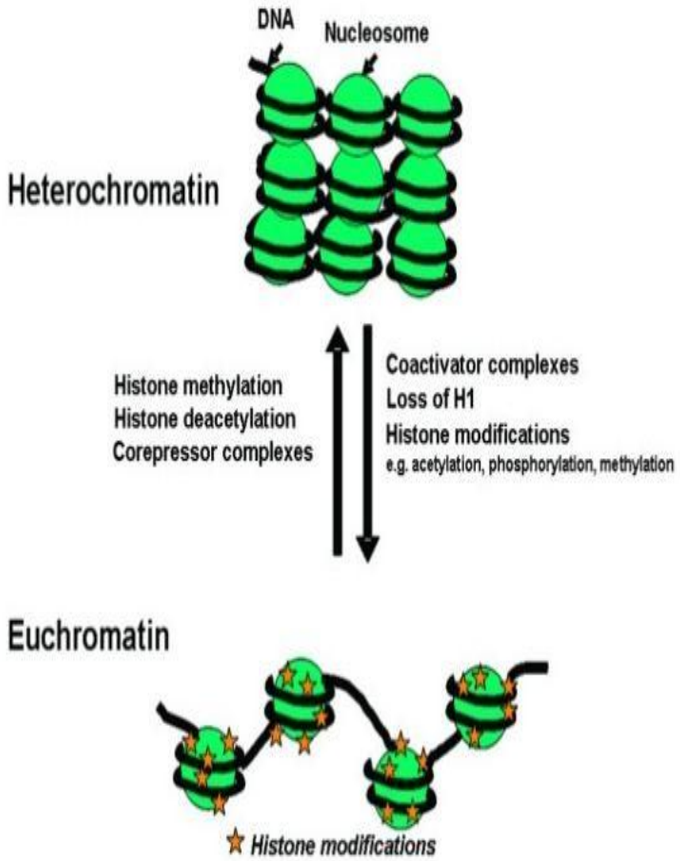
- So, if we look into the chromosome, we can recognize that there are different arms in it, which are known as sister *chromatids* (remember this is just one chromosome but we have two chromatids),
- somewhere in the middle we see a constricted part known as *centromere,* whereas the terminals are known as *telomeres,* (remember these nomenclature while we are discussing the heterochromatin and euchromatin parts).

Staining with dyes:

- So chromosomes if stained with the dyes, they give different coloring patterns, we can come up with the following:
- Dense heterochromatin (dense regions obviously take more color)
- Light Euchromatin (light regions take less color)

Amina Zee

- If we look into the gene expression, the heterochromatic regions are packed so the enzymatic machinery cannot reach there;
- hence these regions are poorly transcribed (expressed). Whereas, the euchromatic regions are highly expressed because they are loosely packed and the enzymatic machinery can easily reach to them.



Here, is the diagram in which we can see the relationship between heterochromatin and euchromatin.

You can look into these nucleosomes (combination of DNA and histones), which are quite jammed packed with one another, so definetly the enzymes cannot access the DNA which is embedded in between.

Amina Zee

- There are different modifications on the DNA or histones that bring about those structures (we can see on the top),
- so for example there are histone methylations (in which methyl groups are added to those histones) in that case the system moves towards downside (as shown in the diagram) so it becomes *euchromatin* and similarly,
- we see that there are some other methylations on some other aminoacids that can move back into the opposite direction also.

Amina Zee

- So, histone methylation, histone deacetylation and there are some other complex proteins which gets attached and give us this *heterochromatin region* and in the reverse process, we get the *euchromatin region*.
- In the Euchromatin region, the histones are quite spaced and DNA can be acessible. So, this is the reason why the euchromatin region is expressed more as compared to the heterochromatin region.

Conclusions:

- We conclude that:
- Eukaryotes are distinguished by the presence of prominent nuclei
- Eukaryotes have larger genomes, tandem repeats and introns in their protein-coding genes (i.e. they are complicated).

Amina Zee

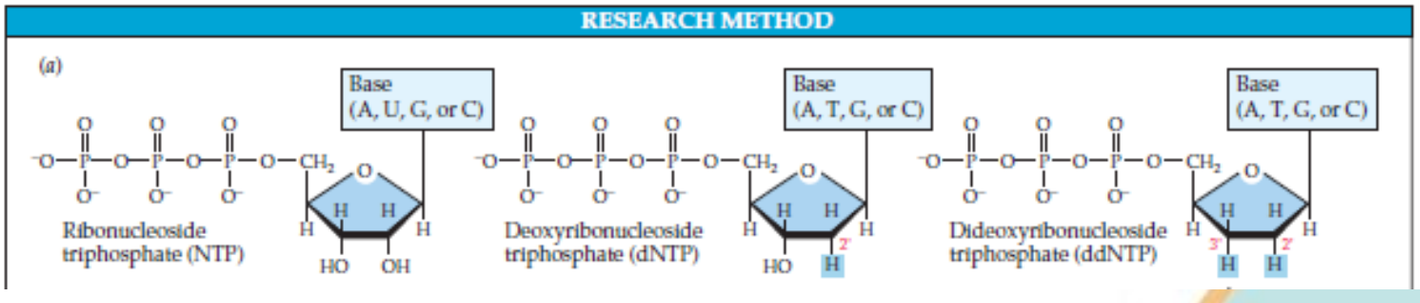
Cell Biology

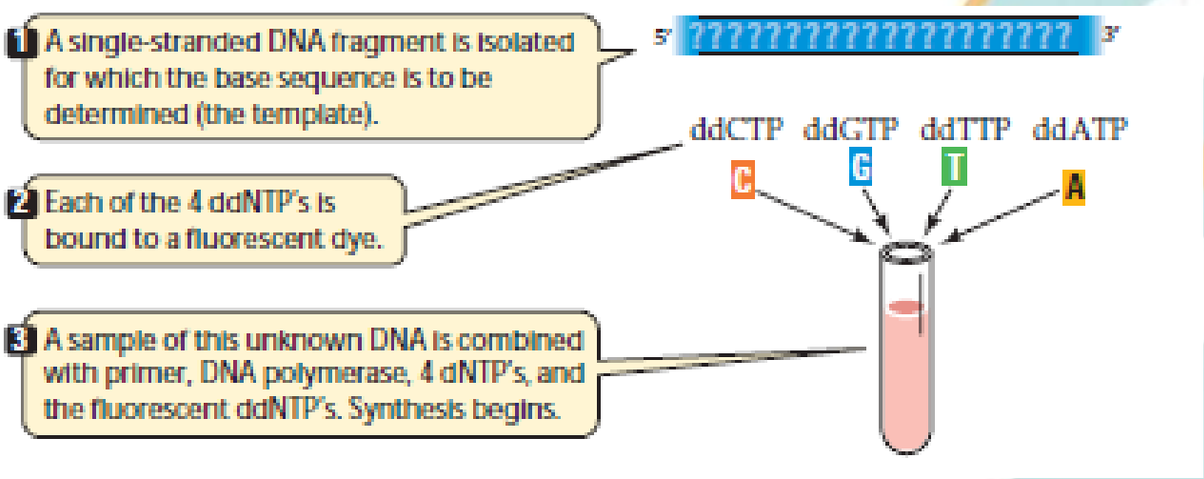
Lec 22

DNA Sequencing

Amina Zee

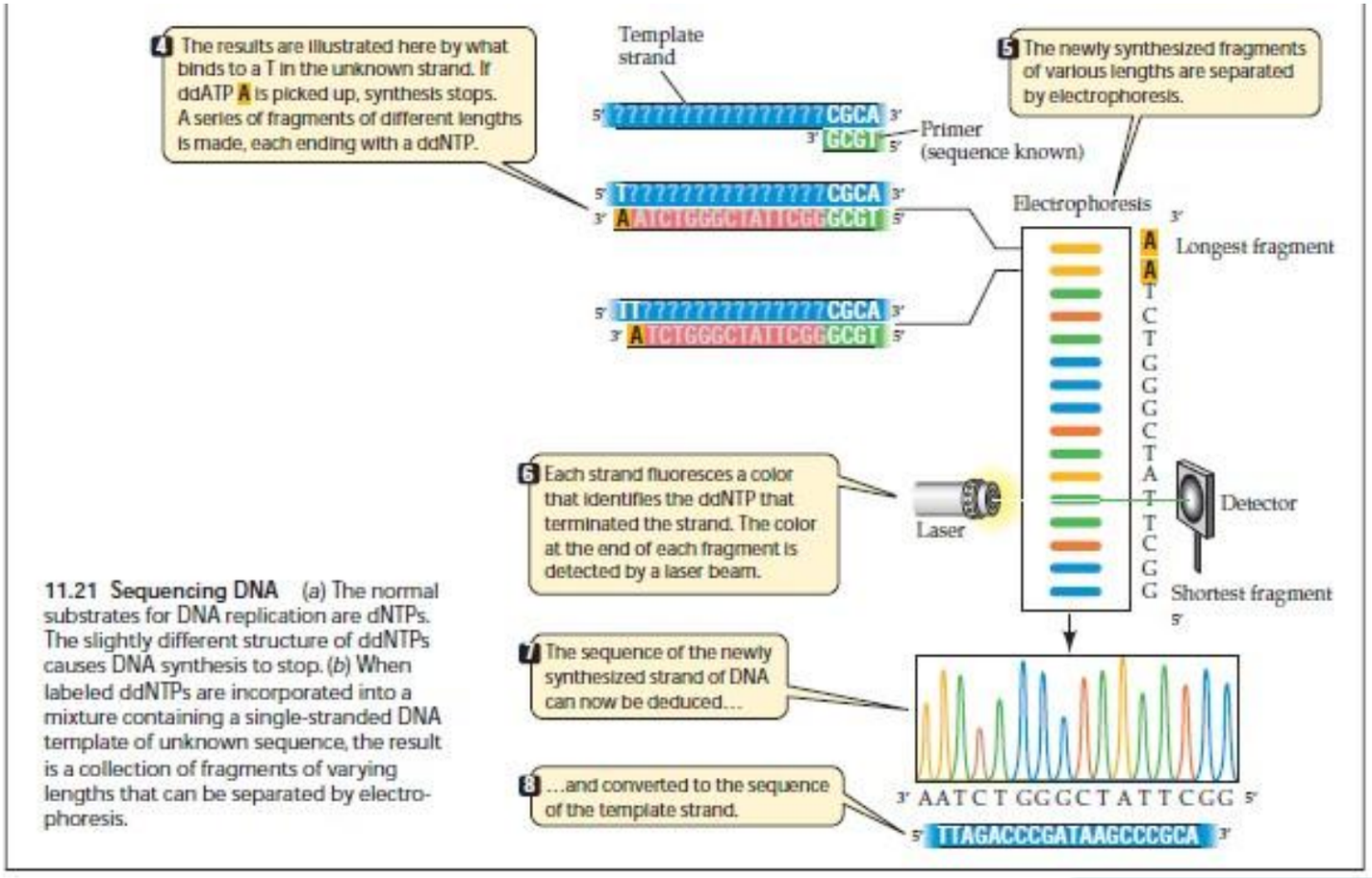
DNA Sequencing





Amina Zee

DNA Sequencing



11.21 Sequencing DNA (*a*) The normal substrates for DNA replication are dNTPs. The slightly different structure of ddNTPs causes DNA synthesis to stop. (*b*) When labeled ddNTPs are incorporated into a mixture containing a single-stranded DNA template of unknown sequence, the result is a collection of fragments of varying lengths that can be separated by electrophoresis.

END

Amina Zee

Cell Biology

Lec 23

Genetic Information is Converted into Protein

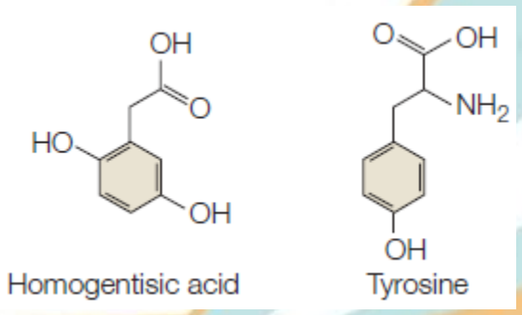
Amina Zee

Archibald Garrod 1908

Alkaptonuria ("black urine")

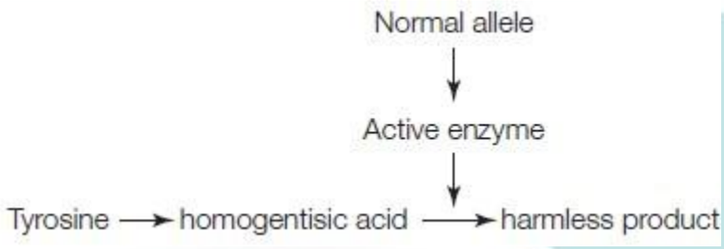
Most common in children whose parents were first cousins 12.5%

The molecular basis of phenotypes was actually discovered before it was known that DNA was the genetic material.



Homogentisic Acid Oxidase

He linked the biochemical phenotype of the disease to an abnormal gene and a missing enzyme.



Amina Zee

Beadle Tatum 1940s

Neurospora

An altered gene resulted in an altered phenotype, associated with an altered enzyme

Neurospora haploid (n) recessive alleles

X- rays, which act as a mutagen prototrophs (original) converted to auxotrophs (increase)

Some mutant strains could no longer grow on the minimal medium. One group could grow if supplemented with the arginine (*arg* mutants)

Amina Zee

Beadle Tatum 1940s

Neurospora

arg **mutants were grown in presence of various compounds** ***suspected intermediates*** **in the synthetic metabolic pathway for arginine,**

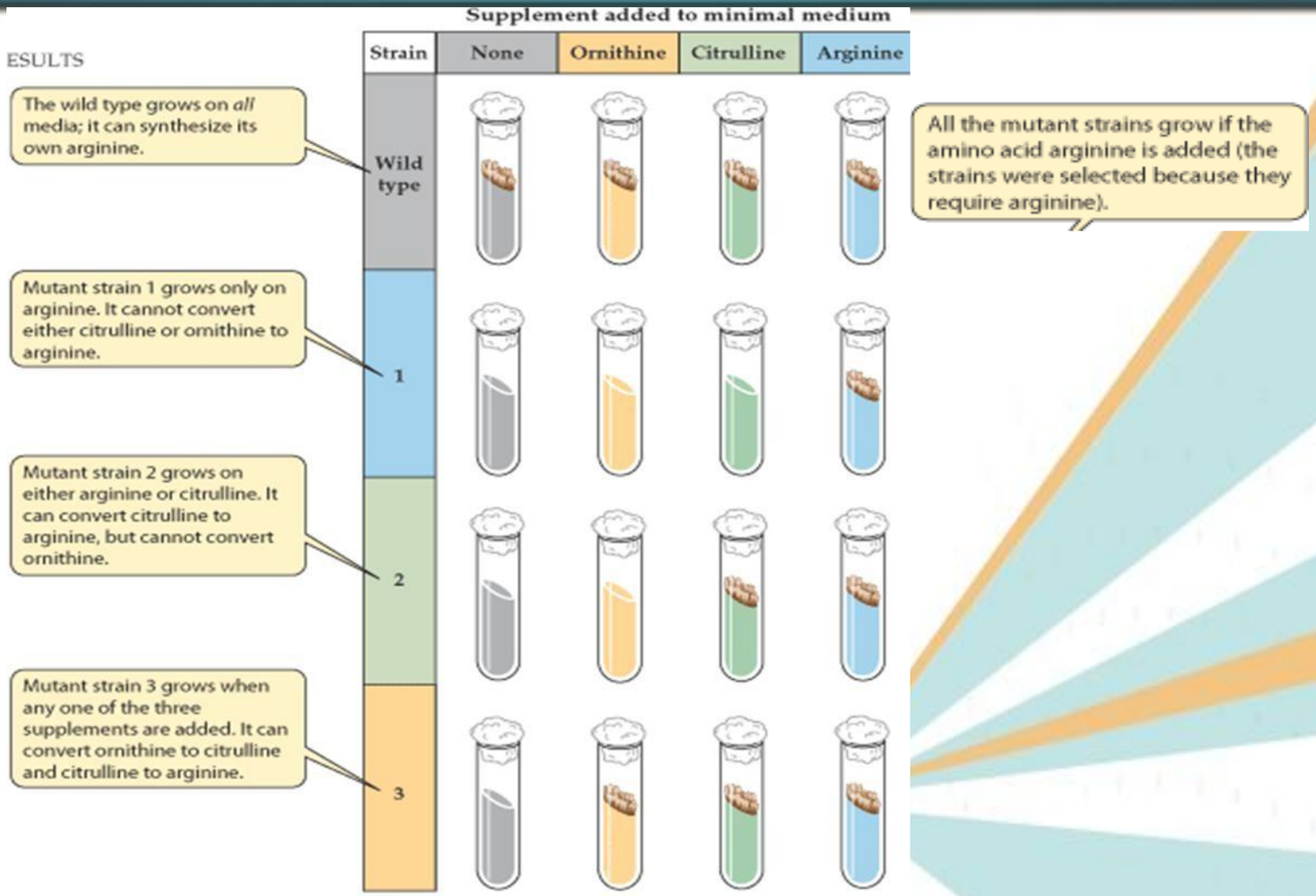
B & T classified each mutation as affecting one enzyme or another.

wild-type and mutant cells examined for enzyme activities.

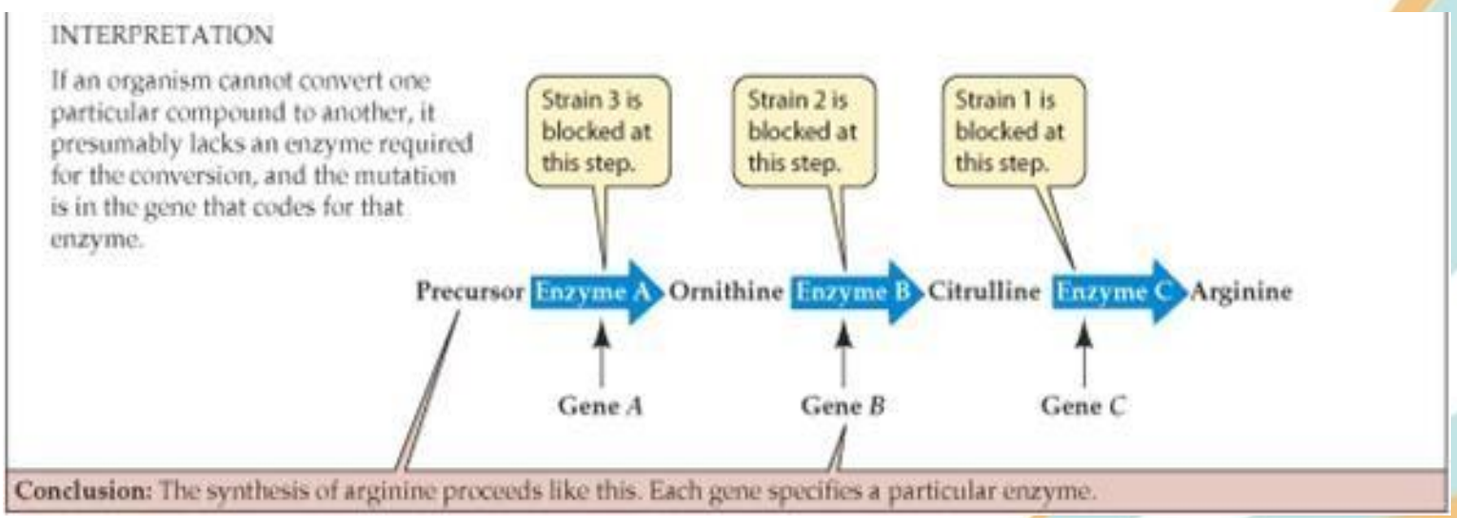
results confirmed: Each mutant strain was indeed missing a single active enzyme in the pathway.

Amina Zee

Beadle Tatum 1940s

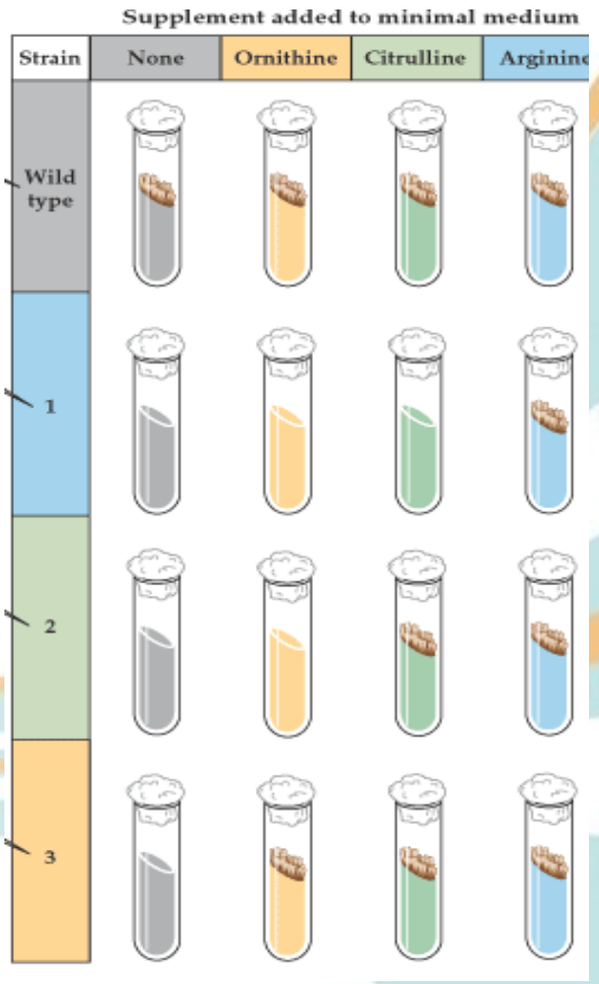


Beadle Tatum 1940s



Amina Zee

Beadle Tatum 1940s



END

Amina Zee

Cell Biology

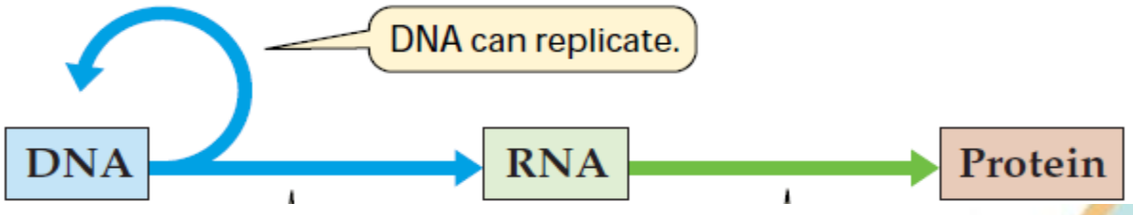
Lec 24

The Dogma

Amina Zee

The Dogma

Info flows from DNA to RNA to proteins



How does genetic information get from the nucleus to the cytoplasm?

What is the relationship between a specific nucleotide sequence in DNA and a specific amino acid sequence in a protein?)

Amina Zee

Crick Hypotheses

THE MESSENGER HYPOTHESIS AND TRANSCRIPTION

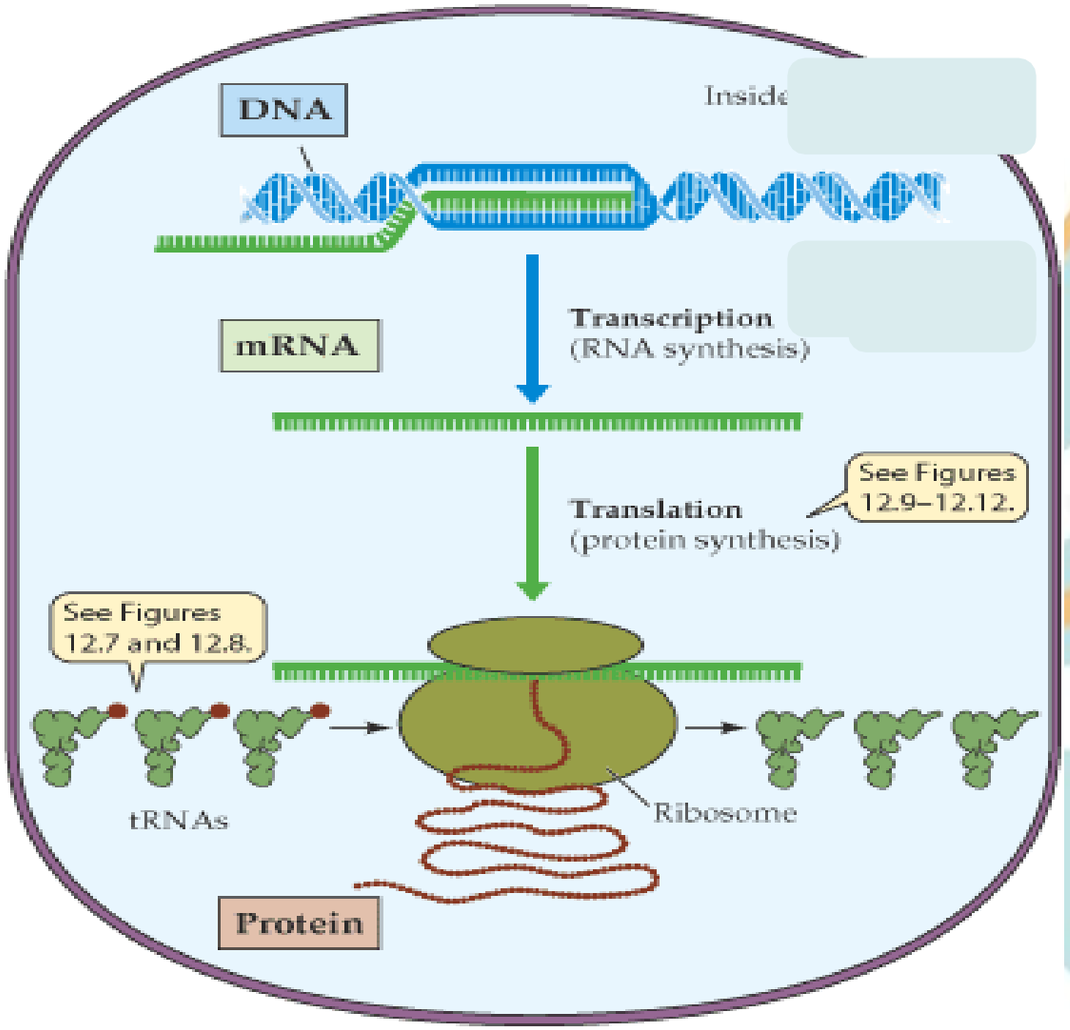
RNA molecule forms complementary copy of one DNA strand, moves to cytoplasm serves as template for protein synthesis.

THE ADAPTER HYPOTHESIS AND TRANSLATION

Adapter molecule binds a specific amino acid with one region and recognize a sequence of nucleotides with another region to translate the language of DNA into the language of proteins

Amina Zee

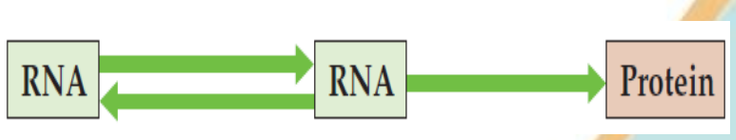
Overview



Amina Zee

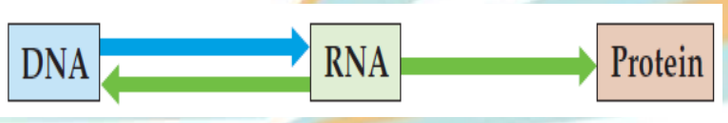
The Viral Exceptions

Influenza virus, & poliovirus



Transcribing from RNA to RNA

HIV



make a DNA copy of their genome

END

Amina Zee

Introduction

Lec 25

Background of Bioinformatics

Amina Zee

Background of Bioinformatics

Modern Biology

- Humans have over 25,000 genes
- The genes may produce > 250,000 different proteins
- Each protein may further take up multiple modifications

Modern Biology

- But then, there are numerous organelles & intermediary molecules as well
- The scale and speed of biomolecular interactions, reactions and transport is simply bewildering!

Background of Bioinformatics

Experiments in Biology

Brisk enhancements in instrumentation:

- Next Generation Sequencers
- High Resolution Mass Spectrometers
- Nuclear Magnetic Resonance Spectroscopy

Amina Zee

Background of Bioinformatics

Digitalization of Biology

- Modern biological experiments produce data which is stored on computers
- The data includes text, numbers, symbols and images

Amina Zee

Background of Bioinformatics

Speed of Data Growth

- Data is being accumulated at an exponentially increasing rate
- E.g. Genome sequences in genome databases are doubling every few years!

Background of Bioinformatics

Conclusion

- Humans are limited in recalling information from their memory
- So, we need the help of computers to store and recall this data!
- Computers can also quickly process this information and help in its manipulation.

END

Amina Zee

Lec 26
Introduction to Bioinformatics

Amina Zee

Motivation

1. An interdisciplinary field
2. New but rapidly developing field
3. Low requirement on infrastructure and research equipment
4. Vast opportunities for scientific discovery

Scope of Bioinformatics

Use of computational algorithms and techniques for:

1. Storage,
2. Organization,
3. Analysis, and
4. Representation of biological information

Amina Zee

Introduction to Bioinformatics

Activities

1. Developing algorithms
2. Writing software
3. Data processing pipelines
4. Statistical evaluation of data
5. Data visualization

Amina Zee

Moving ahead...

- Learn the available bioinformatics concepts, algorithms and tools
- Advance these concepts by developing novel ideas
- Apply these ideas to forward the understanding of biology and life

Introduction to Bioinformatics

Conclusion

- The benefits of studying bioinformatics in Pakistan
- New degree programs in several universities
- Opportunity for cutting edge discoveries in a resource-limited setting

END

Amina Zee

Introduction

Lec 27
Need for Bioinformatics - I

Amina Zee

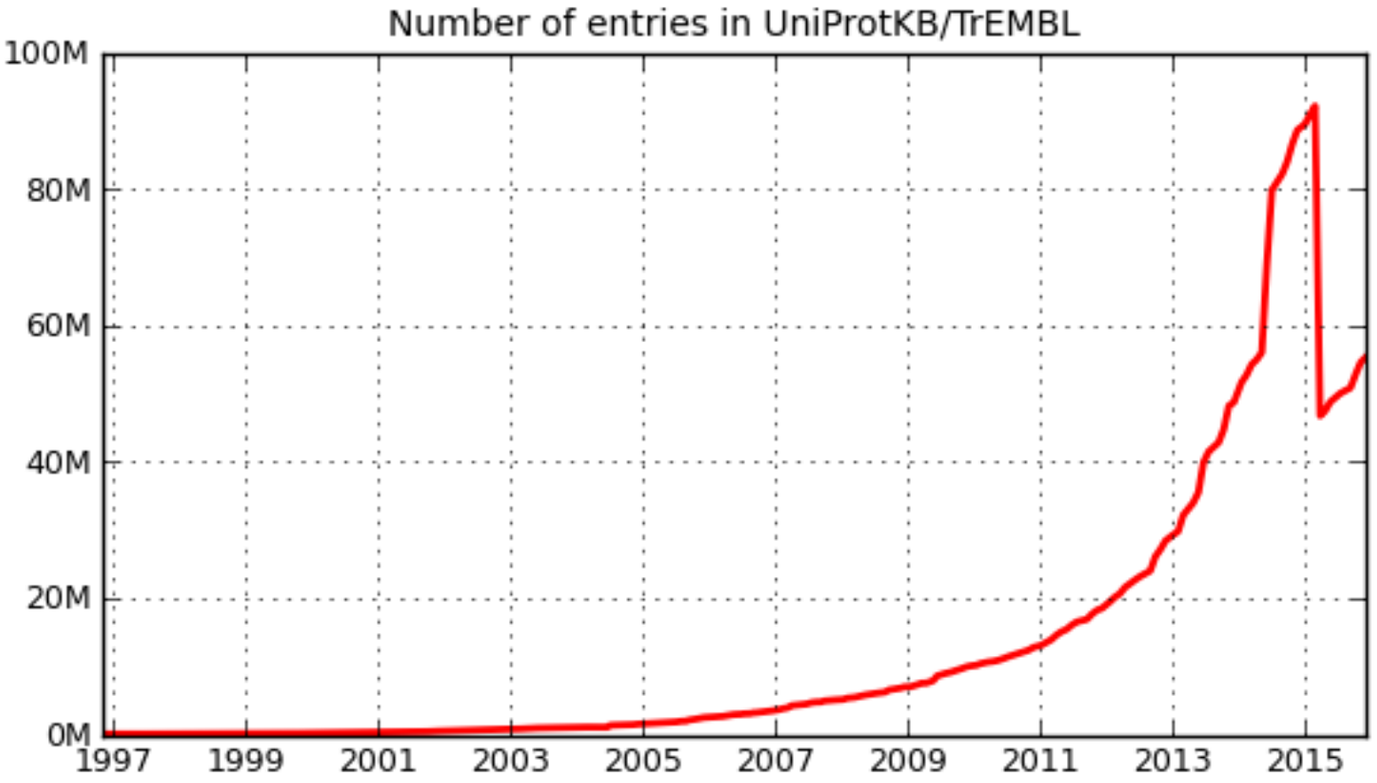
Need for Bioinformatics – I

Motivation

1. An interdisciplinary field
2. New but rapidly developing field
3. Low requirement on infrastructure and research equipment
4. Vast opportunities for scientific discovery

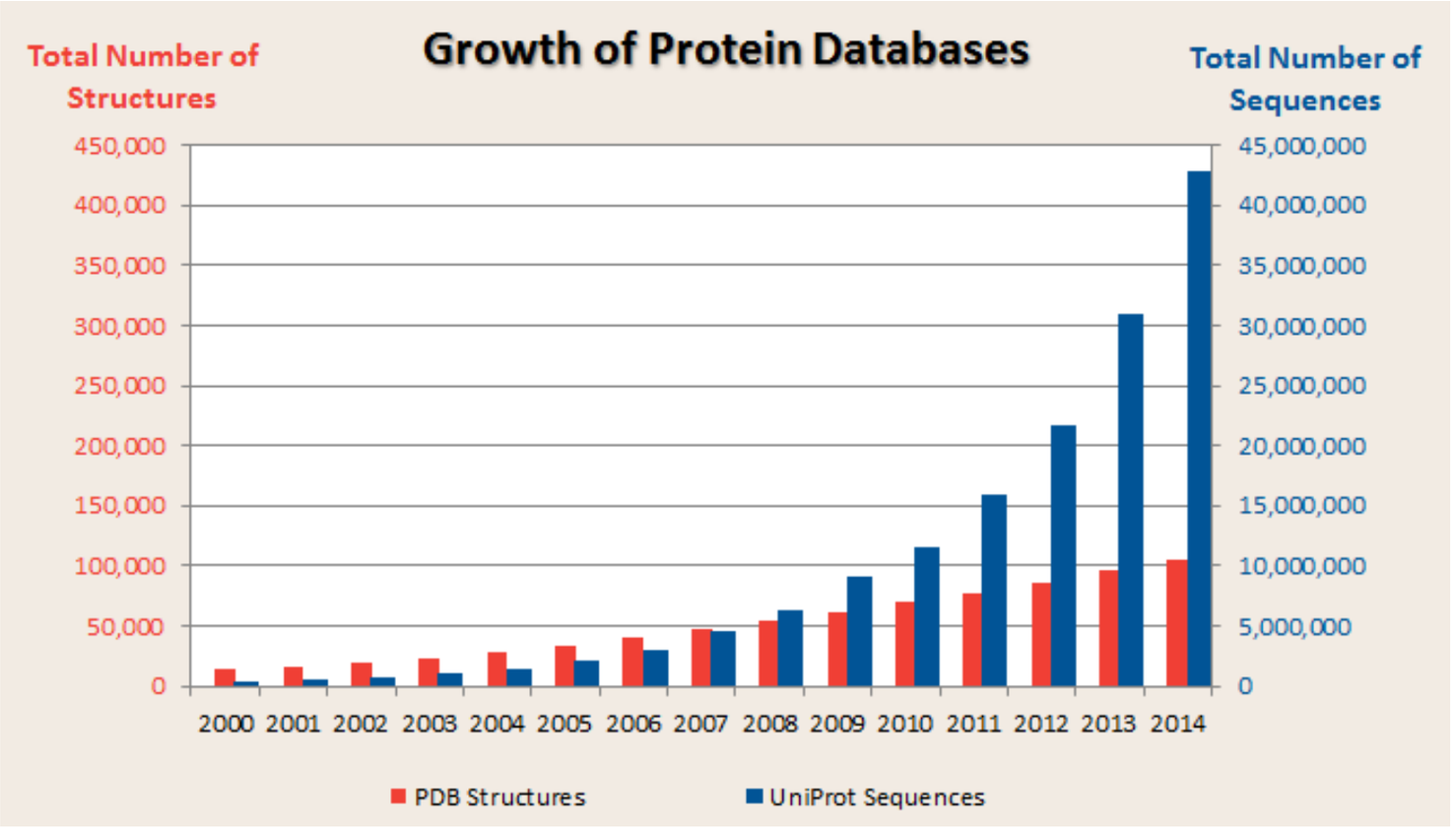
Amina Zee

Need for Bioinformatics – I



www.ebi.ac.uk

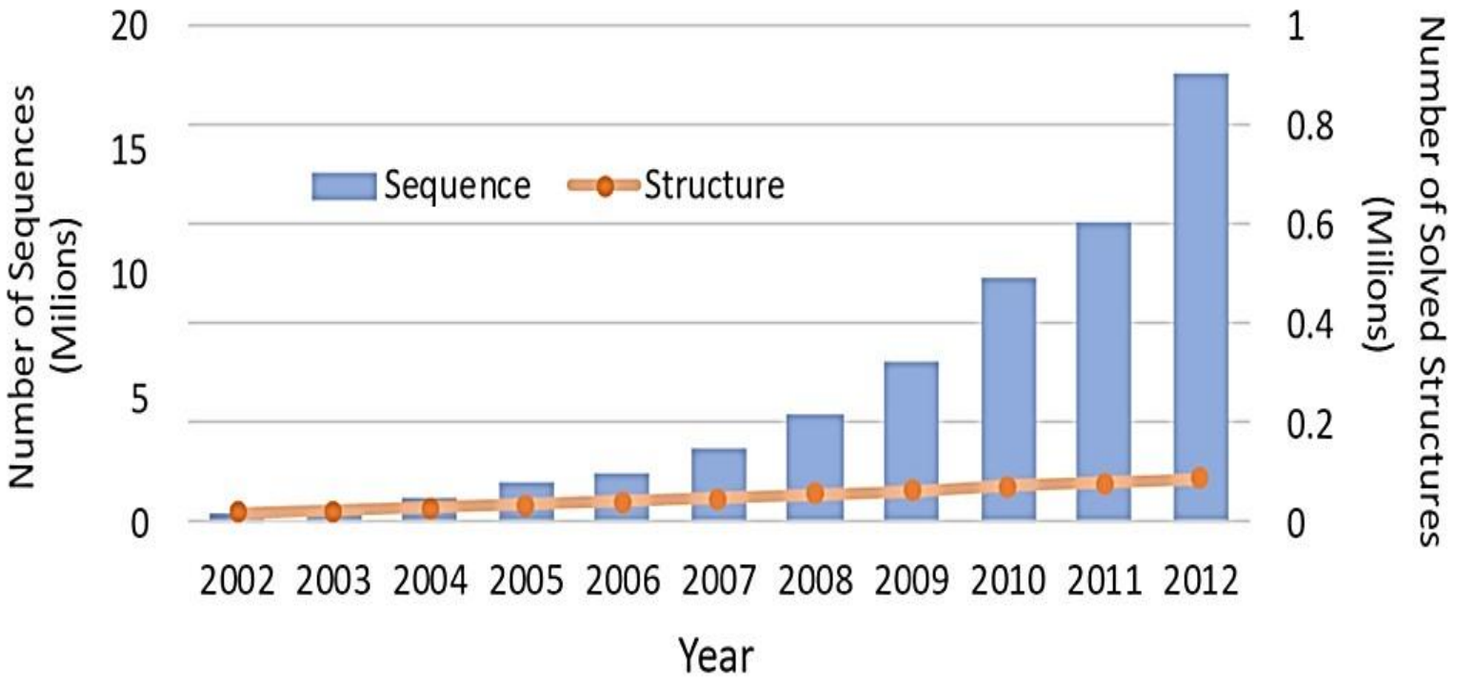
Need for Bioinformatics – I



www.dnastar.com

Amina Zee

Need for Bioinformatics – I



www.mdpi.com

Amina Zee

Need for Bioinformatics – I

Conclusion

- Since genome and proteome information is publically available, you can process this information after downloading it
- If you carefully search this online data, with a little bit of biology background, it is a gold mine!

END

Amina Zee

Introduction

Lec 28
Need for Bioinformatics - II

Amina Zee

Need for Bioinformatics – II

Quote

Biology easily has 500 years of exciting problems to work on.

Donald Knuth, Professor, Stanford University

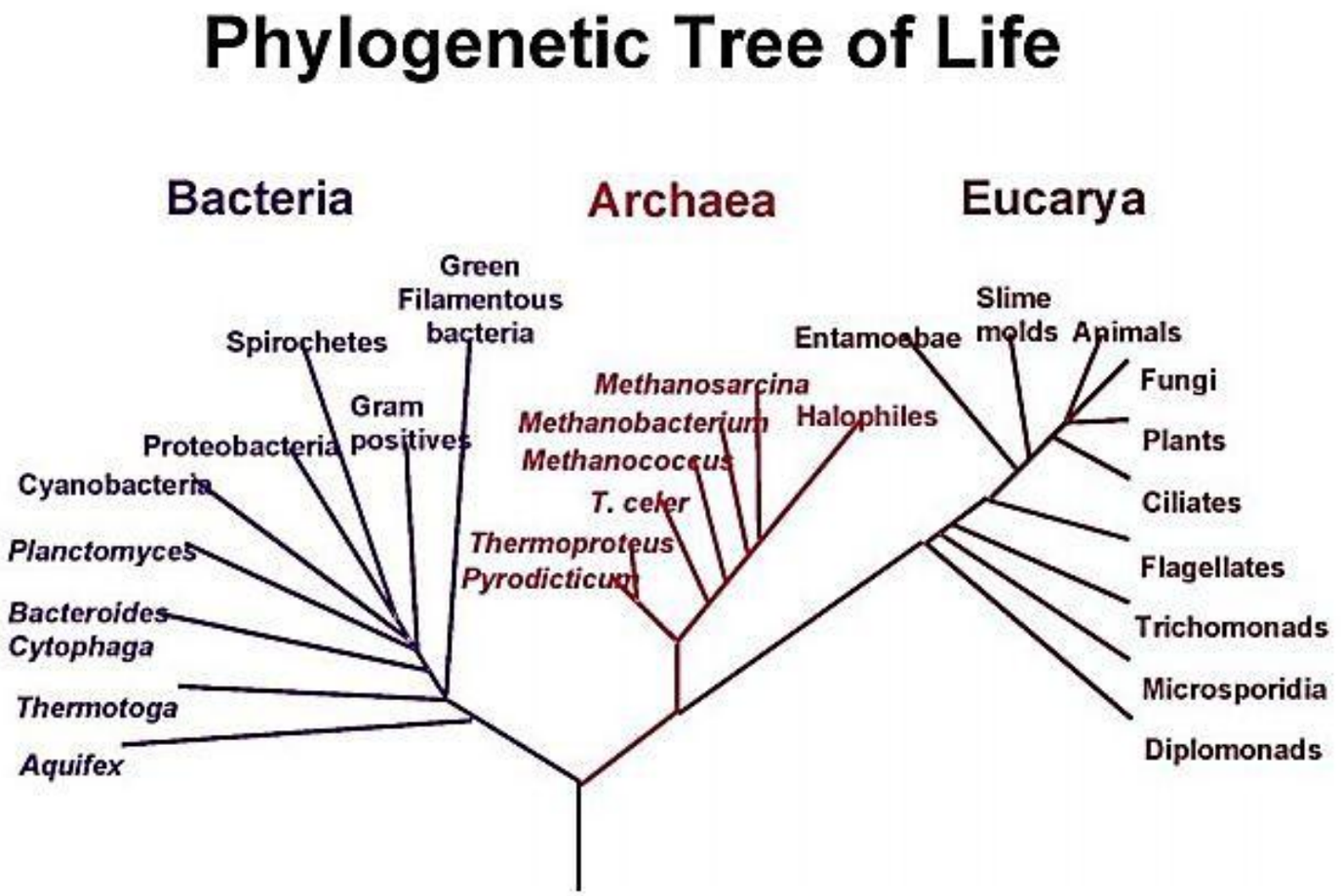
Amina Zee

Need for Bioinformatics – II

http://www.ncbi.nlm.nih.gov/Genbank/genbankstats.html

Amina Zee

Need for Bioinformatics – II



Need for Bioinformatics – II

What can bioinformatics deliver?

- **Life, evolution and disease can be better understood**
- **Drugs can be better developed after understanding molecular basis of life**

Need for Bioinformatics – II

Possible contributions

- Solve important problems related to life
- Handle mammoth quantities of experimental data
- Develop efficient solutions towards these goals

Need for Bioinformatics – II

Conclusion

- **From gene sequences to protein sequences Bioinformatics is the way forward**

- **Protein structure, protein-protein interactions and systems biology are other research areas in bioinformatics**

END

Amina Zee

Introduction

Lec 29
Applications of Bioinformatics - I

Amina Zee

Applications of Bioinformatics - I

Introduction

- Where can bioinformatics be applied specifically?
- What are the avenues in biology that can benefit from this science?
- What benefits can it deliver to the society?

Amina Zee

Applications of Bioinformatics - I

1. Genomics

- DNA Sequencing
- Gene Finding
- Genome Assembly
- Variation in Genomes
- Transcription Data
- Databases

Amina Zee

Applications of Bioinformatics - I

2. Evolutionary Studies

- Evolutionary relationships
- Evolutionary distances
- Phylogenetics
- Tree of life

Amina Zee

Applications of Bioinformatics - I

3. Proteomics

- Protein Sequencing
- Protein Structures
- Post-translational Modifications
- Protein-Protein Interactions
- Database Development

Amina Zee

Applications of Bioinformatics - I

4. Systems Biology

- Model protein and gene interactions
- Dynamical analysis of such models
- Understand system properties
- Predict system level behaviors

Amina Zee

Applications of Bioinformatics - I

Conclusion

- Genomics
- Transcriptomics
- Proteomics
- Metabolomics
- Structural Proteomics
- Drug Design
- Systems Biology
- Personalized Medicine

END

Amina Zee

Introduction

Lec 30
Applications of Bioinformatics - II

Amina Zee

Applications of Bioinformatics - II

Introduction

- Genomics
- Transcriptomics
- Proteomics
- Metabolomics
- Structural Proteomics
- Drug Design
- Systems Biology
- Personalized Medicine

Amina Zee

Applications of Bioinformatics - II

Introduction

- **Bioinformatics techniques have enabled us to generate 'omic' data and its utilization!**
- **Step by step application of bioinformatics from genome level to entire biosystems**

Amina Zee

Applications of Bioinformatics - II

Small to Big

- Gene finding, function prediction
- Structure prediction, modeling
- Predicting Molecular (Protein) interaction
- Biological network building & modeling

Amina Zee

Applications of Bioinformatics - II

Small to Big

- Cell level modeling & simulation
- Cell signaling modelling & simulation
- Tissue & morphology modelling
- Biological network building & modeling

Amina Zee

Applications of Bioinformatics - II

Small to Big

- Models integrate biological data
- Simulations help validate testable hypotheses
- Moreover, these simulations can also predict disease progression & outcomes

Amina Zee

Applications of Bioinformatics - II

Conclusion

- **Bioinformatics not only organizes, stores and analyzes biological data, but can also validate novel hypotheses**
- **Modern day bioinformatics also helps predict disease outcomes as well as drugs to treat them!**

END

Amina Zee

Introduction

Lec 31
Frontiers in Bioinformatics - I

Amina Zee

Frontiers in Bioinformatics - I

Introduction

- **Bioinformatics is a relative young scientific area**
- **It is rapidly expanding to cater for the variety and scale of biological data**
- **Several unresolved challenges exist in this expansion!**

Amina Zee

Frontiers in Bioinformatics - I

Frontiers in Genomics

- Next generation sequencing of whole genomes
- Massive amounts of data (Tera byte files)
- Handling & analysis of this data
- Genome assembly & prediction

Amina Zee

Frontiers in Transcriptomics

- Identification of hitherto unknown types as well as roles of RNA
- Understanding the regulatory dynamics of these RNA molecules
- Interactions of the RNA molecules

Amina Zee

Frontiers in Proteomics

- Identification of low abundance proteins in patient tissue samples
- Large scale protein identification
- Identification of known as well as novel post-translational modifications

Amina Zee

Frontiers in Bioinformatics - I

Conclusion

- **Bioinformatics is full of challenges and opportunities**
- **Amongst other frontiers in bioinformatics, there are protein structures, systems biology and personalized medicine!**

END

Amina Zee

Introduction

Lec 32
Frontiers in Bioinformatics - II

Amina Zee

Frontiers in Bioinformatics - II

Introduction

Frontiers in bioinformatics included:

- **Next generation genomics**
- **Transcriptomics**
- **High throughput proteomics**

Amina Zee

Frontiers in Protein Structure

- Accurate solution of protein structures is one of the toughest problems
- How to determine protein structures?
- How to predict protein structures?

Amina Zee

Frontiers in Protein Structure

- How to simulate the protein folding process?
- How to predict the protein – protein interaction?
- How to model protein-drug interaction?

Amina Zee

Frontiers in Systems Biology

- Cells function as a whole!
- How to integrate genes, proteins and metabolites into a unified system?
- How to simulate these models in real-time?

Amina Zee

Frontiers in Personalized Medicine

- Not all medicines work all the time!
- Some medicines have side effects on certain patients
- How to evaluate drugs using Bioinformatics tools?

Amina Zee

Conclusion

- 21st century is the century of Bioinformatics
- What will it hold for us, we can already guess that!
- An era of personalized medicines and eradication of several common diseases!

END

Amina Zee

Lec 33

Central dogma

Amina Zee

The Central Dogma

- The central dogma outlines the flow of genetic information during growth and division of the cells.
- Genetic information flows from DNA to RNA to protein during cell growth.

Amina Zee

The Central Dogma

- In addition, all living cells must replicate their DNA when they divide.
- During cell division each daughter cell receives a copy of the genome of the parent cell.

Amina Zee

The Central Dogma

- Replication is the process by which two identical copies of DNA are made from an original molecule of DNA.
- So Replication occurs in the cells prior to cell division.

Amina Zee

The Central Dogma

- An important point is that information does not flow from protein to RNA or DNA.
- However, flow of information from RNA “backwards” to DNA is possible in certain special circumstances due to the operation of reverse transcriptase.

Amina Zee

The Central Dogma

- By the end of 1953, the working hypothesis was adopted that chromosomal DNA functions as the template for the synthesis of RNA molecules.

Amina Zee

The Central Dogma

- These RNA molecules, the subsequently move to the cytoplasm, where they determine the arrangement of amino acids within proteins.

Amina Zee

The Central Dogma

- In 1956, Francis Crick referred to this pathway for the flow of genetic information as the Central Dogma.

Amina Zee

The Central Dogma

Amina Zee

The Central Dogma

- An important point in the above equation is that the two arrows are unidirectional which means that RNA sequences are never determined by protein templates nor was DNA then imagined ever to be made on RNA templates.

Amina Zee

The Central Dogma

- The idea that proteins never serve as templates for RNA has stood the test of time.
- However, RNA chains sometimes do act as templates for the synthesis of DNA chains of complementary sequence.

Amina Zee

The Central Dogma

- Such reversals of the normal flow of information are very rare events compared with the enormous number of RNA molecules made on DNA templates.

Amina Zee

The Central Dogma

- Thus, the central dogma as originally proclaimed more than 50 years ago still remains essentially valid.

end

Amina Zee

Sequence Analysis

Lec 34
Gene, mRNA and Protein Sequences

Amina Zee

Gene, mRNA and Protein Sequences

Introduction

- All living things are composed of cells
- How are cells made?
- DNA provides the blueprint for building cells
- DNA dictates the production of cell's proteins, carbohydrates & vitamins

Amina Zee

Gene, mRNA and Protein Sequences

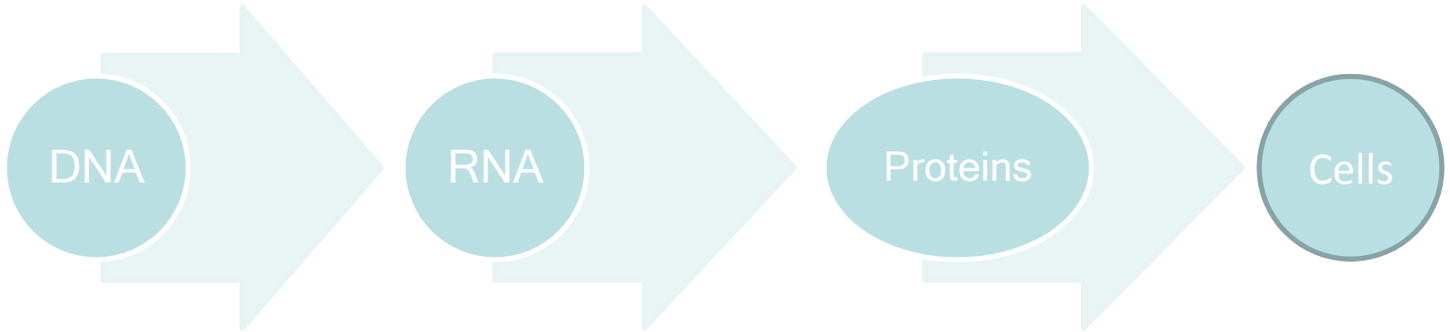
Introduction

- The transformation of information in DNA into these molecules is termed the Central Dogma
- Central dogma: DNA -> RNA -> Proteins
- Proteins are then used in constructing cells

Amina Zee

Gene, mRNA and Protein Sequences

Overview

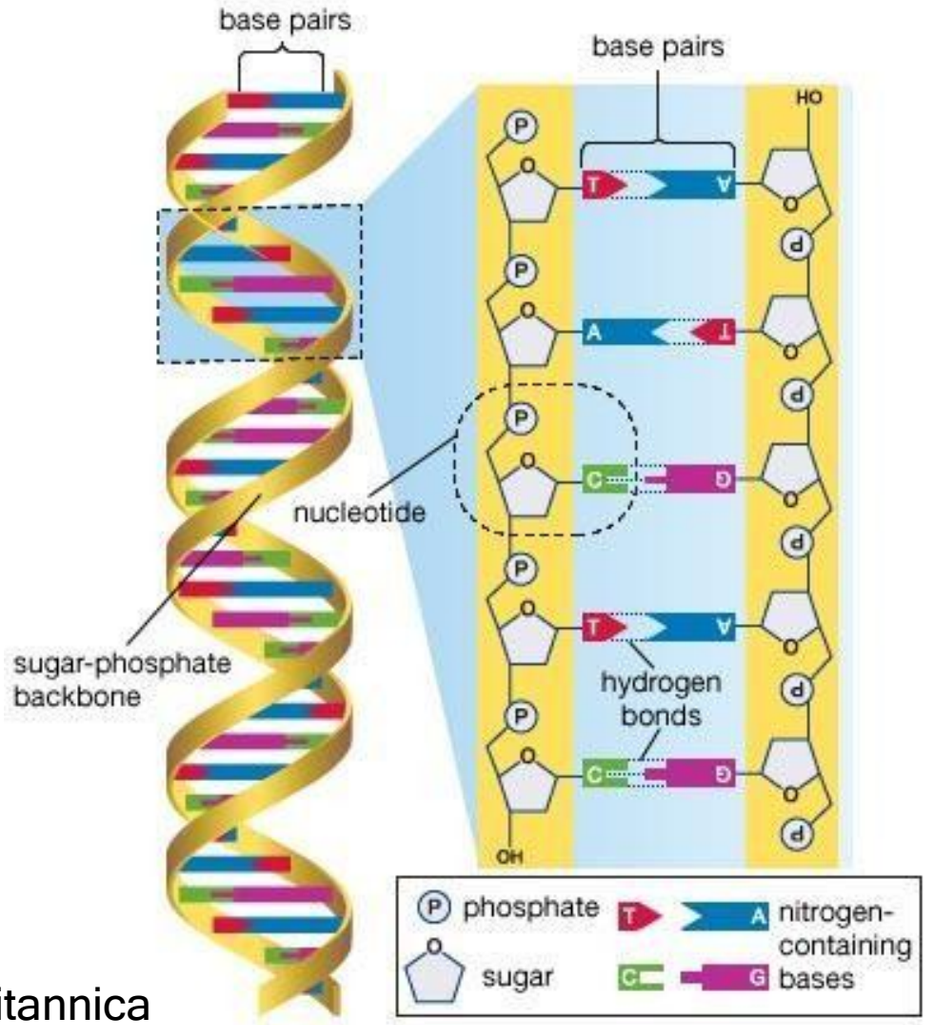


Central Dogma

Amina Zee

Gene, mRNA and Protein Sequences

DNA

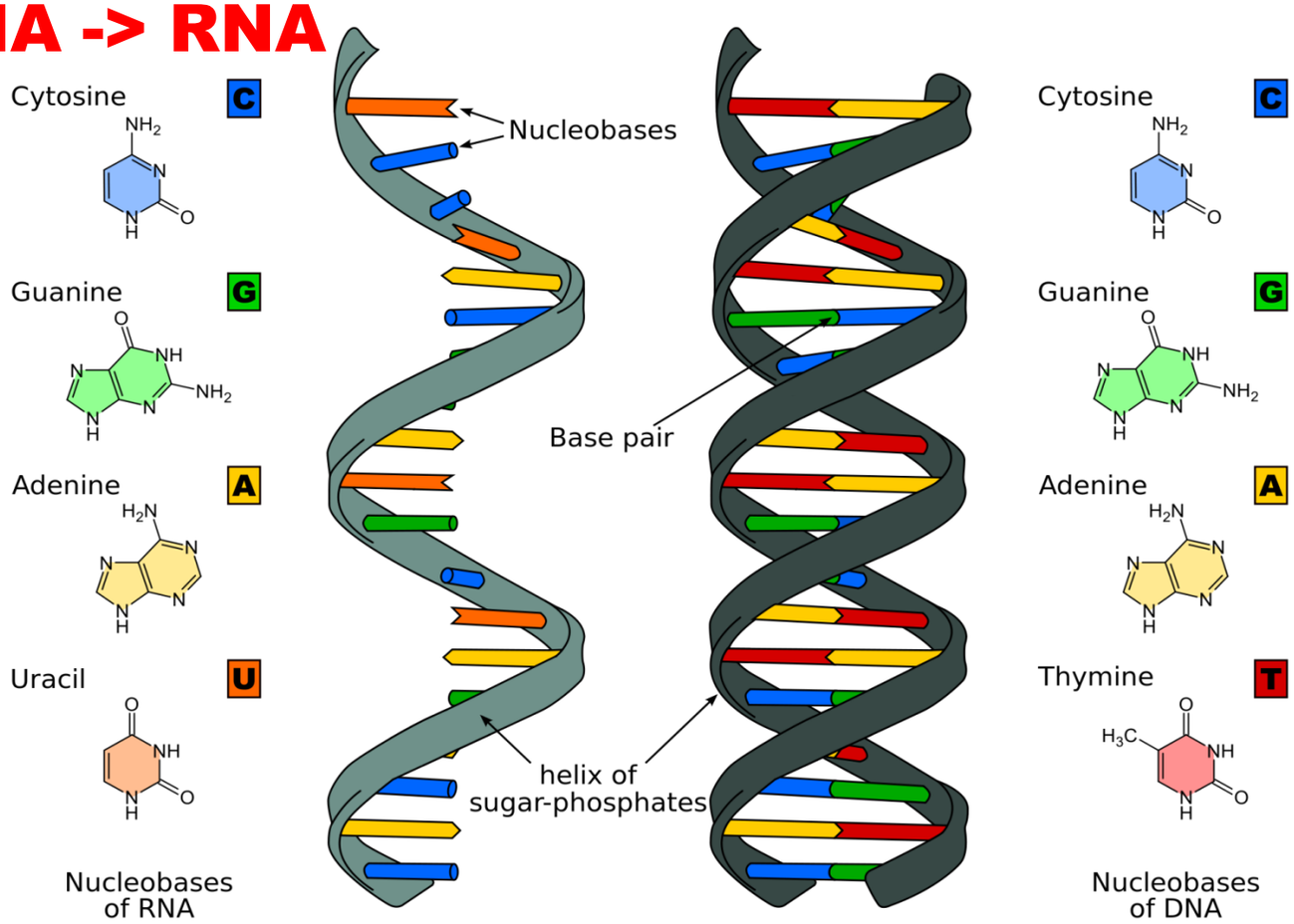


Courtesy Britannica

Amina Zee

Gene, mRNA and Protein Sequences

DNA -> RNA

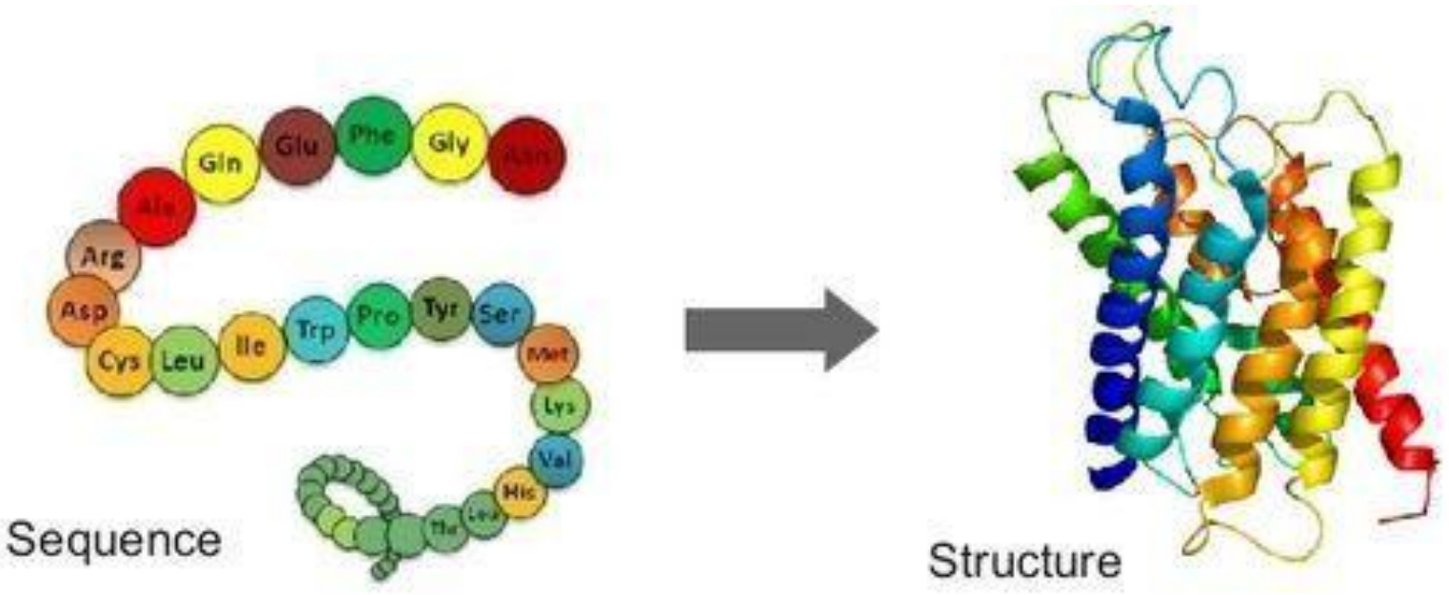


Courtesy Wikipedia

Amina Zee

Gene, mRNA and Protein Sequences

RNA -> Proteins

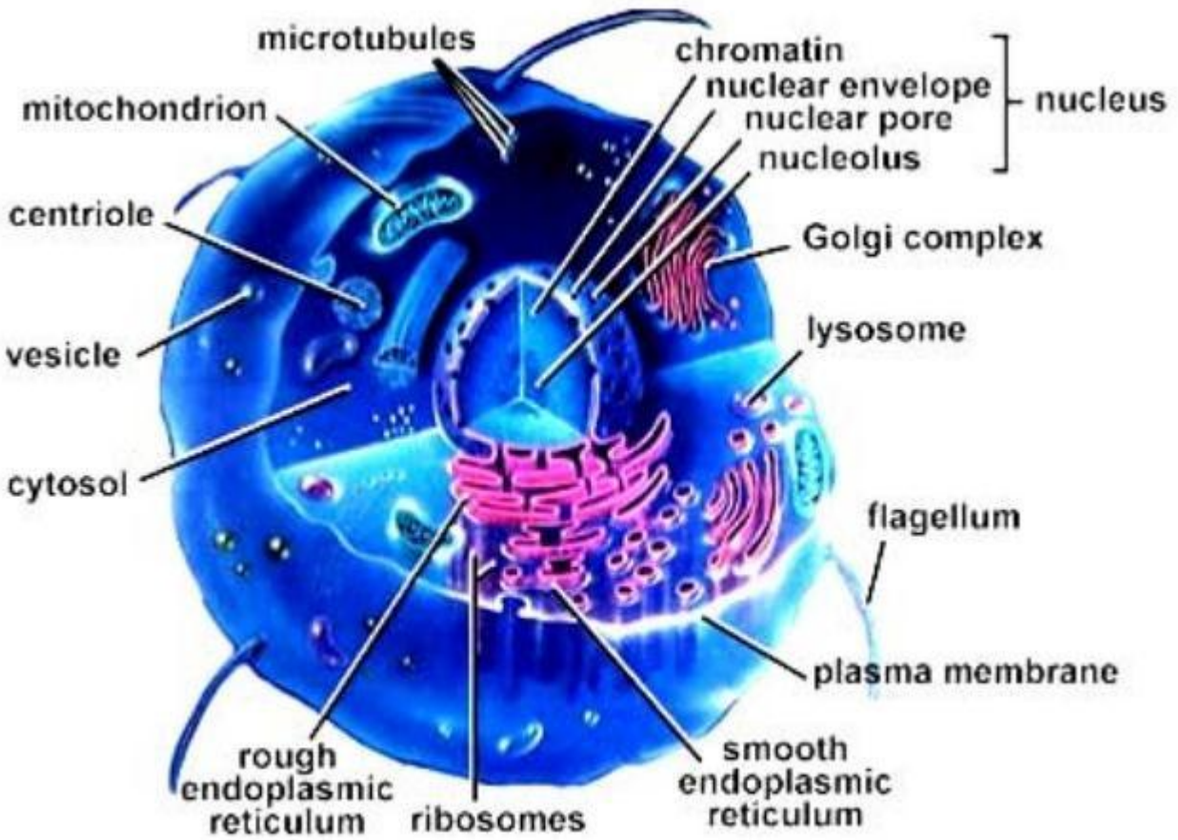


Courtesy TU-Berlin

Amina Zee

Gene, mRNA and Protein Sequences

Proteins -> Cells



Amina Zee

Gene, mRNA and Protein Sequences

Conclusion

- DNA codes for an RNA
- RNA in turn helps produce Proteins
- Proteins along with some other molecules form cells including their organelles and membrane

END

Amina Zee

Sequence Analysis

Lec 35
Nucleotides

Amina Zee

Nucleotides

Background

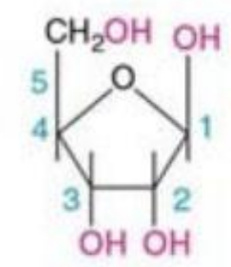
- DNA and RNA molecules are built up from four types of molecules, each
- These molecules are called nucleotides
- These are Adenine (A), Cytosine (C), Thymine (T), Uracil (U) and Guanine (G)

Amina Zee

Nucleotides

Base complementarity

In RNA

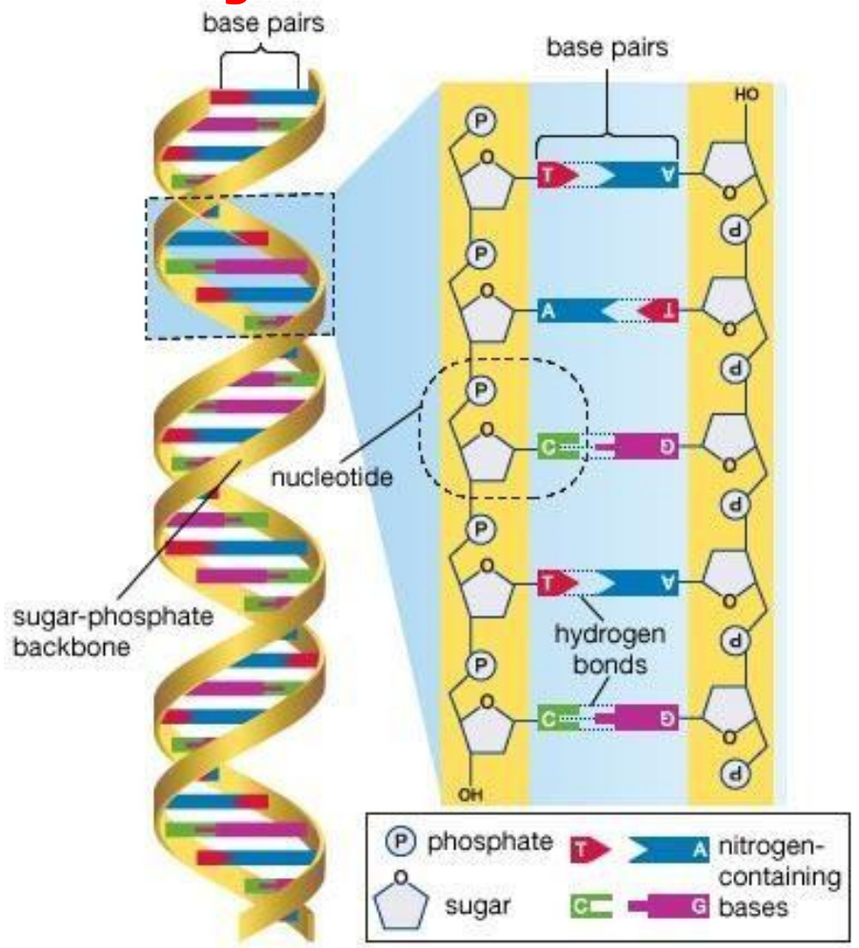


Ribose

In DNA

2-deoxyribose

Sugars

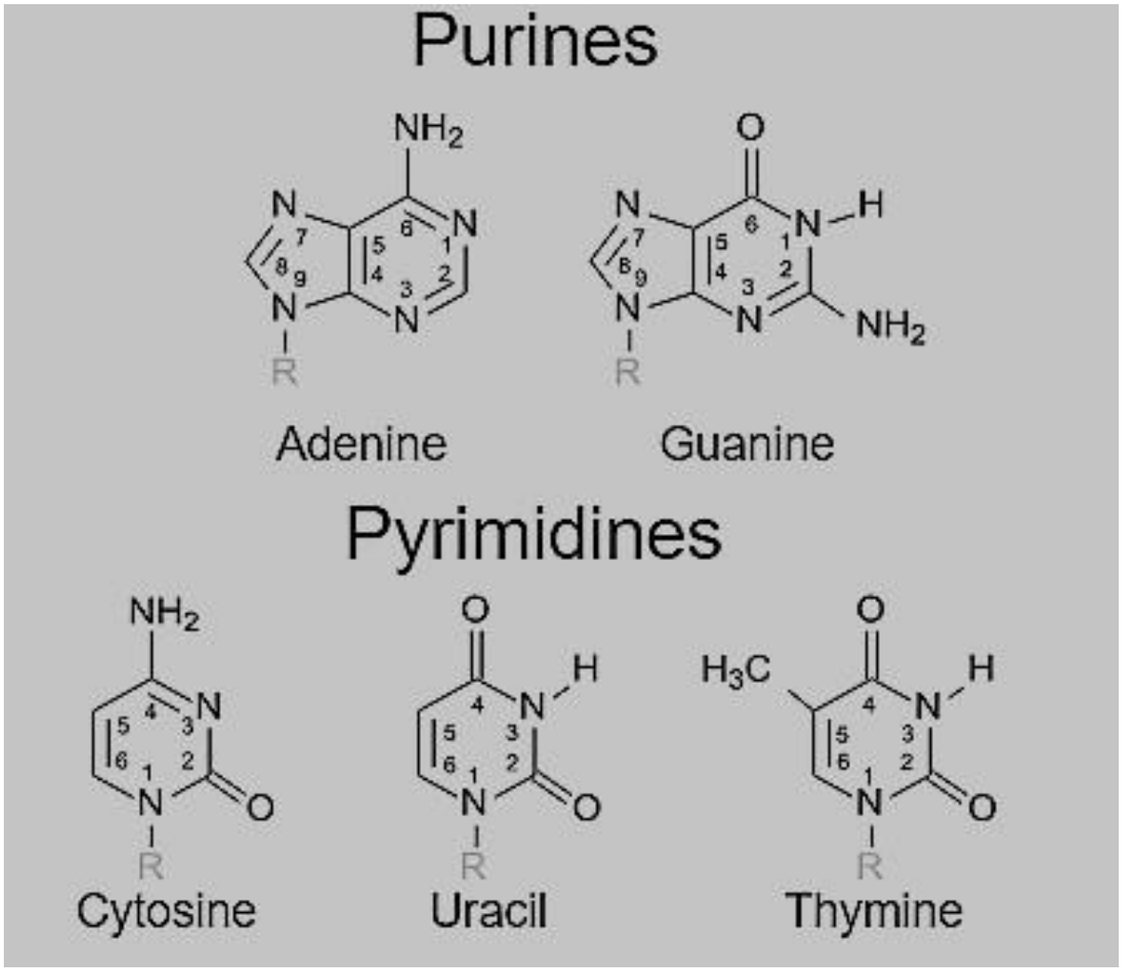


Courtesy Britannica

Amina Zee

Nucleotides

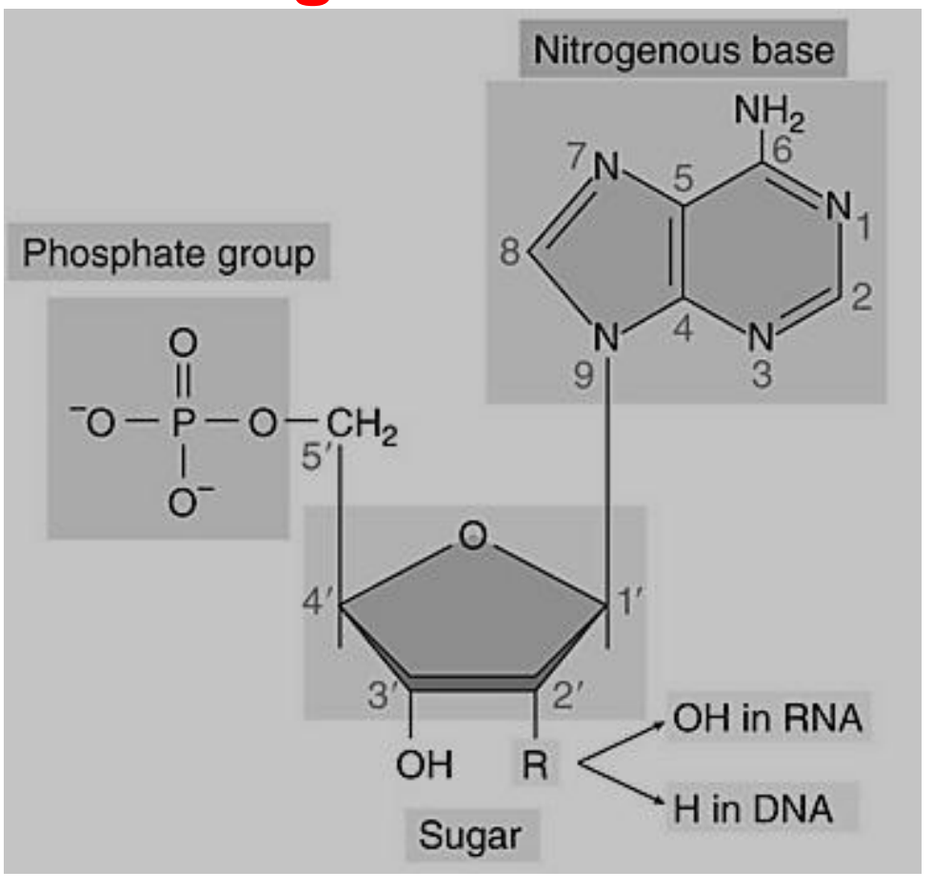
Structures of A, C, T, U and G



Wiki

Nucleotides

Phosphate, Sugar and Nitrogenous bases come together



Amina Zee

Nucleotides

Conclusion

- **DNA gives rise to RNA**
- **DNA has A, T, C & G while RNA has A, U, G & C**
- **DNA is double stranded while RNA is single stranded**
- **RNA helps produce proteins**

END

Amina Zee

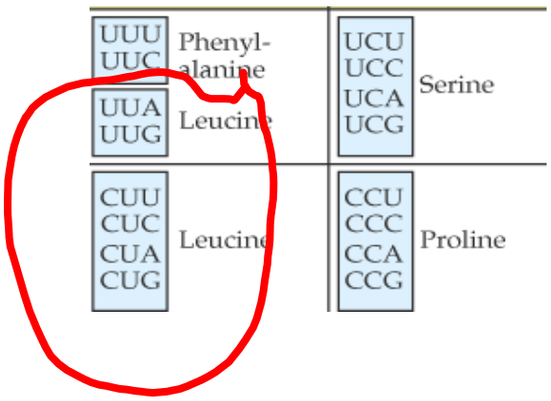
Cell Biology

Lec 36
Genetic Code

Amina Zee

Genetic Code

Three mRNA nucleotides form a codon



which is complementary & antiparallel to corresponding DNA.

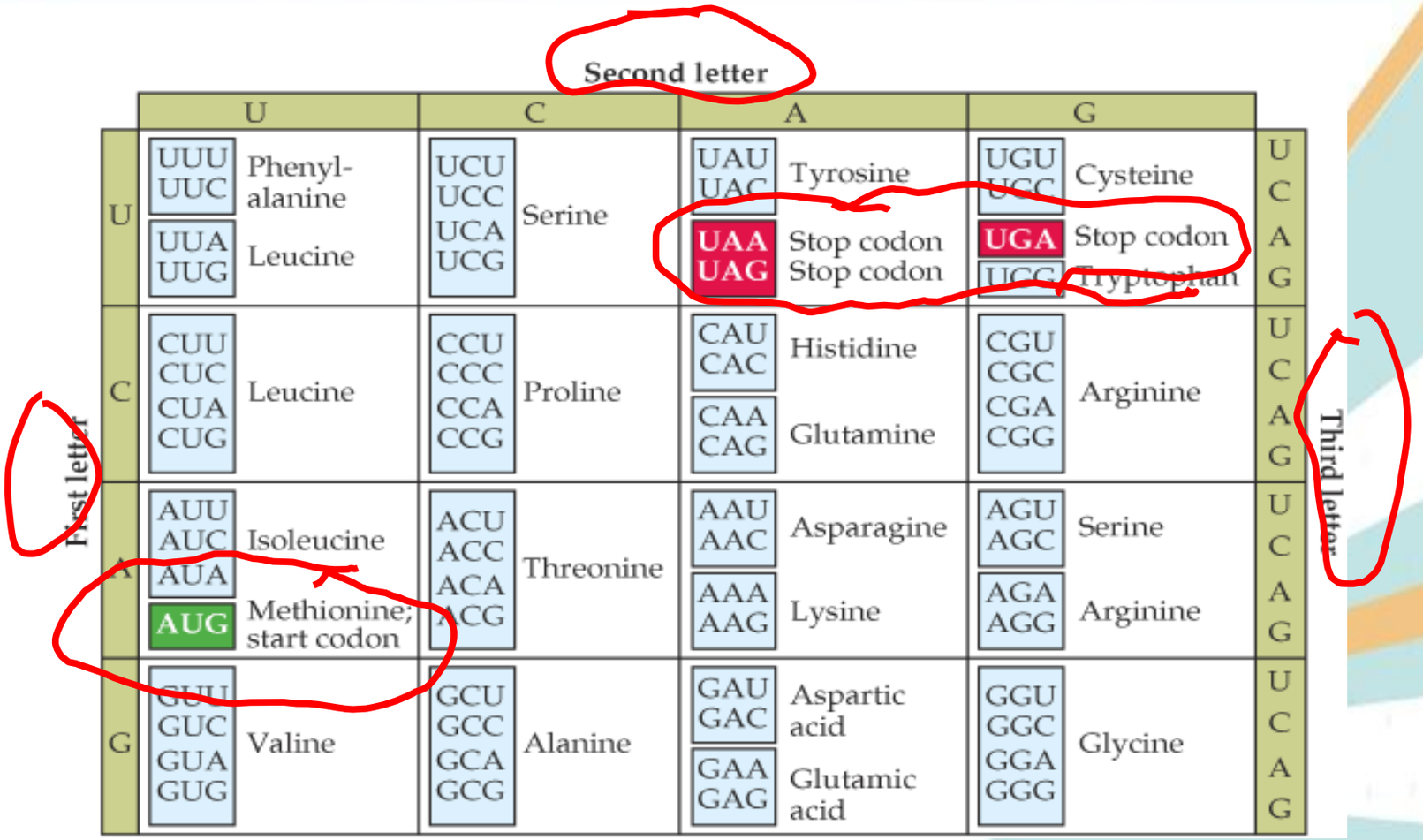
Genetic code relates codons to their specific AA.

~ Universal & Redundant but not ambiguous some exceptions.

Start & Stop

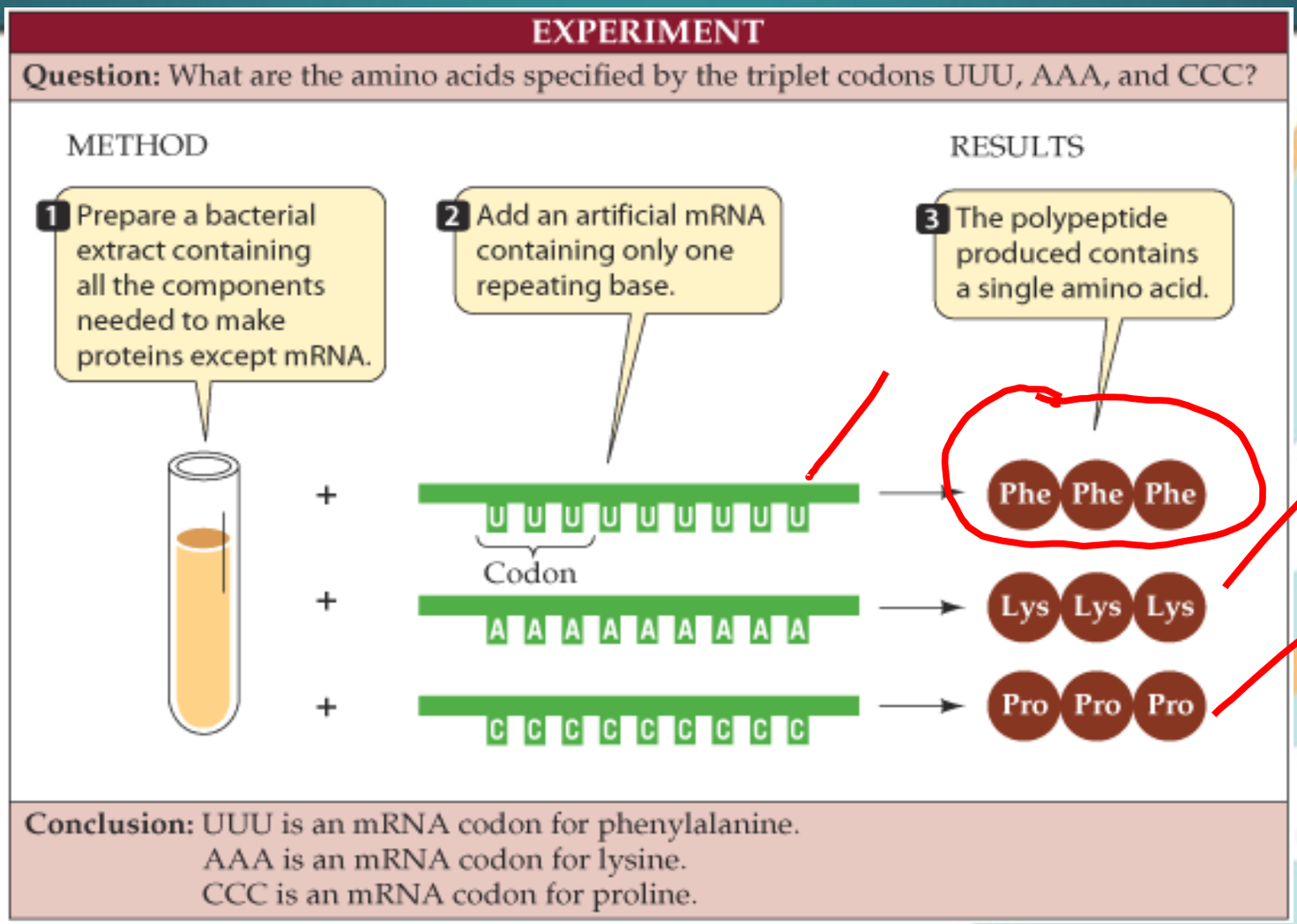
Amina Zee

Genetic Code



Amina Zee

Deciphering the Genetic Code



Amina Zee

The Adapter

tRNA Translates Nucleotide Language into AA Language

1) Carries AA
2) Associates with mRNA
3) Binds ribosome

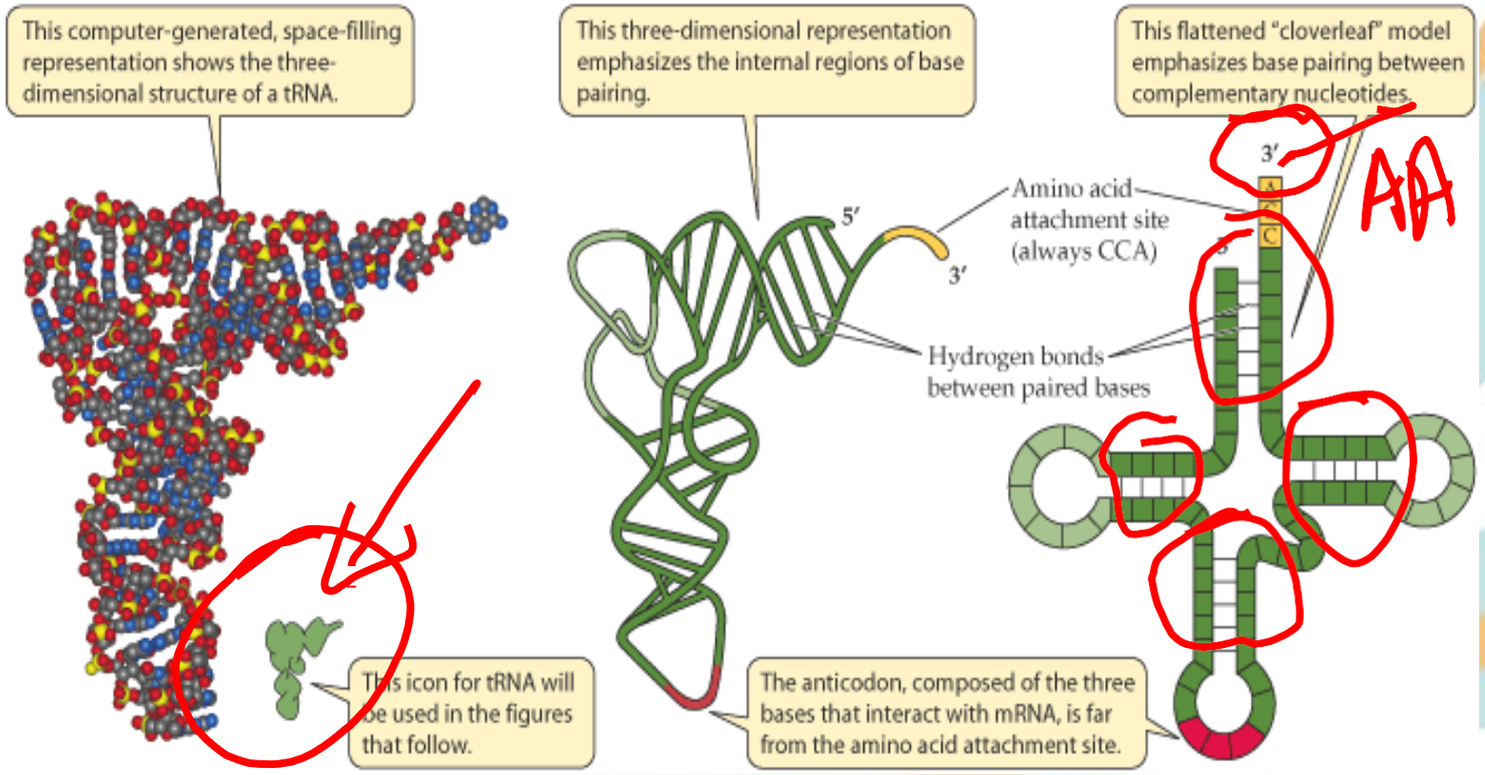
tRNA has 75-80 nucleotides with internal H-bonding.

Conformation of tRNA allows it to bind ribosome.

3′ end of tRNA (CCA) is attached to AA & mid point has anticodon (antiparallel).

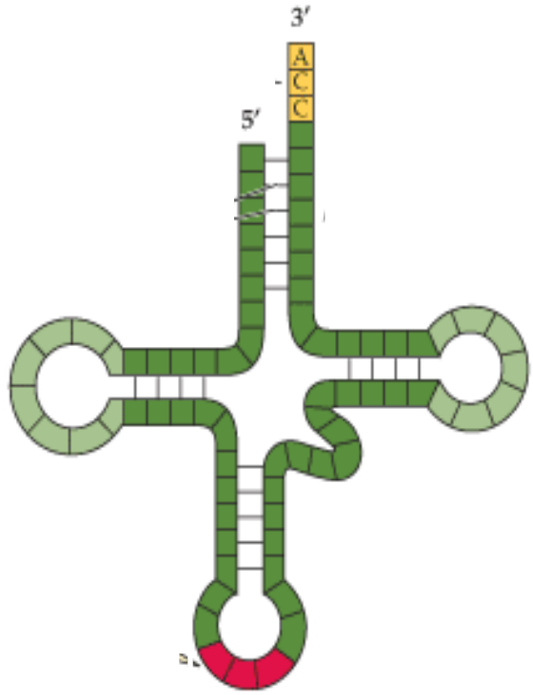
Amina Zee

T-RNA Structure



Amina Zee

tRNA Structure



END

Amina Zee

Cell Biology

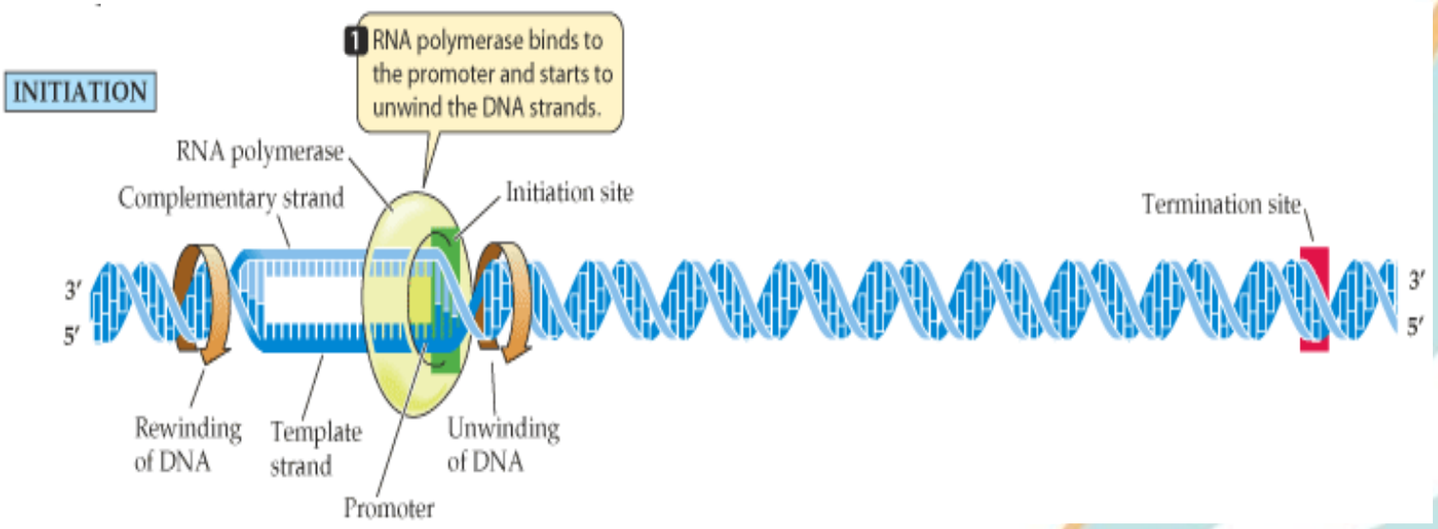
Lec 37

Transcription

Amina Zee

Transcription Initiation

Initiation begins at promoter -tightly binds RNA Pol



1) **Where to start transcription.**
2) **Which strand to read.**
3) **The direction to take start from.**

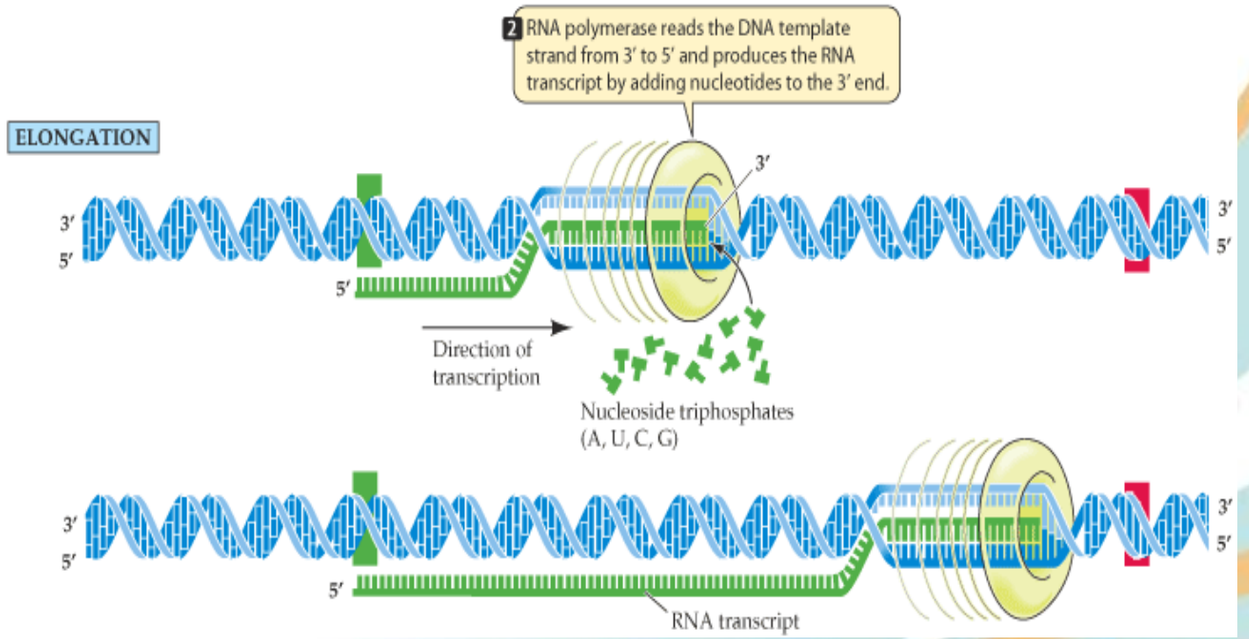
Part of promoter is initiation site (transcription start).

Amina Zee

Transcription Elongation

RNA Pol moves 3′ to 5′, synthesis 5′ to 3′, ~~primer~~

Unwinds 20 bp at a time.



Transcript antiparallel to DNA template strand.

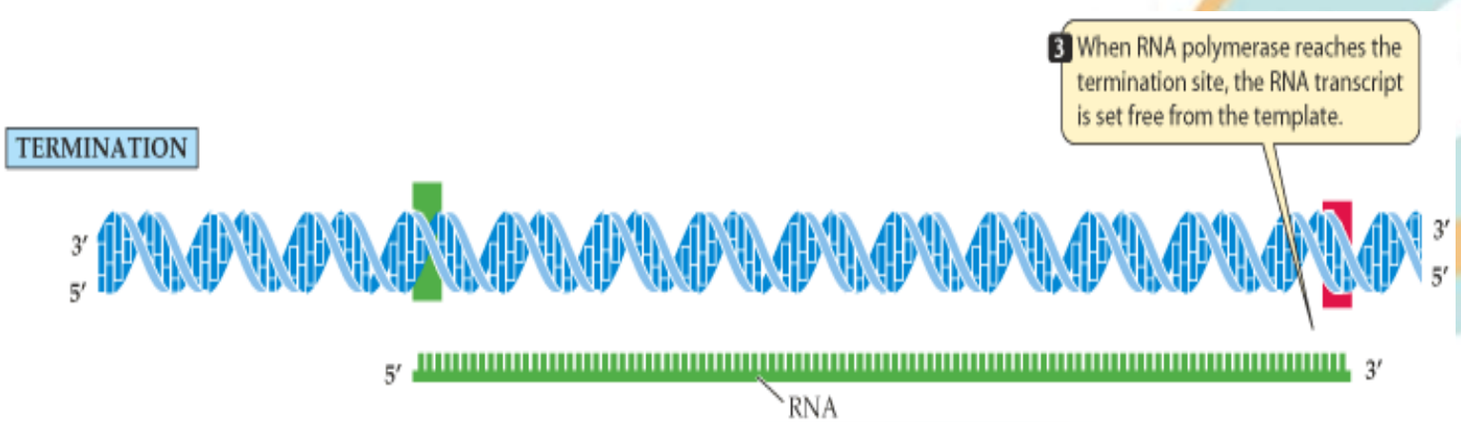
No proof reading error rate 1 in 10,000-100,000

Amina Zee

Transcription Termination

Translation starts before termination in prokaryotes.

Pre-mRNA 1st product of transcription in eukaryotes.



Amina Zee

Transcription

END

Amina Zee

Sequence Analysis

Lec 38
Transcription

Amina Zee

Transcription

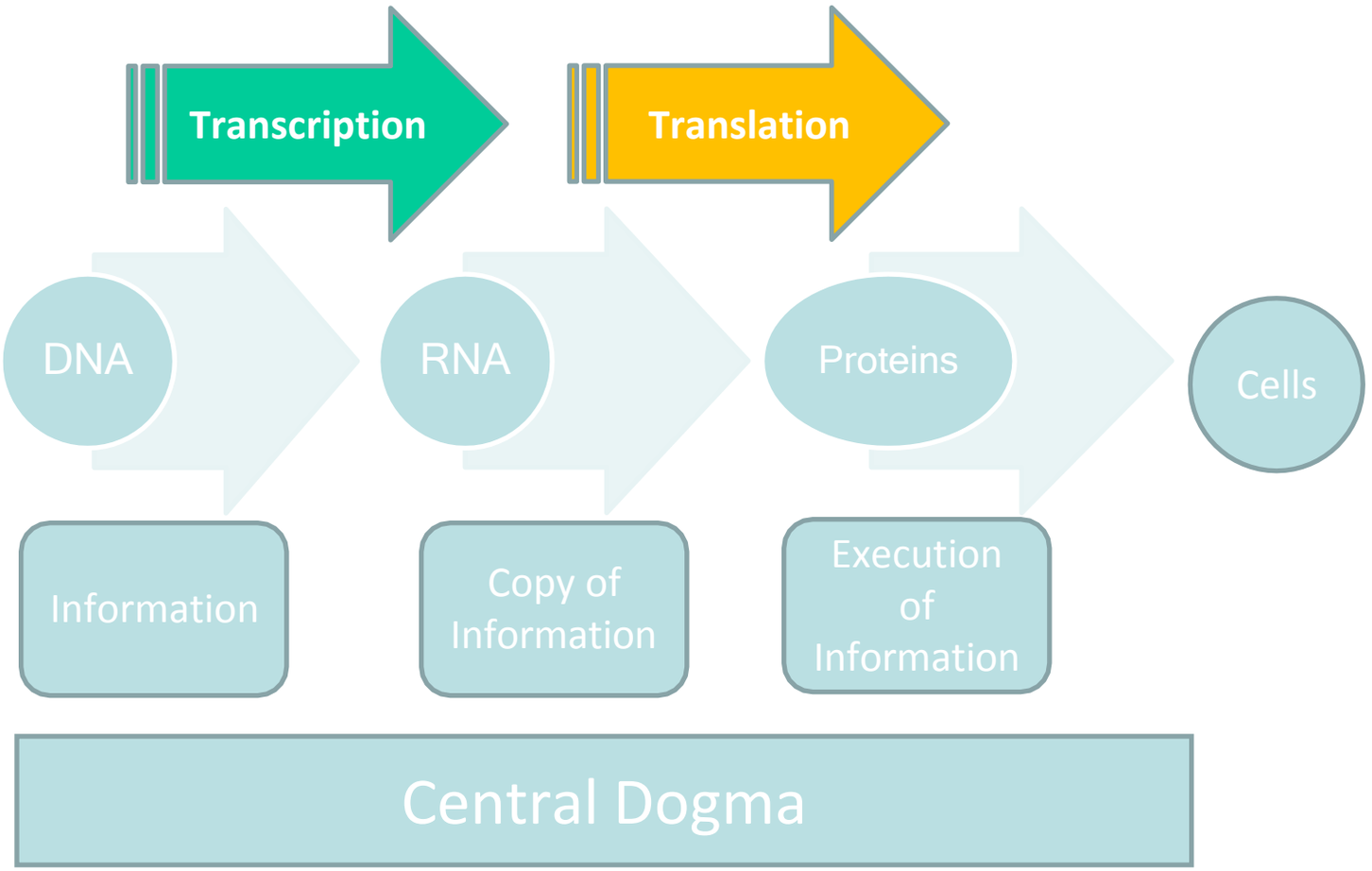
Background

- Cells are built up of proteins and carbohydrates
- These molecules are produced after transformation of DNA into proteins
- What are the salient steps underlining this transformation?

Amina Zee

Transcription

Overview



Amina Zee

Transcription

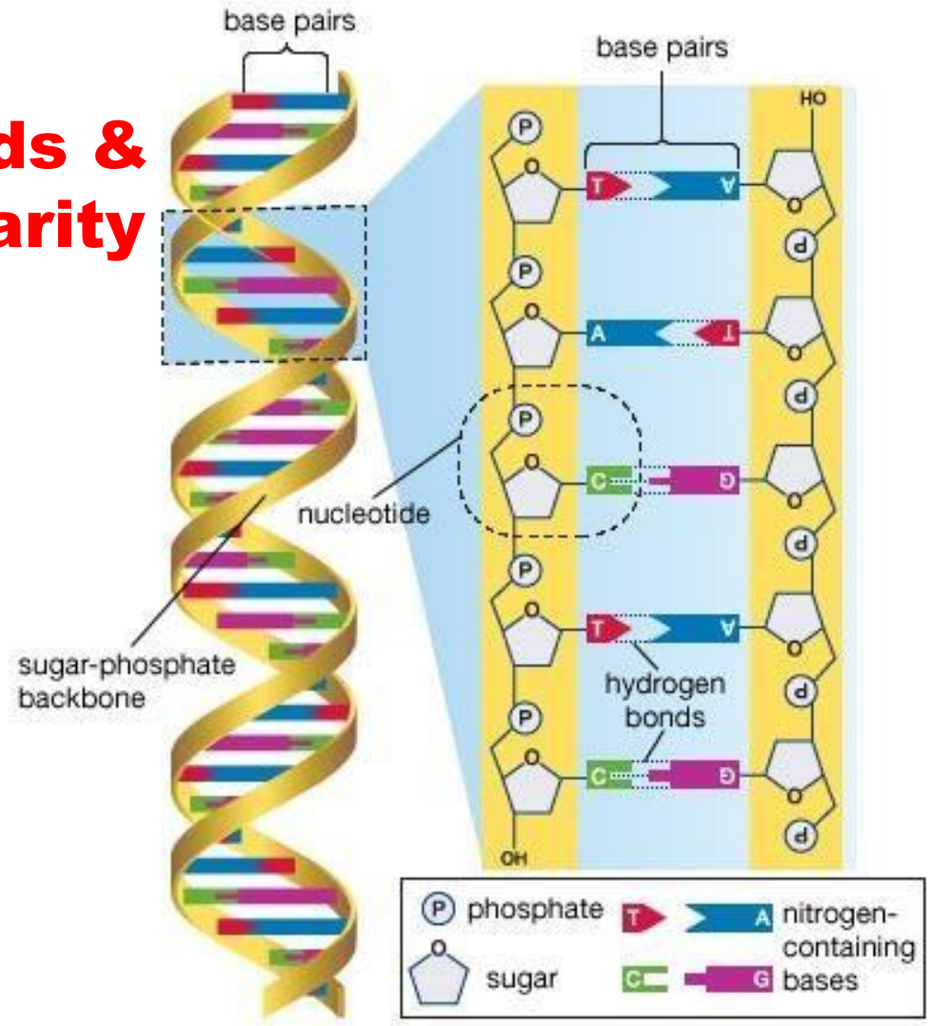
Transcription

- DNA is constituted by four bases
- These are Adenine (A), Cytosine (C), Thymine (T), and Guanine (G)
- DNA molecule may contain thousands of A, C, T and G.
- A pairs with T only and C pairs with G

Amina Zee

Transcription

DNA bases, double strands & complementarity



Courtesy Britannica

Transcription

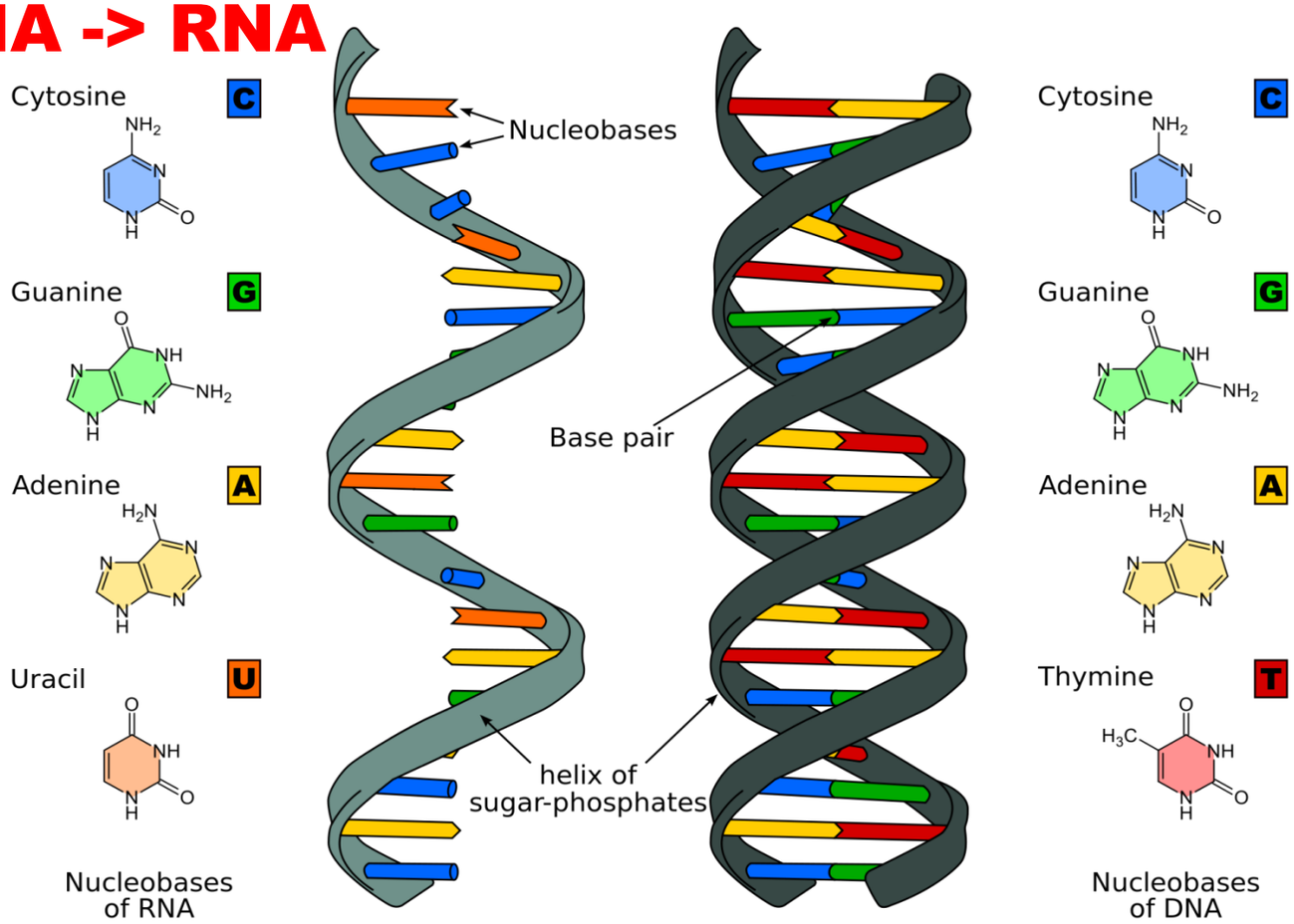
- Process of transcription decodes these bases in the DNA

- Based on the information received after decoding DNA, an RNA molecule is encoded

Amina Zee

Transcription

DNA -> RNA



Courtesy Wikipedia

Amina Zee

Transcription

Transcription

- RNA is constituted by four bases
- These are Adenine (A), Cytosine (C), Uracil (U), and Guanine (G)
- RNA molecule may contain thousands of A, C, U and G.
- A pairs with U only and C pairs with G

Amina Zee

Transcription

Conclusion

- DNA gives rise to RNA
- DNA has A, T, C & G while RNA has A, U, G & C
- DNA is double stranded while RNA is single stranded
- RNA helps produce proteins

END

Amina Zee

Lec 39

Types of Ribonucleotides

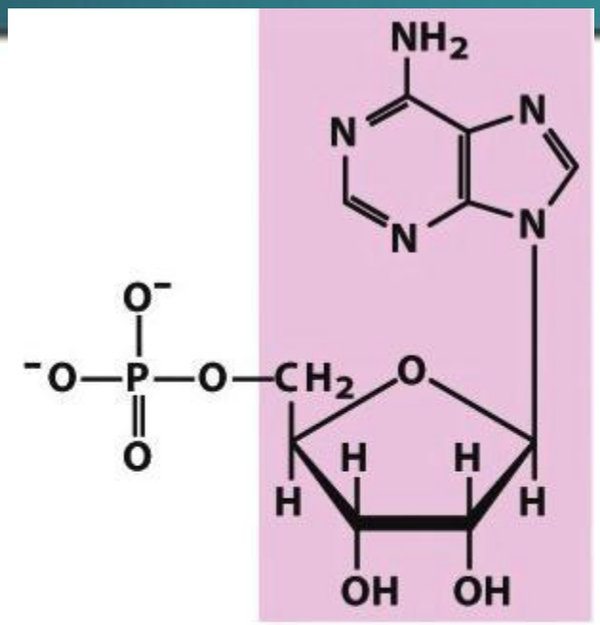
Amina Zee

Types of Ribonucleotides

- There are mainly four types of ribonucleotides depending upon the types of nitrogenous bases present in RNA.

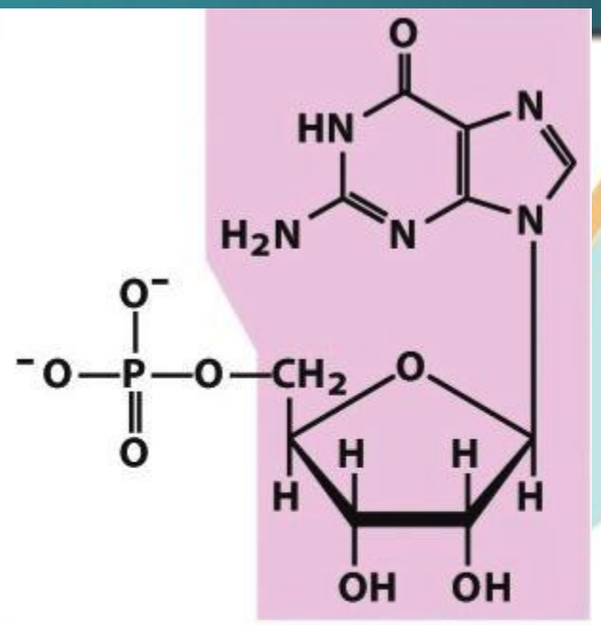
Amina Zee

Types of Ribonucleotides



Adenylate (adenosine 5′-monophosphate)

Adenosine

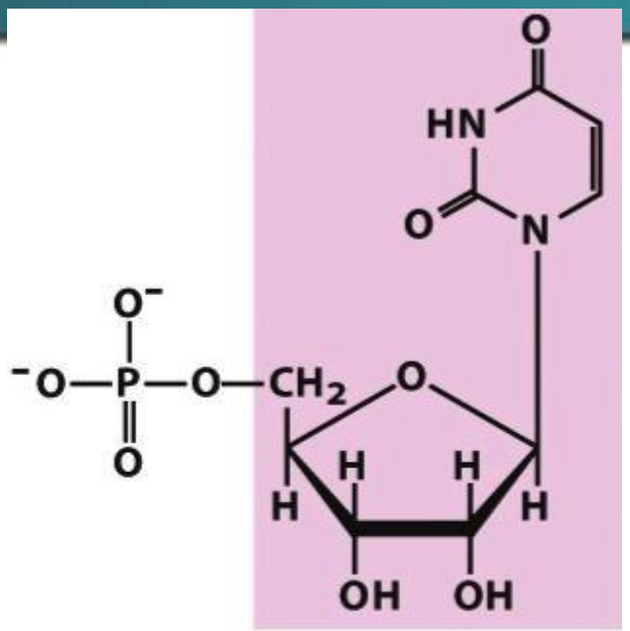


Guanylate (guanosine 5′-monophosphate)

Guanosine

Amina Zee

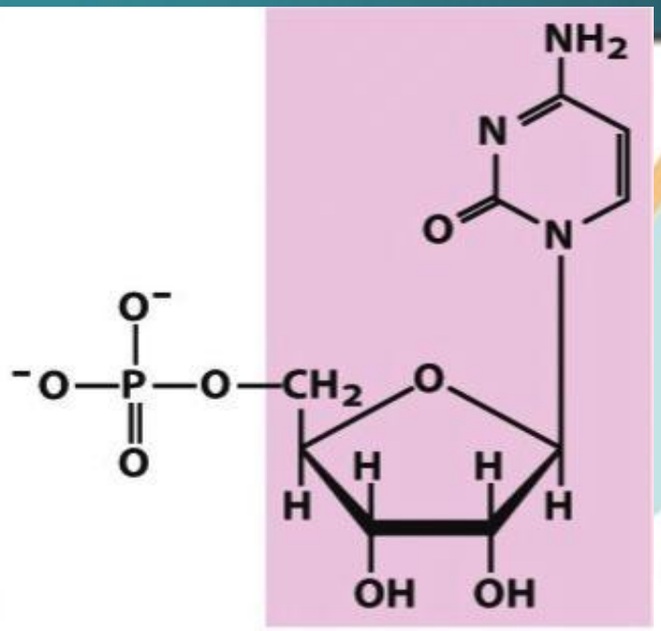
Types of Ribonucleotides



Uridylate (uridine 5′-monophosphate)

U, UMP

Uridine



Cytidylate (cytidine 5′-monophosphate)

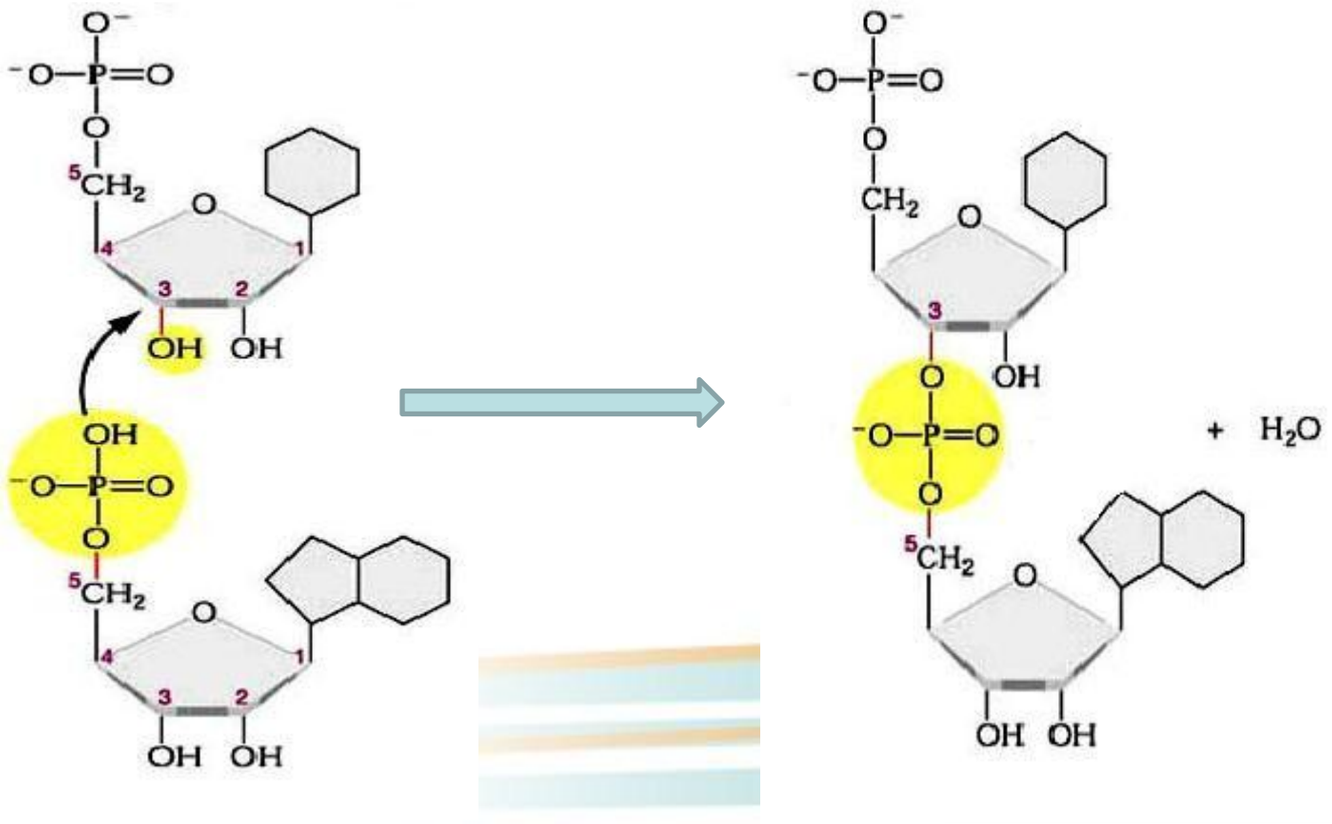
C, CMP

Cytidine

Amina Zee

Types of Ribonucleotides

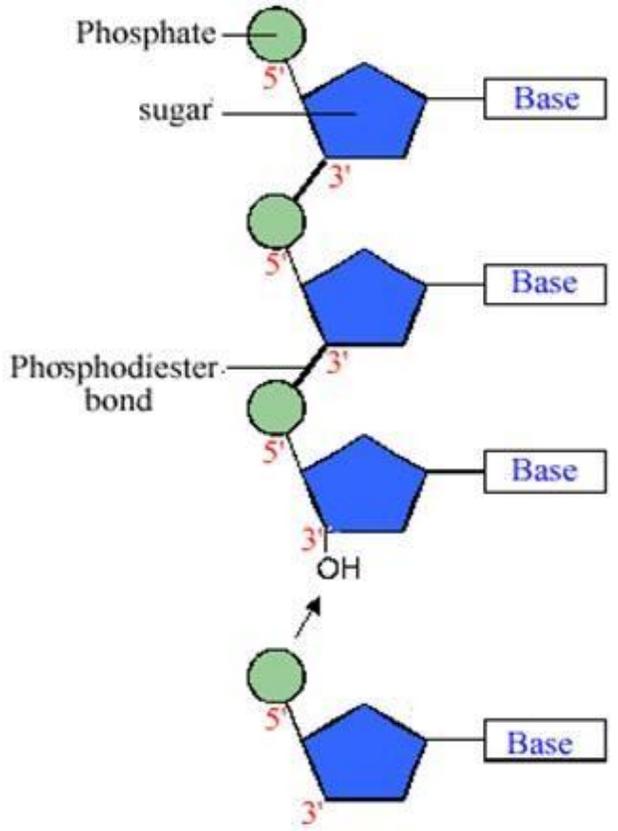
How do Ribonucleotides Join?



Amina Zee

Types of Ribonucleotides

A Poly-Ribonucleotide



Amina Zee

Lec 40

Types of RNA

Amina Zee

Types of RNAs

- There are mainly three types of Ribonucleic acids (RNAs) present in the cells of living organisms.
  - Messenger RNA (mRNA)
  - Transfer RNA (tRNA)
  - Ribosomal RNA (rRNA)

Amina Zee

Types of RNAs

Messenger RNA (mRNA)

- It is the type of RNA that carries genetic information from DNA to the protein biosynthetic machinery of the ribosome.
- It provides the templates that specify amino acid sequences in polypeptide chains.
- The process of forming mRNA on a DNA template is known as transcription.

Amina Zee

Types of RNAs

Messenger RNA (mRNA)

- It may be monocistronic or polycistronic.
- The length of mRNA molecules is variable and it depends on the length of gene.

Amina Zee

Types of RNAs

Transfer RNA (tRNA)

- Transfer RNAs serve as adapter molecules in the process of protein synthesis.
- They are covalently linked to an amino acid at one end.

Amina Zee

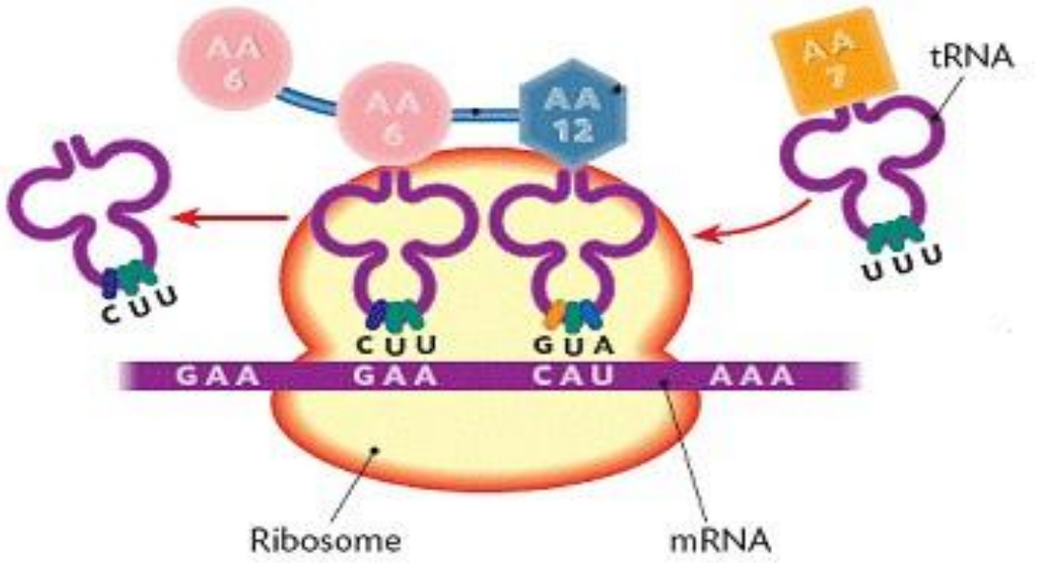
Types of RNAs

Transfer RNA (tRNA)

- They pair with the mRNA in such a way that amino acids are joined to a growing polypeptide in the correct sequence.

Amina Zee

Types of RNAs



Amina Zee

Types of RNAs

Ribosomal RNA (rRNA)

- Ribosomal RNAs are components of ribosomes.
- rRNA is a predominant material in the ribosomes constituting about 60% of its weight.
- It has a number of functions to perform in the ribosomes.

Amina Zee

Lec 41

Structure of RNA

Amina Zee

Structures of RNAs

- mRNA is always single stranded when it is formed from DNA.
- But this single strand assumes a double helical conformation soon after its formation.
- This confirmation is achieved mainly due to base stacking interactions.

Amina Zee

Structures of RNAs

Messenger RNA (mRNA)

Amina Zee

Structures of RNAs

- Self-complementary sequences may occur in the RNA molecules which produce more complex structures.
- So RNA can base-pair with complementary regions of either RNA or DNA.
- RNA has no any regular secondary structure that serves as a reference point. The three-dimensional structures of many RNAs are complex and unique.

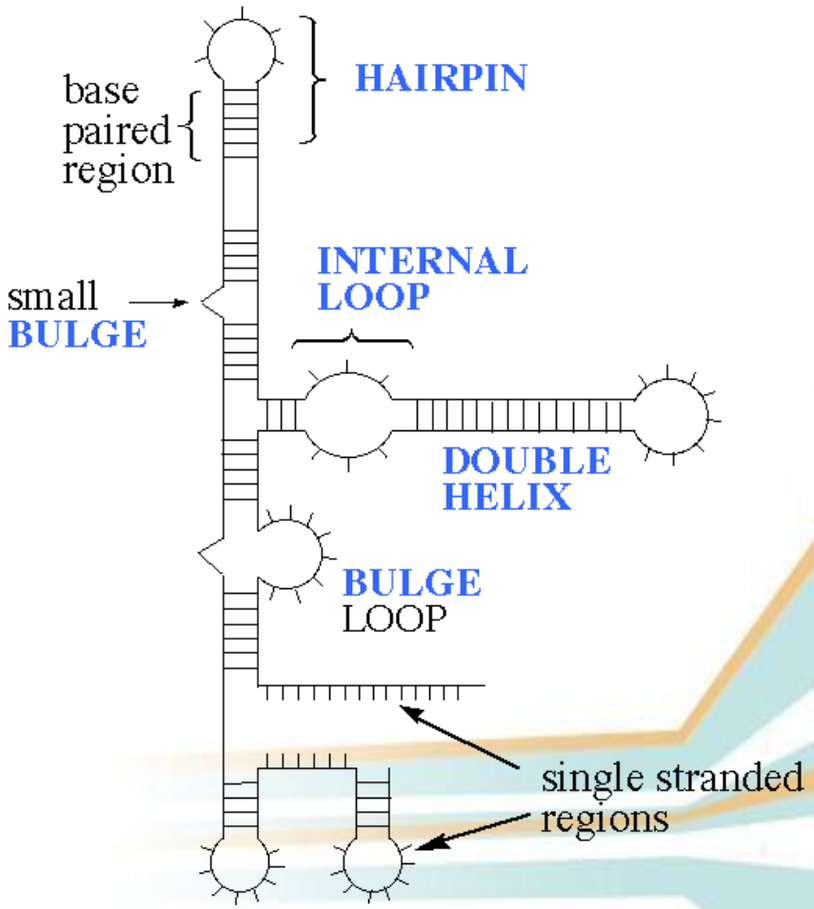
Amina Zee

Structures of RNAs

- Breaks in the helix caused by mismatched or unmatched bases in one or both strands are common and result in bulges or internal loops.
- Hairpin loops form between nearby self-complementary sequences.

Amina Zee

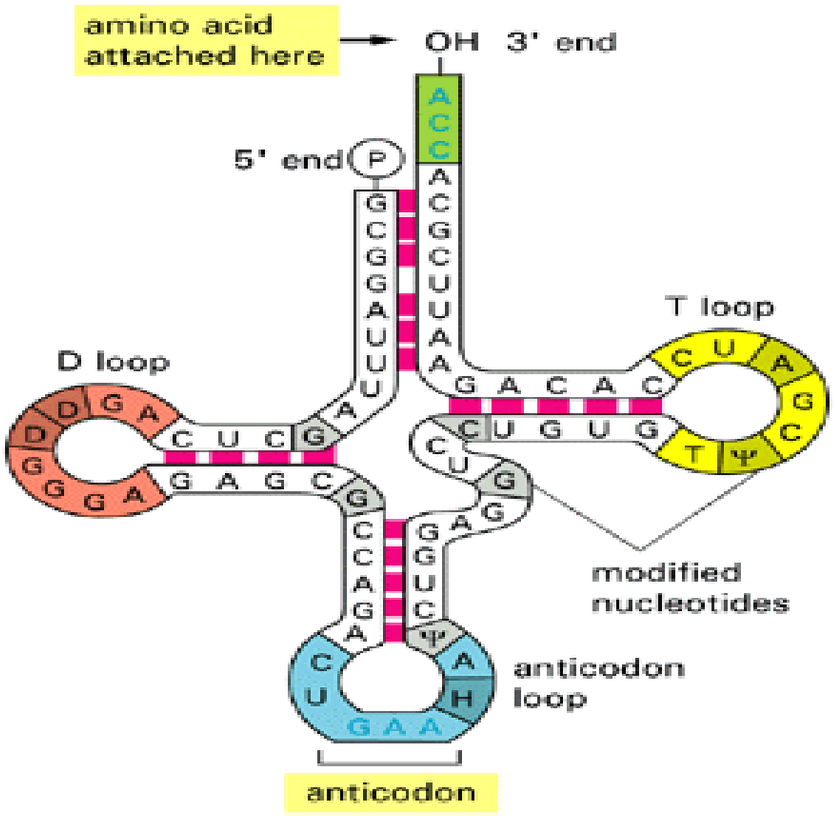
Structures of RNAs



Amina Zee

Structures of RNAs

Transfer RNA (tRNA)



Amina Zee

Lec 42

Messenger RNA

Amina Zee

Messenger RNA

- The protein-coding region(s) of each mRNA is composed of a contiguous, non-overlapping string of codons called an open reading frame (commonly known as an ORF).

Amina Zee

Messenger RNA

- Each ORF specifies a single protein and starts and ends at internal sites within the mRNA.
- That is, the ends of an ORF are distinct from the ends of the mRNA.

Amina Zee

Messenger RNA

- Translation starts at the 5’ end of the ORF and proceeds one codon at a time to the 3’ end. The first and last codons of an ORF are known as the start and stop codons.

Amina Zee

Messenger RNA

- In bacteria, the start codon is usually 5’-AUG-3’, but 5’-GUG-3’ and sometimes even 5’-UUG-3’ are also used.
- Eukaryotic cells always use 5’-AUG-3’ as the start codon.

Amina Zee

Messenger RNA

- The start codon has two important functions.
- First, it specifies the first amino acid to be incorporated into the growing polypeptide chain.
- Second, it defines the reading frame for all subsequent codons.

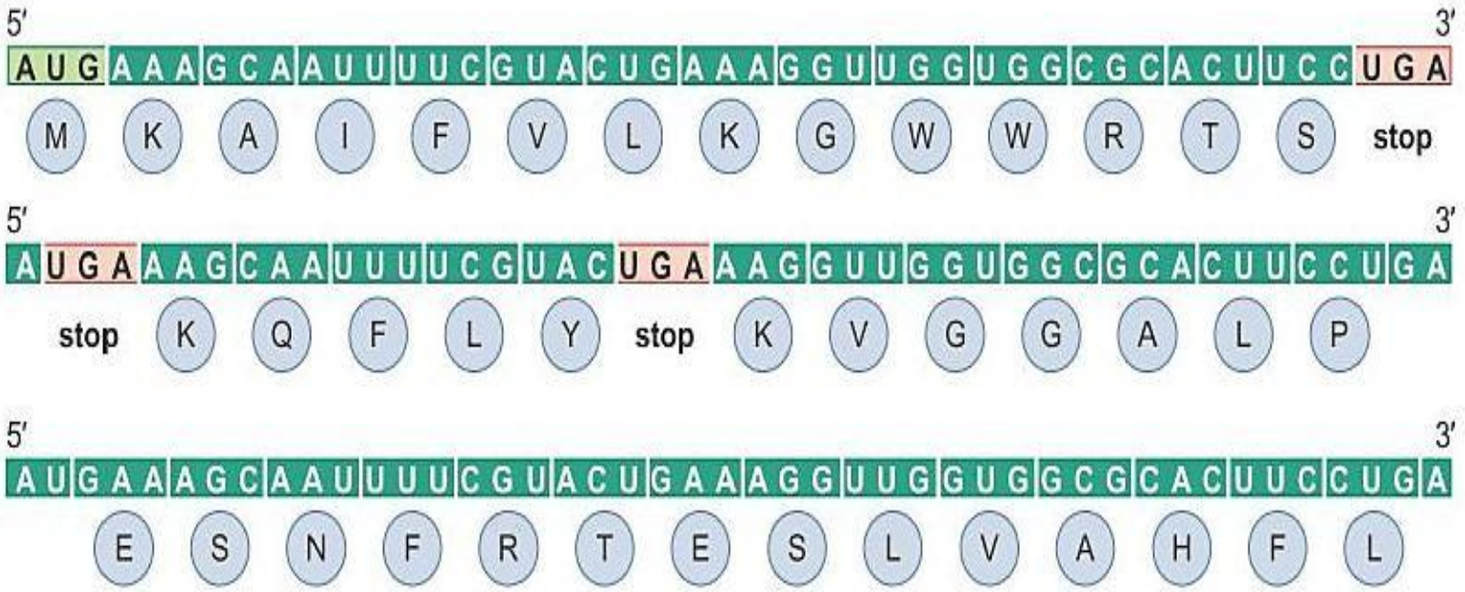
Amina Zee

Messenger RNA

- Because each codon is immediately adjacent to the next codon, and because codons are three nucleotides long, any stretch of mRNA could be translated in three different reading frames.

Amina Zee

Messenger RNA



Three possible reading frames of the *E. coli trp* leader sequence

Amina Zee

Messenger RNA

- Once translation starts, however, the reading frame is determined.
- Thus, by setting the location of the first codon, the start codon determines the location of all following codons.

Amina Zee

Messenger RNA

- Stop codons, of which there are three (5’-UAG-3’, 5’-UGA-3’, and 5’-UAA-3’), define the end of the ORF and signal termination of polypeptide synthesis.

Messenger RNA

- You can now understand the origin of the term open reading frame. It is a contiguous stretch of codons “read” in a particular frame (as set by the first codon) that is “open” to translation because it lacks a stop codon.

Amina Zee

Messenger RNA

- mRNAs contain at least one ORF. The number of ORFs per mRNA is different between eukaryotes and prokaryotes.
- Eukaryotic mRNAs almost always contain a single ORF.

Amina Zee

Messenger RNA

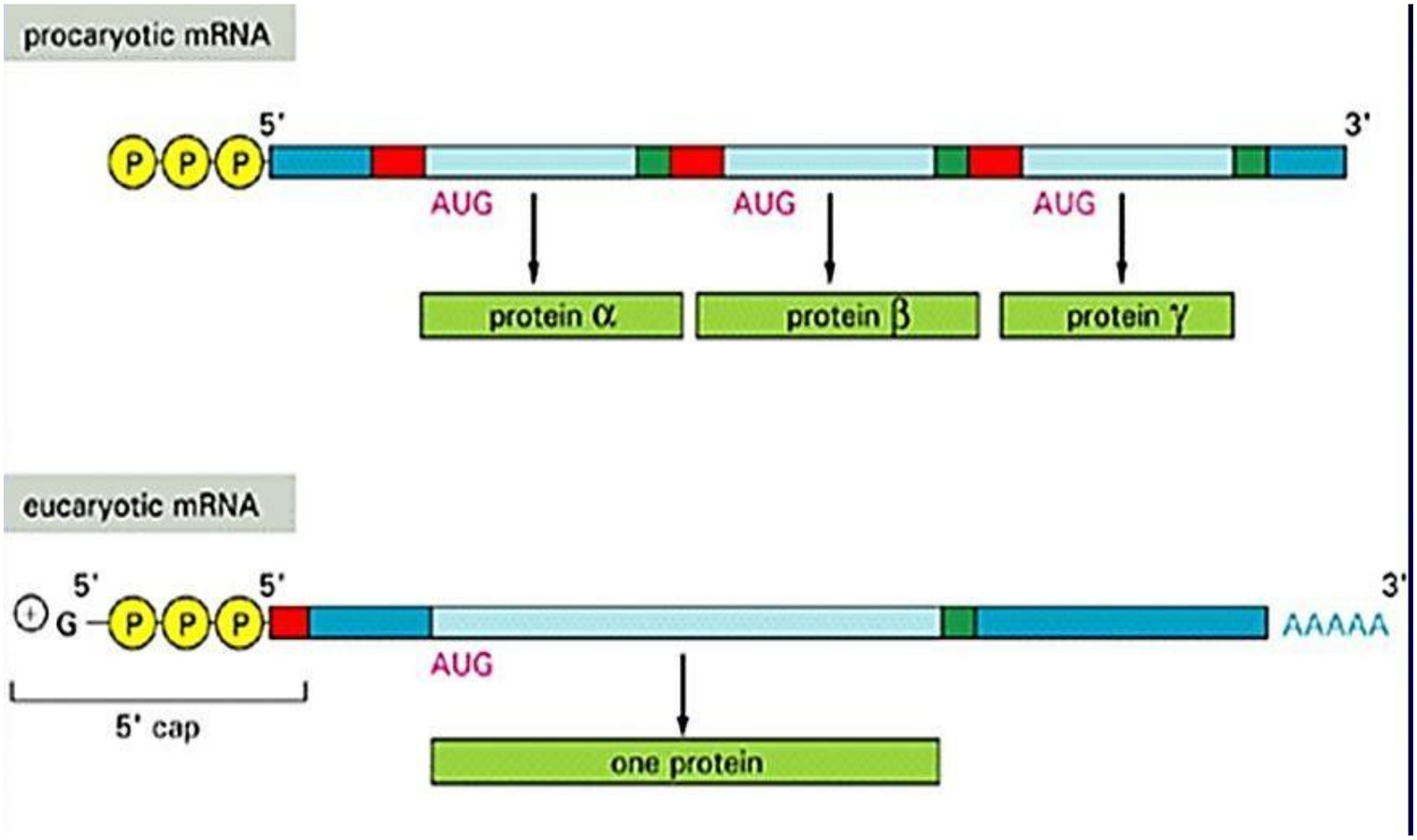
- In contrast, prokaryotic mRNAs frequently contain two or more ORFs.
- mRNAs containing multiple ORFs are known as polycistronic RNAs and those encoding a single ORF are known as monocistronic RNAs.

Messenger RNA

- The polycistronic mRNAs found in bacteria often encode proteins that perform related functions, such as different steps in the biosynthesis of an amino acid or nucleotide.

Amina Zee

Messenger RNA



Amina Zee

Lec 43

Prokaryotic mRNA

Amina Zee

Prokaryotic mRNAs

- For translation to occur, the ribosome must be recruited to the mRNA.
- Prokaryotic mRNAs have a ribosome-binding site that recruits the translational machinery.

Amina Zee

Prokaryotic mRNAs

- To facilitate binding by a ribosome, many prokaryotic ORFs contain a short sequence upstream (on the 5’ side) of the start codon called the ribosome-binding site (RBS).

Amina Zee

Prokaryotic mRNAs

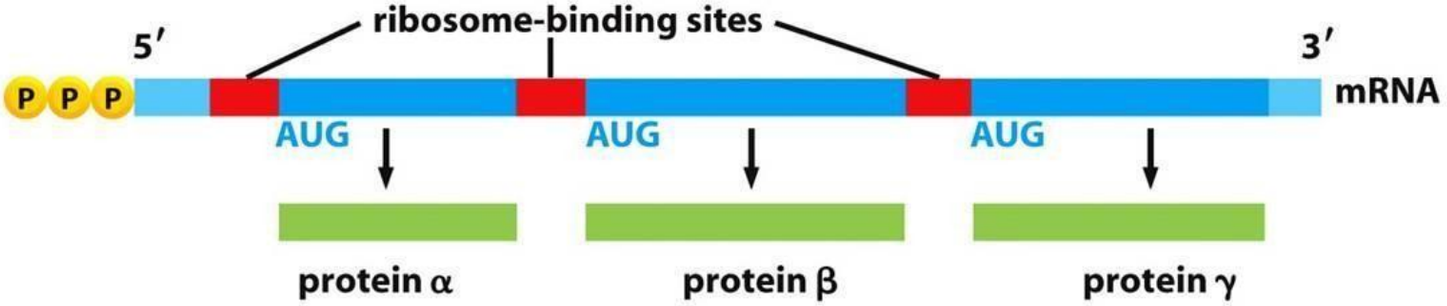
- This element is also referred to as a Shine Őalgarno sequence after the scientists who discovered it by comparing the sequences of multiple mRNAs.

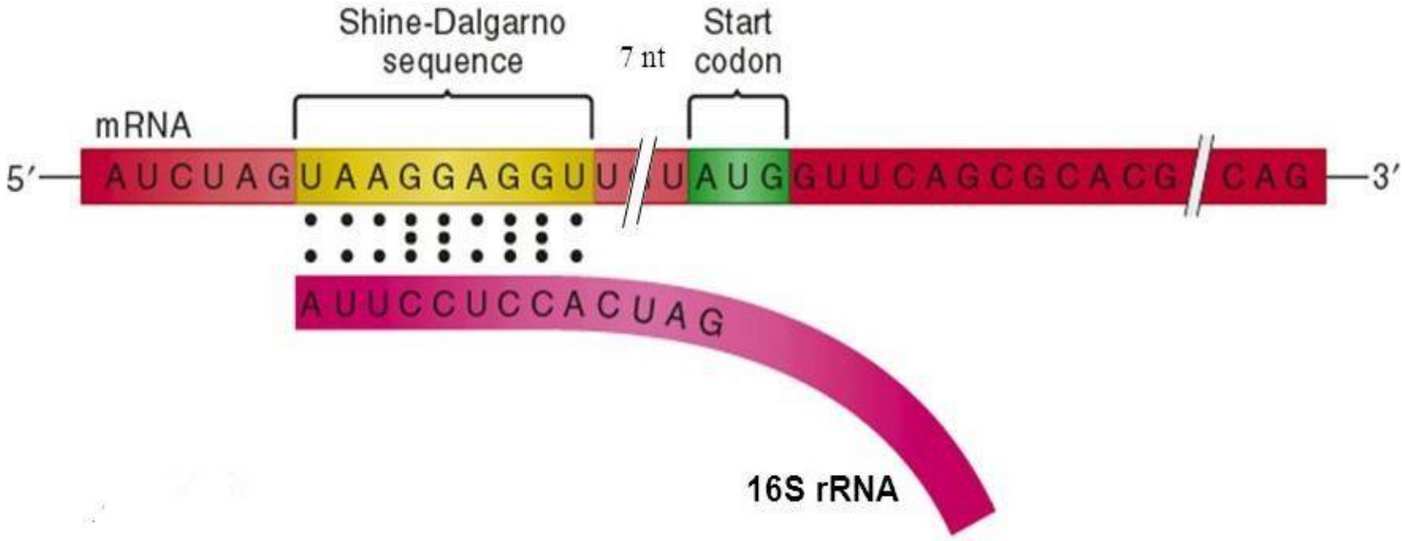
Amina Zee

Prokaryotic mRNAs

- The RBS, typically located 3–9 bp on the 5’ side of the start codon, is complementary to a sequence located near the 3’ end of one of the ribosomal RNA components, the 16S ribosomal RNA (rRNA).

Prokaryotic mRNAs





Amina Zee

Prokaryotic mRNAs

- The RBS base-pairs with this RNA, thereby aligning the ribosome with the beginning of the ORF. The core of this region of the 16S rRNA has the sequence 5’-CCUCCU-3’.

Amina Zee

Prokaryotic mRNAs

- Not surprisingly, prokaryotic RBS are most often a subset of the sequence 5’-AGGAGG-3’.

Prokaryotic mRNAs

- The extent of complementarity and the spacing between the RBS and the start codon has a strong influence on how actively a particular ORF is translated.

Prokaryotic mRNAs

- High complementarity and proper spacing promote active translation, whereas limited complementarity and/or poor spacing generally support lower levels of translation.

Prokaryotic mRNAs

- Some prokaryotic ORFs lack a strong RBS but are nonetheless actively translated.
- These ORFs are not the first ORF in an mRNA but instead are located just after another ORF in a polycistronic message.

Amina Zee

Prokaryotic mRNAs

- In these cases, the start codon of the downstream ORF often overlaps the 3’ end of the upstream ORF.

Prokaryotic mRNAs

- Thus, a ribosome that has just completed translating the upstream ORF is positioned to begin translating from the start codon for the downstream ORF.

Prokaryotic mRNAs

- This phenomenon of linked translation between overlapping ORFs is known as translational coupling.
- So in this situation translation of the downstream ORF requires translation of the upstream ORF.

Amina Zee

Prokaryotic mRNAs

- Indeed, with two translationally coupled genes, a mutation that leads to a premature stop codon in the upstream ORF also prevents translation of the downstream ORF.

END

Amina Zee

Lec 44

Eukaryotic RNA

Amina Zee

Eukaryotic mRNAs

- Unlike their prokaryotic counterparts, eukaryoticmRNAs recruit ribosomes using a specific chemical modification called the 5’ cap, which is located at the extreme 5’ end of the mRNA.

Amina Zee

Eukaryotic mRNAs

- The 5’ cap is a methylated guanine nucleotide that is joined to the 5’ end of the mRNA via an unusual 5’-to-5’ linkage.

Amina Zee

Eukaryotic mRNAs

- Created in three steps, the guanine nucleotide of the 5’ cap is connected to the 5’ end of the mRNA through three phosphate groups.

Amina Zee

Eukaryotic mRNAs

- The resulting 5’ cap is required to recruit the ribosome to the mRNA. Once bound to the mRNA, the ribosome moves in a 5’⟶ 3’ direction until it encounters a 5’-AUG-3’ start codon, a process called scanning.

Eukaryotic mRNAs

- Two other features of eukaryotic mRNAs stimulate translation. One feature is the presence, in some mRNAs, of a purine three bases upstream of the start codon and a guanine immediately downstream (5’-G/ANNAUGG-3’).

Amina Zee

Eukaryotic mRNAs

- This sequence was originally identified by Marilyn Kozak and is referred to as the Kozak sequence. Many eukaryotic mRNA slack these bases, but their presence increases the efficiency of translation.

Amina Zee

Eukaryotic mRNAs

- In contrast to the situation in prokaryotes, these bases are thought to interact with the initiator tRNA, not with an RNA component of the ribosome.

Amina Zee

Eukaryotic mRNAs

- A second feature that contributes to efficient translation is the presence of a poly-A tail at the extreme 3’ end of the mRNA.
- This tail is added enzymatically by the enzyme poly-A polymerase.

Amina Zee

Eukaryotic mRNAs

- Despite its location at the 3’ end of the mRNA, the poly-A tail enhances the level of translation of them RNA by enhancing the recruitment of key translation initiation factors.

Amina Zee

Eukaryotic mRNAs

- Importantly, in addition to their roles in translation, these 5’- and 3’-end modifications also protect eukaryotic mRNAs from rapid degradation.

END

Amina Zee

Lec 45

Transfer RNA

Amina Zee

Transfer RNA

- The heart of protein synthesis is the “translation” of nucleotide sequence information (in the form of codons) into amino acids.

Amina Zee

Transfer RNA

- This is accomplished by tRNA molecules, which act as adaptors between codons and the amino acids they specify.

Transfer RNA

- There are many types of tRNA molecules, but each is attached to a specific amino acid, and each recognizes a particular codon, or codons, in the mRNA (most tRNAs recognize more than one codon).

Amina Zee

Transfer RNA

- tRNA molecules are between 75 and 95 ribonucleotides in length.
- Although the exact sequence varies, all tRNAs have certain features in common.

Amina Zee

Transfer RNA

- First, all tRNAs end at the 3’ terminus with the sequence 5’-CCA-3’.
- Consistent with this absolute conservation, the 3’ end of this sequence is the site that is attached to the cognate amino acid.

Amina Zee

Transfer RNA

- A second striking aspect of tRNAs is the presence of several unusual bases in their primary structure.
- These unusual features are created post-transcriptionally by enzymatic modification of normal bases in the polynucleotide chain.

Amina Zee

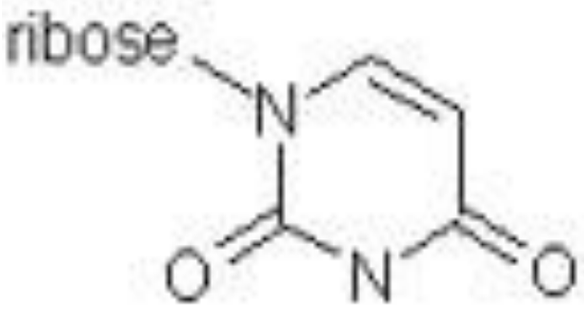
Transfer RNA

- For example, pseudouridine (ψU) is derived from uridine by an isomerization in which the site of attachment of the uracil base to the ribose is switched from the nitrogen at ring position 1 to the carbon at ring position 5.

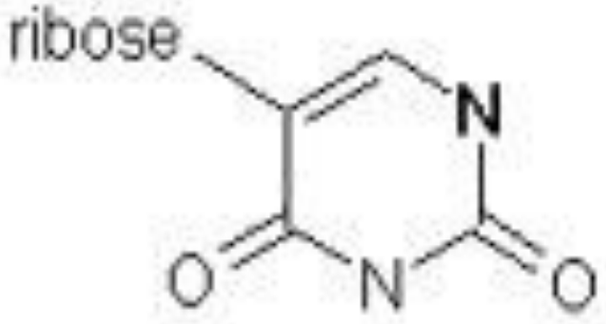
Transfer RNA

- Likewise, dihydrouridine (D) is derived from uridine by enzymatic reduction of the double bond between the carbons at positions 5 and 6.

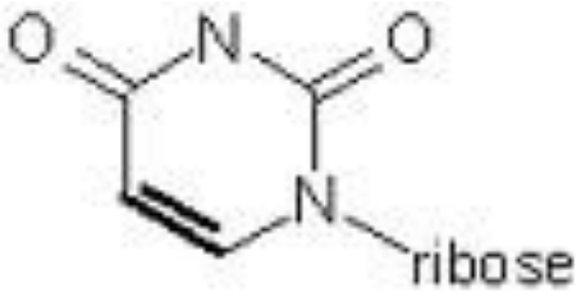
Transfer RNA



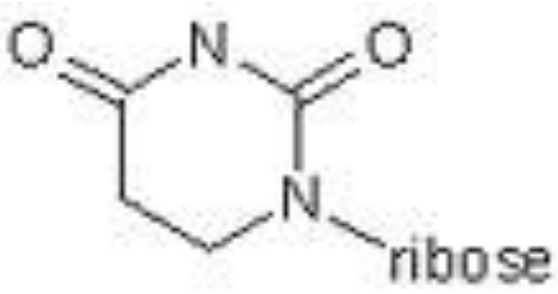
uridine



Pseudouridine



uridine



dihydrouridine

Amina Zee

Transfer RNA

- Other unusual bases found in tRNA include hypoxanthine, thymine, and methylguanine.
- These modified bases are not essential for tRNA function, but cells lacking these modified bases show reduced rates of growth.

Amina Zee

Transfer RNA

- This observation suggests that the modified bases lead to improved tRNA
- function.
- For example, hypoxanthine plays an important role in the process of codon recognition by certain tRNAs.

END

Amina Zee

Lec 46

Secondary structure of RNA

Amina Zee

Transfer RNA

- The heart of protein synthesis is the “translation” of nucleotide sequence information (in the form of codons) into amino acids.

Amina Zee

Transfer RNA

- This is accomplished by tRNA molecules, which act as adaptors between codons and the amino acids they specify.

Transfer RNA

- There are many types of tRNA molecules, but each is attached to a specific amino acid, and each recognizes a particular codon, or codons, in the mRNA (most tRNAs recognize more than one codon).

Amina Zee

Transfer RNA

- tRNA molecules are between 75 and 95 ribonucleotides in length.
- Although the exact sequence varies, all tRNAs have certain features in common.

Amina Zee

Transfer RNA

- First, all tRNAs end at the 3’ terminus with the sequence 5’-CCA-3’.
- Consistent with this absolute conservation, the 3’ end of this sequence is the site that is attached to the cognate amino acid.

Amina Zee

Transfer RNA

- A second striking aspect of tRNAs is the presence of several unusual bases in their primary structure.
- These unusual features are created post-transcriptionally by enzymatic modification of normal bases in the polynucleotide chain.

Amina Zee

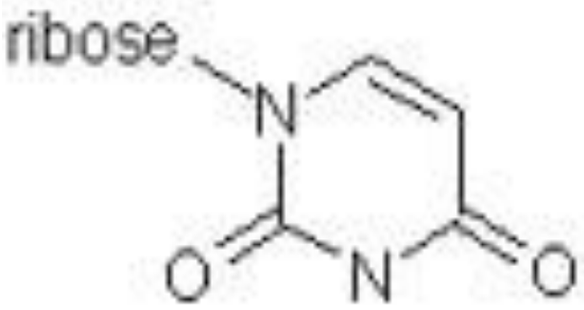
Transfer RNA

- For example, pseudouridine (ψU) is derived from uridine by an isomerization in which the site of attachment of the uracil base to the ribose is switched from the nitrogen at ring position 1 to the carbon at ring position 5.

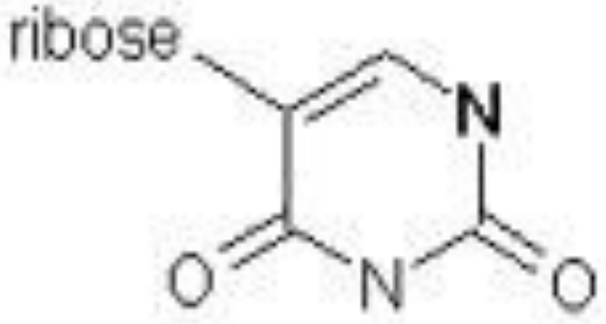
Transfer RNA

- Likewise, dihydrouridine (D) is derived from uridine by enzymatic reduction of the double bond between the carbons at positions 5 and 6.

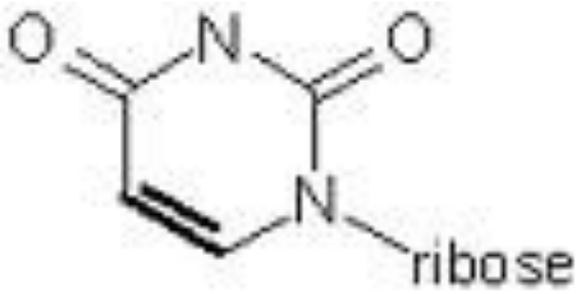
Transfer RNA



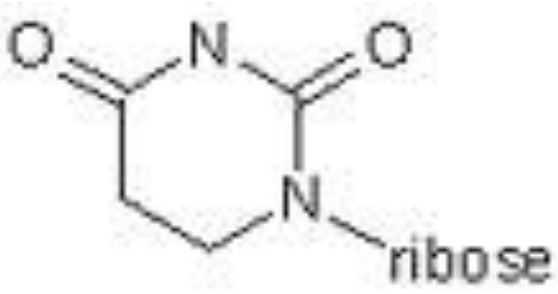
uridine



Pseudouridine



uridine



dihydrouridine

Amina Zee

Transfer RNA

- Other unusual bases found in tRNA include hypoxanthine, thymine, and methylguanine.
- These modified bases are not essential for tRNA function, but cells lacking these modified bases show reduced rates of growth.

Amina Zee

Transfer RNA

- This observation suggests that the modified bases lead to improved tRNA
- function.
- For example, hypoxanthine plays an important role in the process of codon recognition by certain tRNAs.

END

Amina Zee

Lec 47

Attachment of Amino Acids to tRNA

Amina Zee

Attachment of Amino Acids to tRNA

- tRNA molecules to which an amino acid is attached are said to be charged, and tRNAs that lack an amino acid are said to be uncharged.

Amina Zee

Attachment of Amino Acids to tRNA

- Charging requires an acyl linkage between the carboxyl group of the amino acid and the 2’- or 3’-hydroxyl group of the adenosine nucleotide that protrudes from the acceptor stem at the 3’ end of the tRNA.

Amina Zee

Attachment of Amino Acids to tRNA

- This acyl linkage is a high-energy bond because its hydrolysis results in a large change in free energy.

Amina Zee

Attachment of Amino Acids to tRNA

- This is significant for protein synthesis: the energy released when this acyl bond is broken is coupled to the formation of the peptide bonds that link amino acids to each other in polypeptide chains.

Amina Zee

Attachment of Amino Acids to tRNA

- All aminoacyl-tRNA synthetases attach an amino acid to a tRNA in two enzymatic steps:
  - Adenylation
  - tRNA charging

Amina Zee

Attachment of Amino Acids to tRNA

- Step one is adenylation in which the amino acid reacts with ATP to become adenylylated with the concomitant release of pyrophosphate.

Amina Zee

Attachment of Amino Acids to tRNA

- Adenylylation refers to transfer of AMP, as opposed to adenylation, which would indicate the transfer of adenine.

Amina Zee

Attachment of Amino Acids to tRNA

- The principal driving force for the adenylylation reaction is the subsequent hydrolysis of pyrophosphate by pyrophosphatase.

Amina Zee

Attachment of Amino Acids to tRNA

- As a result of adenylylation, the amino acid is attached to adenylic acid via a high-energy ester bond in which the carbonyl group of the amino acid is joined to the phosphoryl group of AMP.

Amina Zee

Attachment of Amino Acids to tRNA

- Step two is tRNA Charging in which the adenylylated amino acid, which remains tightly bound to the synthetase, reacts with tRNA.

Amina Zee

Attachment of Amino Acids to tRNA

- This reaction results in the transfer of the amino acid to the 3’ end of the tRNA via the 2’- or 3’-hydroxyl and the release of AMP.

Attachment of Amino Acids to tRNA

- There are two classes of tRNA synthetases:
- Class I enzymes attach the amino acid to the 20-OH of the tRNA and are generally monomeric.

Amina Zee

Attachment of Amino Acids to tRNA

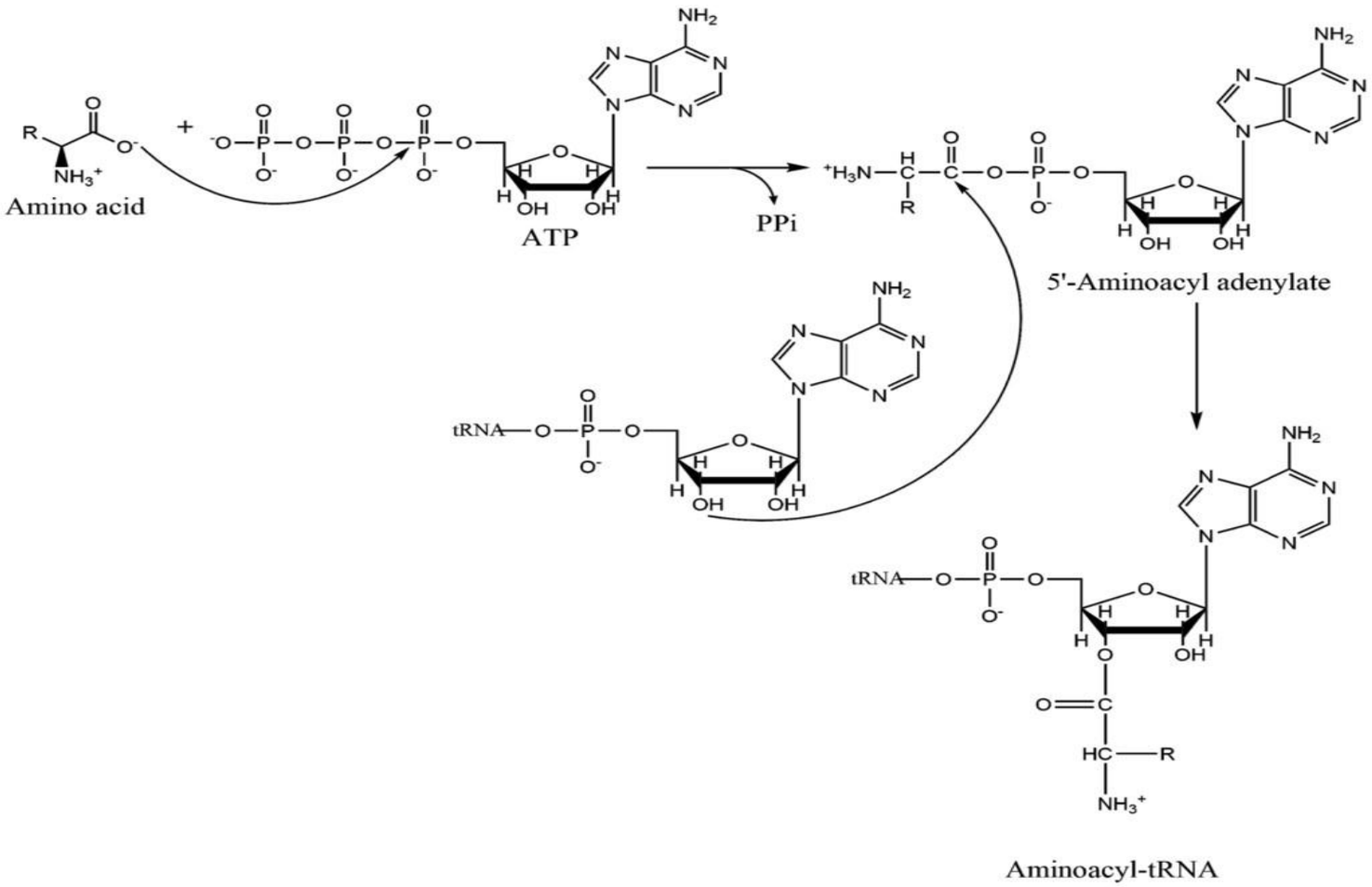
- Class II enzymes attach the amino acid to the 30-OH of the tRNA and are typically dimeric or tetrameric.

Amina Zee

Attachment of Amino Acids to tRNA

- Although the initial coupling between the tRNA and the amino acid is different, once released from the synthetase, the amino acid rapidly equilibrates between attachment at the 3'-OH and the 2'-OH.

Attachment of Amino Acids to tRNA



Amina Zee

Attachment of Amino Acids to tRNA

- Each of the 20 amino acids is attached to the appropriate tRNA by a single, dedicated tRNA synthetase.

Amina Zee

Attachment of Amino Acids to tRNA

- Because most amino acids are specified by more than one codon, it is not uncommon for one synthetase to recognize and charge more than one tRNA (known as isoaccepting tRNAs).

Amina Zee

Attachment of Amino Acids to tRNA

- Nevertheless, the same tRNA synthetase is responsible for charging all tRNAs for a particular amino acid.
- Thus, one and only one tRNA synthetase attaches each amino acid to all of the appropriate tRNAs.

END

Amina Zee

Lec 48

The Ribosomes

Amina Zee

The Ribosomes

- The ribosome is the macromolecular machine that directs the synthesis of proteins.
- The ribosome is larger and more complex than the minimal machinery required for DNA or RNA synthesis.

Amina Zee

The Ribosomes

- The machinery for polymerizing amino acids is composed of at least three RNA molecules and more than 50 different proteins, with an overall molecular mass of >2.5 MDa.

Amina Zee

The Ribosomes

- Compared with the speed of DNA replication i.e., 200 – 1000 nucleotides per second; translation takes place at a rate of only two to 20 amino acids per second.

The Ribosomes

- In prokaryotes, the transcription machinery and the translation machinery are located in the same compartment. Thus, the ribosome can commence translation of the mRNA as it emerges from the RNA polymerase.

Amina Zee

The Ribosomes

- This situation allows the ribosome to proceed in tandem with the RNA polymerase as it elongates the transcript.

Amina Zee

The Ribosomes

- Recall that the 5’ end of an RNA is synthesized first, and thus the ribosome, which begins translation at the 5’ end of the mRNA, can start translating a nascent transcript as soon as it emerges from the RNA polymerase.

END

Amina Zee

Lec 49

Formation of Peptide Bonds

Amina Zee

Formation of Peptide Bonds

- Each new amino acid is added to the carboxyl terminus of the growing polypeptide chain (often referred to as synthesis in the amino- to carboxy-terminal direction).

Amina Zee

Formation of Peptide Bonds

- The ribosome catalyzes a single chemical reaction — the formation of a peptide bond.

Amina Zee

Formation of Peptide Bonds

- This reaction occurs between the amino acid residue at the carboxy-terminal end of the growing polypeptide and the incoming amino acid to be added to the chain.

Amina Zee

Formation of Peptide Bonds

- Both the growing chain and the incoming amino acid are attached to tRNAs; as a result, during peptide-bond formation, the growing polypeptide is continuously attached to a tRNA.

Amina Zee

Formation of Peptide Bonds

- The actual substrates for each round of amino acid addition are two charged species of tRNAs — an aminoacyl-tRNA and a peptidyl-tRNA.

Amina Zee

Formation of Peptide Bonds

- As you know the aminoacyl-tRNA is attached at its 3’ end to the carboxyl group of the amino acid. The peptidyl-tRNA is attached in exactly the same manner (at its 3’ end) to the carboxyl terminus of the growing polypeptide chain.

Amina Zee

Formation of Peptide Bonds

- The bond between the aminoacyl-tRNA and the amino acid is not broken during the formation of the next peptide bond.

Amina Zee

Formation of Peptide Bonds

- Instead, the bond between the peptidyl-tRNA and the growing polypeptide chain is broken as the growing chain is attached to the amino group of the amino acid attached to the aminoacyl-tRNA to form a new peptide bond.

Amina Zee

Formation of Peptide Bonds

- To catalyze peptide-bond formation, the 3’ ends of these two tRNAs are brought into close proximity by the ribosome.

Amina Zee

Formation of Peptide Bonds

- The resulting tRNA positioning allows the amino group of the amino acid attached to aminoacyl-tRNA to attack the carbonyl group of the most carboxy-terminal amino acid attached to the peptidyl-tRNA.

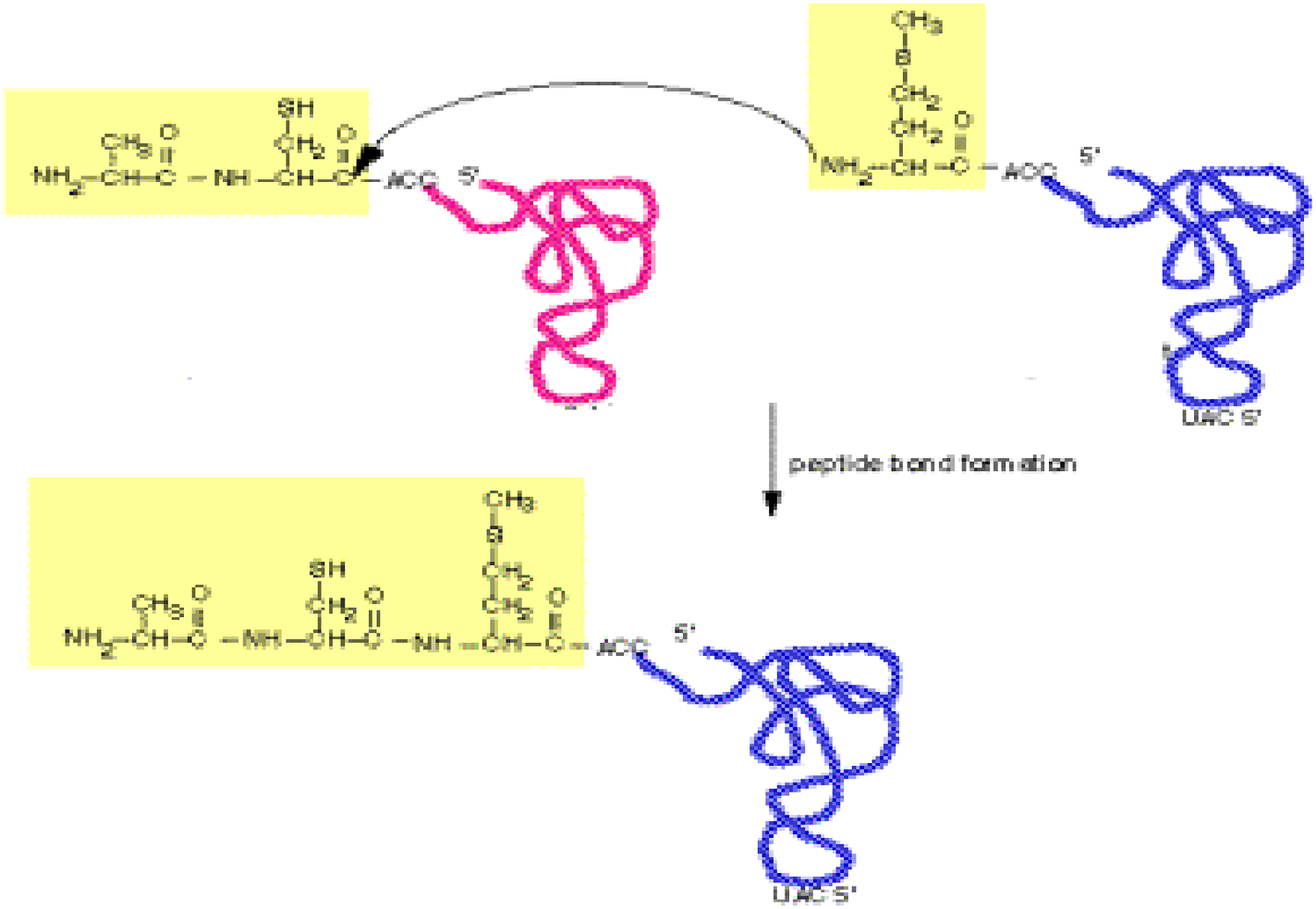
Amina Zee

Formation of Peptide Bonds

- The result of this nucleophilic attack is the formation of a new peptide bond between the amino acids attached to the tRNAs and the release of the polypeptide chain from the peptidyl tRNA.

Amina Zee

Formation of Peptide Bonds



Amina Zee

Formation of Peptide Bonds

- There are two consequences of this method of polypeptide synthesis. First, this mechanism of peptide-bond formation requires that the amino terminus of the protein be synthesized before the carboxyl terminus.

Amina Zee

Formation of Peptide Bonds

- Second, the growing polypeptide chain is transferred from the peptidyl-tRNA to the aminoacyl-tRNA. For this reason, the reaction to form a new peptide bondis called the peptidyl transferase reaction.

Amina Zee

Formation of Peptide Bonds

- Interestingly, peptide-bond formation takes place without the simultaneous hydrolysis of a nucleoside triphosphate.

Amina Zee

Formation of Peptide Bonds

- This is because peptide-bond formation is driven by breaking the high-energy acyl bond that joins the growing polypeptide chain to the tRNA.

Amina Zee

Formation of Peptide Bonds

- Recall that this bond was created during the tRNA synthetase – catalyzed reaction that is responsible for charging tRNA.
- And the charging reaction involves the hydrolysis of a molecule of ATP.

Amina Zee

Formation of Peptide Bonds

- Thus, the energy for peptide-bond formation originates from the molecule of ATP that was hydrolyzed during the tRNA charging reaction.

END

Amina Zee

Lec 50

Binding Sites on Ribosomes for tRNA

Amina Zee

Binding Sites on Ribosomes for tRNA

- The ribosome is composed of two subassemblies of RNA and protein known as the large and small subunits.

Amina Zee

Binding Sites on Ribosomes for tRNA

- The large subunit contains the peptidyl transferase center, which is responsible for the formation of peptide bonds.

Amina Zee

Binding Sites on Ribosomes for tRNA

- The small subunit contains the decoding center in which charged tRNAs read or "decode" the codon units of the mRNA.

Amina Zee

Binding Sites on Ribosomes for tRNA

- Both the decoding center and the peptidyl transferase center are buried within the intact ribosome.

Amina Zee

Binding Sites on Ribosomes for tRNA

- Yet, mRNA must be threaded through the decoding center during translation, and the nascent polypeptide chain must escape from the peptidyl transferase center.

Amina Zee

Binding Sites on Ribosomes for tRNA

- How do these polymers enter and exit the ribosome?
- The answer is provided by the structure of the ribosome, which reveals that there are "tunnels" in and out of the ribosome.

Amina Zee

Binding Sites on Ribosomes for tRNA

- To perform the peptidyl transferase reaction, the ribosome must be able to bind at least two tRNAs simultaneously.
- In fact, the ribosome contains three tRNA-binding sites, called the A-, P-, and E-sites.

Amina Zee

Binding Sites on Ribosomes for tRNA

- The A-site is the binding site for the aminoacylated-tRNA, the P-site is the binding site for the peptidyl-tRNA, and

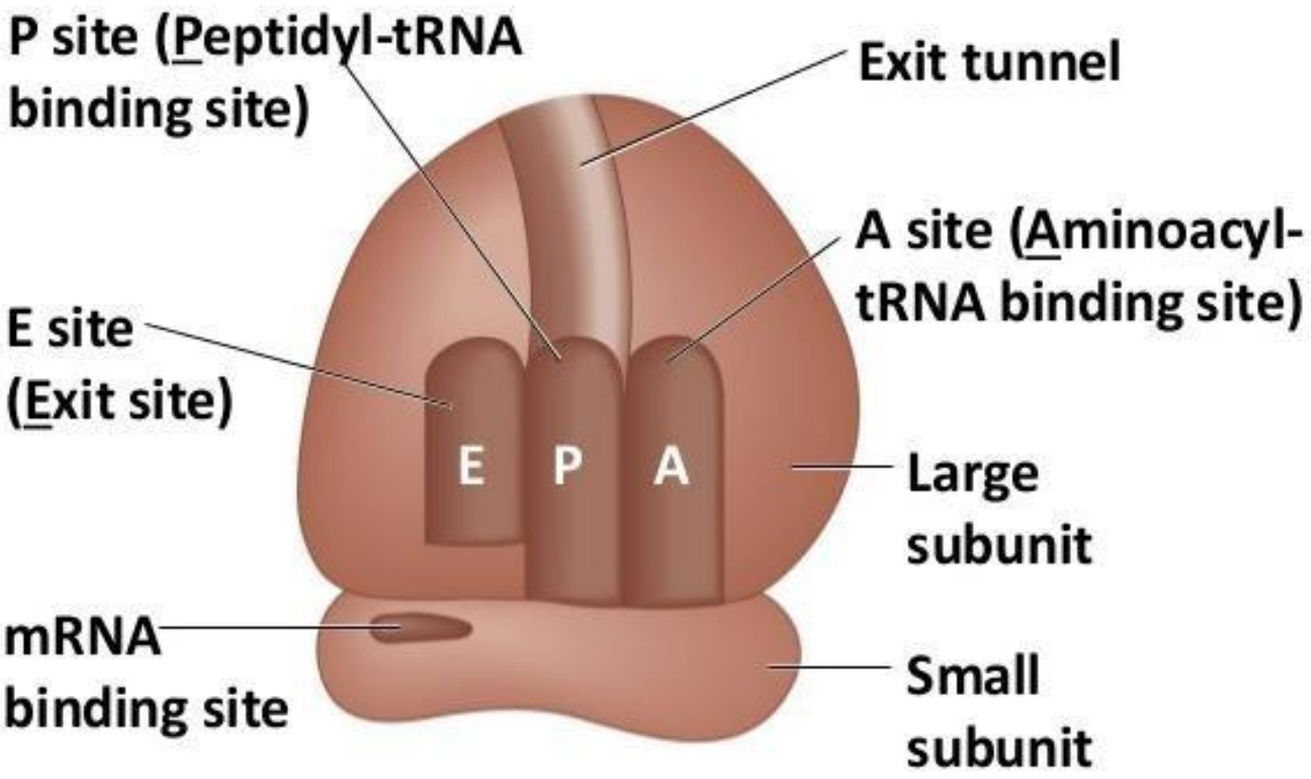
Amina Zee

Binding Sites on Ribosomes for tRNA

- The E-site is the binding site for the tRNA that is released after the growing polypeptide chain has been transferred to the aminoacyl-tRNA (E is for “exiting”).

Amina Zee

Binding Sites on Ribosomes for tRNA



Amina Zee

Binding Sites on Ribosomes for tRNA

- Each tRNA binding site is formed at the interface between the large and the small subunits of the ribosome.

Amina Zee

Binding Sites on Ribosomes for tRNA

- In this way, the bound tRNAs can span the distance between the peptidyl transferase center in the large subunit and the decoding center in the small subunit.

Amina Zee

Binding Sites on Ribosomes for tRNA

- The 3' ends of the tRNAs that are coupled to the amino acid or to the growing peptide chain are adjacent to the large subunit.
- The anticodon loops of the bound tRNAs are located adjacent to the small subunit.

END

Amina Zee

Lec 51

Initiation of Translation

Amina Zee

Initiation of Translation

- For translation to be successfully initiated, three events must occur:-
- i) the ribosome must be recruited to the mRNA.
- ii) a charged tRNA must be placed into the P-site of the ribosome.

Amina Zee

Initiation of Translation

- iii) the ribosome must be precisely positioned over the start codon.

Amina Zee

Initiation of Translation

- The correct positioning of the ribosome over the start codon is critical because this establishes the reading frame for the translation of the mRNA.

Amina Zee

Initiation of Translation

- In prokaryotes, the assembly of the ribosome on an mRNA occurs one subunit at a time. The small subunit associates with the mRNA first.

Amina Zee

Initiation of Translation

- In prokaryotes, the association of the small subunit with the mRNA is mediated by base-pairing interactions between the RBS and the 16S rRNA.

Amina Zee

Initiation of Translation

- For ideally positioned RBSs, the small subunit is positioned on the mRNA such that the start codon will be in the P-site when the large subunit joins the complex.

Amina Zee

Initiation of Translation

- The large subunit joins its partner only at the very end of the initiation process, just before the formation of the first peptide bond.
- Thus, many of the key events of translation initiation occur in the absence of the full ribosome.

Amina Zee

Initiation of Translation

- Translation initiation is the only time a tRNA binds to the P-site without previously occupying the A-site. This event requires a special tRNA known as the initiator tRNA.

Amina Zee

Initiation of Translation

- The initiator tRNA base-pairs with the start codon (AUG or GUG). AUG and GUG have a different meaning when they occur within an ORF, where they are read by tRNAs for methionine and valine, respectively.

Amina Zee

Initiation of Translation

- Although the initiator tRNA is first charged with a methionine, a formyl group is rapidly added to the methionine amino group by a separate enzyme (Met-tRNA transformylase).

Amina Zee

Initiation of Translation

- Thus rather than valine or methionine, the initiator tRNA is coupled to N-formyl methionine. The charged initiator tRNA is referred to as fMet-tRNA^{fMet}.

Amina Zee

Initiation of Translation

- Because N-formyl methionine is the first amino acid to be incorporated into a polypeptide chain, one might think that all prokaryotic proteins have a formyl group at their amino termini.

Amina Zee

Initiation of Translation

- This is not the case, however, because an enzyme known as a deformylase removes the formyl group from the amino terminus during or after the synthesis of the polypeptide chain.

Initiation of Translation

- In fact, many mature prokaryotic proteins do not even start with a methionine; aminopeptidases often remove the amino-terminal methionine as well as one or two additional amino acids.

END

Amina Zee

Lec 52

The Initiation Factors

Amina Zee

The Initiation Factors

- The initiation of prokaryotic translation commences with the small subunit and is catalyzed by three translation initiation factors called IF1, IF2, and IF3.
- Each factor facilitates a key step in the initiation process.

Amina Zee

The Initiation Factors

- **IF1:**
  - It prevents tRNAs from binding to the portion of the small subunit that will become part of the A-site.

The Initiation Factors

- **IF2:**
  - It is a GTPase that interacts with three key components of the initiation machinery: the small subunit, IF1, and the charged initiator tRNA (fMet-tRNA ^{fMet}).

Amina Zee

The Initiation Factors

- By interacting with these components, IF2 facilitates the association of fMet-tRNA ^{fMet} with the small subunit and prevents other charged tRNAs from associating with the small subunit.

Amina Zee

The Initiation Factors

- **IF3:**
  - It binds to the small subunit and blocks it from re-associating with a large subunit. Because initiation requires a free small subunit, the binding of IF3 is critical for a new cycle of translation.

The Initiation Factors

- IF3 becomes associated with the small subunit at the end of a previous round of translation when it helps to dissociate the 70S ribosome into its large and small subunits.

Amina Zee

The Initiation Factors

- Each of the initiation factors binds at, or near, one of the three tRNA binding sites on the small subunit.

Amina Zee

The Initiation Factors

- Consistent with its role in blocking the binding of charged tRNAs to the A-site, IF1 binds directly to the portion of the small subunit that will become the A-site.

Amina Zee

The Initiation Factors

- IF2 binds to IF1 and reaches over the A-site into the P-site to contact the fMet - tRNA ^{fMet}.
- Finally, IF3 occupies the part of the small subunit that will become the E-site.

Amina Zee

The Initiation Factors

- Thus, of the three potential tRNA-binding sites on the small subunit, only the P-site is capable of binding a tRNA in the presence of the initiation factors.

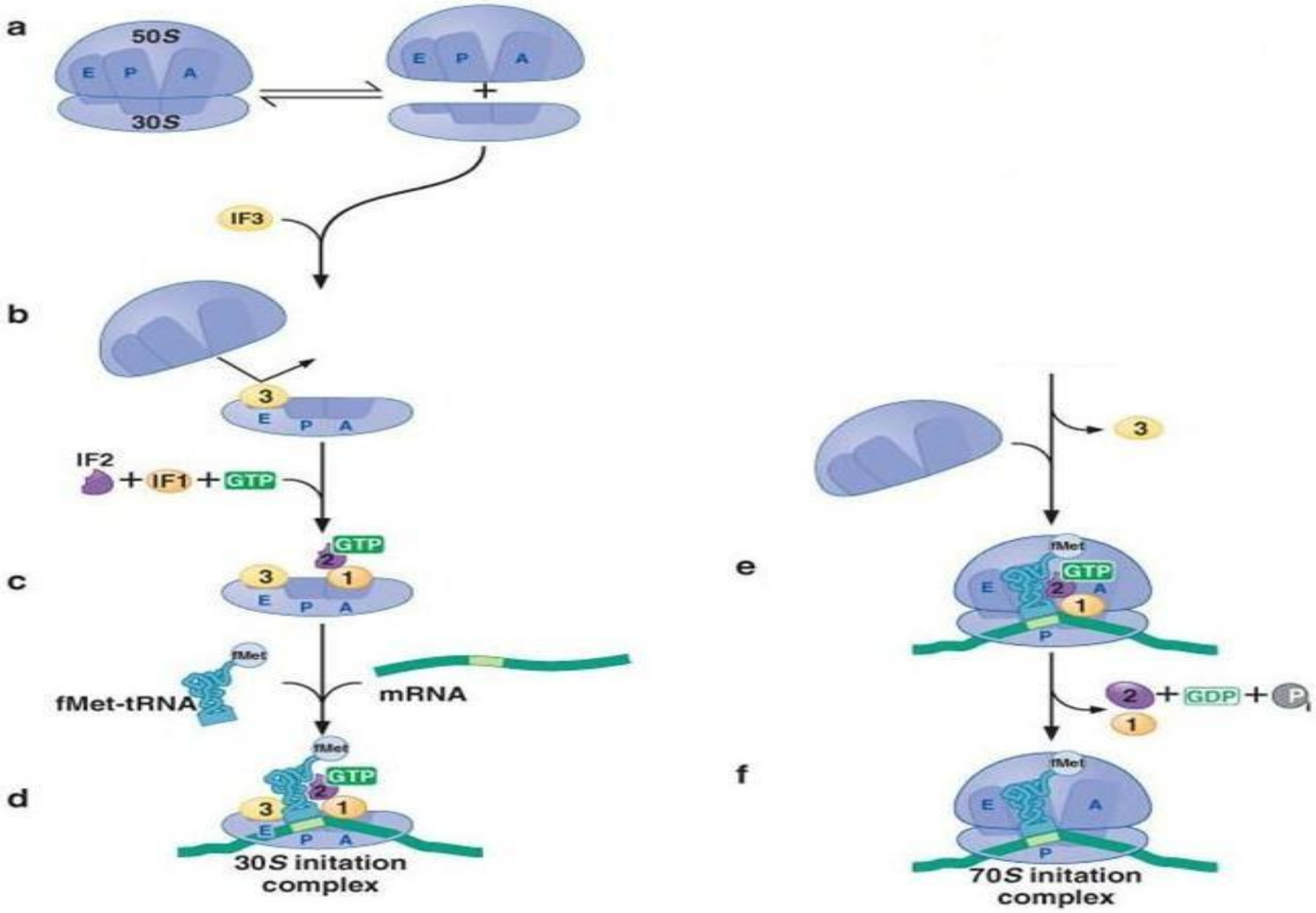
Amina Zee

The Initiation Factors

- With all three initiation factors bound, the small subunit is prepared to bind to the mRNA and the initiator tRNA .
- These two RNAs can bind in either order and independently of each other.

Amina Zee

The Initiation Factors



Amina Zee

The Initiation Factors

- Binding fMet-tRNA ^{fMet} to the small subunit is facilitated by its interactions with IF2 bound to GTP and base pairing between the anticodon and the start codon of the mRNA.

Amina Zee

The Initiation Factors

- Similarly, base pairing between the fMet-tRNA ^{fMet} and the mRNA serves to position the start codon in the P-site.

Amina Zee

The Initiation Factors

- The last step of initiation involves the association of the large subunit to create the 70S initiation complex.
- When the start codon and fMet-tRNA^{fMet} base-pair, the small subunit undergoes a change in conformation.

Amina Zee

The Initiation Factors

- This altered conformation results in the release of IF3.
- In the absence of IF3, the large subunit is free to bind to the small subunit with its cargo of IF1, IF2, mRNA, and fMet-$tRNA^{fMet}$.

Amina Zee

The Initiation Factors

- In particular, IF2 acts as an initial docking site of the large subunit, and this interaction subsequently stimulates the GTPase activity of IF2.GTP.

The Initiation Factors

- IF2 bound to GDP has reduced affinity for the ribosome and the initiator tRNA, leading to the release of IF2.GDP as well as IF1 from the ribosome.

Amina Zee

The Initiation Factors

- Thus, the net result of initiation is the formation of an intact (70S) ribosome assembled at the start site of the mRNA with fMet-$tRNA^{fMet}$ in the P-site and an empty A-site.

Amina Zee

The Initiation Factors

- The ribosome – mRNA complex is now poised to accept a charged tRNA into the A-site and commence polypeptide synthesis.

END

Amina Zee

Lec 53

Translation Elongation

Amina Zee

Translation Elongation

- Once the ribosome is assembled with the charged initiator tRNA in the P site, polypeptide synthesis can begin.
- There are three key events that must occur for the correct addition of each amino acid.

Amina Zee

Translation Elongation

- First, the correct aminoacyl-tRNA is loaded into the A site of the ribosome as dictated by the A-site codon.

Amina Zee

Translation Elongation

- Second, a peptide bond is formed between the aminoacyl-tRNA in the A site and the peptide chain that is attached to the peptidyl-tRNA in the P site.

Amina Zee

Translation Elongation

- This peptidyl transferase reaction results in the transfer of the growing polypeptide from the tRNA in the P site to the amino acid moiety of the charged tRNA in the A site.

Amina Zee

Translation Elongation

- Third, the resulting peptidyl-tRNA in the A site and its associated codon must be translocated to the P site so that the ribosome is poised for another cycle of codon recognition and peptide bond formation.

Amina Zee

Translation Elongation

- As with the original positioning of the mRNA, this shift must occur precisely to maintain the correct reading frame of the message.
- Two auxiliary proteins known as elongation factors control these events.

Amina Zee

Translation Elongation

- Both of these factors use the energy of GTP binding and hydrolysis to enhance the rate and accuracy of ribosome function.

Amina Zee

Translation Elongation

- Unlike the initiation of translation, the mechanism of elongation is highly conserved between prokaryotic and eukaryotic cells.

Amina Zee

Translation Elongation

- Aminoacyl-tRNAs do not bind to the ribosome on their own. Instead, they are "escorted " to the ribosome by the elongation factor EF-Tu.

Amina Zee

Translation Elongation

- Once a tRNA is aminocylated, EF-Tu binds to the tRNA's 3 ' end, masking the coupled amino acid. This interaction prevents the bound aminoacyl-tRNA from participating in peptide bond formation until it is released from EF-Tu.

Amina Zee

Translation Elongation

- Like the initiation factor IF2, the elongation factor EF-Tu binds and hydrolyzes GTP and the type of guanine nucleotide bound governs its function.

Amina Zee

Translation Elongation

- EF-Tu can only bind to an aminoacyl-tRNA when it is associated with GTP. EF-Tu bound to GDP, or lacking any bound nucleotide, shows little affinity for aminoacyl-tRNAs.

Amina Zee

Translation Elongation

- Thus, when EF-Tu hydrolyzes its bound GTP, any associated aminoacyl-tRNA is released.

Amina Zee

Translation Elongation

- The trigger that activates the EF-Tu GTPase is the same domain on the large subunit of the ribosome that activates the IF2 GTPase when the large subunit joins the initiation complex.

Amina Zee

Translation Elongation

- This domain is known as the factor binding center.
- EF-Tu only interacts with the factor binding center after the tRNA is loaded into the A site and a correct codon-anticodon match is made.

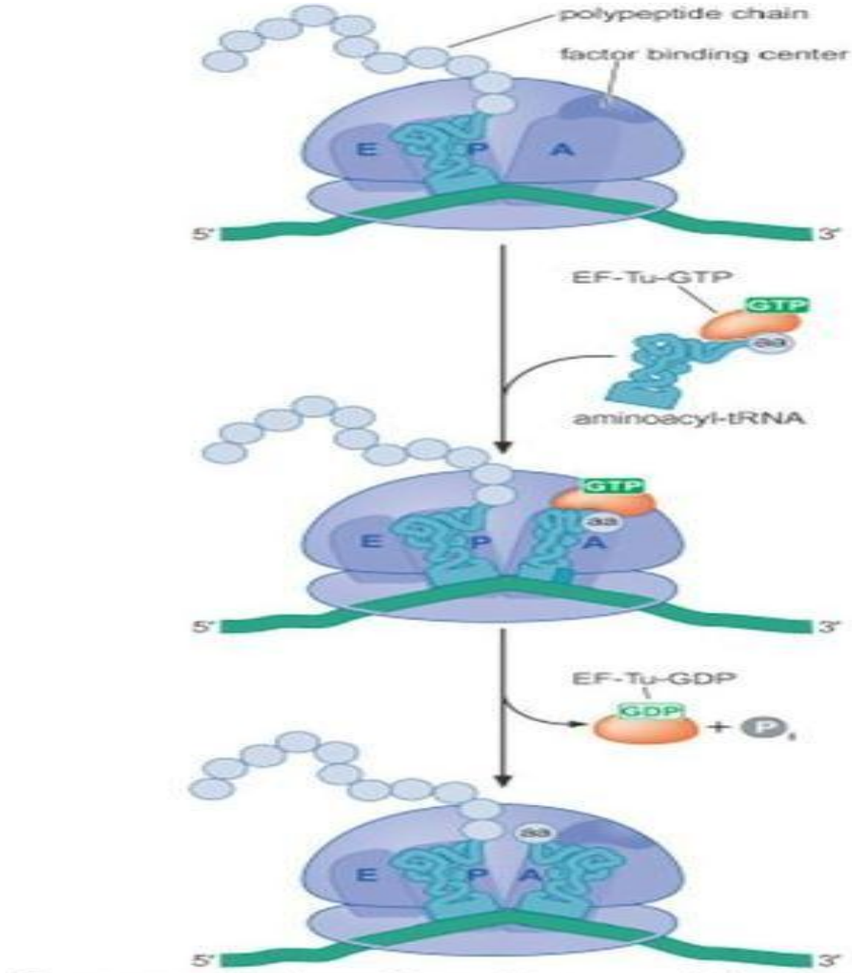
Amina Zee

Translation Elongation

- At this point, EF-Tu hydrolyzes its bound GTP and is released from the ribosome.
- The control of GTP hydrolysis by EF-Tu is critical to the specificity of translation.

Amina Zee

Translation Elongation



Pearson Education, Inc., publishing as Benjamin

Amina Zee

Translation Elongation

- The error rate of translation is between 10^{-3} to 10^{-4}.
- The ultimate basis for the selection of the correct aminoacyl-tRNA is the base pairing between the charged tRNA and the codon displayed in the A site of the ribosome.

Amina Zee

Translation Elongation

- However, in some cases, the base pairing in the anticodon-codon interaction may be mismatched, yet the ribosome rarely allows such mismatched aminoacyl-tRNAs to continue in the translation process.

END

Amina Zee

Lec 54

The Ribosome Is a Ribozyme

Amina Zee

The Ribosome Is a Ribozyme

- Once the correctly charged tRNA has been placed in the A site and has rotated into the peptidyl transferase center, peptide bond formation takes place.

Amina Zee

The Ribosome Is a Ribozyme

- This reaction is catalyzed by RNA, specifically the 23 S rRNA component of the large subunit.

Amina Zee

The Ribosome Is a Ribozyme

- Early evidence for this came from experiments in which it was shown that a large subunit that had been largely stripped of its proteins was still able to carry out peptide bond formation.

Amina Zee

The Ribosome Is a Ribozyme

- Proof that the peptidyl transferase is entirely composed of RNA has come from the high-resolution, three-dimensional structure of the ribosome, which reveals that no amino acid is located closer than 18 Aº from the active site.

Amina Zee

The Ribosome Is a Ribozyme

- Because catalysis requires distances in the 1 - 3 Aº range, it is clear that the peptidyl transferase center is a ribozyme. That is an enzyme composed of RNA.

Amina Zee

The Ribosome Is a Ribozyme

- How does the 23 S rRNA catalyze peptide bond formation?
- The exact mechanism remains to be determined, but some answers to this question are beginning to emerge.

Amina Zee

The Ribosome Is a Ribozyme

- First, base-pairing between the 23 S rRNA and the CCA ends of the tRNAs in the A and the P sites help to position the alpha-amino group of the aminoacyl-tRNA to attack the carbonyl group of the growing polypeptide attached to the peptidyl-tRNA.

Amina Zee

The Ribosome Is a Ribozyme

- These interactions are also likely to stabilize the aminoacyl-tRNA after accommodation.

Amina Zee

The Ribosome Is a Ribozyme

- Because close proximity of substrates is rarely sufficient to generate high levels of catalysis, it is hypothesized that other elements of the ribosomal RNA change the chemical environment of the peptidyl transferase active site.

Amina Zee

The Ribosome Is a Ribozyme

- For example, it has been proposed that nucleotides in the peptidyl transferase center accept a hydrogen from the alpha amino group of the aminoacyl-tRNA, making the associated nitrogen a stronger nucleophile.

Amina Zee

The Ribosome Is a Ribozyme

- This is a common mechanism used by many proteins to stimulate nucleophilic attack of carbonyl groups.

END

Amina Zee

Lec 55

Translocation in the Large Subunit

Amina Zee

Translocation in the Large Subunit

- Once the peptidyl transferase reaction has occurred, the tRNA in the P-site is deacetylated (no longer attached to an amino acid), and the growing polypeptide chain is linked to the tRNA in the A-site.

Amina Zee

Translocation in the Large Subunit

- For a new round of peptide chain elongation to occur, the P-site tRNA must move to the E-site and the A-site tRNA must move to the P-site.
- At the same time, the mRNA must move by three nucleotides to expose the next codon.

Amina Zee

Translocation in the Large Subunit

- These movements are coordinated within the ribosome and are collectively referred to as translocation.
- The initial steps of translocation are coupled to the peptidyl transferase reaction.

Amina Zee

Translocation in the Large Subunit

- Once the growing peptide chain has been transferred to the A-site tRNA, the A- and P-site tRNAs have a preference to occupy new positions in the large subunit.

Amina Zee

Translocation in the Large Subunit

- The 3’ end of the A-site tRNA is bound to the growing polypeptide chain and prefers to bind in the P-site of the large subunit.

Amina Zee

Translocation in the Large Subunit

- The now deacetylated P-site tRNA is no longer attached to the growing polypeptide chain and prefers to bind in the E-site of the large subunit.

Amina Zee

Translocation in the Large Subunit

- In contrast, at this time, the anticodons of these tRNAs remain in their initial location in the small subunit bound to the mRNA.

Amina Zee

Translocation in the Large Subunit

- Thus, translocation is initiated in the large subunit before the small subunit, and the tRNAs are said to be in “hybrid states.”

Amina Zee

Translocation in the Large Subunit

- Their 3’ ends have shifted into a new location, but their anticodon ends are still in their pre-peptidyl transfer position.

Amina Zee

Translocation in the Large Subunit

- Importantly, this change is associated with a counter clockwise rotation of the small subunit relative to the large subunit facilitating interaction of the tRNAs with distinct tRNA-binding sites in the different subunits.

Amina Zee

Translocation in the Large Subunit

- The completion of translocation requires the action of a second elongation factor called EF-G.
- Initial binding of EF-G to the ribosome occurs when associated with GTP.

Amina Zee

Translocation in the Large Subunit

- After the peptidyl transferase reaction, EF-G-GTP binds to and stabilizes the ribosome in the rotated, hybrid state.

Amina Zee

Translocation in the Large Subunit

- When EF-G-GTP binds, it contacts the factor-binding center of the large subunit, which stimulates GTP hydrolysis.
- GTP hydrolysis changes the conformation of EF-G with two consequences.

Amina Zee

Translocation in the Large Subunit

- First, interactions between EF-G-GDP and the ribosome are thought to “unlock” the ribosome.

Amina Zee

Translocation in the Large Subunit

- Structural studies reveal that there are “gates” that separate the A-, P-, and E-sites and EF-G-GDP is said to unlock the ribosome by opening these gates.

Amina Zee

Translocation in the Large Subunit

- Second, the changed EF-G–GDP conformation binds to the A-site of the decoding center.
- This interaction competes with the tRNA for binding to the A-site of the decoding center.

Amina Zee

Translocation in the Large Subunit

- Because the ribosome is unlocked, the formerly A-site tRNA can move into the P-site, allowing EF-G-GDP to bind the A-site.

Amina Zee

Translocation in the Large Subunit

- Completion of translocation is accompanied by a clockwise rotation of the small subunit back to its starting position.
- The resulting ribosome structure has dramatically reduced affinity for EF-G-GDP.

Amina Zee

Translocation in the Large Subunit

- Release of EF-G results in the return of the ribosome to a “locked” state in which the tRNAs and mRNA are once again tightly associated with the small subunit decoding center and the gates between the A-, P- and E-sites are closed.

Amina Zee

Translocation in the Large Subunit

- Together, these events result in the translocation of the A-site tRNA into the P-site, the P-site tRNA into the E-site, and the movement of the mRNA by exactly 3 bp.
- The ribosome is now ready for a new cycle of amino acid addition to begin.

END

Amina Zee

Lec 56,57

Termination of Translation I,II

Amina Zee

Termination of Translation

- The ribosome's cycle of aminoacyl-tRNA binding, peptide-bond formation, and translocation continues until one of the three stop codons enters the A-site.

Amina Zee

Termination of Translation

- It was initially postulated that there would be one or more chain terminating tRNAs that would recognize these codons.
- However, this is not the case.

Termination of Translation

- Instead, stop codons are recognized by proteins called release factors (RFs) that activate the hydrolysis of the polypeptide from the peptidyl-tRNA.

Amina Zee

Termination of Translation

- There are two classes of release factors.
- Class I release factors recognize the stop codons and trigger hydrolysis of the peptide chain from the tRNA in the P-site.

Amina Zee

Termination of Translation

- Prokaryotes have two class I release factors called RF1 and RF2.
- RF1 recognizes the stop codon UAG and RF2 recognizes the stop codon UGA.
- The third stop codon, UAA, is recognized by both RF1 and RF2.

Amina Zee

Termination of Translation

- In eukaryotic cells, there is a single class I release factor called eRF1 that recognizes all three stop codons.
- Class II release factors stimulate the dissociation of the class I factors from the ribosome after release of the polypeptide chain.

Amina Zee

Termination of Translation

- Prokaryotes and eukaryotes have only one class II factor called RF3 and eRF3, respectively.
- Like EF-G, IF2, and EF-Tu, class II release factors are regulated by GTP binding and hydrolysis.

Amina Zee

Termination of Translation

- How do release factors recognize stop codons?
- Because release factors are composed entirely of protein, protein-RNA interaction must mediate stop codon recognition.

Amina Zee

Termination of Translation

- Experiments in which short coding regions were genetically swapped between RF1 and RF2 (having different stop-codon specificity) identified a three-amino-acid sequence that is critical for release factor specificity.

Amina Zee

Termination of Translation

- Exchange of these three amino acids between RF1 and RF2 swaps the stop-codon specificity of the two complexes.

Termination of Translation

- For this reason, this three-amino-acid sequence is called a peptide anticodon and must interact with and recognize stop codons.
- A 3D structure of RF1 bound to the ribosome confirms that RF1 binds to the A-site of the ribosome.

Amina Zee

Termination of Translation

- In this structure, the peptide anticodon is located very near the anticodon, but it is likely that there are additional protein regions that contribute to codon recognition.

Amina Zee

Termination of Translation

- A region of class I release factors that stimulates polypeptide release has also been identified.
- All class I factors share a conserved three-amino-acid sequence (glycine, glycine, glutamine) that is essential for polypeptide release.

Amina Zee

Termination of Translation

- Moreover, the structure of RF1 bound to the ribosome confirms that the GGQ motif is located in close proximity to the peptidyl transferase center.

Amina Zee

Termination of Translation

- It remains unclear whether the GGQ motif is directly involved in the release of polypeptide from the peptidyl-tRNA or it induces a change in the peptidyl transferase center that allows the center itself to catalyze hydrolysis.

Amina Zee

Termination of Translation

- Studies of the conserved bases found adjacent to the CCA ends in the peptidyl transferase center indicate that several of these residues are required for peptide hydrolysis.

Amina Zee

Termination of Translation

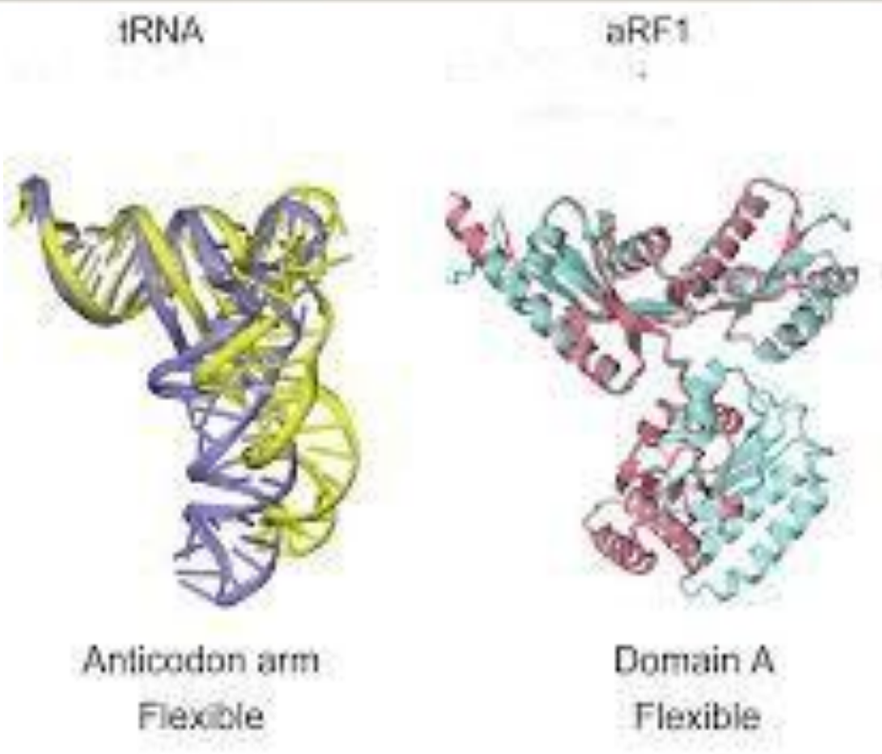
- Indeed, these bases appear to play a more important role in peptide release than they do in peptide-bond formation.

Termination of Translation

- Together, these studies have led to the hypothesis that class I release factors functionally mimic a tRNA, having a peptide anticodon that interacts with the stop codon and a GGQ motif that reaches into the peptidyl transferase center.

Amina Zee

Termination of Translation



Amina Zee

Termination of Translation

- Once the class I release factor has triggered the hydrolysis of the peptidyl tRNA linkage, it must be removed from the ribosome.

Amina Zee

Termination of Translation

- This step is stimulated by the class II release factor, RF3.
- RF3 is a GTP-binding protein but, unlike the other GTP-binding proteins involved in translation, this factor has a higher affinity for GDP than GTP.

Amina Zee

Termination of Translation

- Thus, free RF3 is predominantly in the GDP-bound form.
- RF3-GDP binds to the ribosome in a manner that depends on the presence of a class I release factor.

Amina Zee

Termination of Translation

- After the class I release factor stimulates polypeptide release, a change in the conformation of the ribosome and the class I release factor stimulates RF3 to exchange its bound GDP for a GTP.

Amina Zee

Termination of Translation

- The binding of GTP to RF3 leads to the formation of a high-affinity interaction with the ribosome that favors the rotated hybrid state.
- This change in conformation displaces the class I factor from the ribosome.

Amina Zee

Termination of Translation

- These changes also allow RF3 to associate with the factor-binding center of the large subunit. As with other GTP-binding proteins involved in translation, this interaction stimulates the hydrolysis of GTP.

Amina Zee

Termination of Translation

- In the absence of a bound class I factor, the resulting RF3.GDP has a low affinity for the ribosome and is released.

Amina Zee

Sequence Analysis

Lec 58
Amino Acids

Amina Zee

Amino Acids

Background

- Translation involves coding of proteins by RNAs at ribosomes
- Codons of three nucleotides from the RNA molecules code one amino acid at a time

Amina Zee

Amino Acids

Introduction

- In all, 20 different amino acids can be coded by such codons during translation
- The amino acids polymerize to form proteins
- Next, we will study amino acids and the respective codons!

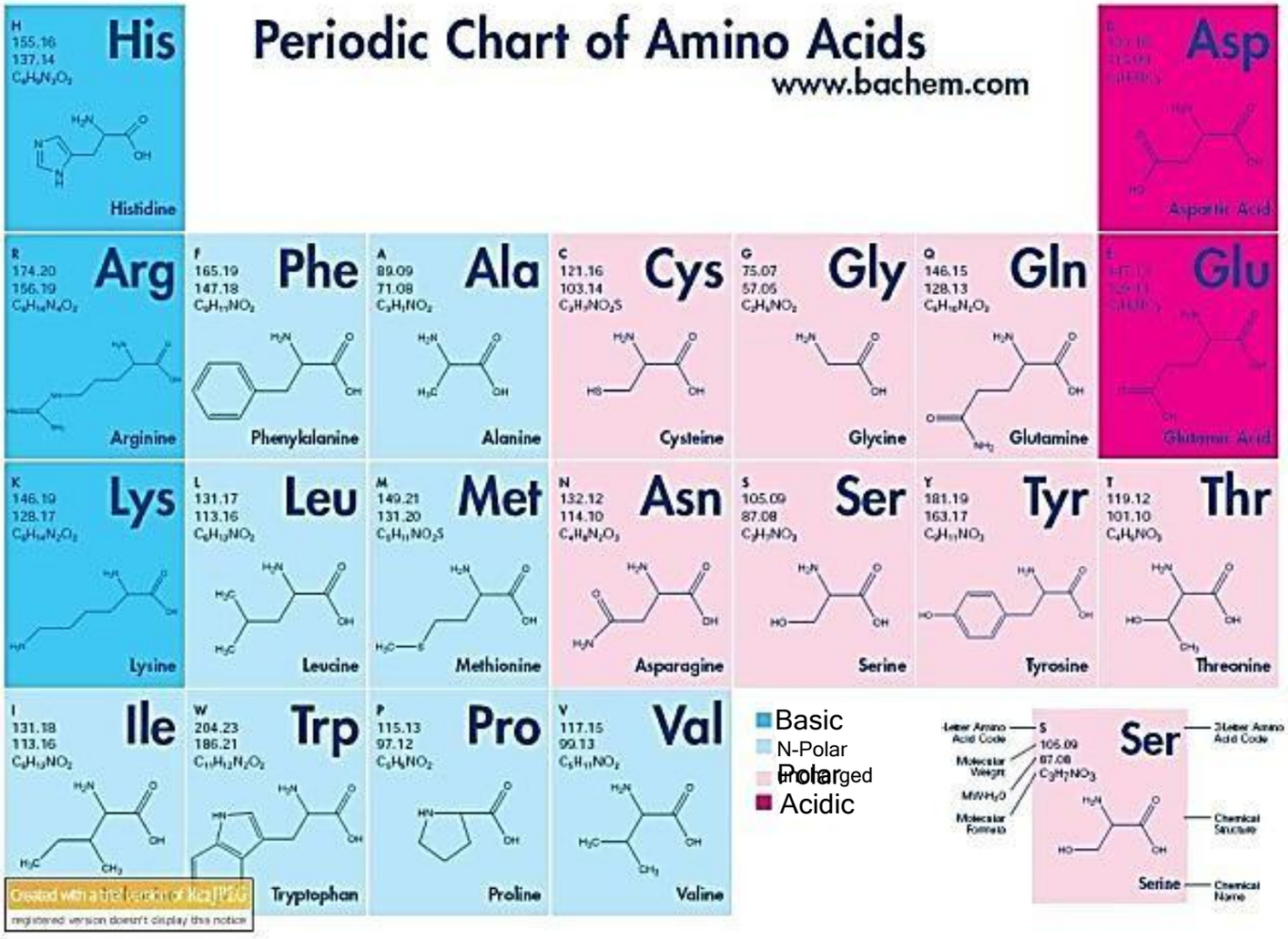
Amina Zee

Amino Acids

First letter	Second letter: U	Second letter: C	Second letter: A	Second letter: G	Third letter
U	UUU, UUC: Phe UUA, UUG: Leu	UCU, UCC, UCA, UCG: Ser	UAU, UAC: Tyr UAA: Stop UAG: Stop	UGU, UGC: Cys UGA: Stop UGG: Trp	U C A G
C	CUU, CUC, CUA, CUG: Leu	CCU, CCC, CCA, CCG: Pro	CAU, CAC: His CAA, CAG: Gln	CGU, CGC, CGA, CGG: Arg	U C A G
A	AUU, AUC, AUA: Ile AUG: Met	ACU, ACC, ACA, ACG: Thr	AAU, AAC: Asn AAA, AAG: Lys	AGU, AGC: Ser AGA, AGG: Arg	U C A G
G	GUU, GUC, GUA, GUG: Val	GCU, GCC, GCA, GCG: Ala	GAU, GAC: Asp GAA, GAG: Glu	GGU, GGC, GGA, GGG: Gly	U C A G

Amina Zee

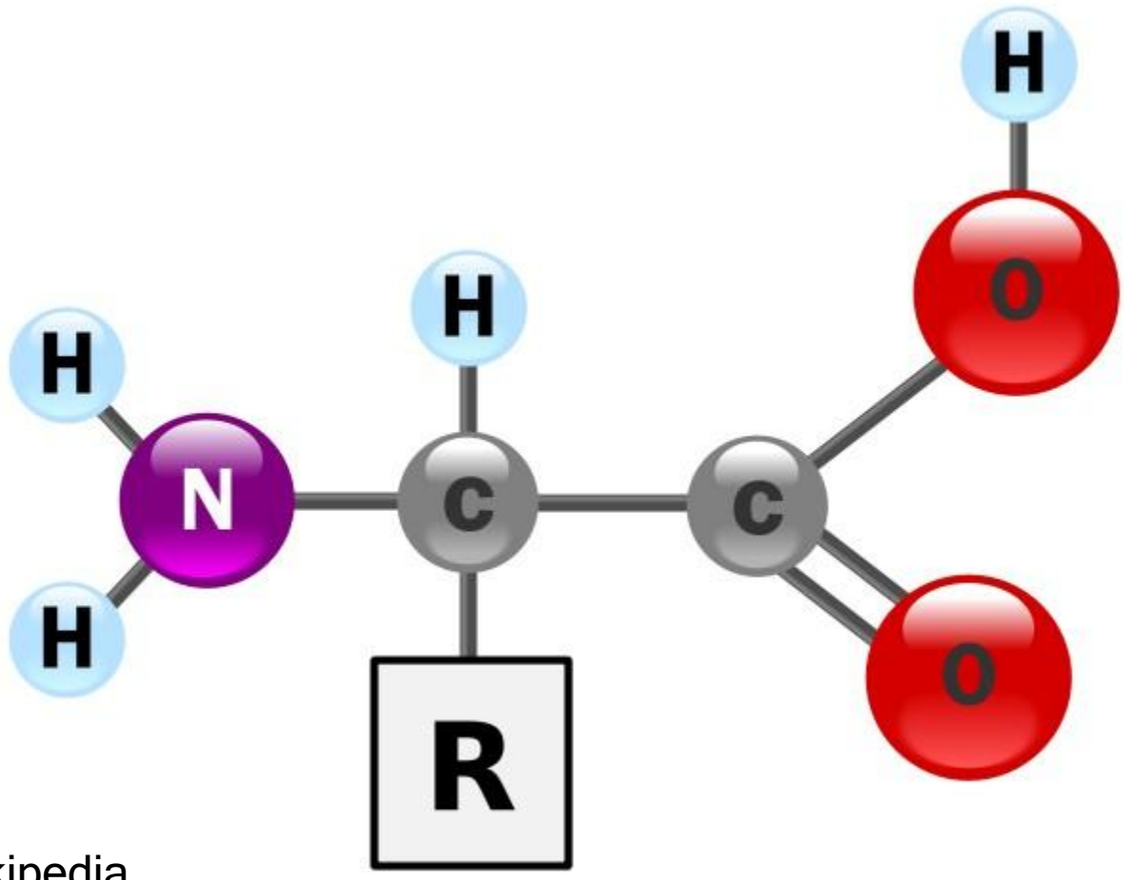
Amino Acids



Amina Zee

Amino Acids

Chemical Structure of an Amino Acid

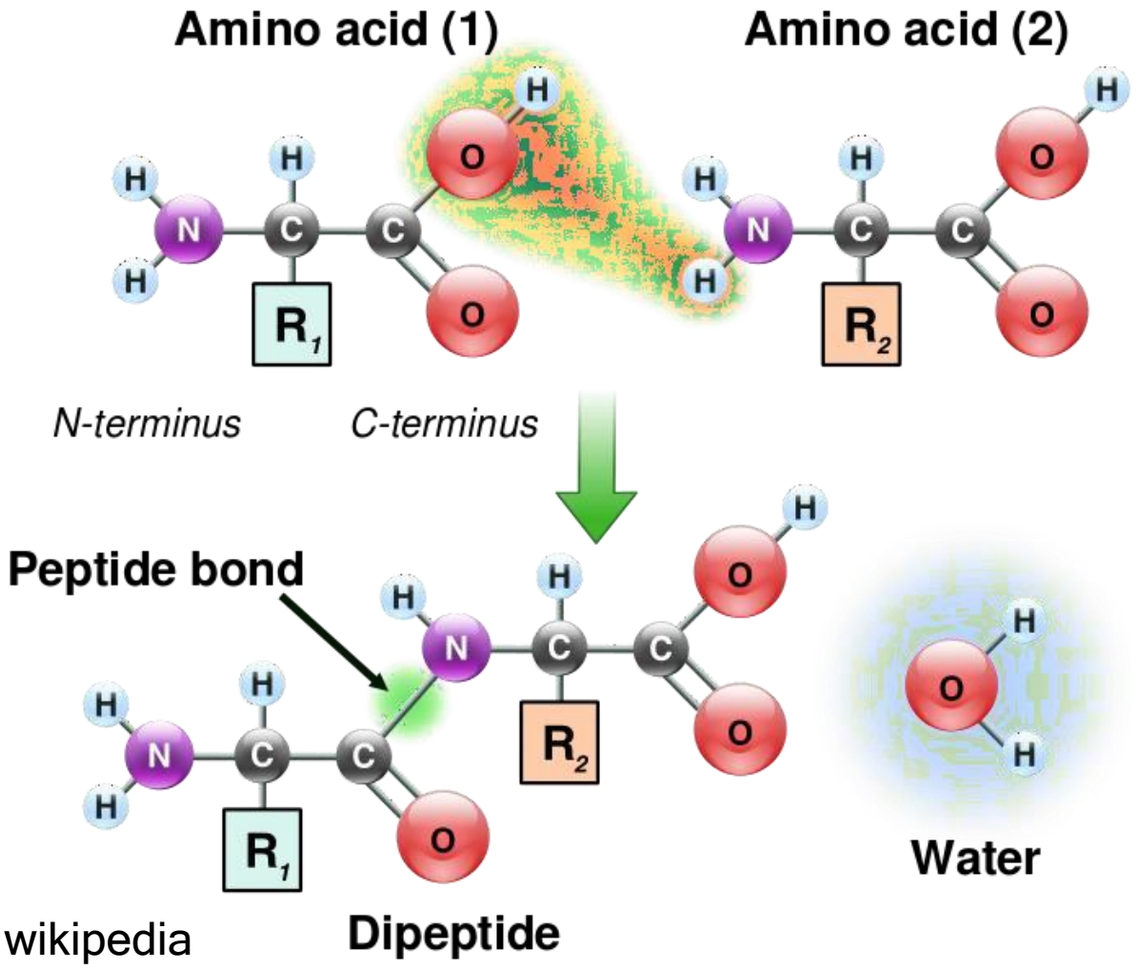


Source: wikipedia

Amina Zee

Amino Acids

Polymerization of Amino Acids by Peptide Bond

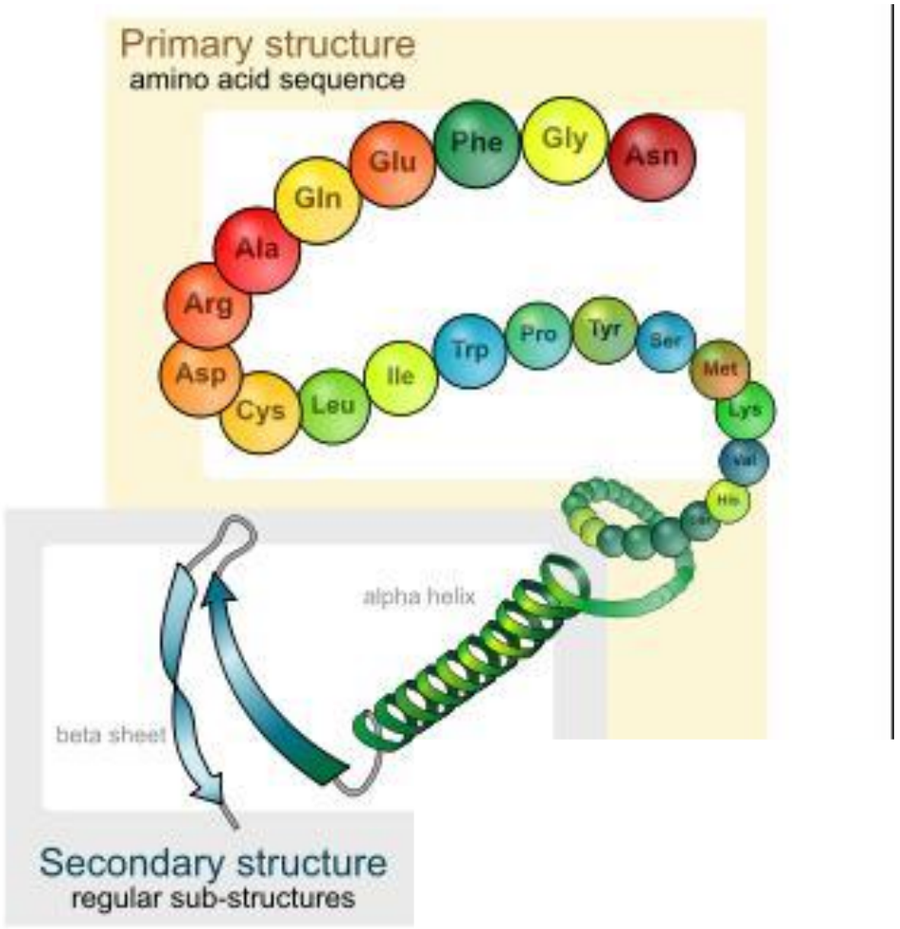


Source: wikipedia

Amina Zee

Amino Acids

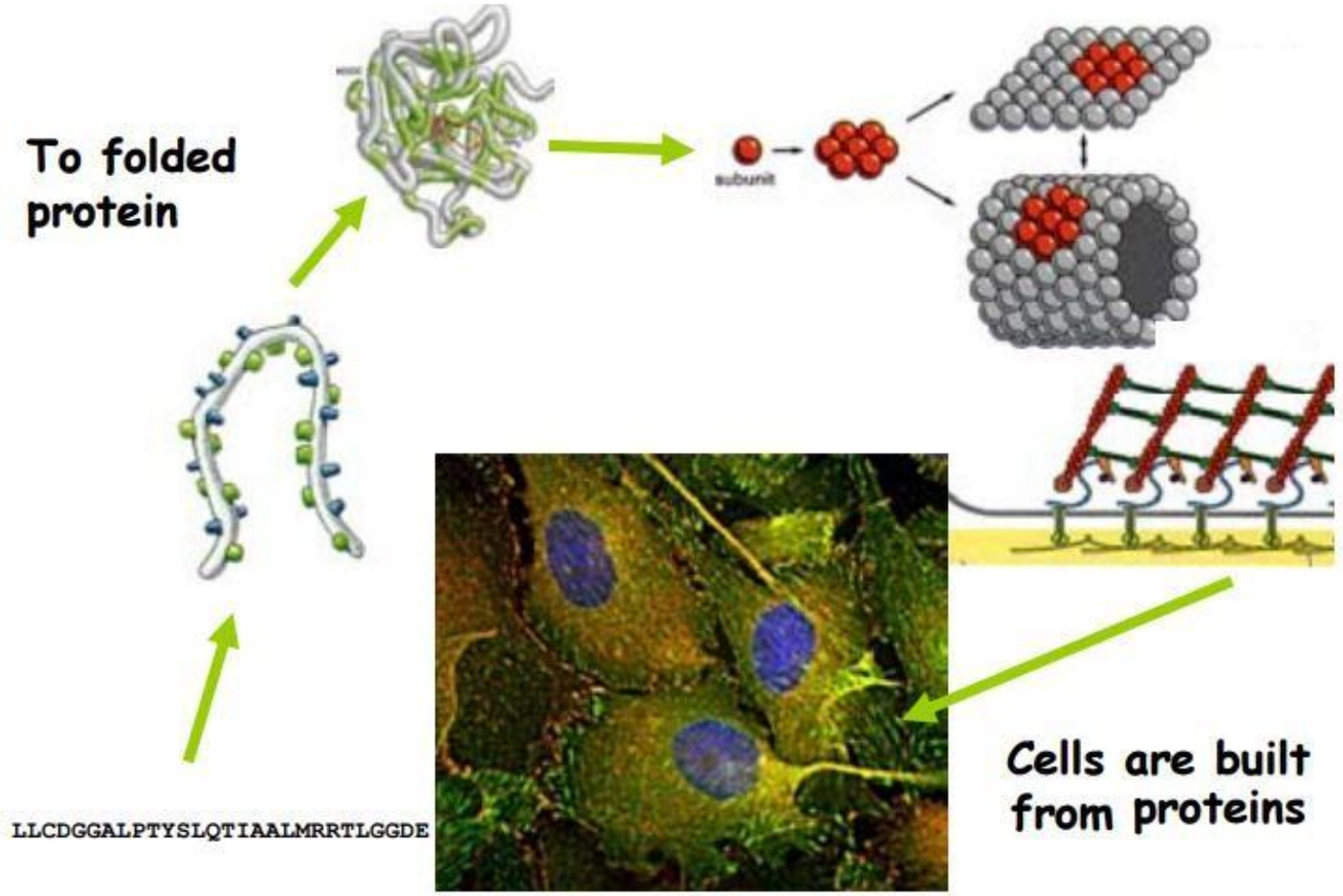
Polymerization of Amino Acids



Source: wikipedia

Amina Zee

Amino Acids



Source: wikipedia

Amina Zee

Amino Acids

Conclusion

- **Proteins are formed after polymerization of amino acids**
- **Amino acids are chained together by peptide bonds**
- **Proteins thus formed fold to take 3D conformations!**

END

Amina Zee

CELL BIOLOGY

Lec 59

Proteins: Polymers of Amino Acids

Amina Zee

Proteins: Polymers of Amino Acids

Protein Functions

Most Abundant Polymer in the cell

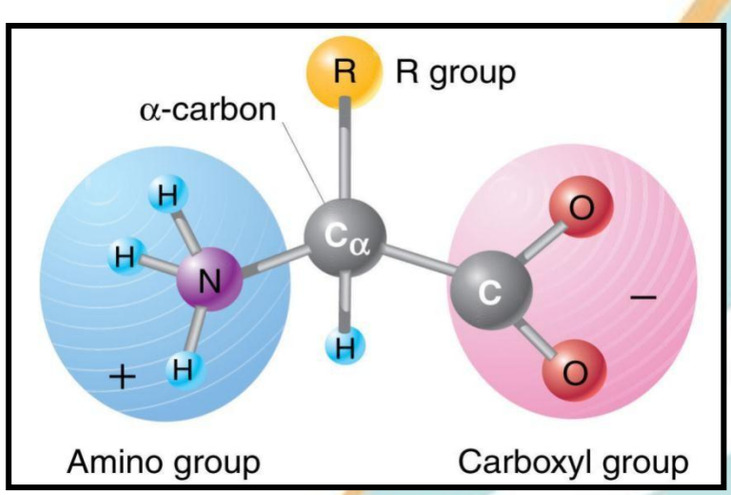
Support, protection, catalysis, transport, defense, regulation & movement

Do not store genetic information

Amina Zee

Proteins: Polymers of Amino Acids

Amino Acid Structure



The side chains or R groups give different properties to each of the amino acids

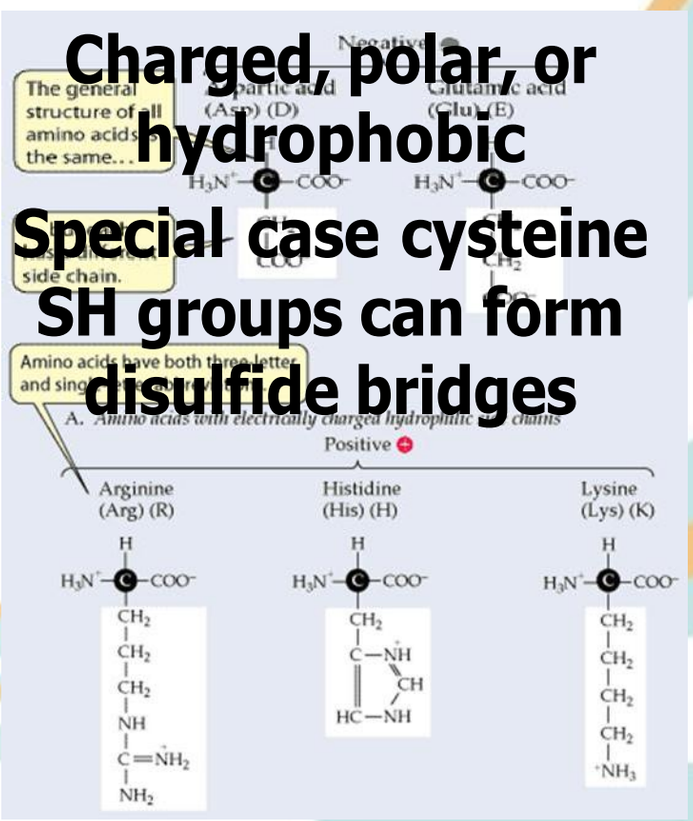
Amina Zee

Proteins: Polymers of Amino Acids

Charged R Groups

Charged, polar, or hydrophobic

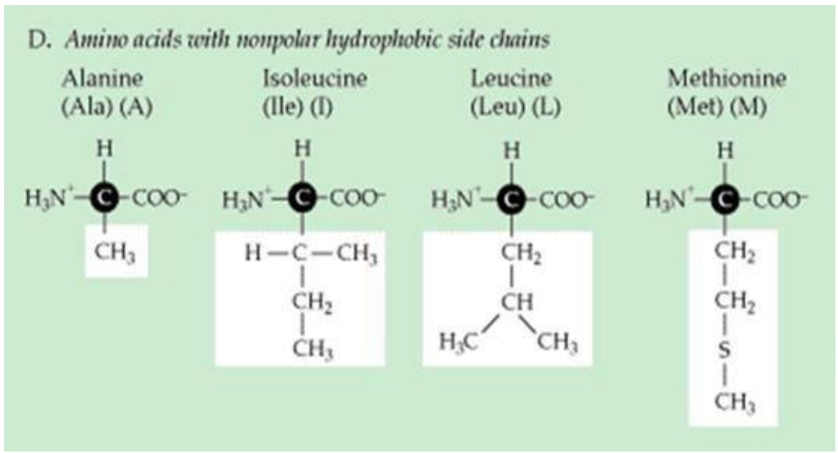
Special case cysteine SH groups can form disulfide bridges

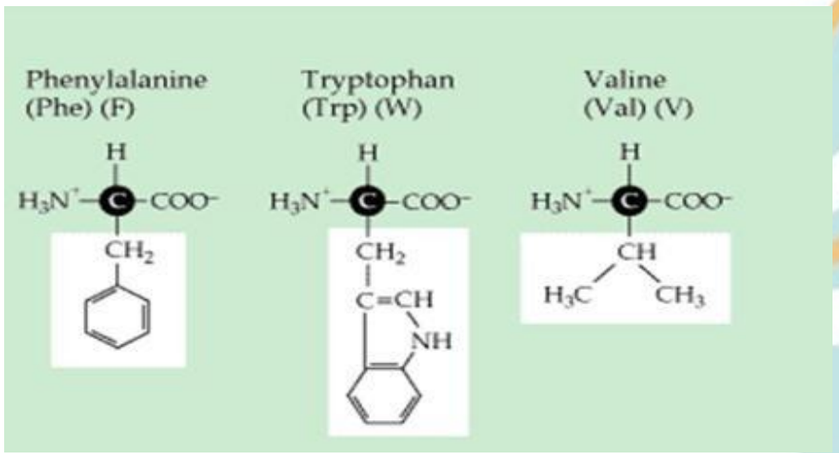


Amina Zee

Proteins: Polymers of Amino Acids

Nonpolar R Groups

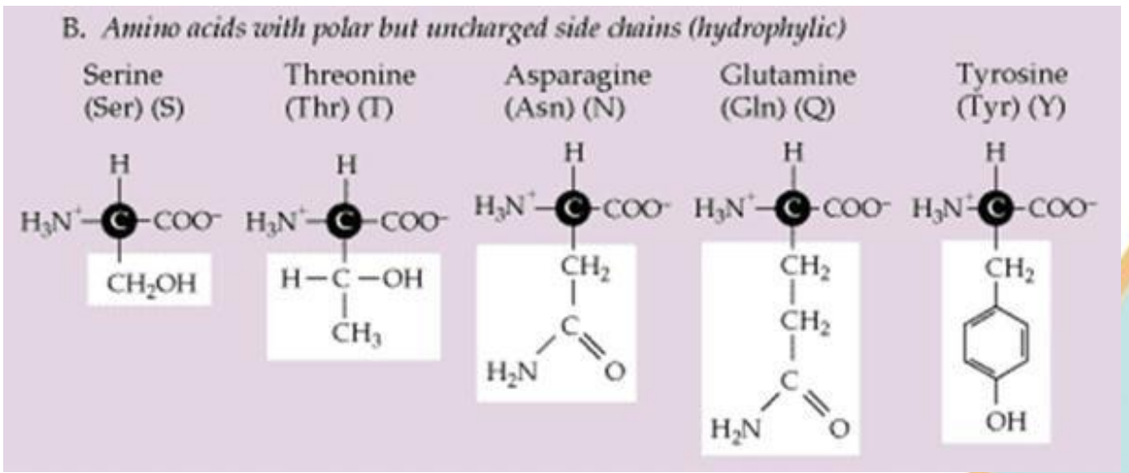


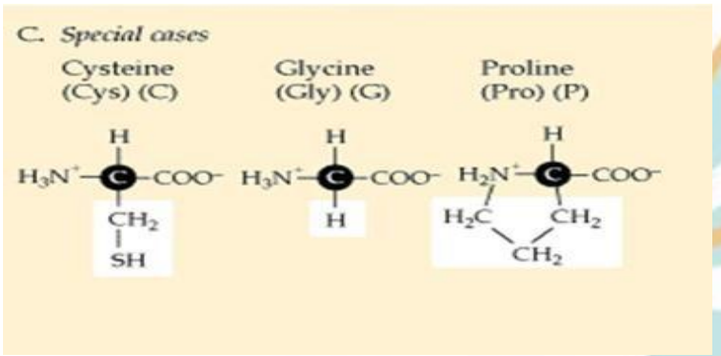


Amina Zee

Proteins: Polymers of Amino Acids

Polar & Special R Groups





Amina Zee

CELL BIOLOGY

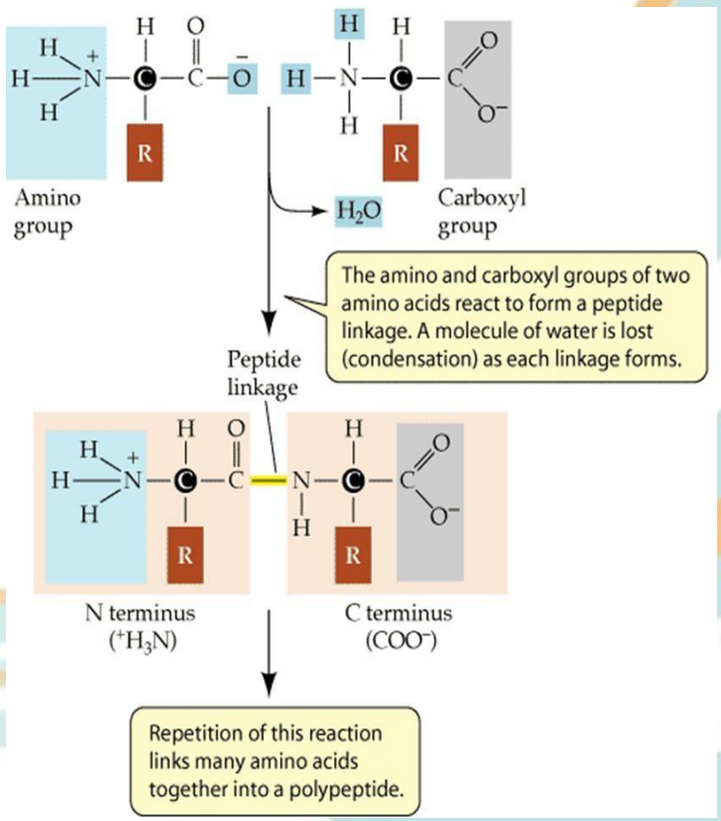
Lec 60

Proteins: Structure

Amina Zee

Proteins: Polymers of Amino Acids

The Peptide Bond

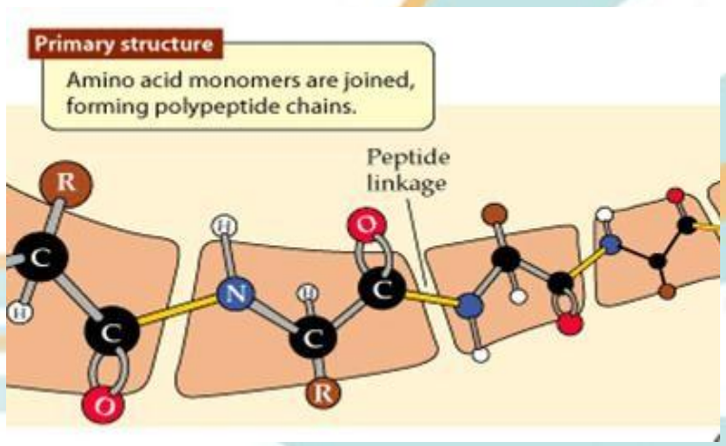


Amina Zee

Proteins: Structure

Primary Structure

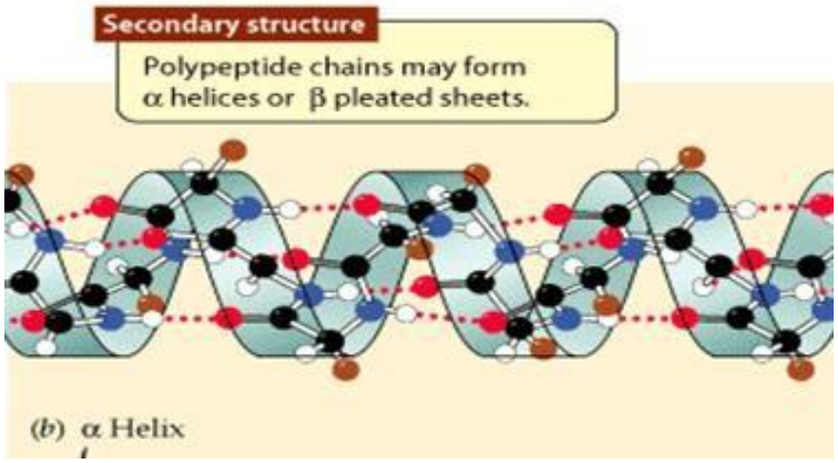
Sequence of amino acids bonded by peptide linkages (Diversity 20n)

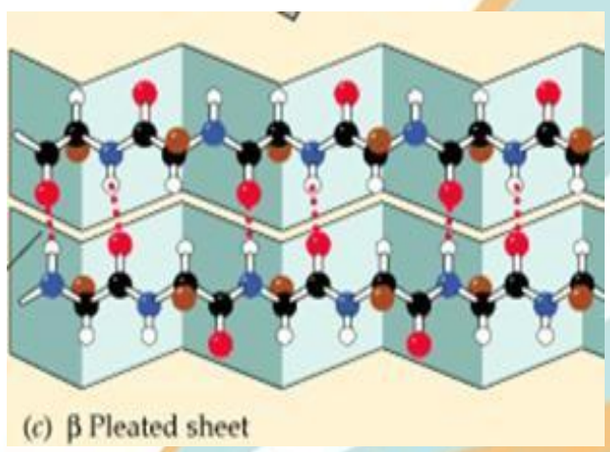


Amina Zee

Proteins: Structure

Secondary Structure
α helices (hydrogen bonds between amino acid residues)



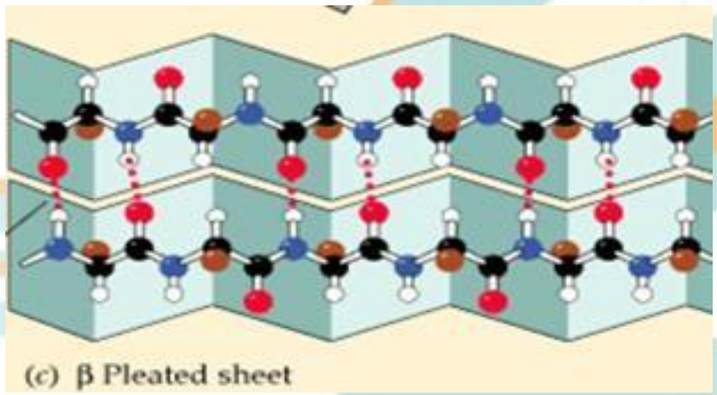


Amina Zee

Proteins: Structure

Secondary Structure

β pleated sheets (hydrogen bonds between amino acid residues)

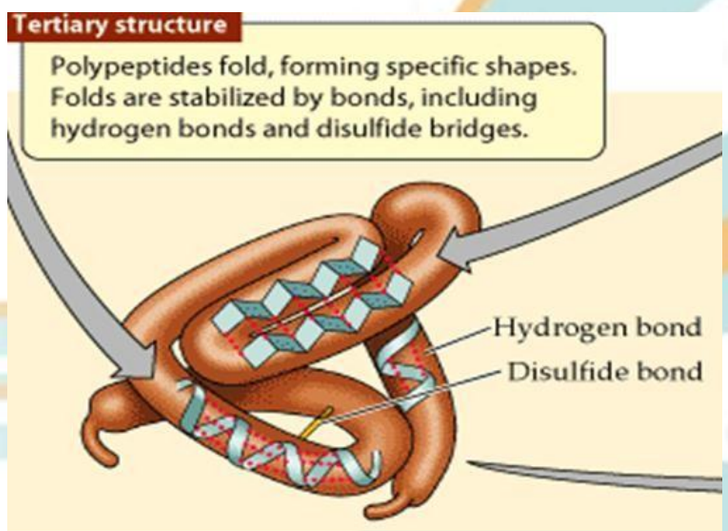


Amina Zee

Proteins: Structure

Tertiary Structure

Generated by bending and folding of the polypeptide chain

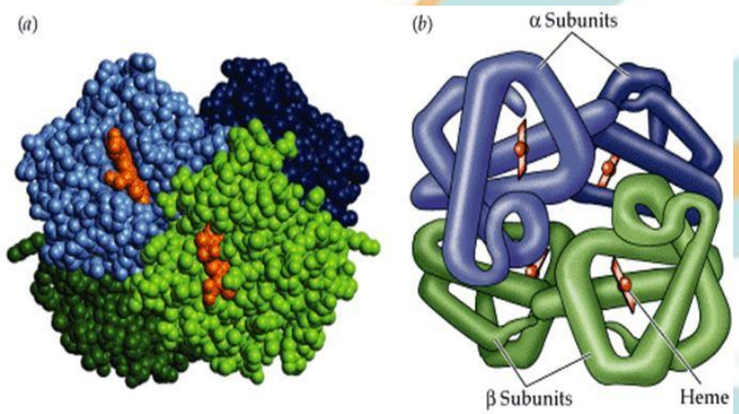


Amina Zee

Proteins: Structure

Quaternary structure

Arrangement of multiple folded protein molecules in a multi-subunit complex.



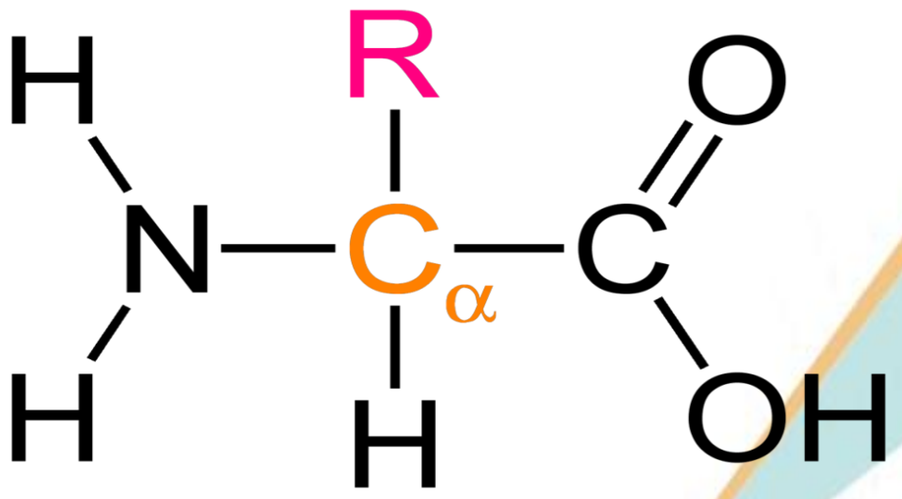
Amina Zee

Lec 62
Chemical composition of proteins

- Proteins are polymers of amino acids.
- They range in size from small to very large.
- All the proteins are made up of Twenty different types of amino acids. So these amino acids are called standard amino acids.

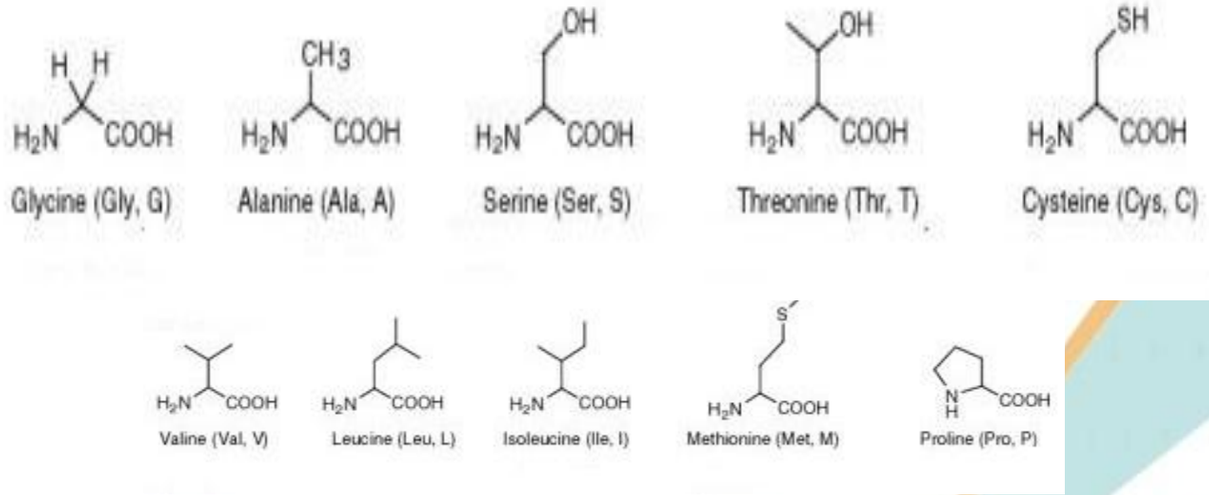
Amina Zee

Chemical composition of proteins



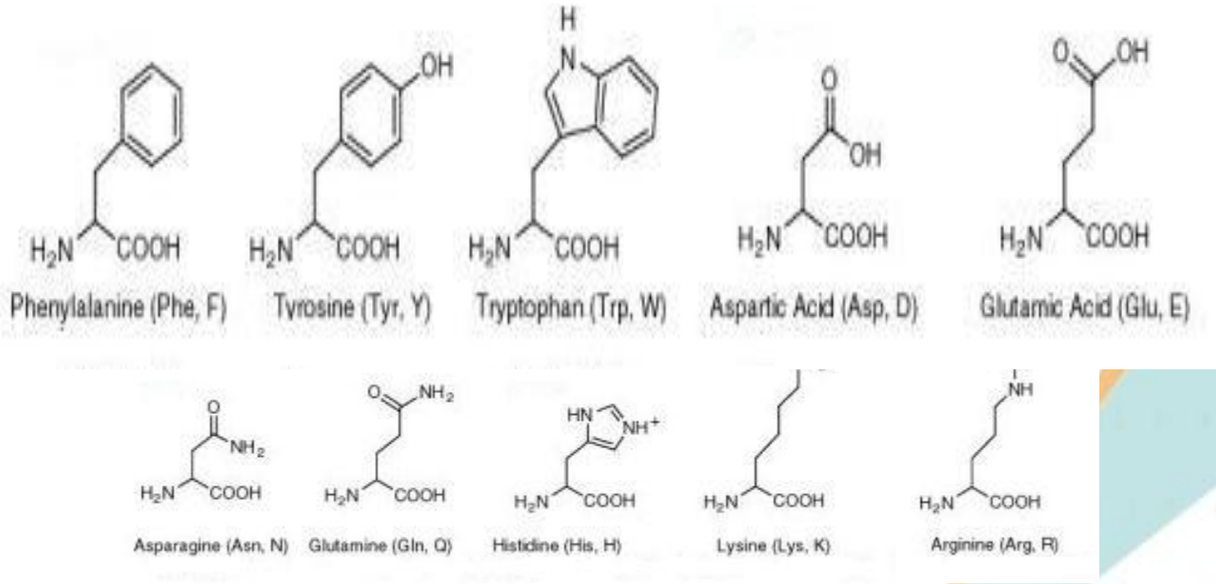
Amina Zee

Chemical composition of proteins



Amina Zee

Chemical composition of proteins

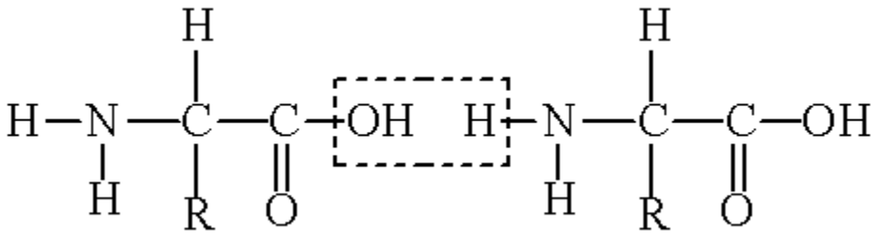


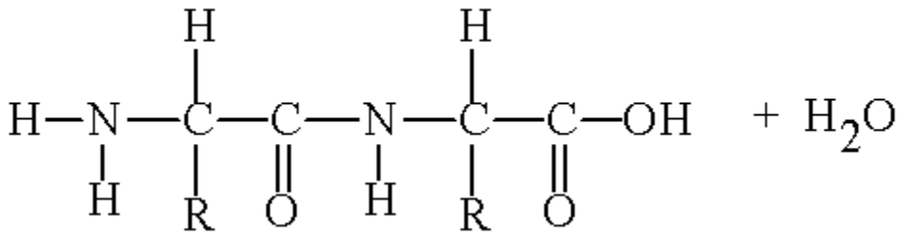
Amina Zee

Chemical composition of proteins

- In a protein molecule, each amino acid residue is joined to its neighbour by a specific type of covalent bond which is called Peptide Bond.

Chemical composition of proteins





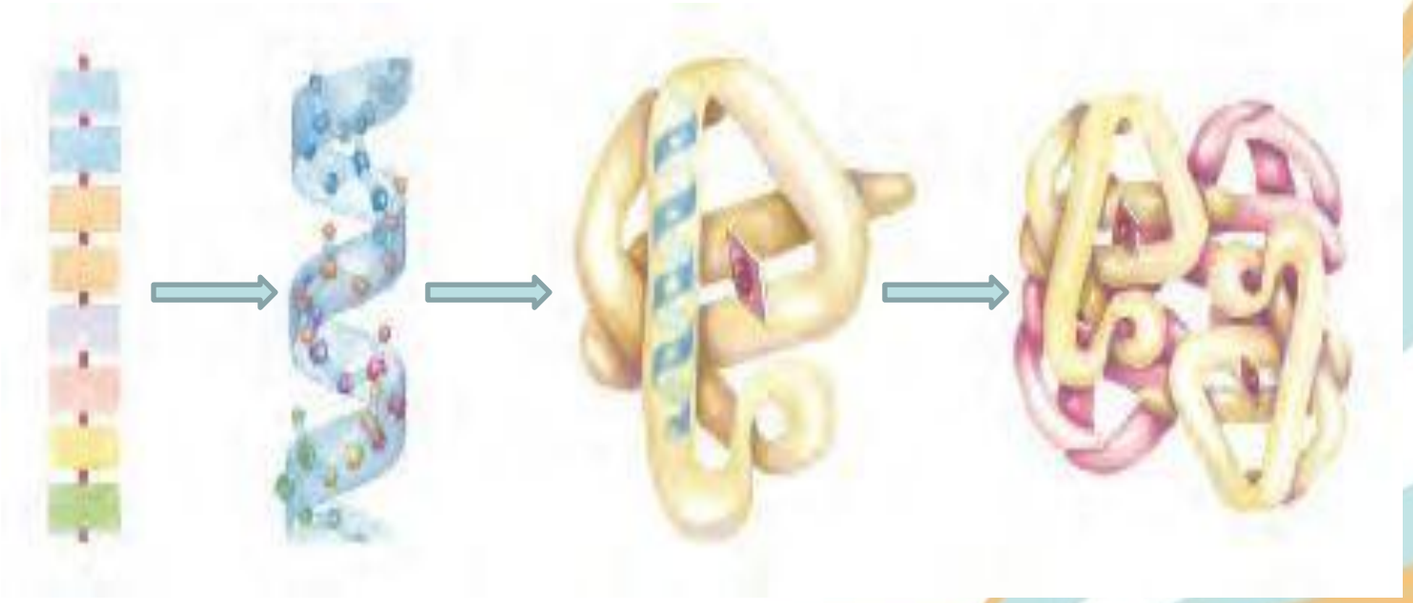
Amina Zee

Chemical composition of proteins

- Amino acids can successively join to form dipeptides, tripeptides, tetrapeptides, oligo peptides and polypeptides.

Amina Zee

Chemical composition of proteins



Amina Zee

Lec 63
Primary structure of proteins

- Primary structure or covalent structure of protein refers to the amino acid sequence of its polypeptide chain.
- Each type of protein has a unique amino acid sequence.

Amina Zee

Primary structure of proteins

Peptide Bond Is Rigid and Planar

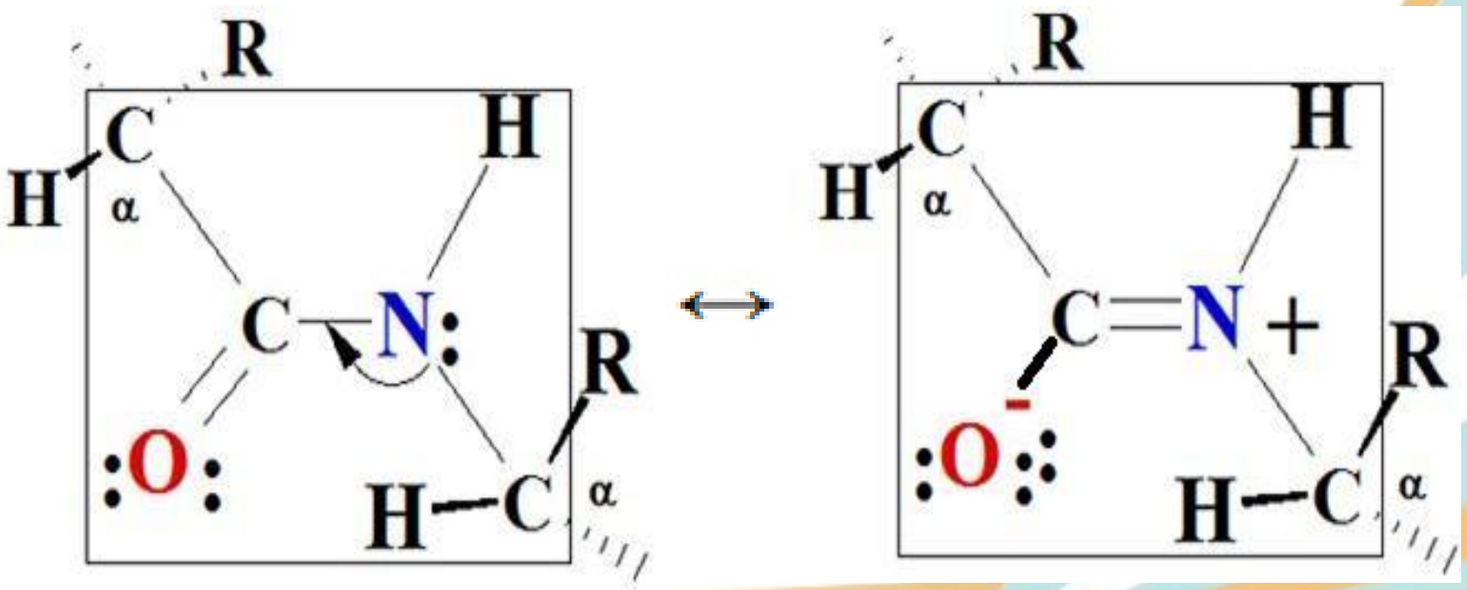
- Linus Pauling and Robert Corey carefully analyzed the peptide bond.
- Their findings laid the foundation for our present understanding of protein structure.

Amina Zee

Primary structure of proteins

- They demonstrated that the peptide C - N bond is somewhat shorter than the C - N bond in a simple amine.

Primary structure of proteins



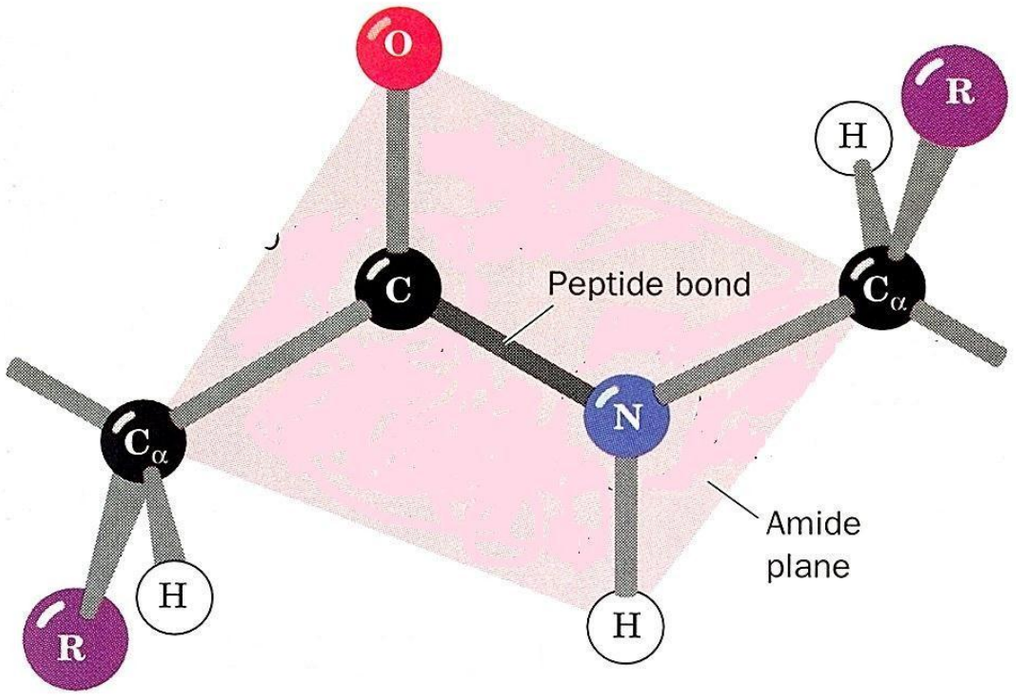
Amina Zee

Primary structure of proteins

- The six atoms of the peptide group are co-planar i.e., lie in a single plane, with the oxygen atom of the carbonyl group and the hydrogen atom of the amide nitrogen trans to each other.

Amina Zee

Primary structure of proteins



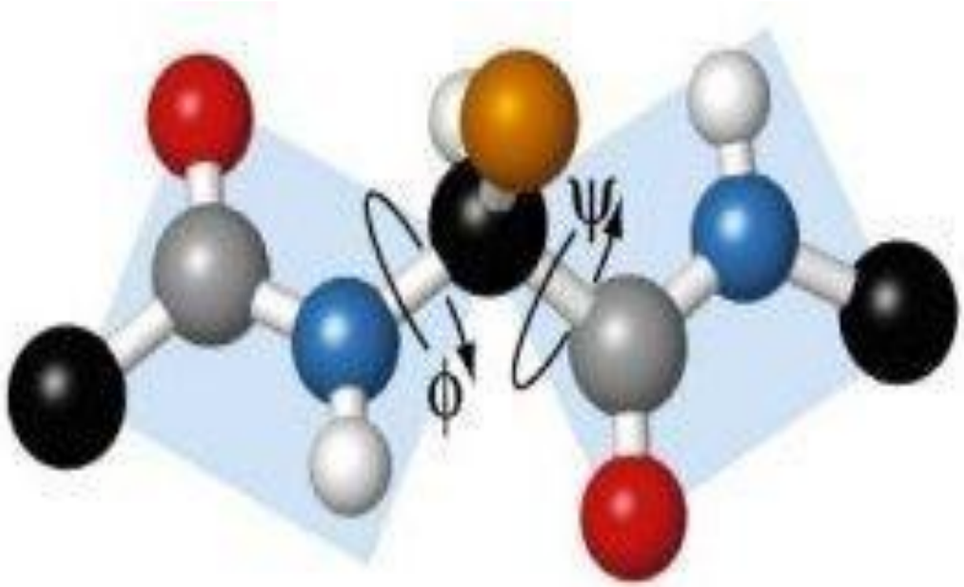
Amina Zee

Primary structure of proteins

- Pauling and Corey concluded that the peptide C - N bonds are unable to rotate freely because of their partial double-bond character.
- Rotation is permitted about the N - αC and the αC - C bonds.

Amina Zee

Primary structure of proteins



Amina Zee

Primary structure of proteins

- The bond angles resulting from rotations at C are labelled ϕ (phi) for the N - αC bond and ψ(psi) for the αC - C bond.
- In principle, ϕ and ψ can have any value between +180 & -180.

Amina Zee

Secondary structure of proteins

- Secondary structure of proteins refers to the local conformation of some part of a polypeptide.

Amina Zee

Lec 64
Secondary structure of proteins

Amina Zee

Secondary structure of proteins

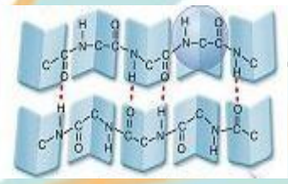
- A few types of secondary structures are particularly stable and occur widely in proteins.

Amina Zee

Secondary structure of proteins

- The most pr[illegible]nent are:-
  - α-helix

  - β- co[illegible]ns.

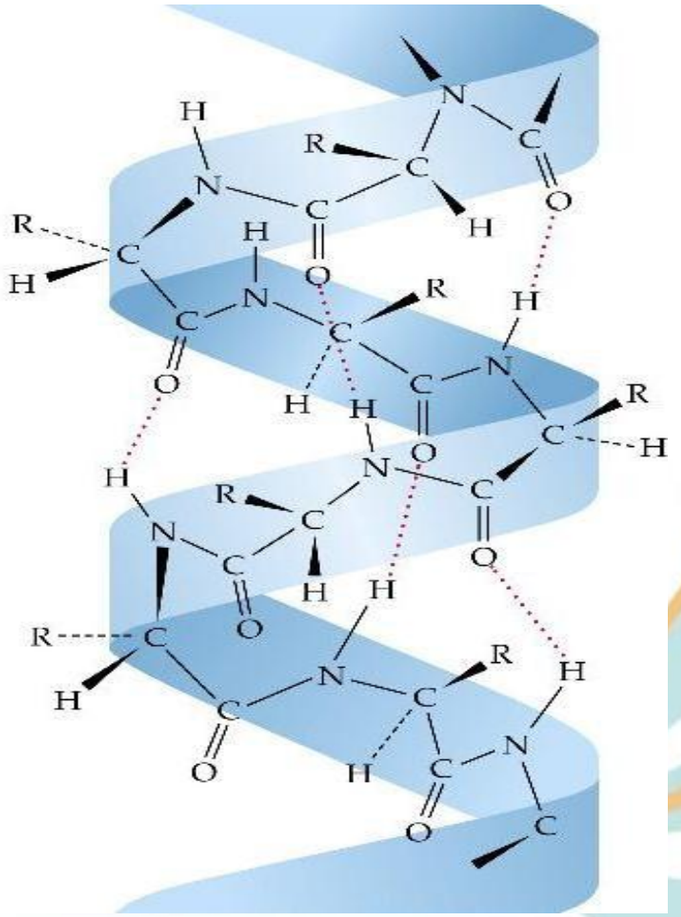


Lec 65
α- Helix

- The simplest arrangement which a polypeptide chain could assume with its rigid peptide bonds is a helical structure, which Pauling and Corey called the **α-helix**.

Amina Zee

α- Helix



Amina Zee

α- Helix

- The helical twist of the *α*-helix found in all proteins is right-handed.
- The repeating unit is a single turn of the helix, which extends about 5.4 Å (includes 3.6 amino acid residues) along the long axis.

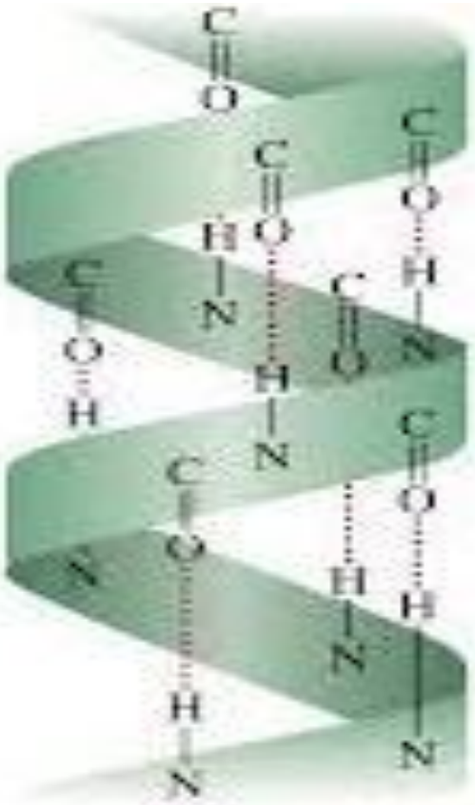
Amina Zee

α- Helix

- The amino acid residues in an helix have conformations with psi = -45 to -50 and phi = - 60.
- An helix makes optimal use of internal hydrogen bonds.

Amina Zee

α- Helix



Amina Zee

α- Helix

- About one-fourth of all amino acid residues in polypeptides are found in α-helices while in some proteins it is the predominant structure.

Lec 66
β- Pleated Sheets

- Pauling and Corey predicted a second type of secondary structure which they called **β-sheets.**
- This is a more extended conformation of polypeptide chains.

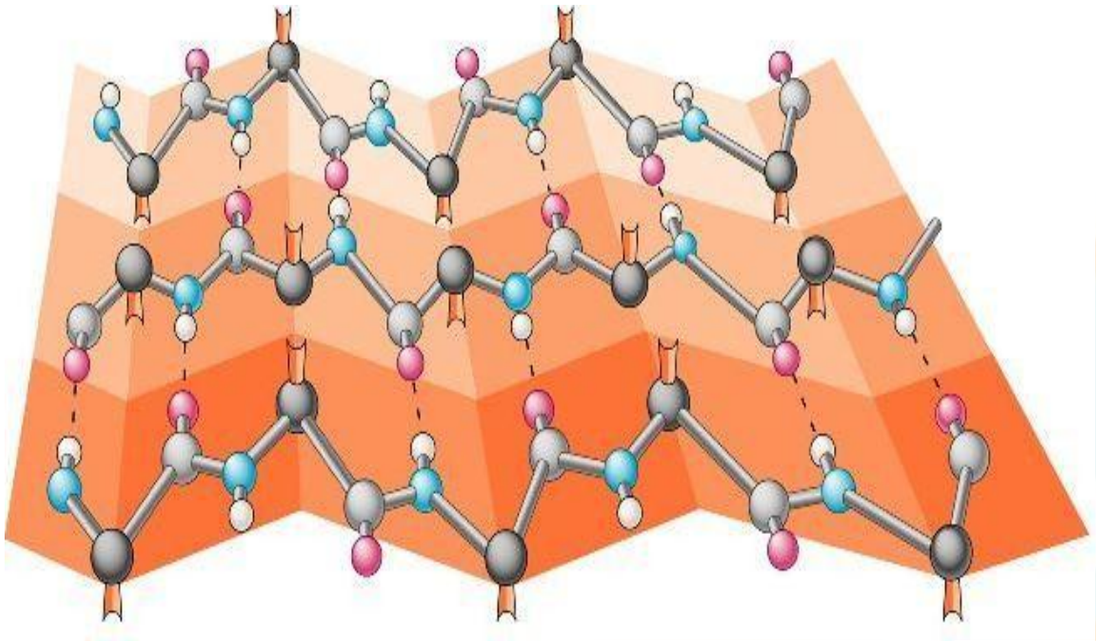
Amina Zee

β- Pleated Sheets

- The backbone of the polypeptide chain is extended into a zigzag structure.
- The zigzag polypeptide chains are arranged side by side to form a structure resembling a series of pleats.

Amina Zee

β- Pleated Sheets



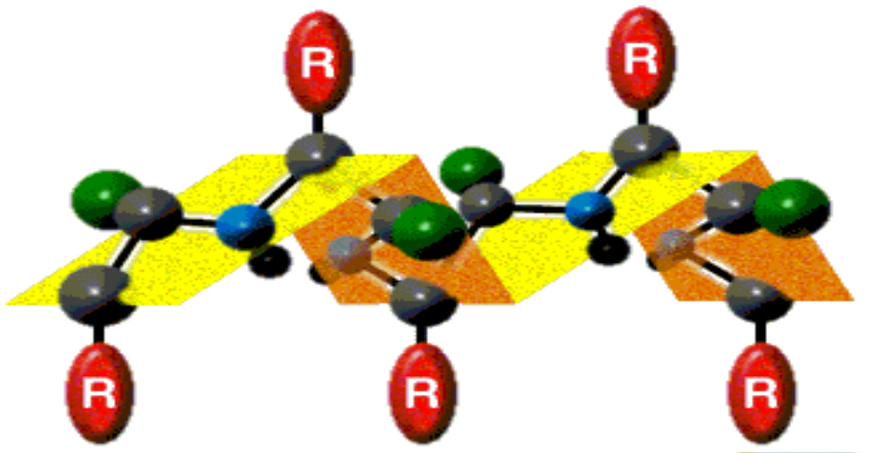
Amina Zee

β- Pleated Sheets

- The R groups of adjacent amino acids protrude from the zigzag structure in opposite directions.

Amina Zee

β- Pleated Sheets



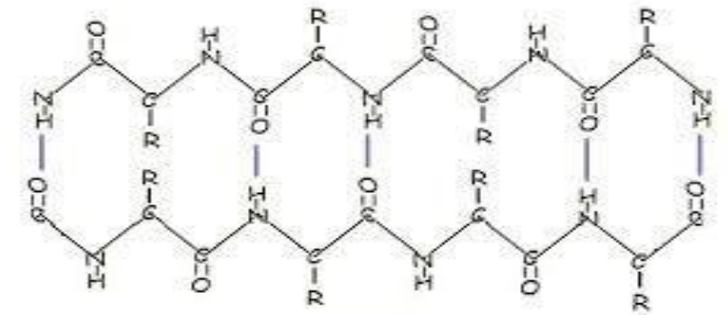
Amina Zee

β- Pleated Sheets

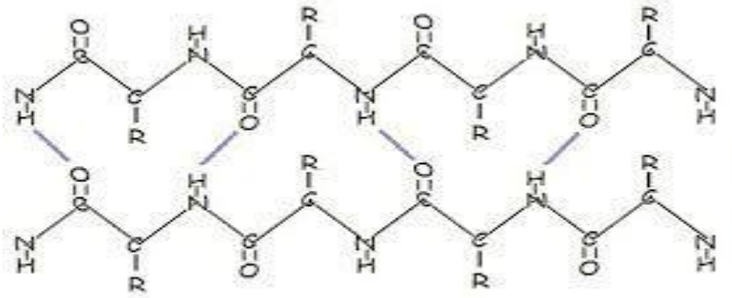
- Hydrogen bonds are formed between adjacent segments of polypeptide chain.
- The adjacent polypeptide chains in a sheet can be either parallel or antiparallel.

Amina Zee

β- Pleated Sheets



Anti-Parallel



Parallel

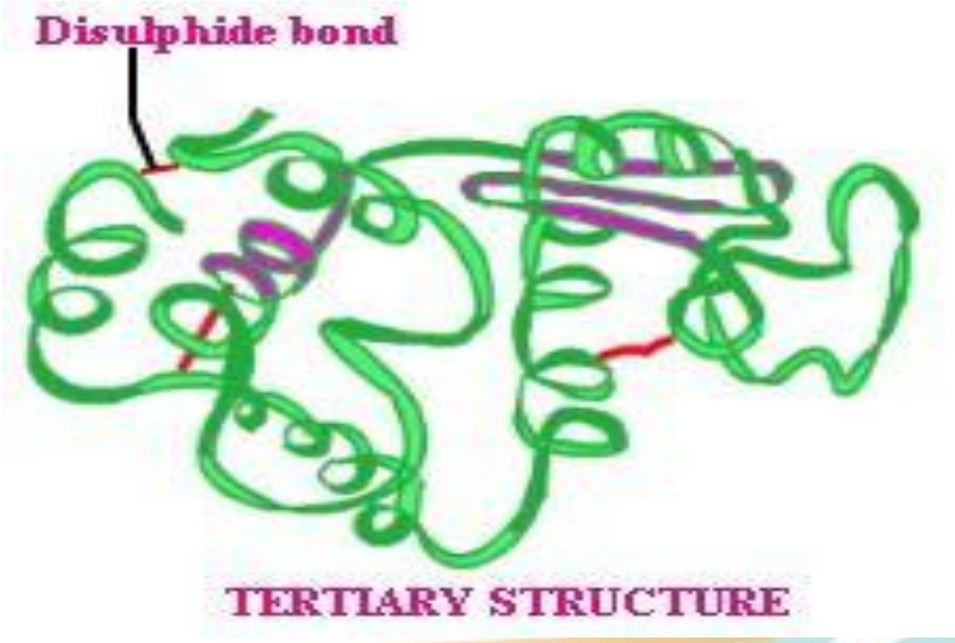
Amina Zee

Lec 67
Tertiary Structure of Proteins

- The overall three-dimensional arrangement of all atoms in a protein is referred to as the protein's **tertiary structure.**

Amina Zee

Tertiary Structure of Proteins



Amina Zee

Tertiary Structure of Proteins

- It includes longer-range aspects of amino acid sequence.
- Amino acids that are far apart in the polypeptide chain may interact within the completely folded structure of a protein.

Amina Zee

Tertiary Structure of Proteins

- Interacting segments of polypeptide chains are held in their characteristic tertiary positions by different kinds of weak interactions (and sometimes by covalent bonds) between the segments.

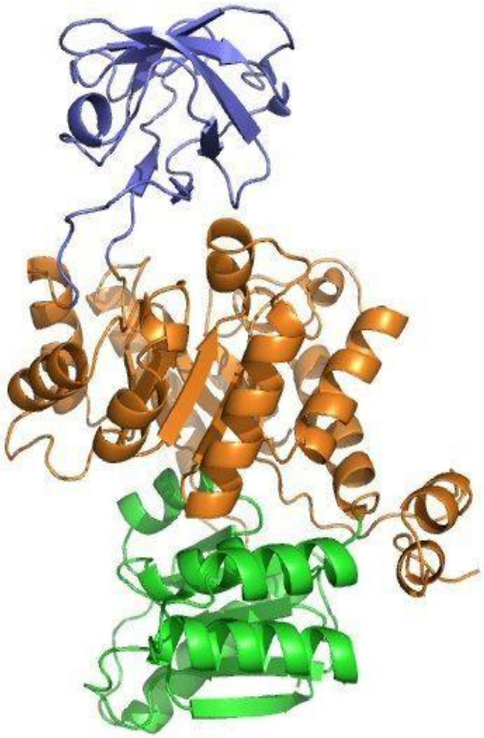
Amina Zee

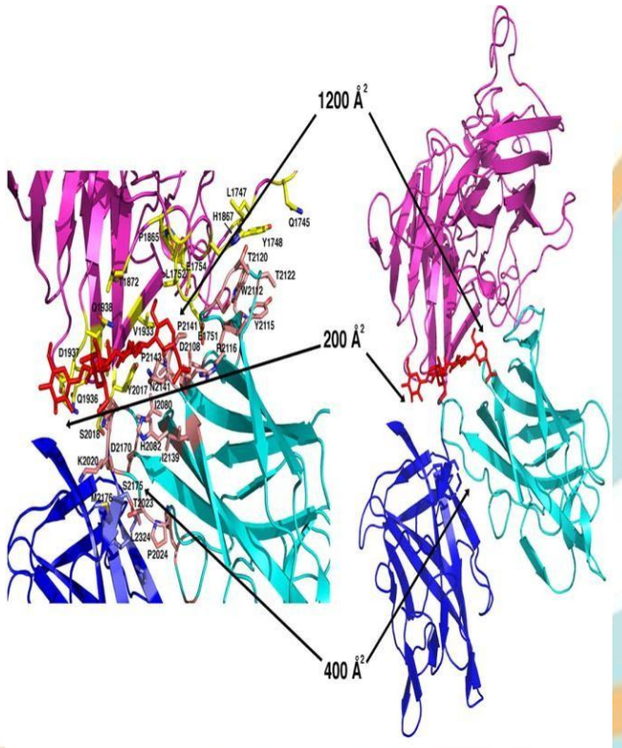
Tertiary Structure of Proteins

- Large polypeptide chains usually fold into two or more globular clusters known as **domains,** which often give these proteins a bi- or multilobal appearance.

Amina Zee

Tertiary Structure of Proteins





Amina Zee

Lec 68
Quaternary Structure of Proteins

- Some proteins contain two or more separate polypeptide chains or subunits, which may be identical or different.
- The spatial arrangement of these subunits is known as a protein's quaternary structure.

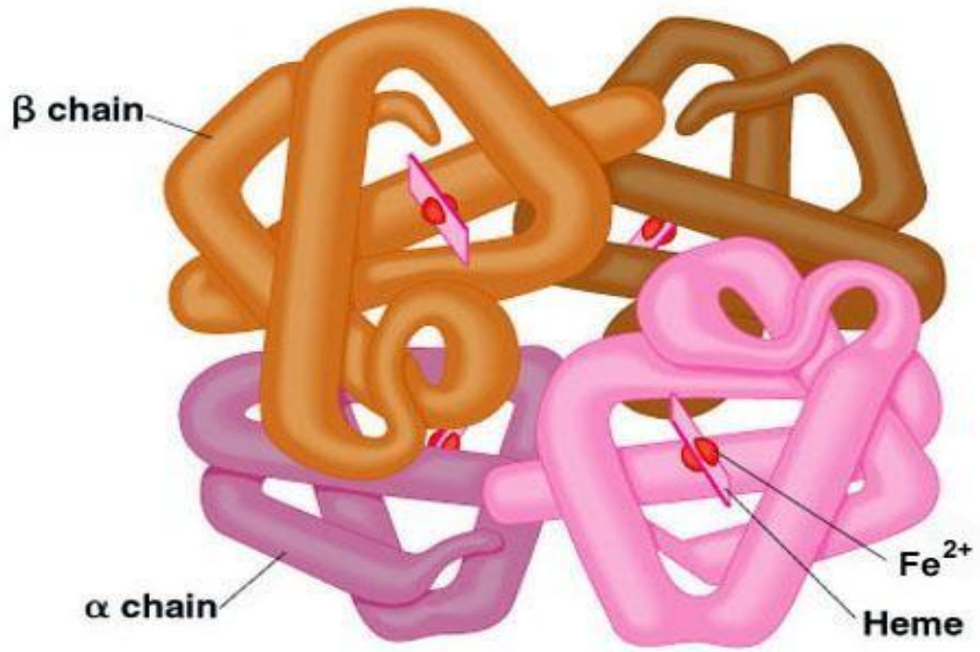
Amina Zee

Quaternary Structure of Proteins

- A multi-subunit protein is also referred to as a multimer.
- A multimer with just a few subunits is called as oligomer and a single subunit or a group of subunits, is called a protomer.

Amina Zee

Quaternary Structure of Proteins



Amina Zee

Quaternary Structure of Proteins

- Identical subunits of multimeric proteins are generally arranged in a symmetric patterns.
- Oligomers can have either rotational symmetry or helical symmetry.

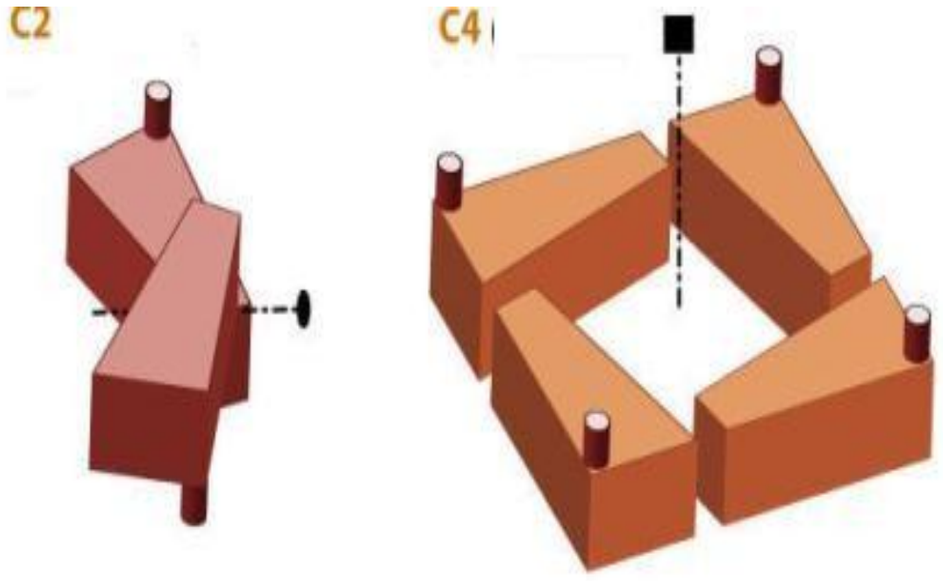
Amina Zee

Quaternary Structure of Proteins

- There are several forms of rotational symmetry. The simplest is cyclic symmetry, involving rotation about a single axis.

Amina Zee

Quaternary Structure of Proteins



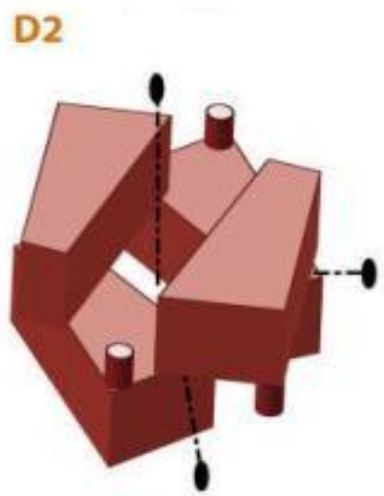
Amina Zee

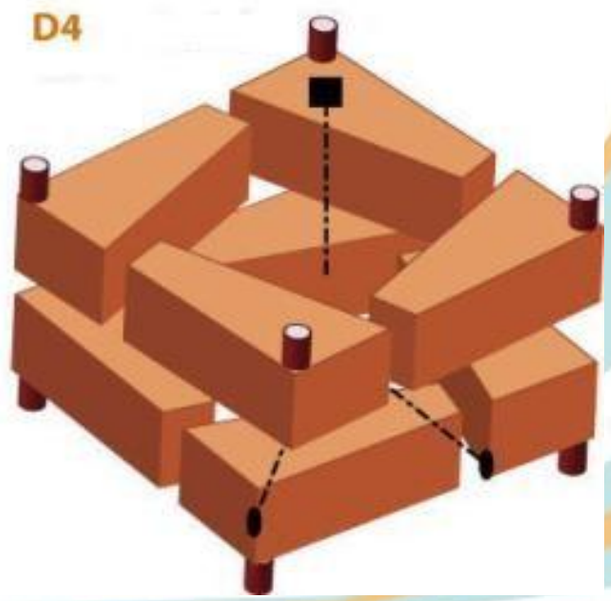
Quaternary Structure of Proteins

- A somewhat more complicated rotational symmetry is dihedral symmetry, in which a twofold rotational axis is present.

Amina Zee

Quaternary Structure of Proteins





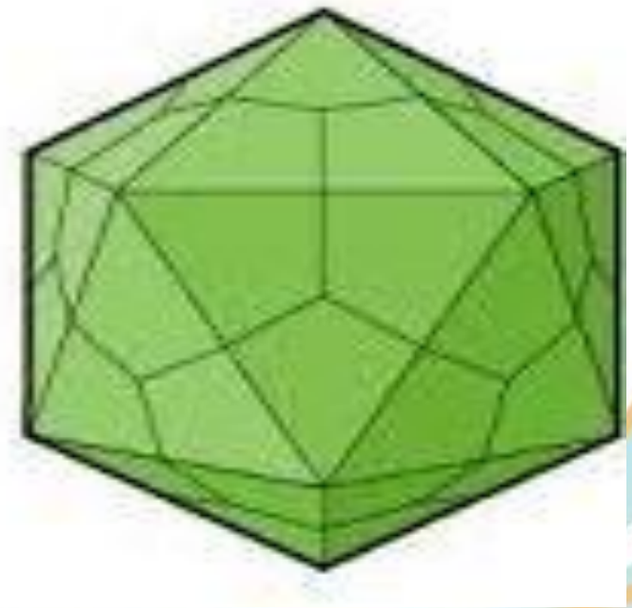
Amina Zee

Quaternary Structure of Proteins

- More complex rotational symmetries include icosahedral symmetry.
- An icosahedron is a regular 12-cornered polyhedron having 20 triangular faces.

Amina Zee

Quaternary Structure of Proteins



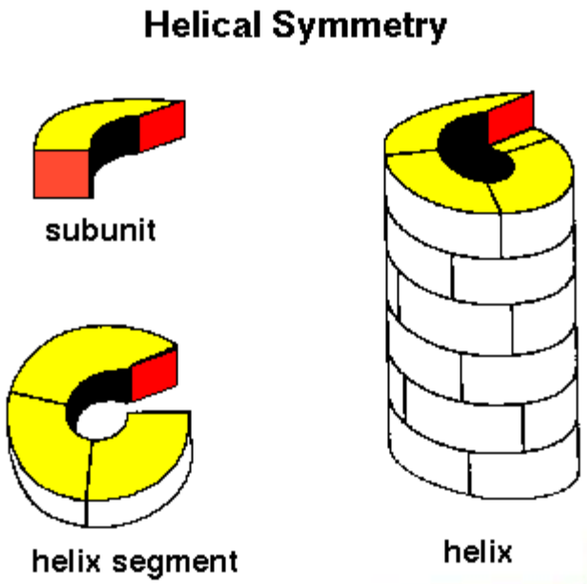
Amina Zee

Quaternary Structure of Proteins

- The other major type of symmetry found in oligomers is helical symmetry.

Amina Zee

Quaternary Structure of Proteins



(adapted from Voet and Voet, 1990)

Amina Zee

Sequence Analysis

Lec 69

Storage of Biological Sequence Information

Amina Zee

Storage of Biological Sequence Information

Background

- DNA sequences contain A, C, T & G
- mRNA sequences contain A, C, U & G
- Protein sequences contain A, R, N, D, C, E, Q, G, H, I, L, K, M, F, P, S, T, W, Y & P
- 20 Amino Acids

Amina Zee

Storage of Biological Sequence Information

Amino Acid	3-Letters	1-Letter
Alanine	Ala	A
Arginine	Arg	R
Asparagine	Asn	N
Aspartic Acid	Asp	D
Cysteine	Cys	C
Glutamic Acid	Glu	E
Glutamine	Gln	Q
Glycine	Gly	G
Histidine	His	H
Isoleucine	Ile	I
Leucine	Leu	L
Lysine	Lys	K
Methionine	Met	M
Phenylalanine	Phe	F
Proline	Pro	P
Serine	Ser	S
Threonine	Thr	T
Tryptophan	Trp	W
Tyrosine	Tyr	Y
Valine	Val	V

Amina Zee

Storage of Biological Sequence Information

Introduction

- DNA, mRNA and Proteins can be sequenced in the laboratory

- We can therefore obtain DNA, mRNA and Protein sequences which are simple lists of alphabets

Amina Zee

Storage of Biological Sequence Information

Storing Sequences

- DNA sequences may contain hundreds of thousands of bases
- RNA sequences can be a little less than the DNA but they are in a much larger variety

Amina Zee

Storage of Biological Sequence Information

Storing Sequences

- Proteins have long sequences and are large in number
- A problem then arises on how to store this information?

Amina Zee

Storage of Biological Sequence Information

Solution: Databases

- The solution lies with public sequence databases
- DNA/RNA-> GenBank (by NIH)
- Proteins->UniProt (by UniProt consortium)

Amina Zee

Storage of Biological Sequence Information

GenBank Overview

What is GenBank?

GenBank ® is the NIH genetic sequence database, an annotated collection of all publicly available DNA sequences (*Nucleic Acids Research*, 2013 Jan:41(D1):D36-42). GenBank is part of the International Nucleotide Sequence Database Collaboration , which comprises the DNA DataBank of Japan (DDBJ), the European Molecular Biology Laboratory (EMBL), and GenBank at NCBI. These three organizations exchange data on a daily basis.

The complete release notes for the current version of GenBank are available on the NCBI ftp site. A new release is made every two months. GenBank growth statistics for both the traditional GenBank divisions and the WGS division are available from each release.

An annotated sample GenBank record for a *Saccharomyces cerevisiae* gene demonstrates many of the features of the GenBank flat file format.

Access to GenBank

There are several ways to search and retrieve data from GenBank.

GenBank Resources

GenBank Home

Submission Types

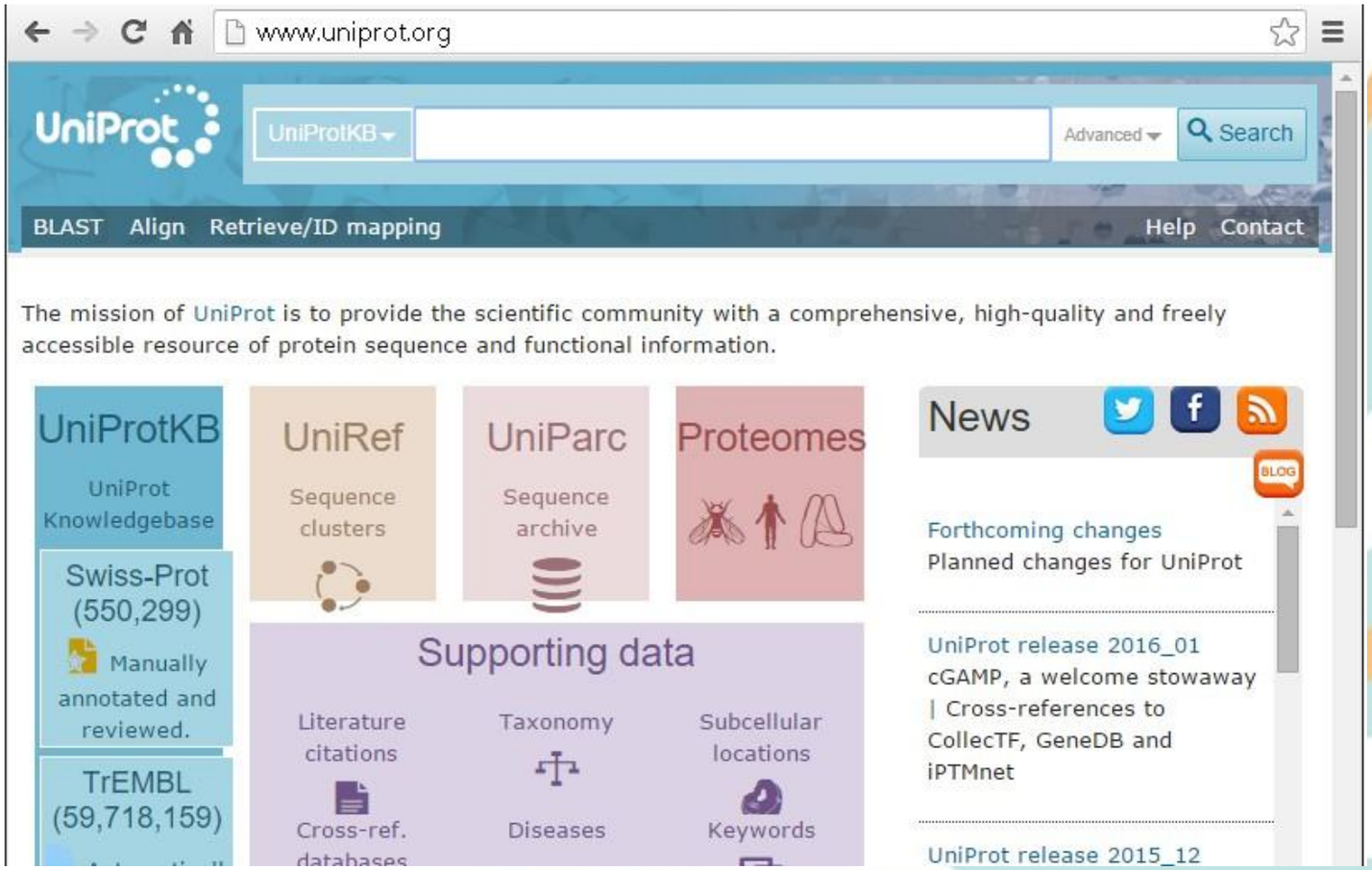
Submission Tools

Search GenBank

Update GenBank Records

Amina Zee

Storage of Biological Sequence Information



Amina Zee

Storage of Biological Sequence Information

Conclusion

- Nucleotide and amino acid sequences are stored in online databases
- DNA & RNA information is available in GenBank
- Protein sequence information is available in UniProt

END

Amina Zee

Sequence Analysis

Lec 70

Comparing Sequences

Amina Zee

Comparing Sequences

Background

- A large number of sequences are available in GenBank and Uniprot, publically
- What if we try to compare them?
- What will be get from such a comparison?

Amina Zee

Comparing Sequences

Introduction

By comparing sequences, we can get:

- Similarity
- Specific Differences
- Relationship
- Evolutionary Insight

Amina Zee

Comparing Sequences

DNA

ACTGGTCAGTCATGTCATGT

Amina Zee

Comparing Sequences

RNA

CAGUACGUAGCUAGGAGUAG

Amina Zee

Comparing Sequences

Protein

MQVKLFTLPHLKIHQLFTLPHLKIH

Amina Zee

Comparing Sequences

Conclusion

Several conclusions can be drawn on a set of sequences in terms of:

- **Similarity**
- **Specific Differences**
- **Relationship**
- **Evolutionary Insight**

END

Amina Zee

Sequence Analysis

Lec 71

Similarities & Differences in Sequences

Amina Zee

Similarities & Differences in Sequences

Background

By comparing two sequences, one can study:

- **Similarity**
- **Specific Differences**
- **Relationship**
- **Evolutionary History**

Amina Zee

Similarities & Differences in Sequences

Protein

M Q V K L F T ....

1. **Similarity**
2. **Specific Differences**
3. **Relationship**
4. **Evolutionary History**

Amina Zee

Similarities & Differences in Sequences

Exact Matching

Not only should the two or more sequences being compared have the exact same number of each nucleotide (for DNA /RNA) or amino acid (for Proteins), but that they should be arranged in the exact same order!

Amina Zee

Similarities & Differences in Sequences

Accommodating Differences – In exact Matching

- Regular Expression
- Signature Sequences
- PROSITE Patterns Database

Amina Zee

Similarities & Differences in Sequences

PROSITE Pattern for MAP Kinase

Haubold, Ch. 3, pg 44

F-x(10)-R-E-x(72,86)-R-D-x-K-x(9)-C

- Hyphens separate the elements of the pattern,
- Letters refer to amino acids,
- X indicates any amino acid,
- Bracketed numbers denote the repeat length of a residue,
- If this repeat length varies, the range of this variation is quoted in the brackets.

Similarities & Differences in Sequences

Conclusion

- **Exact Matches include complete matching in terms of residues and their order**

- **In exact matching includes partial matching in terms of residues and order with room for variations in both.**

END

Amina Zee

Sequence Analysis

Lec 72

Pairwise Sequence Alignment - I

Amina Zee

Pairwise Sequence Alignment - I

Background

- **Exact matches** include complete matching in terms of residues and their order
- **In exact matching** includes partial matching in terms of residues and order with room for variations in both.

Amina Zee

Pairwise Sequence Alignment - I

PROSITE Pattern for MAP Kinase

Haubold, Ch. 3, pg 44

F-x(10)-R-E-x(72,86)-R-D-x-K-x(9)-C

- Hyphens separate the elements of the pattern,
- Letters refer to amino acids,
- X indicates any amino acid,
- Bracketed numbers denote the repeat length of a residue,
- If this repeat length varies, the range of this variation is quoted in the brackets.

Pairwise Sequence Alignment - I

Introduction

- Concept of conserved residues or nucleotides

- Before starting to matching nucleotide or amino acid sequences, we need to find location of conserved residues or nucleotides

Amina Zee

Pairwise Sequence Alignment - I

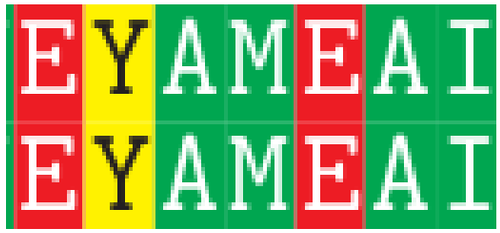
Alignment

- The process of inexact matching while keeping in view the conserved residues is called sequence "Alignment"
- Pairwise sequence alignment is therefore alignment of a pair of two sequences

Amina Zee

Pairwise Sequence Alignment - I

Two Examples of Pairwise Alignment



Amina Zee

Pairwise Sequence Alignment - I

Conclusion

- **Pairwise sequence alignment helps compare two sequences**
- **It takes into consideration inexact matching**
- **Matches are colored and missing nucleotides are denoted by '.'**

END

Amina Zee

Sequence Analysis

Lec 73

Pairwise Sequence Alignment - II

Amina Zee

Pairwise Sequence Alignment - II

Background

- Pairwise sequence alignment compares two amino acid or nucleotide sequences
- It employs inexact matching
- Matches are colored and missing nucleotides are denoted by '.'

Amina Zee

Pairwise Sequence Alignment - II

Salient Points

- Sliding the sequences past each other
- Aim is to maximize matches
- Gaps ('.') could be inserted to account for insertions and deletions

Pairwise Sequence Alignment - II

Salient Points

- **Gaps ('.') may carry a penalty for reducing scores from unreasonable alignments**
- **Without a gap penalty an exact match and match with gaps will get equal scores**

Pairwise Sequence Alignment - II

Types of pairwise alignments

Global	Local
• Maximizing sequence matches over the entire length of two sequences, by introducing gaps. • Used to determine overall similarity, conservation.	• Find regions between two sequences that have the strongest similarity, excluding less similar regions. • Finding similar domains, motifs, detecting distant homology

Pairwise Sequence Alignment - II

Global alignment - maximizes the number of matches between the query and source sequences along the entire length of both the sequences.

Local alignment - gives the highest scoring local match between both query and sequences.

Optimal alignment - one that exhibits the most correspondences between the query and the source sequences. It is the alignment with the **highest score**. Biologically meaningful?

Amina Zee

Pairwise Sequence Alignment - II

Conclusion

- **Gaps are inserted in a sequence being aligned to accommodate missing amino acids or nucleotides**
- **Global and Local alignment is used depending on the required type of pairwise alignment**

END

Amina Zee

Sequence Analysis

Lec 74

Pairwise Sequence Alignment - III

Amina Zee

Pairwise Sequence Alignment - III

Background

- Given a pair of sequences (strings of characters)
- Pairwise alignment tells the similarity between the sequences by maximizing the matches

Amina Zee

Pairwise Sequence Alignment - III

- Sequences can differ from each other in three different ways:
  - Substitutions A**C**GA → A**G**GA
  - Insertions ACGA → AC**C**GA
  - Deletions A**C**GA → AGA

- No change? ***match***
- Insertions or deletions ("indels") result in ***gaps*** in alignments
- Substitutions result in ***mismatches***

Pairwise Sequence Alignment - III

Optimal Alignment

Align 1 & 2

1: THIS IS A RATHER LONGER SENTENCE THAN THE NEXT.
2: THIS IS A SHORT SENTENCE.

1: THIS IS A RATHER LONGER SENTENCE THAN THE NEXT.
2: THIS IS A .....SHORT.....SENTENCE.....

OR

1: THIS IS A RATHER LONGER SENTENCE THAN THE NEXT.
2: THIS IS A ....SHORT.....SENT..EN.....CE.....

Which alignment is optimal in biological context?

Amina Zee

Pairwise Sequence Alignment - III

Conclusion

- **Global and local pairwise alignments are two types of alignment and they can lead to different results**
- **Indels lead to Gaps in alignments**
- **Substitutions lead to mismatches**

END

Amina Zee

Sequence Analysis

Lec 75

Identity vs. Similarity

Amina Zee

Identity vs. Similarity

Background

- How to compare two biological sequences?
- How to measure the degree of match between sequences?
- Two concepts:
  - Identity
  - Similarity

Amina Zee

Identity vs. Similarity

Sequence Identity

"Identity is the number of nucleotides or amino acids which match exactly between two biological sequences"

1: CATGCTT
2: CATGC

Calculate the identity between sequences 1 & 2.

Number of matches = 5

Smaller Length: Length (1) = 7, Length (2) = 5

Identity = Num. of matches/Smaller Legth * 100% = 100%

Amina Zee

Identity vs. Similarity

Points to remember

- Gaps are not counted
- Identity measurement is made on the shorter of the two sequences

Amina Zee

Identity vs. Similarity

Sequence Similarity

"Result of matching and transforming one sequence to the other by finding the smallest set of edit operations (insertions, deletions, and substitutions)."

```
1: CA T GC T .  C
2: CA .  G .  TG C
```

Can be calculated by pairwise sequence

For computing similarity between sequences, you will need to first align the two sequences using pairwise sequence alignment!

Identity vs. Similarity

Conclusion

- **Identity is the count of exact matches between two sequences**
- **Gaps are excluded**
- **Similarity is the comparison between sequences calculated by using alignment approach**

END

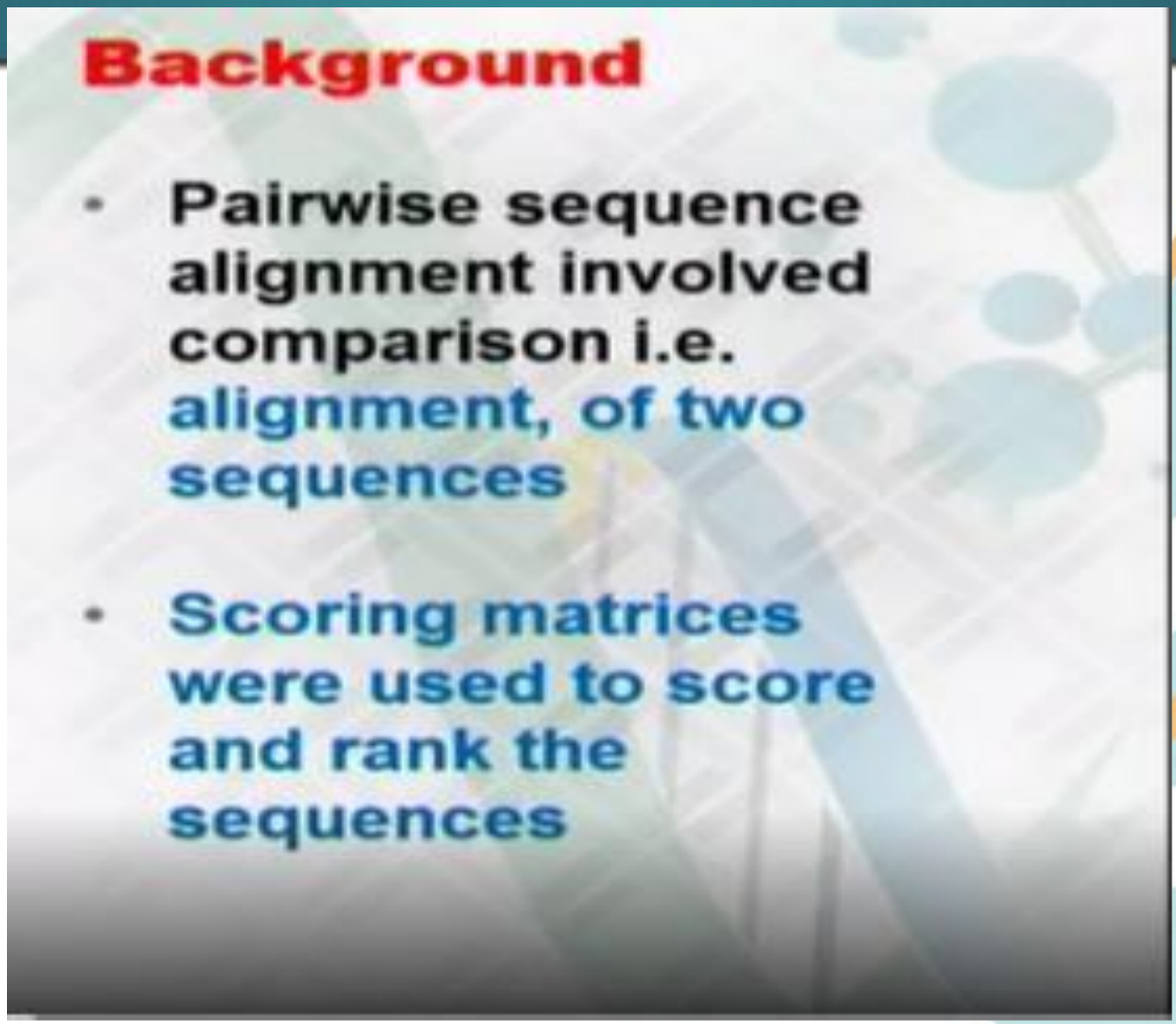
Amina Zee

Lec 76

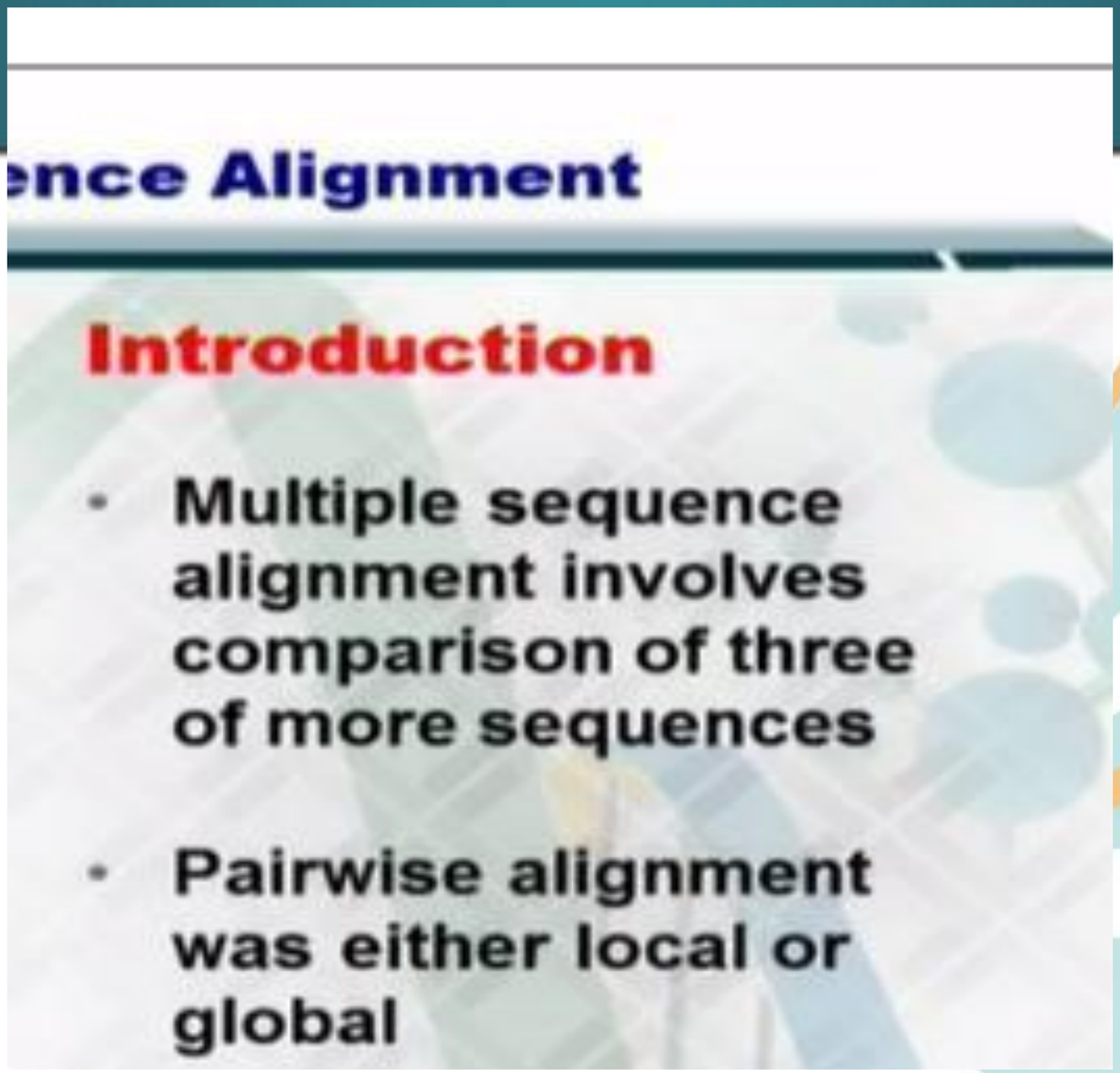
Introduction to multiple sequence Alignment

Amina Zee

Introduction to multiple sequence Alignment



Amina Zee


ence Alignment
Introduction
Multiple sequence alignment involves comparison of three of more sequences
Pairwise alignment was either local or global

Amina Zee

MSA vs Pairwise

- For pairwise alignments, we used Dynamic Programming
- For MSA, Dynamic Programming may get very expensive, computationally.
- Solution: Progressive Alignment

MSA vs Pairwise

- For pairwise alignments, we used Dynamic Programming
- For MSA, Dynamic Programming may get very expensive, computationally.
- Solution: Progressive Alignment

Amina Zee

Conclusion

- MSA can help align multiple sequences
- Dynamic Programming cannot be used as it becomes expensive
- Progressive alignment can help perform MSA!
- CLUSTAL – Online tool

Amina Zee

Lec 77

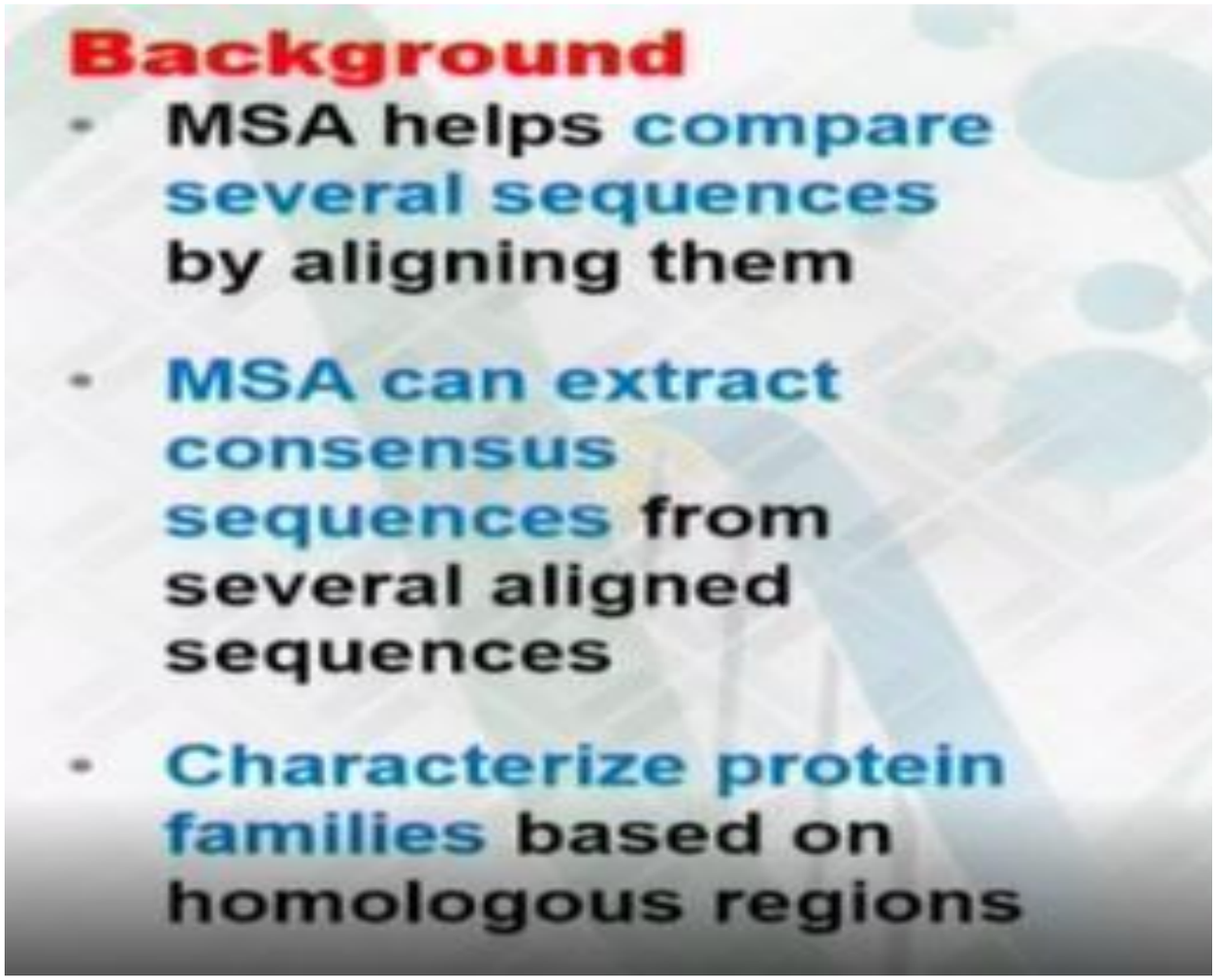
More on Multiple Sequence Alignment

Amina Zee

Background

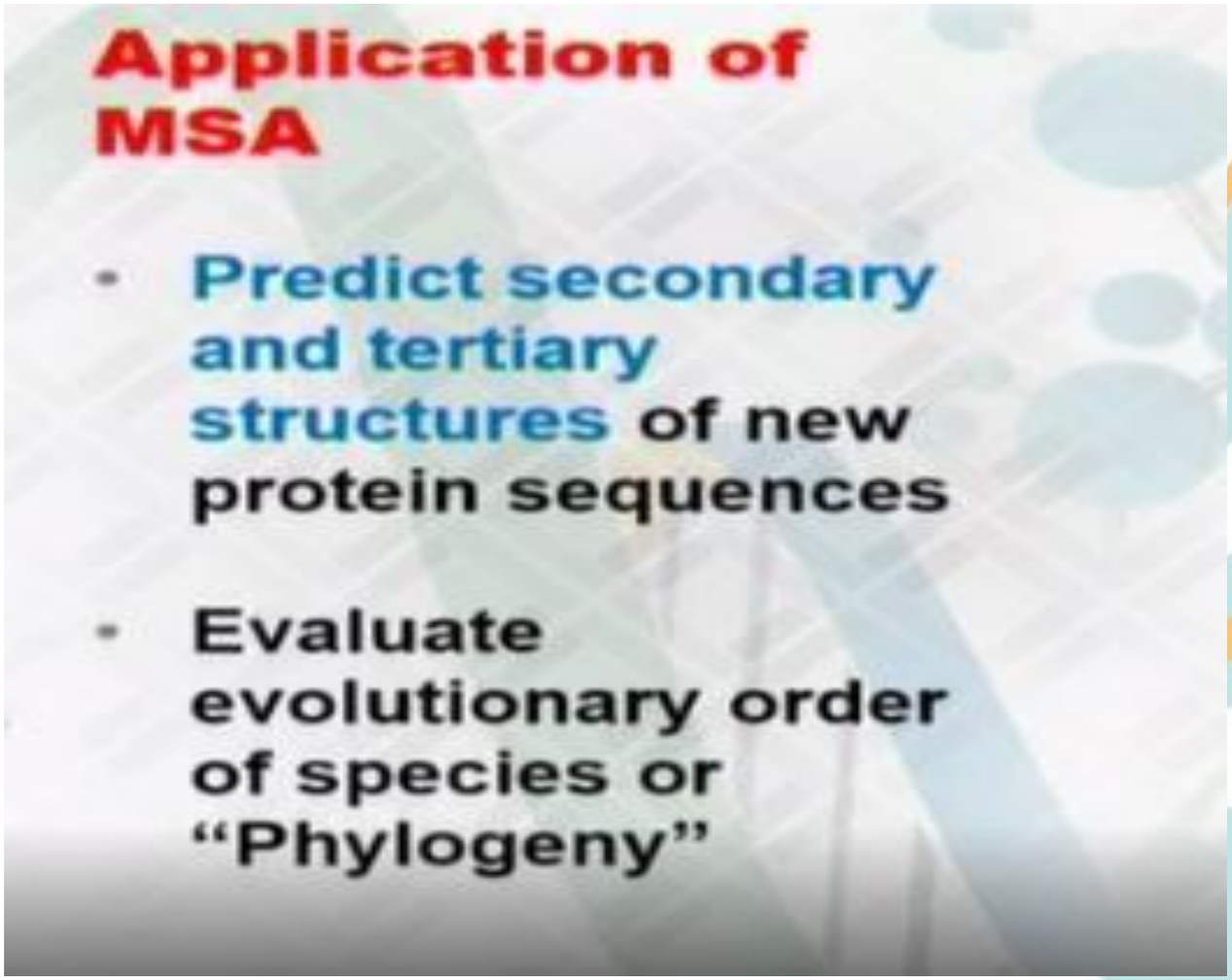
- MSA involves progressive alignment of sequences
- Doing so many progressive alignments can be slow
- How do progressive alignments work?

Amina Zee


Background
MSA helps compare several sequences by aligning them
MSA can extract consensus sequences from several aligned sequences
Characterize protein families based on homologous regions

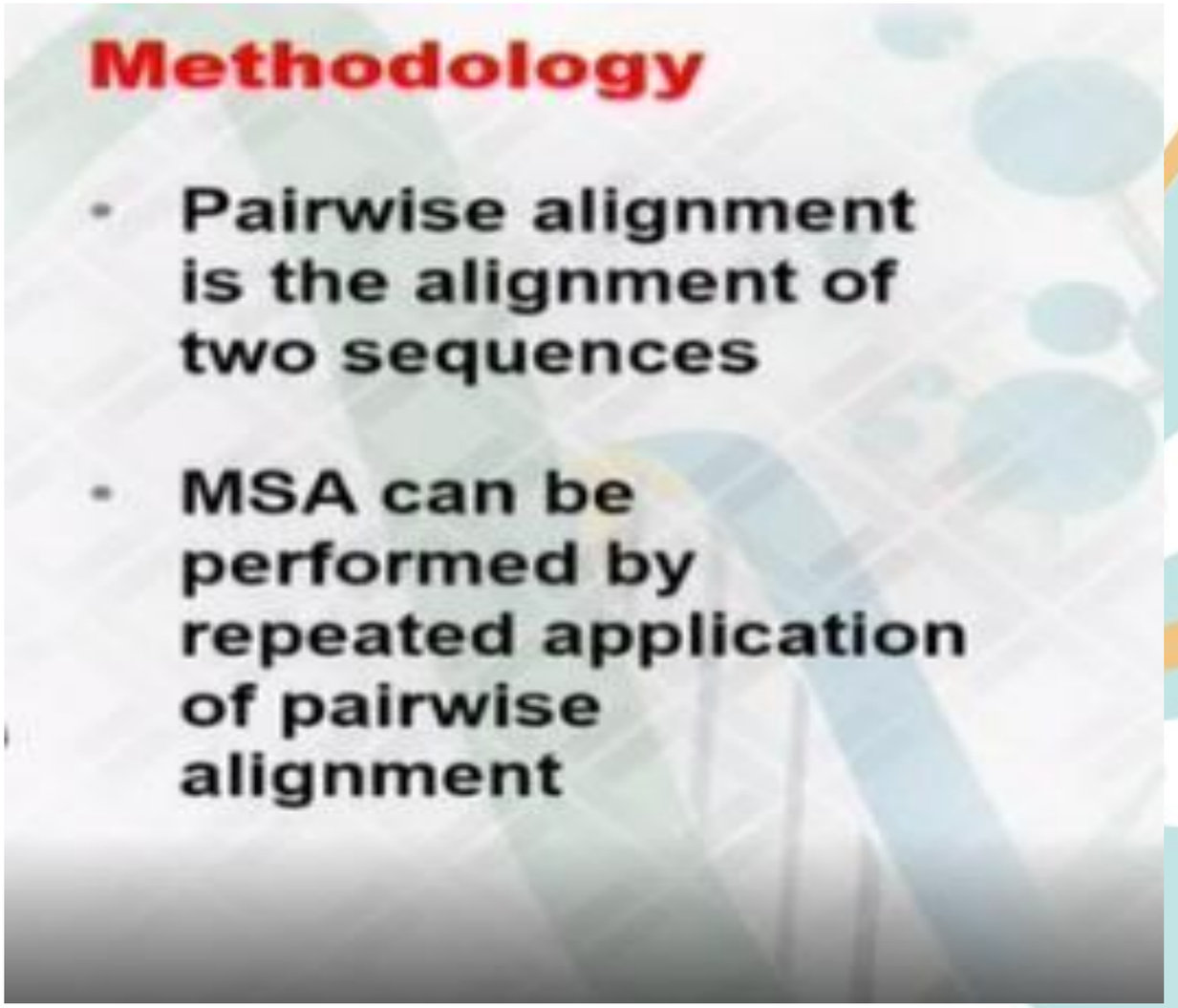
Amina Zee

More on Multiple Sequence Alignment



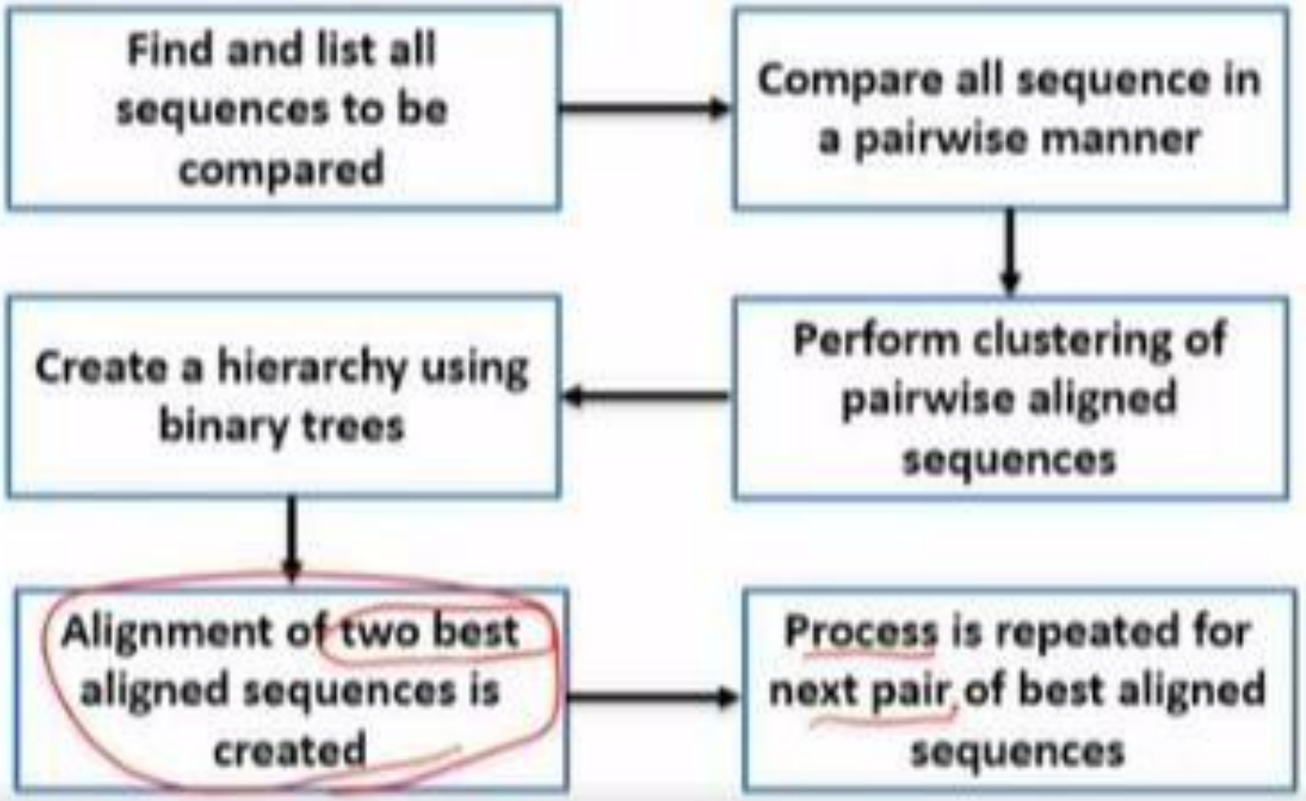
Amina Zee

More on Multiple Sequence Alignment



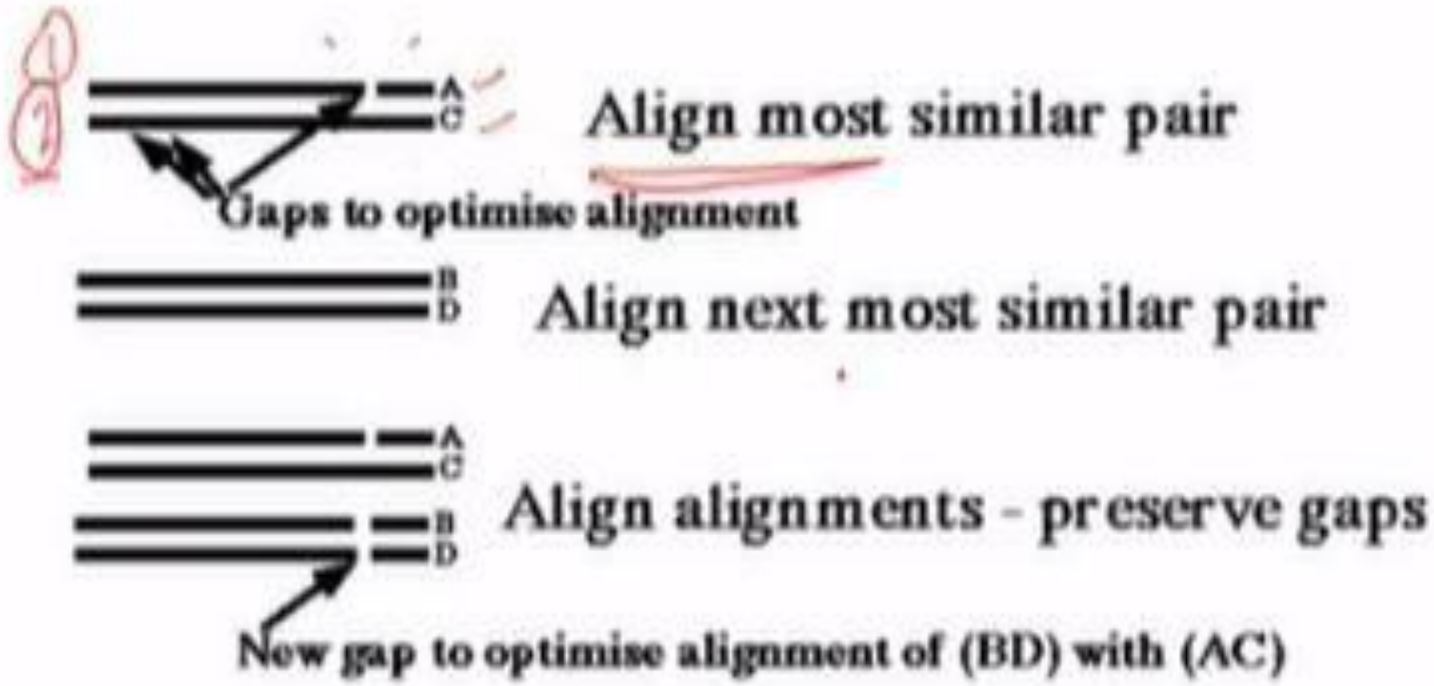
Amina Zee

More on Multiple Sequence Alignment



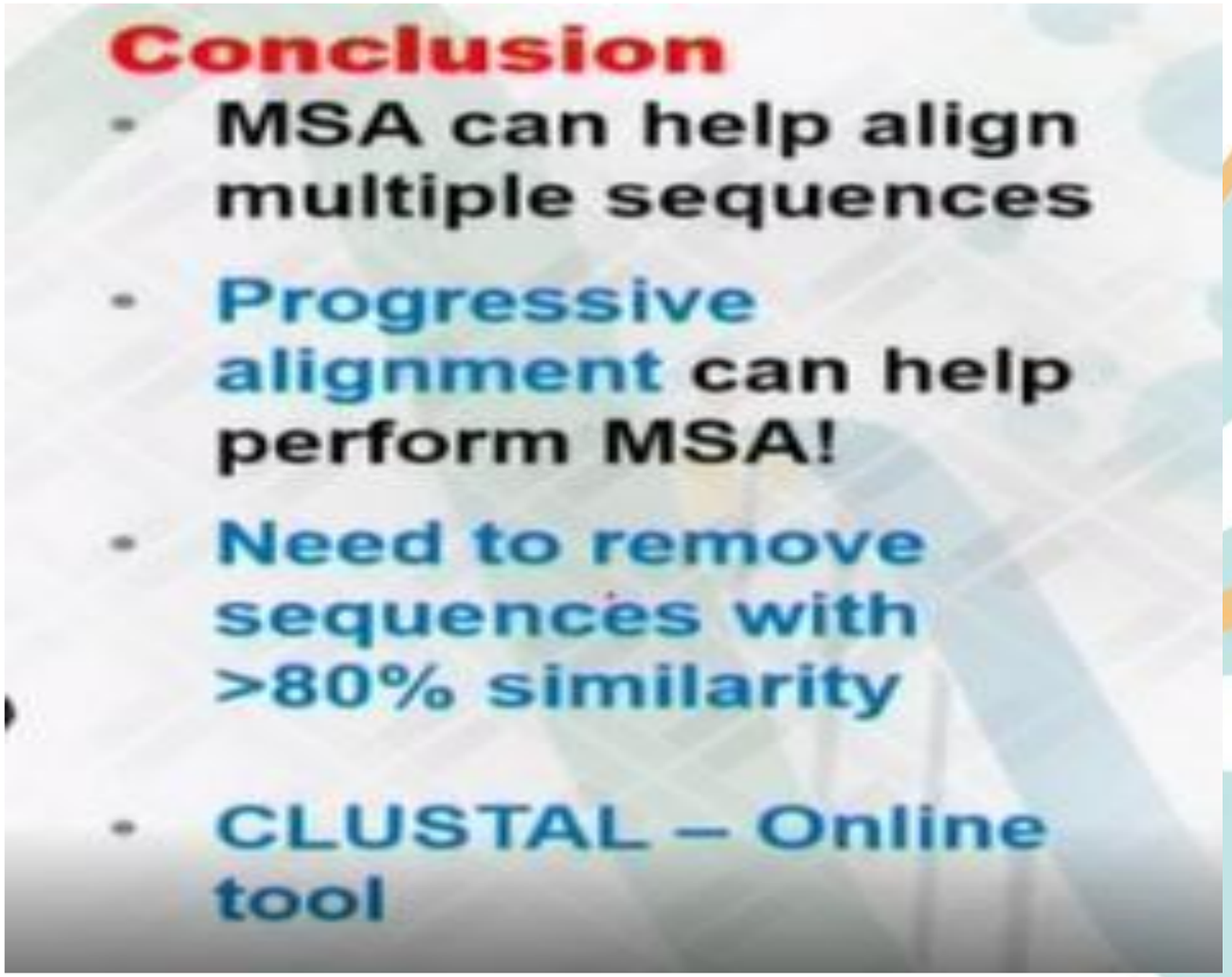
Amina Zee

More on Multiple Sequence Alignment



Amina Zee

More on Multiple Sequence Alignment



Amina Zee

Lesson 78

Progressive Alignment for MSA

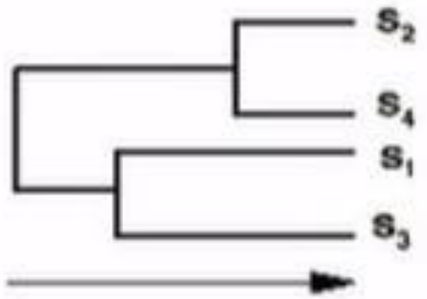
Amina Zee

Progressive Alignment for MSA

Amina Zee

Progressive Alignment for MSA

- Step 1 : Pairwise Alignment of all sequences
  Example: S_1, S_2, S_3, S_4 , so that is 6 pairwise comparisons!
- Step 2: Construct a Guide Tree (Dendogram) using a *Distance Matrix*



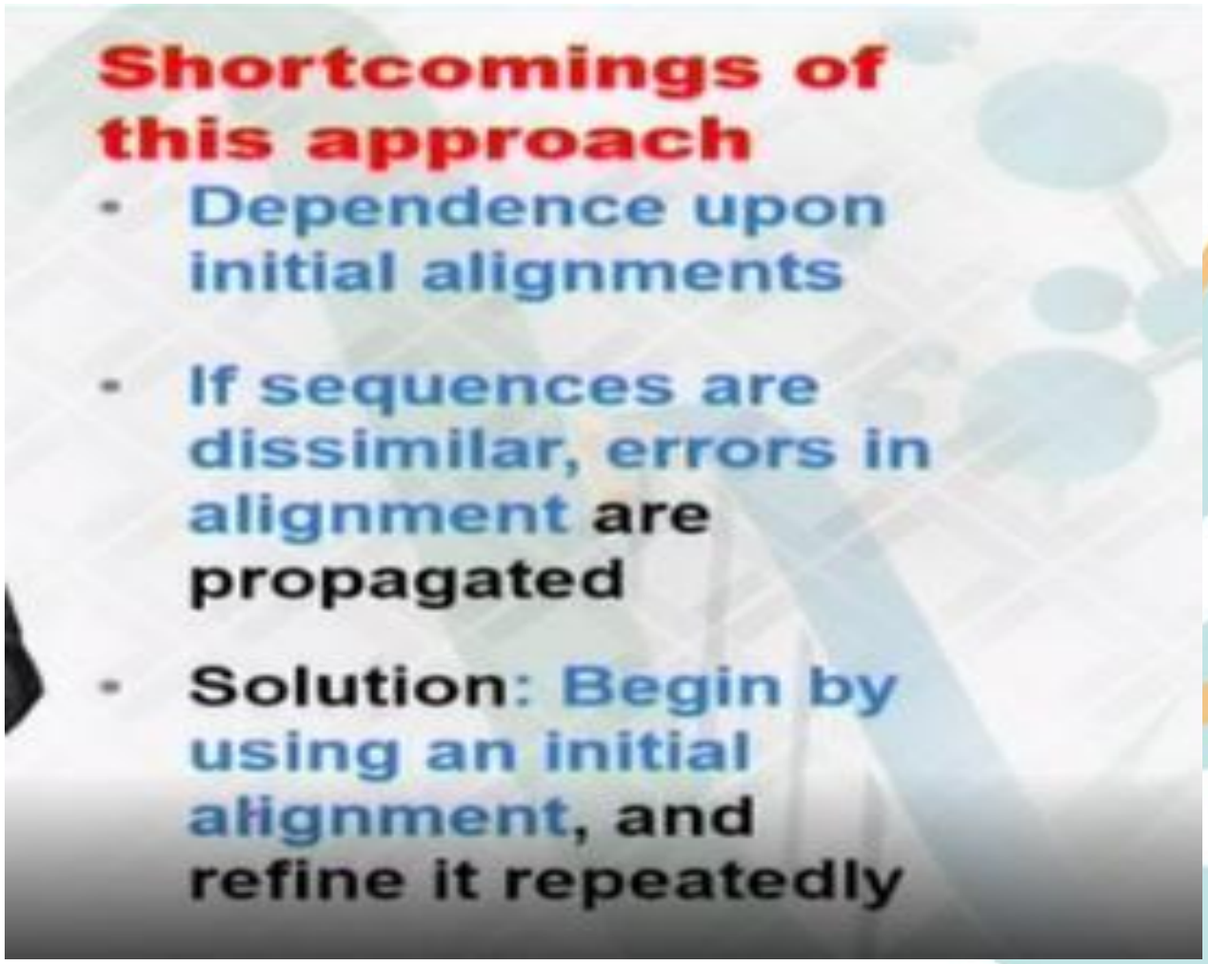
$$Distance = 1 - \frac{Matches}{Comparisons}$$

$$Similarity = \frac{Matches}{Comparisons}$$

- Step 3: Progressive alignment following branching order in tree

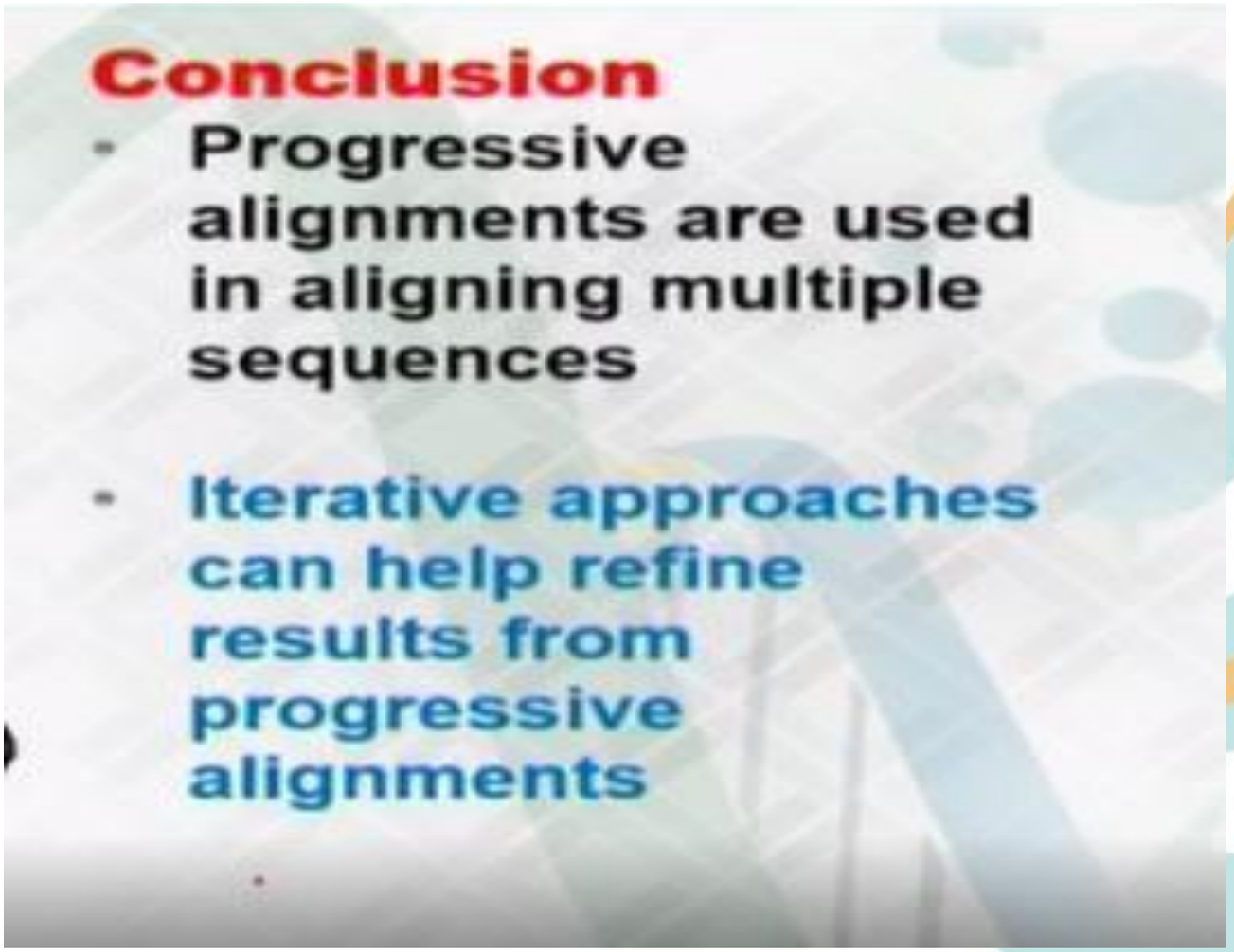
Amina Zee

Progressive Alignment for MSA



Amina Zee

Progressive Alignment for MSA



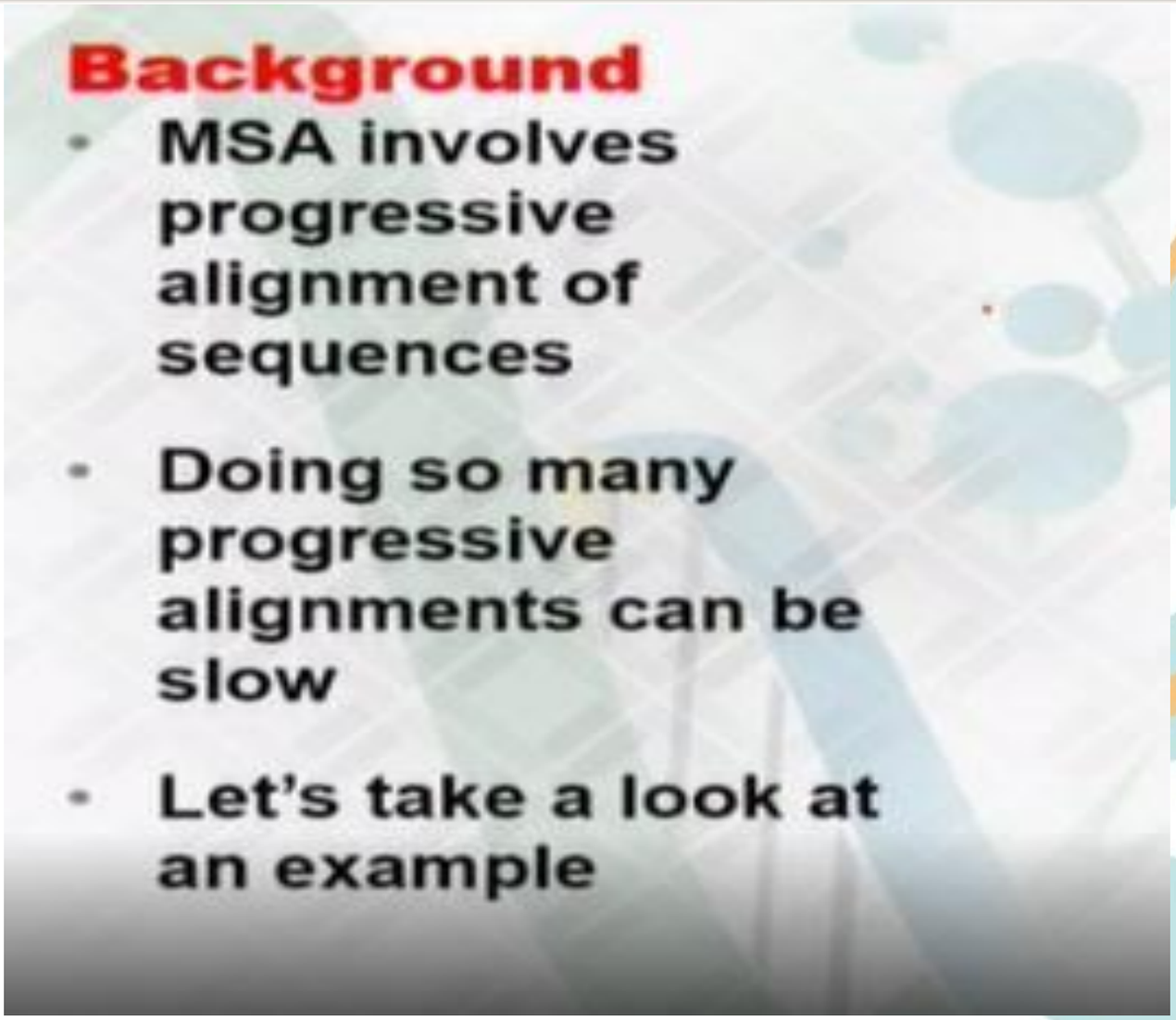
Amina Zee

Lec 79

- **MSA Example**

Amina Zee

MSA Example



Amina Zee

MSA Example

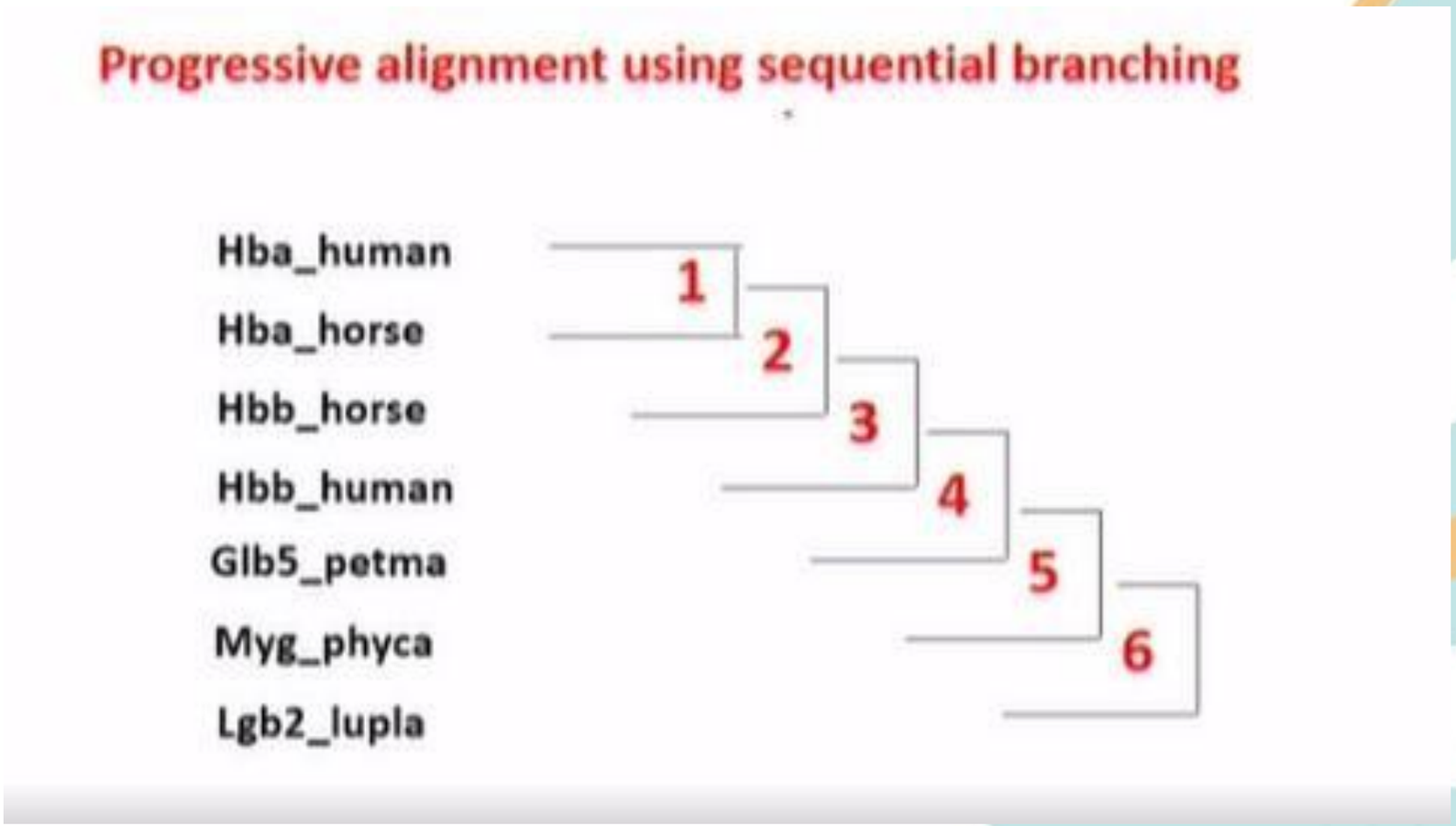
MSA on globin sequences

$$\text{distance between 2 sequences} = 1 - \frac{\text{No. identical residues}}{\text{No. aligned residues}}$$

		Hbb_human	Hbb_horse	Hba_human	Hba_horse	Myg_phyca	Glb5_petma	Lgb2_lupla
Hbb_human	1	-						
Hbb_horse	2	.17	-					
Hba_human	3	.59	.60	-				
Hba_horse	4	.59	.59	.13	-			
Myg_phyca	5	.77	.77	.75	.75	-		
Glb5_petma	6	.81	.82	.73	.74	.80	-	
Lgb2_lupla	7	.87	.86	.86	.88	.93	.90	-

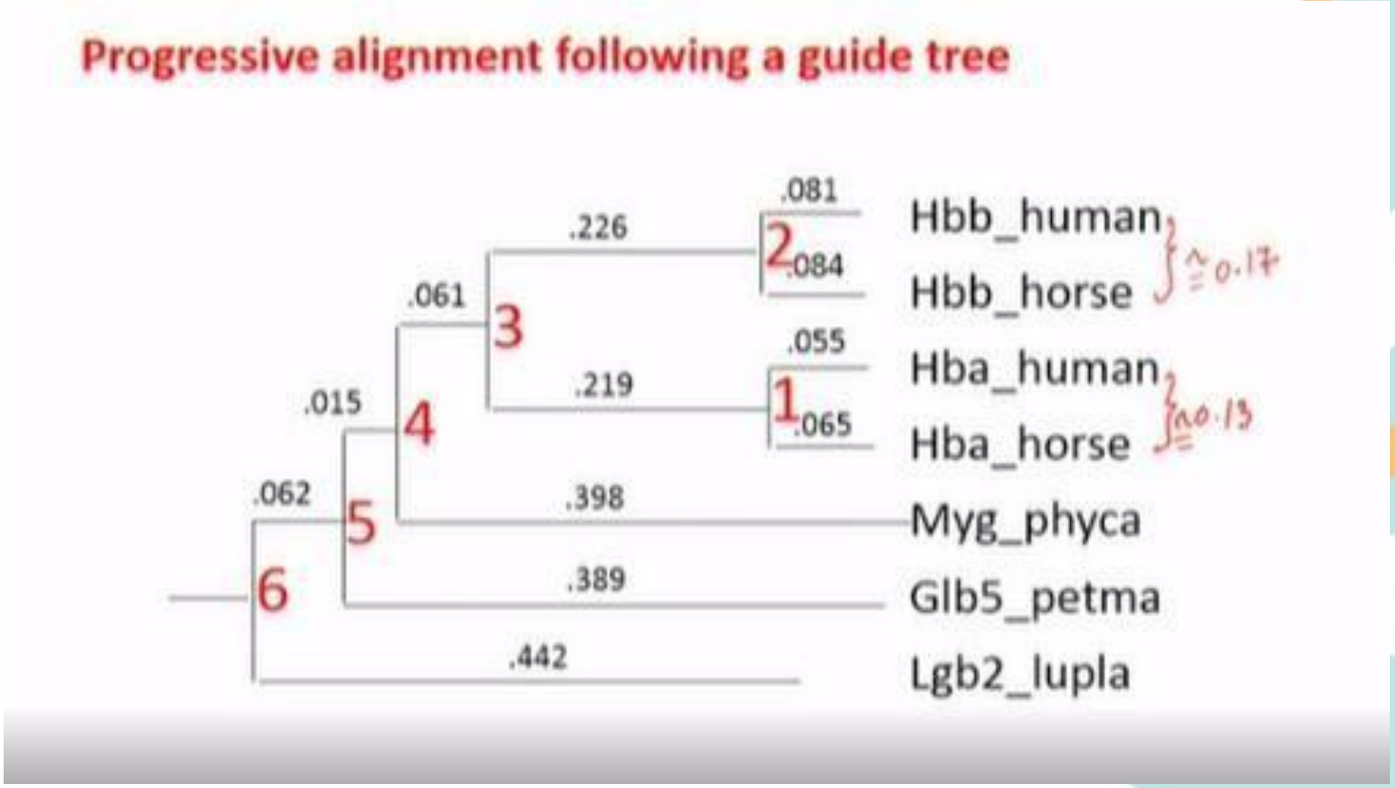
Amina Zee

MSA Example



Amina Zee

MSA Example



Amina Zee

MSA Example

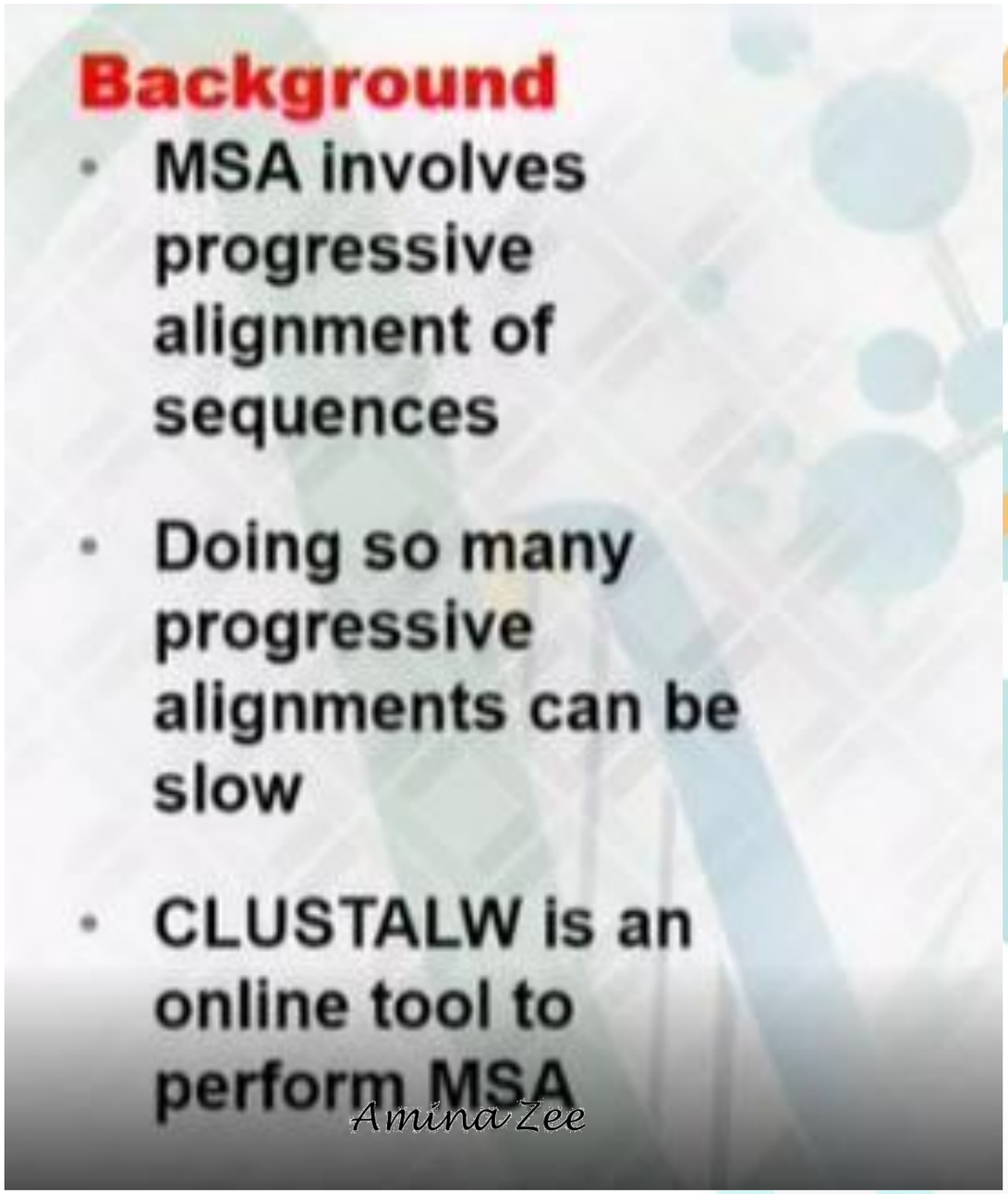
Conclusion
MSA can be better performed using clustering strategies followed by alignment of the alignments later
CLUSTAL is a free online tool that does all of this for us!
Amina Zee

Lec 80

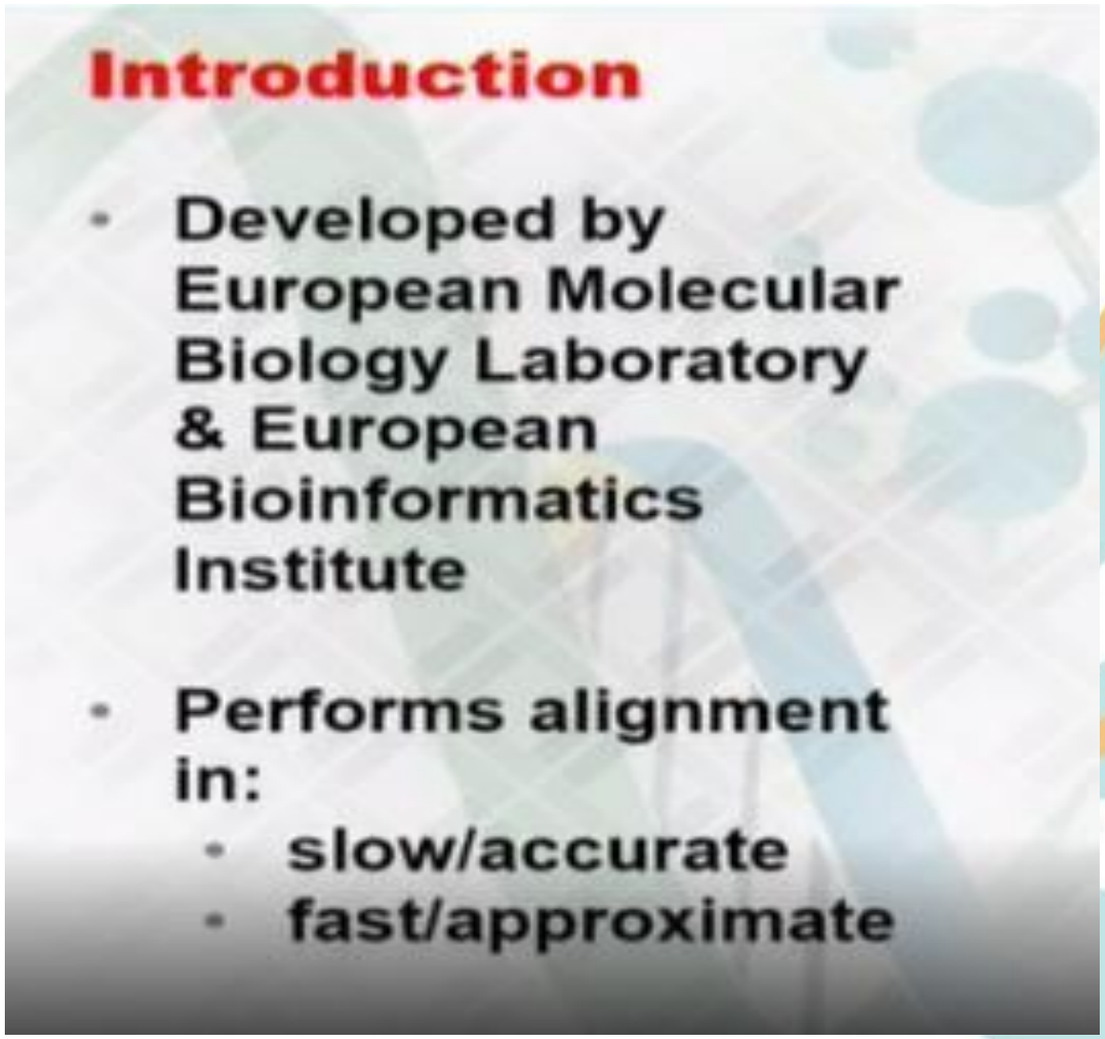
CLUSTAL

Amina Zee

CLUSTAL

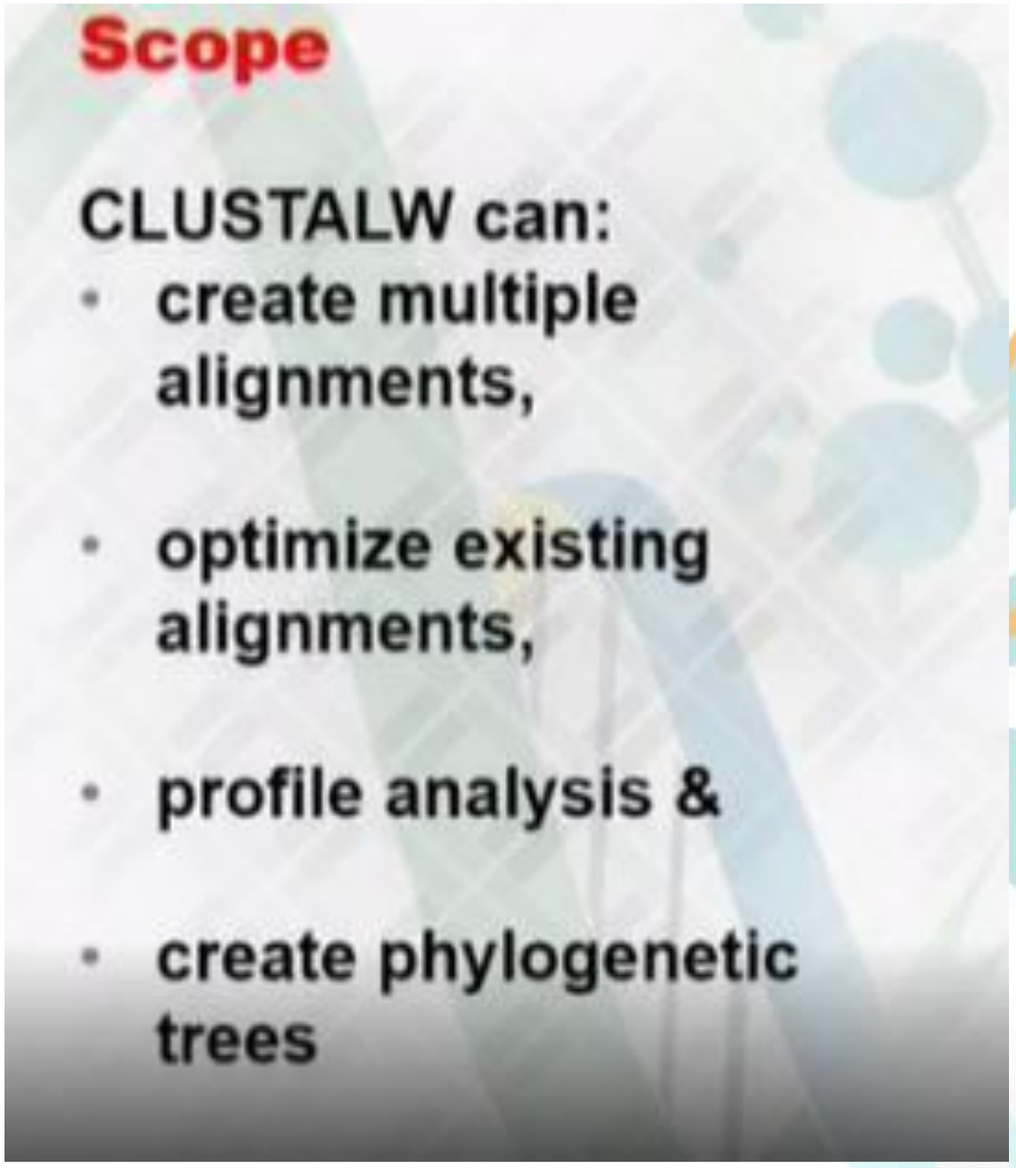

Background
MSA involves progressive alignment of sequences
Doing so many progressive alignments can be slow
CLUSTALW is an online tool to perform MSA
Amina Zee

CLUSTAL



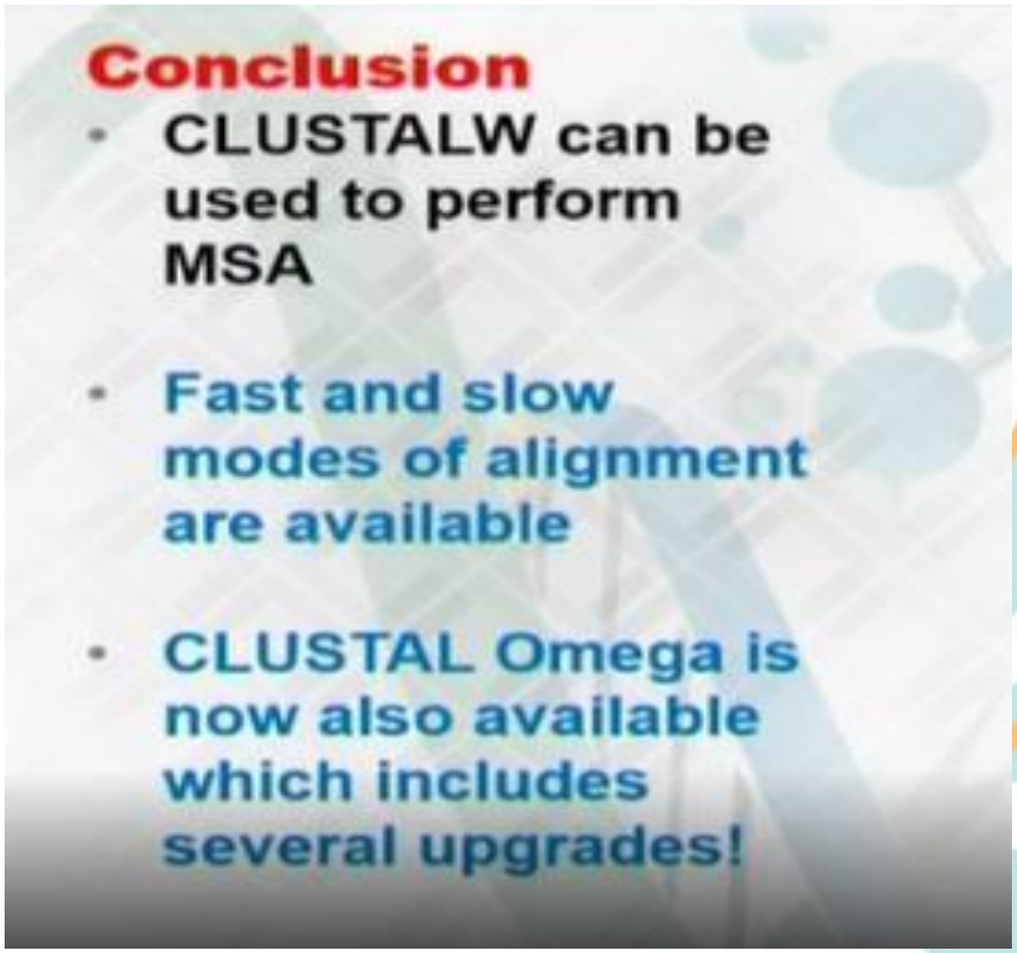
Amina Zee

CLUSTAL



Amina Zee

CLUSTAL



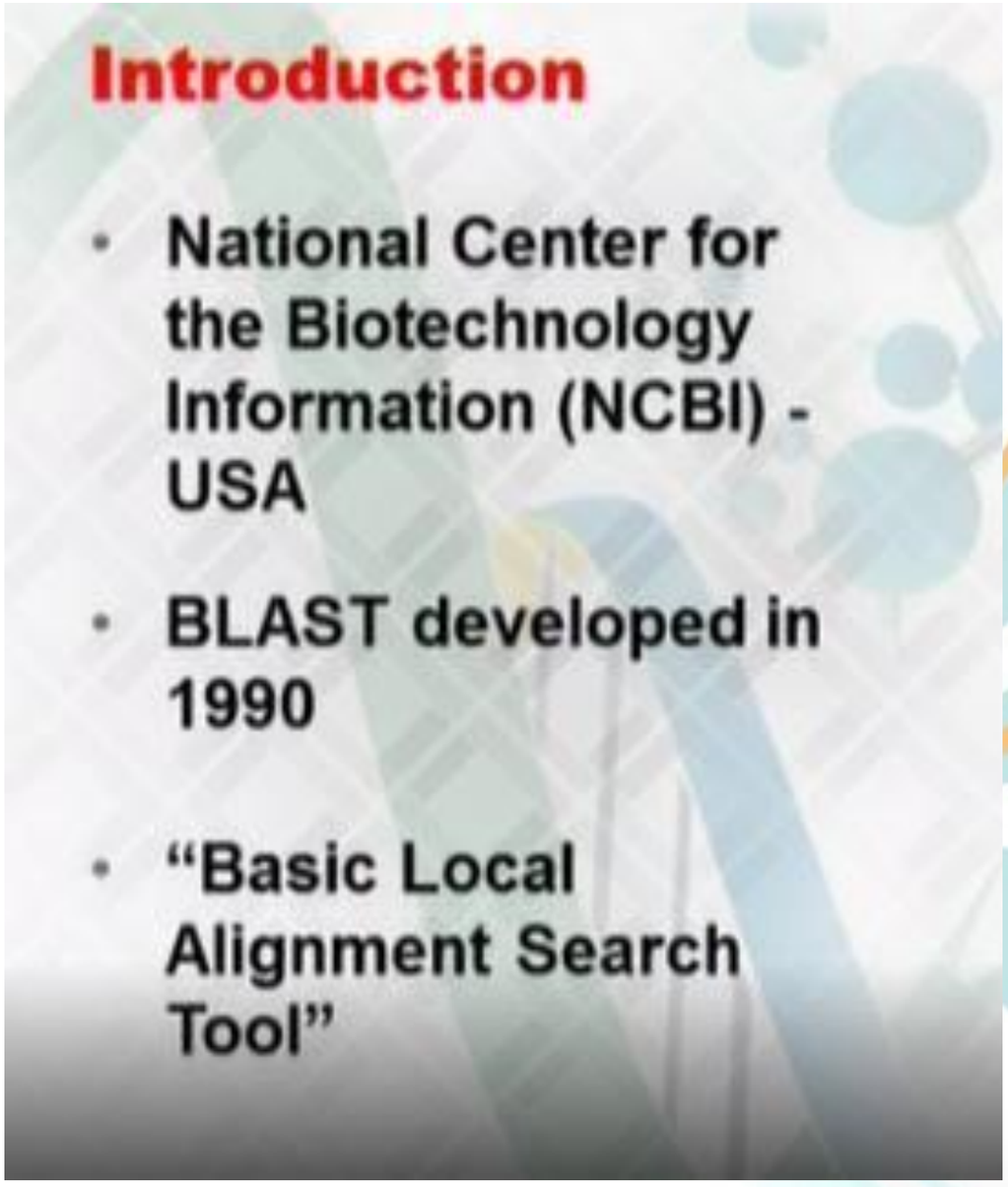
Amina Zee

Lec 81

Introduction to BLAST - I

Amina Zee

Introduction to BLAST - I

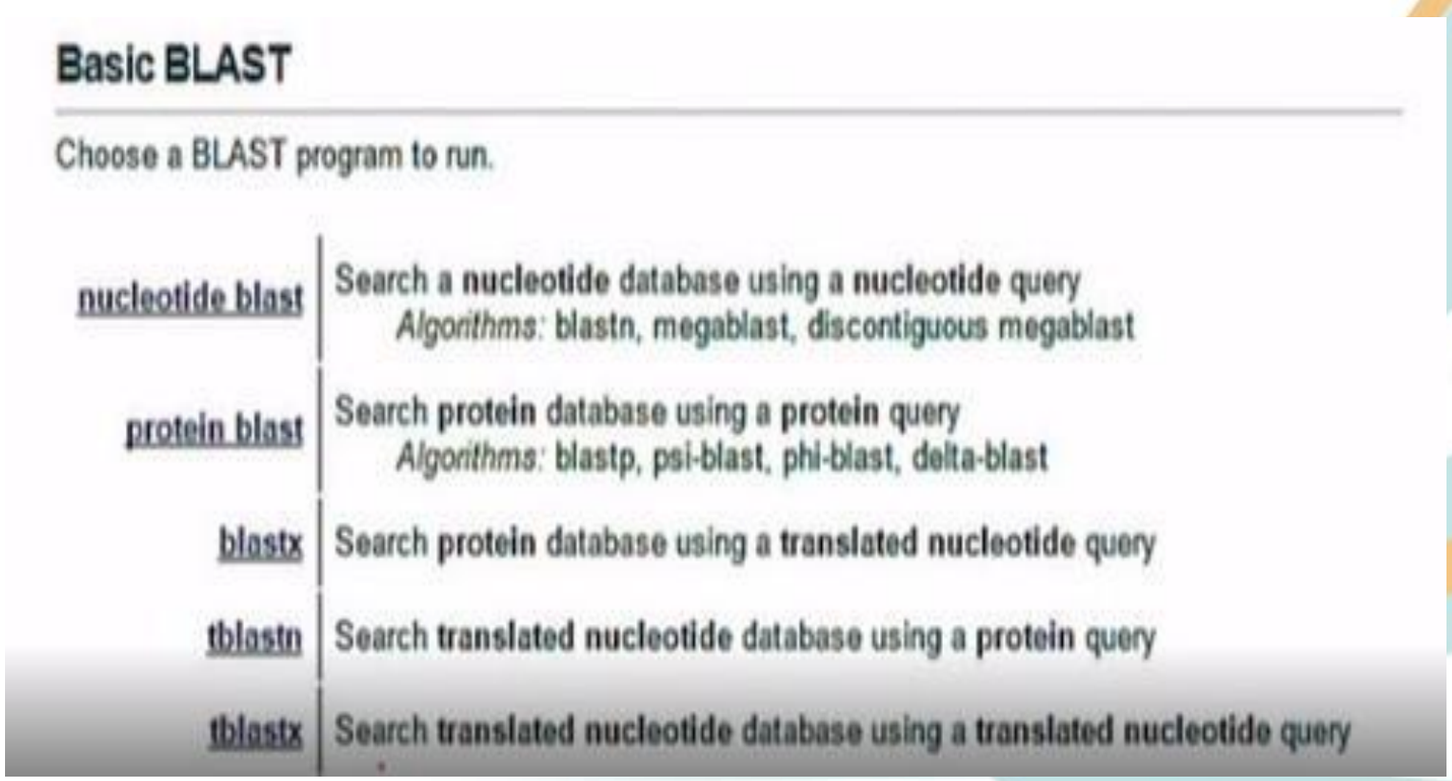


Amina Zee

Background
• Searches databases for query protein and nucleotide sequences
• Also searches for translational products etc
• Online availability
blast.ncbi.nlm.nih.gov/Blast.cgi

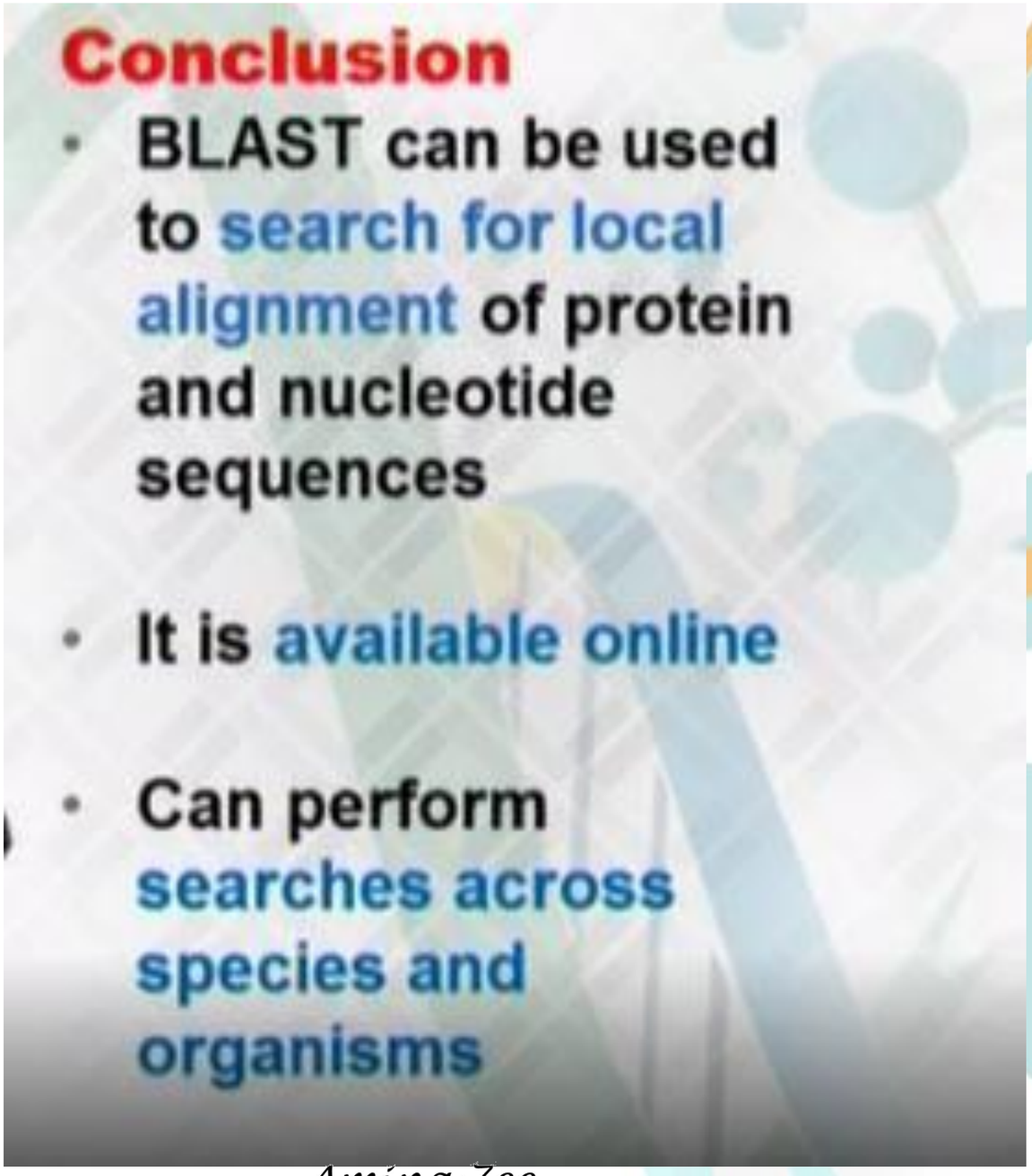
Amina Zee

Introduction to BLAST - I



Amina Zee

Introduction to BLAST - I



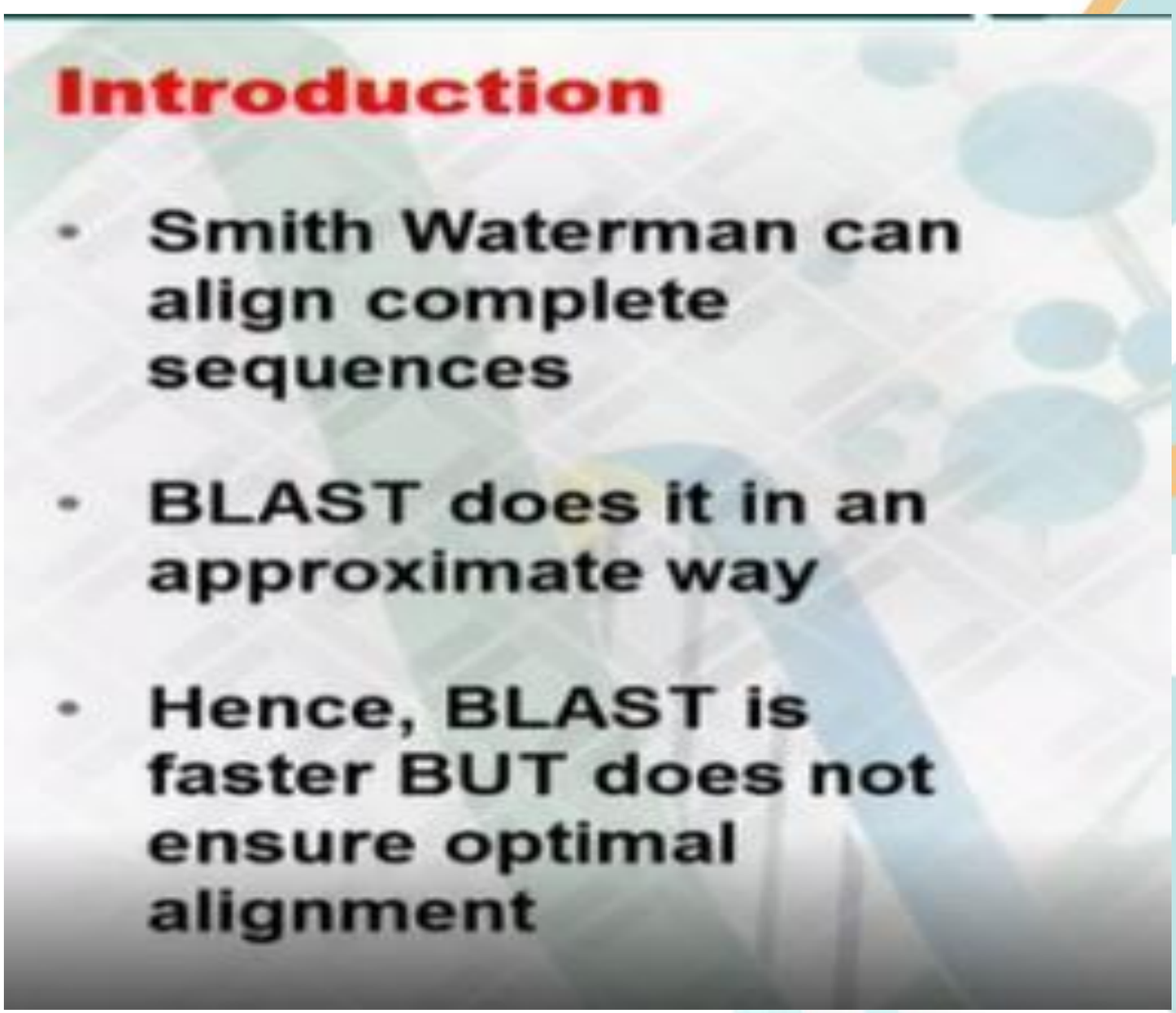
Amina Zee

Lec 82

Introduction to BLAST - II

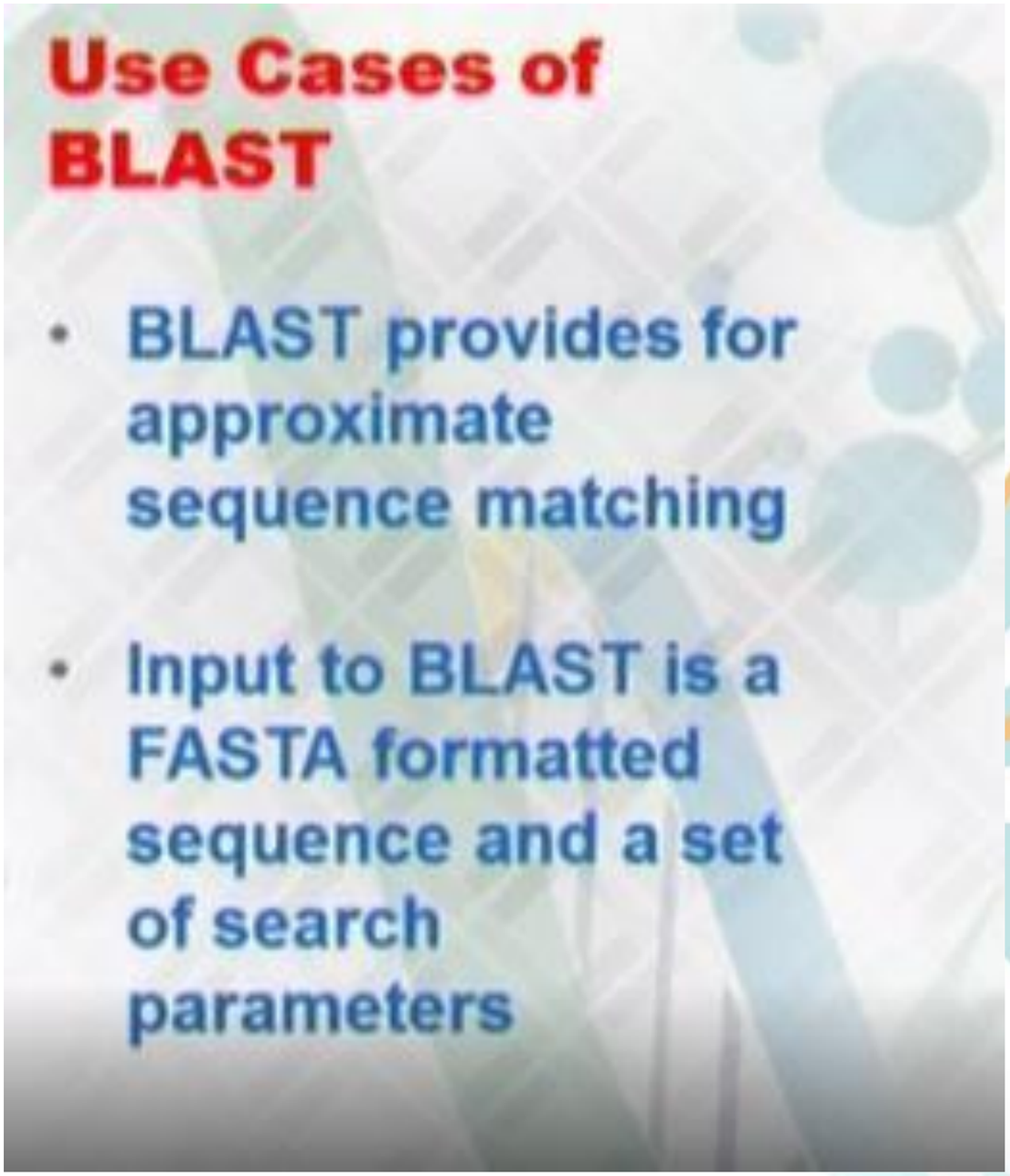
Amina Zee

Introduction to BLAST - II



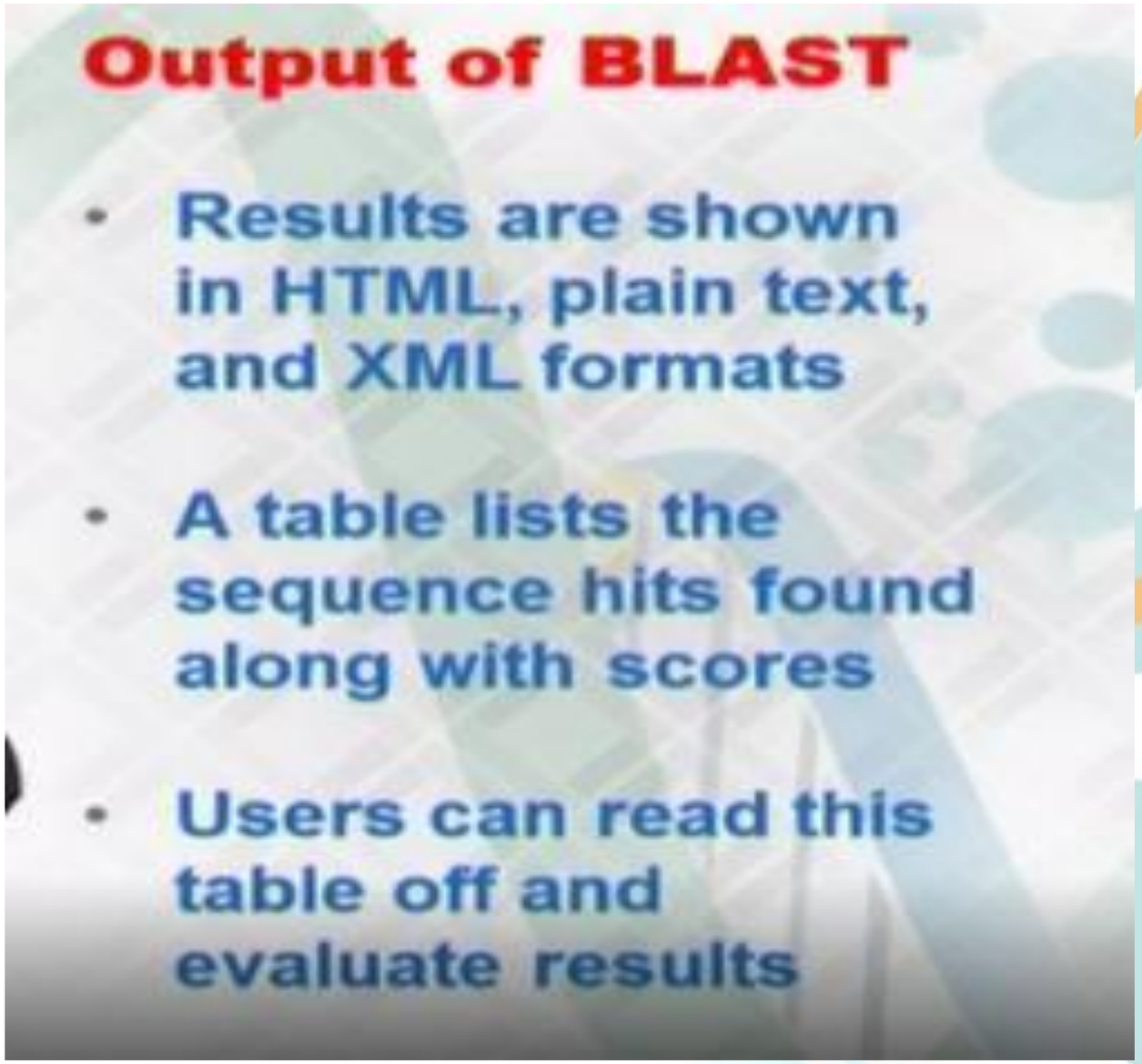
Amina Zee

Introduction to BLAST - II



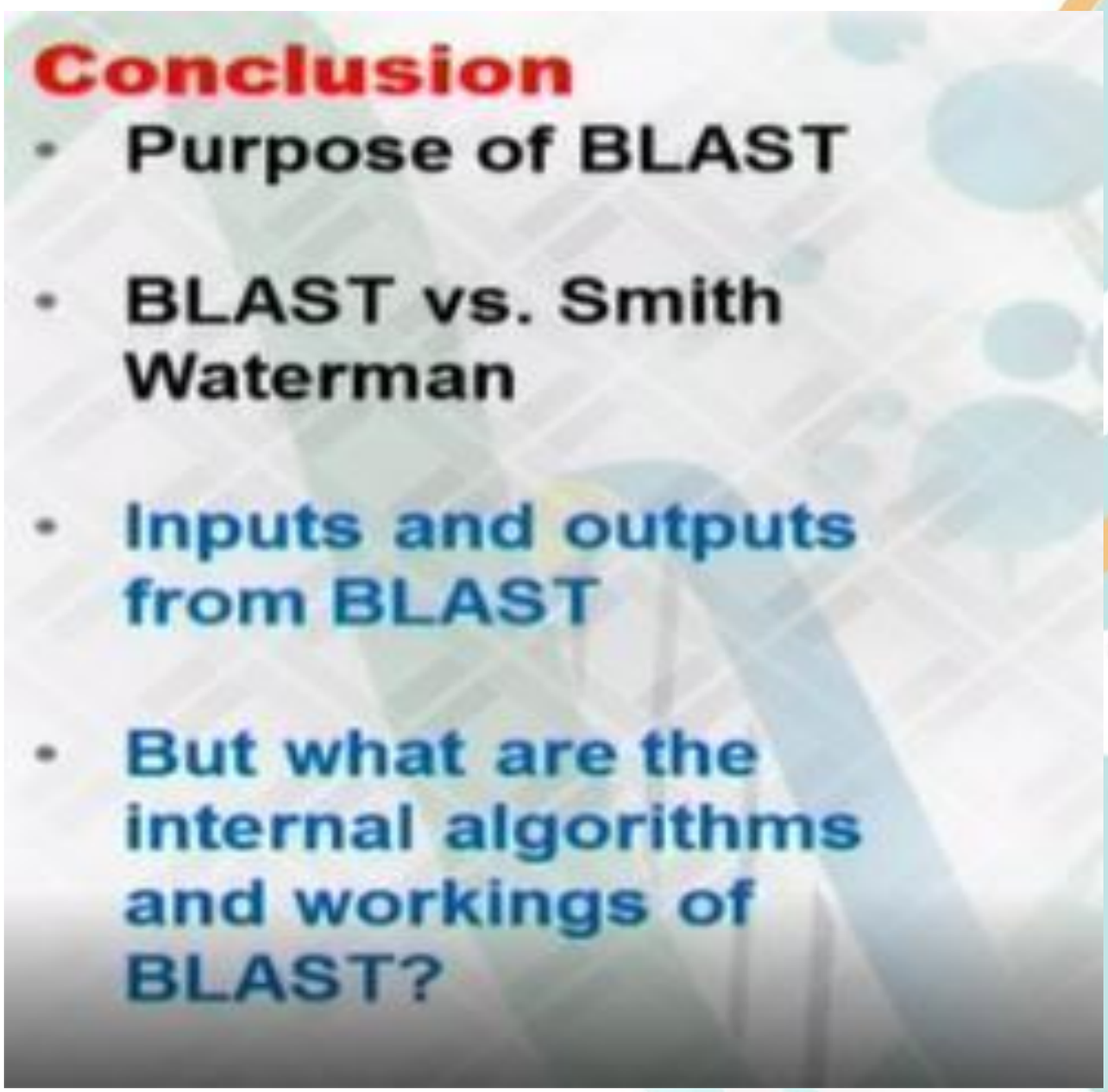
Amina Zee

Introduction to BLAST - II



Amina Zee

Introduction to BLAST - II



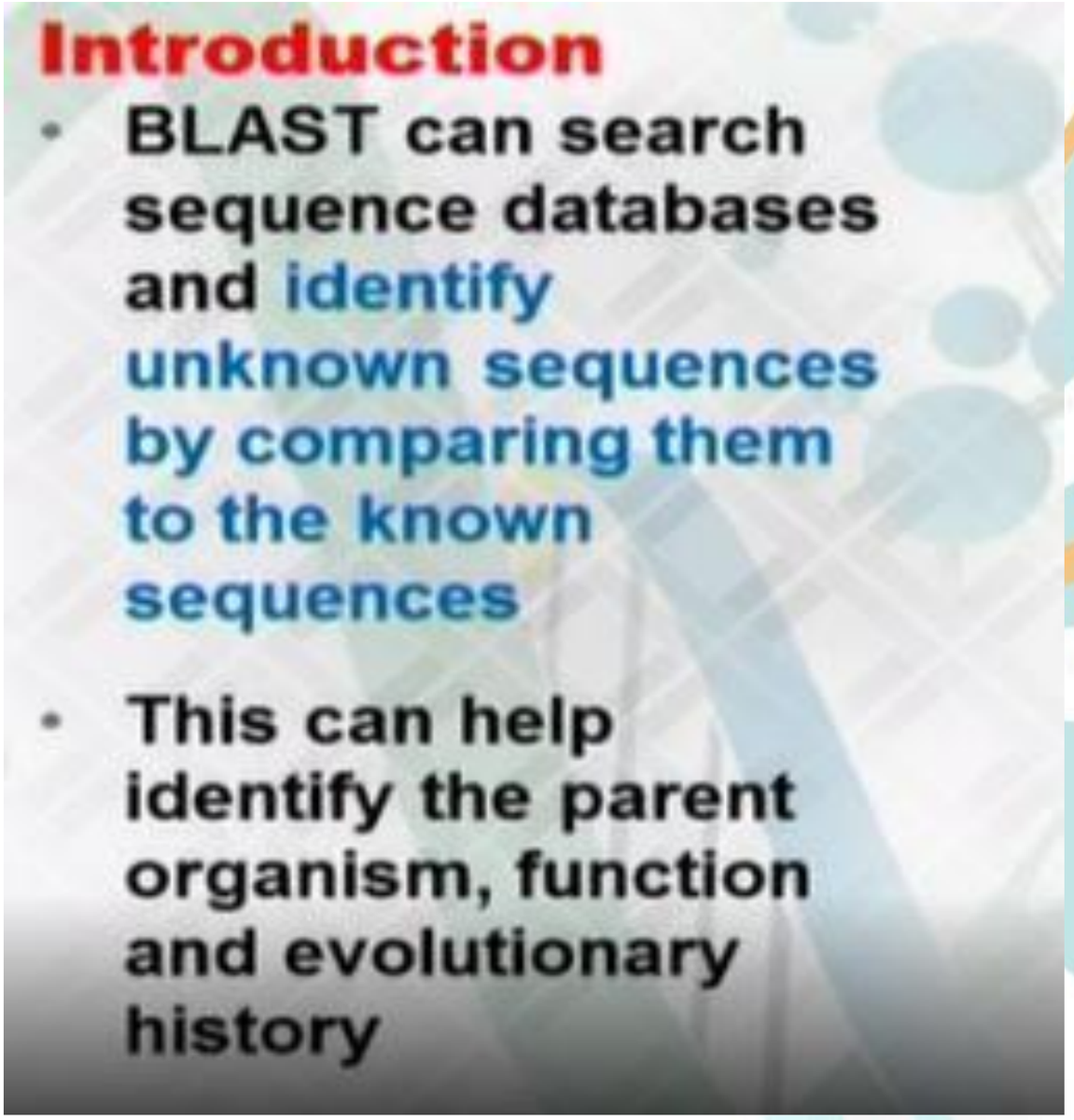
Amina Zee

Lec 83

BLAST Algorithm

Amina Zee

BLAST Algorithm



Amina Zee

BLAST Algorithm

Query sequence: PQGELV

Make list of all possible words (length 3 for proteins)
PQG (score 15)
QGE (score 9)
GEL (score 12)
ELV (score 10)

Assign scores from Blosum62, use those with score> 11 :
PQG & GEL

Mutate words such that score still > 11
PQG (score 15) similar to PEG (score 13)

Amina Zee

BLAST Algorithm

Find all database sequences that have at least 2 matches among our 3 words: PQG, GEL & PEG

Find database hits and extend alignment (High-scoring Segment Pair):

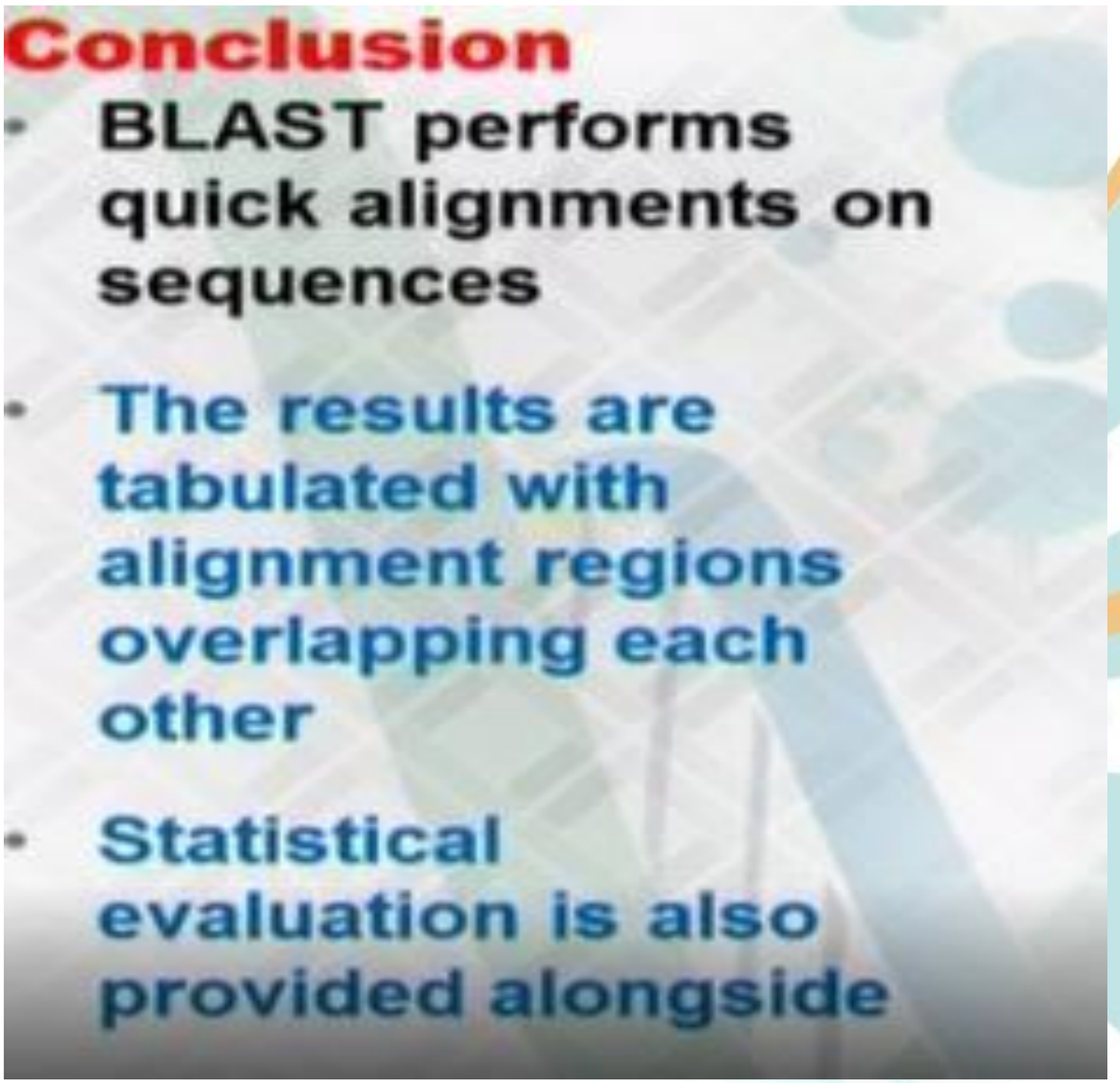
```
Query:     M E T P Q G I A V
Database:  - - - P Q G E L V
           8 5 5 2 0 8
```

High Scoring Pair: PQGI (score 8+5+5+2)

If 2 HSP in query sequence are < 40 positions away
Full dynamic alignment on query and hit sequences

Amina Zee

BLAST Algorithm



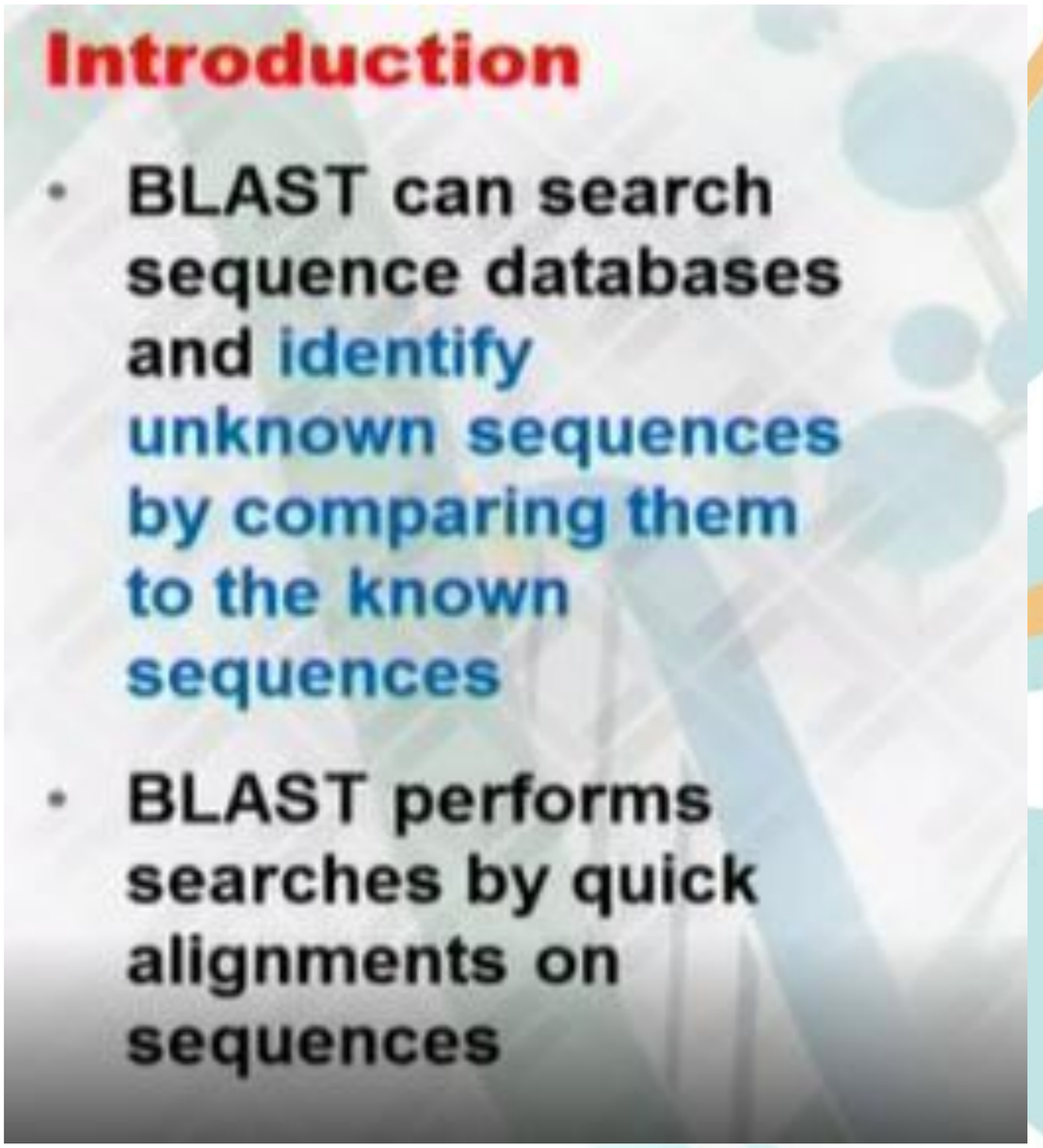
Amina Zee

Lec 84

Types of BLAST

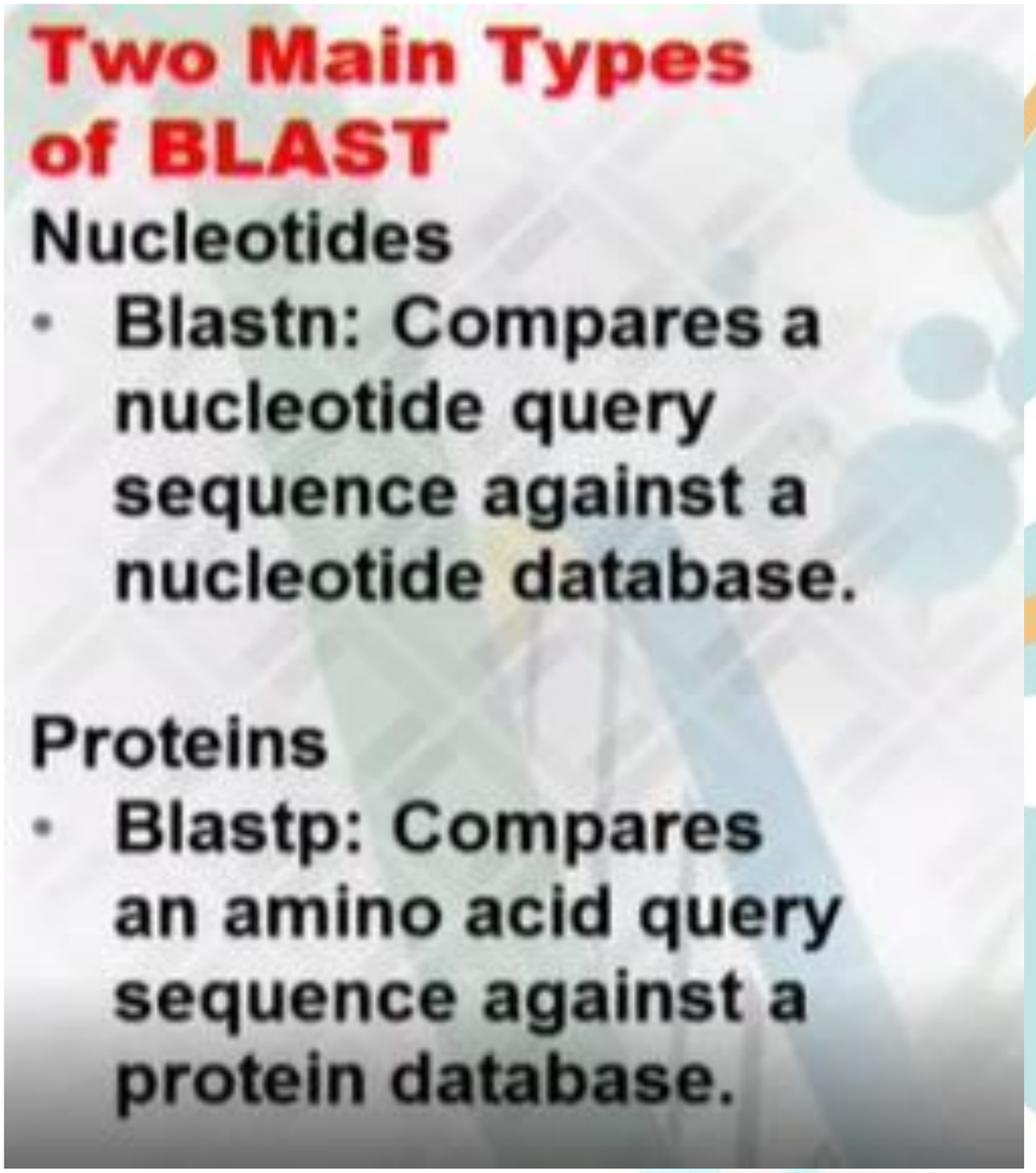
Amina Zee

Types of BLAST



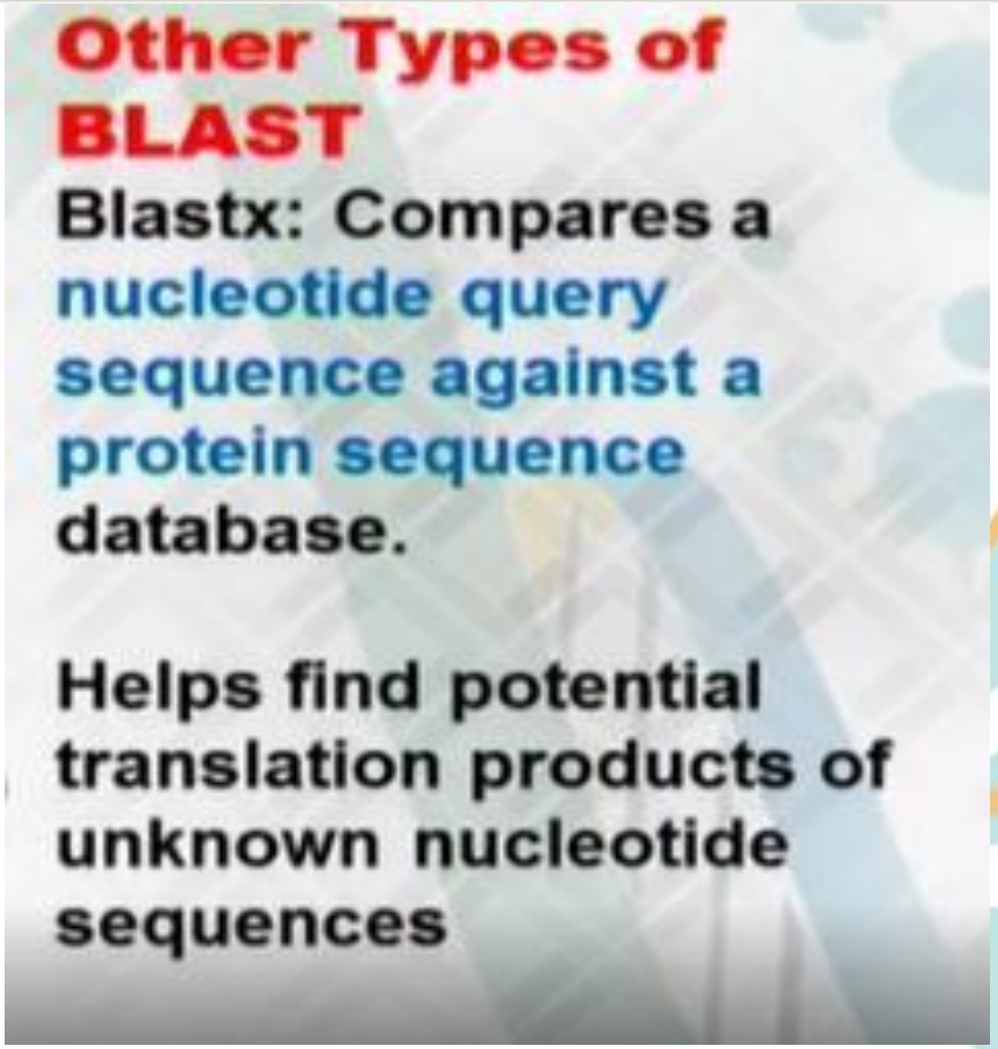
Amina Zee

Types of BLAST



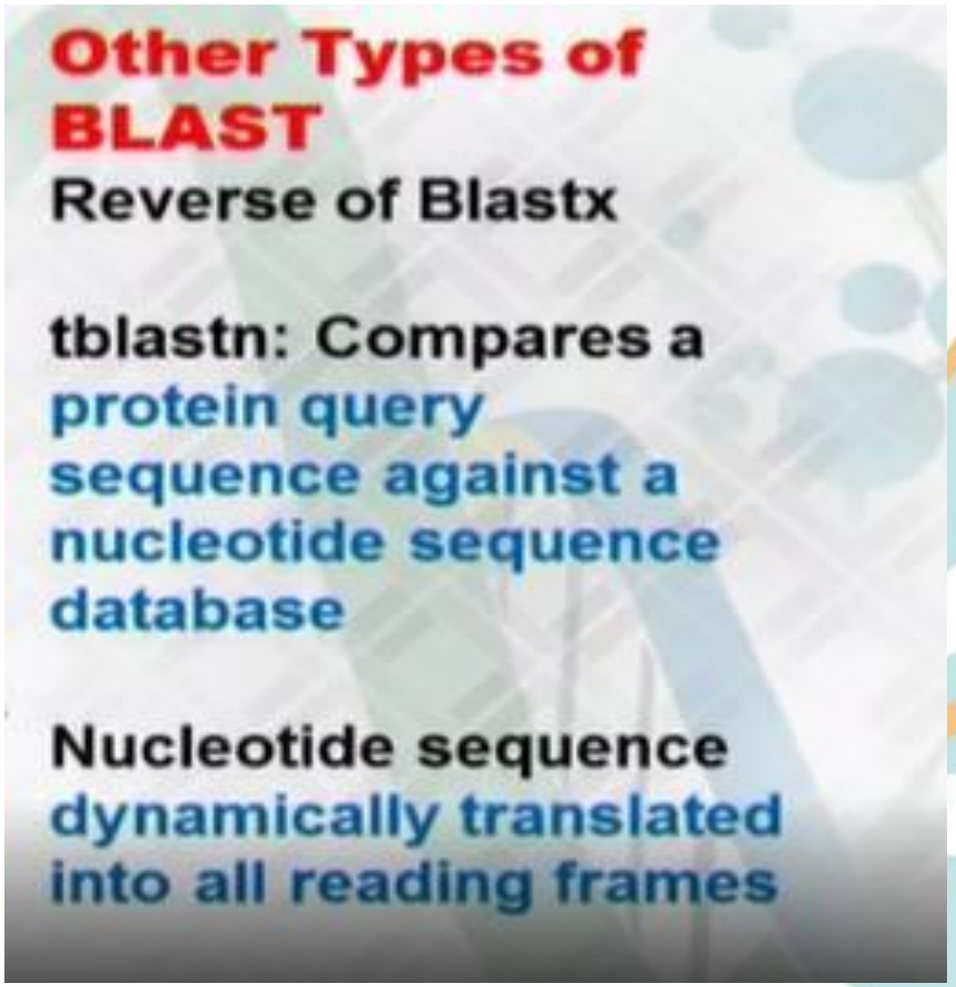
Amina Zee

Types of BLAST



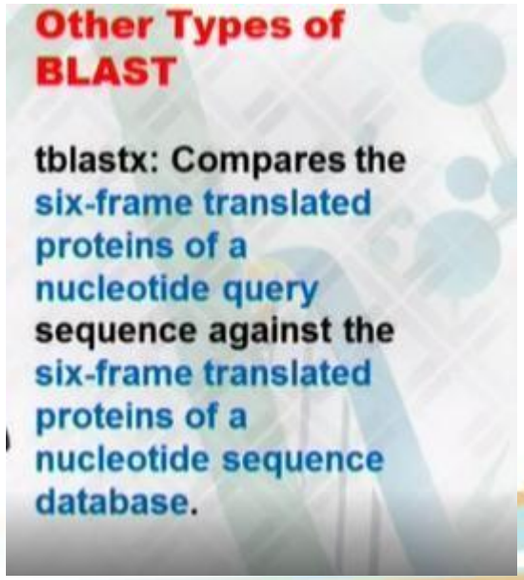
Amina Zee

Types of BLAST



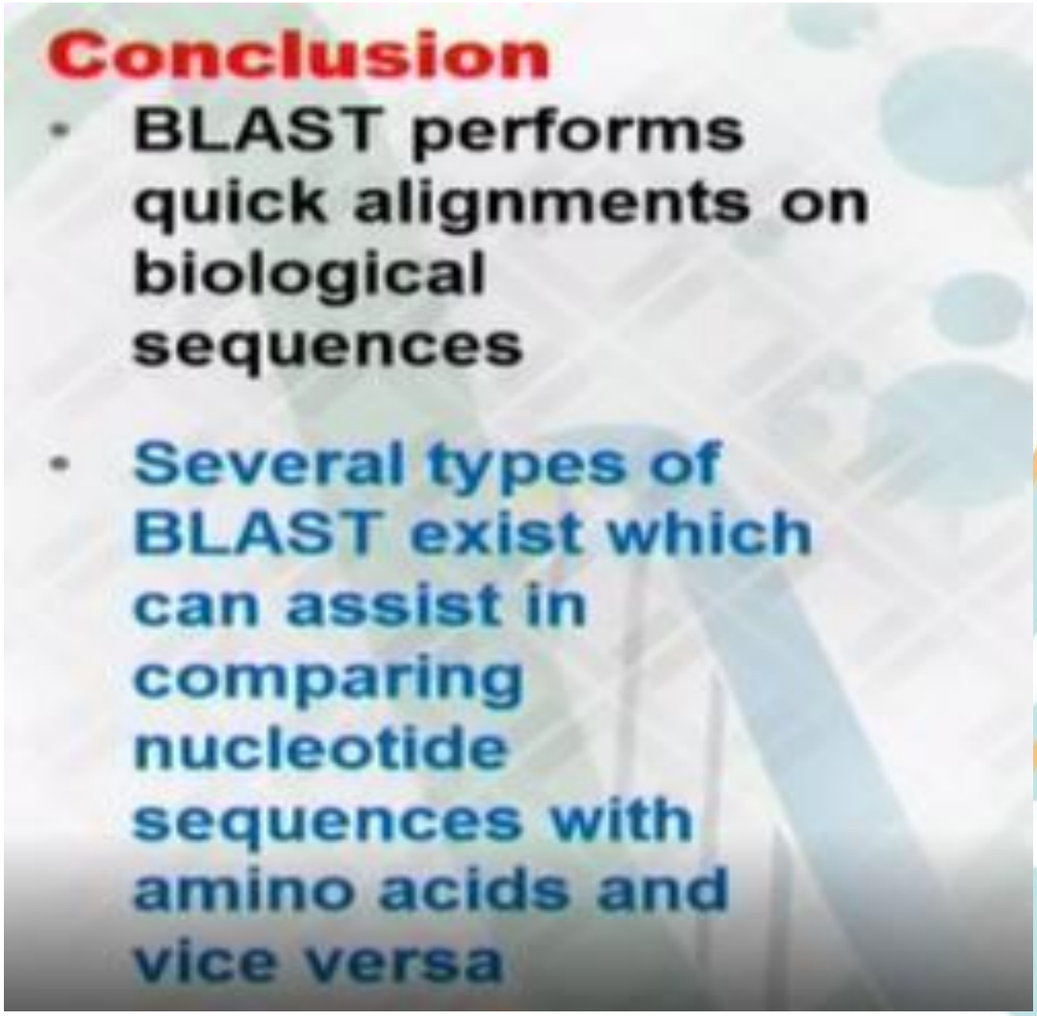
Amina Zee

Types of BLAST



Amina Zee

Types of BLAST



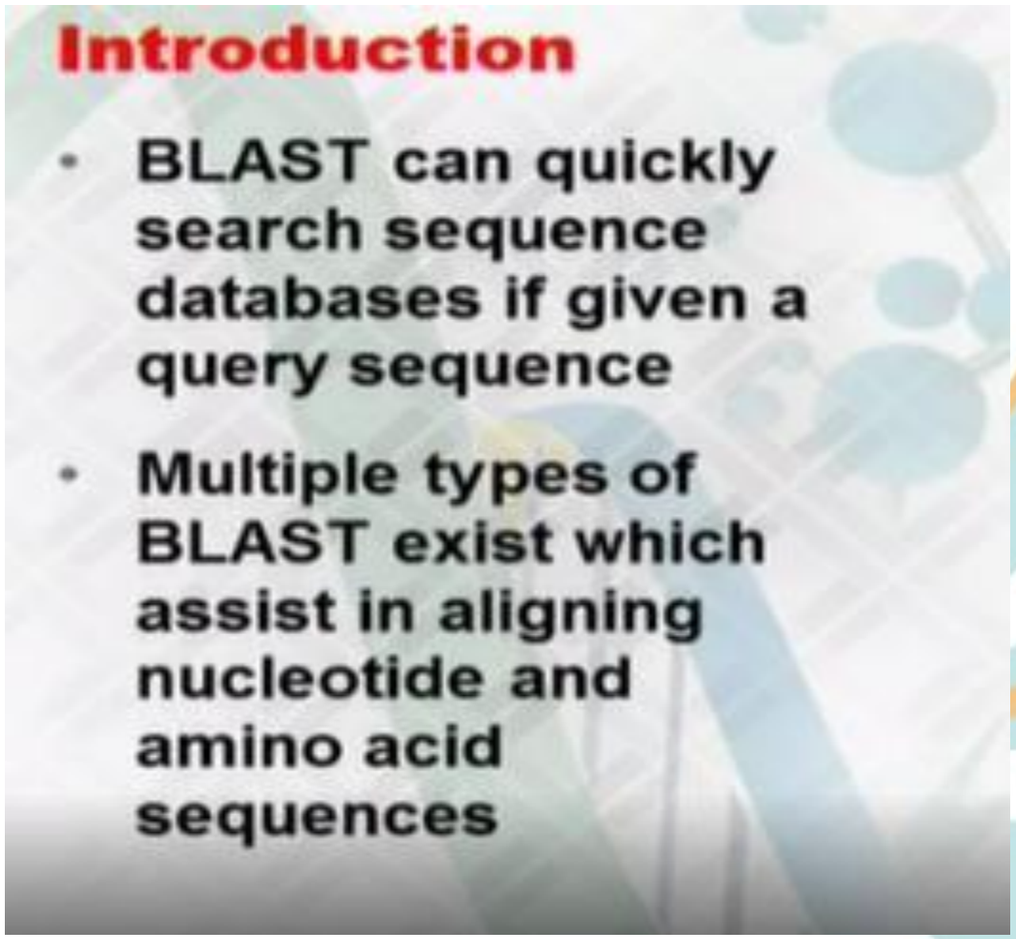
Amina Zee

Lec 85

Summary of BLAST

Amina Zee

Summary of BLAST



Amina Zee

Summary of BLAST

Step 1: Obtain a query sequence

For known sequences: Use NCBI, UCSC etc..
For unknown sequences: Use NGS or Mass Spectrometry

Step 2: Choose a type of BLAST

Basic BLAST

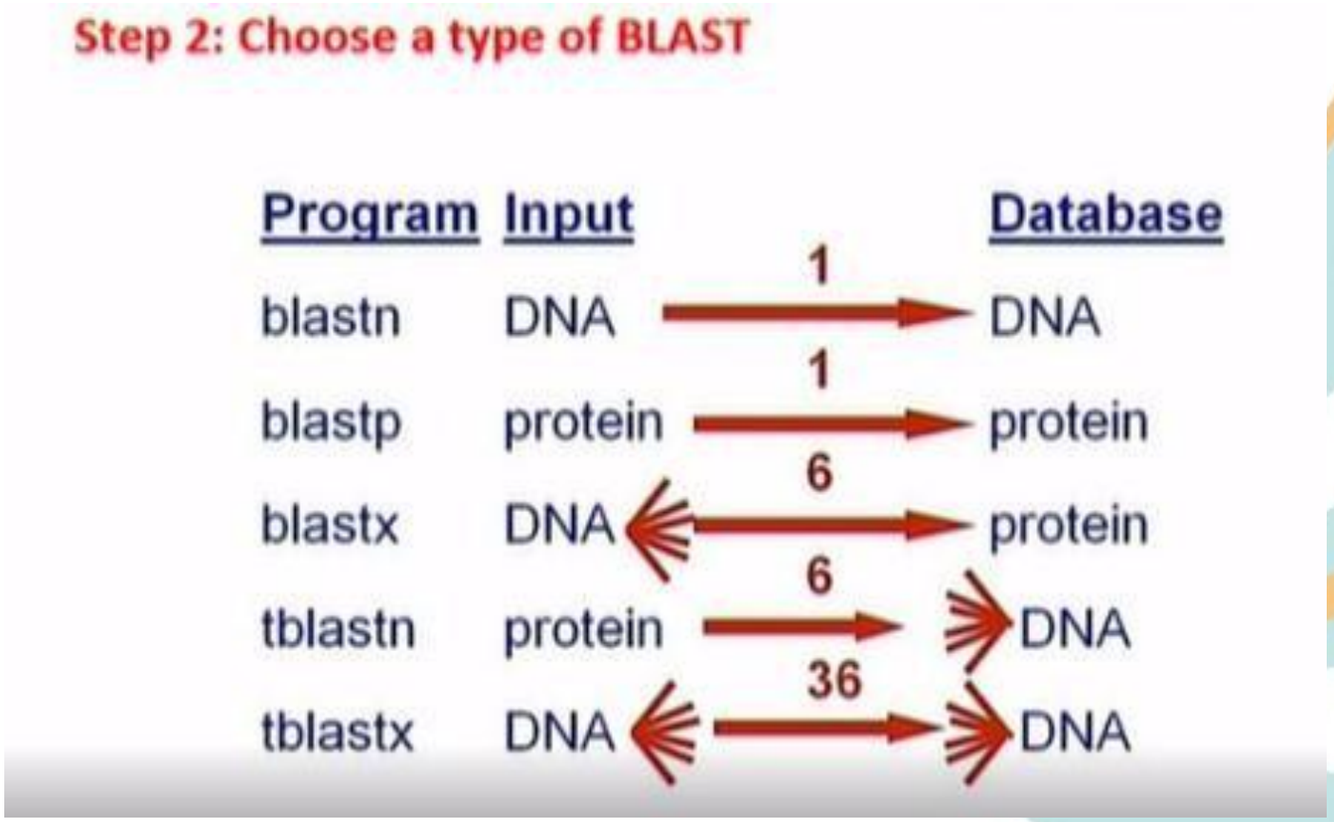
Choose a BLAST program to run.

nucleotide blast	Search a **nucleotide** database using a **nucleotide** query *Algorithms:* blastn, megablast, discontiguous megablast
protein blast	Search **protein** database using a **protein** query *Algorithms:* blastp, psi-blast, phi-blast, delta-blast
blastx	Search **protein** database using a **translated nucleotide** query
tblastn	Search **translated nucleotide** database using a **protein** query
tblastx	Search **translated nucleotide** database using a **translated nucleotide** query

- **blastn (nucleotide BLAST)**
- **blastp (protein BLAST)**
- **blastx (translated BLAST)**
- **tblastn (translated BLAST)**
- **tblastx (translated BLAST)**

Amina Zee

Summary of BLAST



Amina Zee

Summary of BLAST

Step 3: Setup Search Parameters

Step 4: Tabulated Search Results

Conclusion

- BLAST is an online tool which performs quick alignments on biological sequences
- Depending on your need you can choose a specific type of BLAST to compare and score alignments

Amina Zee

Sequence Analysis

Lec 86
Introduction to FASTA - I

Amina Zee

Introduction to FASTA - I

Background

- Comparison between two sequences can be performed by pairwise sequence alignment
- For comparing multiple sequences, multiple sequence alignment (MSA) is used!

Amina Zee

Introduction to FASTA - I

Background

- **MSA is a progressive pairwise alignment so as to handle multiple sequences**
- **Smith-Waterman performs local alignment**
- **Needleman-Wunsch calculates global alignment**

Amina Zee

Introduction to FASTA - I

Background

- The local and global alignment algorithms use Dynamic Programming Approaches
- But, this takes time if there are several sequences that need comparison!

Amina Zee

Introduction to FASTA - I

Solution?

- **We used BLAST which is an approximate local alignment search tool**
- **BLAST compares a large number of sequences, quickly!**
- **FASTA took a similar approach!**

Amina Zee

Introduction to FASTA - I

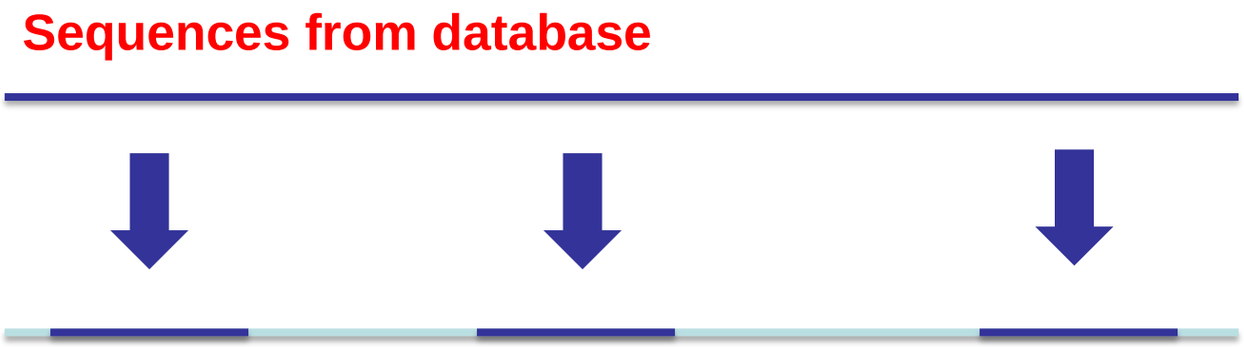
FASTA

- Developed in 1988
- “Fast Alignment”
- Searches databases for query protein and nucleotide sequences
- Was later improved upon in BLAST

Amina Zee

Introduction to FASTA - I

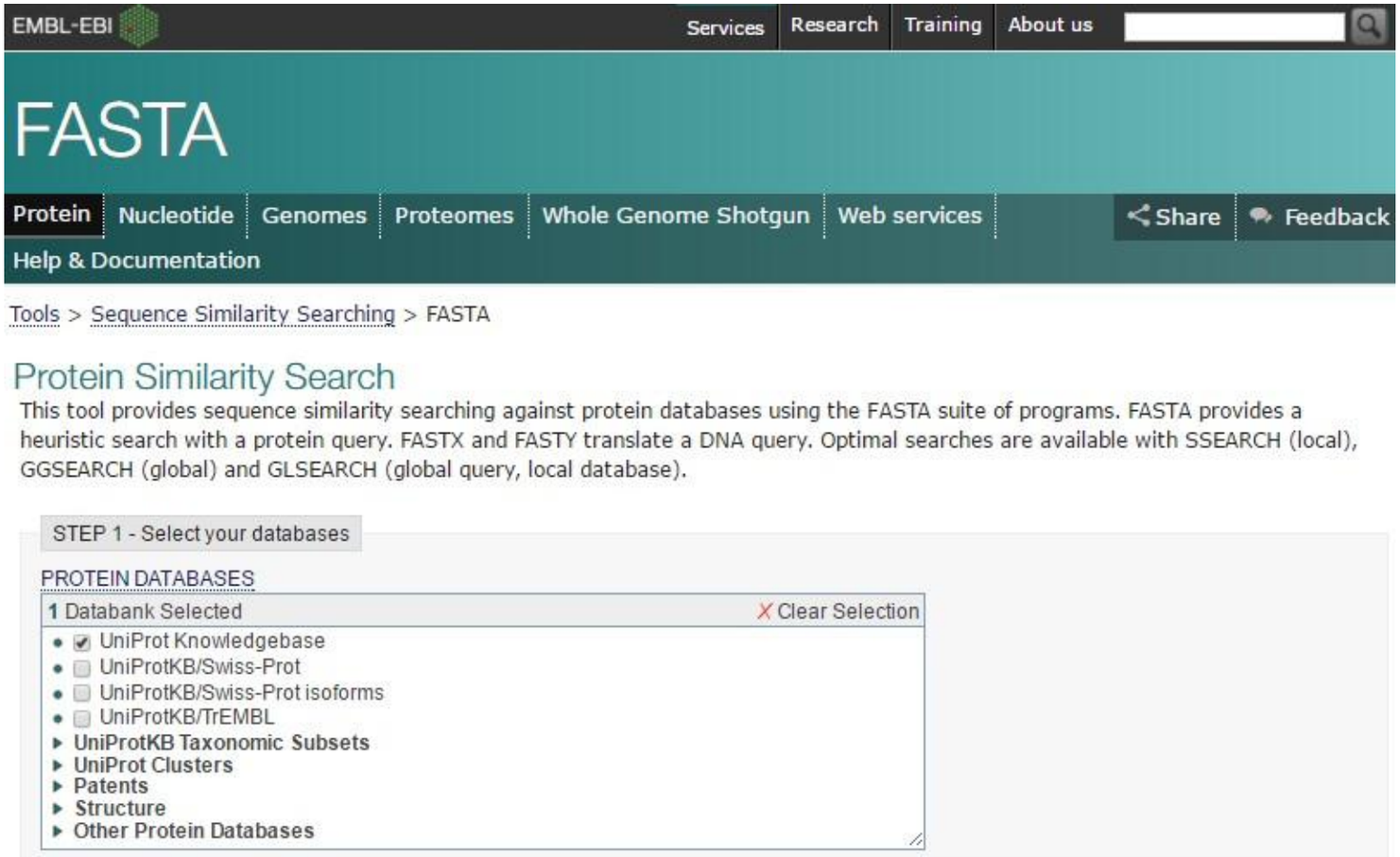
Regions of absolute identity



Amina Zee

Introduction to FASTA - I

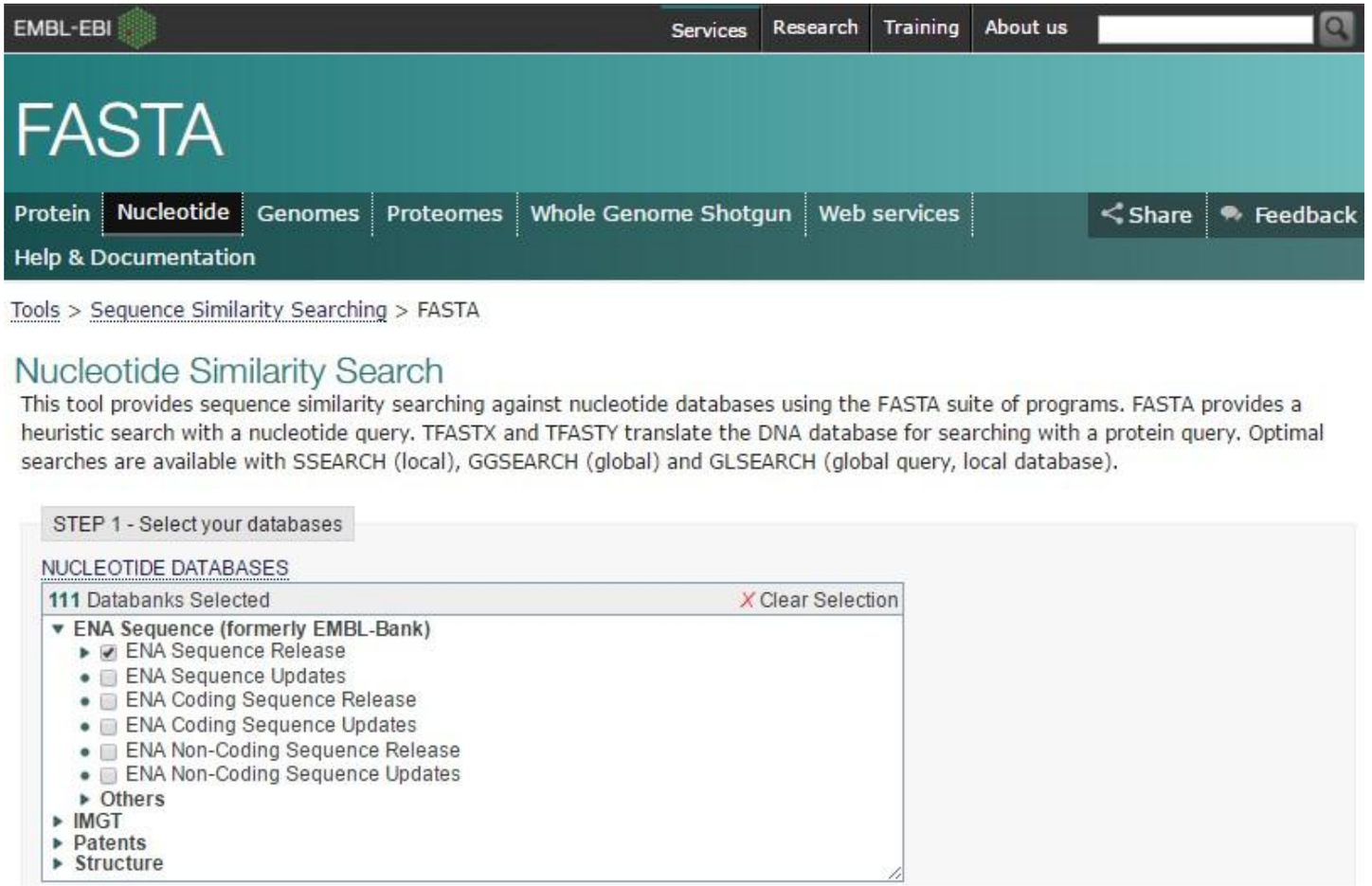
http://www.ebi.ac.uk/Tools/sss/fasta/



Amina Zee

Introduction to FASTA - I

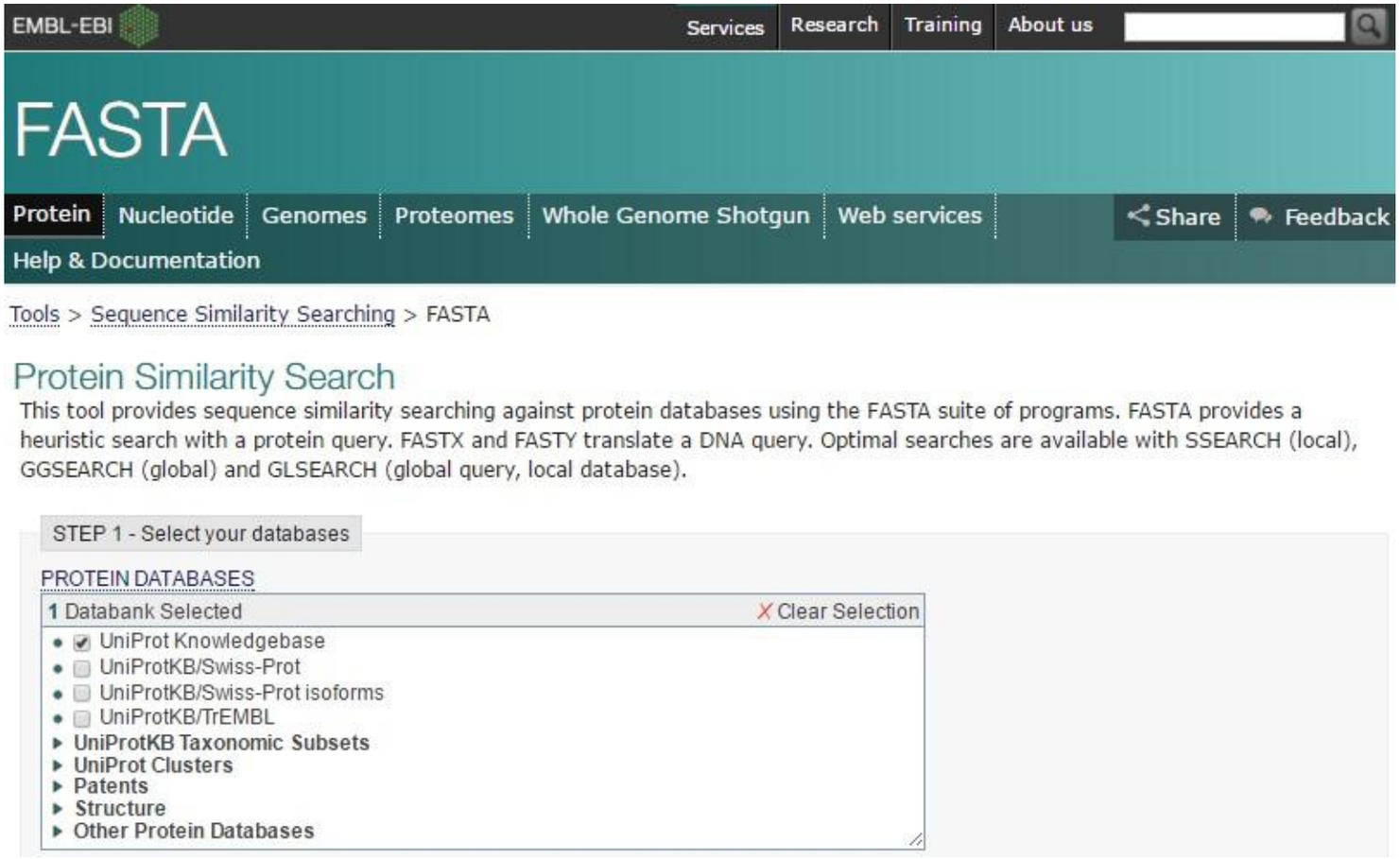
Nucleotide FASTA



Amina Zee

Introduction to FASTA - I

Protein FASTA



Amina Zee

Introduction to FASTA - I

Conclusion

- **FASTA can perform quick comparison of protein and nucleotide sequences**
- **Can also perform genome and proteome similarity search**
- **It is available online**

END

Amina Zee

Sequence Analysis

Lec 87
Introduction to FASTA - II

Amina Zee

Introduction to FASTA- II

Introduction

- FASTA – Fast Alignment Algorithm
- Classical global and local alignment algorithms are time consuming
- FASTA achieves alignment by using short lengths of exact matches

Amina Zee

Introduction to FASTA- II

Use Cases of FASTA

- FASTA relies on aligning subsequences of absolute identity

- Input to FASTA search can be in FASTA, EMBL, GenBank, PIR, NBRF, PHYLIP or UniProt formats

Amina Zee

Introduction to FASTA- II

Output of BLAST

- Results are output in visual format along with functional prediction
- A table lists the sequence hits found along with scores
- Users can click on each reported match to look at the details

Amina Zee

Introduction to FASTA- II

Input to FASTA: Gene IDs

Nucleotide Similarity Search

This tool provides sequence similarity searching against nucleotide databases using the FASTA suite of programs. FASTA provides a heuristic search with a nucleotide query. TFASTX and TFASTY translate the DNA database for searching with a protein query. Optimal searches are available with SSEARCH (local), GGSEARCH (global) and GLSEARCH (global query, local database).

STEP 1 - Select your databases

NUCLEOTIDE DATABASES

111 Databanks Selected — ✗ Clear Selection

- **ENA Sequence (formerly EMBL-Bank)**
  - ☑ ENA Sequence Release
  - ☐ ENA Sequence Updates
  - ☐ ENA Coding Sequence Release
  - ☐ ENA Coding Sequence Updates
  - ☐ ENA Non-Coding Sequence Release
  - ☐ ENA Non-Coding Sequence Updates
  - **Others**
- **IMGT**
- **Patents**
- **Structure**

STEP 2 - Enter your input sequence

Enter or paste a DNA sequence in any supported format:

```
CAGTCAGTCATAGTCGTAGATGTACGTAGCTAGTAGTGATGTAGTCAGTGATGCTAGTAGTGTTAGTAGTGATGTAGTCAGTGATGCT
AGTAGTGTTAGTAGTGATGTAGTCAGTGATGCTAGTAGTGTTAGTAGTGATGTAGTCAGTGATGCTAGTAGTGTTAGTAGTGATGTAG
TCAGTGATGCTAGTAGTGTTAGTAGTGATGTAGTCAGTGATGCTAGTAGTGTTAGTAGTGATGTAGTCAGTGATGCTAGTAGTGTTAG
TAGTGATGTAGTCAGTGATGCTAGTAGTGTTAGTAGTGATGTAGTCAGTGATGCTAGTAGTGTTAGT
```

or upload a file: Choose File No file chosen

Amina Zee

Introduction to FASTA- II

Input to FASTA: Protein Sequence

Tools > Sequence Similarity Searching > FASTA

Protein Similarity Search

This tool provides sequence similarity searching against protein databases using the FASTA suite of programs. FASTA provides a heuristic search with a protein query. FASTX and FASTY translate a DNA query. Optimal searches are available with SSEARCH (local), GGSEARCH (global) and GLSEARCH (global query, local database).

STEP 1 - Select your databases

PROTEIN DATABASES

1 Databank Selected — ✗ Clear Selection

- ☑ UniProt Knowledgebase
- ☐ UniProtKB/Swiss-Prot
- ☐ UniProtKB/Swiss-Prot isoforms
- ☐ UniProtKB/TrEMBL
- ▸ **UniProtKB Taxonomic Subsets**
- ▸ **UniProt Clusters**
- ▸ **Patents**
- ▸ **Structure**
- ▸ **Other Protein Databases**

STEP 2 - Enter your input sequence

Enter or paste a PROTEIN sequence in any supported format:

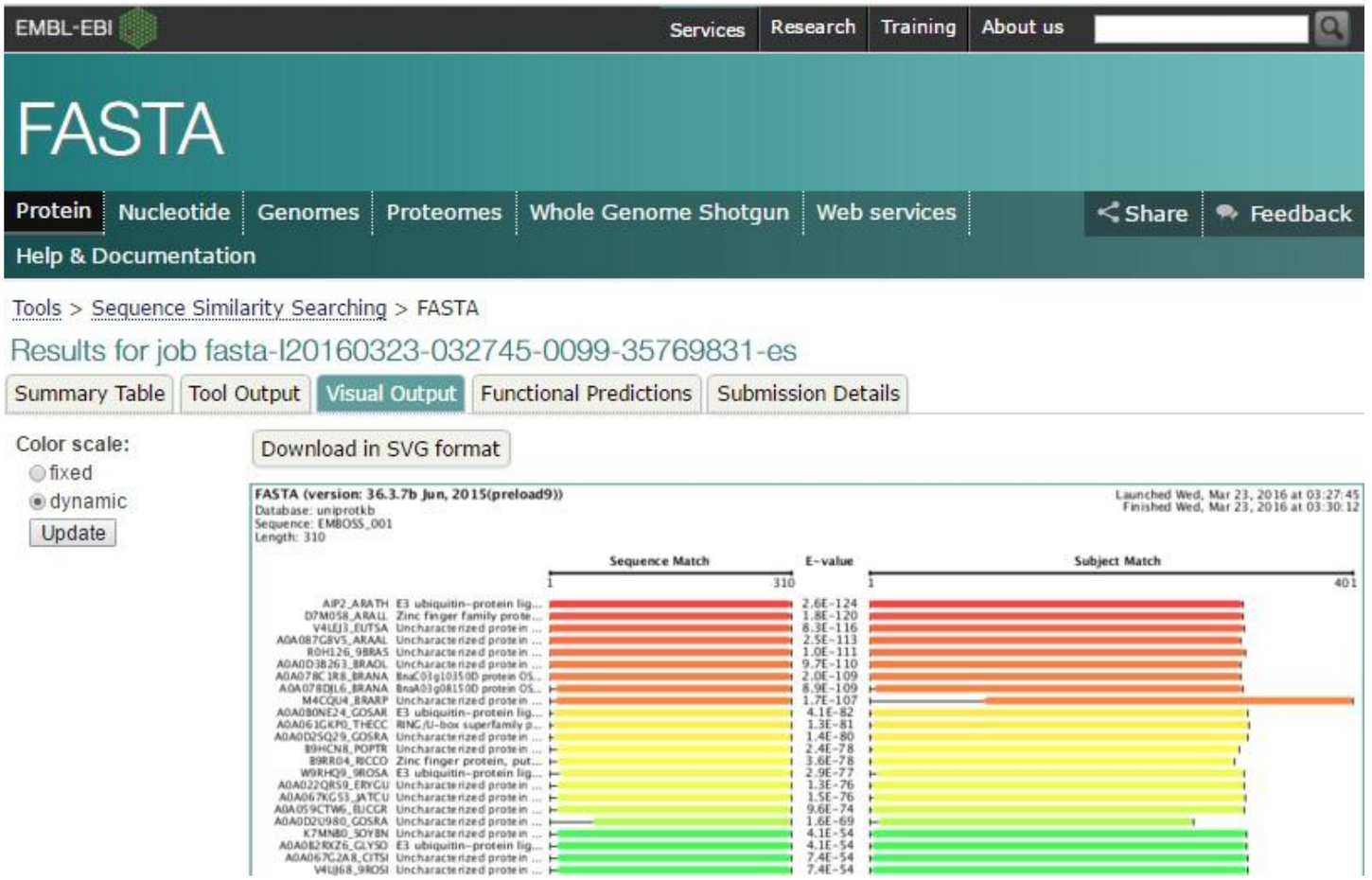
```
MDASSSPSPSEESLKLELDDLQKQLNKKLRFEASVCSIHNLLRDHYSSSSPSLRKQFYIV
VSRVATVLKTRYTATGFWVAGLSLFEEAERLVSDASEKKHLKSCVAQAKEQLSEVDNQPT
ESSQGYLFEGHLTVDREPPQPQWLVQQNLMSAFASIVGGESSNGPTENTIGETANLMQEL
INGLDMIIPDILDDGGPPRAPPASKEVVEKLPVIIFTEELLKKFGAEAECCICKENLVIG
DKMQELPCKHTFHPPCLKPWLDEHNSCPICRHELPTDDQKYENWKEREKEAEEERKGAEN
AVRGGEYMYV
```

or Upload a file: Choose File No file chosen

Amina Zee

Introduction to FASTA- II

Results from FASTA



Amina Zee

Introduction to FASTA- II

Conclusion

- Purpose of FASTA
- FASTA vs. Smith Waterman
- Inputs and outputs from FASTA
- Internal working of FASTA?

END

Amina Zee

Sequence Analysis

Lec 88
FASTA Algorithm

Amina Zee

FASTA Algorithm

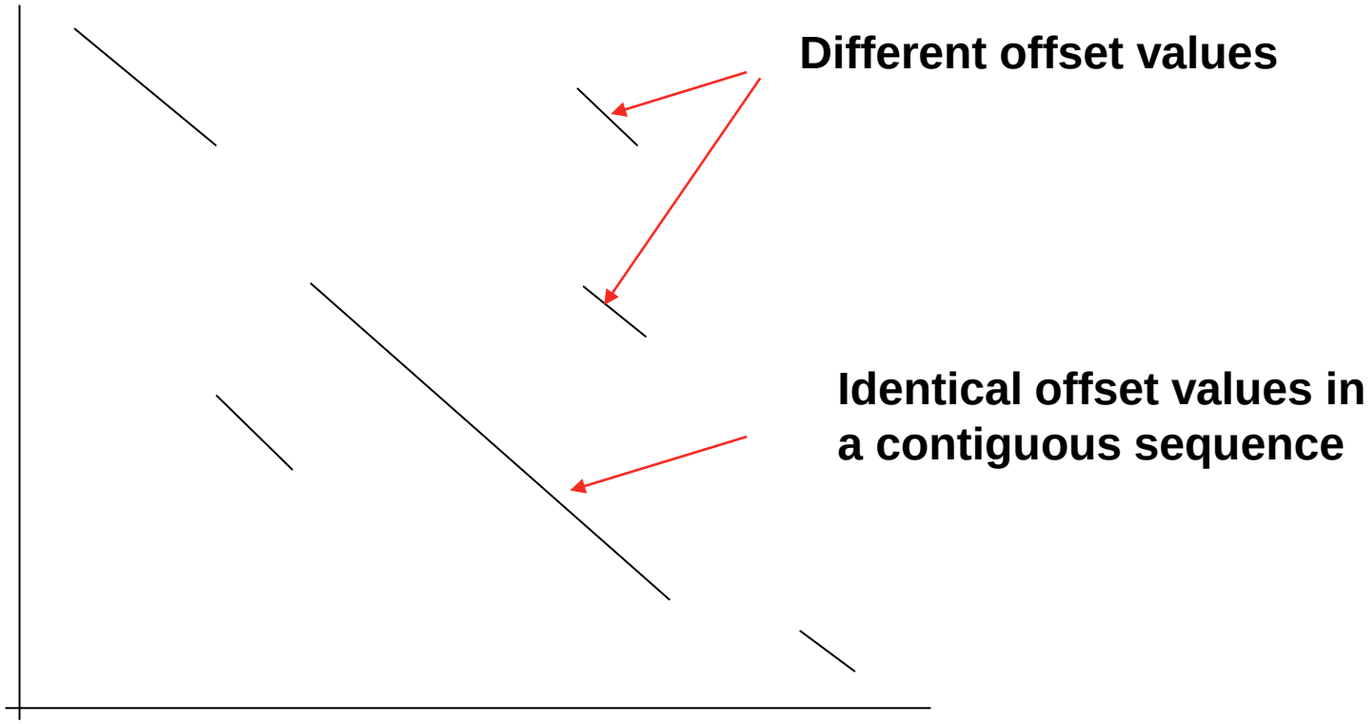
Introduction

- FASTA can search sequence databases and identify unknown sequences by comparing them to the known sequence databases
- This can help obtain information on the parent organism, function and evolutionary history

Amina Zee

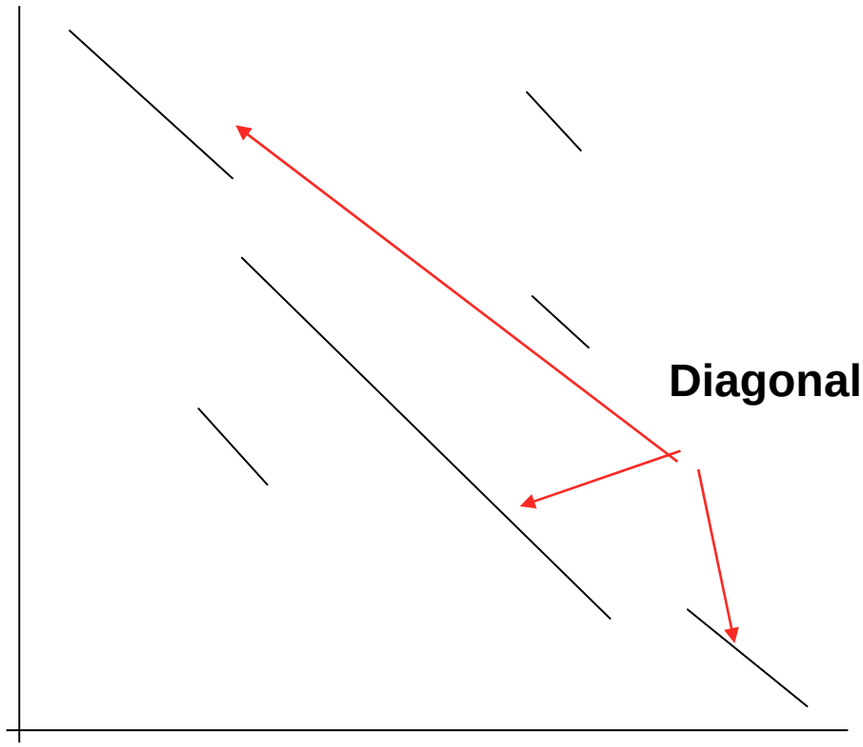
FASTA Algorithm

Step 1: Local regions of identity are found



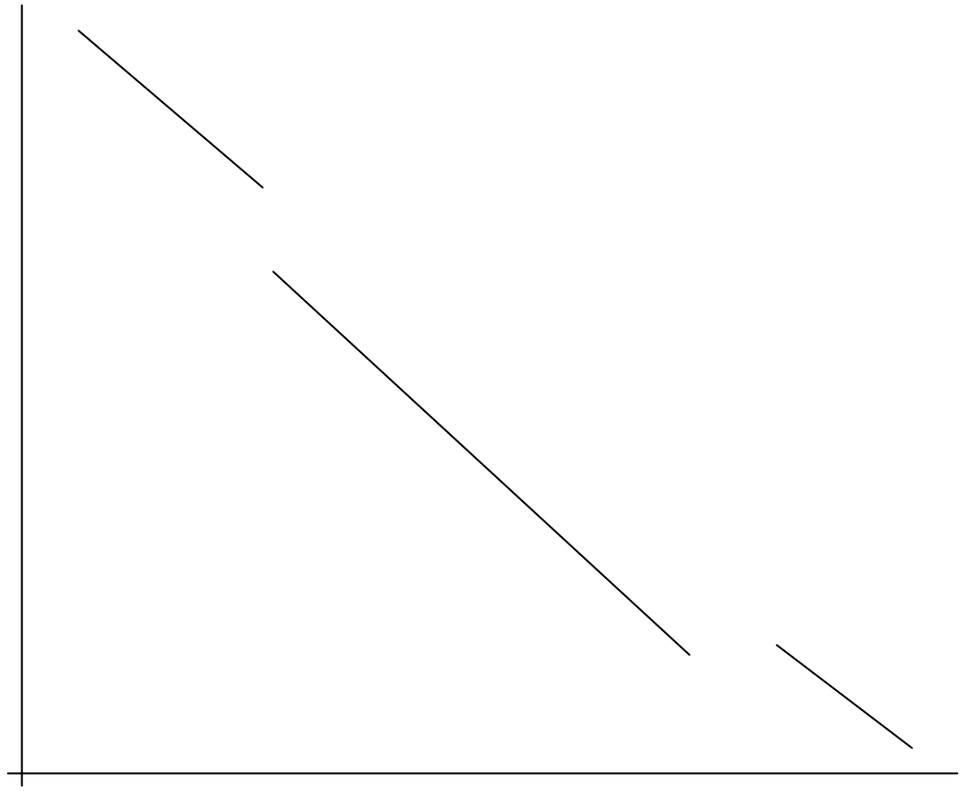
FASTA Algorithm

Step 2: Rescore the local regions using PAM or BLOSUM matrix



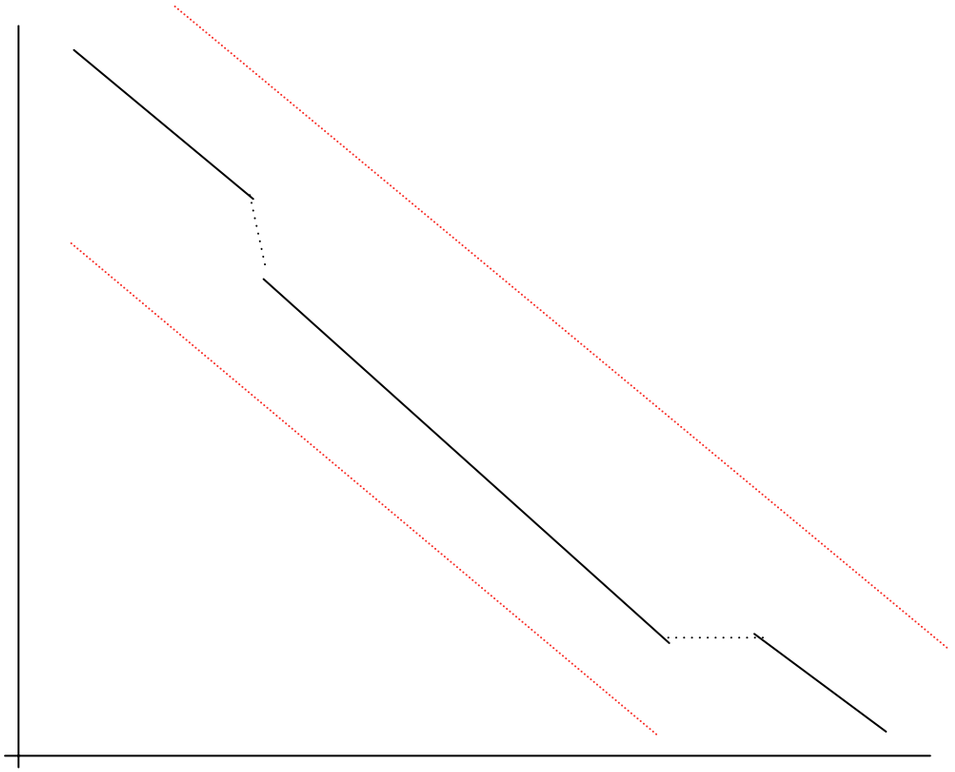
FASTA Algorithm

Step 3: Eliminate short diagonals below a cutoff score



FASTA Algorithm

Step 4: Create a gapped alignment in a narrow segment and then perform Smith Watermann alignment



FASTA Algorithm

Conclusion

- **FASTA performs preliminary alignments quickly followed by Smith-Waterman**
- **Results are output in visual format along with statistical scores**

END

Amina Zee

Sequence Analysis

Lec 89
Types of FASTA

Amina Zee

Types of FASTA

Introduction

- FASTA can search sequence databases and compare sequences to the known sequences in DNA/RNA/Protein Databases
- It performs search on matching words followed by application Smith-Waterman

Amina Zee

Types of FASTA

Fasta35

- **Scan a protein or DNA sequence library for similar sequences**

Fastx35

- **Compare a translated DNA sequence (6 ORFs) to a protein sequence database**

Amina Zee

Types of FASTA

tfastx35

- **Compare a protein sequence to a DNA sequence database (6 ORFs)**

fasty35

- **Compare a DNA sequence (6ORFs) to a protein sequence database**

Amina Zee

Types of FASTA

fasts35

- **Compare unordered peptides to a protein sequence database**

fastm35

- **Compare ordered peptides (or short DNA sequences) to a protein (DNA) sequence database**

Amina Zee

Types of FASTA

Conclusion

- **FASTA performs quick alignments on biological sequences**
- **Several types of FASTA exist which can assist in comparing DNA/RNA/Protein sequences with each other**

END

Amina Zee

Sequence Analysis

Lec 90
Summary of FASTA

Amina Zee

Summary of FASTA

Introduction

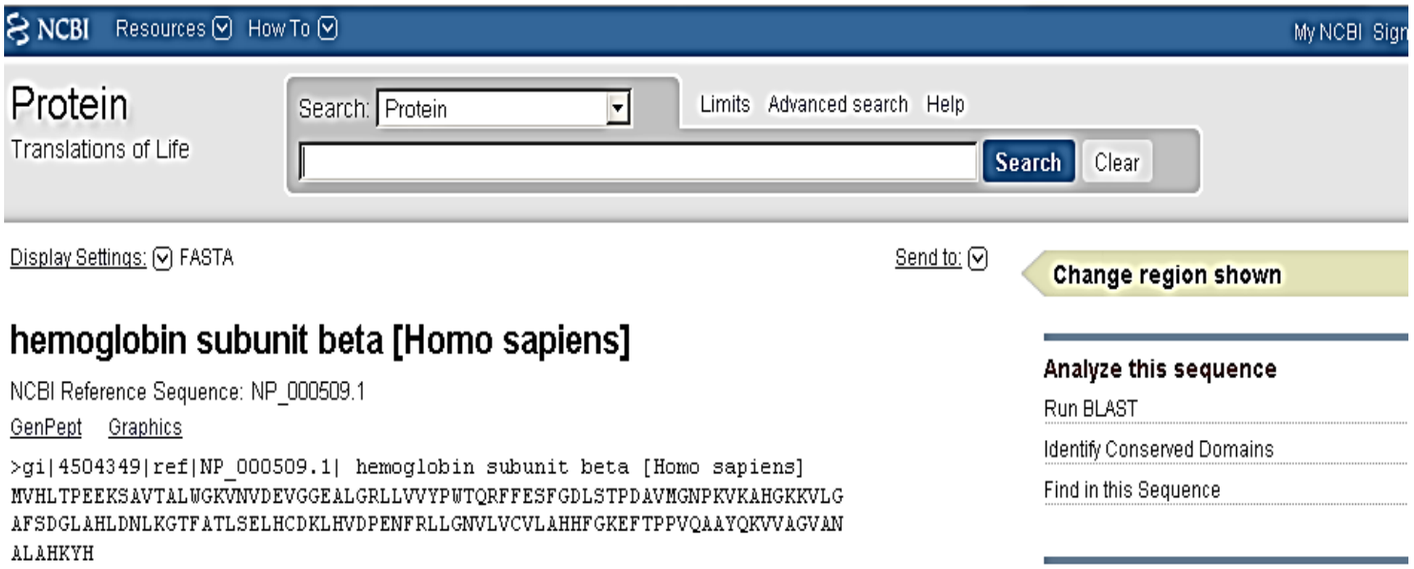
- FASTA can briskly perform sequence search databases if given a query sequence
- Multiple types of FASTA exist which assist in aligning DNA/RNA/Protein sequences

Amina Zee

Summary of FASTA

Step 1: Obtain a query sequence

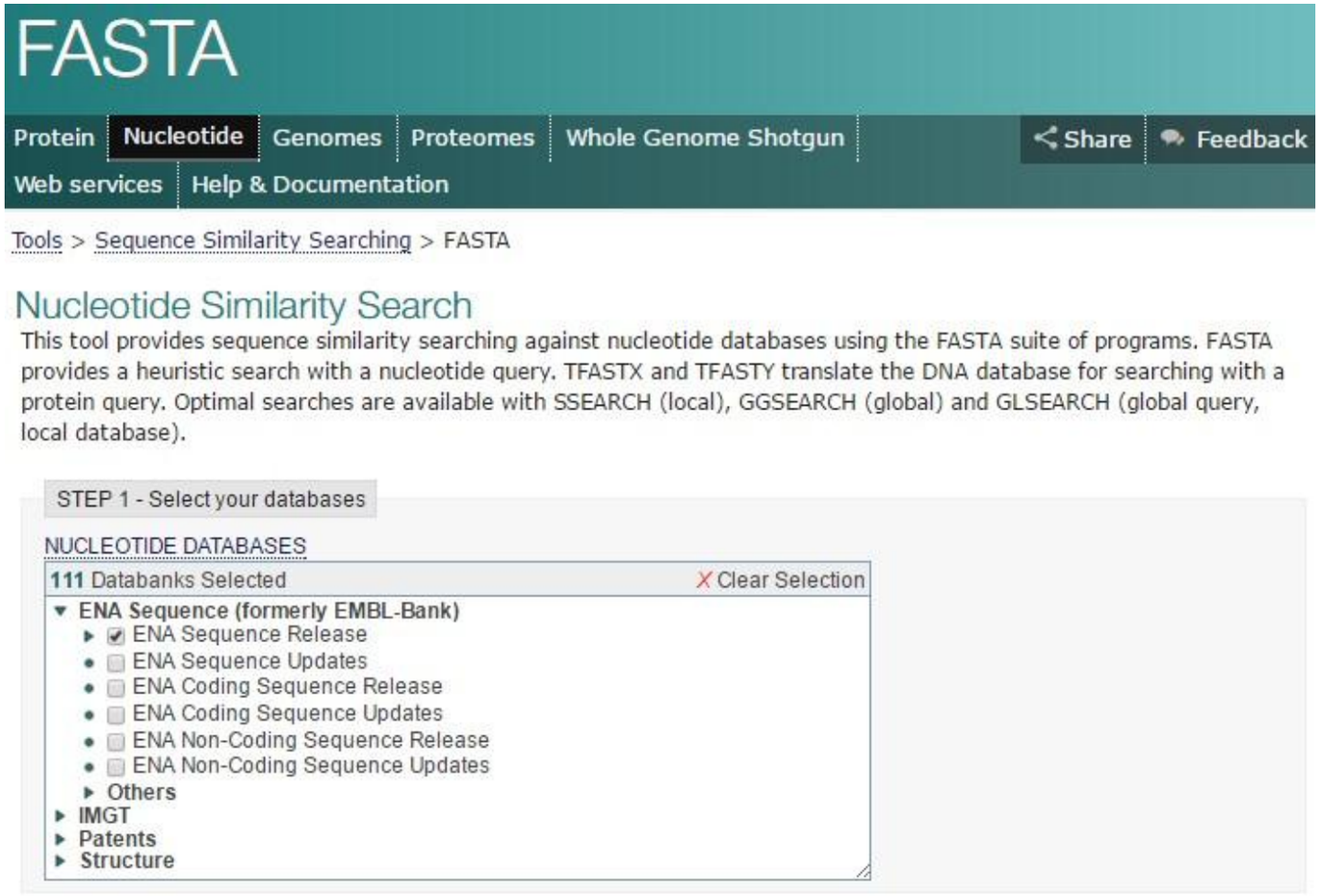
For known sequences: Use NCBI, UCSC etc..
For unknown sequences: Use NGS or Mass Spectrometry



Amina Zee

Summary of FASTA

Step 2: Choose a type of FASTA



Amina Zee

Summary of FASTA

Step 2: Type of FASTA

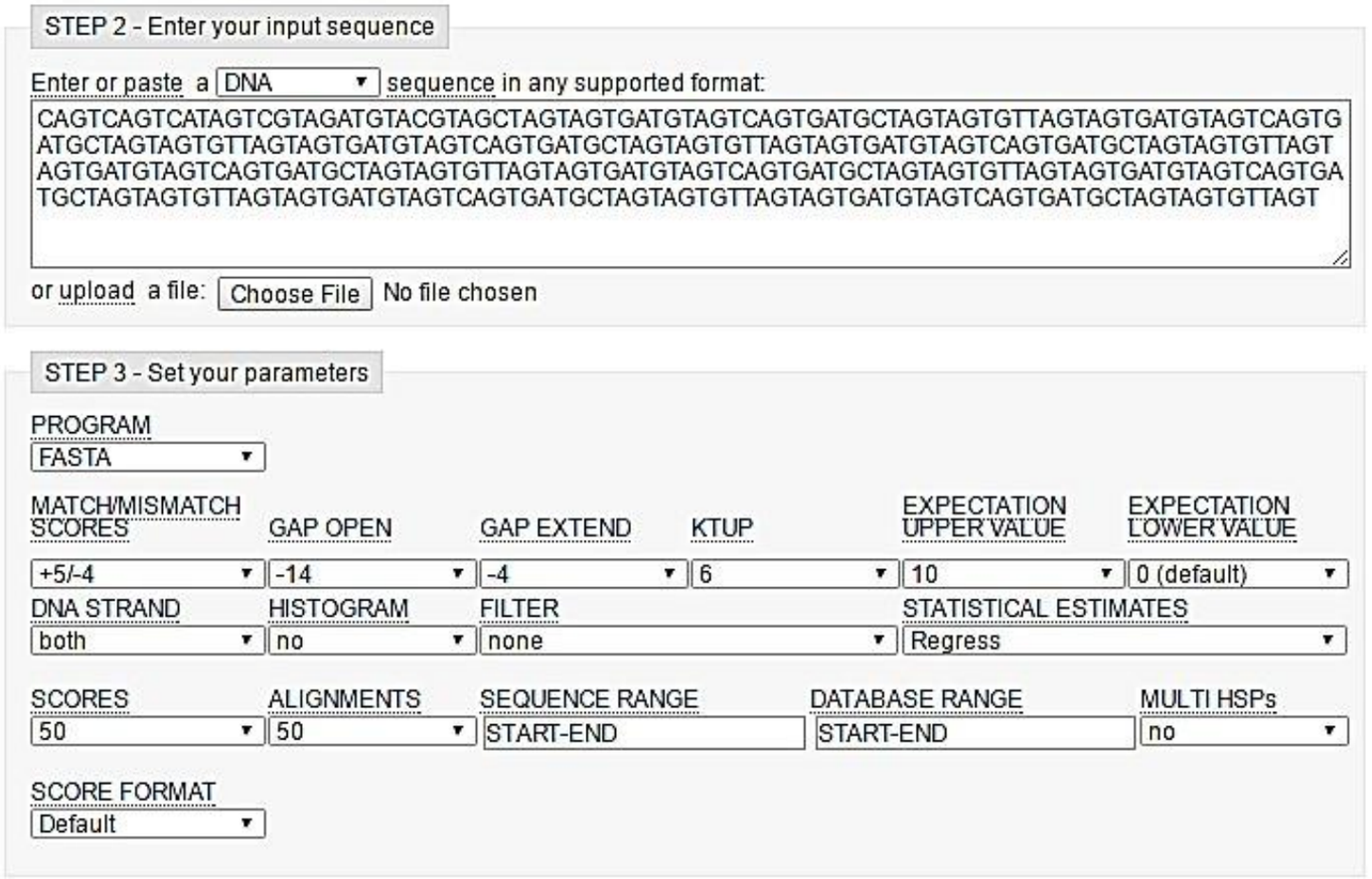
http://fasta.bioch.virginia.edu/fasta_docs/fasta35.shtml

`fasta35, fasta35_t*`	scan a protein or DNA sequence library for similar sequences
`fastx35, fastx35_t`	compare a DNA sequence to a protein sequence database, comparing the translated DNA sequence in forward and reverse frames.
`tfastx35, tfastx35_t`	compare a protein sequence to a DNA sequence database, calculating similarities with frameshifts to the forward and reverse orientations.
`fasty35, fasty35_t`	compare a DNA sequence to a protein sequence database, comparing the translated DNA sequence in forward and reverse frames.
`tfasty35, tfasty35_t`	compare a protein sequence to a DNA sequence database, calculating similarities with frameshifts to the forward and reverse orientations.
`fasts35, fasts35_t`	compare unordered peptides to a protein sequence database
`fastm35, fastm35_t`	compare ordered peptides (or short DNA sequences) to a protein (DNA) sequence database
`tfasts35, tfasts35_t`	compare unordered peptides to a translated DNA sequence database
`fastf35, fastf35_t`	compare mixed peptides to a protein sequence database
`tfastf35, tfastf35_t`	compare mixed peptides to a translated DNA sequence database
`ssearch35, ssearch35_t`	compare a protein or DNA sequence to a sequence database using the Smith-Waterman algorithm.

Amina Zee

Summary of FASTA

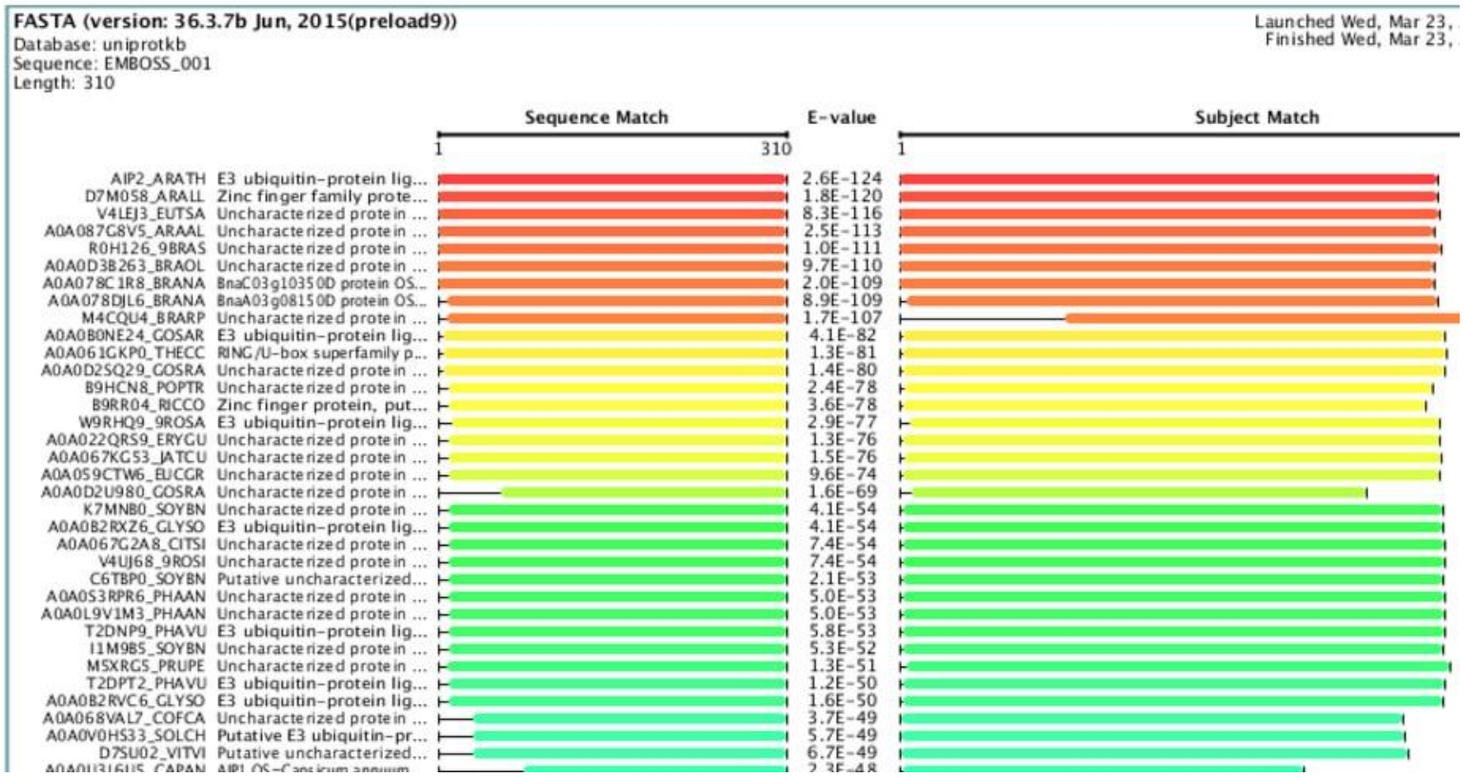
Step 3: Setup Search Parameters



Amina Zee

Summary of FASTA

Step 4: Tabulated Search Results



Amina Zee

Summary of FASTA

Step 4: Tabulated Search Results

Align.	DB:ID	Source	Length	Score (Bits)	Identities %	Positives %	E()
☑ 1	SP:AIP2_ARATH	E3 ubiquitin-protein ligase AIP2 OS=Arabidopsis thaliana GN=AIP2 PE=1 SV=1 *Cross-references and related information in:* ▶ Gene expression ▶ Small molecules ▶ Nucleotide sequences ▶ Genomes & metagenomes ▶ Enzymes ▶ Samples & ontologies ▶ Molecular interactions ▶ Protein families ▶ Literature ▶ Protein sequences	310	453.0	100.0	100.0	2.6E-124
☑ 2	TR:D7M058_ARALL	Zinc finger family protein OS=Arabidopsis lyrata subsp. lyrata GN=ARALYDRAFT_488997 PE=4 SV=1 *Cross-references and related information in:* ▶ Nucleotide sequences ▶ Genomes & metagenomes ▶ Samples & ontologies	310	440.3	96.5	99.7	1.8E-120

Amina Zee

Summary of FASTA

Conclusion

- **FASTA is an online tool which performs quick alignments on biological sequences**
- **Depending on your need you can choose a specific type of FASTA to compare and score alignments**

END

Amina Zee

Lec 91

Database

Amina Zee

)

Lecture Handout

Database Management System

Lecture No. 01

Reading Material

Overview of Lecture

- Introduction to the course
- Database definitions
- Importance of databases
- Introduction to File Processing Systems
- Advantages of the Database Approach

Introduction to thecourse

This course is first (fundamental) course on database management systems. The course discusses different topics of the databases. We will be covering both the theoretical and practical aspects of databases. As a student to have a better understanding of the subject, it is very necessary that you concentrate on the concepts discussed in the course.

Areas to be covered in this Course:

- Database design and application development: How do we represent a real- world system in the form of a database? This is one major topic covered in this course. It comprises of different stages, we will discuss all these stages one by one.

- Concurrency and robustness: How does a DBMS allow many users to access data concurrently, and how does it protect against failures?

- Efficiency and Scalability: How does the database cope with large amounts of data?

Amina Zee

Database definitions:

Definitions are important, especially in technical subjects because definition describes very comprehensively the purpose and the core idea behind the thing. Databases have been defined differently in literature. We are discussing different definitions here, if we concentrate on these definitions, we find that they support each other and as a result of the understanding of these definitions, we establish a better understanding of use, working and to some extent the components of a database.

Def 1: A shared collection of logically related data, designed to meet the information needs of multiple users in an organization. The term database is often erroneously referred to as a synonym for a "database management system (DBMS)". They are not equivalent and it will be explained in the next section.

Def 2: A collection of data: part numbers, product codes, customer information, etc. It usually refers to data organized and stored on a computer that can be searched and retrieved by a computer program.

Def 3: A data structure that stores metadata, i.e. data about data. More generally we can say an organized collection of information.

Def 4: A collection of information organized and presented to serve a specific purpose. (A telephone book is a common database.) A computerized database is an updated, organized file of machine readable information that is rapidly searched and retrieved by computer.

Def 5: An organized collection of information in computerized format.

Def 6: A collection of related information about a subject organized in a useful manner that provides a base or foundation for procedures such as retrieving information, drawing conclusions, and making decisions.

Def 7: A Computerized representation of any organizations flow of information and storage of data.

Each of the above given definition is correct, and describe database from slightly variant perspectives. From exam point of view, anyone will do. However, within this course, we will be referring first of the above definitions more frequently, and concepts discussed in the definition like, logically related data, shared collection should be clear. Another important thing that you should be very clear about is the difference between database and the database management system (DBMS). See, the database is the collection of data about anything, could be anything. Like cricket teams, students, busses, movies, personalities, stars, seas, buildings, furniture, lab equipment, hobbies, hotels, pets, countries, and many more anything about which you

want to store data. What we mean by data; simply the facts or figures. Following table shows the things and the data that we may want to store about them:

Thing	Data (Facts or figures)
Cricket Player	Country, name, date of birth, specialty, matches played, runs etc.
Scholars	Name, data of birth, age, country, field, books published etc.
Movies	Name, director, language (Punjabi is default in case of Pakistan) etc.
Food	Name, ingredients, taste, preferred time, origin, etc.
Vehicle	Registration number, make, owner, type, price, etc.

There could be infinite examples, and please note that the data that is listed about different things in the above table is not the only data that can be defined or stored about these things. As has been explained in the definition one above, there could be so many facts about each thing that we are storing data about; what exactly we will store depends on the perspective of the person or organization who wants to store the data. For example, if you consider food, data required to be stored about the food from the perspective of a cook is different from that of a person eating it. Think of a food, like, Karhahi Ghost, the facts about Karhahi ghosht that a cook will like to store may be, quantity of salt, green and red chilies, garlic, water, time required to cook and like that. Where as the customer is interested in chicken or meat, then black or red chilies, then weight, then price and like that. Well, definitely there are some things common but some are different as well. The thing is that the perspective or point of view creates the difference in what we store; however, the main thing is that the database stores the data.

The database management system (DBMS), on the other hand is the software or tool that is used to manage the database and its users. A DBMS consist of different components or subsystem that we will study about later. Each subsystem or component of the DBMS performs different function(s), so a DBMS is collection of different programs but they all work jointly to manage the data stored in the database and its users. In many books and may be in this course sometimes database and database management system are used interchangeably but there is a clear difference and we should be clear about them. Sometimes another term is used, that is, the database system, again, this term has been used differently by different people, however in this course we use the term database system as a combination of database and the database management system. So database is collection of data, DBMS is tool to manage this data, and both jointly are called database system.

Importance of the Databases

Databases are important; why? Traditionally computer applications are divided into commercial and scientific (or engineering) ones. Scientific applications involve more computations, that is, different type of calculations that vary from simple to very complex. Today such applications exist, like in the fields of space, nuclear, medicine that take hours or days of computations on even computers of the modern age. On the other hand, the applications that are termed as commercial or business applications do not involve much computations, rather minor computation but mainly they perform the input/output operations. That is, these applications mainly store the data in the computer storage, then access and present it to the users in different formats (also termed as data processing) for example, banks, shopping, production, utilities billing,

customer services and many others. As is clear from the example systems mentioned, the commercial applications exist in the day to day life and are related directly with the lives of common people. In order to manage the commercial applications more efficiently databases are the ultimate choice because efficient management of data is the sole objective of the databases. So such applications are being managed by databases even in a developing country like Pakistan, yet to talk about the developed countries. This way databases are related directly or indirectly almost every person in society.

Databases are not only being used in the commercial applications rather today many of the scientific/engineering application are also using databases less or more. databases are concern of the effectively latter form of applications are more Commercial applications involve The goal of this course is to present an in-depth introduction to databases, with an emphasis on how to organize information in the database and to maintain it and retrieve it efficiently, that is, how to design a database and use it effectively.

Databases and Traditional File Processing Systems

Traditional file processing system or simple file processing system refers to the first computer-based approach of handling the commercial or business applications. That is why it is also called a replacement of the manual file system. Before the use computers, the data in the offices or business was maintained in the files (well in that perspective some offices may still be considered in the pre-computer age). Obviously, it was laborious, time consuming, inefficient, especially in case of large organizations. Computers, initially designed for the engineering purposes were though of as blessing, since they helped efficient management but file processing environment simply transformed manual file work to computers. So processing became very fast and efficient, but as file processing systems were used, their problems were also realized and some of them were very severe as discussed later.

It is not necessary that we understand the working of the file processing environment for the understanding of the database and its working. However, a comparison between the characteristics of the two definitely helps to understand the advantages of the databases and their working approach. That is why the characteristics of the traditional file processing system environment have been discussed briefly here.

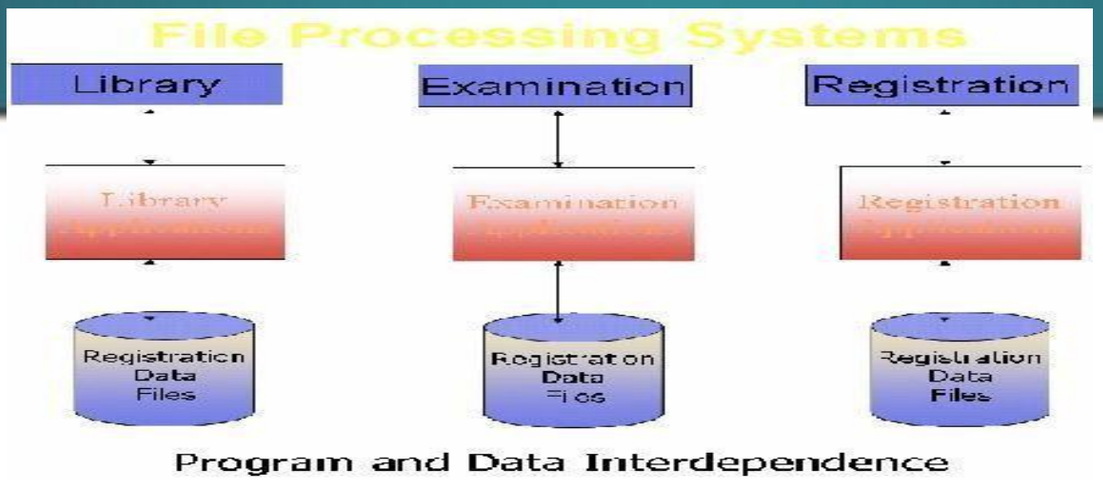


Fig. 1: A typical file processing environment

The diagram presents a typical traditional file processing environment. The main point being highlighted is the program and data interdependence, that is, program and data depend on each other, well they depend too much on each other. As a result any change in one affects the other as well. This is something that makes a change very painful or problematic for the designers or developers of the system. What do we mean by change and why do we need to change the system at all. These things are explained in the following.

The systems (even the file processing systems) are created after a very detailed analysis of the requirements of the organizations. But it is not possible to develop a system that does not need a change afterwards. There could be many reasons, mainly being that the users get the real taste of the system when it is established. That is, users tell the analysts or designers their requirements, the designers design and later develop the system based on those requirements, but when system is developed and presented to the users, it is only then they realize the outcome of the effort. Now it could be slightly and (unfortunately) sometimes very different from what they expected or wanted it to be. So the users ask changes, minor or major. Another reason for the change is the change in the requirements. For example, previously the billing was performed in an organization on the monthly basis, now company has decided to bill the customers after every ten days. Since the bills are being generated from the computer (using file processing system), this change has to be incorporated in the system. Yet another example is that initially bills did not contain the address of the customer, now the company wants the address to be placed on the bill, so here is change. There could be many more examples, and it is so common that we can say that almost all systems need changes, so system development is always an on-going process.

PDF Creator: PDF4U Pro DEMO Version. If you want to remove this line, please purchase the full version.

Amina Zee

So we need changes in the system, but due to program-data interdependence these changes in the systems were very hard to make. A change in one will affect the other whether related or not. For example, suppose data about the customer bills is stored in the file, and different programs use this file for different purposes, like adding data into the bills file, to compute the bill and to print the bill. Now the company asks to add the customers' address in the bills, for this we have to change the structure of the bill file and also the program that prints the bill. Well, this was necessary, but the painful thing is that the other programs that are using these bills files but are not concerned with the printing of the bills or the change in the bill will also have to be changed, well; this is needless and causes effort.

Another major drawback in the traditional file system environment is the non-sharing of data. It means if different systems of an organization are using some common data then rather than storing it once and sharing it, each system stores data in separate files. This creates the problem of redundancy or wastage of storage and on the other hand the problem on inconsistency. The change in the data in one system sometimes is not reflected in the same data stored in other system. So different systems in organization;

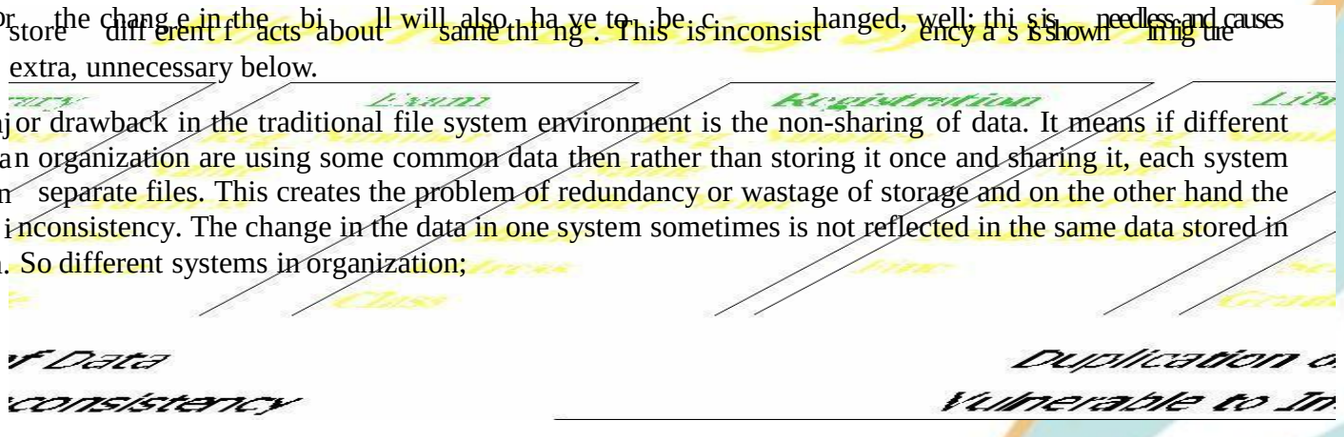


Fig. 2: Some more problems in File System Environment Previous section highlighted the file processing system environment and major problems found there. The following section presents the benefits of the database systems.

Advantages of Databases

It will be helpful to reiterate our database definition here, that is, database is a shared collection of logically related data, designed to meet the information needs of multiple users in an organization. A typical database system environment is shown in the figure 3 below:

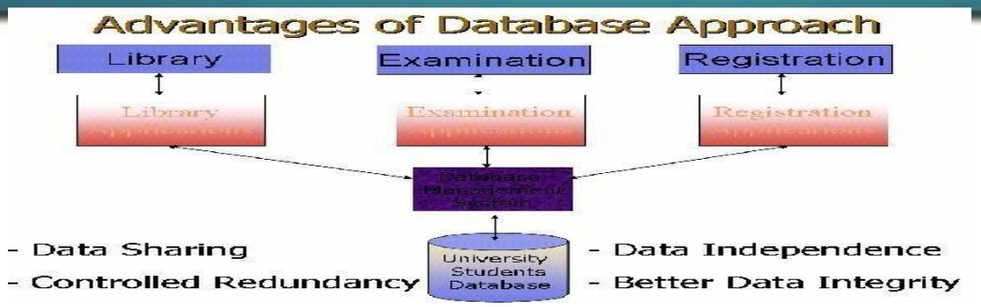


Fig. 3: A typical Database System environment

The figure shows different subsystem or applications in an educational institution, like library system, examination system, and registration system. There are separate, different application programs for every application or subsystem. However, the data for all applications is stored at the same place in the database and all application programs, relevant data and users are being managed by the DBMS. This is a typical database system environment and it introduces the following advantages:

- **Data Sharing:**
  The data for different applications or subsystems is placed at the same place. This introduces the major benefit of data sharing. That is, data that is common among different applications need not to be stored repeatedly, as was the case in the file processing environment. For example, all three systems of an educational institution shown in figure 3 need to store the data about students. The example data can be seen from figure 2. Now the data like registration number, name, address, father name that is common among different applications is being stored repeatedly in the file processing system environment, where as it is being stored just once in database system environment and is being shared by all applications. The interesting thing is that the individual applications do not know that the data is being shared and they do not need to. Each application gets the impression as if the data is being for stored for it. This brings the advantage of saving the storage along with others discussed later.
- **Data Independence:**
  Data and programs are independent of each other, so change is once has no or minimum effect on other. Data and its structure is stored in the database where as application programs manipulating this data are stored separately, the change in one does not unnecessarily effect other

.

- Controlled Redundancy
  Means that we do not need to duplicate data unnecessarily; we do duplicate data in the databases, however, this duplication is deliberate and controlled.

- Better Data Integrity
  Very important feature; means the validity of the data being entered in the database. Since the data is being placed at a central place and being managed by the DBMS, so it provides a very conducive to check or ensure that the data being entered into the database is actually valid. Integrity of data is very important, since all the processing and the information produced in return are based on the data. Now if the data entered is not valid, how can we be sure that the processing in the database is correct and the results or the information produced is valid? The businesses make decisions on the basis of information produced from the database and the wrong information leads to wrong decisions, and business collapse. In the database system environment, DBMS provides many features to ensure the data integrity, hence provides more reliable data processing environment.

Dear students, that is all for this lecture. Today we got the introduction of the course, importance of the databases. Then we saw different definitions of database and studied what is data processing then studied different features of the traditional file processing environment and database (DB) system environment. At the end of lecture we were discussing the advantages of the DB approach. There some others to be studied in the next lecture. Suggestions are welcome.

Exercises

- Think about the data that you may want to store about different things around you
- List the changes that may arise during the working of any system, lets say Railway Reservation System

Thanks and good luck

Amina Zee

Database Management Systems

lec 92

database Advantages

Amina Zee

Overview of lecture

- Database definitions
- File processing systems
- Advantages of database

Amina Zee

Today's Lecture

- More advantages
- Some costs
- Levels of data
- Database users

Data & Information

Company: Super Soft Dept: Sales

Emp Name	Age	Salary
Malik Sharif	23	55
Sh. M. Akmal	24	55
M. A. Butt	20	40
Malik Junaid	19	20

Amina Zee

- Schema
- Database Applications
- Database Management System (DBMS)

Other Advantages

- Data consistency
- Better data security
- Faster development of new applications

They also provide

- Economy of scale
- Better concurrency control
- Better backup and recovery procedures
- BUT

Its not always just the SUGAR

Disadvantages

- Higher costs
- Conversion cost
- More difficult recovery

Data As Resource

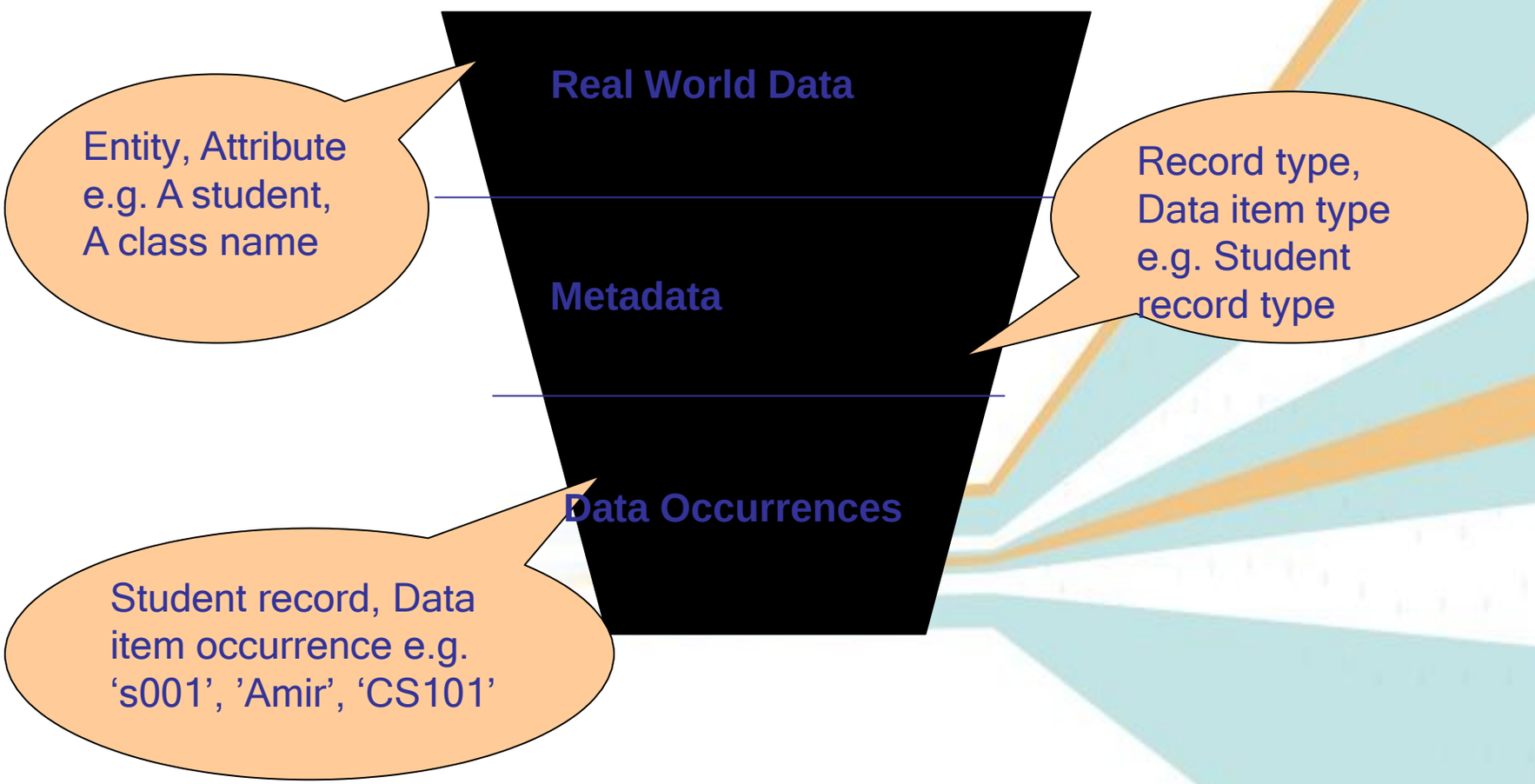
Resource

Any asset that is of value to an organization and that incurs cost

Is data a resource ?

YES

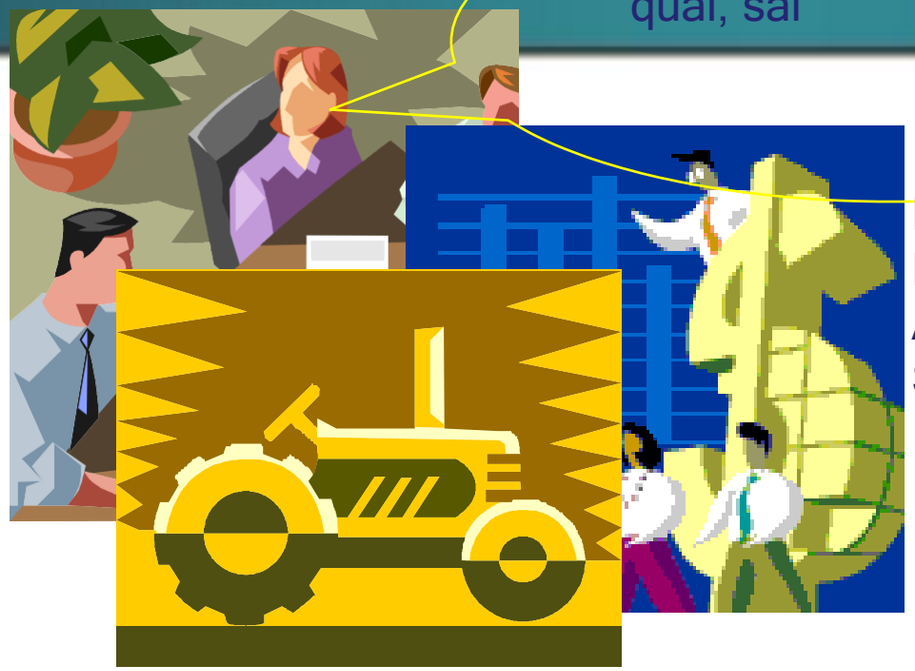
Levels of Data,
Rizwan prepare ppt for front



Amina Zee

Employee

name, age,
qual, sal



Emp
Name text
Age number
Sal number

Malik Sharif	23	55
Sh. M. Akmal	24	55
M. A. Butt	20	40
Malik Junaid	19	20

Amina Zee

Levels of Data

- Real-world data
- Metadata
- Data Occurrence

Amina Zee

Database Users

- Application Programmers
- End Users
  - Naïve
  - Sophisticated

Database Users

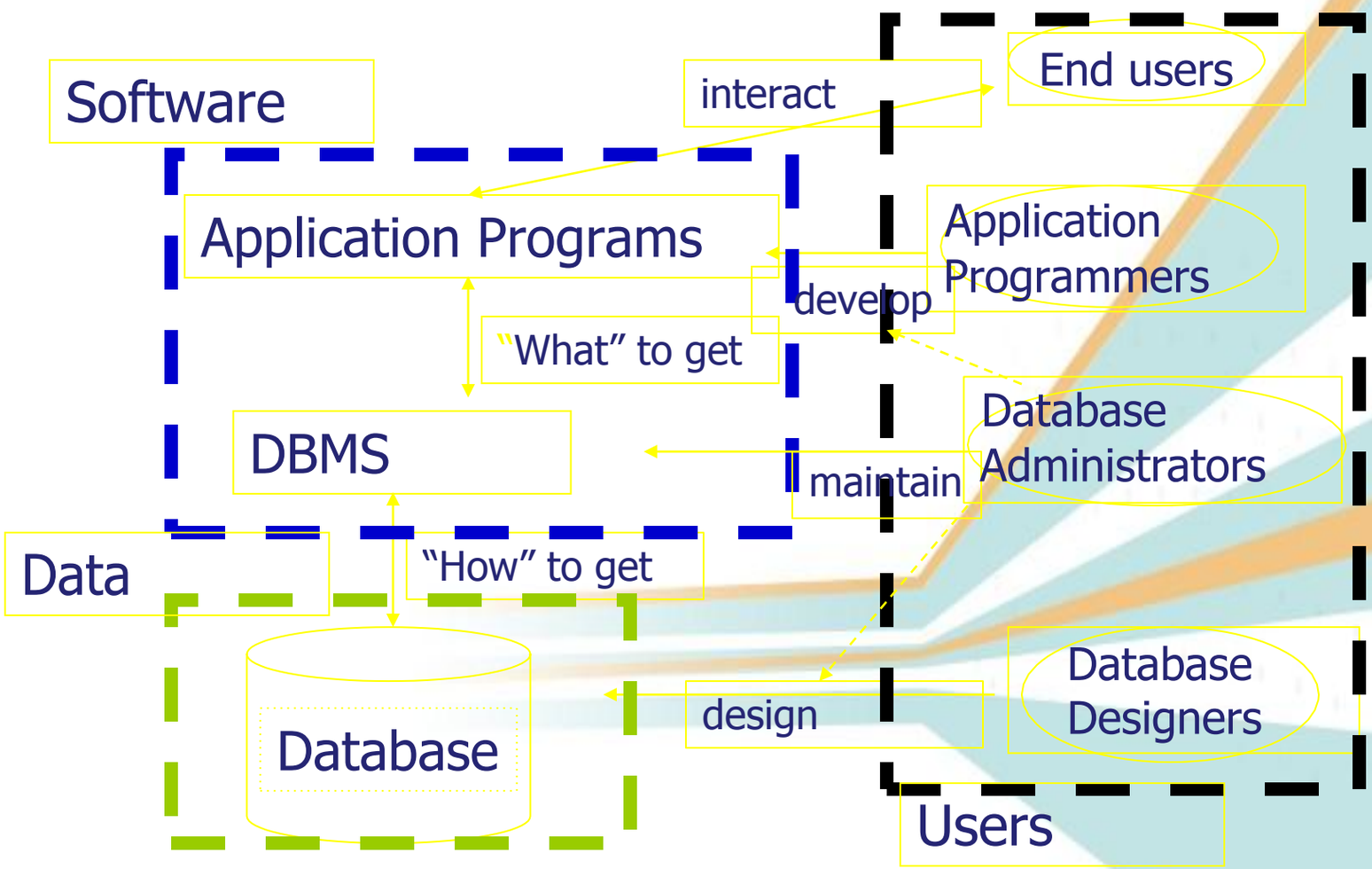
- Database Administrator (DBA)

  A person who has central control over data and programs that access this data

Functions of DBA

- Schema definition
- Granting data access
- Routine Maintenance
  - **Backups**
  - **Monitoring disk space**
  - **Monitoring jobs running**

Typical Components



Database Architecture

Amina Zee

History

- Initiates in 1960s; IMS, IDS
- Conference on DAta Systems Languages (CODASYL)
- Data Base Task Group (DBTG)
- General architecture in 1971

History

- DBTG's two layered architecture having
  - System View- Schema
  - User View- Sub rschemas

History

- American National Standards Institute (ANSI)
- Standards Planning and Requirements Committee (SPARC)
- ANSI-SPARC's 3-layered Architecture

Amina Zee

Summary

- Basic Terminology
- Pros n Cons of DB Env
- Levels of Data
- Types of Users
- History of 3-L Architecture

Amina Zee

Lec 93

Data Management

Amina Zee

During the last lecture … (Intelligent Systems)

- We looked at the distinguishing features of intelligent systems w.r.t. other software systems
- We looked at the role of intelligent systems in scientific, business, consumer and other applications
- We discussed several techniques for designing intelligent systems

(Artificial) Intelligent Systems

- SW programs or SW/HW systems designed to perform *complex* tasks employing strategies that mimic some aspect of human thought

Not a Suitable Hammer for All Nails!

if the nature of computations required in a task is not well understood

or there are too many exceptions to the rules

or known algorithms are too complex or inefficient

then AI has the potential of offering an acceptable solution

Selected Applications

- Games: Chess, SimCity
- Image recognition
- Medical diagnosis
- Robots
- Business intelligence

Neural Networks (1)

- Original inspiration was the human brain; emphasis now on usefulness as a computational tool

Genetic Algorithms (1)

- Based on Darwin's evolutionary principle of 'survival of the fittest'

- GAs require the ability to recognize a good solution, but not how to get to that solution

Rulebased Systems (1)

- Based on the principles of the logical reasoning ability of humans

Fuzzy Logic (1)

- Based on the principles of the approximate reasoning faculty that humans use when faced with linguistic ambiguity

The Right Technique

- Selection of the right AI technique requires intimate knowledge about the problem as well as the techniques under consideration

- Real problems may require a combination of techniques (AI and/or nonAI) for an optimal solution

Three exciting areas areas of AI applications

Amina Zee

Robotics

- Automatic machines that perform various tasks that were previously done by humans

Autonomous Web Agents (1)

- Computer program that performs various actions continuously, autonomously on behalf of their principal!

Decision Support Systems

- Interactive software designed to improve the decision-making capability of their users

- The do not make decisions - just assist in the process

Today's Goals:
(Data Management)

- First of a two-lecture sequence
- Today we will become familiar with the issues and problems related to data-intensive computing
- We will find out about flat-files, the simpleast databases
- Next time, in our 4th lecture on productivity software, we will discuss relational databases and implement a simple relational database

Data Management

- Keeping track of a few dozen data items is straight forward
- However, dealing with situations that involve significant number of data items, requires more attention to the data handling process
- Dealing with millions - even billions - of inter-related data items requires even more careful thought

BholiBooks.com (1)

- Consider the situation of a large, online bookstore
- They have an inventory of millions of books, with new titles constantly arriving, and old ones being phased out on a regular basis
- The price for a book is not a static feature; it varies every once in a while

BholiBooks.com(2)

- Thousands of books are shipped each day, changing the inventory constantly
- Some are returned, again changing the inventory situation constantly
- The cost of each shipped order depends on:
  - Prices of individual books
  - Size of the order
  - Location of the customer
  - Mode of shipment

BholiBooks.com(3)

- For each order, the customer's particulars – name, address, phone number, credit card number – are required

- Generally, that data is not deleted after the completion of the transaction; instead, it is kept for future reference

BholiBooks.com(4)

- All the transaction activity and the inventory changes result in:
  - Thousands of data items changing every day
  - Thousands of additional data items being added everyday

- Keeping track & taking care (i.e. management) of all that constantly changing and expanding data is not a trivial task and requires disciplined attention and actions for ensuring the smooth & profitable operation of the bookstore

Issues in Data Management

- Data entry
- Data updates
- Data integrity
- Data security
- Data accessibility

Data Entry

- New titles are added every day
- New customers are being added every day

- Some of the above *may* require manual entry of new data into the computer systems
- That new data needs to be added accurately
- That can be achieved, for one, by user-interfaces that prevent the input of invalid data

Data Updates (1)

- Old titles are deleted on a regular basis
- Inventory changes every instant
- Book prices change
- Shipping costs change
- Customers' personal data change
- Various discount schemes are always commencing and concluding

Data Updates (2)

- All those actions require updates to existing data
- Those changes need to be entered accurately
- That can also be achieved by user-interfaces that prevent the input of invalid data

Data Security (1)

- All the data that BholiBooks has in its computer systems is quite critical to its operation
- The security of the customers' personal data is of utmost importance. Hackers are always looking for that type of data, especially for credit card numbers
- Enough leaks of that type, and customers will stop doing business with BholiBooks

Data Security (2)

- This problem can be managed by using appropriate security mechanisms that provide access to authorized persons/computers only

- Security can also be improved through:
  - Encryption
  - Private or virtual-private networks
  - Firewalls
  - Intrusion detectors
  - Virus detectors

Data Integrity

- Integrity refers to maintaining the correctness and consistency of the data
  - Correctness: Free from errors
  - Consistency: No conflict among related data items
- Integrity can be compromised in many ways:
  - Typing errors
  - Transmission errors
  - Hardware malfunctions
  - Program bugs
  - Viruses
  - Fire, flood, etc.

Ensuring Data Integrity (1)

- *Type Integrity* is implemented by specifying the type of a data item:
  - Example: A credit card number consists of 12 digits. An update attempting to assign a value with more or fewer digits or one including a non-numeral should be rejected

- *Limit Integrity* is enforced by limiting the values of data items to specified ranges to prevent illegal values
  - Example: Age of person should not be negative

Ensuring Data Integrity (2)

- *Referential Integrity* requires that an item referenced by the data for some other item must itself exist in the database
  - Example: If an airline reservation is requested for a particular flight, then the corresponding flight number must actually exist

- *Physical Integrity* is ensured through hardware redundancy, backups, etc

Data Accessibility (1)

- If the transaction and inventory data is placed in a disorganized fashion on a hard disk, it becomes very difficult to later search for a stored data item

- What is required is that:
  - Data be stored in an organized manner
  - Additional info about the data be stored

  so that the data access times are minimized

Data Accessibility (2)

- What if two customers check on the availability of a certain title simultaneously?
- On seeing its availability, they both order the title – for which, unfortunately, only a single copy is available
- Same is the case when two airline customers try booking the only available seat

Data Accessibility (3)

- A solution to this *concurrency control* problem: Lock access to data while someone is using it

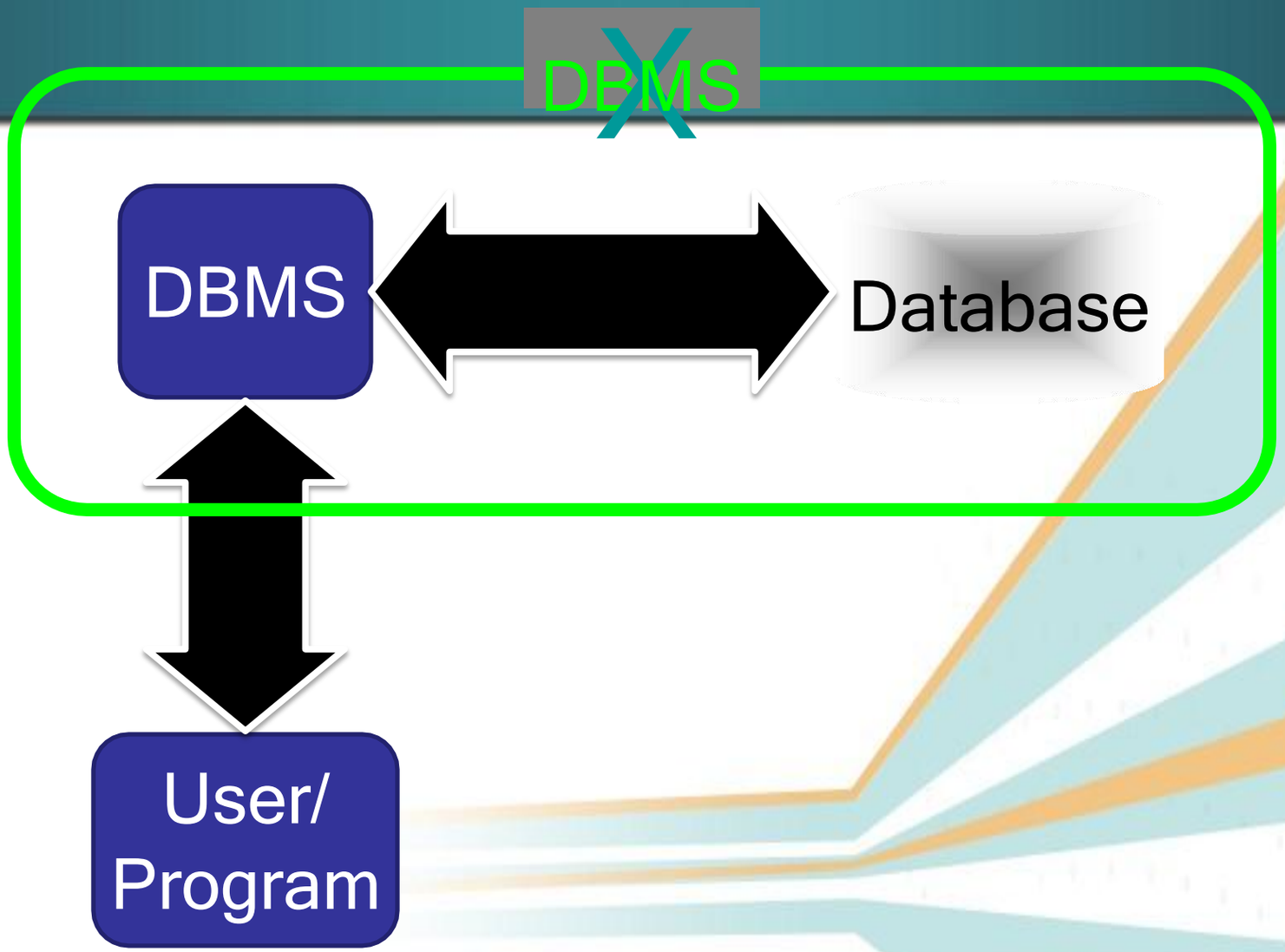
We can write our own SW that can take care of all the issues that we just discussed

OR

We can save ourselves lots of time, cost, and effort by buying ourselves a Database Management System (DBMS) that takes care of most, if not all, of the issues

DBMS (1)

- DBMSes are popularly, but incorrectly, also known as ‘Databases’
- A DBMS is the SW system that operates a database, and is not the database itself
- Some people even consider the database to be a component of the DBMS, and not an entity outside the DBMS


DBMS
DBMS
Database
User/
Program

DBMS (2)

- A DBMS takes care of the storage, retrieval, and management of large data sets on a database
- It provides SW tools needed to organize & manipulate that data in a flexible manner
- It includes facilities for:
  - Adding, deleting, and modifying data
  - Making queries about the stored data
  - Producing reports summarizing the required contents

Database (1)

- A collection of data organized in such a fashion that the computer can quickly search for a desired data item

- All data items in it are generally related to each other and share a single domain

Database (2)

- They allow for easy manipulation of the data
- They are designed for easy modification & reorganization of the information they contain
- They generally consist of a collection of interrelated computer files

Example: VU Student Database

- Student's name
- Student's photograph
- Father's name
- Phone number
- Street address
- eMail address
- Courses being taken
- Courses already taken & grades
- Pre-VU educational record

Example: BholiBooks' Customer DB

- Name, address, phone & fax, eMail
- Credit card type, number, expiration date
- Shipping preference
- Books on order
- All books that were ever shipped to the customer
- Book preference

Example: BholiBooks' Inventory DB

- Book title, author, publisher, binding, date of publication, price
- Book summary, table of contents
- Customers', editors', newspaper reviews
- Number in stock
- Number on order
- Special offer details

OS Independence (1)

- DBMS stores data in a database, which is a collection of interrelated files
- Storage of files on the computer is managed by the computer OS's file system
- Intimate knowledge of the OS & its file system is required to provide rapid access to the data

OS Independence (2)

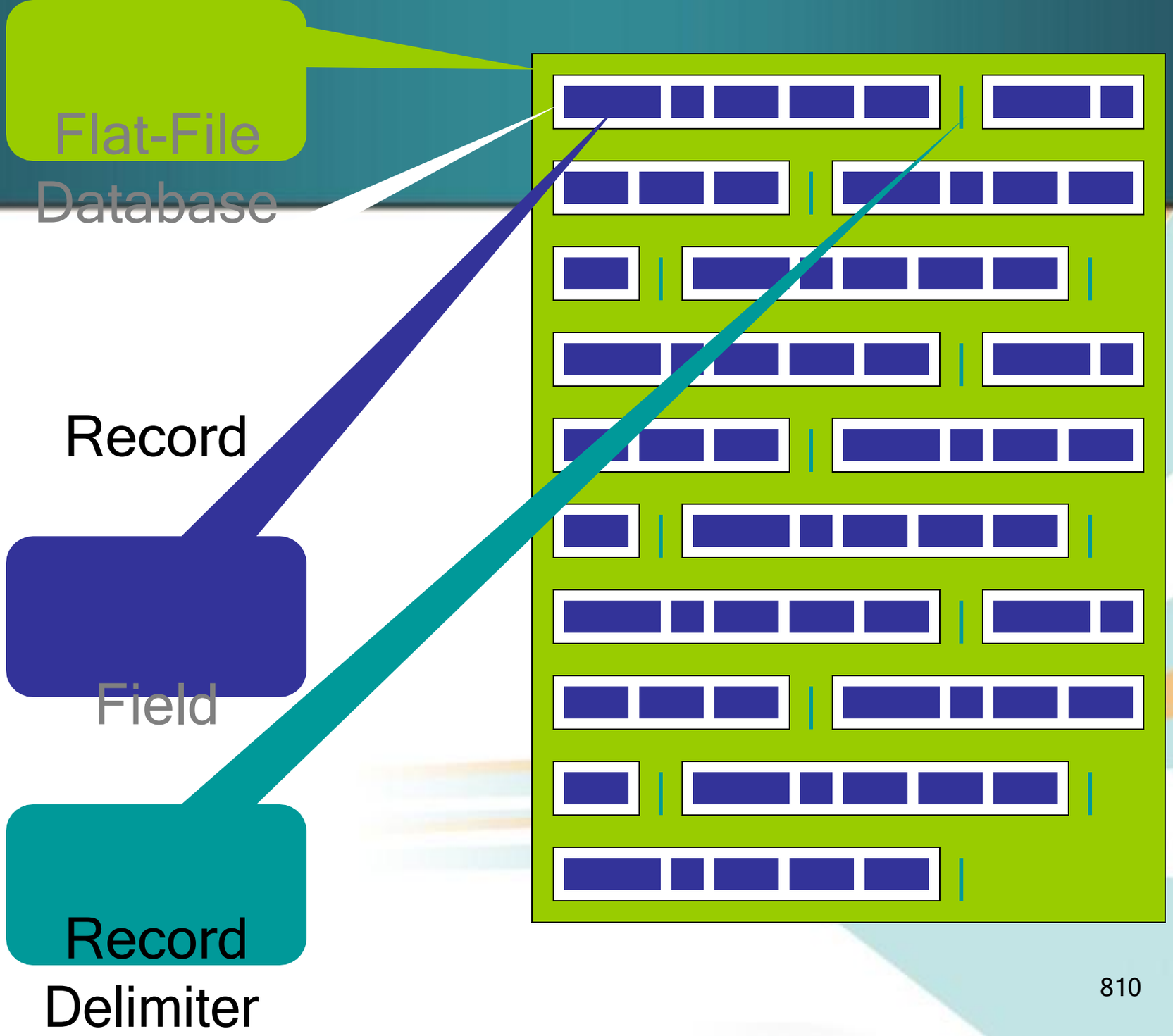
- The DBMS takes care of those details
- It hides the actual storage details of data files from the user
- It provides an OS-independent view of the data to the user, making data manipulation and management much more convenient

What can be stored in a database?

- In the old days, databases were limited to numbers, Booleans, and text
- These days, anything goes
- As long as it is digital data, it can be stored:
  - Numbers, Booleans, text
  - Sounds
  - Images
  - Video

In the very, very old days …

- Even large amounts of data was stored in text files, known as *flat-file databases*
- All related info was stored in a single long, tab- or comma-delimited text file
- Each group of info – called a *record* - in that file was separated by a special character; vertical bar '|' was a popular option
- Each record consisted of a group of *fields*, each field containing some distinct data item


Flat-File Database
Record
Field
Record Delimiter

The Trouble with Flat-File Databases

- The text file format makes it hard to search for specific information or to create reports that include only certain fields from each record
- Reason: One has to search sequentially through the entire file to gather desired info, such as 'all books by a certain author'
- However, for small sets of data – say, consisting of several tens of kB – they can provide reasonable performance

Consider this tabular approach ...

(same records, same fields, but in a different format)

Title	Author	Publisher	Price	InStock
Good Bye Mr. Bhola	Altaf Khan	BholiBooks	1000	Y
The Terrible Twins	Bhola Champion	BholiBooks	199	Y
Calculus & Analytical Geometry	Smith Sahib	Good Publishers	325	N
Accounting Secrets	Zamin Geoffry	Sung-e-Kilometer Publishers	29	Y[812]

Amina Zee

Tabular Storage: Features & Possibilities

1. Similar items of data form a column

2. *Fields* placed in a particular row – same as a flat-file *record* – are strongly interrelated

3. One can sort the table w.r.t. any column

4. That makes searching – e.g., for all the books written by a certain author – straight forward

Tabular Storage: Features & Possibilities

5. Similarly, searching for the 10 cheapest/most expensive books can be easily accomplished through a sort

6. Effort required for adding a new field to all the records of a flat-file is much greater than adding a new column to the table

CONCLUSION: Tabular storage is better than flat-file storage

We will continue on this theme next time

Today's Summary: (Data Management)

- First of a two-lecture sequence
- Today we became familiar with the issues and problems related to data-intensive computing
- We also found out about flat-file and tabular storage

Next Lecture: (Database SW)

- Next time, in our 4th lecture on productivity SW, we will continue our discussion on data management
- We will find out about relational databases
- We will also implement a simple relational database

Lec 94

Database Software

Amina Zee

Focus of the last Lecture was on Data Management

- First of a two-lecture sequence
- We became familiar with the issues and problems related to data-intensive computing
- We also found out about flat-file and tabular storage

Data Management

- Keeping track of a few dozen data items is straight forward
- However, dealing with situations that involve significant number of data items, requires more attention to the data handling process
- Dealing with millions - even billions - of inter-related data items requires even more careful thought

Issues in Data Management

821

Amina Zee

Data Entry

- New titles are added every day
- New customers are being added every day

- That new data needs to be added accurately

Data Updates (2)

- All those actions require updates to existing data
- Those changes need to be entered accurately

Data Security (1)

- All the data that BholiBooks has in its computer systems is quite critical to its operation

- The security of the customers' personal data is of utmost importance. Hackers are always looking for that type of data, especially for credit card numbers

Data Security (2)

- This problem can be managed by using appropriate security mechanisms that provide access to authorized persons/computers only

- Security can also be improved through:
  - Encryption
  - Private or virtual-private networks
  - Firewalls
  - Intrusion detectors
  - Virus detectors

Data Integrity

- Integrity refers to maintaining the correctness and consistency of the data
  - Correctness: Free from errors
  - Consistency: No conflict among related data items
- Integrity can be compromised in many ways:
  - Typing errors
  - Transmission errors
  - Hardware malfunctions
  - Program bugs
  - Viruses
  - Fire, flood, etc.

Ensuring Data Integrity (1)

- *Type Integrity*
- *Limit Integrity*

Ensuring Data Integrity (2)

- *Referential Integrity*
- *Physical Integrity*

Data Accessibility (1)

- What is required is that:
  - Data be stored in an organized manner
  - Additional info about the data be stored

  so that the data access times are minimized

Data Accessibility (3)

- A solution to this *concurrency control* problem: Lock access to data while someone is using it

DBMS (2)

- A DBMS takes care of the storage, retrieval, and management of large data sets on a database
- It provides SW tools needed to organize & manipulate that data in a flexible manner
- It includes facilities for:
  - Adding, deleting, and modifying data
  - Making queries about the stored data
  - Producing reports summarizing the required contents

Database (1)

- A collection of data organized in such a fashion that the computer can quickly search for a desired data item

OS Independence (2)

- It provides an OS-independent view of the data to the user, making data manipulation and management much more convenient

What can be stored in a database?

- As long as it is digital data, it can be stored:
  - Numbers, Booleans, text
  - Sounds
  - Images
  - Video

In the very, very old days …

- Even large amounts of data was stored in text files, known as *flat-file databases*
- All related info was stored in a single long, tab- or comma-delimited text file
- Each group of info – called a *record* - in that file was separated by a special character; vertical bar '|' was a popular option
- Each record consisted of a group of *fields*, each field containing some distinct data item

The Trouble with Flat-File Databases

- The text file format makes it hard to search for specific info or to create reports that include only certain fields from each record
- Reason: One has to search sequentially through the entire file to gather desired info, such as 'all books by a certain author'
- However, for small sets of data – say, consisting of several tens of kB – they can provide reasonable performance

Tabular Storage: Features & Possibilities

1. Similar items of data form a column

2. *Fields* placed in a particular row – same as a flat-file *record* – are strongly interrelated

3. One can sort the table w.r.t. any column

4. That makes searching – e.g., for all the books written by a certain author – straight forward

Tabular Storage: Features & Possibilities

5. Similarly, searching for the 10 cheapest/most expensive books can be easily accomplished through a sort

6. Effort required for adding a new field to all the records of a flat-file is much greater than adding a new column to the table

CONCLUSION: Tabular storage is better than flat-file storage

We will continue on with tables' theme today

Today's Lecture: Database SW

- In our 4th & final lecture on productivity software, we will continue our discussion from last week on data management
- We will find out about relational databases
- We will also implement a simple relational database

Let’s continue on with the tabular approach

We stored data in a table last time, and liked it

Let’s revisit that table and then put together another one

Table from the Last Lecture

Title	Author	Publisher	Price	InStock
Good Bye Mr. Bhola	Altaf Khan	BholiBooks	1000	Y
The Terrible Twins	Bhola Champion	BholiBooks	199	Y
Calculus & Analytical Geometry	Smith Sahib	Good Publishers	325	N
Accounting Secrets	Zamin Geoffry	Sung-e-Kilometer Publishers	29	Y

Another table ...

Customer	Title	Shipment	Type
Aadil Ali	Good Bye Mr. Bhola	2002.12.26	Air
Aadil Ali	The Terrible Twins	2002.12.26	Air
Miftah Muslim	Calculus & Analytical Geometry	2002.12.25	Surface
Karen Kaur	Good Bye Mr. Bhola	2002.12.24	Air

This & the previous table are related

- They share a column, & are related through it
- A program can match info from a field in one table with info in a corresponding field of another table to *generate* a 3rd table that combines requested data from both tables
- That is, a program can use matching values in 2 tables to *relate* info in one to info in the other

Q: Who is BholiBooks' best customer?

- That is, who has spent the most money on the online bookstore?

- To answer that question, one can process the inventory and the shipment tables to generate a third table listing the customer names and the prices of the books that they have ordered

The *generated* table

Customer	Price
Aadil Ali	1000
Aadil Ali	199
Miftah Muslim	325
Karen Kaur	1000

Can you now process this table to find the answer to our question

Relational Databases (1)

- Databases consisting of two or more related tables are called *relational databases*
- A typical relational database may have anywhere from 10 to over a thousand tables
- Each column of those tables can contain only a single type of data (contrast this with spreadsheet columns!)
- Table rows are called records; row elements are called fields

Relational Databases (2)

- A relational database stores all its data inside tables, and nowhere else
- All operations on data are done on those tables or those that are generated by table operations
- Tables, tables, and nothing but tables!

RDBMS

- Relational DBMS software
- Contains facilities for creating, populating, modifying, and querying relational databases
- Examples:
  - Access
  - FileMaker Pro
  - SQL Server
  - Oracle
  - DB2
  - Objectivity/DB
  - MySQL
  - Postgres

The Trouble with Relational DBs (1)

- Much of current SW development is done using the object-oriented methodology

- When we want to store the object-oriented data into an RDBMS, it needs to be translated into a form suitable for RDBMS

The Trouble with Relational DBs (2)

- Then when we need to read the data back from the RDBMS, the data needs to be translated back into an object-oriented form before use

- These two processing delays, the associated processing, and time spent in writing and maintaining the translation code are the key disadvantages of the current RDBMSes

Solution?

- Don’t have time to discuss that, but try searching the Web on the following terms:
  - Object-oriented databases
  - Object-relational databases

Classification of DBMS w.r.t. Size

- Personal/Desktop/Single-user (MB-GB)
  - Examples: Tech. papers' list; Methai shop inventory
  - Typical DMBS: Access

- Server-based/Multi-user/Enterprise (GB-TB)
  - Examples: HBL; Amazon.com
  - Typical DMBS: Oracle, DB2

- Seriously-huge databases (TB-PB-XB)
  - Examples: 2002 - BaBar experiment at Stanford (500TB); 2005 - LHC database at CERN (1XB)
  - Typical DMBS: Objectivity/DB

Some Terminology (1)

- *Primary Key* is a field that uniquely identifies each record stored in a table

- *Queries* are used to view, change, and analyze data. They can be used to:
  - Combine data from different tables, efficiently
  - Extract the exact data that is desired

- *Forms* can be used for entering, editing, or viewing data, one record at a time

Some Terminology (2)

- *Reports* are an effective, user-friendly way of presenting data. All DBMSes provide tools for producing custom reports.

- *Data normalization* is the process of efficiently organizing data in a database. There are two goals of the normalization process:
  - Eliminate redundant data
  - Storing only related data in a table

Before we do a demo, let me just mention my favorite database application: Data Mining

Data Mining

- The process of analyzing large databases to identify patterns
- Example: Mining the sales records from a BholiBooks could identify interesting shopping patterns like “53% of customers who bought book A also bought book B”. This pattern can be put to good use!
- Data mining often utilizes intelligent systems’ techniques

Let's now demonstrate the use of a desktop RDBMS

- We will create a new relational database
- It will consist of two tables
- We will populate those tables
- We will generate a report after combining the data from the two tables

Assignment # 13

Develop a database by designing two tables, populate them, and then generate a report

Further information on this assignment will be provided to you on the CS101 Web site

Today's Lecture:

- In this final lecture on productivity software, we continued our discussion from last week on data management
- We found out about relational databases
- We also implemented a simple relational database

Next Lecture' Goals (Cyber Crime)

- To know the different types of computer crimes that occur over cyber space
- To familiarize ourselves with with several methods that can be used to minimize the effect of these crimes
- To get familiar with a few policies and legislation designed to tackle cyber crime

Lec 95

Nucleotide Sequence databases

Amina Zee

Nucleotide Sequence Databases

Biological Databases

Biological databases in general store biological data and their main goals are

- **Data storage**
- **Information retrieval**
- **Knowledge discovery**

Amina Zee

Nucleotide Sequence Databases

Classification

Biological databases can be classified as

- **Primary databases**
- **Secondary databases**
- **Specialized Databases**

Amina Zee

Nucleotide Sequence Databases

Classification

Biological databases can also be classified on the bases of types of data as

- **Nucleotide databases**
- **Protein databases**
- **RNA databases**
- **Genome databases**
- **Expression databases**

Amina Zee

Nucleotide Sequence Databases

Issues

Due to limited **Q/A**

- **Redundancy**
- **Inconsistency**
- **Incompatibility (format, terminology, data types, etc.)**

Amina Zee

Nucleotide Sequence Databases

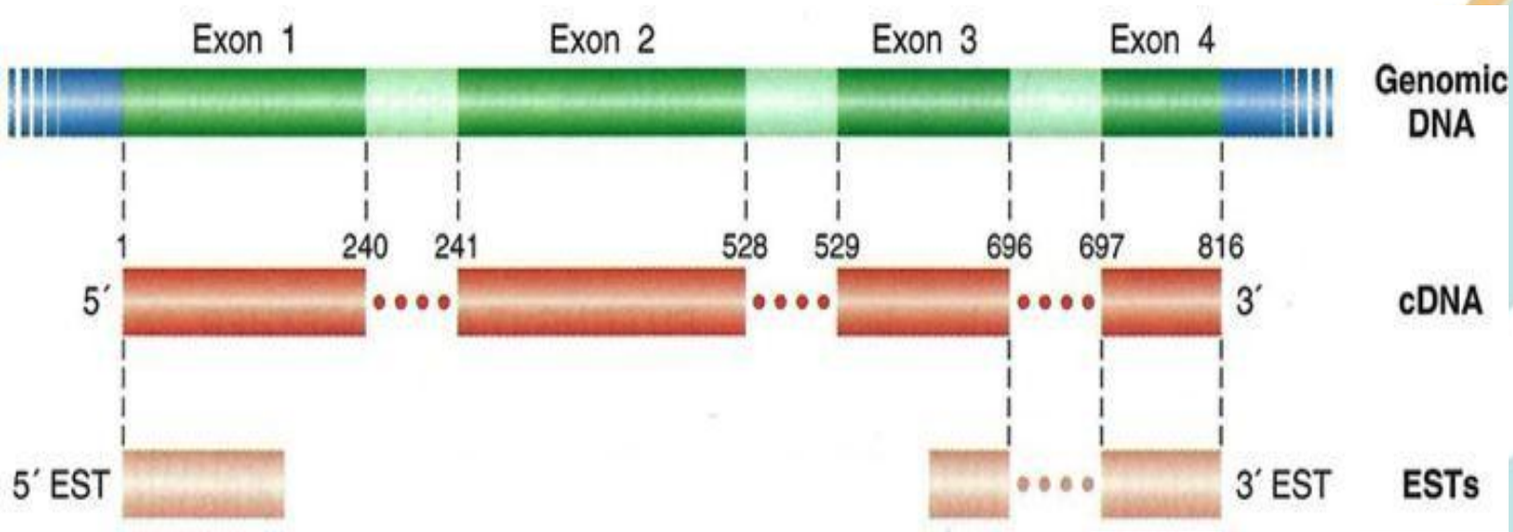
Nucleotide Sequence databases

Nucleotide Sequence Databases store DNA and cDNA or EST sequences.

Amina Zee

Nucleotide Sequence Databases

Nucleotide Sequence databases



Amina Zee

Nucleotide Sequence Databases

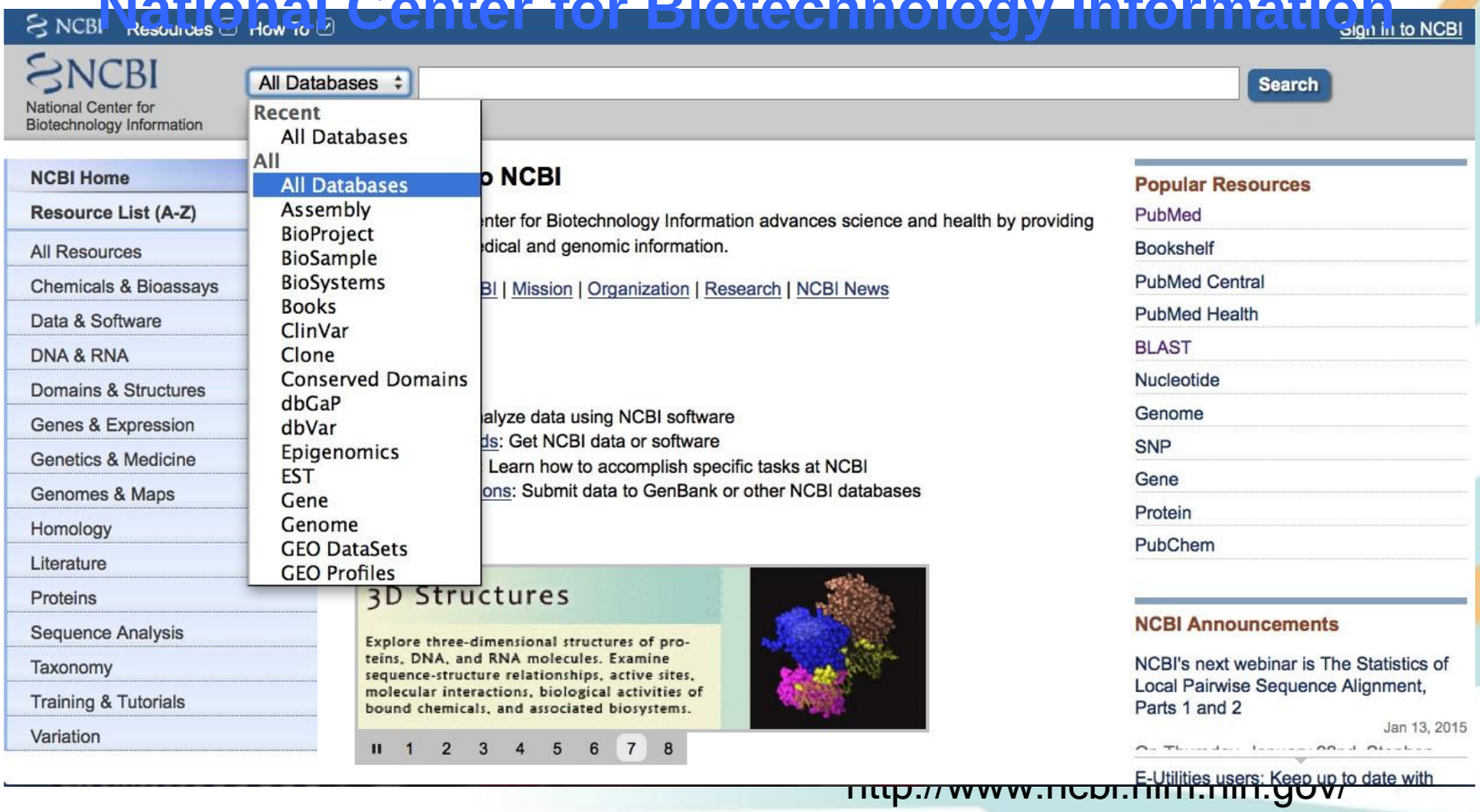
Origin

- First assembled into Genebank (1982) at Los Alamos National Laboratory (LANL), New Maxico by Walter Goad et al
- **GeneBank** is now under NCBI

Amina Zee

Nucleotide Sequence Databases

National Center for Biotechnology Information

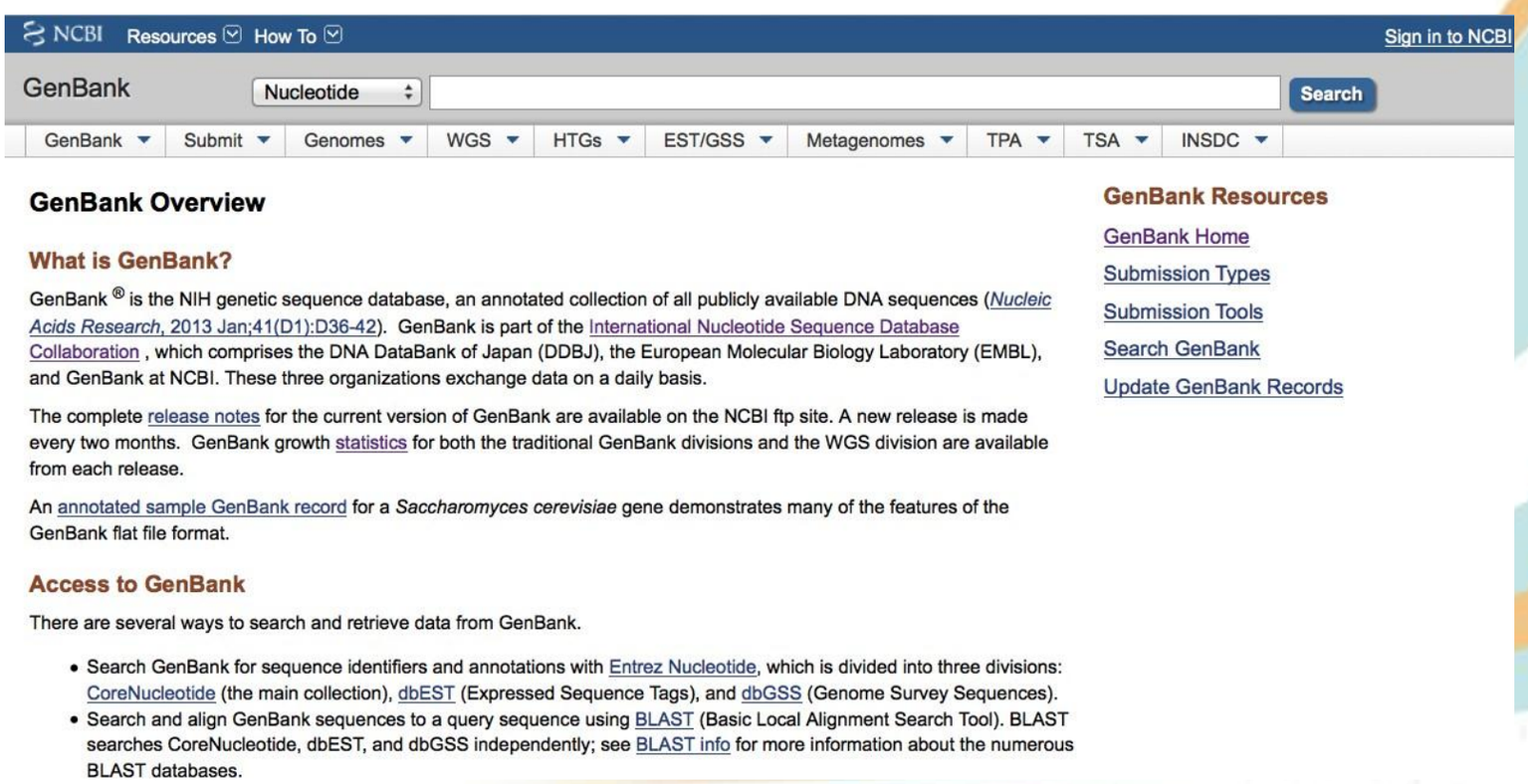


http://www.ncbi.nlm.nih.gov/

Amina Zee

Nucleotide Sequence Databases

GeneBank



http://www.ncbi.nlm.nih.gov/genbank/

Amina Zee

Nucleotide Sequence Databases

EMBL and DDBJ

- European Molecular biology Lab (**EMBL** established1980)
- DNA databank of Japan **(DDBJ)** established in Mishima japan (1984)

Amina Zee

Nucleotide Sequence Databases

INSDC

- Genebank, DDBJ and EMBL joined together in **I**nternational **N**ucleotide **S**equence **D**atabase **C**ollaboration (INSDC)

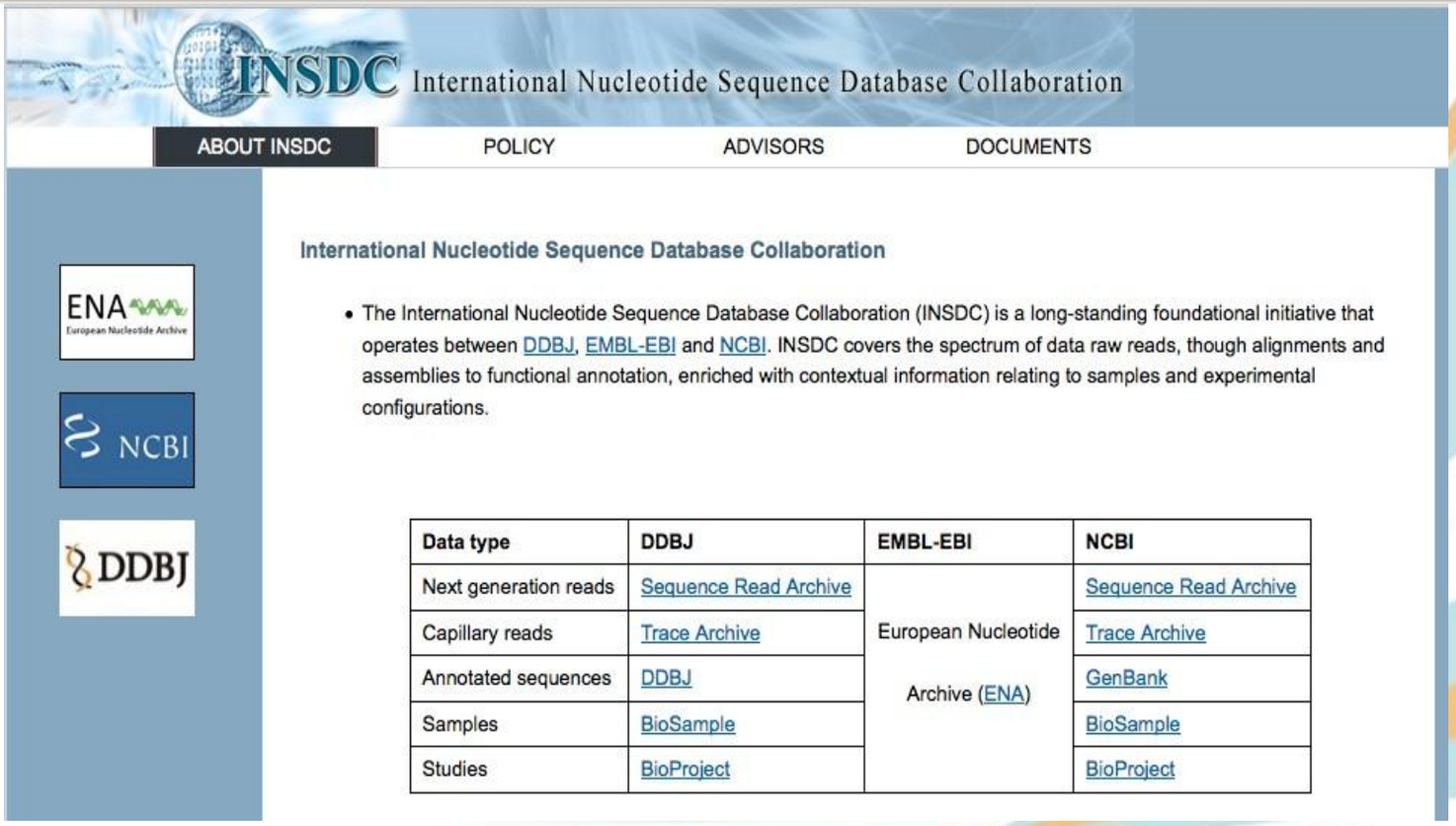
Amina Zee

Nucleotide Sequence Databases

INSDC

Amina Zee

Nucleotide Sequence Databases



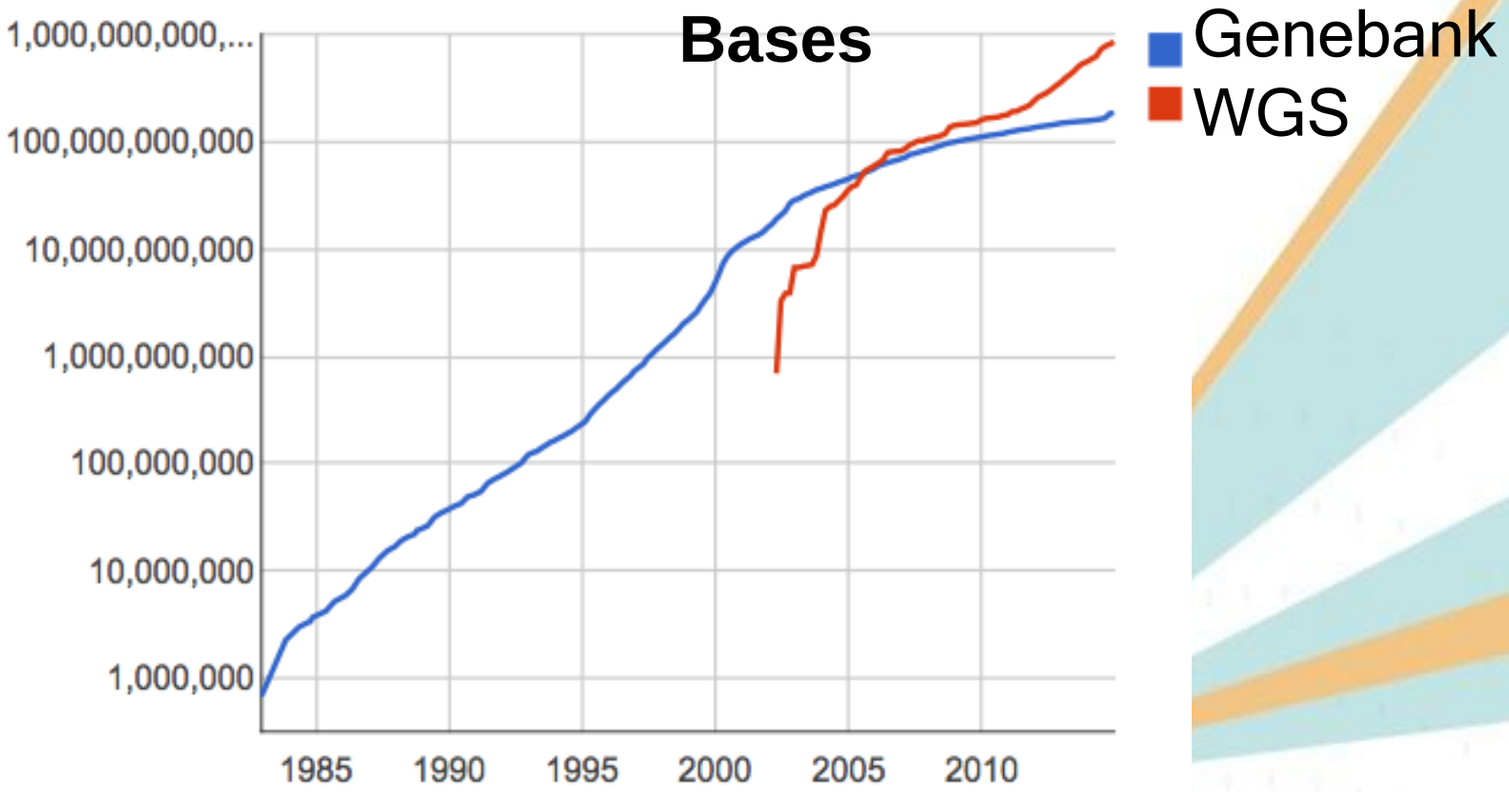
Data type	DDBJ	EMBL-EBI	NCBI
Next generation reads	Sequence Read Archive	European Nucleotide Archive (ENA)	Sequence Read Archive
Capillary reads	Trace Archive		Trace Archive
Annotated sequences	DDBJ		GenBank
Samples	BioSample		BioSample
Studies	BioProject		BioProject

http://www.insdc.org/

Amina Zee

Nucleotide Sequence Databases

Growth of Genebank

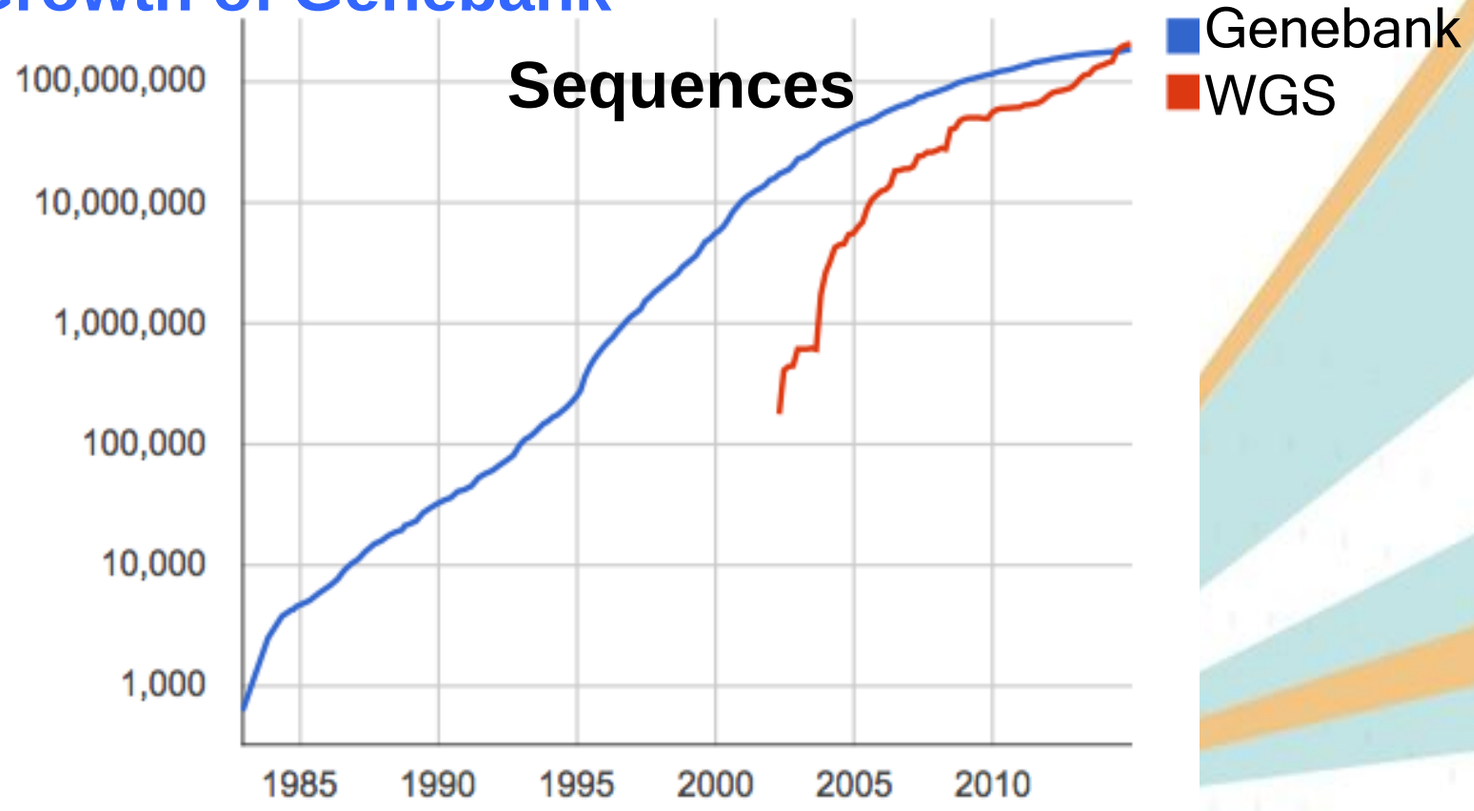


http://www.ncbi.nlm.nih.gov/genbank/statistics

Amina Zee

Nucleotide Sequence Databases

Growth of Genebank



http://www.ncbi.nlm.nih.gov/genbank/statistics

Amina Zee

Nucleotide Sequence Databases

Conclusions

- Biological databases store biological data
- **INSDC** is joint venture of NCBI, EMBL and DDBJ
- Growth of bases in Genebank is exponential, doubbling every 18 months

Amina Zee

Lec 96

Protein databases

Amina Zee

Protein Databases

Introduction

Protein databases store

- **Protein sequences**
- **Motif**
- **Structure**
- **Structure alignments**

Amina Zee

Protein Databases

Origin

First sequences to be collected were Proteins using **Sanger and Tupy's** methods (1951) where Common protein families like cytochromes were sequenced

Amina Zee

Protein Databases

Origin

Atlas of protein sequences (mainly cytochromes) was assembled by Margret Dayhoff and her collaborators at **N**ational **B**iomedical **R**esearch **F**oundation (NBRF) in 1960s

Amina Zee

Protein Databases

PIR ((Protein Information Resource)

The collection of Dayhoff and co became **PIR**

which is now a collaboration of **NBRF**, **M**unich **C**enter for **P**rotein **S**equences (**MIPS**) and **J**apan **I**nternational **P**rotein **I**nformation **D**atabase (**JIPID**)

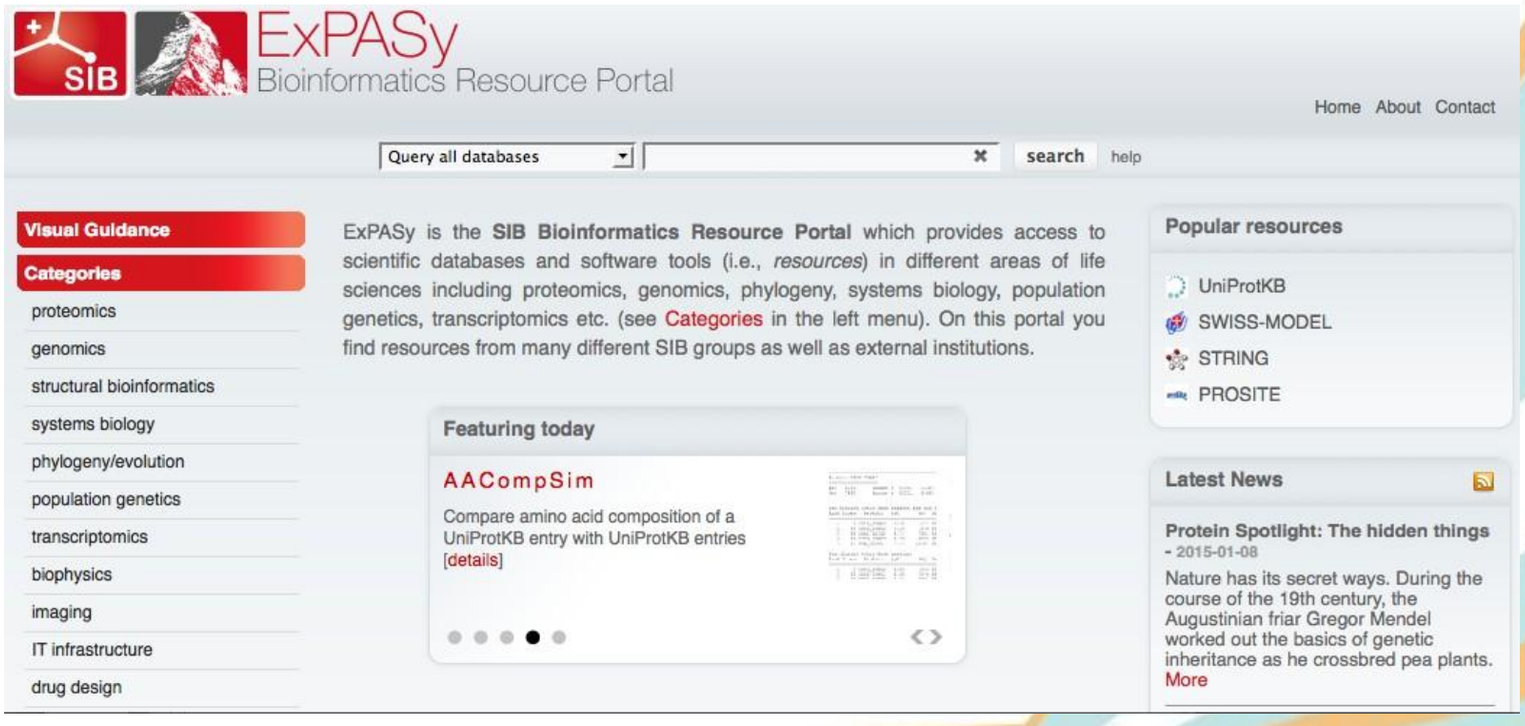
Amina Zee

Protein Databases

Protein Sequences

- **Swiss-Prot** is Collaboration between the SIB and EBI
- Weekly releases from about 50 servers across the world, the main source being ExPASy in Geneva

Protein Databases



http://www.expasy.org/

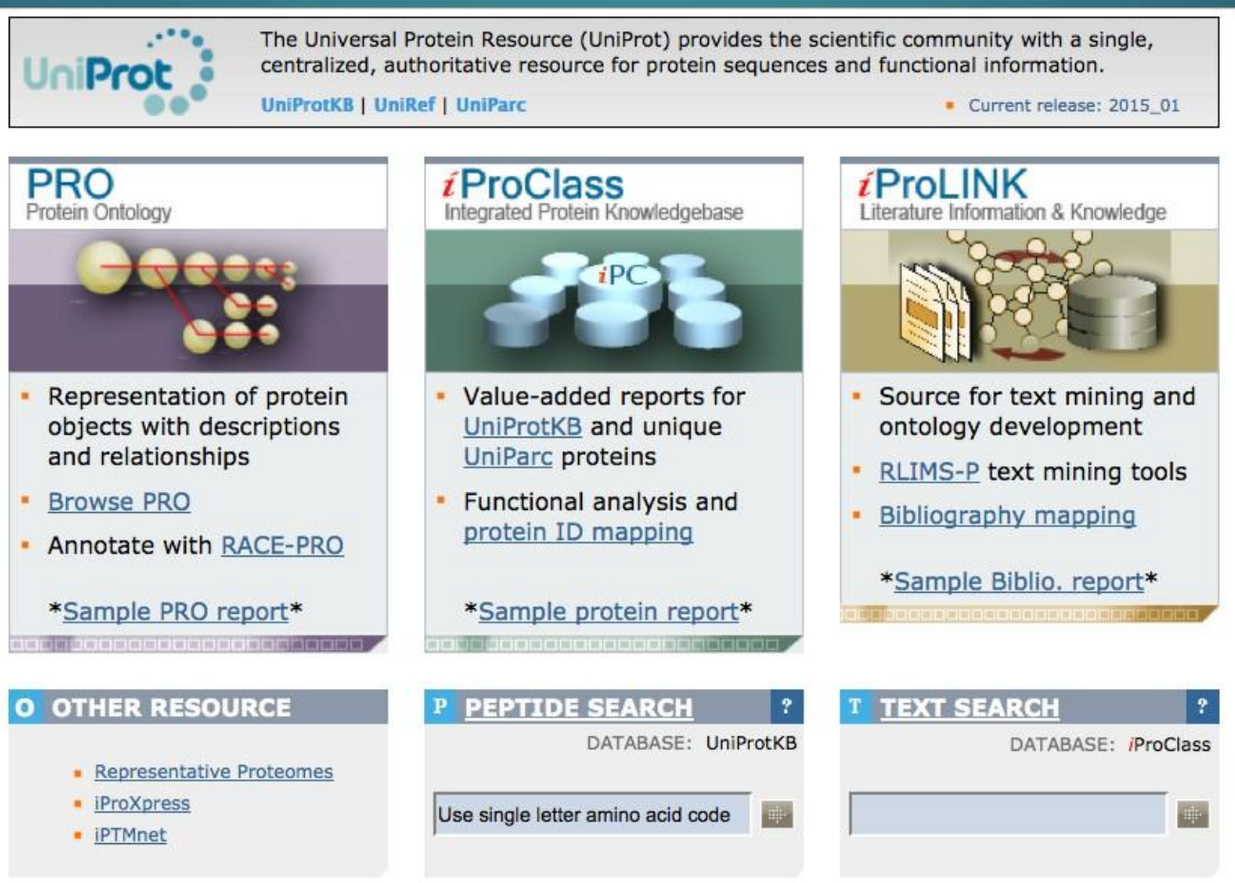
Amina Zee

Protein Databases

Protein Sequences

- International partnership between PIR, EBI and SIB
- Created **UniProt**, by unifying the PIR-PSD, Swiss-Prot, and TrEMBL databases

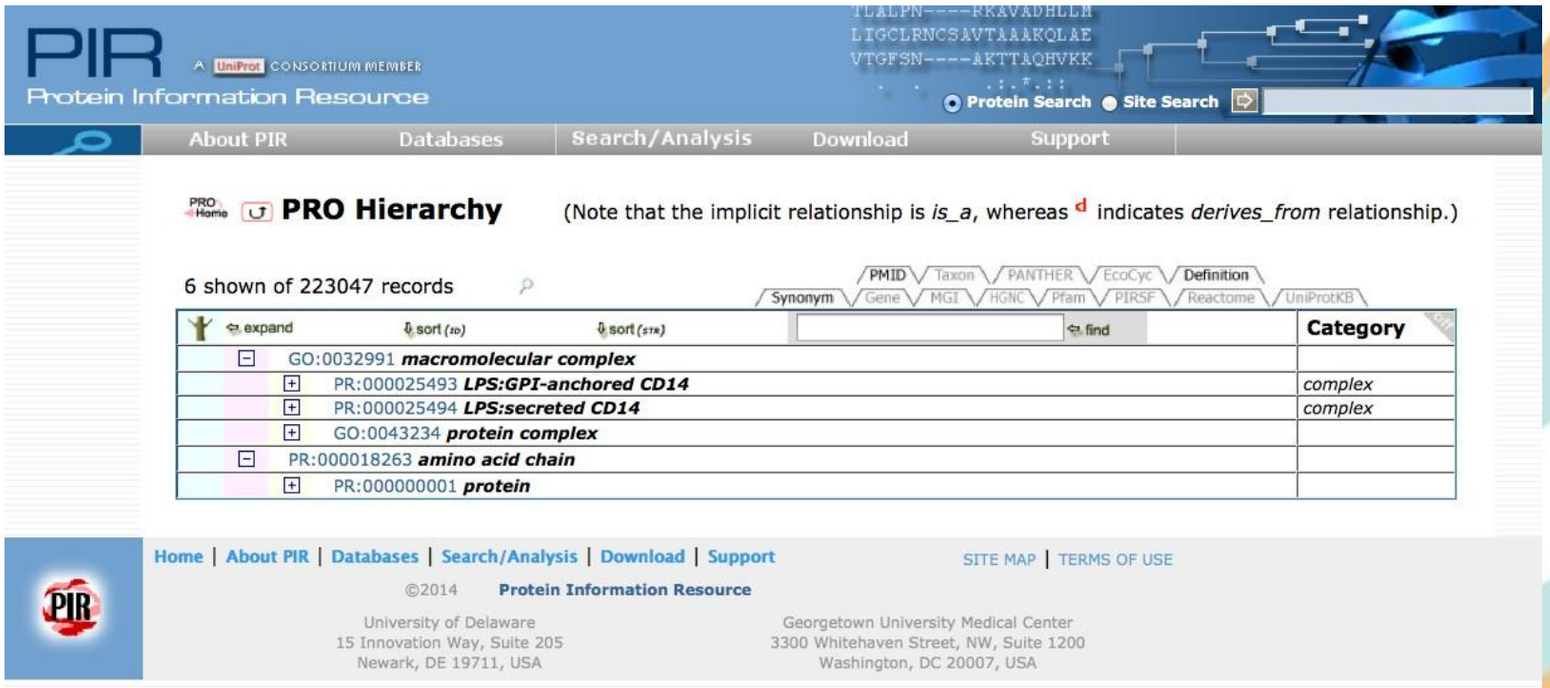
Protein Databases



http://www.uniprot.org/

Amina Zee

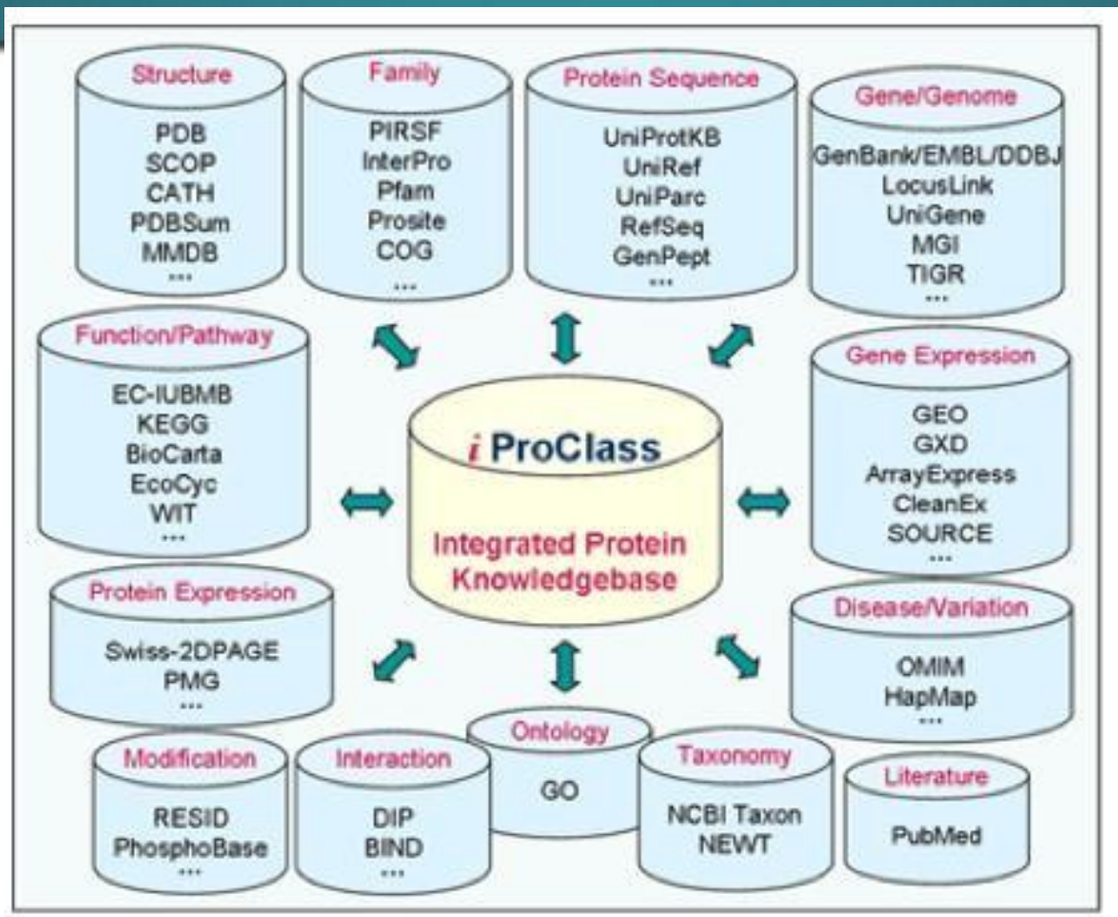
Protein Databases



http://pir.georgetown.edu/

Amina Zee

Protein Databases



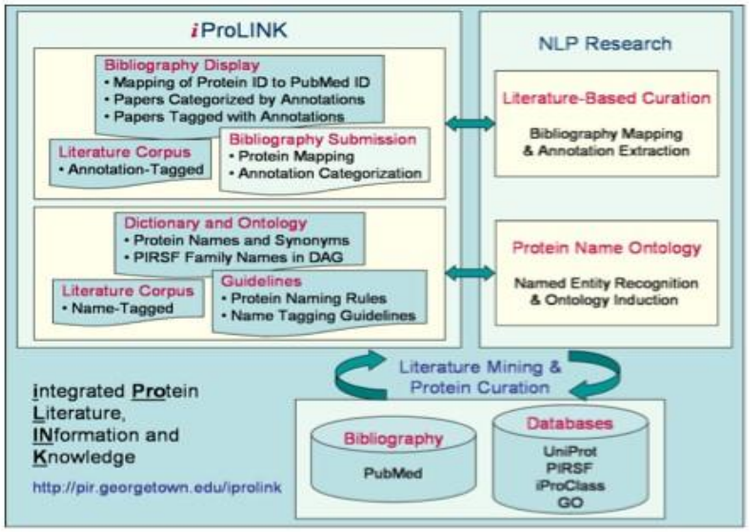
http://www.uniprot.org/

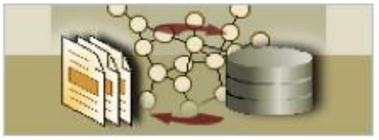
Amina Zee

Protein Databases

iProLINK Tutorial

*i*ProLINK (*i*ntegrated **P**rotein **L**iterature, **IN**formation and **K**nowledge) has been developed as a resource to facilitate text mining in the area of literature-based database curation, named entity recognition, and protein ontology development. The collection of data sources can be utilized by computational and biological researchers to explore literature information on proteins and their features or properties (Hu *et al.*, 2004).





Bibliography Mapping/ Annotation Extraction

- Bibliography Display/Submission
- Annotation-Tagged Corpora
- RLIMS-P Text Mining Tool
- eFIP Text Mining Tool
- eGIFT Text Mining Tool
- iSimp Sentence Simplification System

Entity Recognition/ Ontology Development

- Name Tagging Guidelines/Corpora
- Protein Ontology Development

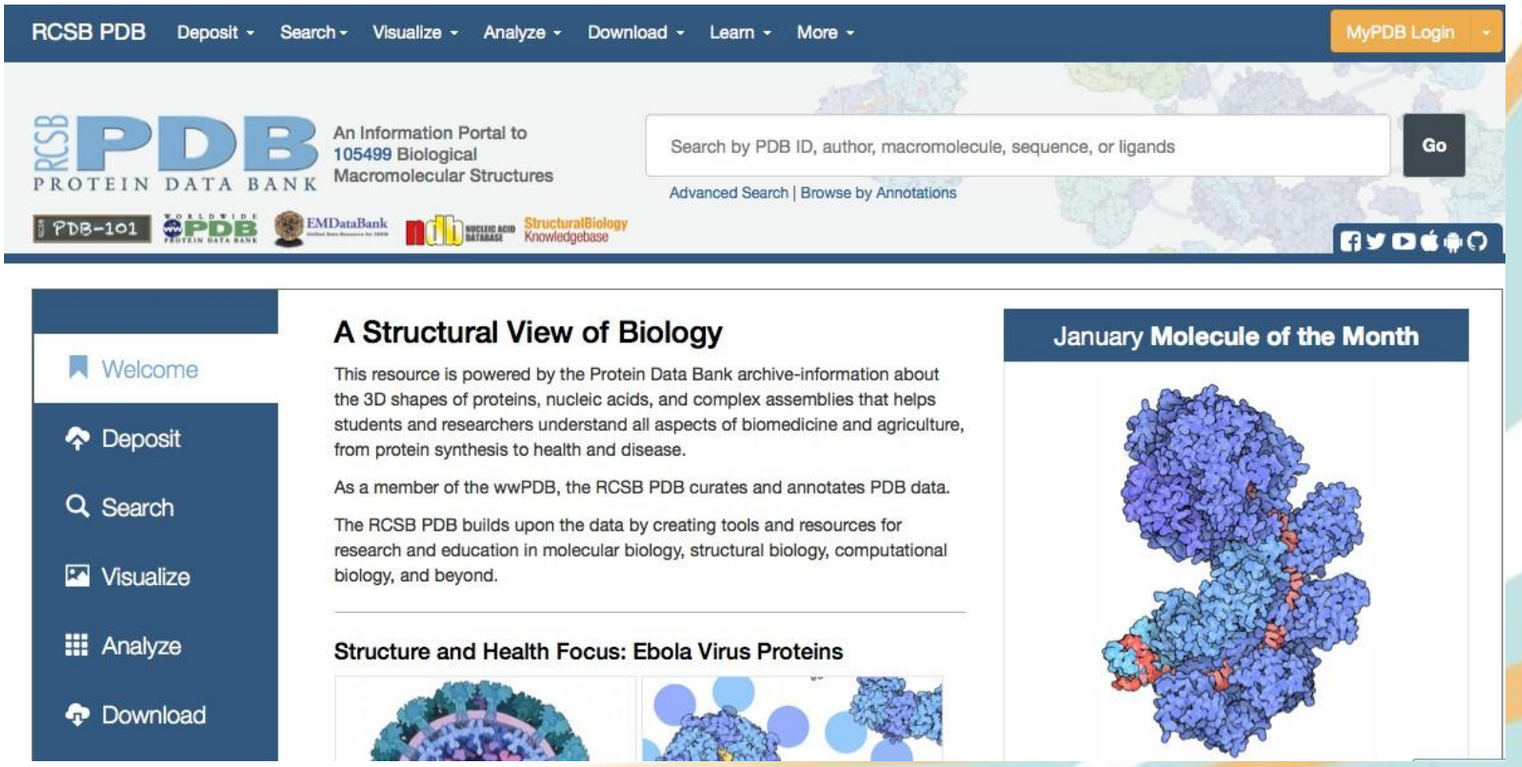
Collaborators

***i*ProLINK Paper**

http://www.uniprot.org/

Amina Zee

Protein Databases



http://www.rcsb.org/pdb/home/home.do

Amina Zee

Protein Databases

SCOPe: Structural Classification of Proteins — extended. Release 2.04 (July 2014, new entries added 2014-12-18)

Browse | Stats & History | ASTRAL Subsets | Downloads | Related Resources | References | Help | About

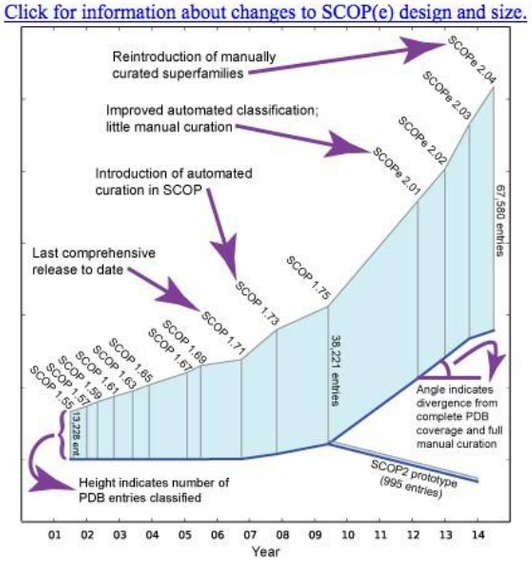
Welcome to SCOPe!
SCOPe is a database developed at the Berkeley Lab and UC Berkeley to extend the development and maintenance of SCOP.
SCOP was conceived at the MRC Laboratory of Molecular Biology, and developed in collaboration with researchers in Berkeley.
Work on SCOP (version 1) concluded in June 2009 with the release of SCOP 1.75.
SCOPe classifies many newer structures through a combination of automation and manual curation, and corrects some errors in SCOP,
aiming to have the same accuracy as the hand-curated SCOP releases. **SCOPe** also incorporates and updates the ASTRAL database.
For prior releases, click on the Stats & History tab above. For more info, click on the About tab above.
New PDB entries were last added on **2014-12-18**; for more info on periodic updates click on the Help tab above.

Search **SCOPe** (example): Search

Classes in SCOPe 2.04:

1. a: All alpha proteins [46456] (285 folds)
2. b: All beta proteins [48724] (176 folds)
3. c: Alpha and beta proteins (a/b) [51349] (148 folds)
4. d: Alpha and beta proteins (a+b) [53931] (380 folds)
5. e: Multi-domain proteins (alpha and beta) [56572] (68 folds)
6. f: Membrane and cell surface proteins and peptides [56835] (57 folds)
7. g: Small proteins [56992] (91 folds)
8. h: Coiled coil proteins [57942] (7 folds)
9. i: Low resolution protein structures [58117] (25 folds)

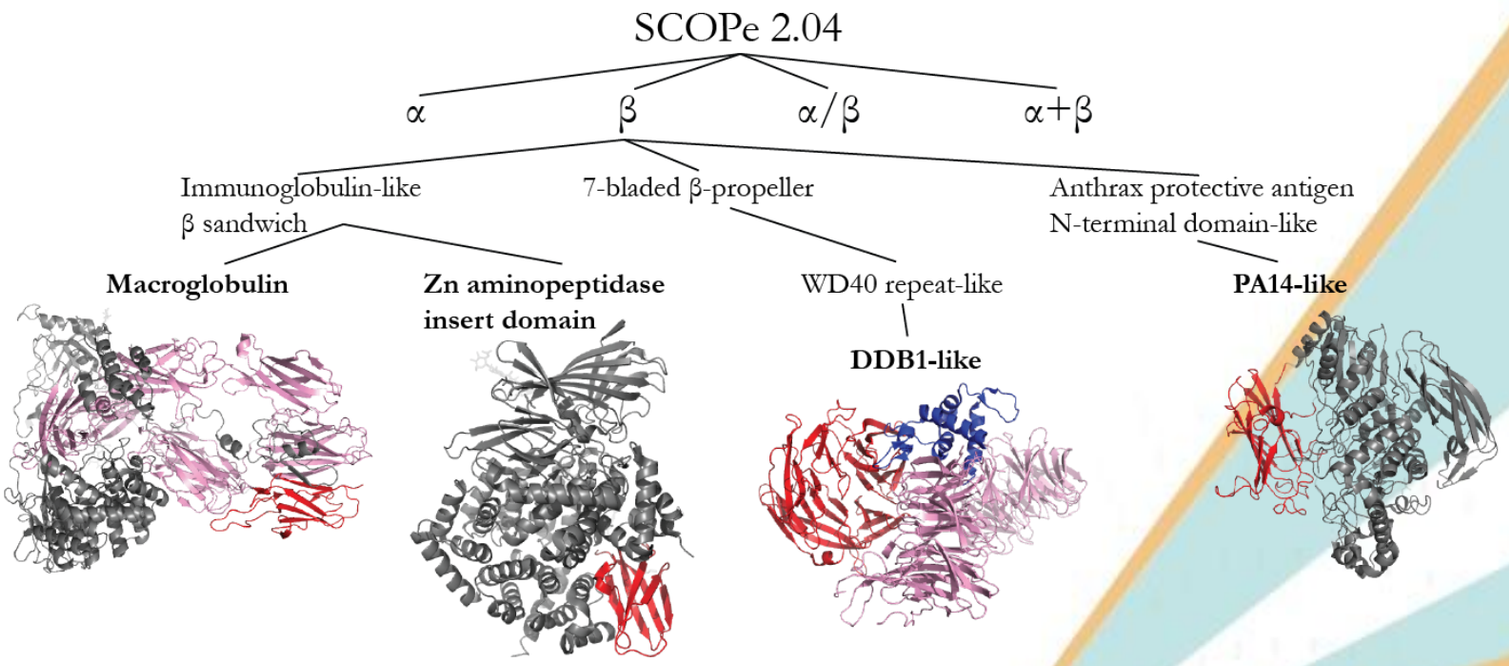
Click for information about changes to SCOP(e) design and size.



http://scop.mrc-lmb.cam.ac.uk/scop/

Amina Zee

Protein Databases



http://scop.mrc-lmb.cam.ac.uk/scop/

Amina Zee

Protein Databases

Conclusions

- First sequences to be collected were Protein sequences
- Protein databases are classified on the basis of sequences, motifs and structures
- Growth of Sequence in Databases is exponential

Amina Zee

Lec 97

Genome and organism specific databases

Amina Zee

Genome & Organism Specific Databases

Origin

- First attempt to sequence free living organism was launched in late 1990's
  - (Blattner et al. 1997)
- Viruses had already been sequenced

Genome & Organism Specific Databases

Origin

- ***Haemophilus influenzae*** was the first published genome
  - (Fleischmann et al. 1995)
- The project initiated at **T**he **I**nstitute of **G**enome **R**esearch **(TIGR)** under leadership of **Craig Ventor**

Amina Zee

Genome & Organism Specific Databases

Origin

- Use of **shotgun sequencing method** speeded up the process
- 1.8 million bp
- Took 9-months and cost was 1 million
- Paved the way for sequencing of many other organisms

Amina Zee

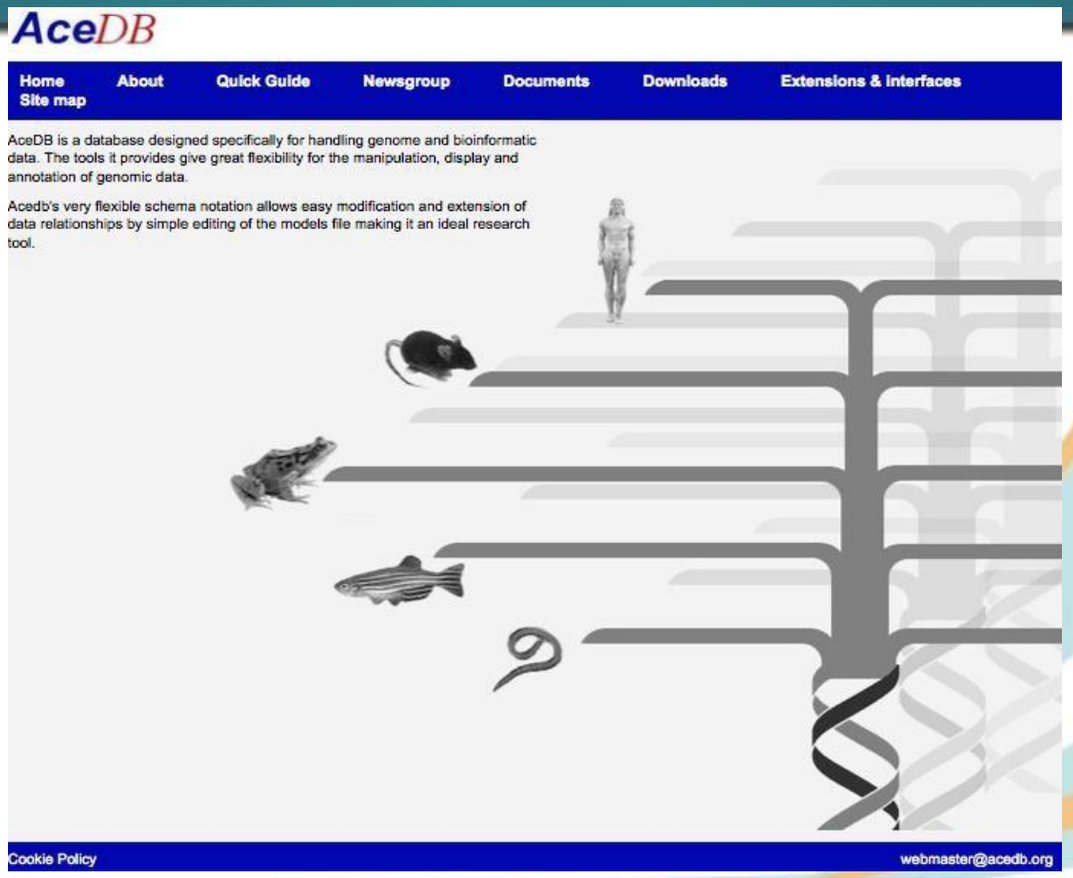
Genome & Organism Specific Databases

Examples

- **AceDB** (**A** ***C. e****legans* **D**ata**B**ase) was the first genome database developed in 1989
  - by **Richard durbin** and **Thierry-Miegi**

Amina Zee

Genome & Organism Specific Databases



http://www.acedb.org/

Amina Zee

Genome & Organism Specific Databases

Examples

- **TAIR** (**T**he **A**rabidopsis **I**nformation **R**esource) http://www.arabidopsis.org/ and **SGB** (**S**accharomyces **G**enome **D**atabase) http://www.yeastgenome.org/ utilized **AceDB** system

Amina Zee

Genome & Organism Specific Databases

Human Genome Project

- Pilot project, the **Human Genome Initiative** begun by **D**epartment **O**f **E**nergy (**DOE**) in 1986

Amina Zee

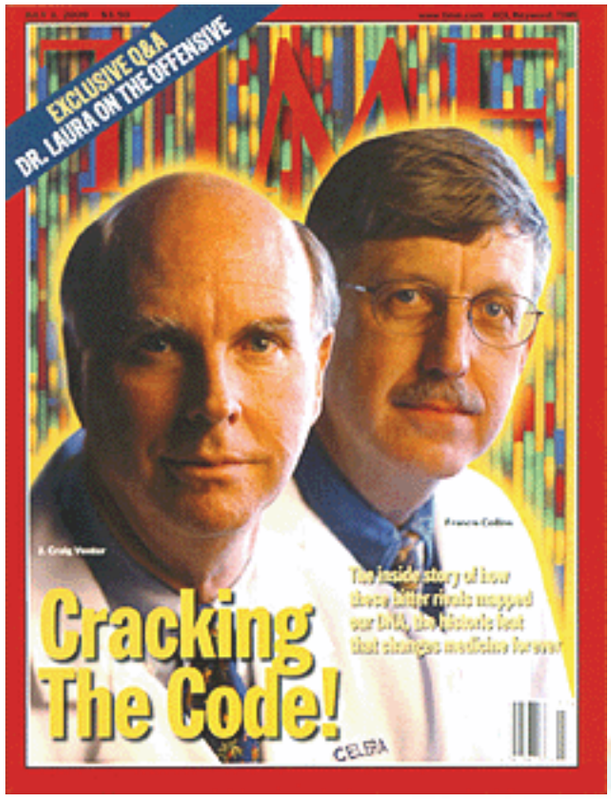
Genome & Organism Specific Databases

Human Genome Project

- **N**ational **H**uman **G**enome **R**esearch **I**nitiative (**NHGRI**), a federally funded organization (NIH) started in 1988 **(Francis Collin)**
- Celera, a commercial vendor **(Craig Venter)** joined the venture in 1998

Amina Zee

Genome & Organism Specific Databases

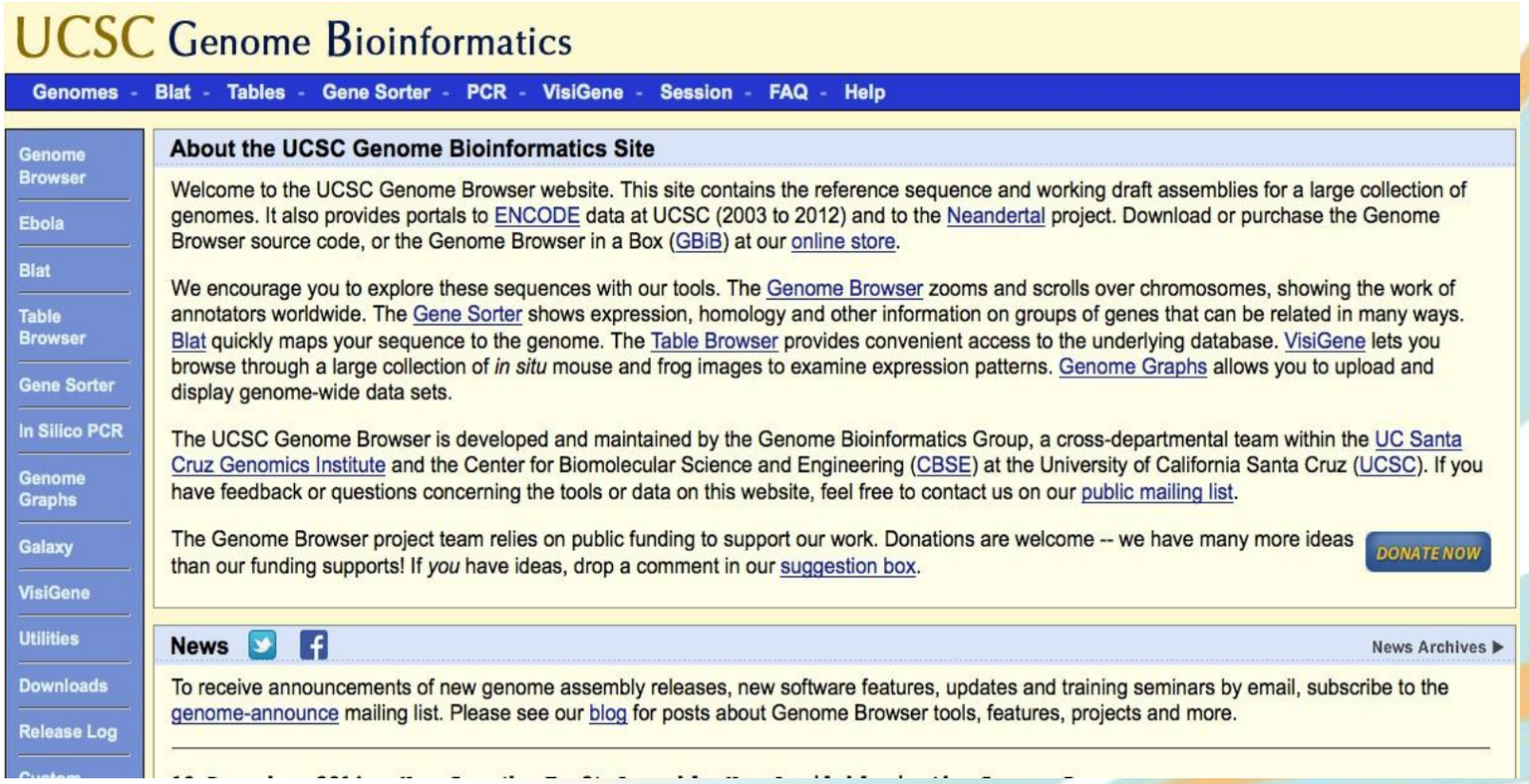


Human Genome Project

- Both simultaneously announced the completion of the project in 2000
- Total 3.4 billion bases sequenced at a cost of $1/base

Amina Zee

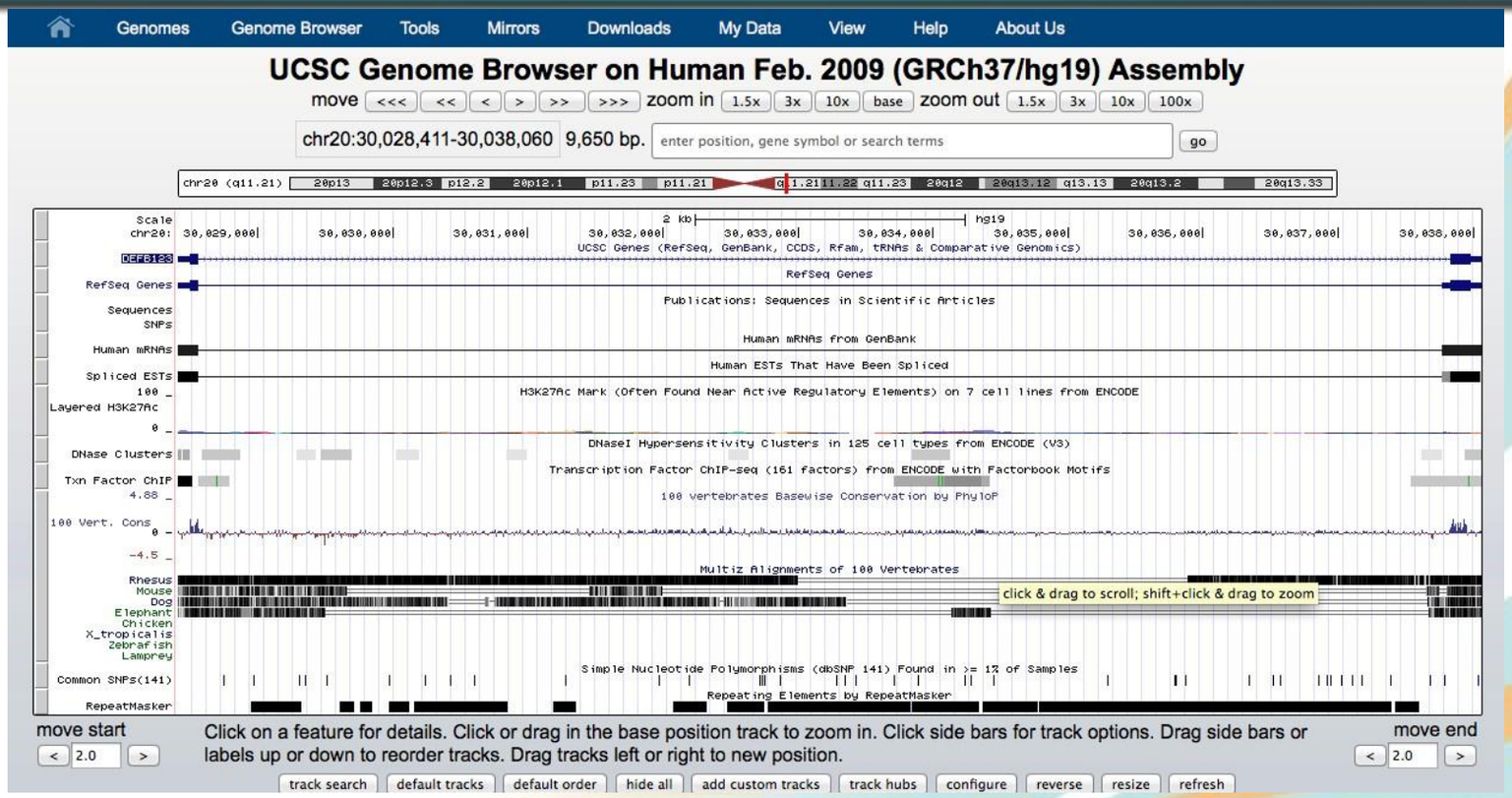
Genome & Organism Specific Databases



http://genome.ucsc.edu/

Amina Zee

Genome & Organism Specific Databases



http://genome.ucsc.edu/

Amina Zee

Genome & Organism Specific Databases

Conclusion

- Success of ***Haemophilus influenzae*** paved the way for other genome sequencing projects
- Human Genome Project was accomplished by NHGRI and Celera
- Genome browsers help viewing Genome features

Amina Zee

Lec 98

Gene expression databases

Amina Zee

Gene Expression Databases

Gene Expression Omnibus (GEO)

- A public repository for the archiving and distribution of gene expression data submitted by the scientific community

Amina Zee

Gene Expression Databases

Gene Expression Omnibus (GEO)

- MIAME compliant data
  - Minimum Information About a Microarray Experiment http://www.mged.org/Workgroups/MIAME/miame.html

Amina Zee

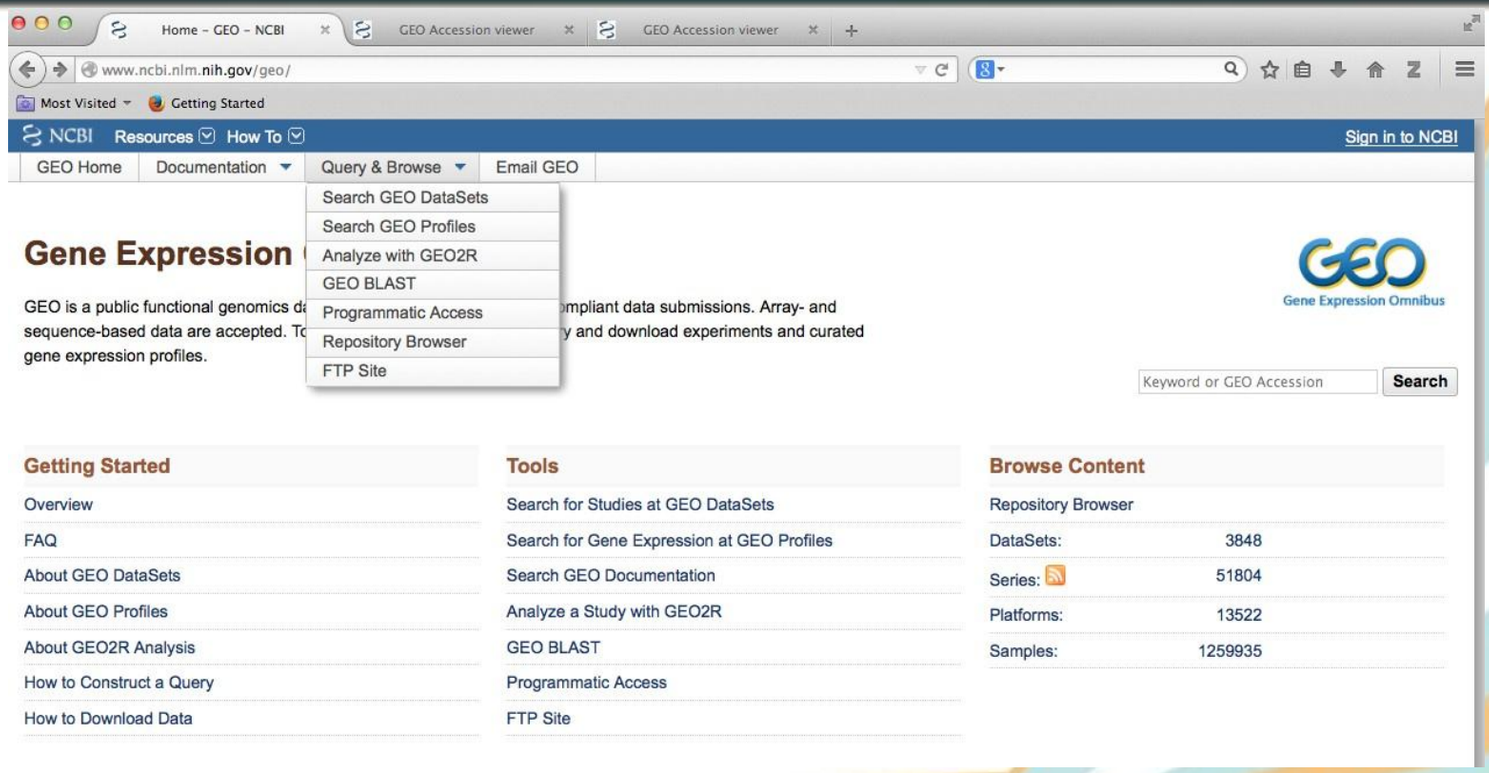
Gene Expression Databases

Gene Expression Omnibus (GEO)

- Convenient for deposition of gene expression data, as required by funding agencies and journals
- Curated resource for gene expression data
- Browsing, querying, analysis and retrieval

Amina Zee

Gene Expression Databases



http://www.ncbi.nlm.nih.gov/geo/

Amina Zee

Gene Expression Databases

Gene Architecture

GEO has four kinds of records

- **Sample (GSM)**
  preparation and description of the samples
- **Platform (GPL)**
  technology used and the features detected
  – microarray or RNASeq

Amina Zee

Gene Expression Databases

Gene Architecture

GEO has four kinds of records

- **Series (GSE)**

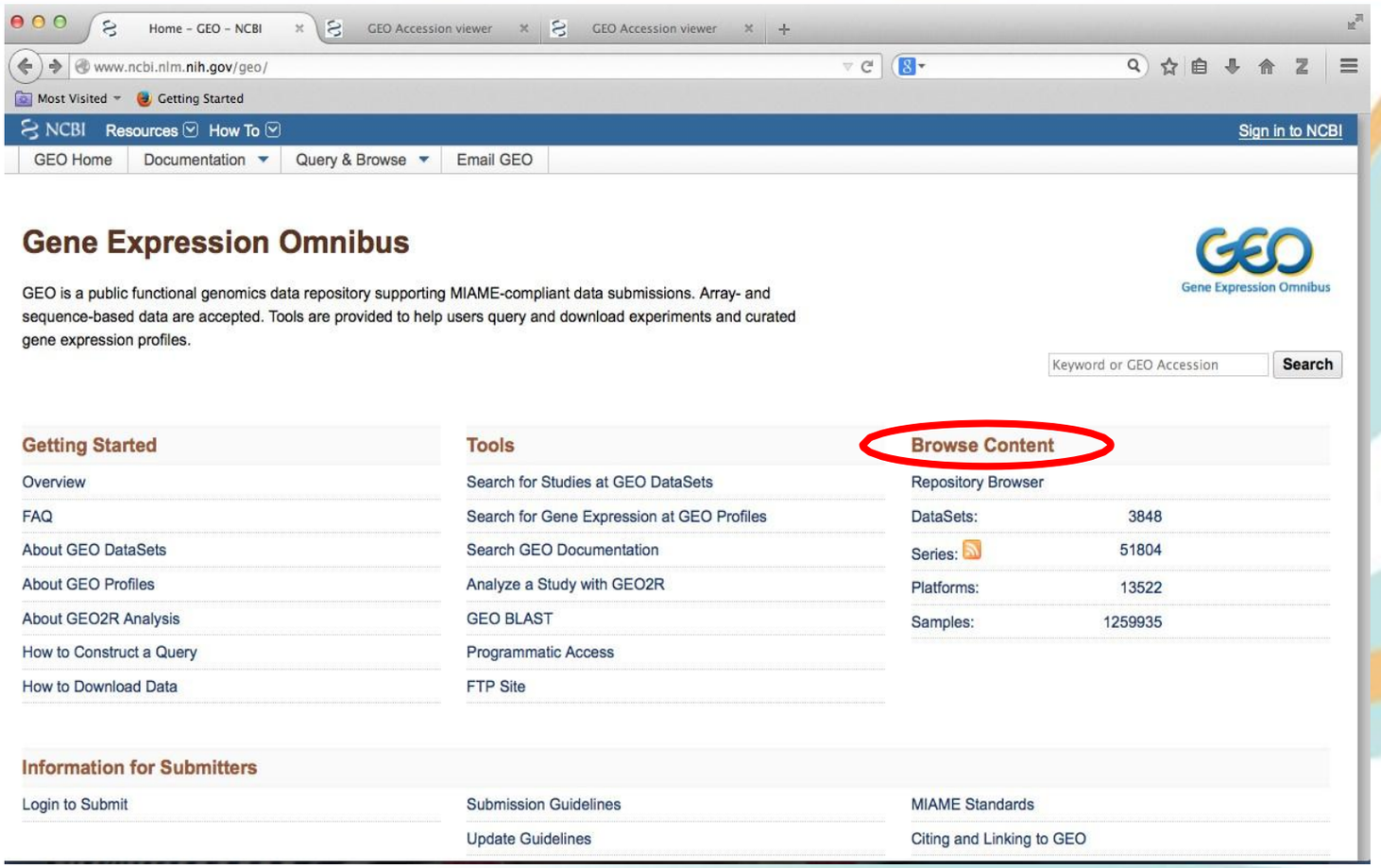
  defines a set of samples and how they are related

- **Datasets (GDS)**

  sample data collections assembled by GEO

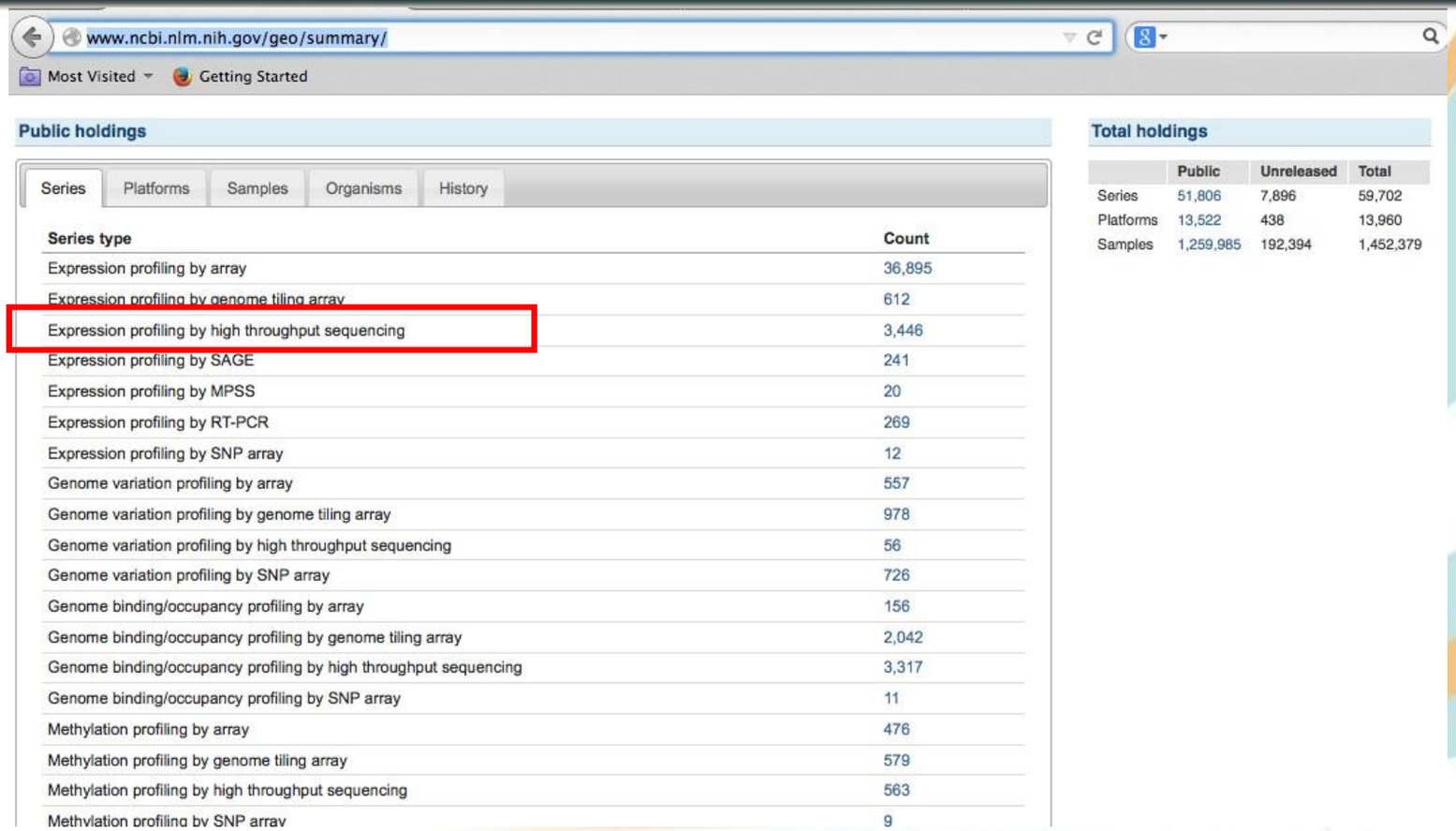
Amina Zee

Gene Expression Databases



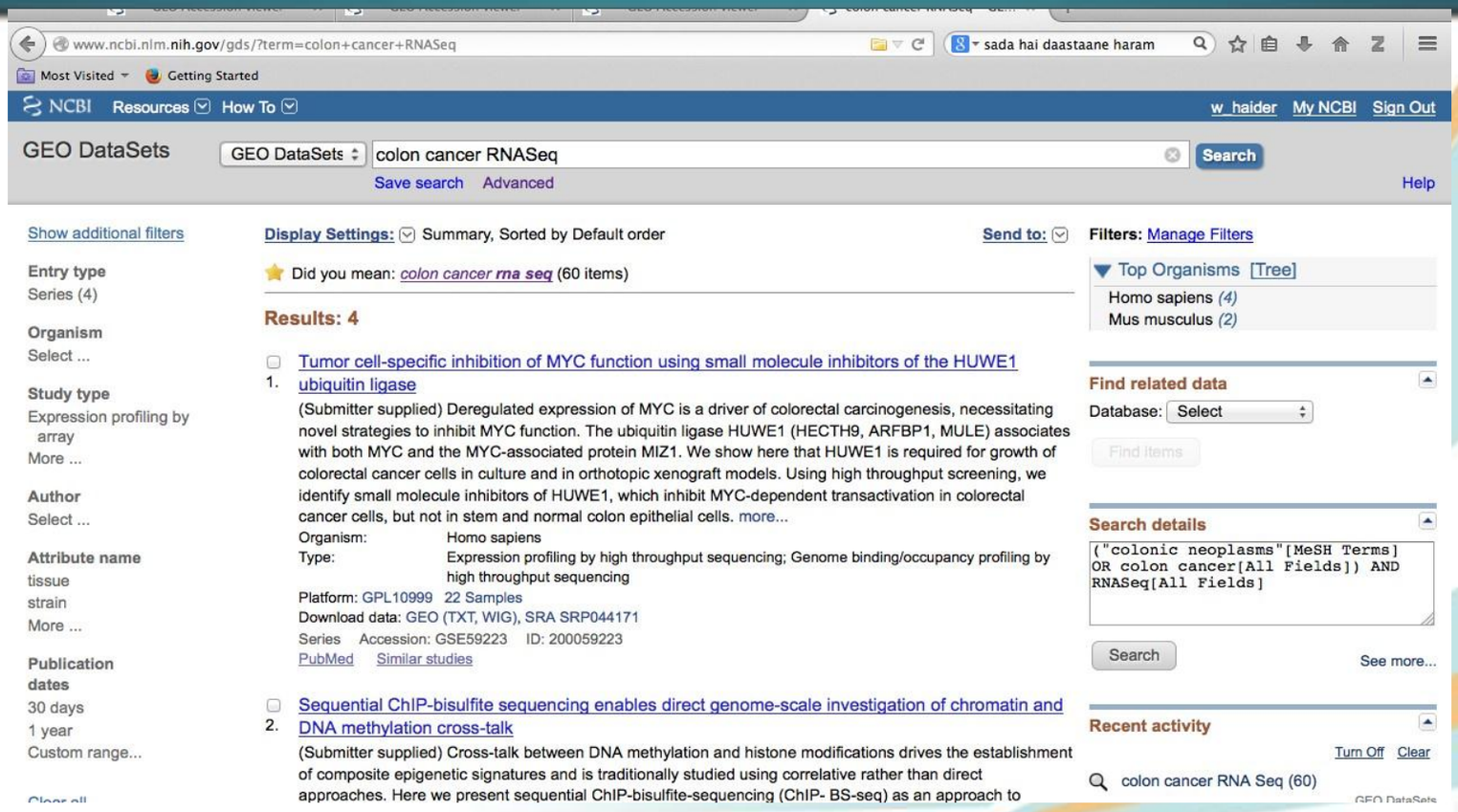
Amina Zee

Gene Expression Databases



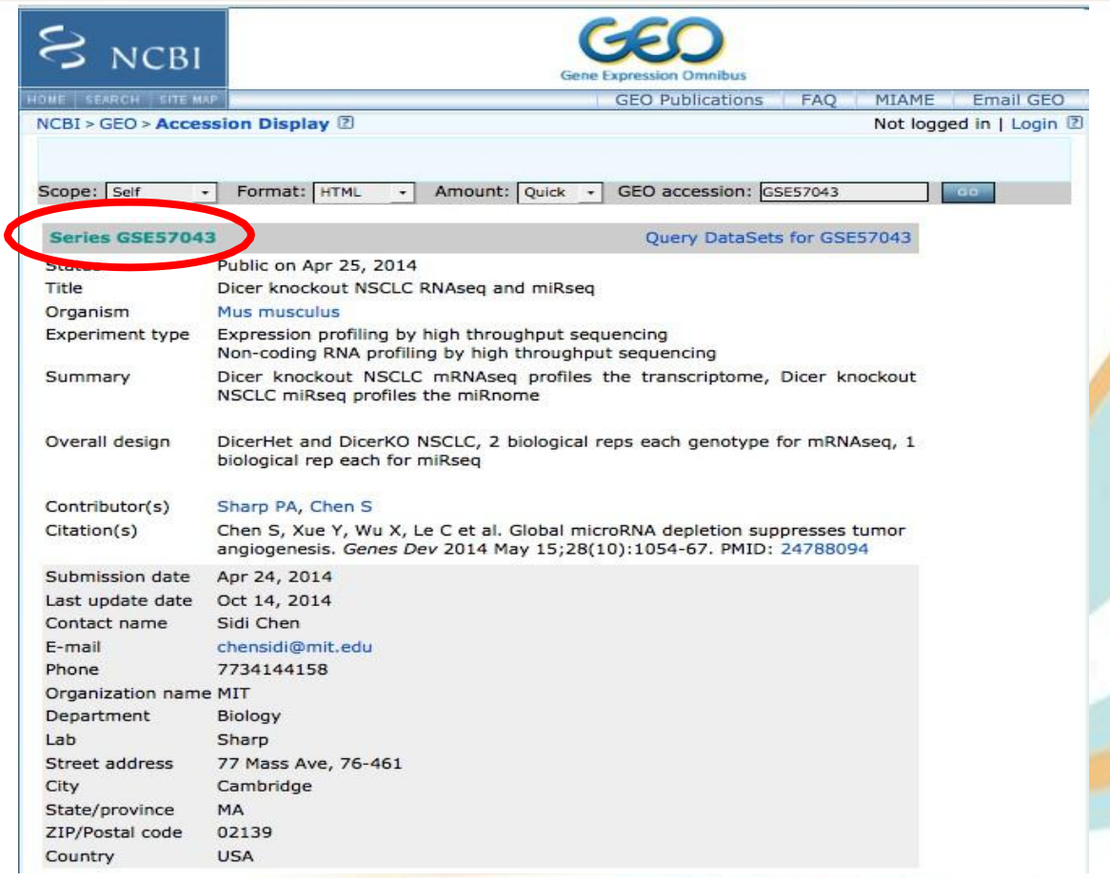
Amina Zee

Gene Expression Databases



Amina Zee

Gene Expression Databases



Data is submitted to GEO as a Series, which represents the experiment design

Amina Zee

Gene Expression Databases

Platforms (1)	GPL13112 Illumina HiSeq 2000 (Mus musculus)
Samples (6)	GSM1373652 DcrHet_1_mRNA
Less...	GSM1373653 DcrHet_2_mRNA
	GSM1373654 DcrKO_1_mRNA
	GSM1373655 DcrKO_2_mRNA
	GSM1373656 DcrHet_1_miR
	GSM1373657 DcrKO_1_miR

Individual samples in a series

All GEO submissions need to be associated with a platform file.

Relations

BioProject	PRJNA245291
SRA	SRP041414

Download family	Format
SOFT formatted family file(s)	SOFT
MINiML formatted family file(s)	MINiML
Series Matrix File(s)	TXT

Click ? One by one

Supplementary file	Size	Download	File type/resource
GSE57043_dicerko_fpkm.txt.gz	875.5 Kb	(ftp)(http)	TXT
GSE57043_dicerko_hairpin_rpm.txt.gz	4.1 Kb	(ftp)(http)	TXT
GSE57043_dicerko_mature_rpm.txt.gz	1.6 Kb	(ftp)(http)	TXT
SRP/SRP041/SRP041414		(ftp)	SRA Study

Raw data provided as supplementary file

Processed data is available on Series record

Normalized counts

Raw Reads

Amina Zee

Gene Expression Databases

PNAS

An expression signature for p53 status in human breast cancer predicts mutation status, transcriptional effects, and patient survival

Lance D. Miller*†, Johanna Smeds‡, Joshy George*, Vinsensius B. Vega*, Liza Vergara*, Alexander Ploner§, Yudi Pawitan§, Per Hall§, Sigrid Klaar‡, Edison T. Liu*†, and Jonas Bergh‡

*Genome Institute of Singapore, 60 Biopolis Street, #02-01, Singapore 138672; ‡Department of Oncology and Pathology, Radiumhemmet, Karolinska Institute and Hospital, S-17176 Stockholm, Sweden; and §Department of Medical Epidemiology and Biostatistics, Karolinska Institutet, 17177 Stockholm, Sweden

GENETICS. For the article "An expression signature for p53 status in human breast cancer predicts mutation status, transcriptional effects, and patient survival," by Lance D. Miller, Johanna Smeds, Joshy George, Vinsensius B. Vega, Liza Vergara, Alexander Ploner, Yudi Pawitan, Per Hall, Sigrid Klaar, Edison T. Liu, and Jonas Bergh, which appeared in issue 38, September 20, 2005, of *Proc. Natl. Acad. Sci. USA* (**102,** 13550-13555; first published September 2, 2005; 10.1073/pnas.0506230102), the breast cancer microarray data discussed in this publication have been deposited in the National Center for Biotechnology Information's Gene Expression Omnibus database (GEO, www.ncbi.nlm.nih.gov/geo/) and are accessible through GEO Series accession no. GSE3494 [NCBI GEO] .

Amina Zee

Gene Expression Databases

Conclusion

- **GEO** is a public repository for the archiving and distribution of gene expression data
- Best resource to get microarray and Next Generation Sequencing (RNASeq) data

Amina Zee

Lec 99

Medical databases

Amina Zee

Medical Databases

Introduction

Informatics in health care may be called as health informatics

- It deals with the resources, devices, and methods required to optimize the acquisition, storage, retrieval, and use of information in health and biomedicine.

(wiki)

Amina Zee

Medical Databases

Introduction

Medical databases store and provide medical information

- The premier database for biomedical literature is the National Library of Medicine (NLM)'s MEDLINE, accessible through **PubMed**

Amina Zee

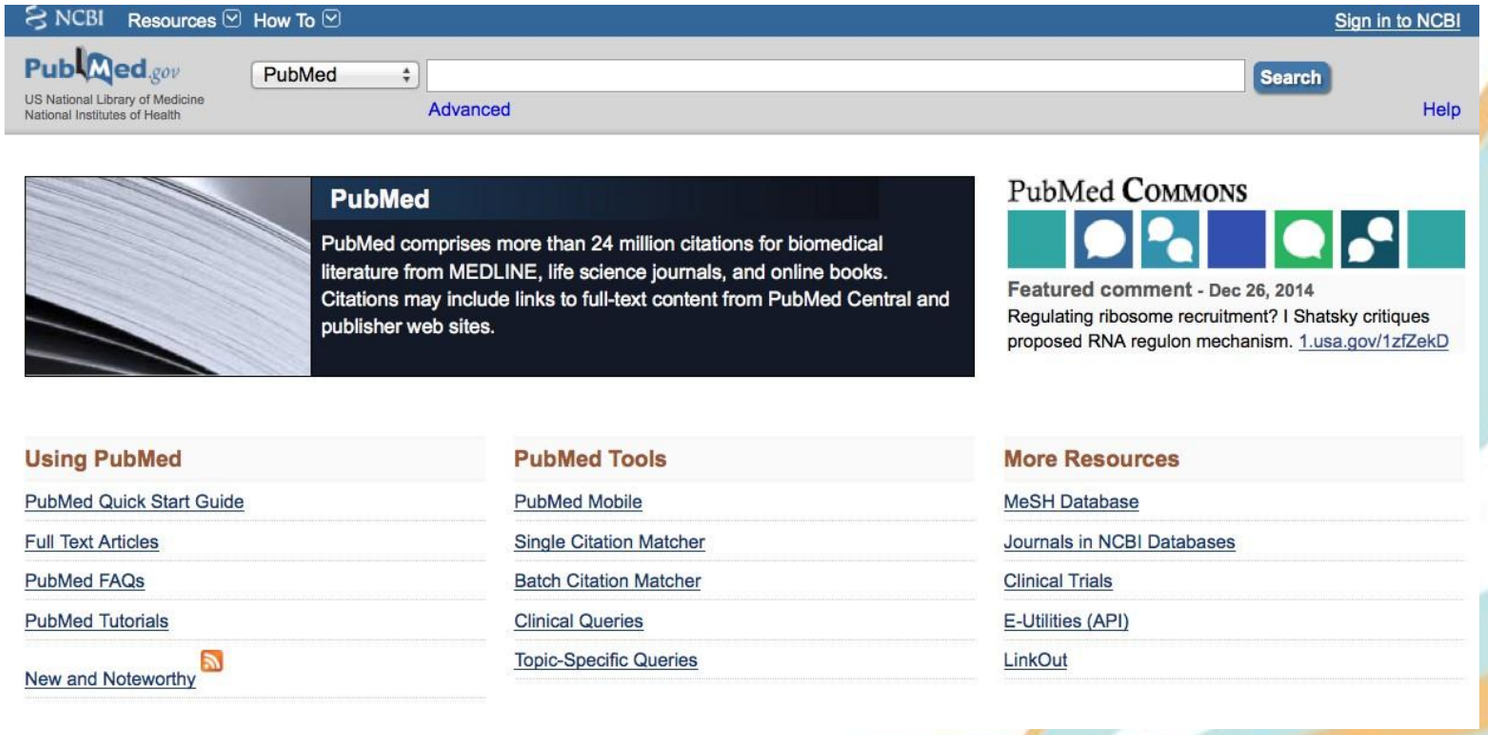
Medical Databases

PUBMED

- Comprises of more than 24 million citations for biomedical literature from MEDLINE, life science journals, and online books
- Citations may include links to full-text content from PubMed Central and publisher web sites

Amina Zee

Medical Databases



Amina Zee

Medical Databases

MEDLINE

- MEDLINE is the primary resource for biomedical journal articles
- Millions of citations to articles in biomedical journals
- MEDLINE uses the MeSH vocabulary

Amina Zee

Medical Databases

Other Databases

MEDLINE is the primary resource, but other databases may also be helpful

- Academic OneFile
- CINAHL (Cumulated Index of Nursing and Allied Health Literature)
- PsycINFO
- Web of Knowledge

Amina Zee

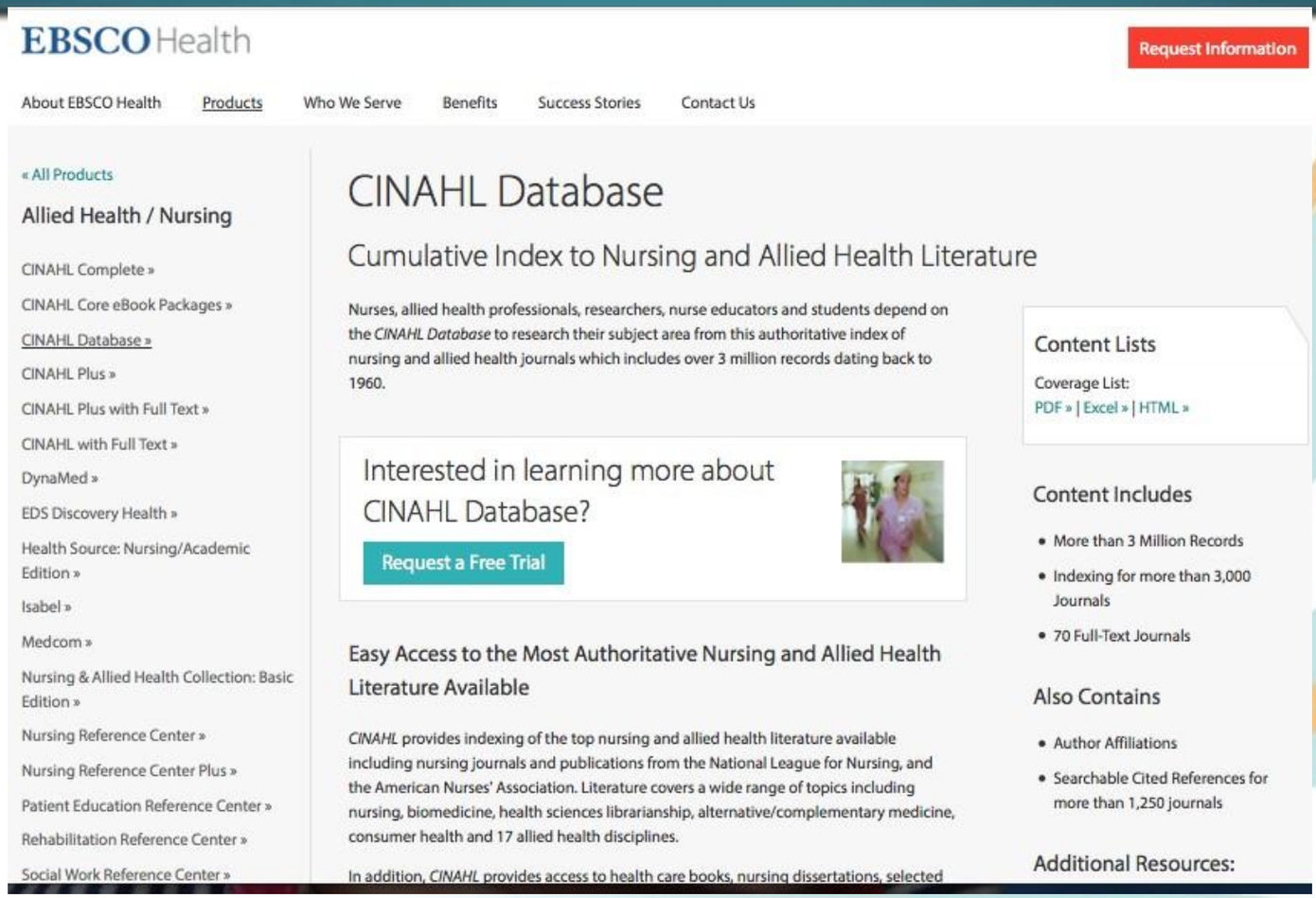
Medical Databases

Academic OneFile

- Academic OneFile lists articles from journals covering a broad range of subjects
- While it does not primarily focus on medical topics, useful articles can still be found here

Amina Zee

Medical Databases



EBSCO Health

Request Information

About EBSCO Health | Products | Who We Serve | Benefits | Success Stories | Contact Us

« All Products

Allied Health / Nursing

- CINAHL Complete »
- CINAHL Core eBook Packages »
- CINAHL Database »
- CINAHL Plus »
- CINAHL Plus with Full Text »
- CINAHL with Full Text »
- DynaMed »
- EDS Discovery Health »
- Health Source: Nursing/Academic Edition »
- Isabel »
- Medcom »
- Nursing & Allied Health Collection: Basic Edition »
- Nursing Reference Center »
- Nursing Reference Center Plus »
- Patient Education Reference Center »
- Rehabilitation Reference Center »
- Social Work Reference Center »

CINAHL Database

Cumulative Index to Nursing and Allied Health Literature

Nurses, allied health professionals, researchers, nurse educators and students depend on the *CINAHL Database* to research their subject area from this authoritative index of nursing and allied health journals which includes over 3 million records dating back to 1960.

Interested in learning more about CINAHL Database?

Request a Free Trial

Easy Access to the Most Authoritative Nursing and Allied Health Literature Available

CINAHL provides indexing of the top nursing and allied health literature available including nursing journals and publications from the National League for Nursing, and the American Nurses' Association. Literature covers a wide range of topics including nursing, biomedicine, health sciences librarianship, alternative/complementary medicine, consumer health and 17 allied health disciplines.

In addition, *CINAHL* provides access to health care books, nursing dissertations, selected

Content Lists

Coverage List:
PDF » | Excel » | HTML »

Content Includes

- More than 3 Million Records
- Indexing for more than 3,000 Journals
- 70 Full-Text Journals

Also Contains

- Author Affiliations
- Searchable Cited References for more than 1,250 journals

Additional Resources:

Amina Zee

Medical Databases

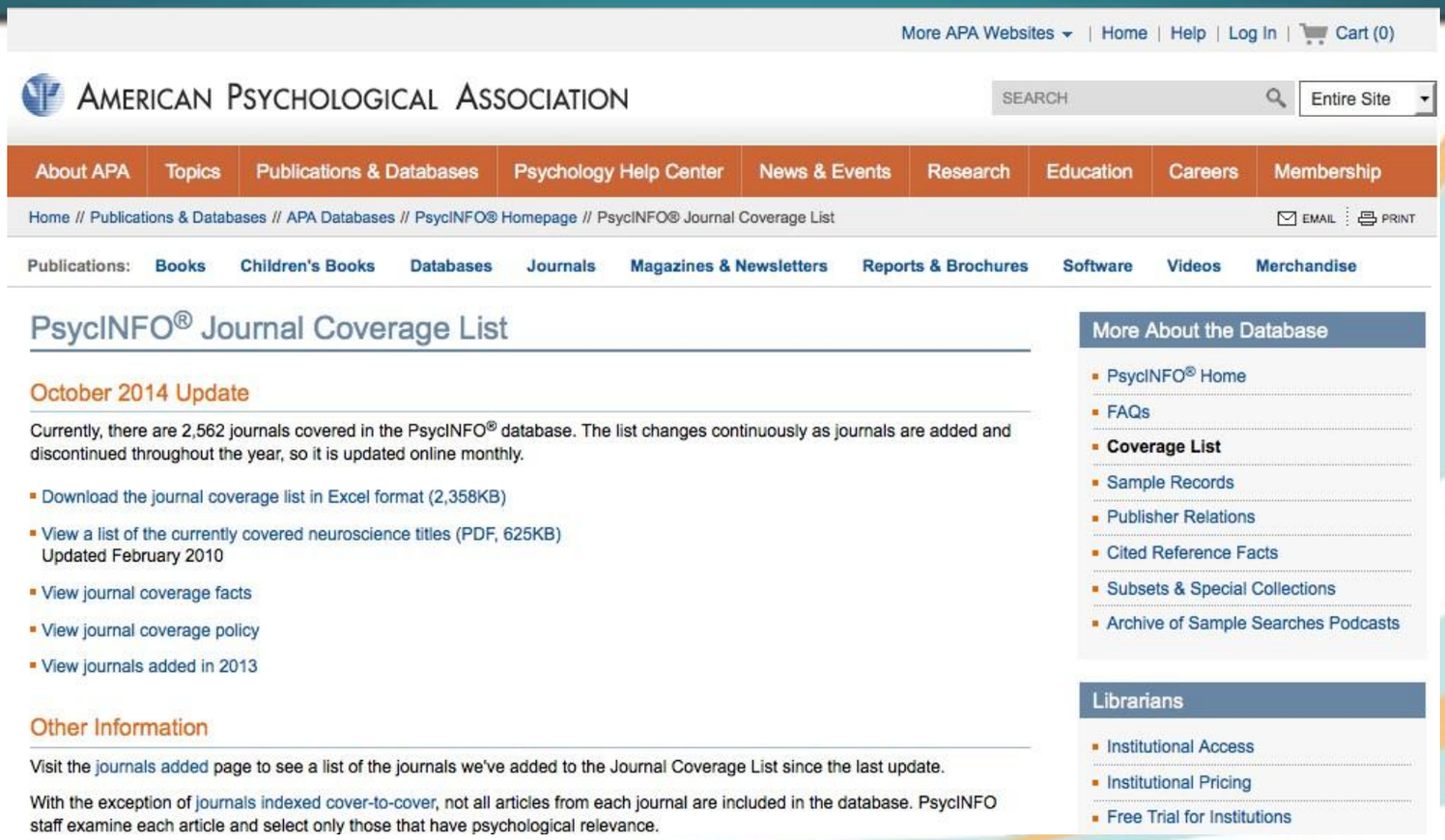
PsycINFO

- PsycINFO searches the psychological literature
- While it does not primarily focus on medical topics, useful articles can still be found here

http://www.apa.org/pubs/databases/psycinfo/coverage.aspx

Amina Zee

Medical Databases



http://www.apa.org/pubs/databases/psycinfo/coverage.aspx

Amina Zee

Medical Databases

Web of Science

- Major source for articles in a wide range of fields, including the sciences, social sciences, and humanities.
- Excellent place to find articles from scientific journals that may not be included in MEDLINE

Amina Zee

Medical Databases

Conclusions

Informatics in health care may be called as health informatics

- Medical databases deal with the acquisition, storage, retrieval, and use of information in health and biomedicine.

Amina Zee

Lec 100

Sequence submission

Amina Zee

Sequence Submission

Introduction

- Sequences are submitted to the databases in order to share them with the scientific community
- Generally sequences are submitted at the time of publication and are reviewed by peers

Amina Zee

Sequence Submission

Caution

- It is important to ensure that sequence files do not contain any special characters
- Control characters in addition to standard ASCII characters must be removed else they might mess up the analysis or data transfer

Sequence Submission

Table 2.1. *Base–nucleic acid codes*

Symbol	Meaning	Explanation
G	G	Guanine
A	A	Adenine
T	T	Thymine
C	C	Cytosine
R	A or G	puRine
Y	C or T	pYrimidine
M	A or C	aMino
K	G or T	Keto
S	C or G	Strong interactions 3 h bonds
W	A or T	Weak interactions 2 h bonds
H	A, C or T not G	H follows G in alphabet
B	C, G or T not A	B follows A in alphabet
V	A, C or G not T (not U)	V follows U in alphabet
D	A, G or T not C	D follows C in alphabet
N	A,C,G or T	Any base

Adapted from NC-IUB (1984).

Mount, pg 28

- Nomenclature committee of International Union of Biochemistry (IUB) has established standard codes to represent ambiguous bases or aminoacids

Amina Zee

Sequence Submission

Table 2.2. *Table of standard amino acid code letters*

1-letter code	3-letter code	Amino acid
A[a]	Ala	alanine
C	Cys	cysteine
D	Asp	aspartic acid
E	Glu	glutamic acid
F	Phe	phenylalanine
G	Gly	glycine
H	His	histidine
I	Ile	isoleucine
K	Lys	lysine
L	Leu	leucine
M	Met	methionine
N	Asn	asparagine
P	Pro	proline
Q	Gln	glutamine
R	Arg	arginine
S	Ser	serine
T	Thr	threonine
V	Val	valine
W	Trp	tryptophan
X	Xxx	undetermined amino acid
Y	Tyr	tyrosine
Z[b]	Glx	either glutamic acid or glutamine

Adapted from IUPAC-IUB (1969, 1972, 1983).

[a] Letters not shown are not commonly used.

[b] Note that sometimes when computer programs translate DNA sequences, they will put a "Z" at the end to indicate the termination codon. This character should be deleted from the sequence.

Mount, pg 28

Amina Zee

Sequence Submission

NCBI

NCBI has two options

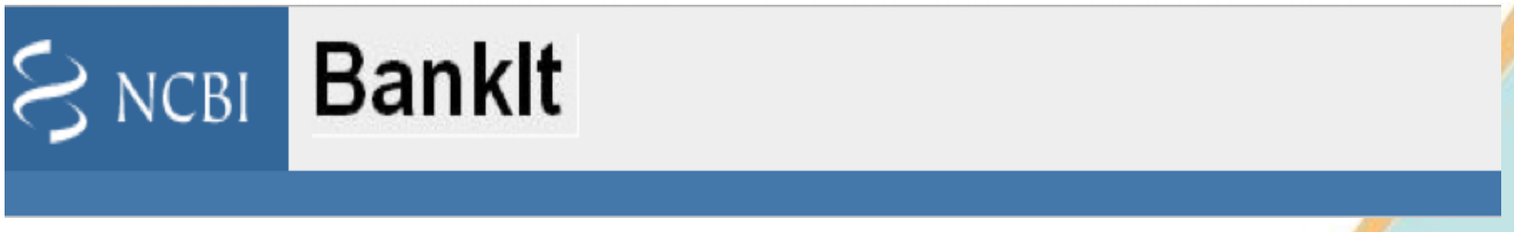
- **BANKIt**
  - For simple sequences and annotations
  - For submission through web
  - Do not require advanced tools

Sequence Submission

NCBI

- **Sequin**
  - For Complex sequences and annotations
  - For off-line submissions
  - Do require advanced tools and graphical reports

Sequence Submission



http://www.ncbi.nlm.nih.gov/WebSub/?tool=genbank

Amina Zee

Sequence Submission

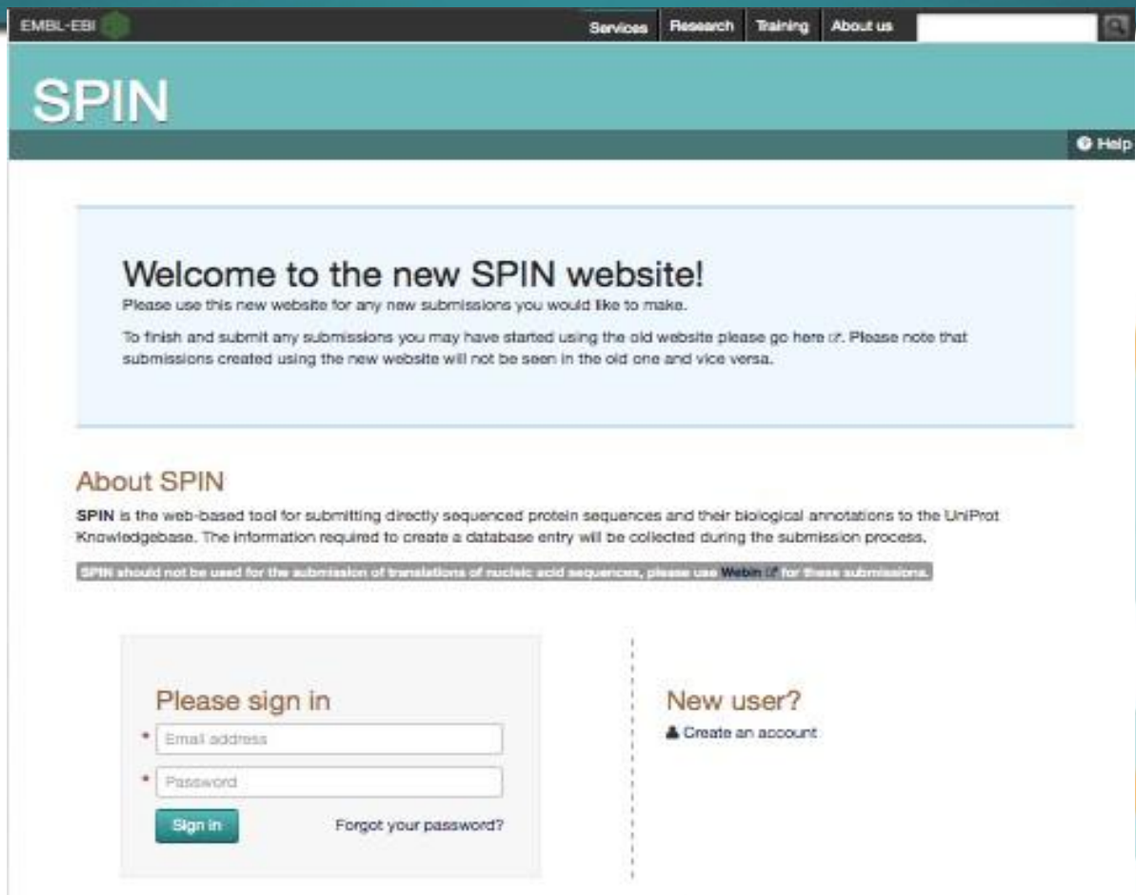
UniProt

SPIN web-based tool for submitting

- directly sequenced protein sequences
- biological annotations to the knowledgebase

Amina Zee

Sequence Submission



https://www.ebi.ac.uk/swissprot/Submissions/spin

Amina Zee

Sequence Submission

Conclusion

- Sequences are stored in databases in specific format

Amina Zee

Lec 101

DNA Sequence Retrieval

Amina Zee

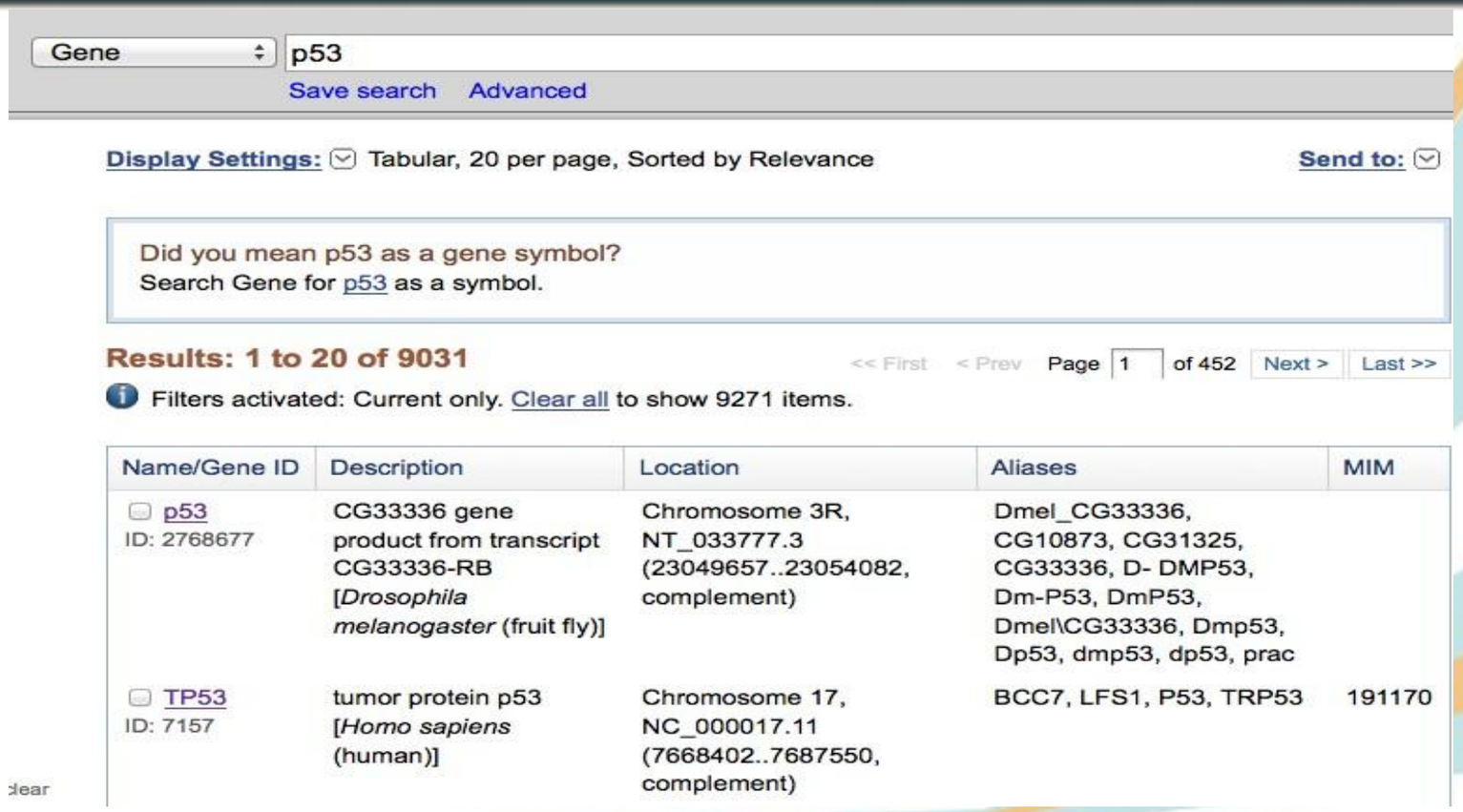
DNA Sequence Retrieval

Introduction

- Databases not merely collect and organize data but allow intelligent data retrieval
- And/or analysis

Amina Zee

DNA Sequence Retrieval



Gene p53

Save search Advanced

Display Settings: Tabular, 20 per page, Sorted by Relevance Send to:

Did you mean p53 as a gene symbol?
Search Gene for p53 as a symbol.

Results: 1 to 20 of 9031

<< First < Prev Page 1 of 452 Next > Last >>

Filters activated: Current only. Clear all to show 9271 items.

Name/Gene ID	Description	Location	Aliases	MIM
p53 ID: 2768677	CG33336 gene product from transcript CG33336-RB [*Drosophila melanogaster* (fruit fly)]	Chromosome 3R, NT_033777.3 (23049657..23054082, complement)	Dmel_CG33336, CG10873, CG31325, CG33336, D- DMP53, Dm-P53, DmP53, Dmel\CG33336, Dmp53, Dp53, dmp53, dp53, prac	
TP53 ID: 7157	tumor protein p53 [*Homo sapiens* (human)]	Chromosome 17, NC_000017.11 (7668402..7687550, complement)	BCC7, LFS1, P53, TRP53	191170

http://www.ncbi.nlm.nih.gov/

Amina Zee

DNA Sequence Retrieval

p53 [*Drosophila melanogaster* (fruit fly)]

Gene ID: 2768677, updated on 4-Jan-2015

Summary

Official Symbol	p53 provided by FlyBase
Primary source	FLYBASE:FBgn0039044
Locus tag	Dmel_CG33336
Gene type	protein coding
RefSeq status	REVIEWED
Organism	Drosophila melanogaster (old-lineage: Eukaryota; Metazoa; Arthropoda; Hexapoda; Insecta; Pterygota; Neoptera; Endopterygota; Diptera; Brachycera; Muscomorpha; Ephydroidea; Drosophilidae; Drosophila; Sophophora)
Lineage	Eukaryota; Metazoa; Ecdysozoa; Arthropoda; Hexapoda; Insecta; Pterygota; Neoptera; Endopterygota; Diptera; Brachycera; Muscomorpha; Ephydroidea; Drosophilidae; Drosophila; Sophophora
Also known as	CG10873; CG31325; CG33336; D-p53; Dm-P53; Dmel\CG33336; dmp53; Dmp53; DmP53; DMP53; dp53; Dp53; prac

http://www.ncbi.nlm.nih.gov/

Amina Zee

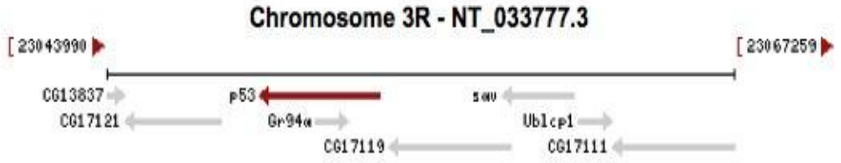
DNA Sequence Retrieval

Genomic context

Location: 94D10-94D10 See p53 in Epigenomics, MapViewer

Exon count: 10

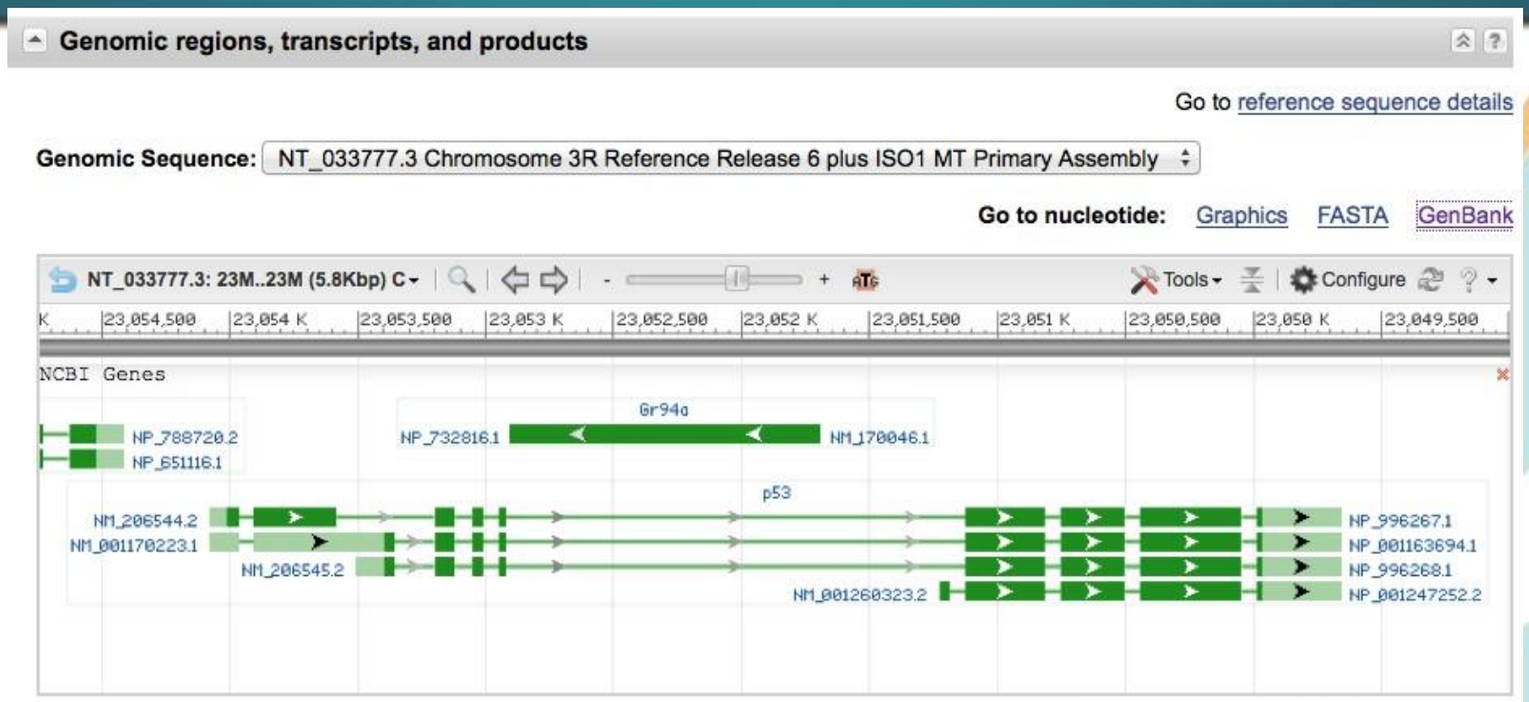
Annotation release	Status	Assembly	Chr	Location
Release 6.01	current	Release 6 plus ISO1 MT (GCF_000001215.4)	3R	NT_033777.3 (23049657..23054082, complement)
Release 5.57	previous assembly	Release 5 (GCF_000001215.2)	3R	NT_033777.2 (18875379..18879804, complement)



http://www.ncbi.nlm.nih.gov/

Amina Zee

DNA Sequence Retrieval



http://www.ncbi.nlm.nih.gov/

Amina Zee

DNA Sequence Retrieval

Drosophila melanogaster chromosome 3R

NCBI Reference Sequence: NT_033777.3

FASTA Graphics

```
LOCUS       NT_033777               4426 bp    DNA     linear   INV 05-AUG-2014
DEFINITION  Drosophila melanogaster chromosome 3R.
ACCESSION   NT_033777 REGION: complement(23049657..23054082)
VERSION     NT_033777.3  GI:671162122
DBLINK      BioProject: PRJNA164
            BioSample: SAMN02803731
KEYWORDS    RefSeq.
SOURCE      Drosophila melanogaster (fruit fly)
  ORGANISM  Drosophila melanogaster
            Eukaryota; Metazoa; Ecdysozoa; Arthropoda; Hexapoda; Insecta;
            Pterygota; Neoptera; Endopterygota; Diptera; Brachycera;
            Muscomorpha; Ephydroidea; Drosophilidae; Drosophila; Sophophora.
REFERENCE   1  (bases 1 to 4426)
  AUTHORS   Hoskins,R.A., Carlson,J.W., Kennedy,C., Acevedo,D., Evans-Holm,M.,
            Frise,E., Wan,K.H., Park,S., Mendez-Lago,M., Rossi,F.,
            Villasante,A., Dimitri,P., Karpen,G.H. and Celniker,S.E.
  TITLE     Sequence finishing and mapping of Drosophila melanogaster
            heterochromatin
  JOURNAL   Science 316 (5831), 1625-1628 (2007)
   PUBMED   17569867
```

http://www.ncbi.nlm.nih.gov/

Amina Zee

DNA Sequence Retrieval

```
FEATURES             Location/Qualifiers
     source          1..4426
                     /organism="Drosophila melanogaster"
                     /mol_type="genomic DNA"
                     /db_xref="taxon:7227"
                     /chromosome="3R"
                     /genotype="y[1]; Gr22b[1] Gr22d[1] cn[1] CG33964[R4.2]
                     bw[1] sp[1]; LysC[1] MstProx[1] GstD5[1] Rh6[1]"
     gene            1..4426
                     /gene="p53"
                     /locus_tag="Dmel_CG33336"
                     /gene_synonym="CG10873; CG31325; CG33336; D-p53; Dm-P53;
                     Dmel\CG33336; dmp53; Dmp53; DmP53; DMP53; dp53; Dp53;
                     prac"
                     /map="94D10-94D10"
                     /db_xref="FLYBASE:FBgn0039044"
                     /db_xref="GeneID:2768677"
     mRNA            join(1..118,178..501,884..964,1035..1071,1135..1161,
                     2959..3268,3333..3579,3642..4036,4096..4426)
                     /gene="p53"
                     /locus_tag="Dmel_CG33336"
                     /gene_synonym="CG10873; CG31325; CG33336; D-p53; Dm-P53;
                     Dmel\CG33336; dmp53; Dmp53; DmP53; DMP53; dp53; Dp53;
                     prac"
                     /product="p53, transcript variant B"
                     /note="p53-RB; Dmel\p53-RB; CG33336-RB; Dmel\CG33336-RB"
                     /transcript_id="NM_206544.2"
                     /db_xref="GI:281362333"
                     /db_xref="FLYBASE:FBtr0084360"
                     /db_xref="FLYBASE:FBgn0039044"
                     /db_xref="GeneID:2768677"
```

http://www.ncbi.nlm.nih.gov/

Amina Zee

DNA Sequence Retrieval

```
CDS             join(75..118,178..501,884..964,1035..1071,1135..1161,
                2959..3268,3333..3579,3642..4036,4096..4118)
                /gene="p53"
                /locus_tag="Dmel_CG33336"
                /gene_synonym="CG10873; CG31325; CG33336; D-p53; Dm-P53;
                Dmel\CG33336; dmp53; Dmp53; DmP53; DMP53; dp53; Dp53;
                prac"
                /note="CG33336 gene product from transcript CG33336-RB;
                CG33336-PB; p53-PB; p53-like regulator of apoptosis and
                cell cycle; Dmp53; protein 53; drosophila p53"
                /codon_start=1
                /product="p53, isoform B"
                /protein_id="NP_996267.1"
                /db_xref="GI:45553461"
                /db_xref="FLYBASE:FBpp0083753"
                /db_xref="FLYBASE:FBgn0039044"
                /db_xref="GeneID:2768677"
                /translation="MSLHKSASFSLTFNQNTSIVSRSNSRTIFEAFKEFLDFWDIGNE
                VSAESAVRVSSNGAFNLPQSFGNESNEYAHLATPVDPAYGGNNTNNMMQFTNNLEILA
                NNNSDGNNKINACNKFVCHKGTDSEDDSTEVDIKEDIPKTVEVSGSELTTEPMAFLQG
                LNSGNLMQFSQQSVLREMMLQDIQIQANTLPKLENHNIGGYCFSMVLDEPPKSLWMYS
                IPLNKLYIRMNKAFNVDVQFKSKMPIQPLNLRVFLCFSNDVSAPVVRCQNHLSVEPLT
                ANNAKMRESLLRSENPNSVYCGNAQGKGISERFSVVVPLNMSRSVTRSGLTRQTLAFK
                FVCQNSCIGRKETSLVFCLEKACGDIVGQHVIHVKICTCPKRDRIQDERQLNSKKRKS
                VPEAAEEDEPSKVRRCIAIKTEDTESNDSRDCDDSAAEWNVSRTPDGDYRLAITCPNK
                EWLLQSIEGMIKEAAAEVLRNPNQENLRRHANKLLSLKKRAYELP"
```

http://www.ncbi.nlm.nih.gov/

Amina Zee

DNA Sequence Retrieval

```
ORIGIN
        1 cctggagcac ggaagattct tgcggacaca aatcgcaact gctaaataaa atttatttat
       61 ttgagtgcac agccatgagt cttcacaagt ccgcgtcgtt tagcttgact tttaaccagt
      121 gagcggagat attttattcg gtcttaccca acaaaataat gttgcgcctt tttgcagaaa
      181 cacttcgatt gtttcgcgta gcaatagtcg cacaattttt gaagctttca aggagttcct
      241 ggatttttgg gatatcggca acgaagtttc tgcagagtca gcagttcggg tctccagcaa
      301 cggagctttc aacttgccgc agagttttgg caacgaatcc aacgaatatg cccacctggc
      361 tacgcctgtg gatccagcct acggaggcaa caacacgaac aacatgatgc agttcacgaa
      421 caatctggaa attttggcca acaataattc cgatggcaat aacaaaatta atgcatgcaa
      481 caaattcgtc tgccacaagg ggtgagcaaa ttcaaaacac gcgctccaat cgataaacat
      541 tggctacggc gattgttcgc gctgcgtggc gaatggcaaa atccaaatag tcggtggcca
      601 ctacgattct gtagtttttt gttagcgaat ttttaatatt tagcctcctt ccccaacaag
      661 atcgcttgat cagatatagc cgactaagat gtatatatca cagccaatgt cgtggcacaa
      721 agaaaggtac agtgcggcaa caaattgatg atcgaacagt agaaaccttg catgtagcaa
     4261 ggcatgttcg atggccgaaa agaaaacatt tttatatttt tgatagtata ctgttgttaa
     4321 ctgcagttct atgtgactac gtaacttttg tctaccacaa caaacatact ctgtacaaaa
     4381 aagccaaaag tgaatttatt aaagagttgt catattttgc aaacat
//
```

http://www.ncbi.nlm.nih.gov/

Amina Zee

DNA Sequence Retrieval

Conclusions

- DNA Sequences are stored in DNA sequence databases in specified formats
- Genebank format is a standard format

Amina Zee

Lec 102

Protein Sequence Retrieval

Amina Zee

Lecture no.10

Protein Sequence Retrieval

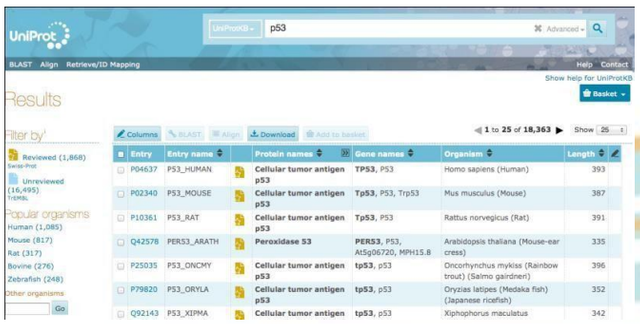
Protein Sequences:

Now we talk about the data retrieval and first we'll talk about the protein sequence retrievals and structures. Protein data is of the following types:

- Actual sequences (from the proteomic data or some other experimental techniques) or translated sequences (sometimes, we go to nucleotides databases, we get those nucleotides and then we translate them by using some softwares, so these are kind of predicted protein sequences) .
- Structures (we can also make structures from those proteins that maybe predicted or the real structures coming from various X-ray Crystallography Techniques).
- Annotations (sometimes, we are interested in the functions of the proteins so those are stored as annotations).

UniProt (*It is an international partnership between PIR, EBI and SIB*):

Now as far as the resources are concerned, we have multiple resources for protein sequences but **UniProt**claims to be the biggest and integrated resource whereas for the structures **PDB** seems like a good resource.



As shown in his figure, is the webpage for data retrieval from UniProt, so we want to search a protein, say p53, where we put it into the search box and press enter which gives us the output. And we see that there are

© Copyright Virtual University of Pakistan

Amina Zee

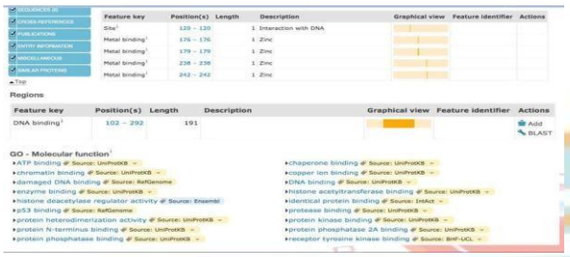
18,000 different records and it is showing us the first 25 out of them.

We can have different columns for the output on this webpage so we can have *entry; it's ID, entry name* (the Suffix Human is written so it's coming from Human, it can be from mouse, rat and Arabidopsis), the *protein name* is Cellular tumour antigen, then *gene name* which is TP53 (where TP stands for Tumour Protein), the *organism* is obviously the human (here) and in the end we have it's length i.e. 393 bp (base pairs).

The link to this webpage is **http://www.uniprot.org/uniprot/.**

So, let's check the first one and here we reach on the record for this protein (shown in the figure on the left) which is Cellular Tumour Antigen p53 protein, commonly known as TP53.

We can have different tabs, showing us the outputs. We can look into the *functions*, its *taxonomy*, and lot many other characteristics so if we look into the function so it gives us some description about what it's doing.



After scrolling the same webpage (shown in the figure on the left), we can see the *feature key* and in some *site* written (there are unique sites in different proteins which are having some specific properties in them so this is just one amino-acid present in this protein that *interacts with the DNA*). Similarly, there are different *metal binding sites* and we can see that it's

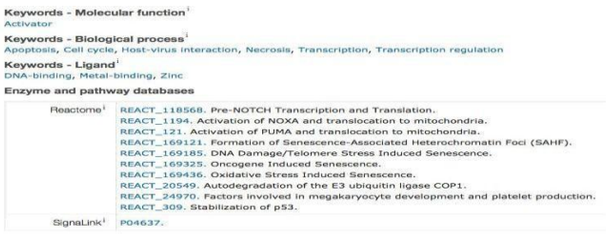
© Copyright Virtual University of Pakistan

Amina Zee

mainly binding to the *Zinc*metal.The number of amino-acids is shown here so these are the regions where it interacts with the metal.

Down below, we can also see the *DNA binding* region, for example here, the amino acids are from 102 to 292 and that is also shown in the *Graphical view* as well.

GO-Molecular function or GO-Gene Ontologies, so gene ontologies are the different functional annotation term, there they define different functions, so amongst them we have molecular functions, biological processes, and we have cellular components. So here we just see a *Molecular function*, so it tells us that it performs the functions as shown in the figure , mainly it's a *ATP binding, it's p53 binding* with various other functions like *DNA binding.* So all those functions related to these proteins are present in the heading of GO-Molecular Function.



Next, we move on to some other functions, in the *Biological process* category (shown in the figure) we see that it is related to *Apoptosis* (which is a cell death and it is related to cell-cycle and some other components).

In the below section, we see some *enzymes and pathway databases,* and*Reactome*is a database in which we have a group of reactions which are categorized so these are the list of those reactions with which it is related.

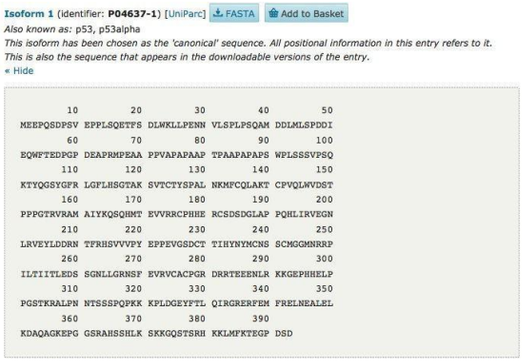
When we move further (as shown in the figure on the left) till we reach its *Taxonomy.*

On the top, we can see something written as *Protein family* or *group databases* whichis TCDB. Basically, there is another classification in which the proteins are classified on the basis of being as transporter proteins so it is associated with the transportation across the membranes and there is 5-digit number, so there is a specific classification code which is given to each protein, and this protein has the specific code as shown in the figure.

© Copyright Virtual University of Pakistan

Amina Zee

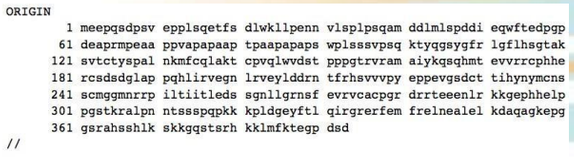
So then we have the *names and taxonomies*, where there are *protein names*, and thetaxonomyof the individual can be seen in the *Taxonomic lineage* row. Let's see how we reach to its sequence and is shown in the figure below:



In this figure, we can see the sequence of the protein which is found to be at the end of the page.

Here, it says *Isoform 1*, so different proteins have different isoforms, different alternative splice variants so this is Isoform 1 as exhibited by its name which is P04637-1, and is the kind of first isoform. We can see the sequence of the protein and starts with a methionine (always a first amino acid in those proteins) and ending at 390TH amino acid. So, it's a 393 aa long protein and the sequence is right here. You can click on the FASTA button on the top and then you can get this output in FASTA format (we'll discuss it later).

NCBI:



We can also get the same protein from NCBI (as shown in the figure on the left)

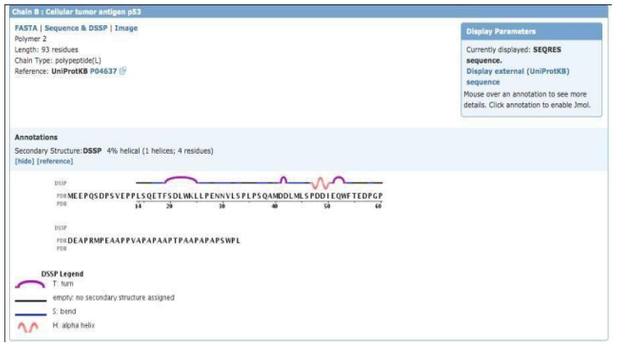
In NCBI, obviously the sequence is pretty similar and the arrangement is slightly different so it is *ORIGIN*, where the sequence starts and sequence ends at those two

© Copyright Virtual University of Pakistan

Amina Zee

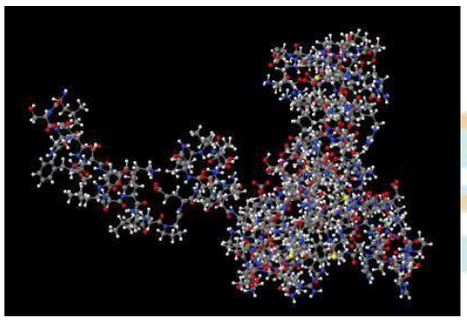
slashes (//). So, we can get the protein sequence from NCBI as well and the link to this website is https://www.ncbi.nlm.nih.gov/.

PDB:



PDB gives us the structures, so we can go to PDB webpage (as shown in the figure on the left) and search for the same ID i.e. P04637 and it gives us the sections or the regions from where it can make up some specific structures.

You can see the *turns* in Annotations section, the black ones are the empty lines where no secondary structure can be formed, blue ones show those bends and the orange ones are designated as alpha helices regions. So in PDB, we can have structures in this format as well as the 3D-Structures as shown in the figure below:



© Copyright Virtual University of Pakistan

Amina Zee

The link towards PDB is:
http://www.rcsb.org/pdb/explore/remediatedSequence.do?structureId=2UX8&bionumber=1

Conclusions:

We conclude that:

- UniProt is the integrated resource between PIR, EBI and SIB and
- PDB is a good resource to get the protein structure.

© Copyright Virtual University of Pakistan

Amina Zee

Lec 103

Sequence Formats

Amina Zee

Sequence Formats

FASTA Sequence Format

- DNA sequences are stored in specified formats
- Different softwares need sequences in different formats

Amina Zee

Sequence Formats

FASTA Sequence Format

- FASTA is the most frequently used format to present DNA and Protein sequences
- It is recommended that all lines of text be shorter than 80 characters in length

Amina Zee

Sequence Formats

>gi|568815581:c7687550-7668402 Homo sapiens chromosome 17, GRCh38 Primary Assembly
GATGGGATTGGGGTTTTCCCCTCCCATGTG
CTCATCTAGAGCCACCGTCCAGGGAGCAG
GTAGCTGCTGGGCTCTCCACGACGGTGAC
ACGC--------

>gi|120407068|ref|NP_000537.3| cellular tumor antigen p53 isoform a [Homo sapiens]
MEEPQSDPSVEPPLSQETFSDLWKLLPENNV
LSPLAPPVLGFLHSGTAKSVTCTYSPALNKMF
CQLAKT--*

Sometimes a * might be present in the end

AmiraZee

Sequence Formats

Genebank Format

- A sequence file in GenBank format can contain several sequences
- One sequence starts with a line containing the word LOCUS and a number of annotation lines

Amina Zee

Sequence Formats

Genebank Format

- The start of the sequence is marked by a line containing "ORIGIN" and the end of the sequence is marked by two slashes ("//")

Sequence Formats

Genebank Format

LOCUS AAU03518 237 bp DNA PRI 04-FEB-1995
DEFINITION Aspergillus awamori internal transcribed spacer 1 (ITS1) and

18S rRNA and 5.8S rRNA genes, partial sequence.
ACCESSION U03518
BASE COUNT 41 a 77 c 67 g 52 t
ORIGIN
1 aacctgcgga aggatcatta ccgagtgcgg gtcctttggg cccaacctcc catccgtgtc
61 tattgtaccc tgttgcttcg gcgggcccgc cgcttgtcgg ccgccggggg ggcgcctctg
121 ccccccgggc ccgtgcccgc cggagacccc aacacgaaca ctgtctgaaa gcgtgcagtc
181 tgagttgatt gaatgcaatc agttaaaact ttcaacaatg gatctcttgg ttccggc
//

Amina Zee

Sequence Formats

EMBLFormat

- An example sequence in EMBL format is:

```
ID   AA03518    standard; DNA; FUN; 237 BP.
XX
AC   U03518;
XX
DE   Aspergillus awamori internal transcribed spacer 1 (ITS1) and 18S
DE   rRNA and 5.8S rRNA genes, partial sequence.
XX
SQ   Sequence 237 BP; 41 A; 77 C; 67 G; 52 T; 0 other;
     aacctgcgga aggatcatta ccgagtgcgg gtcctttggg cccaacctcc catccgtgtc        60
     tattgtaccc tgttgcttcg gcgggcccgc cgcttgtcgg ccgccggggg ggcgcctctg       120
     ccccccgggc ccgtgcccgc cggagacccc aacacgaaca ctgtctgaaa gcgtgcagtc       180
     tgagttgatt gaatgcaatc agttaaaact ttcaacaatg gatctcttgg ttccggc          237
//
```

Amina Zee

Sequence Formats

SwissProt Format

- SwissProt protein sequence format is similar to EMBL format
- There is considerably more information about physical and biochemical properties of a protein is provided

Sequence Formats

SwissProt Format

ID - Identification.

AC - Accession number(s).
DT - Date.
DE - Description.
GN - Gene name(s).
OS - Organism species.
OG - Organelle.
OC - Organism classification.
RN - Reference number.

RP - Reference position.
RC - Reference comments.
RX - Reference cross-references.
RA - Reference authors.
RL - Reference location.
CC - Comments or notes.
DR - Database cross-references.
KW - Keywords.
FT - Feature table data.
SQ - Sequence header.
// - Termination line.

Amina Zee

Sequence Formats

XML Format

- Extensible Markup Language
- Readable by both man and machine
- Becoming standard data format for transferring genome data

Amina Zee

Sequence Formats

XML Format

```
<xsd:annotation>
    <xsd:documentation>
      XML Schema for SBOL core data model
compatible with RDF/XML serialization.
      <dc:date>2012-01-19</dc:date>
      <dc:creator>Evren Sirin</dc:creator>
      <dc:contributor>Michal
Galdzicki</dc:contributor>
    </xsd:documentation>
  </xsd:annotation>
```

Sequence Formats

Sequence converters

- READSEQ is a useful sequence converter developed by D.G.Gilbert at Indiana University, USA
- Recognizes DNA or Protein sequence file and interconvert them

Sequence Formats

Conclusions

- Databases store sequences in specified formats
- Genebank, DDBJ and EMBL has similar formats
- Different softwares need sequences in different formats

Amina Zee

Lec 104

Data Retrieval

Amina Zee

Data Retrieval

Data Retrieval

- Nearly all biological databases are available for download as simple text (flat) files
- A local version of the database allows one greater freedom in processing the data

Data Retrieval

Entrez

- is an integrated search engine which allows users to search and retrieve different data from the NCBI
- It can be accessed from the site www.ncbi.nlm.nih.gov/Entrez/

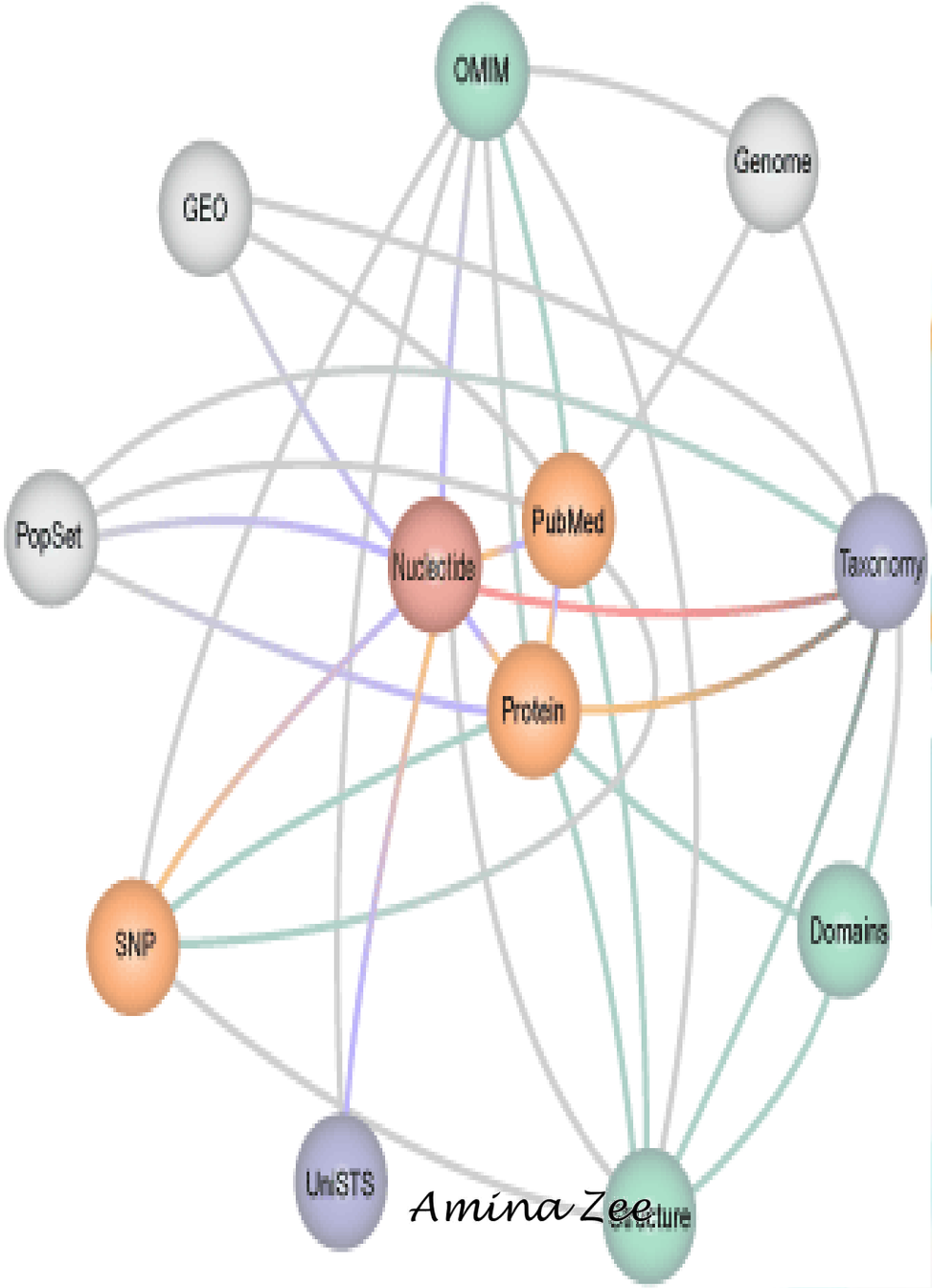
Amina Zee

Data Retrieval

Entrez

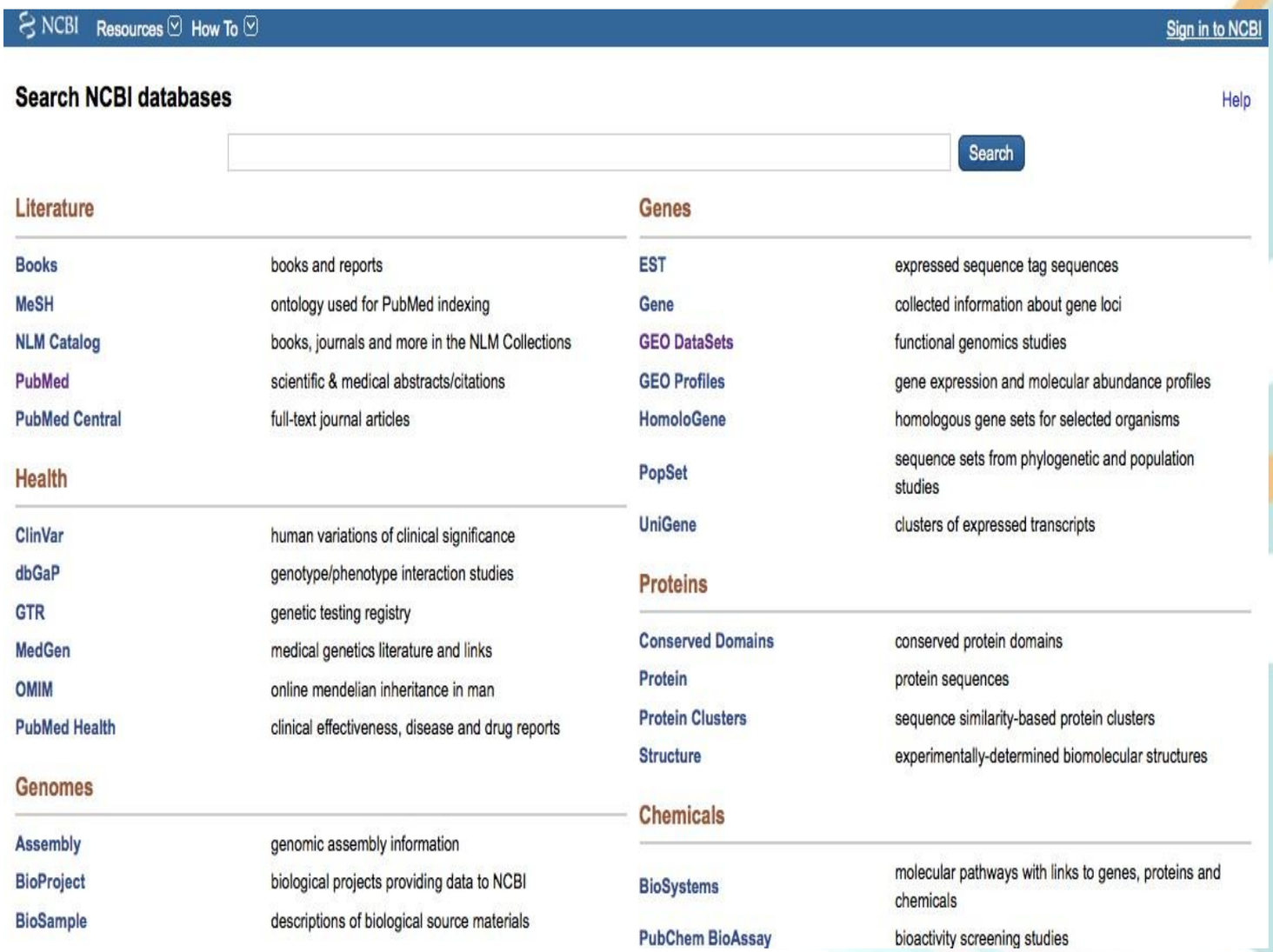
- integrates PubMed and 39 other scientific literatures, nucleotide and protein databases
- protein domain data, population studies, expression data, pathways, genome details and taxonomic information

Data Retrieval



Amina Zee

Data Retrieval



Amina Zee

Data Retrieval

Bulk Data Retrieval

- The best option is to use ftp (File transfer protocol)
- The File Transfer Protocol (**FTP**) is a standard network protocol used to transfer files
- Via command line or application programs like FTP clients

Data Retrieval

Bulk Data Retrieval

- Data needs to be transformed or processed using programming languages
- PERL and Python are good for processing Biological data

Amina Zee

Data Retrieval

Conclusions

- Data is transferred over the internet
- Data needs to be transformed or processed

Lec 105

Why proteomics

Amina Zee

why Proteomics

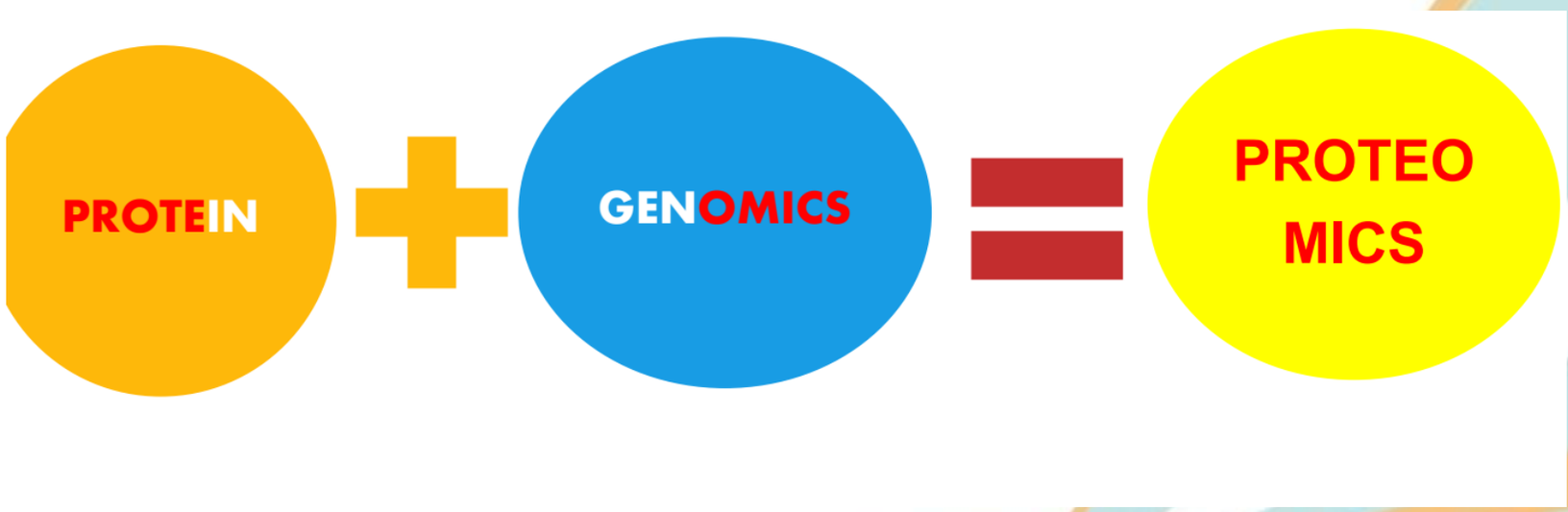
Amina Zee

Lecture-2

What do mean by proteomics?

- Proteomics is the large-scale study of proteins, usually by biochemical methods. The word proteomics has been associated
- traditionally with displaying a large number of proteins from a given
- cell line or organism on two -dimensional polyacrylamide gels

Amina Zee



Amina Zee

Scope of Proteomics

The identification, characterization and quantification of all proteins involved in a particular pathway, organelle, cell, tissue, organ or organism that can be studied in concert to provide accurate and comprehensive data about that system.

Or - A complete description of proteins expressed in any given cell at any given time

Amina Zee

Why should we study Proteomics?

- directly contributes to drug development
- Verification of a gene product by proteomic methods
- Modifications of the proteins
- Protein expression level d

Amina Zee

WHY PROTEOMICS?

Many types of information cannot be obtained from the study of genes alone. For example, proteins, not genes, are responsible for the phenotypes of cells. It is impossible to elucidate mechanisms of disease, aging, and effects of the environment solely by studying the genome.

Amina Zee

Advantages of study of proteomics

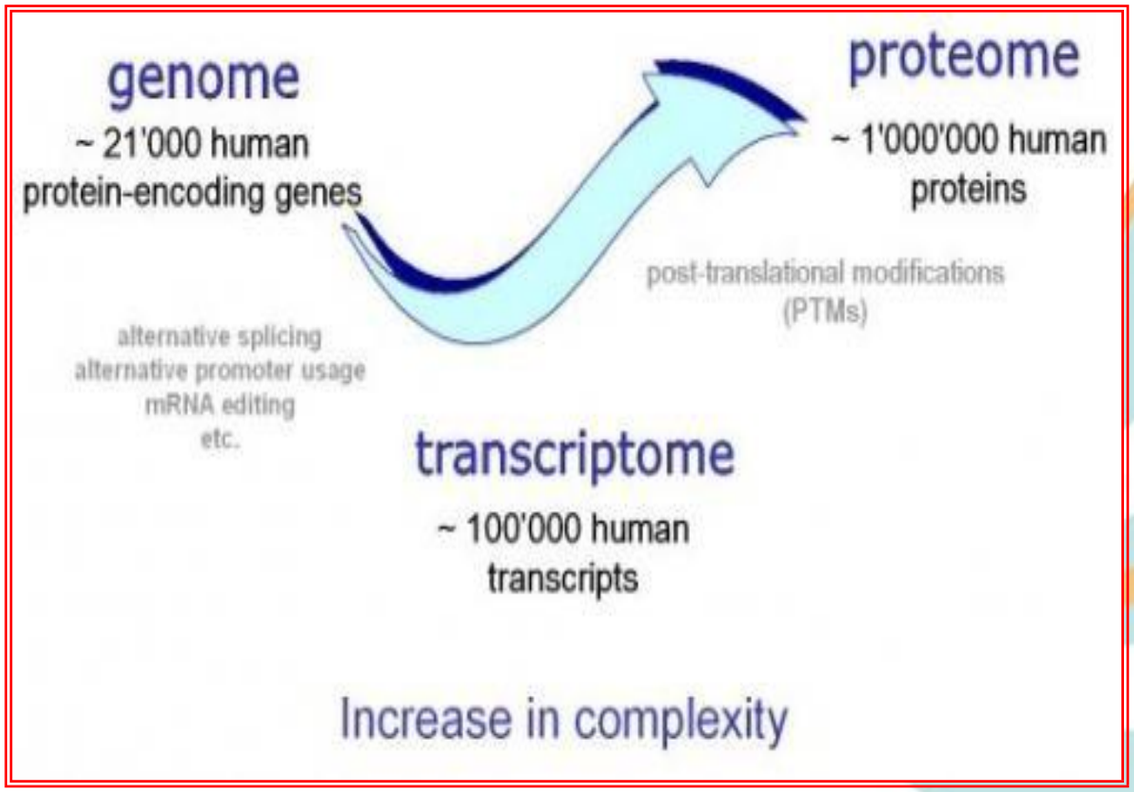
- Shows that genetic alterations are not the reaso for all types of diseases
- Helps in determining the proper treatment of diseases
- With the help of three dimensional analysis of proteins we have found that HIV protease is the enzyme which is responsible for AIDS.
- One of the most important use of proteomics in diagnosis is the identification of biomarkers. The study of drugs in proteomics is called pharmacoproteomics.

Amina Zee

Proteomics aims

- Genomics integrated strategies
- Study of post-translational modifications
- Identification of novel protein targets for drugs
- Analysis of tumor tissues
- Comparison between normal and diseased tissues
- Comparison between diseased and pharmacologically treated tissues

Amina Zee


genome
~ 21'000 human
protein-encoding genes
proteome
~ 1'000'000 human
proteins
post-translational modifications
(PTMs)
alternative splicing
alternative promoter usage
mRNA editing
etc.
transcriptome
~ 100'000 human
transcripts
Increase in complexity

Amina Zee

Limitations of Genomics Challenge of Proteomics

- co-translational modifications
  - differential mRNA splicing
- post-translational modifications (PTMs)
  - C-terminal GPI anchor
  - phosphorylation
  - sulfation
  - glycosylation
  - N-myristoylation
  - hydroxylation
  - N-methylation
  - carboxymethylation
  - signal peptidase site.....

Amina Zee

Lec 106

Central dogma

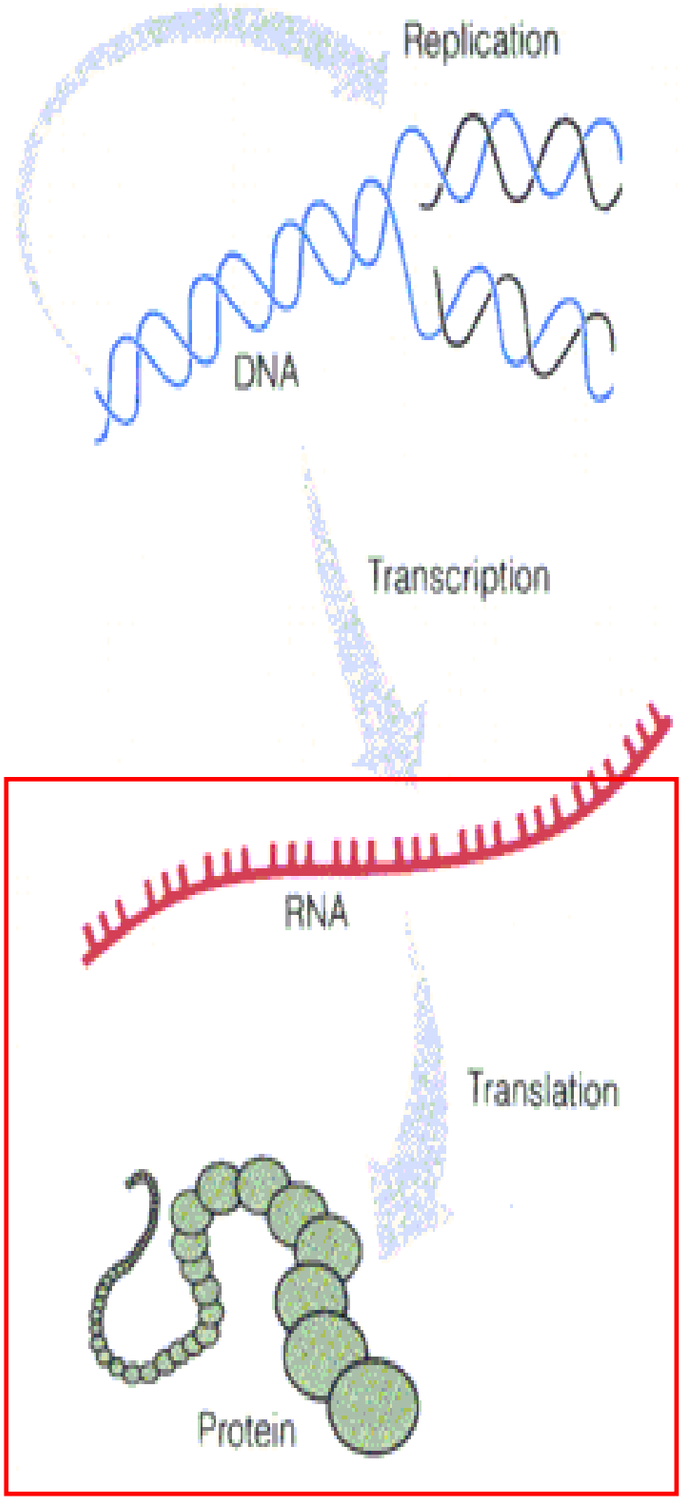
Amina Zee

Proteomics Introducción

Lecture-1

Amina Zee

The Central Dogma



The central dogma states that information in nucleic acid can be perpetuated or transferred, but the transfer of information form into protein is irreversible.

Amina Zee

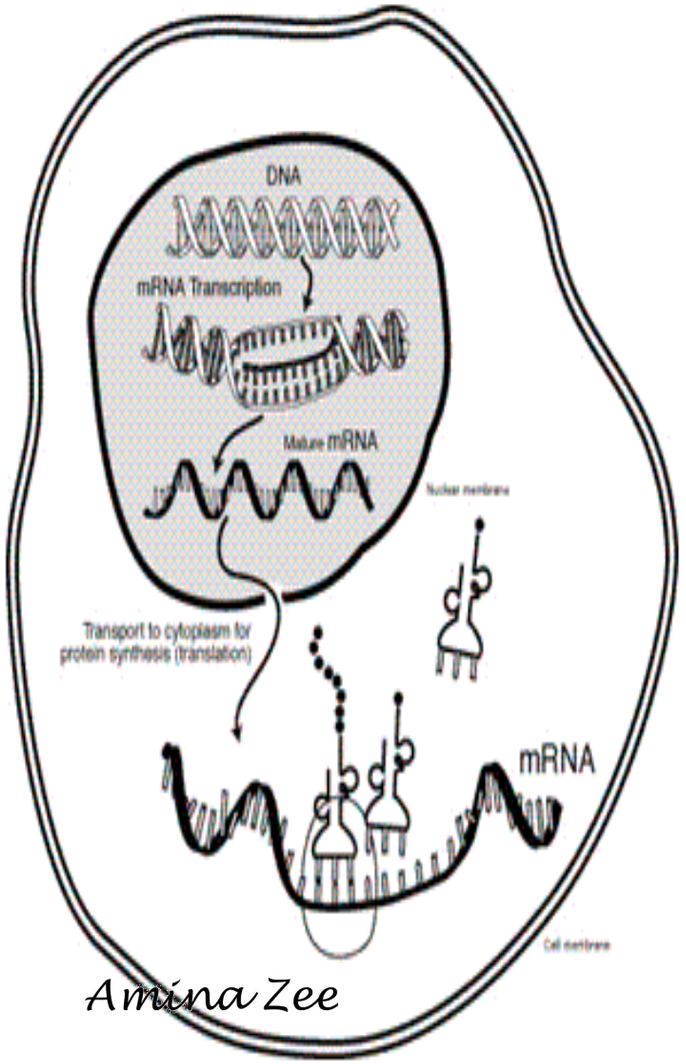
Central Dogma of Biology

- DNA -> RNA -> Protein Synthesis
- Transcription:
  - Process of DNA serving as a template for RNA synthesis
- Translation:
  - Process of RNA serving as a template for protein synthesis

Amina Zee

How do DNA and genes relate to proteins?

- DNA provides the genes, or genetic code, for protein synthesis
- Genes are expressed because DNA codes for RNA which then codes for ALL of our proteins



Background: 3 Types of RNA

- mRNA: Messenger RNA
  - 1st RNA's made DIRECTLY from DNA template
  - Travel from nucleus to ribosome
- rRNA: Ribosomal RNA
  - Helps form ribosomes in cytoplasm
- tRNA: Transfer RNA
  - Brings amino acids from cytoplasm to ribosome so proteins can be made

Amina Zee

Step 1: Transcription

- INSIDE of the nucleus DNA is used to make mRNA
- DNA is unzipped then RNA polymerase makes an mRNA strand from the DNA template
- New mRNA strand then leaves the nucleus and travels into the cytoplasm
- DNA is ALWAYS left protected in the nucleus

Amina Zee

Step 1: Transcription

- DNA: 5’ AAA TTT GGG CCC ATC GCA 3’
- mRNA: 3’ UUU AAA CCC GGG UAG CGU 5’

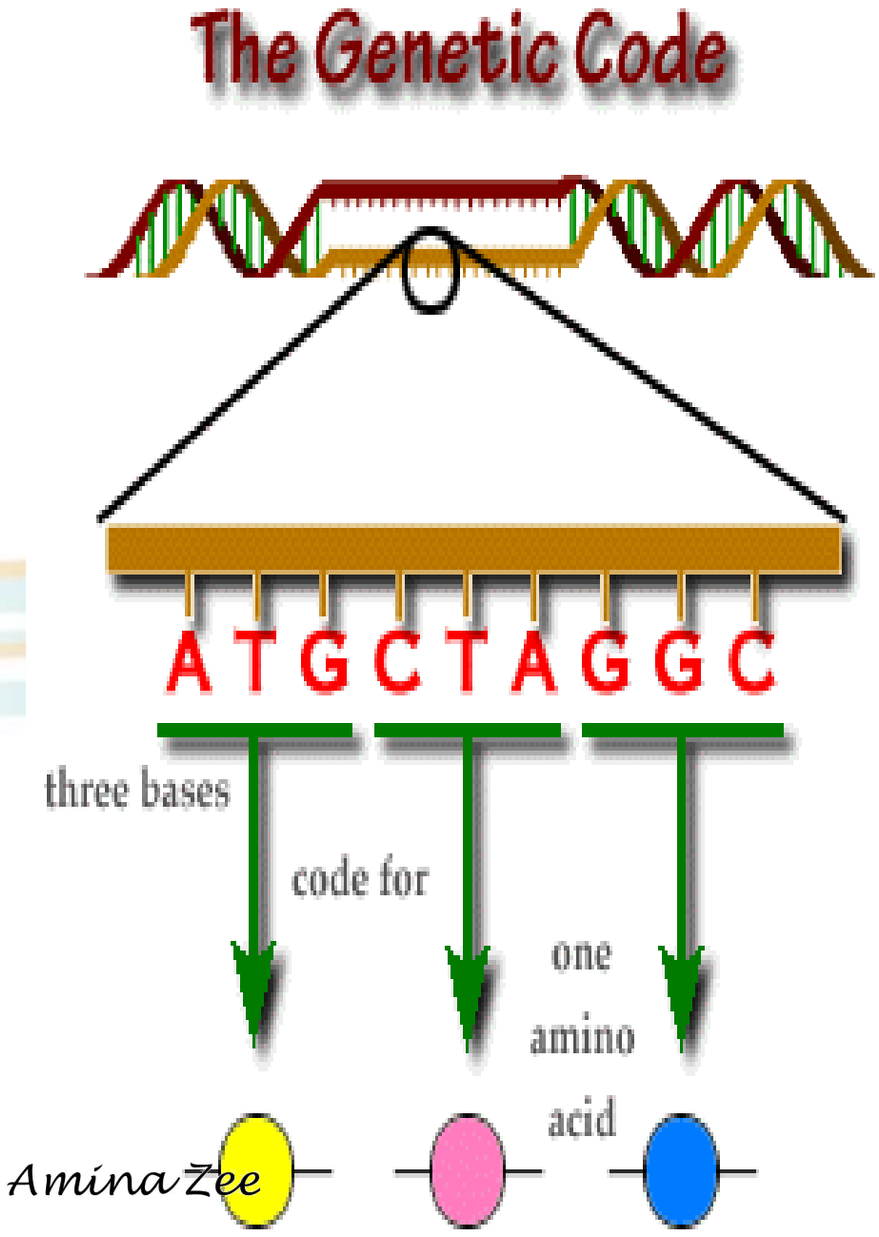
- DNA: CTA GTT CCC TAA AAG GAG
- mRNA: GAU CAA GGG AUU UUC CUC

- DNA: TAC CGA GGT TTA ACT
- mRNA: AUG UGA CCA AAU UGA

Amina Zee

Step 2: Translation

- Each nucleotide sequence serves as a code for what amino acid will be added to the protein being made
- Nucleotides read in triplets, or codons



Step 2: Translation

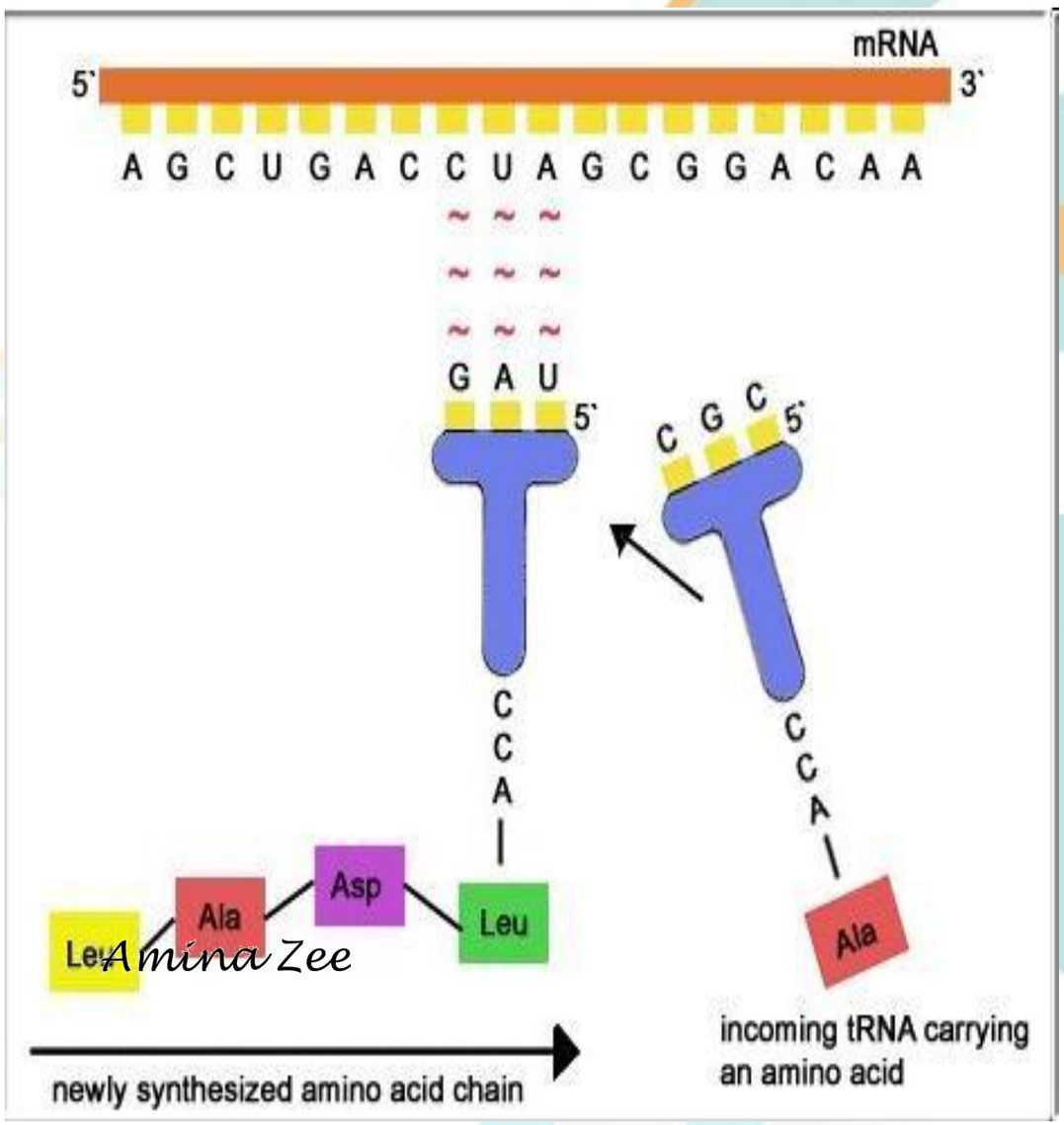
Second base in codon

First base in codon	U	C	A	G	Third base in codon
U	Phe	Ser	Tyr	Cys	U
	Phe	Ser	Tyr	Cys	C
	Leu	Ser	STOP	STOP	A
	Leu	Ser	STOP	Trp	G
C	Leu	Pro	His	Arg	U
	Leu	Pro	His	Arg	C
	Leu	Pro	Gln	Arg	A
	Leu	Pro	Gln	Arg	G
A	Ile	Thr	Asn	Ser	U
	Ile	Thr	Asn	Ser	C
	Ile	Thr	Lys	Arg	A
	Met	Thr	Lys	Arg	G
G	Val	Ala	Asp	Gly	U
	Val	Ala	Asp	Gly	C
	Val	Ala	Glu	Gly	A
	Val	Ala	Glu	Gly	G

Amina Zee

Step 2: Translation

- mRNA is now connected to the ribosome
- tRNA has a corresponding anti-codon and brings over the corresponding amino acid



The end result...

- An amino acid sequence that makes a protein
- GENES code for proteins/enzymes
- We NEED proteins to function
- The shape of the protein determines its function

Amina Zee

Lec 107

Types of proteomics

Amina Zee

TYPES OF PROTEOMICS

Amina Zee

Lecture-3

Scope of Proteomics

- Expression proteomics
- Structural proteomics
- Functional proteomics

Amina Zee

EXPRESSION PROTEOMICS

- Expression proteomics is used to study the qualitative and quantitative expression of total proteins under two different conditions.
  - Normal and diseased state.
  - E.g. :tumor or normal cell.
  - It studied that protein is over expressed or under expressed.
  - 2-D electrophorasis.

Amina Zee

STRUCTURAL PROTEOMICS

- Structural proteomics helps to understand three dimensional shape and structural complexities of functional proteins.
- It determine either by amino acid sequence in protei from a gene this process is known as **homology modeling**.
- It identify all the protein present in complex system protein-protein interaction.
- Mass spectroscopy is used for structure determinati

Amina Zee

FUNCTIONAL PROTEOMICS

 Functional proteomics explains understanding the protein functions well as unrevealing molecular mechanisms within the cell that depend on identification of the interacting protein partners.

 So that detailed description of the cellular signaling pathways mi greatly benefit from the elucidation of protein- protein interacti

Amina Zee

Limitations of Genomics Challenge of Proteomics

- co-translational modifications
  - differential mRNA splicing
- post-translational modifications (PTMs)
  - C-terminal GPI anchor
  - phosphorylation
  - sulfation
  - glycosylation
  - N-myristoylation
  - hydroxylation
  - N-methylation
  - carboxymethylation
  - signal peptidase site.....

Amina Zee

Lec 108
Structural Proteomic

Amina Zee

Lecture-1

What is structural proteomics/genomics?

- High-throughput determination of the 3D structure of proteins
- Goal: to be able to determine or predict the structure of every protein.
  - Direct determination - X-ray crystallography and nuclear magnetic resonance (NMR).
  - Prediction
    - Comparative modeling -
    - Threading/Fold recognition
    - Ab initio

Amina Zee

What is structural proteomics/genomics?

- High-throughput determination of the 3D structure of proteins
- Goal: to be able to determine or predict the structure of every protein.
  - Direct determination - X-ray crystallography and nuclear magentic resonance (NMR).
  - Prediction
    - Comparative modeling -
    - Threading/Fold recognition
    - Ab initio

Why structural proteomics?

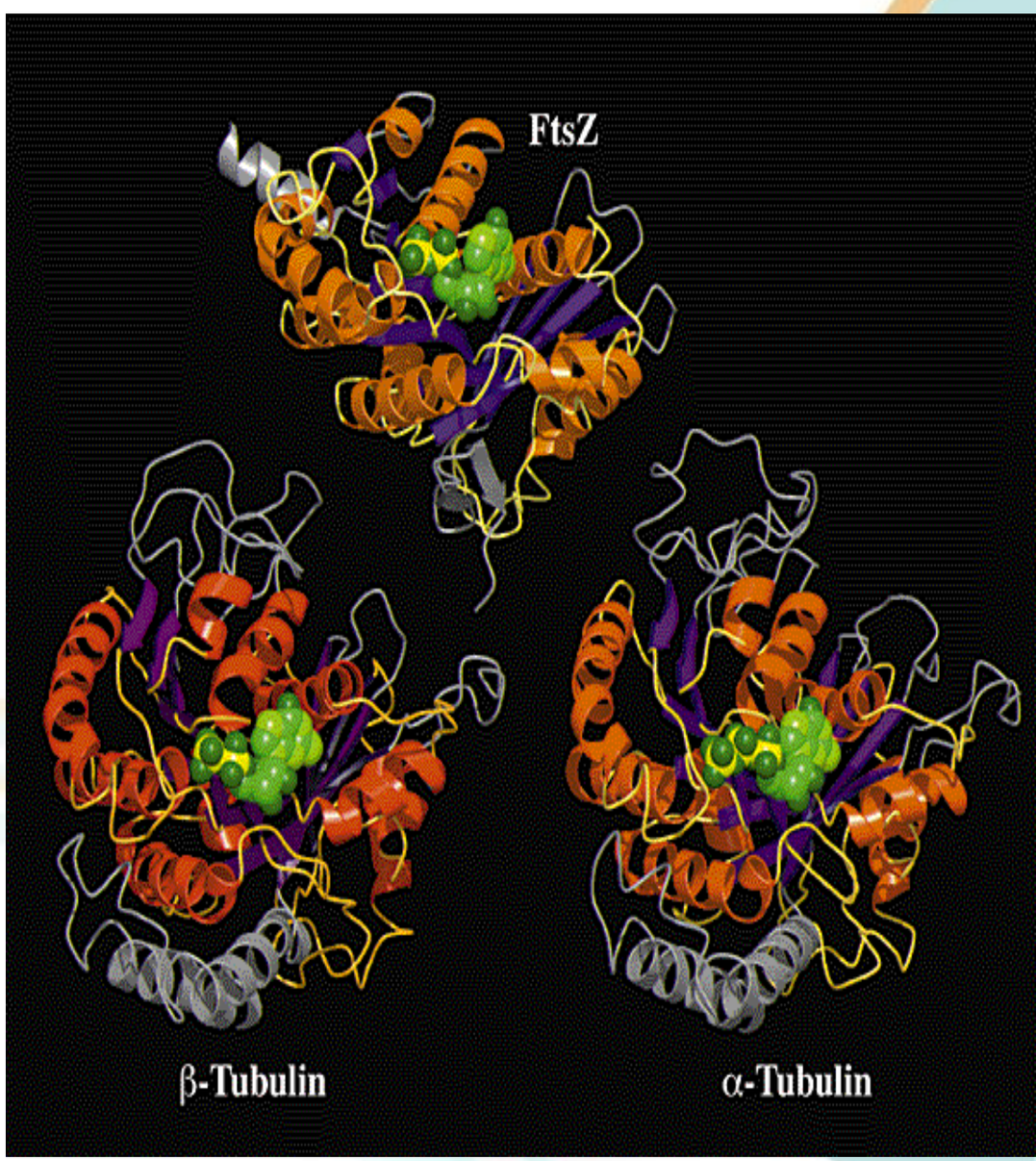
- To study proteins in their active conformation.
  - Study protein:drug interactions
  - Protein engineering
- Proteins that show little or no similarity at the primary sequence level can have strikingly similar structures.

Amina Zee

An example

- FtsZ - protein required for cell division in prokaryotes, mitochondria, and chloroplasts.
- Tubulin - structural component of microtubules - important for intracellular trafficking and cell division.
- FtsZ and Tubulin have limited sequence similarity and would not be identified as homologous proteins by sequence analysis.

Amina Zee



FtsZ and tubulin have little similarity at the amino acid sequence level

Amina Zee

Burns, R., Nature **391**:121-123
Picture from E. Nogales

Are FtsZ and tubulin homologous?

- Yes! Proteins that have conserved secondary structure can be derived from a common ancestor even if the primary sequence has diverged to the point that no similarity is detected.

Amina Zee

Current state of structural proteomics

- As of Feb. 2002 - 16,500 structures
  - Only 1600 non-redundant structures
- To identify all possible folds - predicted another 16,000 novel sequences needed for 90% coverage.
  - Of the 2300 structures deposited in 2000, only 11% contained previously unidentified folds.
- Overall goal - directly solve enough structures directly to be able to computationally model all future proteins.

Protein domains - structure

- “clearly recognizable portion of a protein that folds into a defined structure”
  - Doesn’t have to be the same as the domains we have been investigating with CDD.
  - RbsB proteins as an example.

Main secondary structure elements

α-helix - right handed helical structure

β-sheet - composed of two or more β-strands, conformation is more “zig-zag” than helical.

Amina Zee

Folds/motifs

- How these secondary structure elements come together to form structure.
  - Helix-turn-helix
- Determining the structure of nearly all folds is the goal of structural biology

X-Ray Crystallography

- Make crystals of your protein
  - 0.3-1.0mm in size
  - Proteins must be in an ordered, repeating pattern.
- X-ray beam is aimed at crystal and data is collected.
- Structure is determined from the diffraction data.

Amina Zee

Lec 111
Origin of proteomic

Amina Zee

Lecture-1

Proteomics Origins

- In 1975, the introduction of the 2D gel by O'Farrell who began mapping proteins from *E. coli.*
- Although many proteins could be separated and visualized, they could not be identified.

Amina Zee

Human protein index

- Despite these limitations, shortly thereafter a large-scale analysis of all human proteins was proposed.
- The goal of this project, termed the human protein index, was to use two-dimensional protein electrophoresis (2-DE) and other methods to catalog all human proteins.
- However, lack of funding and technical limitations prevented this project from continuing.

Amina Zee

Proteomics Origins

- The first major technology to emerge for the identification of proteins was the sequencing of proteins by Edman degradation.
- A major breakthrough was the development of microsequencing techniques for electroblotted proteins.

Amina Zee

Proteomics Origins

- Microsequencing technique was used for the identification of proteins from 2-D gels to create the first 2-D databases.
- Improvements in microsequencing technology resulted in increased sensitivity of Edman sequencing in the 1990s to high picomole amounts.

Amina Zee

Proteomics Origins

- One of the most important developments in protein identification has been the development of MS technology.
- The sensitivity of analysis and accuracy of results for protein identification by MS have increased by several orders of magnitude.
- It is now estimated that proteins in the femtomolar range can be identified in gels.
- Because MS is more sensitive, can tolerate protein mixtures, and is amenable to high-throughput operations

Amina Zee

Lec 112
Genomics vs. proteomic

Amina Zee

Lecture-1

Genomics vs. proteomics

- Genomics and proteomics are closely-related fields.
- The main difference between genomics and proteomics is that genomics is the study of the entire set of genes in the genome of a cell whereas proteomics is the study of the entire set of proteins produced by the cell.

Amina Zee

Nature of study material

Genomics: The genome is constant. Every cell of an organism has the same set of genes.

Proteomics: The proteome is dynamic and varies. The set of proteins produced in different tissues varies according to the gene expression.

Amina Zee

Use of High throughput techniques

Genomics: High throughput techniques are used in the genomics to map, sequence, and analyze genomes.

Proteomics: In proteomics, characterization of the 3D structure and the function of proteins is carried out by the use of high throughput methods.

Amina Zee

Techniques involved

Genomics: The techniques involved in genomics include gene sequencing strategies such as directed gene sequencing, whole-genome shotgun sequencing, construction of expressed sequence tags (ESTs), identification of single nucleotide polymorphisms (SNPs)

Amina Zee

Techniques involved

Proteomics:

- Extraction and electrophoretic separation of proteins.
- Digestion of proteins with the use of trypsin into small fragments.
- Determination of the amino acid sequence by mass spectrometry.
- Identification of proteins using the information in the protein databases.

Amina Zee

Importance

Genomics: Genomic studies are important to understand the structure, function, location, regulation of the genes of an organism.

Proteomics: The study of the entire set of proteins produced by a cell type is done in order to understand its structure and function.

Amina Zee

Lec 113
Life and death of a protein

Amina Zee

Lecture-1

Life and Death of a Protein

Proteins are synthesized by the translation of mRNAs into polypeptides on ribosomes.

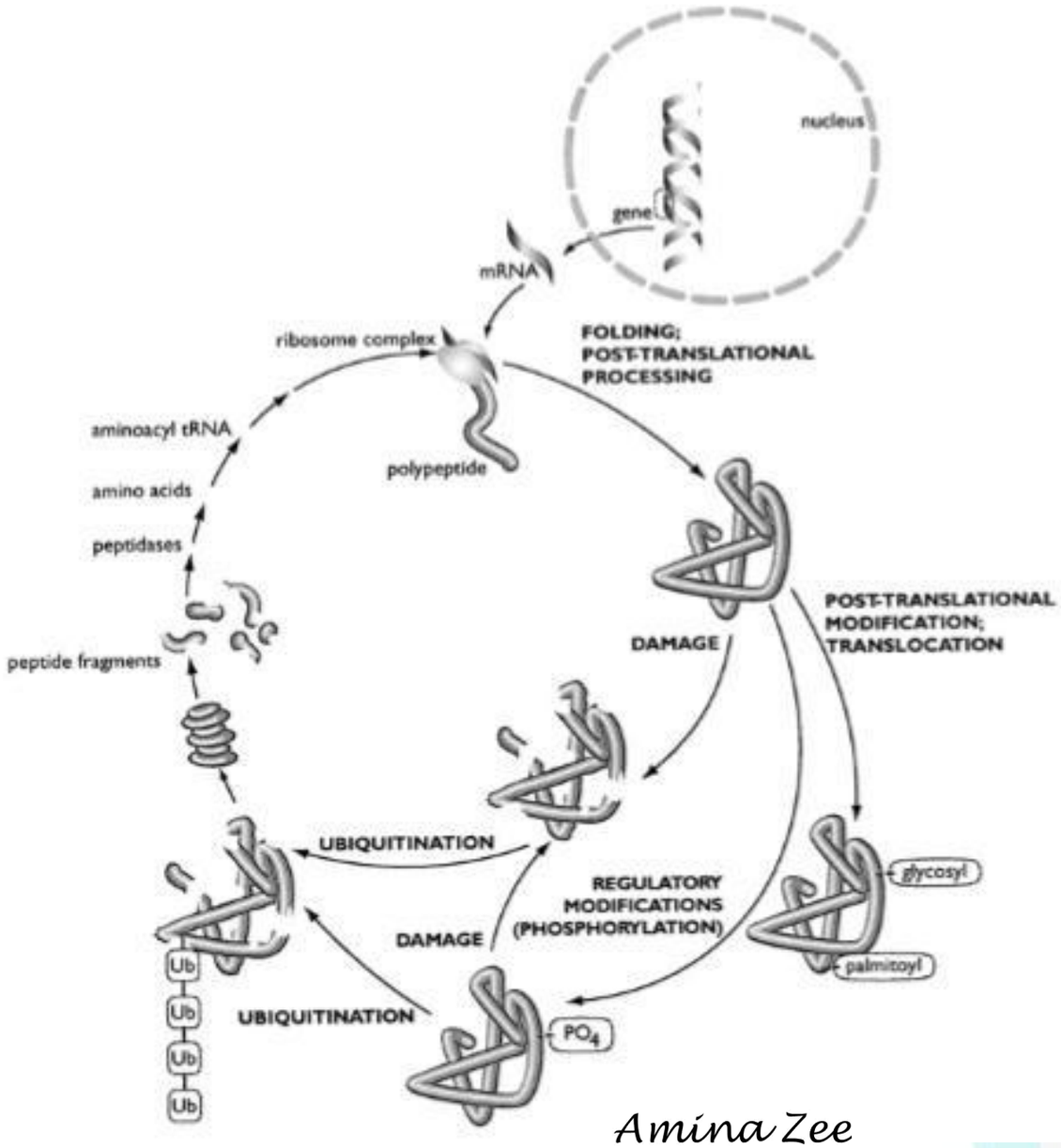
In most cases, the initial polypeptide-translation product undergoes some type of modification before it assumes its functional role in a living system.

These changes are broadly termed “posttranslational modifications” and encompass a wide variety of reversible and irreversible chemical reactions.

Approximately 200 different types of posttranslational modifications have been reported. Some of these are summarized in Fig. 1, which depicts the life cycle of a prototypical protein.

Amina Zee

Life Cycle of the Cell



Amina Zee

Modifications during Protein Cycle

Modifications those occur early in the life of the protein

- Carboxylation of glutamate residues
- Removal of the N-terminal methionine
- Glycosylation
- Addition of Prosthetic groups
- Formation of multisubunit complexes
- Prenylation of cysteine residues assists anchoring of proteins in or on membranes.

These more or less “permanent” modifications and transport ultimately result in the delivery of functional proteins to specific locations in cells.

Amina Zee

Cont…

- The activities of many proteins are then controlled by posttranslational modifications.
- The most prominent and best-understood of these is phosphorylation of serine, threonine, or tyrosine residues.
- Phosphorylation may activate or inactivate enzymes, alter proteinprotein interactions and associations, change protein structures, and target proteins for degradation.
- Protein phosphorylation regulates protein function in diverse contexts and appears to be a key switch for rapid on-off control of signaling cascades, cell-cycle control, and other key cellular functions.

Degradation of Proteins

- Protein modifications appear to be critical to initiating processes that ultimately degrade proteins.
- Phosphorylation of some proteins is rapidly followed by conjugation with ubiquitin, which leads to degradation by the 26S proteasomal complex.
- There evidently are other stimuli for protein ubiquitination and turnover, including oxidative damage and other protein modifications.
- Proteins also undergo degradation by lysosomal enzymes.
- Any protein may be present in many forms at any one time in a cell.
- Collectively, the proteome of a cell comprises all of these many forms of all expressed proteins. This certainly makes the proteome bewilderingly complex.

Amina Zee

Lec 114
Proteins as Modular Stru

Amina Zee

Lecture-1

Proteins as Modular Structures

Segments of amino acid sequences can be considered as functional building blocks or modules.

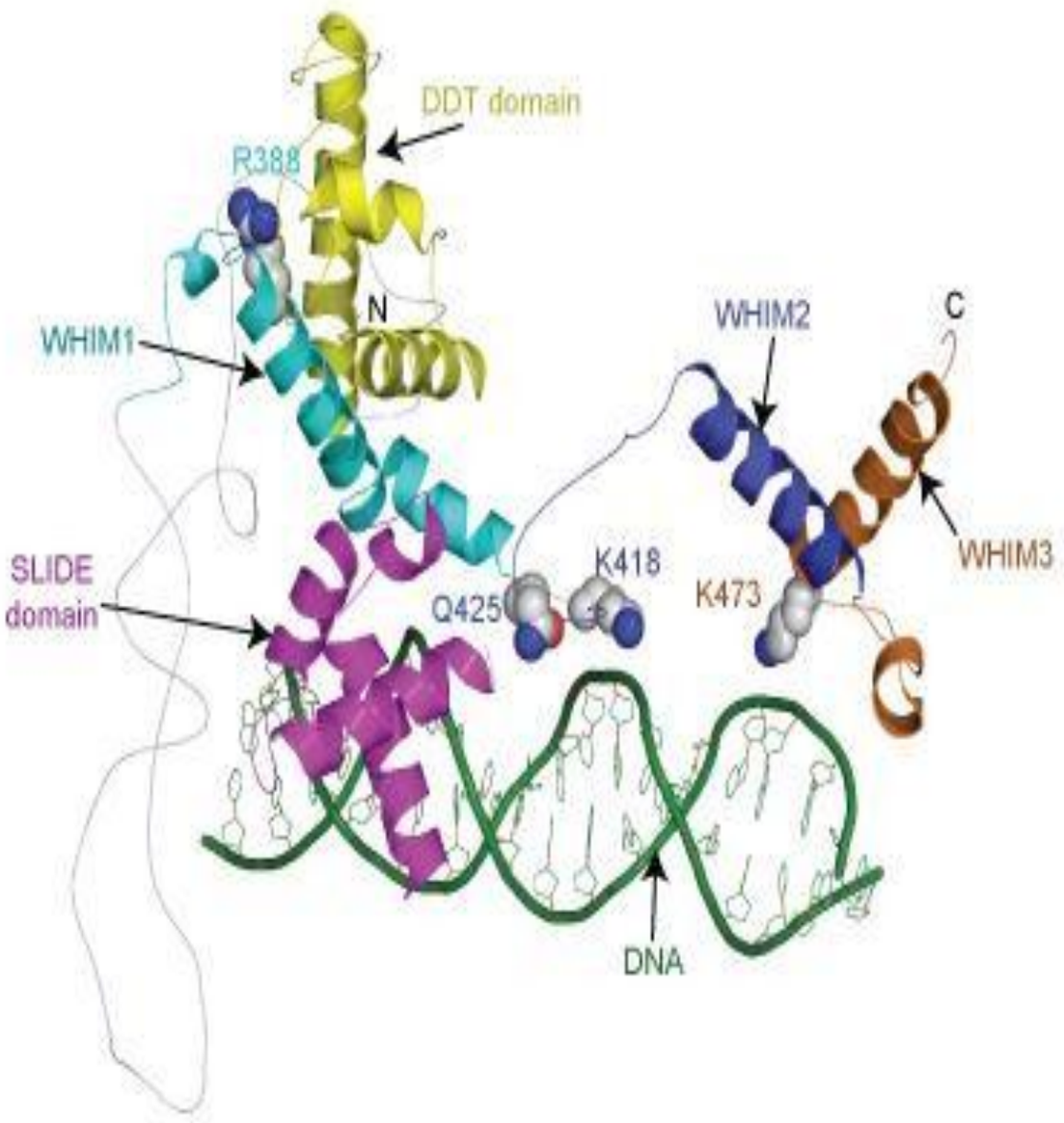
The modular units in proteins that confer specific properties and functions are referred to as “motifs” or “domains”.

Motifs and domains are recognizable sequences that confer similar properties or functions when they occur in a variety of proteins.

In some cases, amino acid sequences within motifs and domains are highly conserved and do not vary from protein to protein.

In other cases, some key amino acids occur in a reproducible relationship to each other in a sequence, even though various substitutions in other amino acids occur

Amina Zee


DDT domain
R388
N
WHIM1
WHIM2
C
WHIM3
SLIDE
domain
K418
Q425
K473
DNA

Amina Zee

Proteins as Modular Structures

Longer amino acid sequences often form domains, which confer specific properties or functions on a protein.

Some domain structures refer simply to sequences that confer a bulk physical property to a segment of the polypeptide, such as transmembrane domains, which simply form helices that span a lipid bilayer membrane.

Other domain structures provide hydrogen bonding or other contacts for key enzyme substrates or prosthetic groups.

In many cases, domains are made up of combinations of units of secondary structure, such as helix-loop-helix domains.

Amina Zee

Lec 115
Proteomics Introducción

Amina Zee

Lecture-1

Protein Localization

- Any process in which a protein is transported to, or maintained in, a specific location.
- Cells are organized into many different compartments such as the cytosol, nucleus, endoplasmic reticulum (ER), and mitochondria. Almost all proteins are made in the cytosol, yet each cellular compartment requires a specific set of proteins.
- How does the cell regulate protein localization to be sure that proteins end up where they should?

Amina Zee

Compartments of an Animal Cell

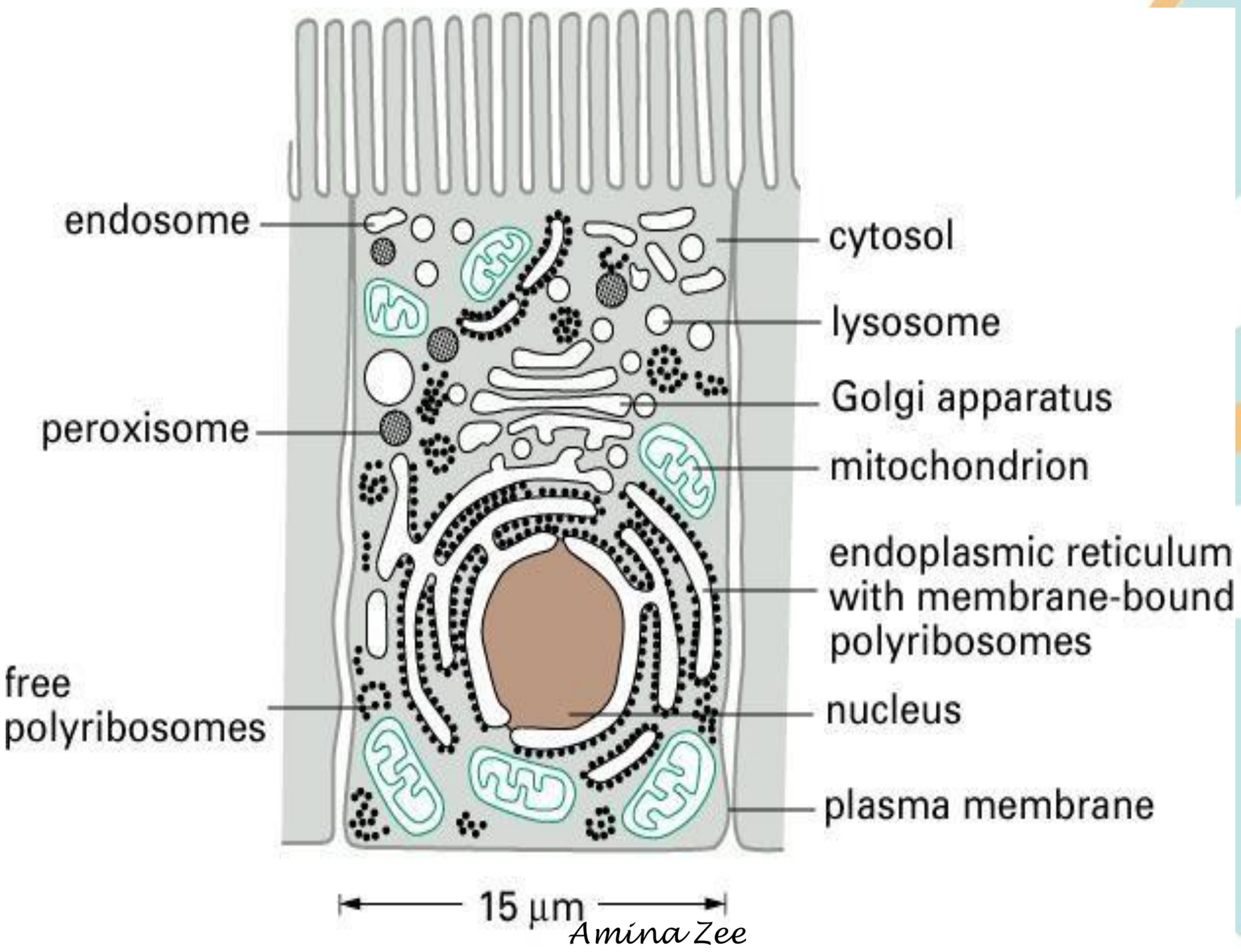


Figure 12–1. Molecular Biology of the Cell, 4th Edition.

Functions of major intracellular compartments:

- Nucleus - contains main genome, DNA and RNA synthesis.
- Cytosol - most protein synthesis, glycolysis and metabolic pathways synthesizing amino acids, nucleotides.
- Endoplasmic reticulum - synthesis of membrane proteins, lipid synthesis.
- Golgi apparatus - covalent modification of proteins from ER, sorting of proteins for transport to other parts of the cell.

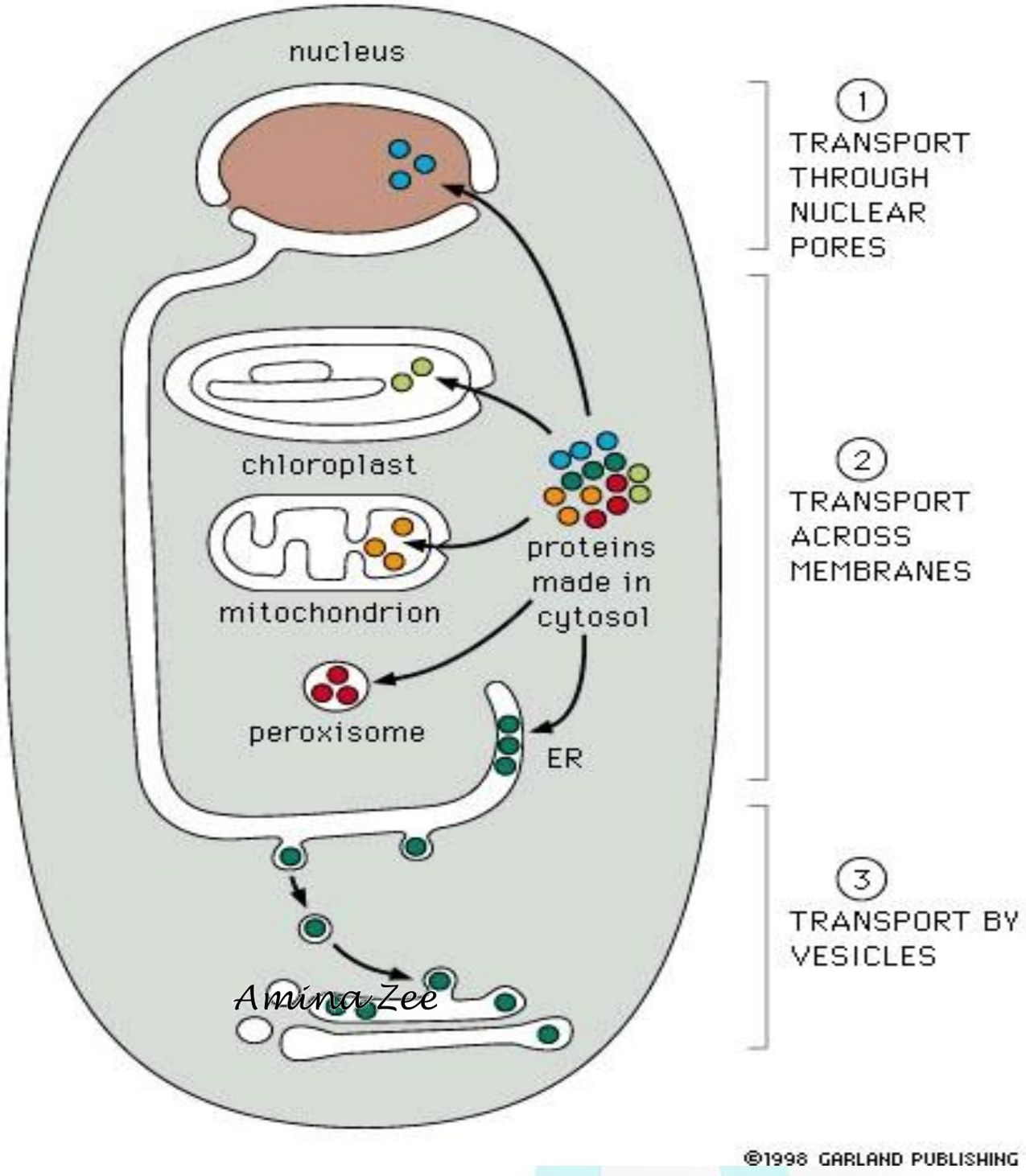
Amina Zee

Cont...

- Mitochondria and chloroplasts (plants) - ATP synthesis.
- Lysosomes - degradation of defunct intracellular organelles and material taken in from the outside of the cell by endocytosis.
- Endosomes - sorts proteins received from both the endocytic pathway and from the Golgi apparatus.
- Peroxisomes - oxidize a variety of small molecules.

Three basic modes of protein Localization

1. Gated transport
2. Transmembrane transport
3. Vesicular transpo



Roadmap of protein traffic

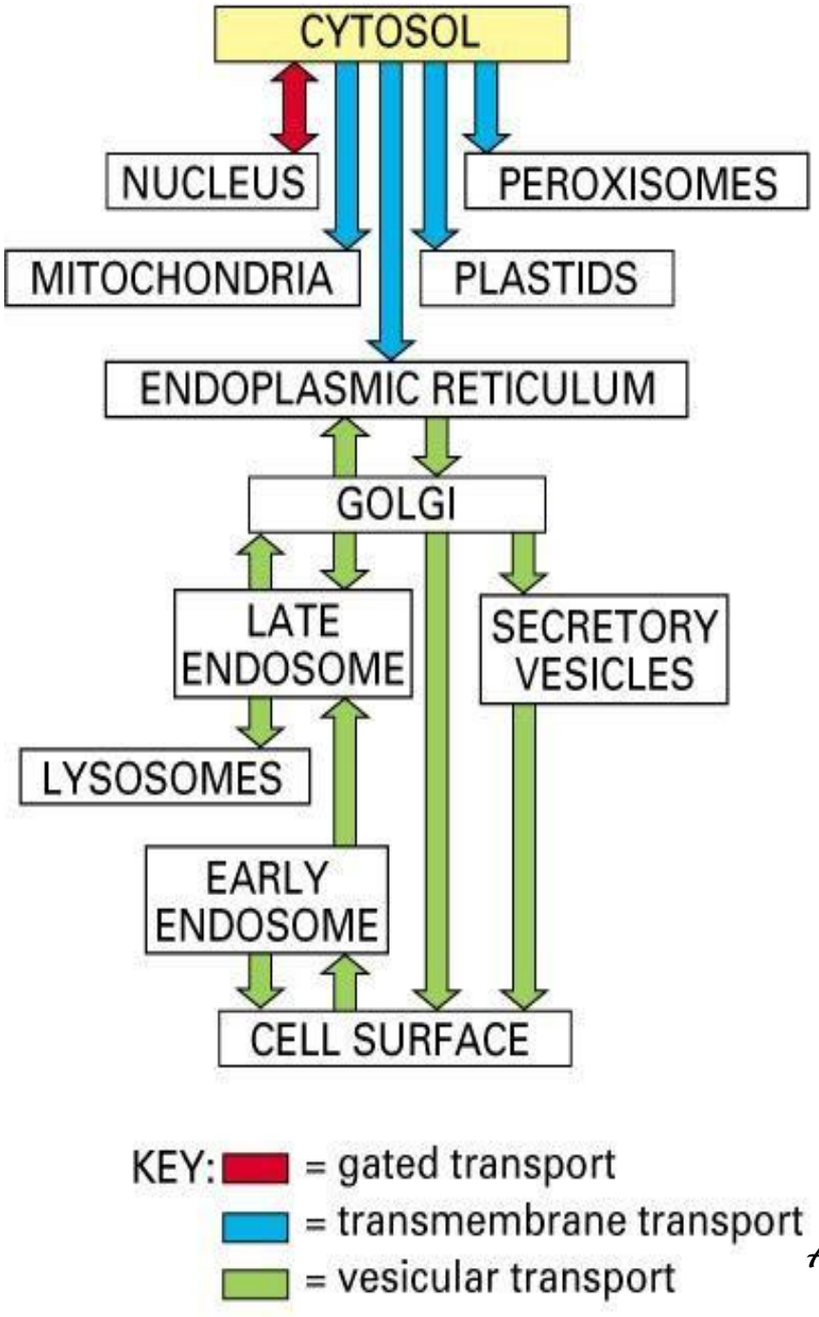


Figure 12–6. Molecular Biology of the Cell, 4th Edition.

The genesis and function of internal compartments depends on the appropriate targeting of proteins

Some of the green route illustrated.

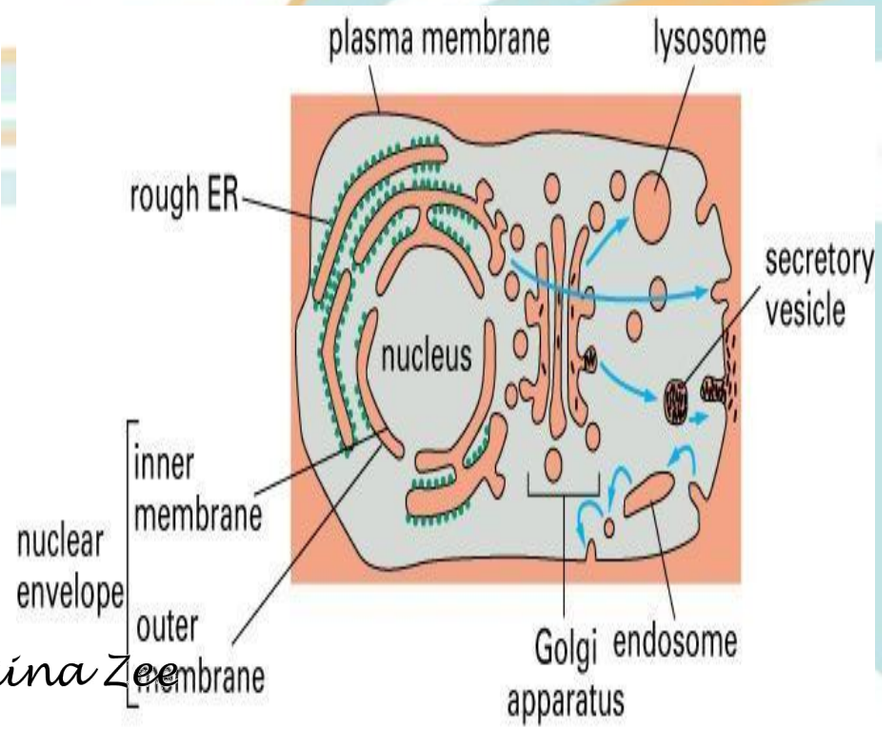


Figure 12–5. Molecular Biology of the Cell, 4th Edition.

Amina Zee

Lec 116
Chemical Compositio of Protein

Amina Zee

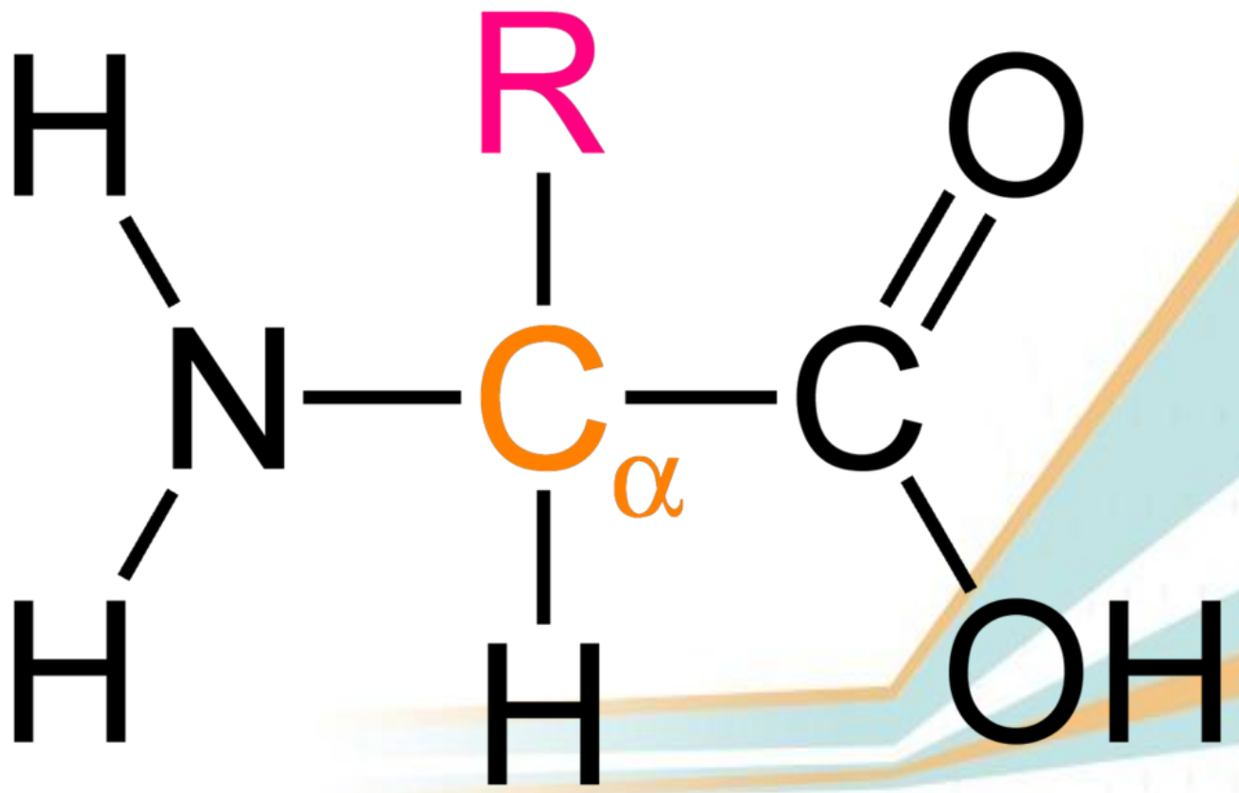
Lecture-

Chemical composition of proteins

- Proteins are polymers of amino acids.
- They range in size from small to very large.
- All the proteins are made up of Twenty different types of amino acids. So these amino acids are called standard amino acids.

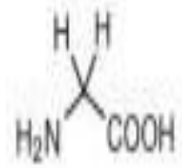
Amina Zee

Chemical composition of Proteins

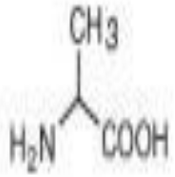


Amina Zee

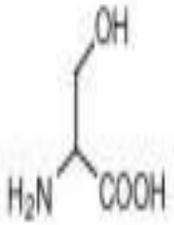
Chemical composition of proteins



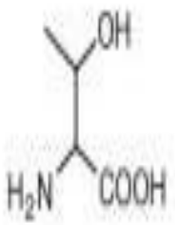
Glycine (Gly, G)



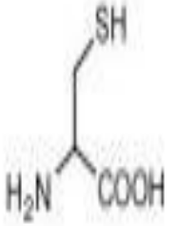
Alanine (Ala, A)



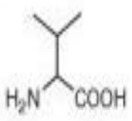
Serine (Ser, S)



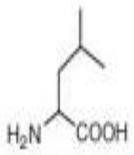
Threonine (Thr, T)



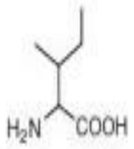
Cysteine (Cys, C)



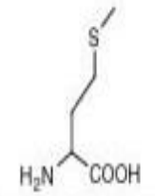
Valine (Val, V)



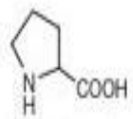
Leucine (Leu, L)



Isoleucine (Ile, I)



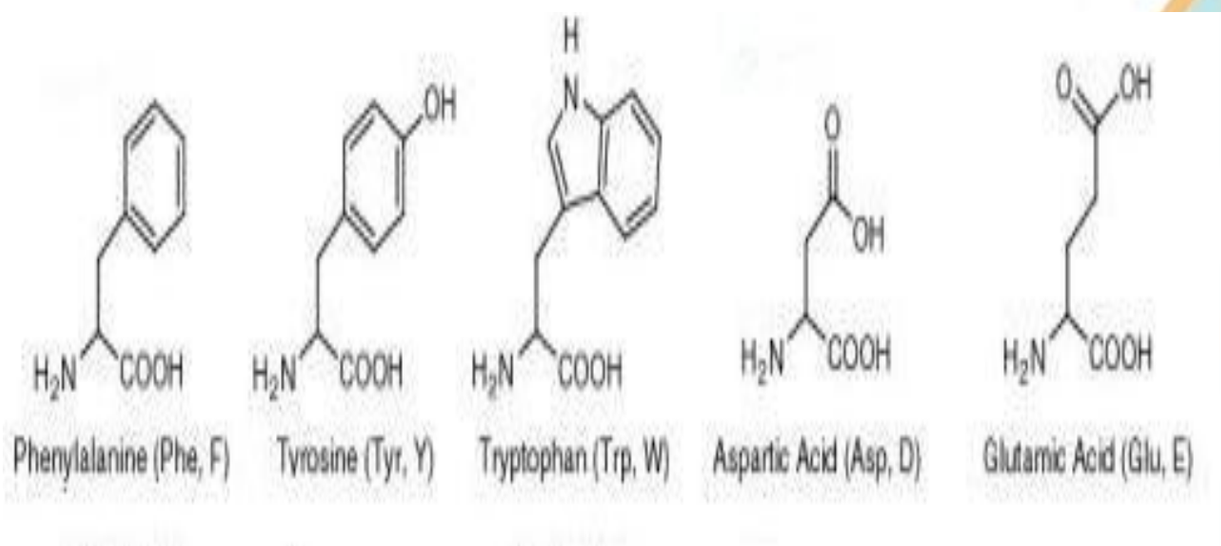
Methionine (Met, M)

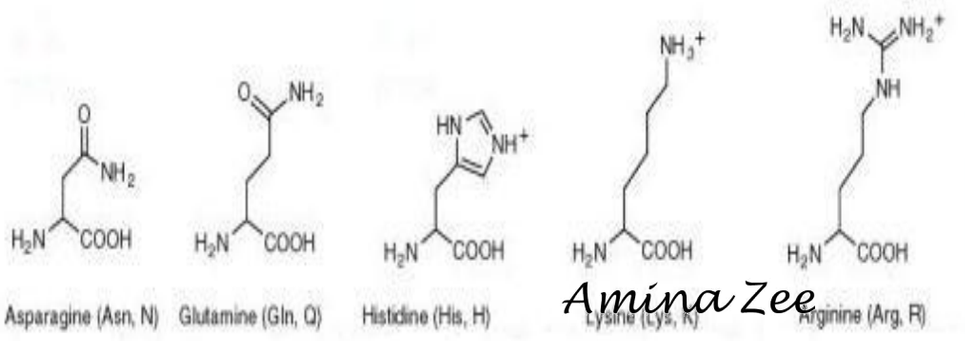


Proline (Pro, P)

Amina Zee

Chemical composition of proteins



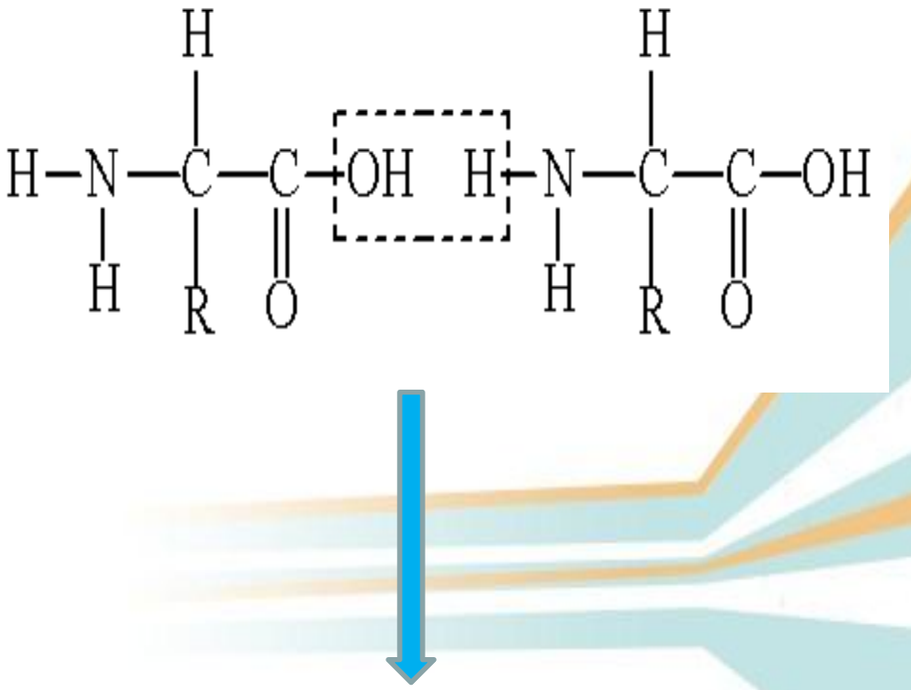


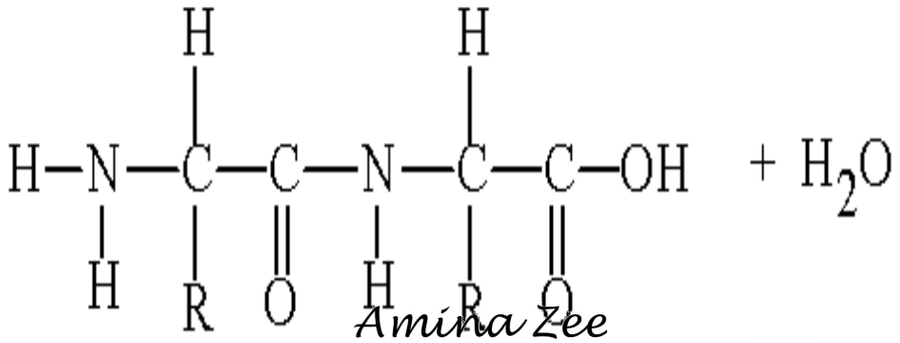
Amina Zee

Chemical composition of proteins

- In a protein molecule, each amino acid residue is joined to its neighbour by a specific type of covalent bond which is called Peptide Bond.

Chemical composition of proteins





Amina Zee

Chemical composition of proteins

- Amino acids can successively join to form dipeptides, tripeptides, tetrapeptides, oligo peptides and polypeptides.

Chemical composition of proteins

Amina Zee

Lec 117
Introduction to Homology Modelling

Amina Zee

Introduction to Homology Modelling

Background

- Proteins are 3D molecules with their own unique structures

- Protein structure is reflective of the protein function

- Protein structure includes 1', 2', 3' and 4' structures

Amina Zee

Introduction to Homology Modelling

Background

- 1’ structure of proteins is the sequence of proteins and can be obtained by mass spectrometry

- 2’ structures formed by proteins are the helices, beta sheets, loops and coils

Amina Zee

Introduction to Homology Modelling

Background

- 3’ structure of proteins is the combination of 2’ structures such that the overall protein structure is formed

- 4’ protein structure is formed when two or more proteins complex together

Amina Zee

Introduction to Homology Modelling

Background

- X-Ray Crystallography and NMR Spectroscopy are used to find the structures of proteins
- However, these methods are difficult and expensive
- Solution: Prediction of structures

Amina Zee

Introduction to Homology Modelling

Introduction

- Protein sequence gives rise to its structure
- If another protein which has a similar sequence also has its structure known, the structure of an unknown protein can be predicted based on that similar protein

Amina Zee

Introduction

- So, it is then possible to identify unknown protein structures by just examining the homologous protein sequences

Introduction to Homology Modelling

Conclusions

- **Sequence Identity**
- **Alignment Length**

Which combination of identity and alignment length is suitable for best for structure prediction?

Amina Zee

Homology Modelling

Lec 118
Homology, Paralogy and Orthology

Amina Zee

Homology, Paralogy and Orthology

Background

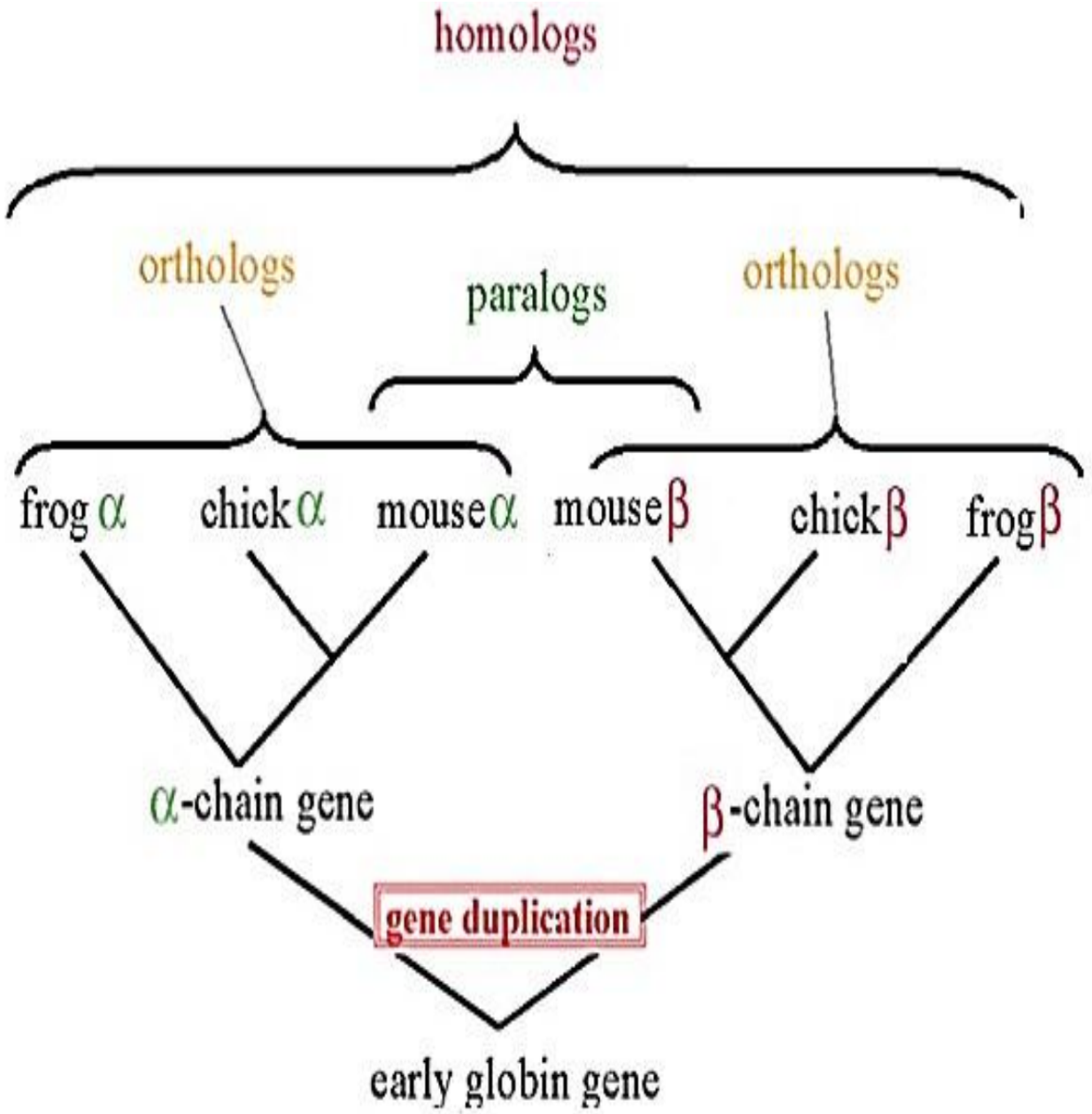
- In homology modelling, proteins with similar 1’ sequences are considered

- Given that one of them has its 3’ structure known, then the 3’ structure of other protein can be predicted

Amina Zee

Homology, Paralogy and Orthology

Homology: Paralogy vs. Orthology

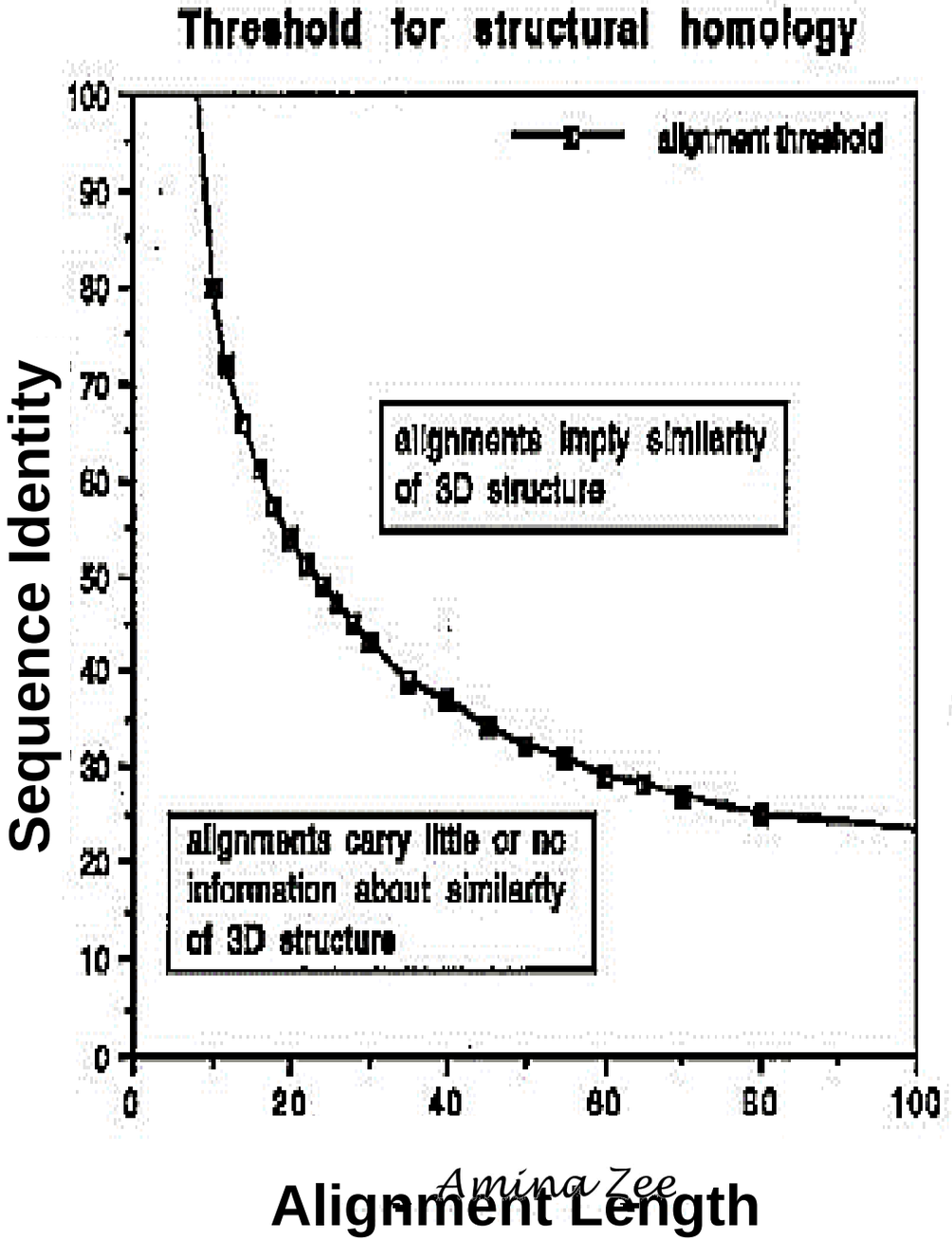


Amina Zee

http://bioweb.uwlax.edu/GenWeb/Molecular/Bioinformatics/Unit_4/Lab_4-2/lab_4-2.htm

Homology, Paralogy and Orthology

How much homology is required or better?



Amina Zee

http://www.cmbi.ru.nl/gvteach/astra/lectures/homology_modelling.ppt

Homology, Paralogy and Orthology

Conclusions

- Good sequence alignment and identity ensures that homology modelling will give accurate results

- Next, what is the workflow for homology modelling?

Homology Modelling

Lec 119
Workflow of Structural Modelling

Amina Zee

Workflow of Structural Modelling

Background

- Homology modelling is used to predict structures of proteins having high sequence similarity with other proteins with known structures!

- Let's consider the workflow of homology modelling

Amina Zee

Workflow of Structural Modelling

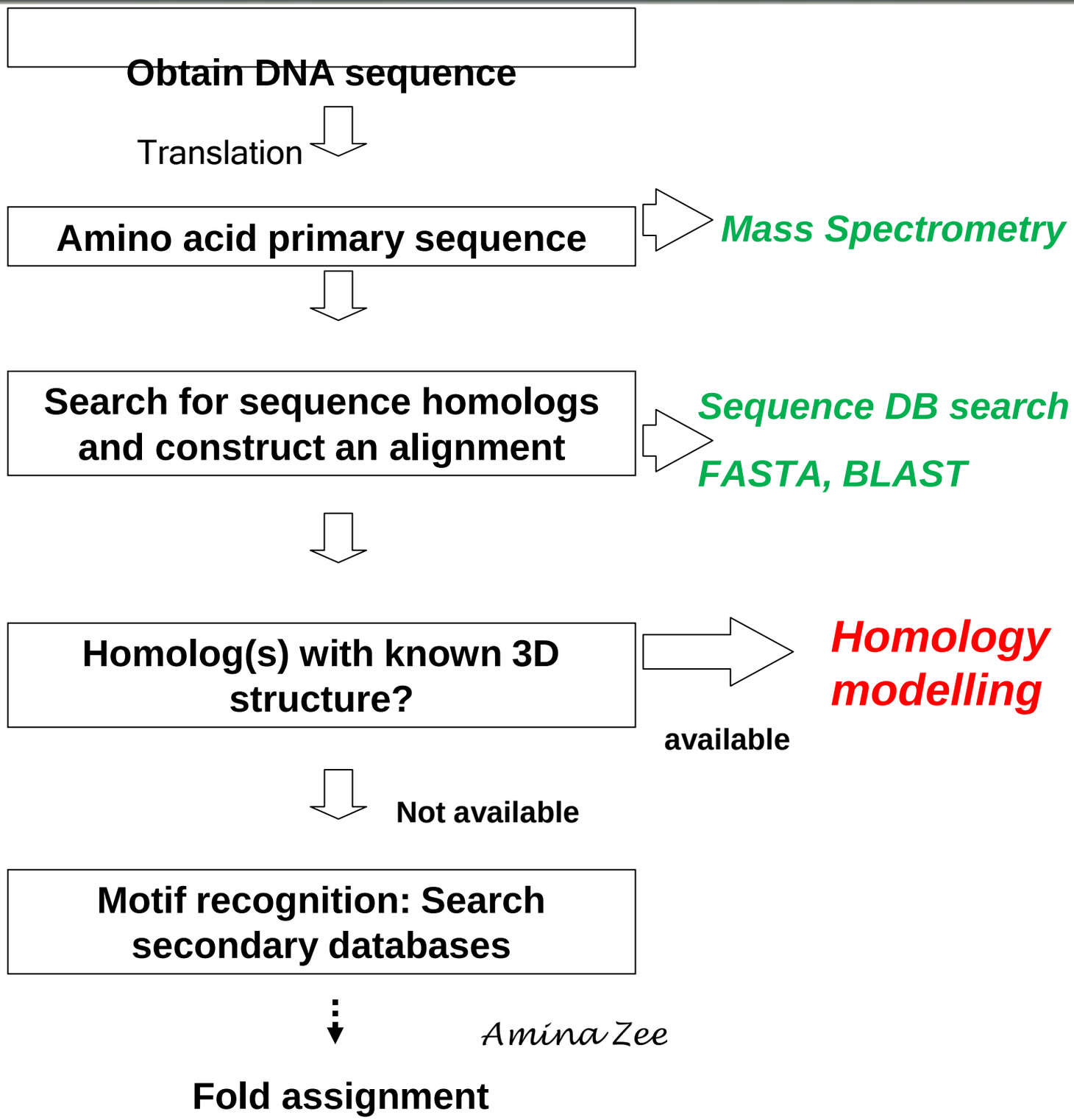
Introduction

Overall, there are three different strategies for structure prediction

1. **Homology Modelling**
2. **Threading/Fold Recognition**
3. **Ab Initio Modelling**

Amina Zee

Workflow of Structural Modelling



Workflow of Structural Modelling

Conclusions

- Next, we will proceed to perform homology modelling
- For that there is a seven step procedure which we will see in the next module.

Amina Zee

Lec 120
Seven Steps to Homology Modelling – I

Amina Zee

Seven Steps to Homology Modelling - I

Background

Protein structure can be predicted by 3 methods:

1. Homology Modelling

2. Fold Recognition / Threading

3. Ab Initio Modelling

Amina Zee

Seven Steps to Homology Modelling - I

Introduction

- Let's start by looking at Homology Modelling
- There are seven salient steps in any Homology Modelling pipeline
- Definition of Template (known) & Target (unknown)

Amina Zee

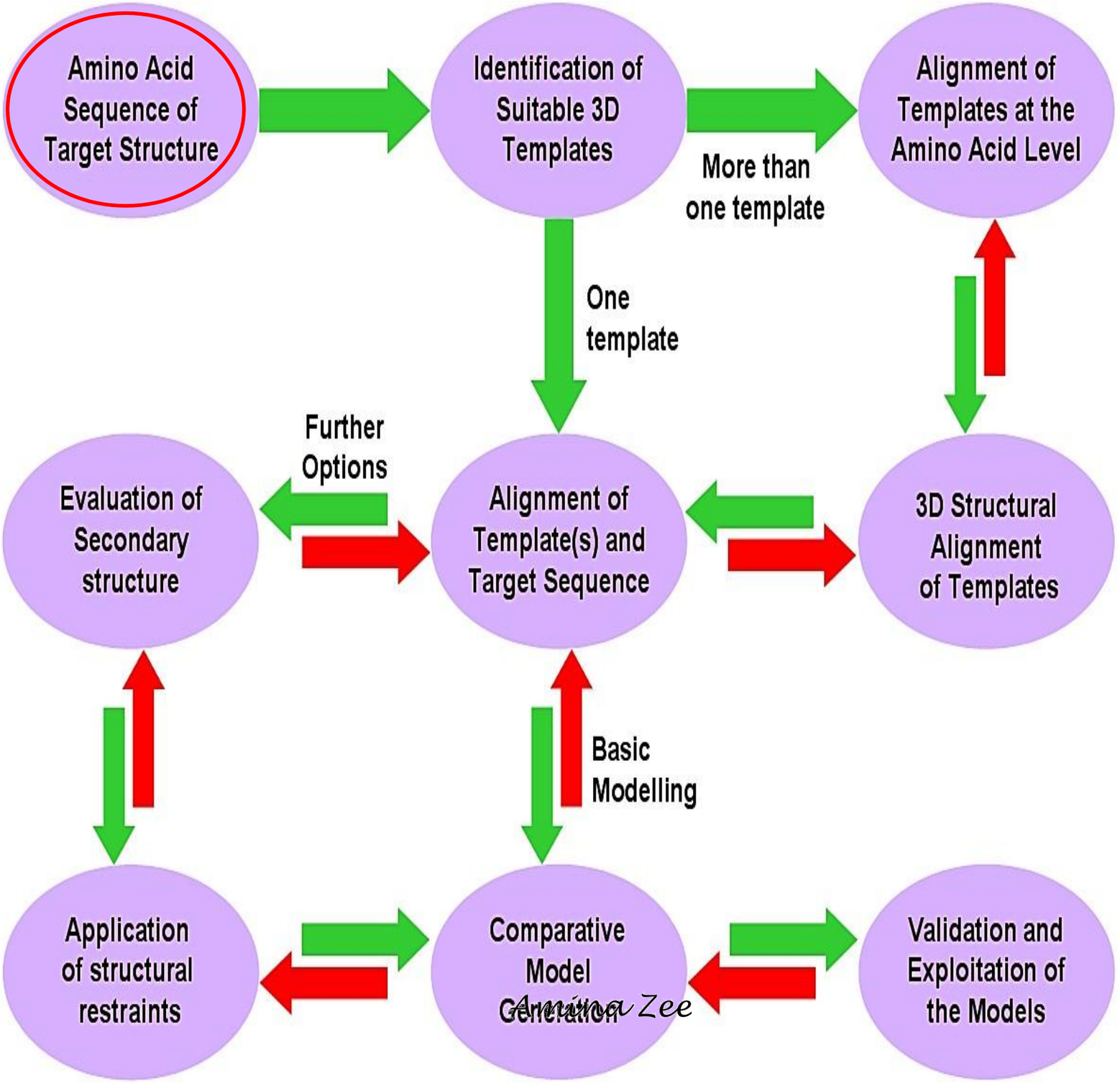
Seven Steps to Homology Modelling - I

Homology modeling of the target structure can be done as follows:

1. Template recognition and initial alignment
2. Alignment correction
3. Backbone generation
4. Loop modeling
5. Side-chain modeling
6. Model optimization
7. Model validation

Amina Zee

Seven Steps to Homology Modelling - I



Seven Steps to Homology Modelling - I

Conclusions

- Homology modelling works in seven steps
- It is a repetitive process
- Next, we will look at each step in detail!

Amina Zee

Homology Modelling

Lec 121

Seven Steps to Homology Modelling – II

Amina Zee

Seven Steps to Homology Modelling - II

Background

Template recognition and initial alignment

Alignment correction

Backbone generation

Loop modeling

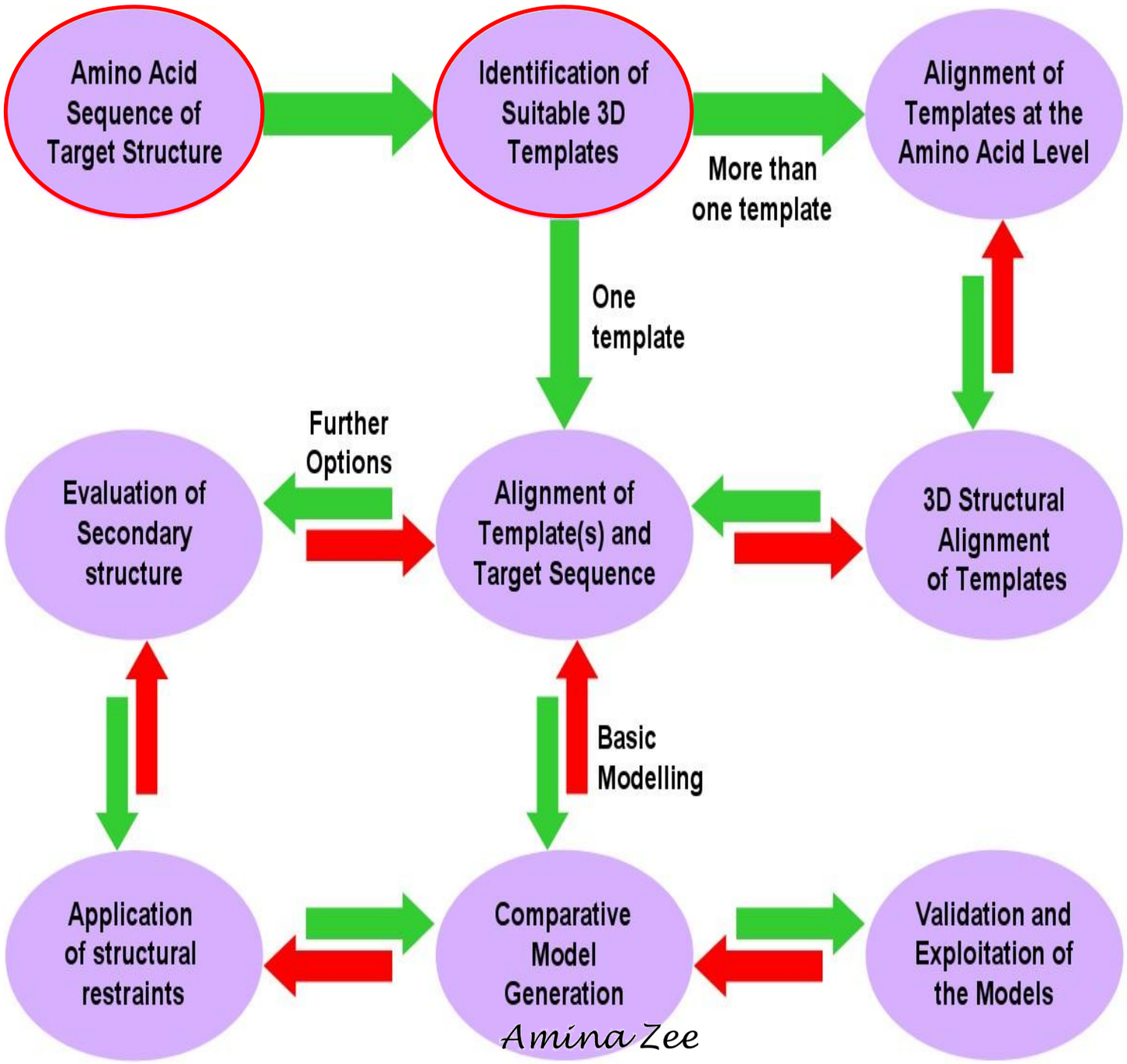
Side-chain modeling

Model optimization

Model validation

Amina Zee

Seven Steps to Homology Modelling - II



Seven Steps to Homology Modelling - II

Compare the sequence of the unknown protein with all the sequences of known structures stored in the Protein Data Bank (PDB).

BLAST this sequence against PDB sequences - Obtain a list of known protein structures that match the sequence.

BLAST uses a residue exchange scoring matrix.
Residues that are easily exchanged (e.g. Ile to Leu) get a better score than residues that have different properties (for example Glu to Trp).
Function specific conserved residues get the best score (e.g. Cys to Cys).

Amina Zee

Seven Steps to Homology Modelling - II

BLAST will provide a list of possible templates for the unknown structure
To make the best initial alignment, BLAST uses an alignment-matrix based on the residue exchange matrix and adds extra penalties for opening and extension of a gap between residues.

The target-sequence is sent to a BLAST server, which searches the PDB to obtain a list of possible templates and their alignments.
The best hit has to be chosen, which is not necessarily the first one.

Amina Zee

Seven Steps to Homology Modelling - II

Conclusions

- Now the template and target are selected
- Next, we perform fine-tuning of alignment and introduce corrections to ready the mismatches and gaps

Amina Zee

Homology Modelling

Lec 122

Seven Steps to Homology Modelling – III

Amina Zee

Seven Steps to Homology Modelling - III

Background

Template recognition and initial alignment

Alignment correction

Backbone generation

Loop modeling

Side-chain modeling

Model optimization

Model validation

Amina Zee

Seven Steps to Homology Modelling - III

Fine tune and adjust the BLAST alignments

Example: Ala -> Glu is possible but unlikely in a hydrophobic core, so these residues should not be aligned.

Use MSA tools (e.g. ClustalW), to find the residues and properties that need to be conserved.

Examine the template structure to check which residues are in the core hence less likely to change than the residues at the outside.

Amina Zee

Seven Steps to Homology Modelling - III

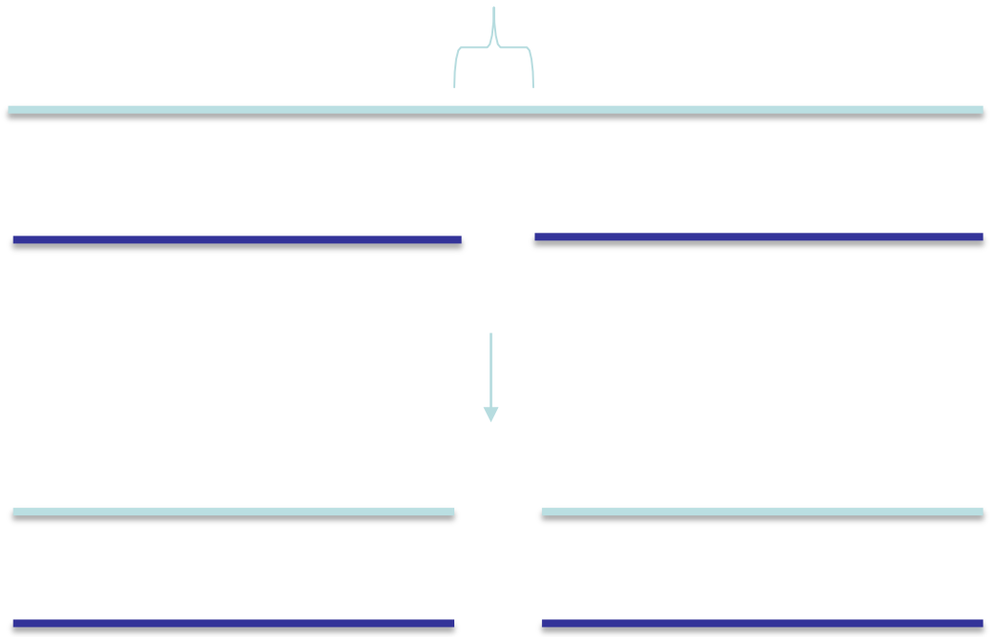
Insertions and deletions can be made in those parts of the sequence which are highly variable .

Note: MSA can be helpful to find these places.

Gaps have to be shifted around until they are as small as possible.

Amina Zee

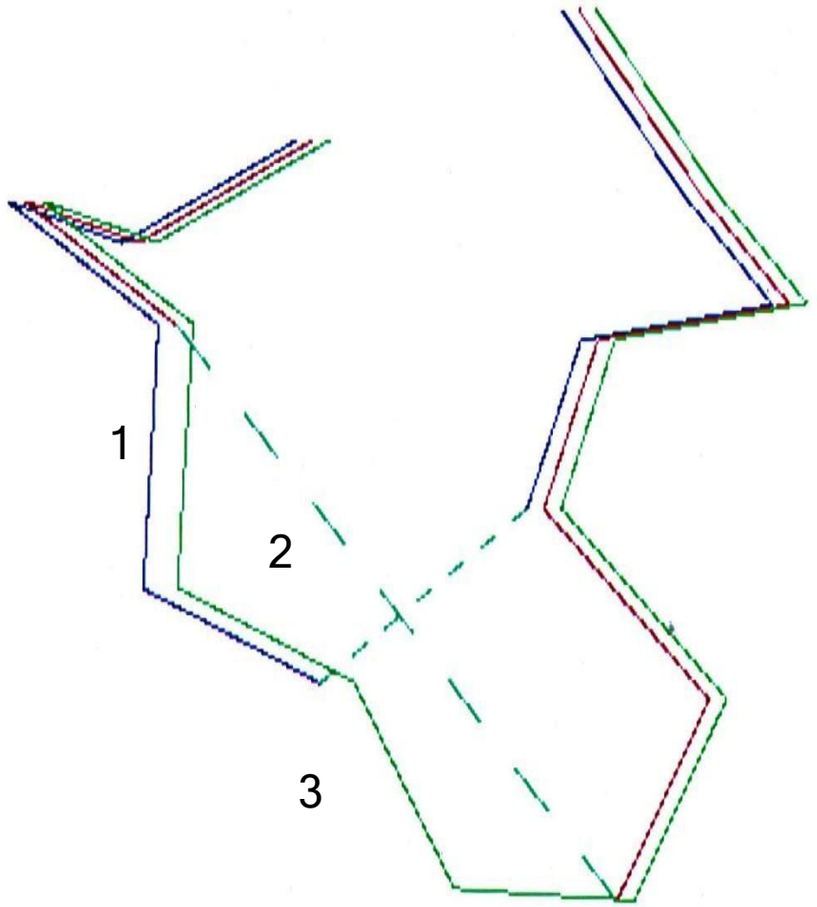
Seven Steps to Homology Modelling - III



Note: After a deletion of 3 residues a big gap occurs in the RED structure, which was the best alignment.

Amina Zee

Seven Steps to Homology Modelling - III



Remember that deletion of 3 residues left a big gap in the RED structure, which was the best alignment.

After shifting several residues, the gap is much smaller (blue structure) and more likely to be correct.

Amina Zee

Seven Steps to Homology Modelling - III

Conclusions

- The alignment now stands fine tuned and corrected
- Gaps and mismatches have been evaluated and adjusted
- Next step, using this alignment, assemble the backbone

Amina Zee

Homology Modelling

Lec 123

Seven Steps to Homology Modelling – IV

Amina Zee

Seven Steps to Homology Modelling - IV

Background

Template recognition and initial alignment

Alignment correction

Backbone generation

Loop modeling

Side-chain modeling

Model optimization

Model validation

Amina Zee

Seven Steps to Homology Modelling - IV

The coordinates of the template-backbone are copied to the target.

When the residues are identical, the side-chain coordinates are also copied.

Note that PDB-file may contain small offsets / errors, so try to use multiple similar templates.

Amina Zee

Seven Steps to Homology Modelling - IV

Next....

Template recognition and initial alignment

Alignment correction

Backbone generation

Loop modeling

Side-chain modeling

Model optimization

Model validation

Amina Zee

Seven Steps to Homology Modelling - IV

When the target (unknown structure) sequence contains a gap, **one option is to delete the corresponding residues in template.**

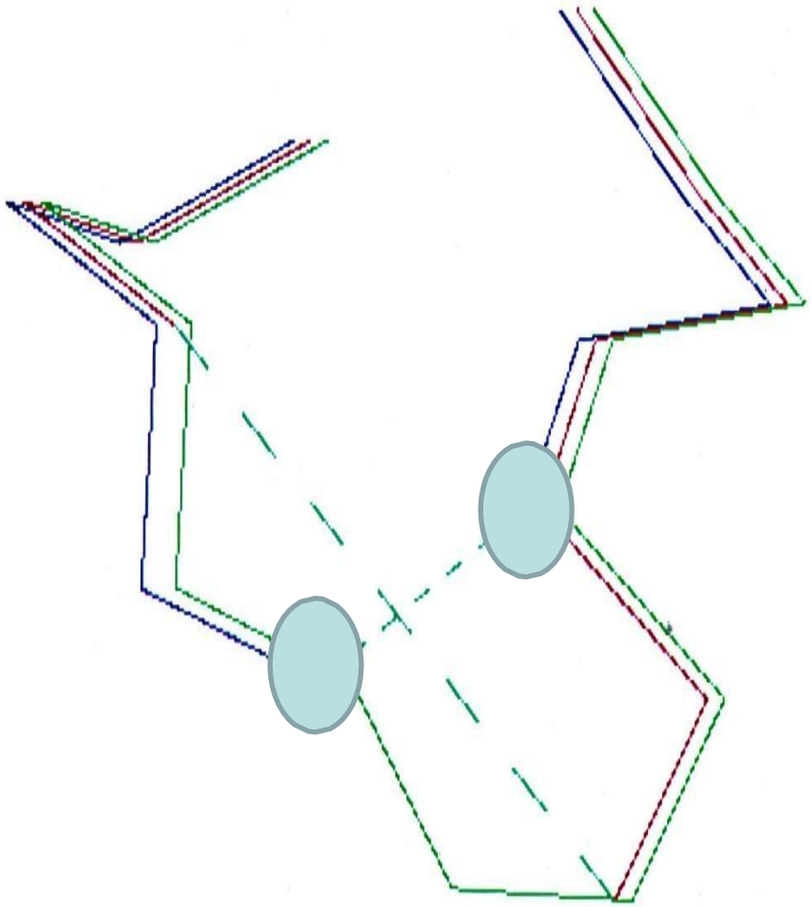
- But this creates a fracture in the template!

When the template (known structure) sequence contains a gap, there are no backbone coordinates known for these residues in model.
The target backbone has to be cut to insert newer residues.

These major changes cannot be modeled in secondary structure elements
Hence, place them in loops and strands.
Therefore, surface loops are flexible and difficult to predict.

Amina Zee

Seven Steps to Homology Modelling - IV



One way to handle loops is to take some residues before and after the insertion as "anchor" residues and search the PDB for loops with the same anchor-residues.
The best loop is simply copied in the model.

Amina Zee

Seven Steps to Homology Modelling - IV

Conclusions

- **The protein backbone is ready!**
- **Next, loops were modelled and used to bridge gaps**
- **Next step, using this backbone and loop choices, place the side-chains**

Amina Zee

Lec 124

Seven Steps to Homology Modelling – V

Amina Zee

Seven Steps to Homology Modelling - V

Background

Template recognition and initial alignment

Alignment correction

Backbone generation

Loop modeling

Side-chain modeling

Model optimization

Model validation

Amina Zee

Seven Steps to Homology Modelling - V

Note that the conserved residues were already copied!
Now, we just need to place the side chains

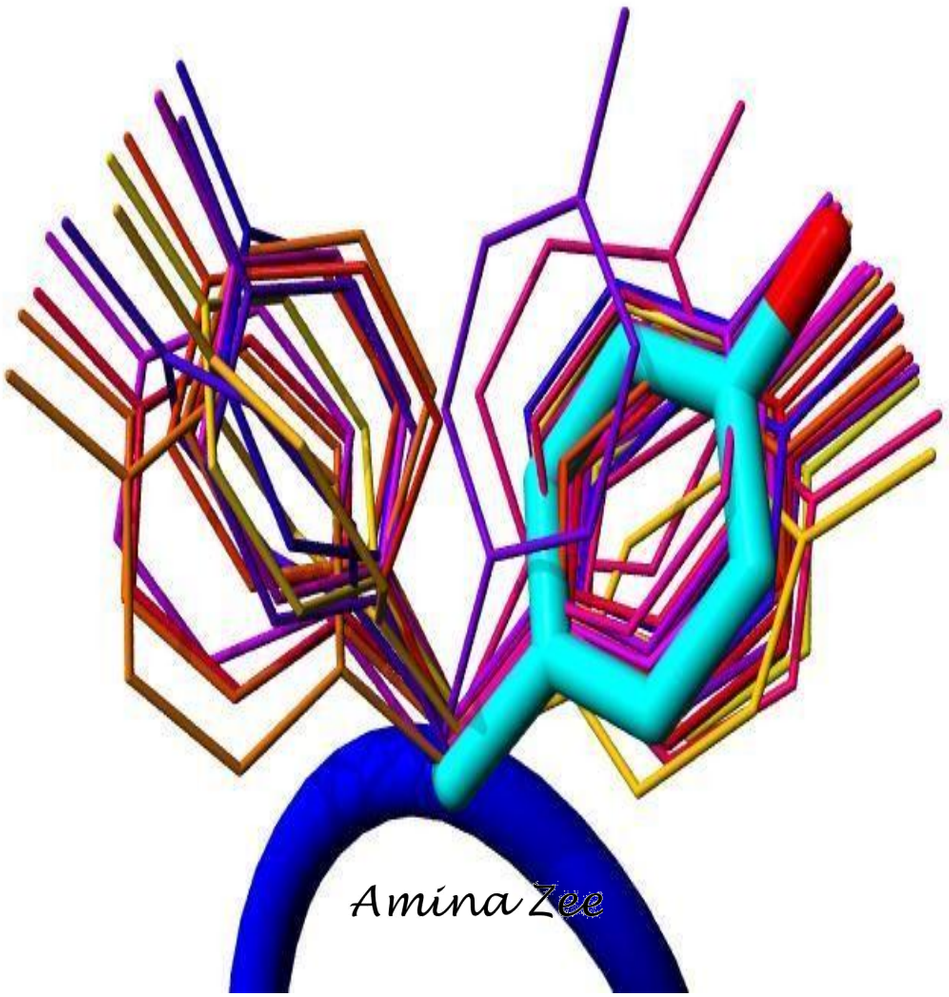
Copy the torsion angles C-alpha/beta to the target!
Rotamers tend to be conserved in homologous proteins and can be predicted as backbone configurations strongly prefer a specific rotamer.

Moreover, libraries of flanking residues can also help estimate the side chain positioning.

Amina Zee

Seven Steps to Homology Modelling - V

The backbone of tyrosine strongly prefers two rotamers and the real side-chain may fit one of them!



Seven Steps to Homology Modelling - V

Next....

Template recognition and initial alignment

Alignment correction

Backbone generation

Loop modeling

Side-chain modeling

Model optimization

Model validation

Amina Zee

Seven Steps to Homology Modelling - V

What is the need for further optimization?

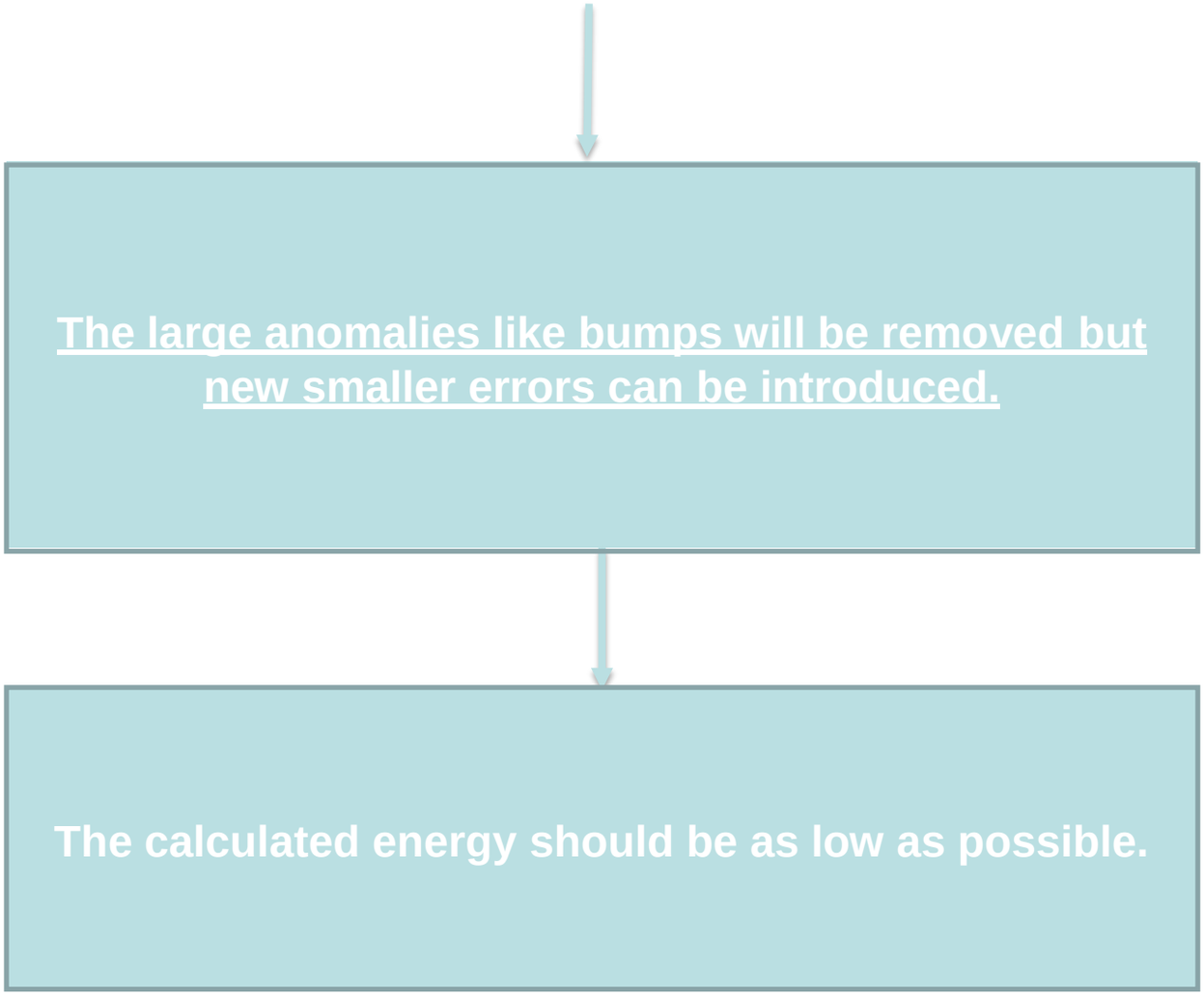
Because the updated side-chains can effect the backbone, and this can effect the structure prediction.

Optimization can be done by performing refinements using Molecular Dynamics simulations of the model.

The model is placed in a force-field and the movements of the molecules are followed in time, this mimics the folding of the protein.

Amina Zee

Seven Steps to Homology Modelling - V

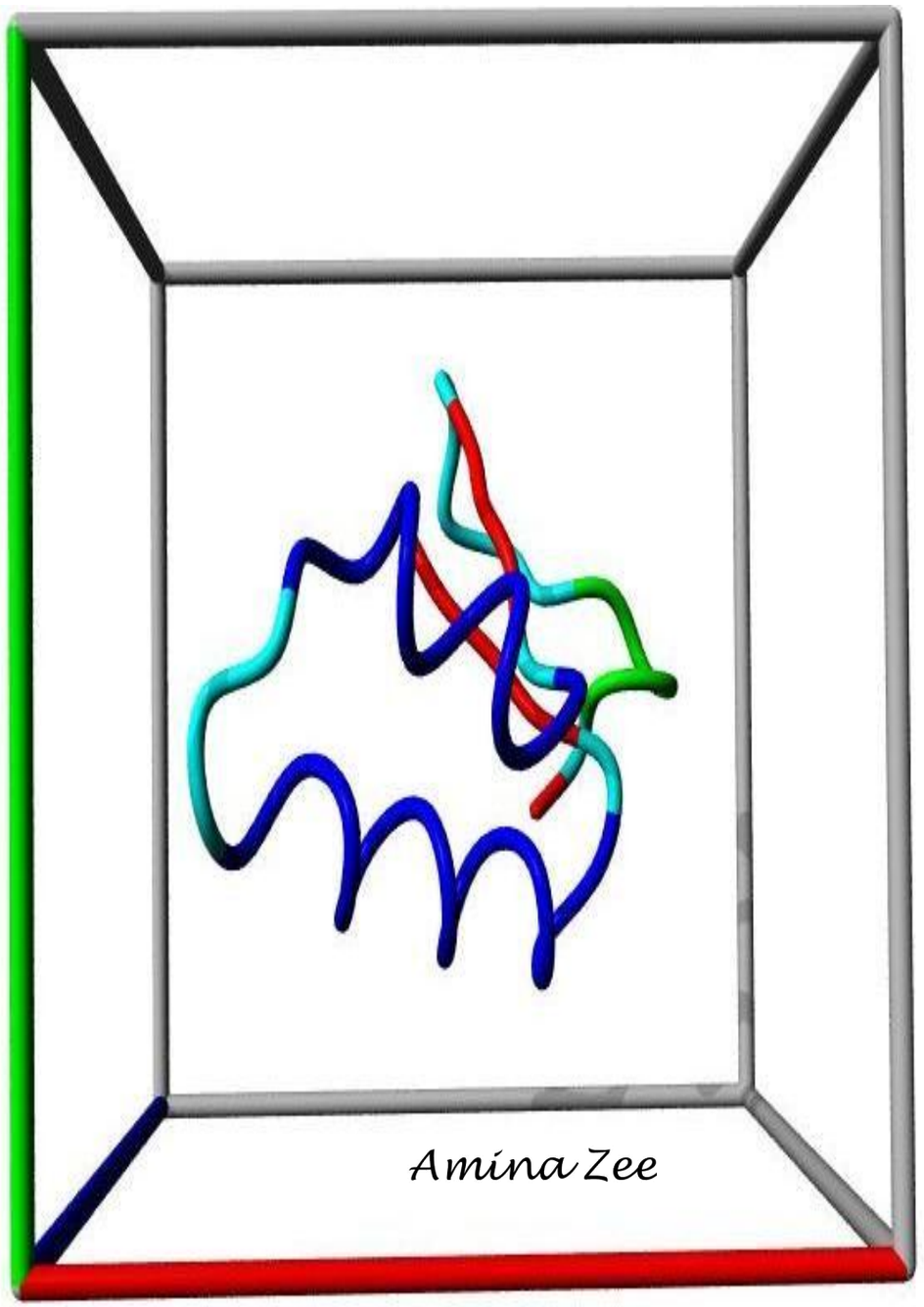


Amina Zee

Seven Steps to Homology Modelling - V

MD Example Sim: Crambin

(Ethiopian Cabbage Protein)



Seven Steps to Homology Modelling - V

Conclusions

- Now we have minimized large errors
- However, smaller errors may still exist
- Next step, validate the model that we have constructed!

Amina Zee

Lec 125

Seven Steps to Homology Modelling – VI

Amina Zee

Seven Steps to Homology Modelling - VI

Background

Template recognition and initial alignment

Alignment correction

Backbone generation

Loop modeling

Side-chain modeling

Model optimization

Model validation

Amina Zee

Seven Steps to Homology Modelling - VI

The model with the lowest energy might still be folded completely wrong.

So, the model should be checked again for normal ranges of:
Bumps, Bond angles, Torsion angles, Bond lengths
Other properties, like the distribution of polar/apolar residues, can be compared with real structures.

Important Points:
An error occurs far away from the active site, is tolerable.
But when an error occurs in the active site, one should reconsider the template and/or alignment.

Amina Zee

Seven Steps to Homology Modelling - VI

Limitations of Homology Modelling

Large Bias towards structure of template

Cannot study conformational changes

Cannot elicit new catalytic/binding sites

Amina Zee

Seven Steps to Homology Modelling - VI

Conclusions

- So how can we overcome such limitations?

- Other strategies include: Threading, and Ab Initio Modelling

- We will also examine online tools for each

Lec 126

MODELLER for Homology Modelling

Amina Zee

MODELLER for Homology Modelling

Background

Template recognition and initial alignment

Alignment correction

Backbone generation

Loop modeling

Side-chain modeling

Model optimization

Model validation

Amina Zee

MODELLER for Homology Modelling

Background

- Modeller is a software for homology modelling
- salilab.org/modeller
- Inputs:
  Python script file,
  Sequence alignment
  & Template (PDB)

Amina Zee

MODELLER for Homology Modelling

```
from modeller import *
from modeller.automodel import *
log.verbose()
env = environ()
env.io.atom_files_directory = './'

a = automodel(
  env,
  alnfile = 'herg.ali',
  knowns = '1q5o',
  sequence = 'herg'
)

a.starting_model= 1
a.ending_model = 1
a.make()
```

Input Python Script (*.py)

```
>P1;1q5o
structureX: 1q5o : 443 : A : 644 : A ::::
DSSRRQYQEKYKQVEQYMSFHKLPADFRQKIHDYYEHRYQ-GKMFDEDSILGELNGPLRE
EIVNFNCRKLVASMPLFANADPNFVTAMLTKLKFEVFQPGDYIIREGTIGKKMYFIQHGV
VSVLTKGNKEMKLSDGSYFGEICLL--TRGRRTASVRADTYCRLYSLSVDNFNEVLEEYP
MMRRAFETVAIDRLDRIGKKNSIL.*

>P1;herg
sequence: herg : 1 :::::::
YSGTARYHTQMLRVREFIRFHQIPNPLRQRLEEYFQHAWSYTNGIDMNAVLKGFPECLQA
DICLHLNRSLLQHCKPFRGATKGCLRALAMKFKTTHAPPGDTLVHAGDLLTALYFISRGS
IEILRGDVVVAILGKNDIFGEPLNLYARPGKSNGDVRALTYCDLHKIHRDDLLEVLDMYP
EFSDHFWSSLEITFNLRDTN-MIP.*
```

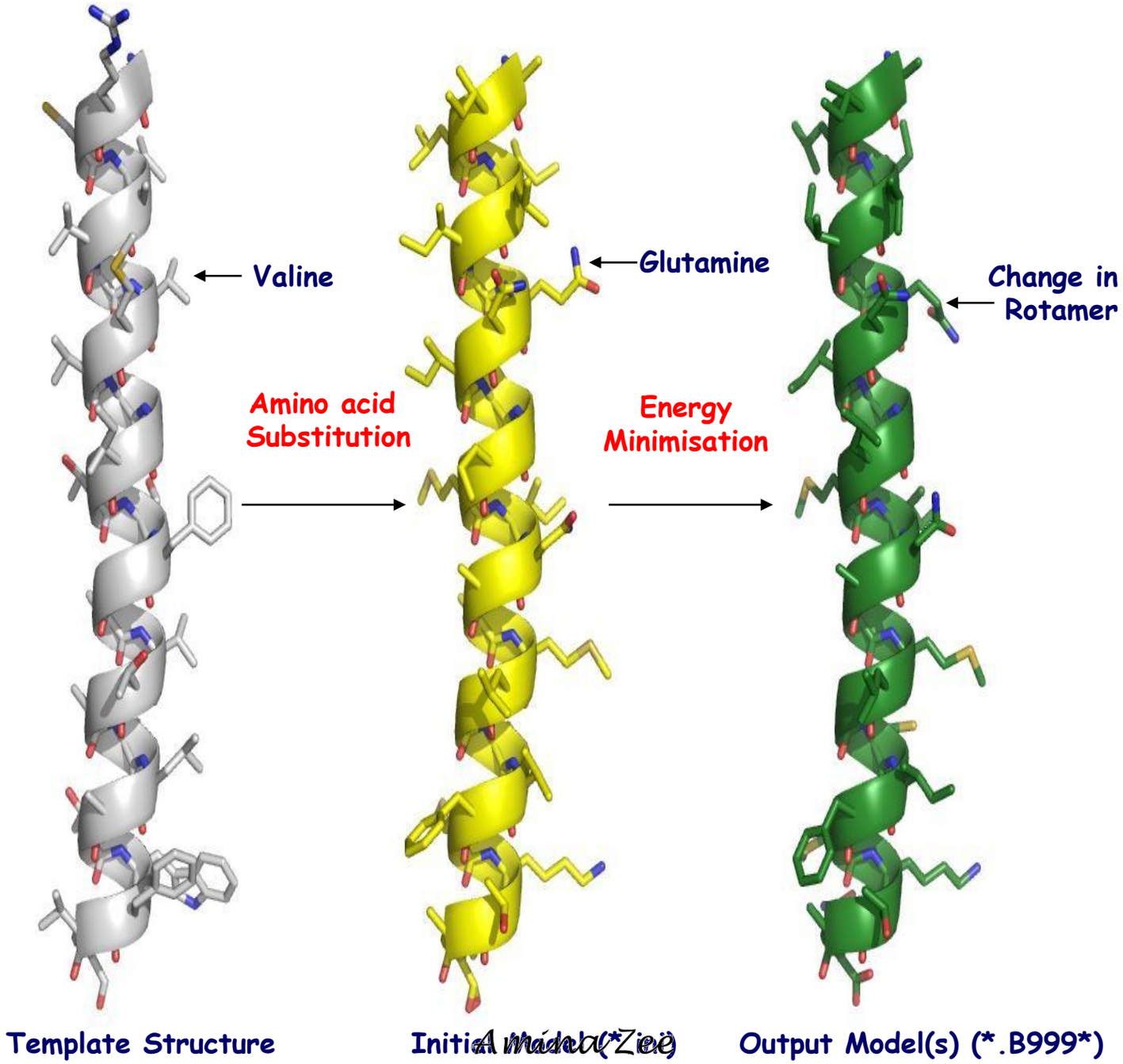
Sequence Alignment (*.ali)

```
ATOM   1  N   ASP A 443    -15.943  41.425  44.702  1.00 44.68
ATOM   2  CA  ASP A 443    -15.424  42.618  45.447  1.00 43.15
ATOM   3  C   ASP A 443    -14.310  43.306  44.686  1.00 41.81
ATOM   4  O   ASP A 443    -14.298  44.528  44.539  1.00 42.61
                          etc...
```

Template Structure (*.pdb)

Amina Zee

MODELLER for Homology Modelling



MODELLER for Homology Modelling

- **.log** : log output from the run.
- **.B*** : model generated in the PDB format.
- **.D*** : progress of optimisation.
- **.V*** : violation profile.
- **.ini** : initial model that is generated.
- **.rsr** : restraints in user format.
- **.sch** : schedule file for the optimisation process.

Amina Zee

MODELLER for Homology Modelling

Automated Modelling Servers

Swiss Model
http://swissmodel.expasy.org//SWISS-MODEL.html

Robetta
http://robetta.bakerlab.org/

3D Jigsaw
http://www.bmm.icnet.uk/servers/3djigsaw/

Phyre
http://www.sbg.bio.ic.ac.uk/phyre/

Amina Zee

MODELLER for Homology Modelling

Conclusions

- Homology modelling helps predict protein structures by using prior structural information

- Several tools are available to perform homology modelling in a programmatic or automated way!

Amina Zee

Homology Modelling

Lec 127

Fold Recognition – Threading I

Amina Zee

Fold Recognition – Threading I

Background

Template recognition and initial alignment

Alignment correction

Backbone generation

Loop modeling

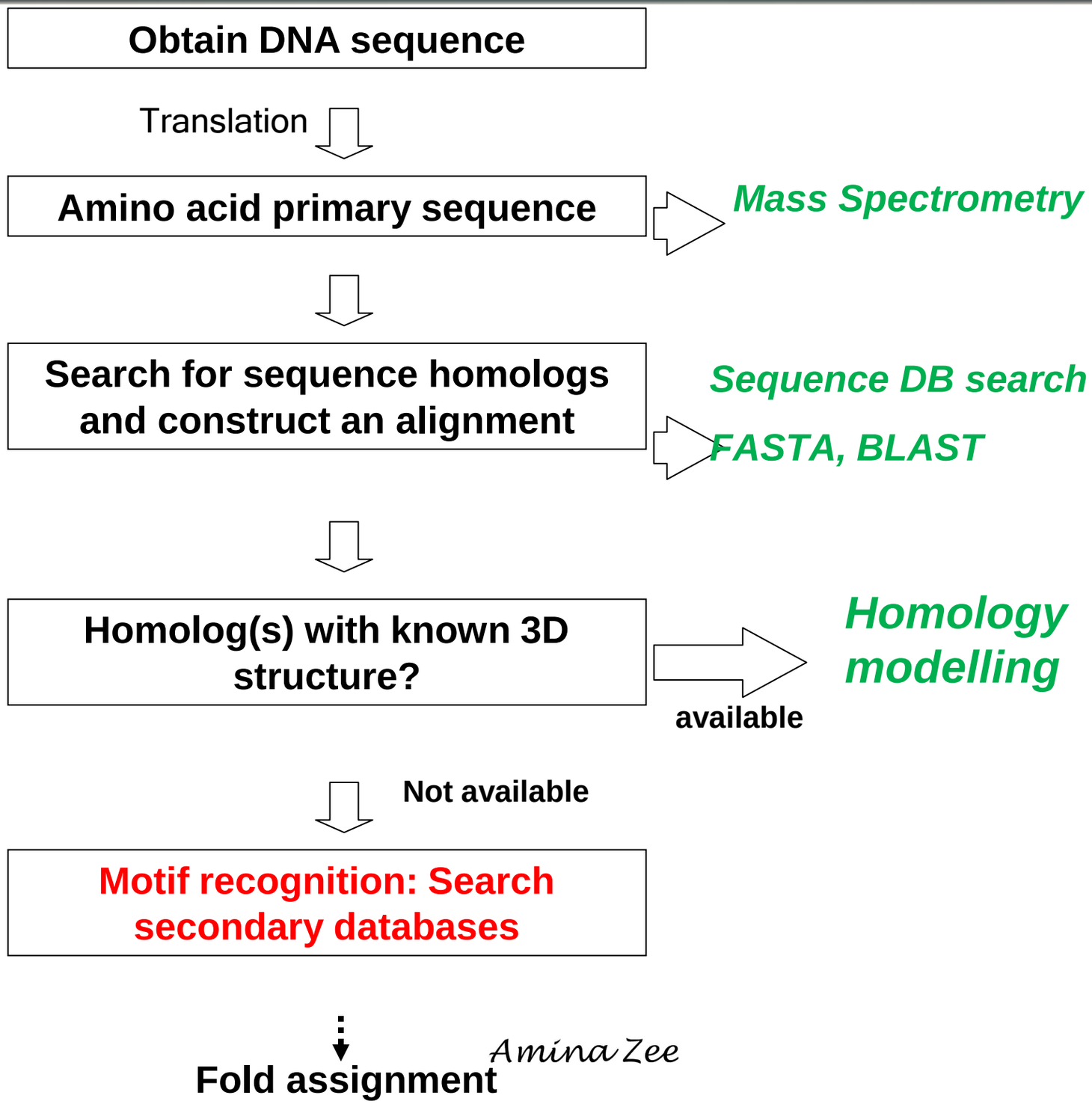
Side-chain modeling

Model optimization

Model validation

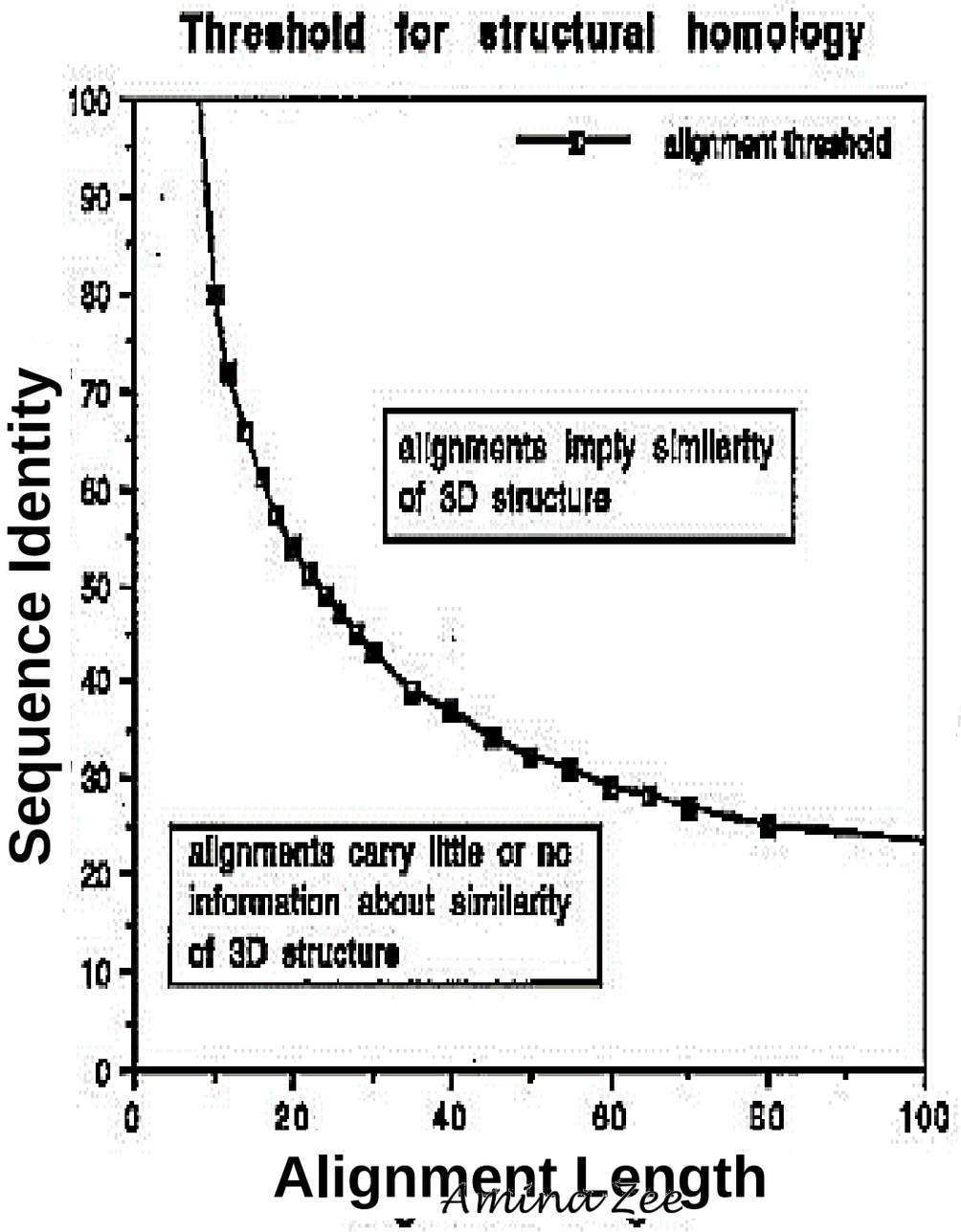
Amina Zee

Fold Recognition – Threading I



Fold Recognition – Threading I

When should we use Fold Recognition?



http://www.cmbi.ru.nl/gvteach/astra/lectures/homology_modelling.ppt

Fold Recognition – Threading I

Introduction

- A protein fold is defined by the way the secondary structure elements of the structure are arranged relative to each other in space.

- Common folds include 4-helix bundle and the TIM barrel.

Amina Zee

Fold Recognition – Threading I

Introduction

- **5,000 stable folds in nature**

- **Fold recognition: Finding the best fit of a sequence to a set of candidate folds**

Amina Zee

Fold Recognition – Threading I

Conclusions

- **Fold recognition or Threading is a technique for predicting protein structures**
- **It is useful in cases where homology modelling fails to predict quality structures**

Amina Zee

Homology Modelling

Lec 128

Fold Recognition – Threading II

Amina Zee

Fold Recognition – Threading II

Background

- Fold recognition is also called Threading
- Technique for predicting protein structures
- Employed when homology modelling cannot predict quality structures

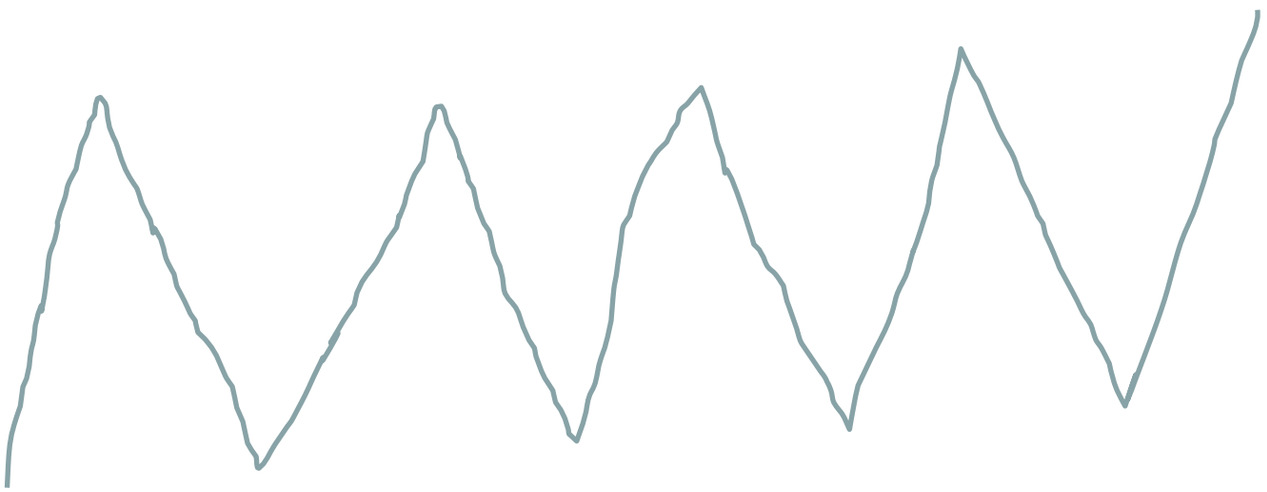
Amina Zee

Fold Recognition – Threading II

Find the best way to

“mount” the residue sequence of one protein

onto a known protein structure!



Amina Zee

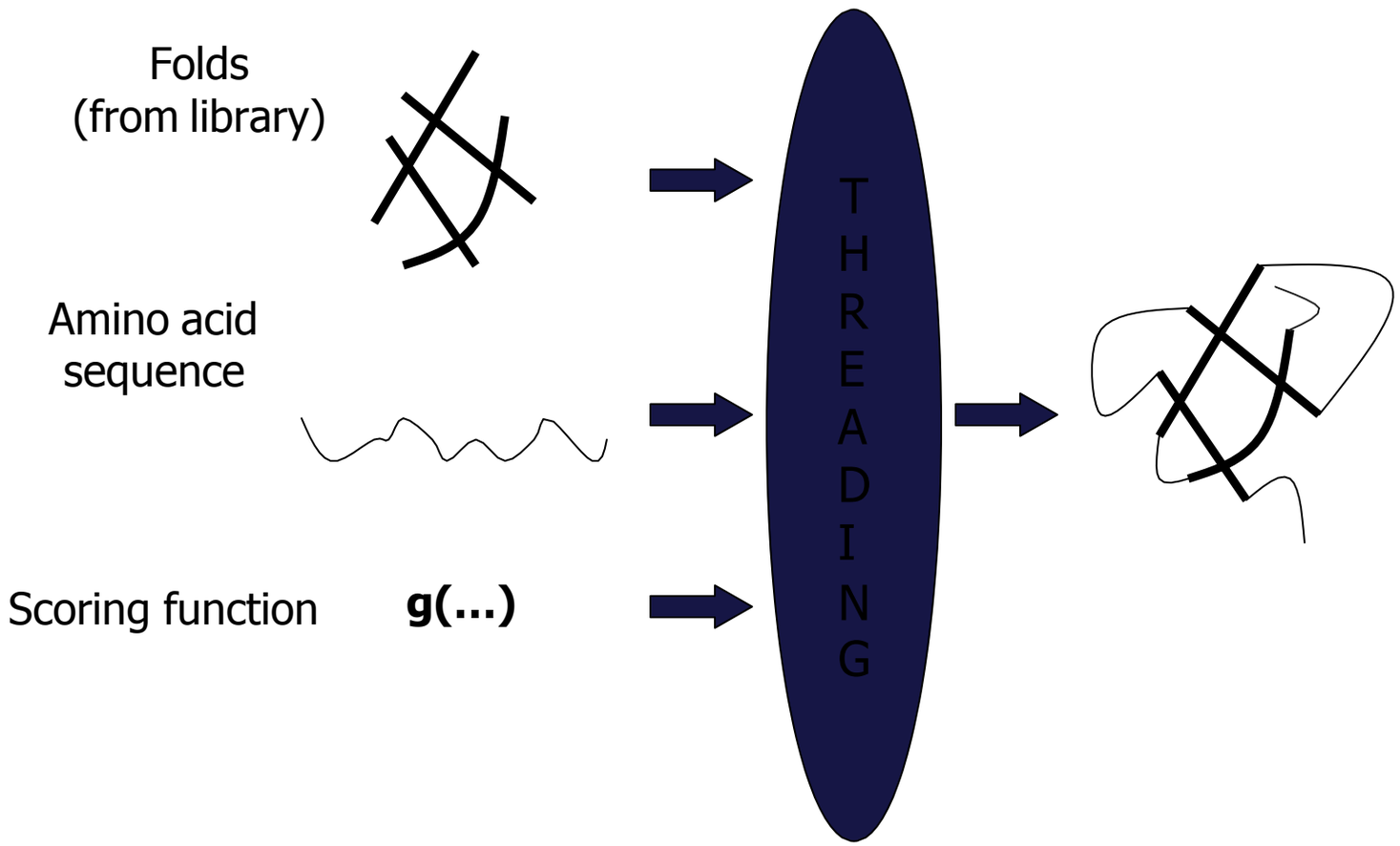
Fold Recognition – Threading II

The process of threading

- In the process of “Threading”, we mount an amino acid sequence on to the backbone of template structures in a folds library
- Each step is “drag” along the sequence (MQVKLFTY...) through each location of each template fold
- Then, for each fold, we must compute the fitness of sequence matching that fold!

Fold Recognition – Threading II

Inputs and outputs of threading



Amina Zee

Fold Recognition – Threading II

Conclusions

- Threading involves “passing” the amino acid sequence through each fold in the database
- The best match is computed using a scoring function

Amina Zee

Homology Modelling

Lec 129

Fold Recognition – Threading III

Amina Zee

Fold Recognition – Threading III

Background

- Threading involves “passing” the amino acid sequence through each fold in the database

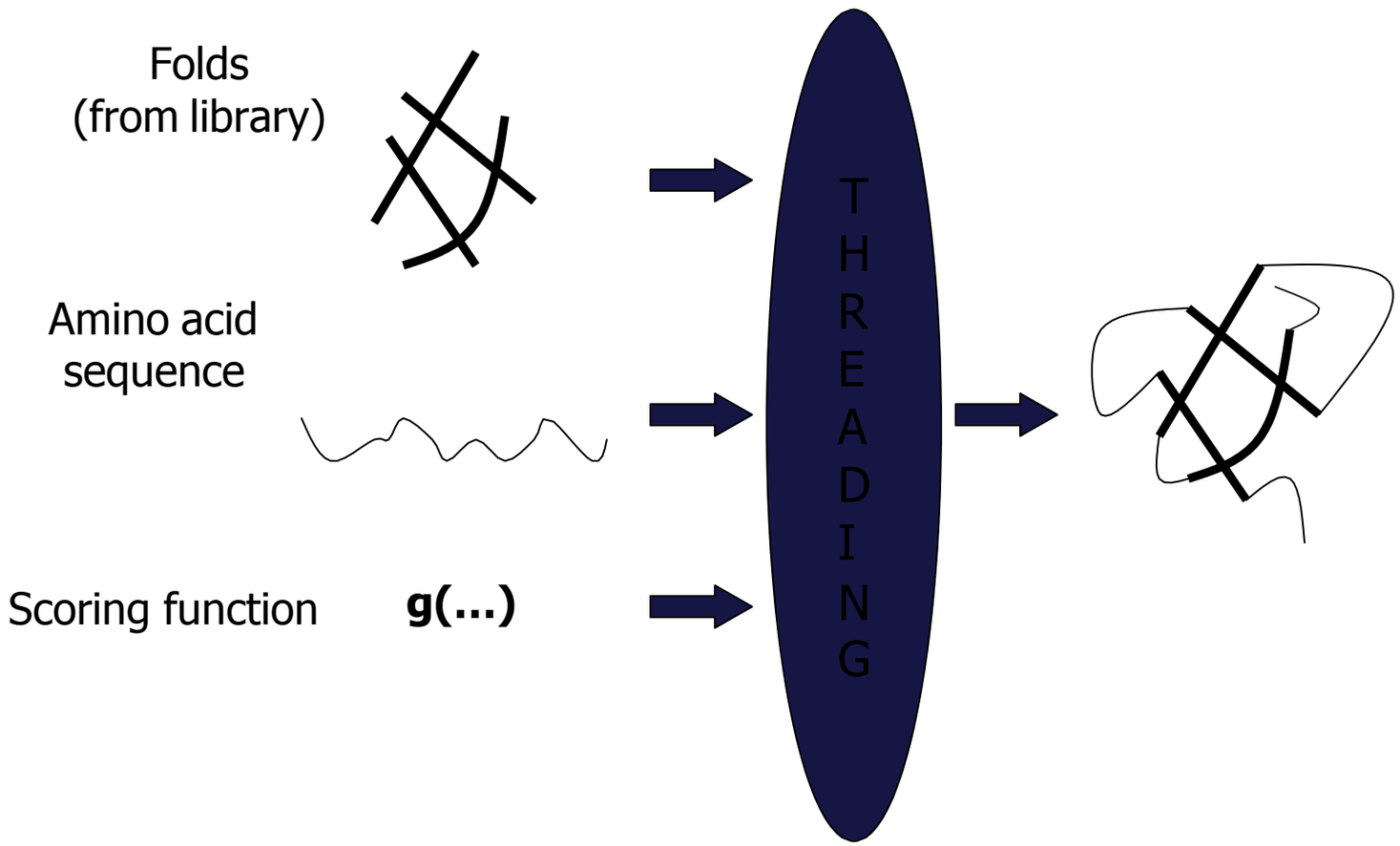
- The best match is computed using a scoring function

- Flowchart

Amina Zee

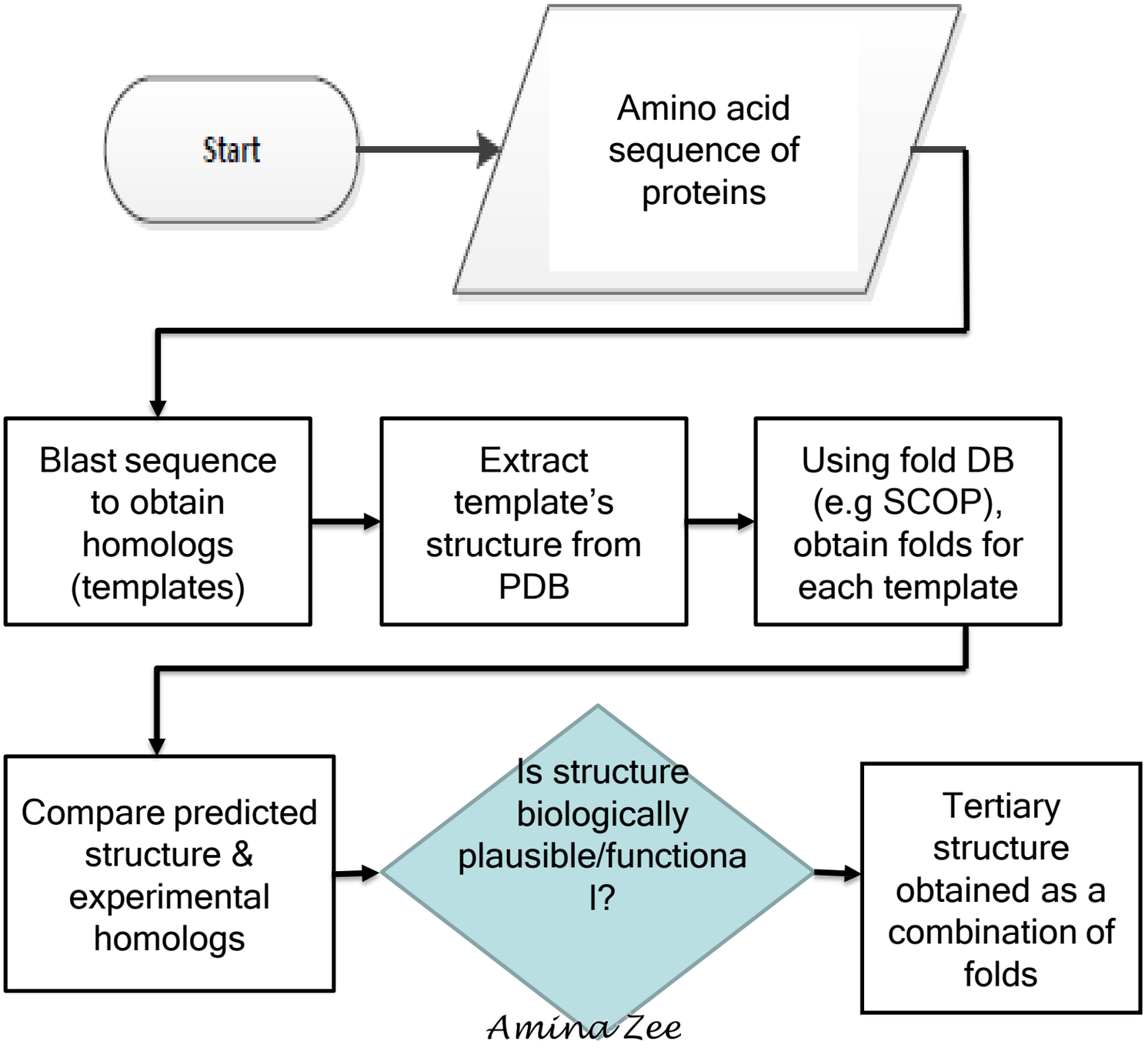
Fold Recognition – Threading III

Inputs and outputs of threading

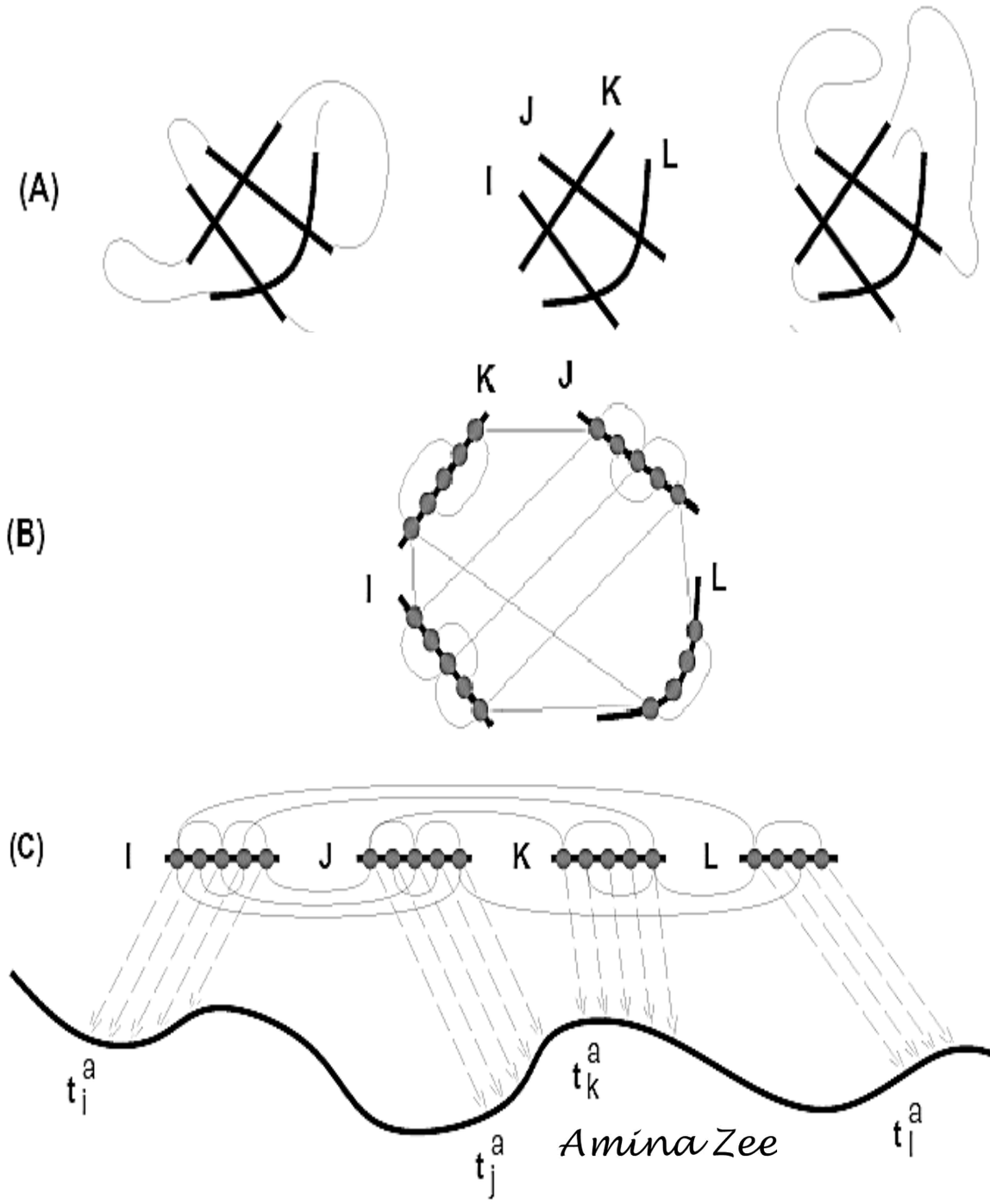


Amina Zee

Fold Recognition – Threading III



Fold Recognition – Threading III



Two structurally similar proteins

Spatial adjacencies (interactions)

Possible threading with a sequence

Fold Recognition – Threading III

Conclusions

- Combinations of secondary structures come together to form the best prediction
- Scoring typically involves using a Z-Score function based on energy of a molecule

Homology Modelling

Lec 130

Online Tools for Threading - iTasser

Amina Zee

Online Tools for Threading - iTasser

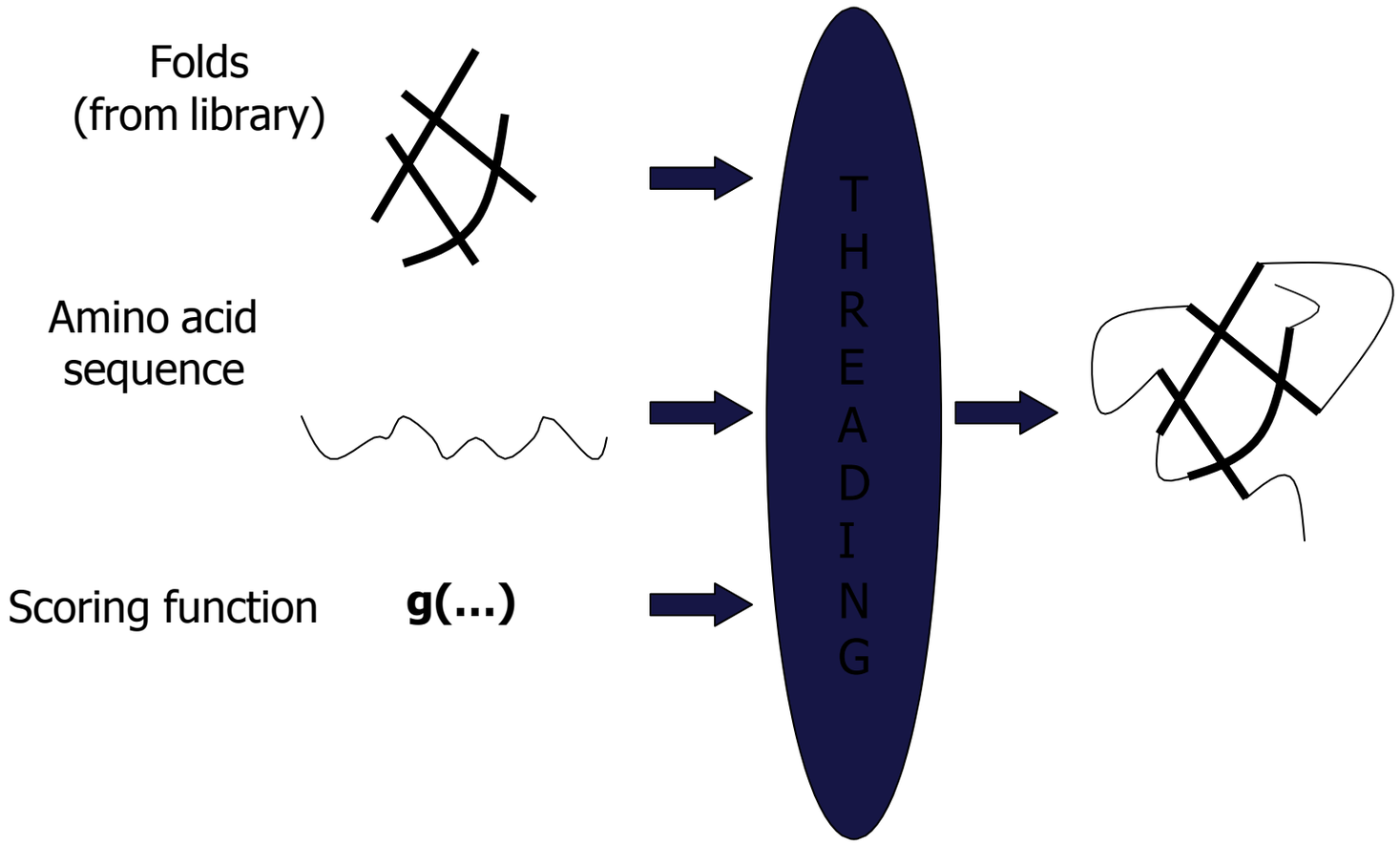
Background

- Threading involves “passing” the amino acid sequence through each fold in the database
- The best match is computed using a scoring function
- iTASSER

Amina Zee

Online Tools for Threading - iTasser

Inputs and outputs of threading



Amina Zee

Online Tools for Threading - iTasser

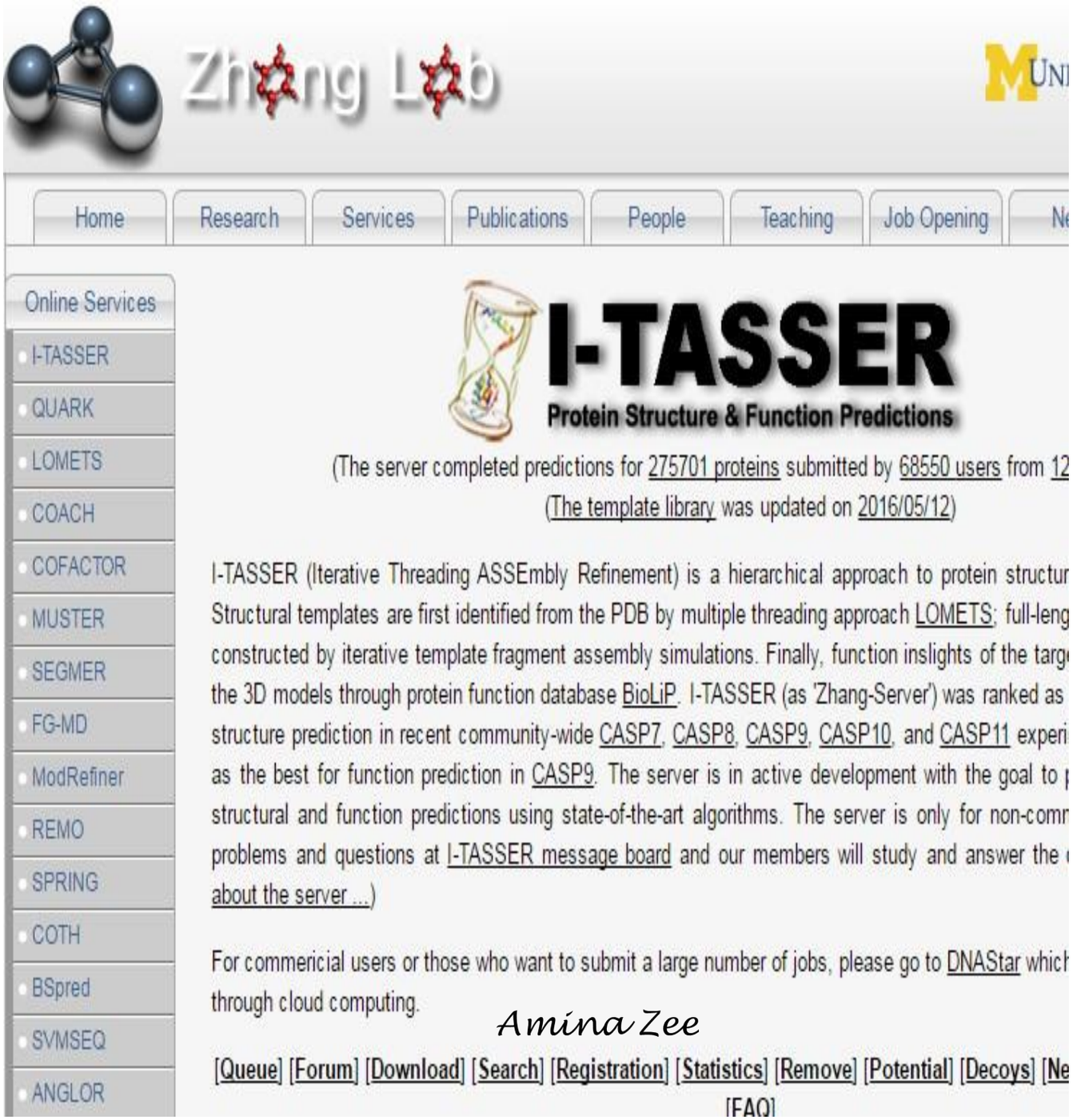
Iterative threading assembly refinement (I-TASSER) server

- Software for automated protein structure &function prediction based on the sequence-to-structure-to-function.
- Steps:
  - Starts from amino acid sequence
  - i-TASSER first generates 3D atomic models from multiple threading alignments and iterative structural assembly simulations.
  - The function of the protein is then inferred by structurally matching the 3D models with other known proteins.
  - Outputs full-length secondary & tertiary structures and functional annotations on ligand-binding sites
  - An estimate of accuracy of the predictions is provided based on the confidence score of the modeling

Amina Zee

Online Tools for Threading - iTasser

Link: http://zhanglab.ccmb.med.umich.edu/I-TASSER/



Online Tools for Threading - iTasser

Link: http://zhanglab.ccmb.med.umich.edu/I-TASSER/

Copy and paste your sequence below ([10, 1500] residues in FASTA format) Click here for a sample input:

Or upload the sequence from your local computer:

Choose File No file chosen

Email: (mandatory, where results will be sent to)

Password: (mandatory, please click here if you do not have a password)

ID: (optional, your given name of the protein)

▶ **Option I:** Assign additional restraints & templates to guide I-TASSER modeling.

▶ **Option II:** Exclude some templates from I-TASSER template library.

▶ **Option III:** Specify secondary structure for specific residues.

☑ Keep my results public (uncheck this box if you want to keep your job private. A key will be assigned for you to access t results)

Amina Zee

Run I-TASSER Clear form

(Please submit a new job only after your old job is completed)

(If you want to submit multiple urgent jobs, please go to DNAStar which implements I-TASSER through cloud computing)

Online Tools for Threading - iTasser

Conclusions

- **iTASSER helps thread amino acid sequences on fold and secondary structure databases**
- **It also helps predict function of structures output.**

Amina Zee

Homology Modelling

Lec 131

Advantages and Disadvantages of Threading

Amina Zee

Advantages and Disadvantages of Threading

Background

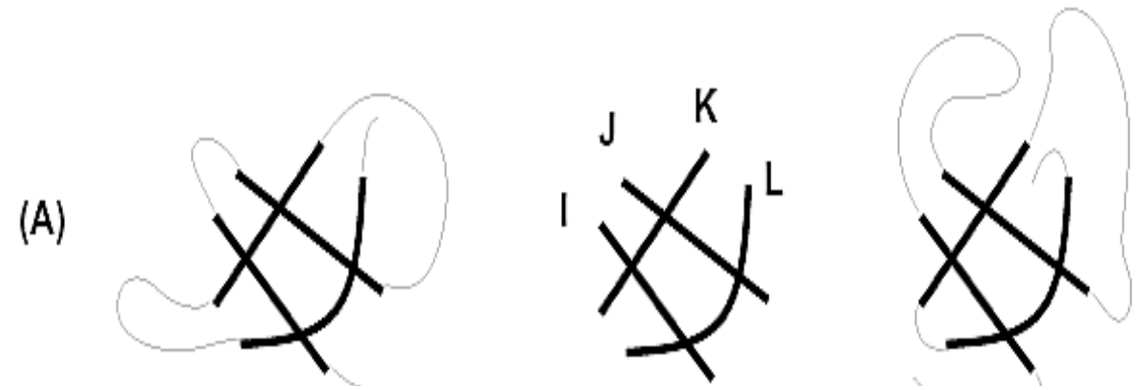
- **Fold recognition or Threading is a technique for predicting protein structures**

- **It is useful in cases where homology modelling fails to predict quality structures**

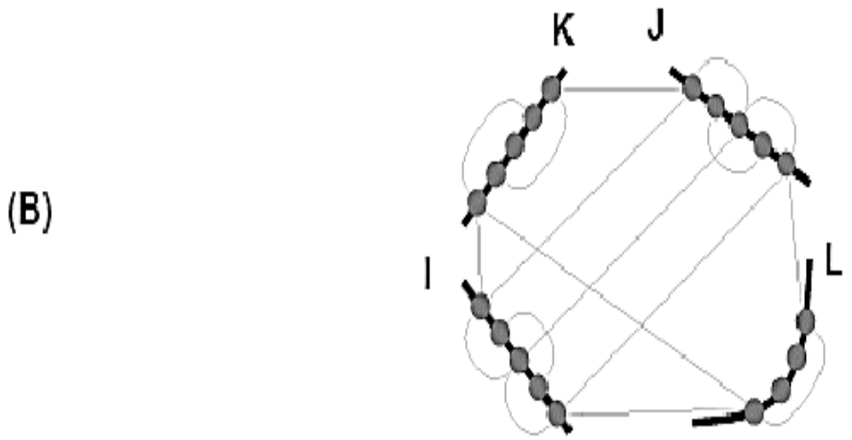
Amina Zee

Advantages and Disadvantages of Threading

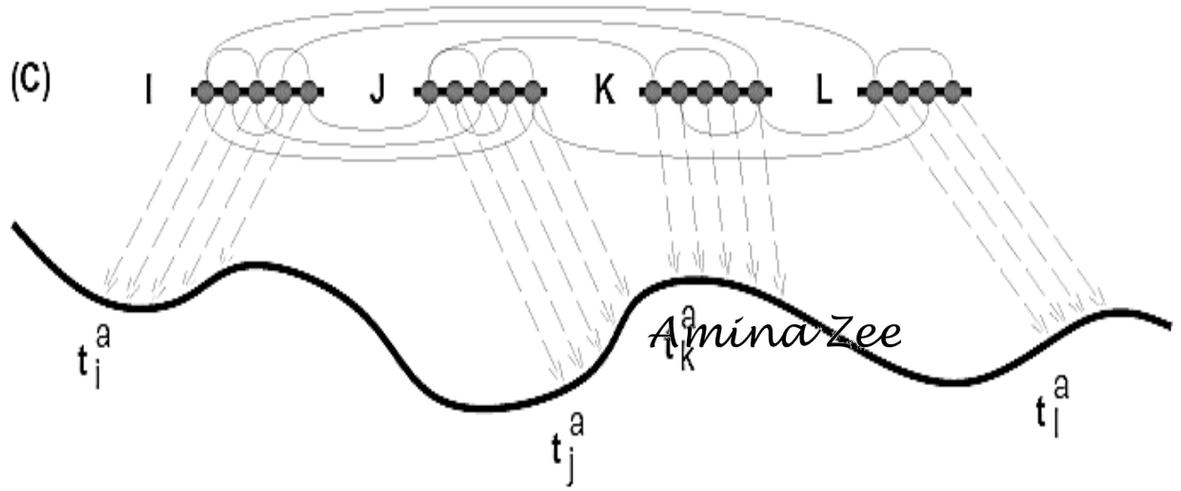
Inputs and outputs of threading



Two structurally similar proteins



Spatial adjacencies (interactions)



Possible threading with a sequence

Advantages and Disadvantages of Threading

Advantages

- Threading helps predict secondary structures of proteins towards tertiary structure prediction

- For the “Twilight Zone” with low alignment quality and identity, threading is useful

Amina Zee

Advantages and Disadvantages of Threading

Disadvantages

- Novel proteins cannot be predicted using threading
- Fewer than 30% of the predicted first hits are true remote homologues
- Validation of each result is necessary

Amina Zee

Homology Modelling

Lec 132

3D-1D Bowie Algorithm

Amina Zee

3D-1D Bowie Algorithm

Background

- Homology employed high alignment scores
- Threading worked by creating combinations of primary sequences and corresponding secondary structures

Amina Zee

3D-1D Bowie Algorithm

Introduction

- **Proposed by Bowie et al in 1991**
- **Converts 3D structure into a 1-D string profile for each structure in the fold library**
- **Align the target sequence to these profiles**

Amina Zee

3D-1D Bowie Algorithm

Inputs and outputs of 3D-1D

- **Identify amino acids based on: protein core, side chain positioning, solubility etc. (6 in all)**
- **Part of secondary structure including α-helix, β-sheet etc (3 in all)**
- **Total of 3 x 6 = 18 distinct states**

3D-1D Bowie Algorithm

Inputs and outputs of 3D-1D

$P_{a:j}$= prob. of finding amino acid (a) in environment (j)

P_a=probability of finding (a) anywhere

Maximize sum of scores for the fold:

$$S_{aj} = \log\left(\frac{P_{a:j}}{P_a}\right)$$

Amina Zee

Advantages and Disadvantages of Threading

Conclusion

- 3D-1D methods convert structure and environment information into “profiles”
- Score for each amino acid is computed for each profile

Homology Modelling

Lec 133

Introduction to Ab Initio Modelling

Amina Zee

Introduction to Ab Initio Modelling

Background

- Ab initio methods have Anfinsen's thermodynamic hypothesis at the center
- These methods attempt to identify the structure with minimum free energy

Amina Zee

Introduction to Ab Initio Modelling

Need for Ab Initio Modelling

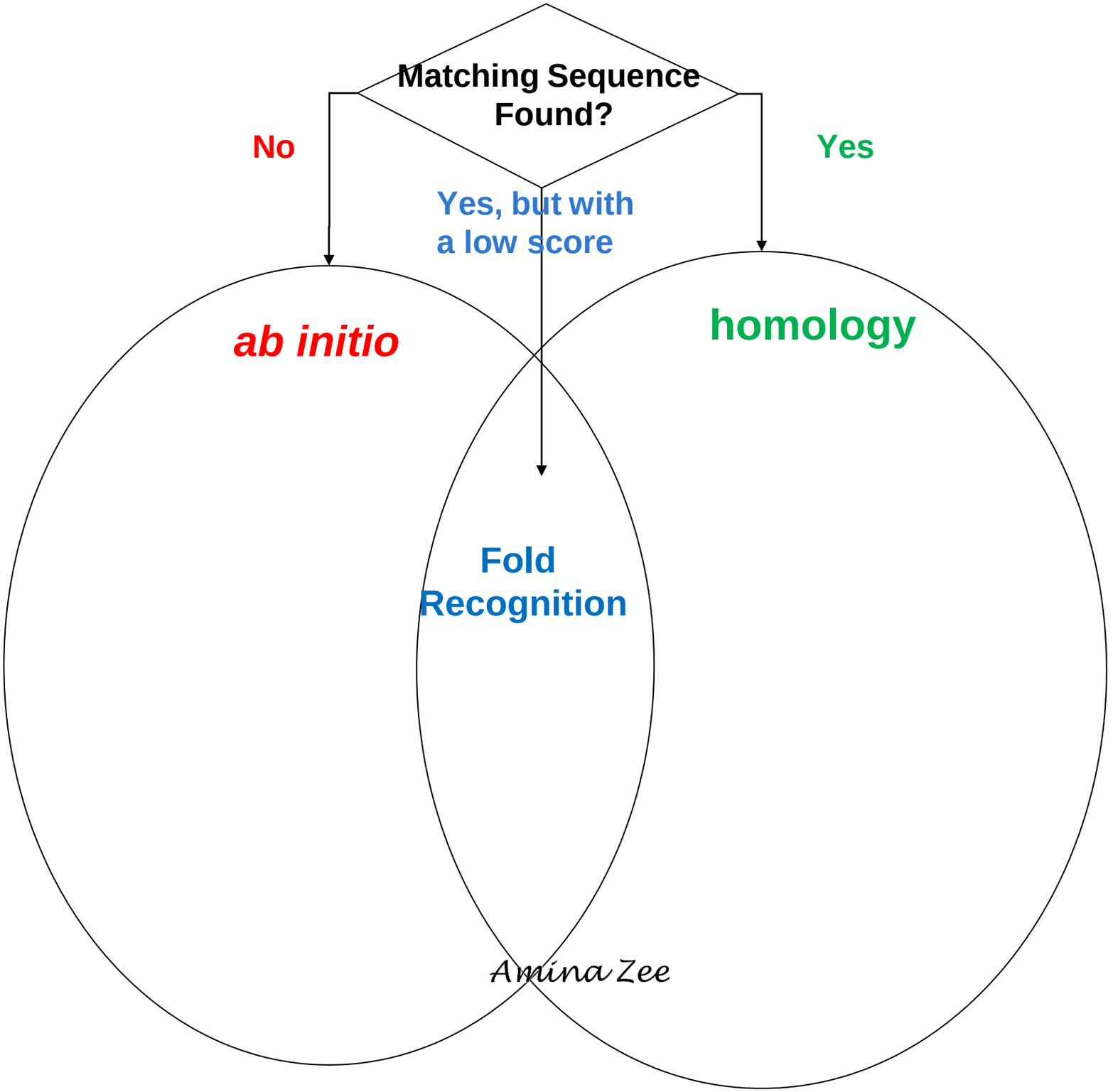
- Applicable to any sequence
- Not very accurate biologically
- Accuracy and applicability are limited by our understanding of the protein folding problem

Amina Zee

Limitation

- Computationally expensive
- Suitable for proteins with less than 100 residues

Introduction to Ab Initio Modelling



Introduction to Ab Initio Modelling

Conclusion

- Ab initio methods rely on computing the energies of folded proteins

- The protein structures with the lowest energy are deemed as plausible predictions

Amina Zee

Homology Modelling

Lec 134

Rationale of Ab Initio Modelling

Amina Zee

Rationale of Ab Initio Modelling

Background

- Ab initio methods rely on computing the energies of folded proteins

- The protein structures with the lowest energy are declared as plausible predictions

Amina Zee

Rationale of Ab Initio Modelling

Rationale

- **Sometimes it so happens that even slightly homologous proteins may not be available.**

- **This renders homology modelling and threading/fold recognition as futile**

Amina Zee

Rationale of Ab Initio Modelling

Rationale

- Also, newer protein structures continue to be discovered every day

- These could not have been identified by methods which only rely on matching with available structures

Rationale of Ab Initio Modelling

Rationale

- **Lastly, homology / fold recognition predict protein structures without computing fundamental physical/chemical properties of the mechanisms and driving forces in structure formation**

Amina Zee

Rationale of Ab Initio Modelling

Conclusion

- Ab initio methods, in contrast, base their predictions on physical models for these mechanisms
- Energy released during the folding process is computed for predicting structure

Amina Zee

Homology Modelling

Lec 135

Strategies for Ab Initio Modelling

Amina Zee

Strategies for Ab Initio Modelling

Background

- **Ab initio methods base their predictions on physical models of folding mechanisms**
- **Stabilization is measured by energy released during the folding process**

Amina Zee

Strategies for Ab Initio Modelling

Energy Optimization in Ab Initio Modelling

1. **Start with a rough initial model.**

2. **Define an energy function mapping structures to energy values. We have to minimize this later!!**

3. **Solve the computational problem of finding the global minimum.**

Amina Zee

Strategies for Ab Initio Modelling

Simulation of the Folding Process

1. **Build an accurate initial model (including energy and forces).**

2. **Accurately simulate the dynamics of the protein folding process.**

3. **The native structure will steadily emerge.**

Amina Zee

Strategies for Ab Initio Modelling

Conclusion

- Start with an energy function
- Fold structures in order to obtain the most stable structure
- This structure will have the minimum energy

Homology Modelling

Lec 136

Energy States of Folded Proteins

Amina Zee

Energy States of Folded Proteins

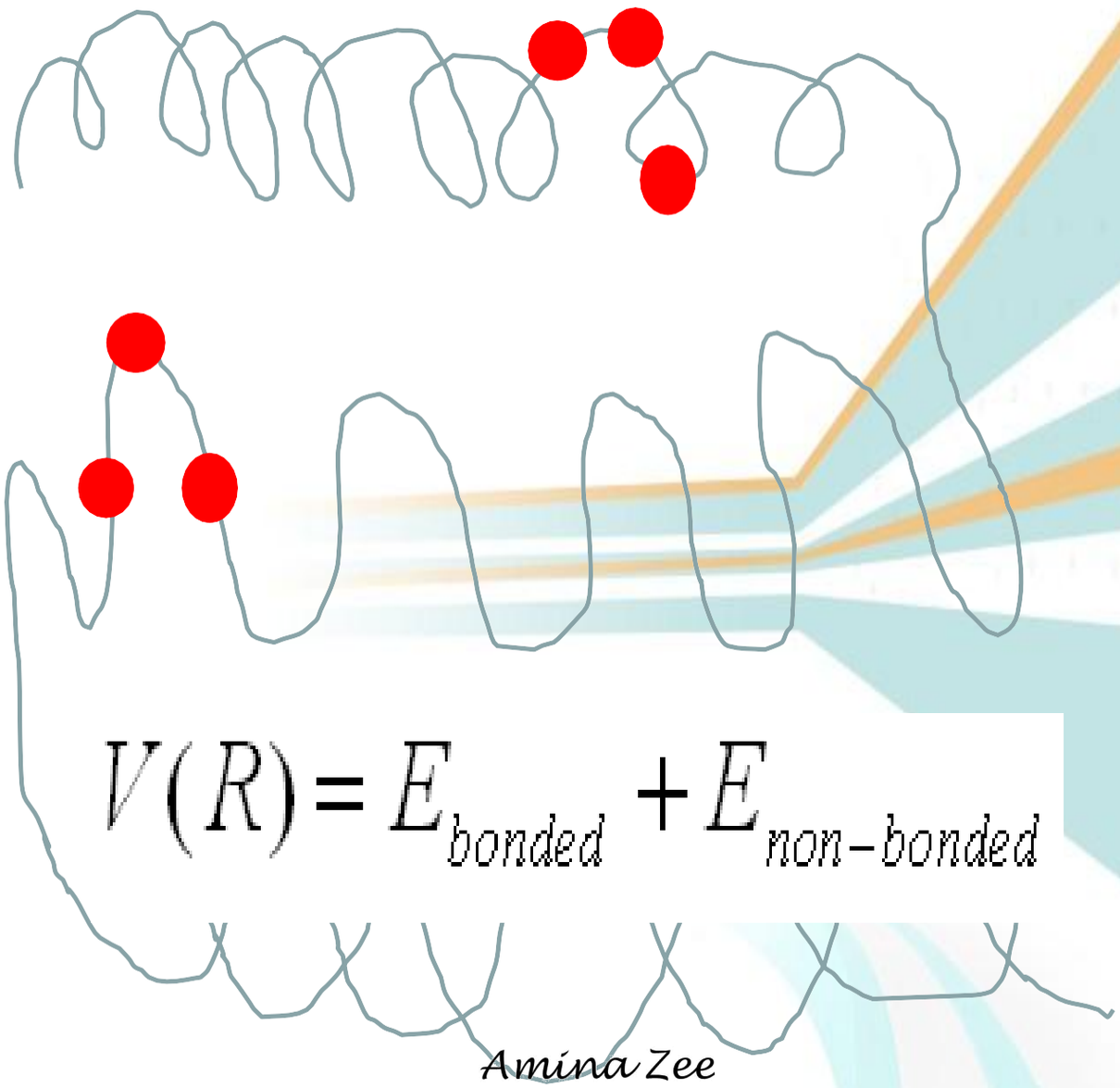
Background

- **Ab initio methods predict protein structures by folding proteins based on each constituent atom's volume, charge, mass etc.**

Amina Zee

Energy States of Folded Proteins

Energies of Bonded Atoms vs. Nonbonded Atoms

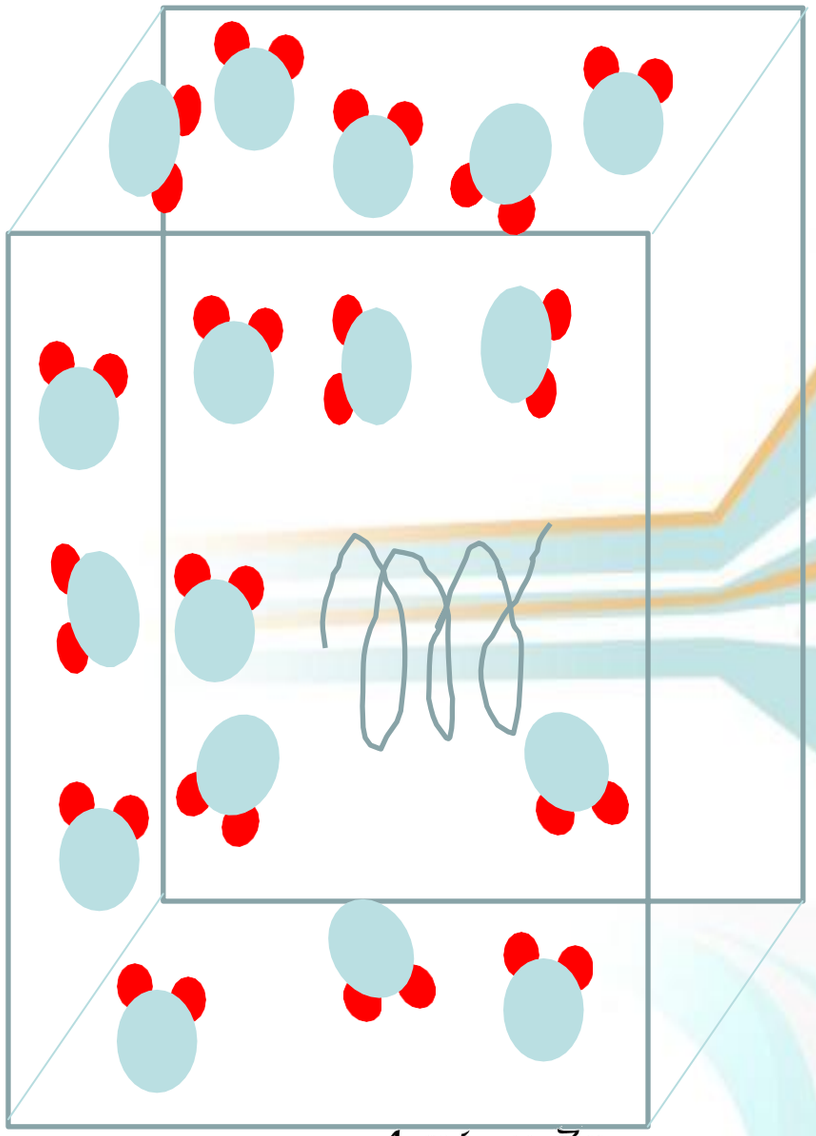


$$V(R) = E_{bonded} + E_{non-bonded}$$

Amina Zee

Energy States of Folded Proteins

Force Fields for Energy Calculations
->Start with an initial structure

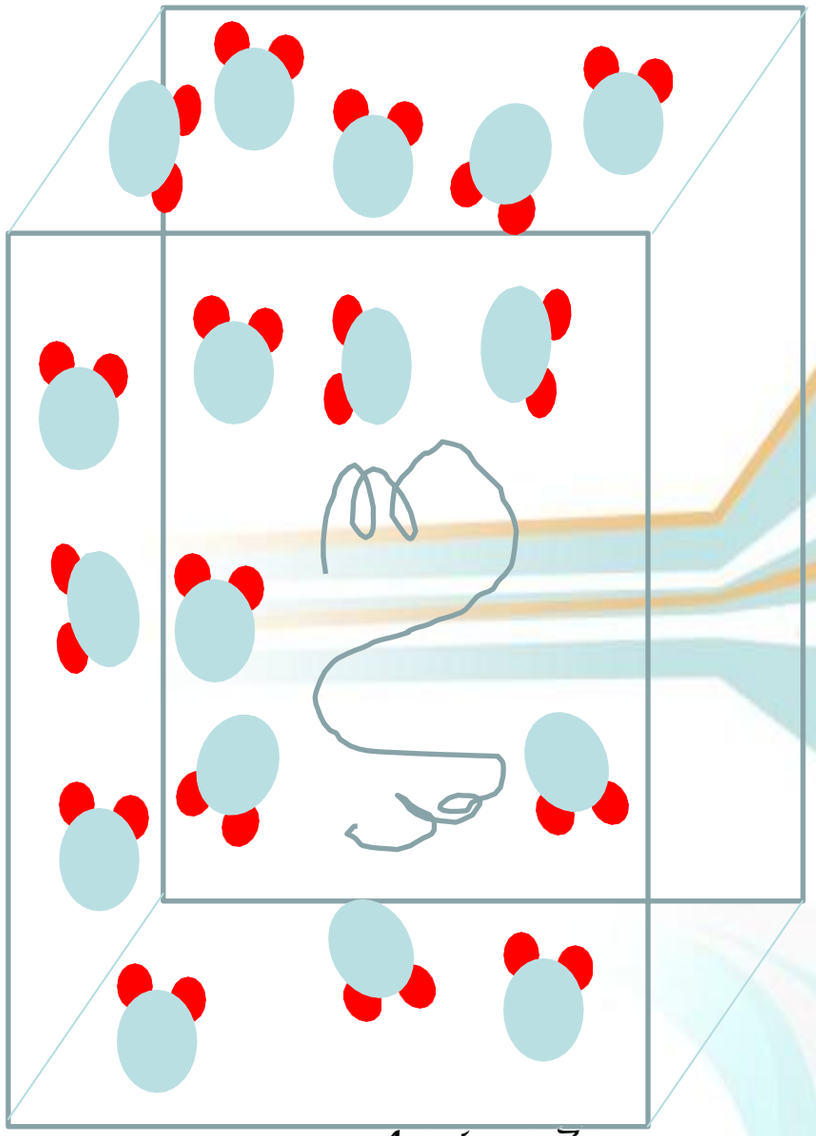


Amina Zee

$$V(R) = E_{bonded} + E_{non-bonded}$$

Energy States of Folded Proteins

Force Fields for Energy Calculations



Amina Zee

$$V(R) = E_{bonded} + E_{non-bonded}$$

Energy States of Folded Proteins

Conclusion

- The protein structure reporting lowest energy is selected to be the optimal structure
- How easy is it to compute the "really" lowest energy of a folded protein?

Homology Modelling

Lec 137

Local versus Global Minima

Amina Zee

Local versus Global Minima

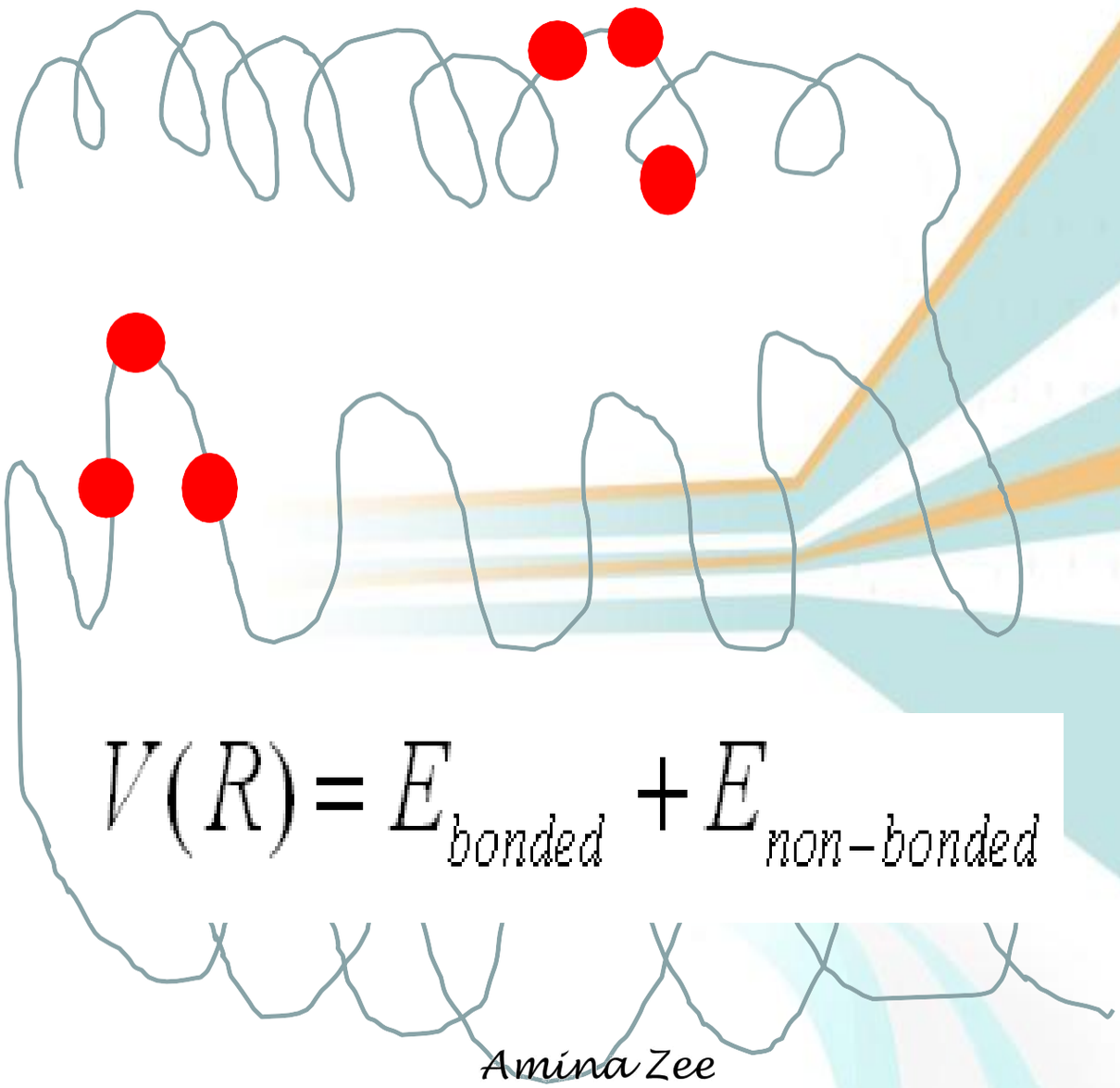
Background

- The protein structure reporting lowest energy is selected to be the optimal structure
- How easy is it to compute the “really” lowest energy of a folded protein?

Amina Zee

Local versus Global Minima

Energies of Bonded Atoms vs. Nonbonded Atoms

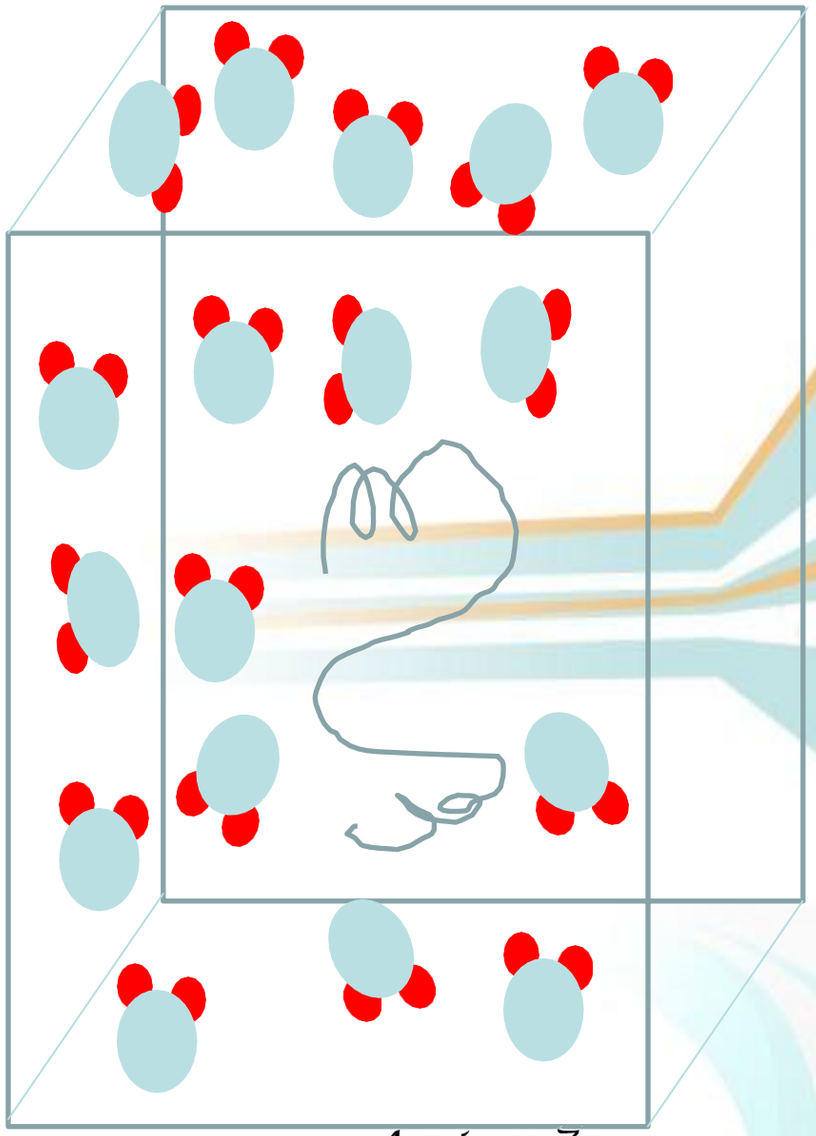


$$V(R) = E_{bonded} + E_{non-bonded}$$

Amina Zee

Local versus Global Minima

Force Fields for Energy Calculations



Amina Zee

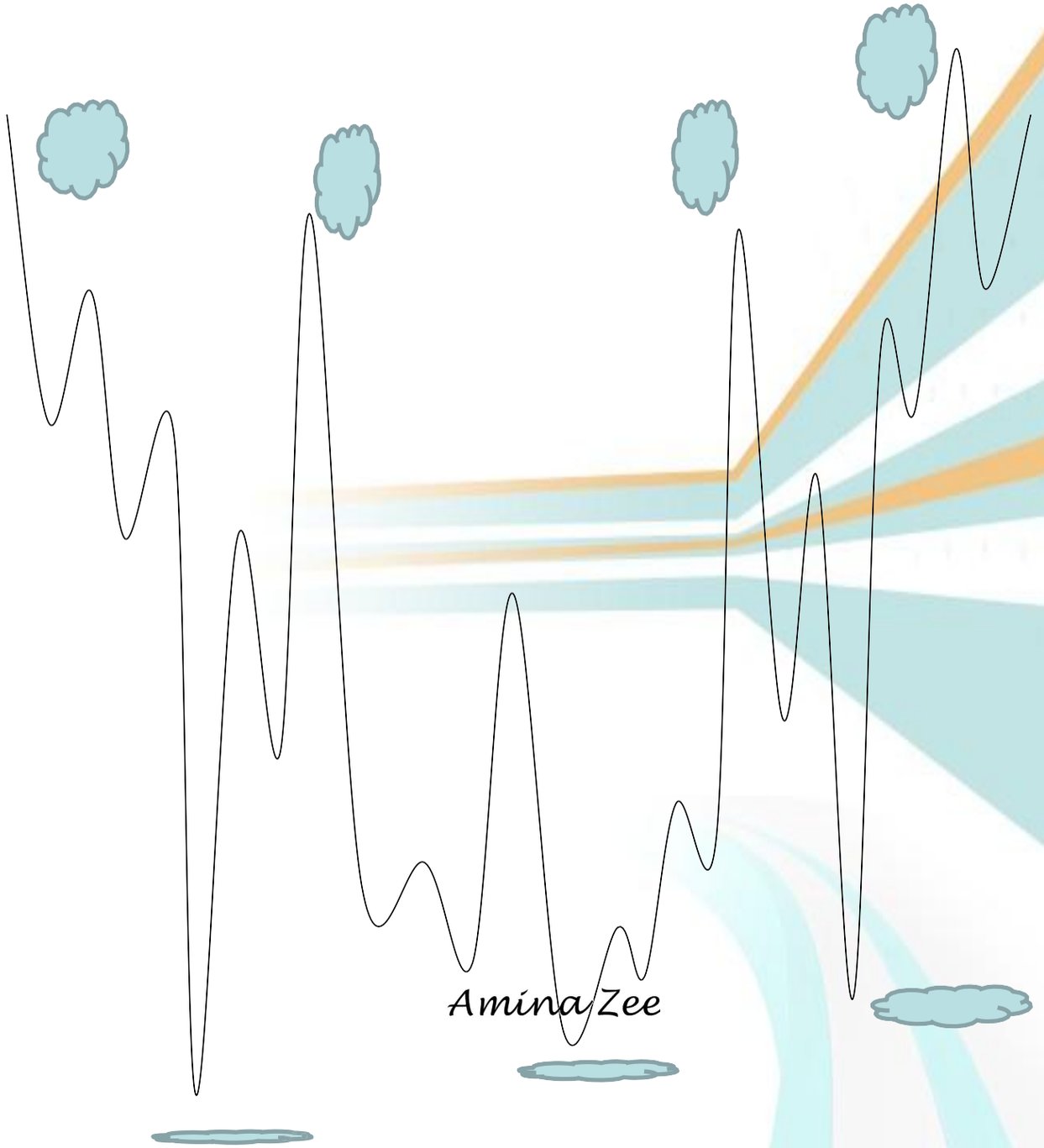
$$V(R) = E_{bonded} + E_{non-bonded}$$

Local versus Global Minima

Global Energy Optimization helps find global minimum

Local versus Global Minima

Global Energy Optimization helps find global minimum



Local versus Global Minima

Best Case Energy Function

- Clear energy minimum in the native structure
- Viable path towards this minimum
- Global optimization finds the most stable structure

Amina Zee

Local versus Global Minima

Optimal Energy Function

- Easier to design and compute
- Native structure not always at the global minimum
- No clear way of choosing among alternative structures that are generated

Homology Modelling

Lec 138

Pros and Cons of Ab Initio Modelling

Amina Zee

Pros and Cons of Ab Initio Modelling

Background

- Native structure not always at the global minimum
- No clear way of choosing among alternative structures that are generated

Amina Zee

Pros and Cons of Ab Initio Modelling

Advantages

- **Ab Initio methods can fold any target sequence using only physical atomic properties**

- **Predictions are mostly accurate and correctly describe the natural folding process**

Amina Zee

Pros and Cons of Ab Initio Modelling

Disadvantages

- Ab initio methods are the very difficult to design (energy function)
- These methods are slow due to the huge possibilities

Amina Zee

Pros and Cons of Ab Initio Modelling

Disadvantages

- An order of 10^{12} steps are needed to simulate protein folding for medium sized protein structures

Amina Zee

Pros and Cons of Ab Initio Modelling

Challenges in Ab Initio Modelling

- Very hard to accurately describe energy functions that can reliably discriminate native and non-native structures.
- Enormous amount of computations.

Amina Zee

Homology Modelling

Lec 139

Summary of Structural Modelling – I

Amina Zee

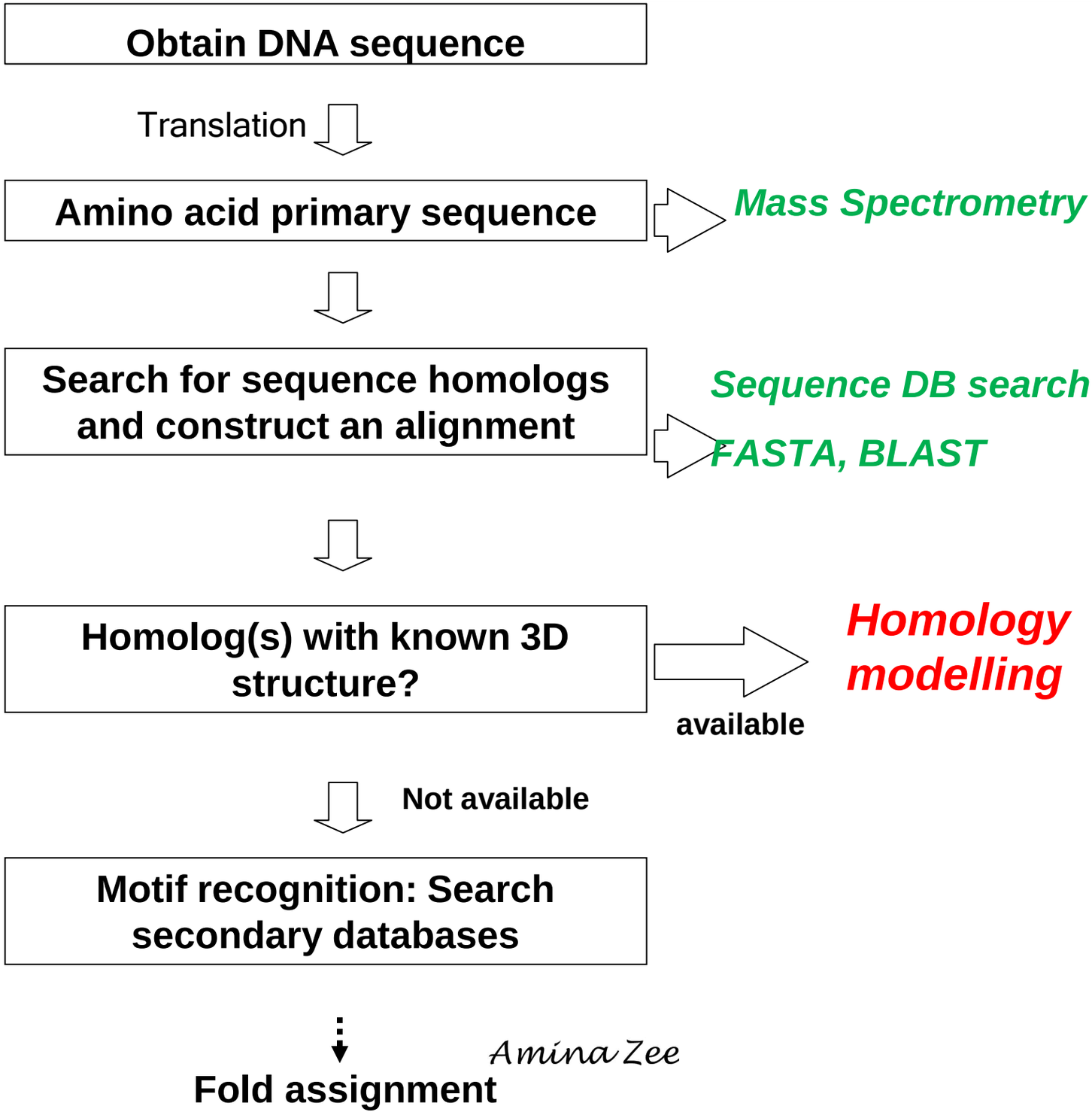
Summary of Structural Modelling - I

Strategies for Structural Modelling

- Homology Modelling
- Fold Recognition
- Ab Initio Modelling

Amina Zee

Summary of Structural Modelling - I



Summary of Structural Modelling - I

Homology modeling of the target structure can be done as follows:

1. Template recognition and initial alignment
2. Alignment correction
3. Backbone generation
4. Loop modeling
5. Side-chain modeling
6. Model optimization
7. Model validation

Amina Zee

Summary of Structural Modelling - I

Conclusion

- Homology modelling is performed in cases of high identity and alignment score
- For the “Twilight zone”, other strategies are employed

Amina Zee

Homology Modelling

Lec 140

Summary of Structural Modelling – II

Amina Zee

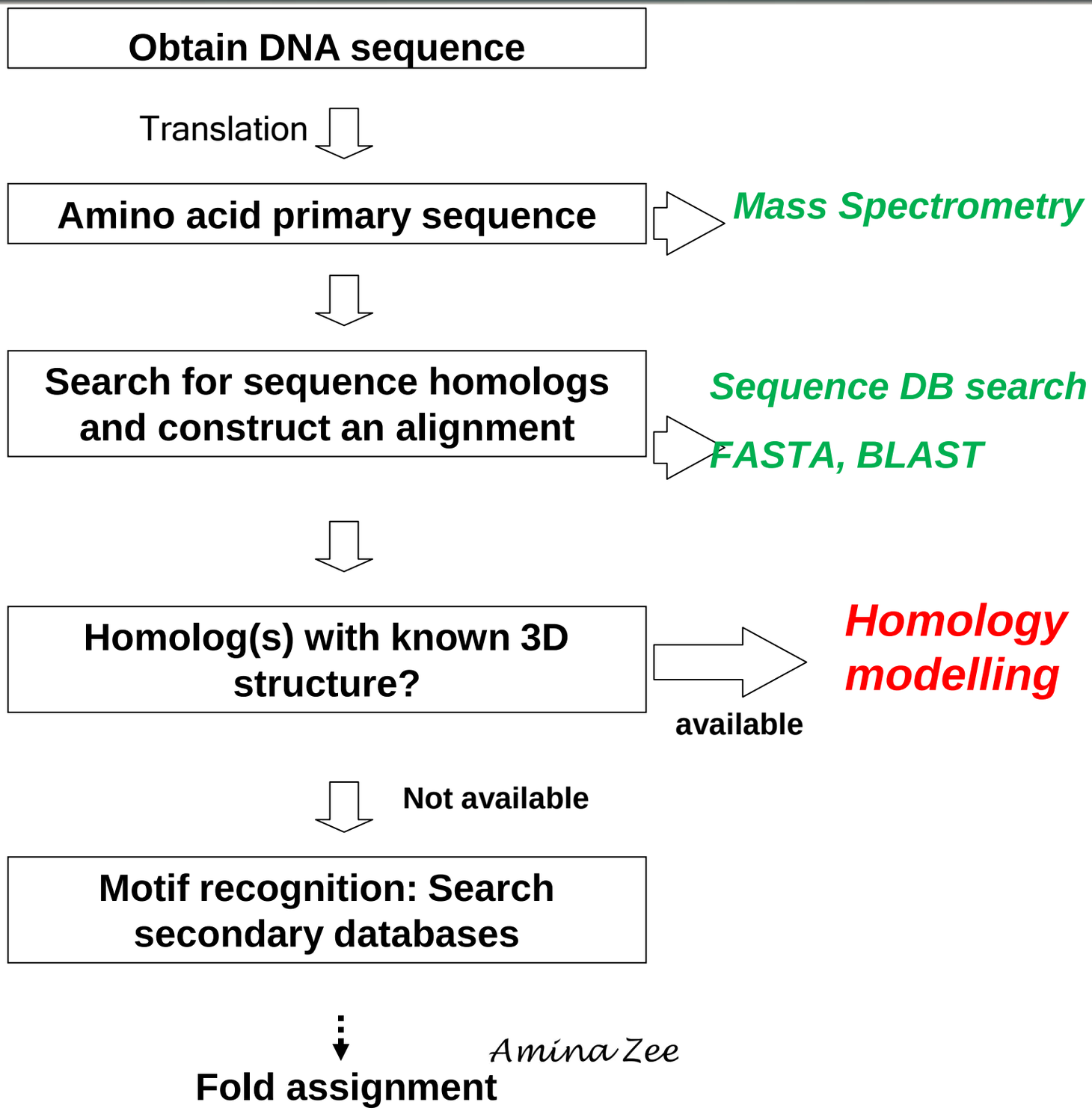
Summary of Structural Modelling - II

Strategies for Structural Modelling

- Homology Modelling
- Fold Recognition
- Ab Initio Modelling

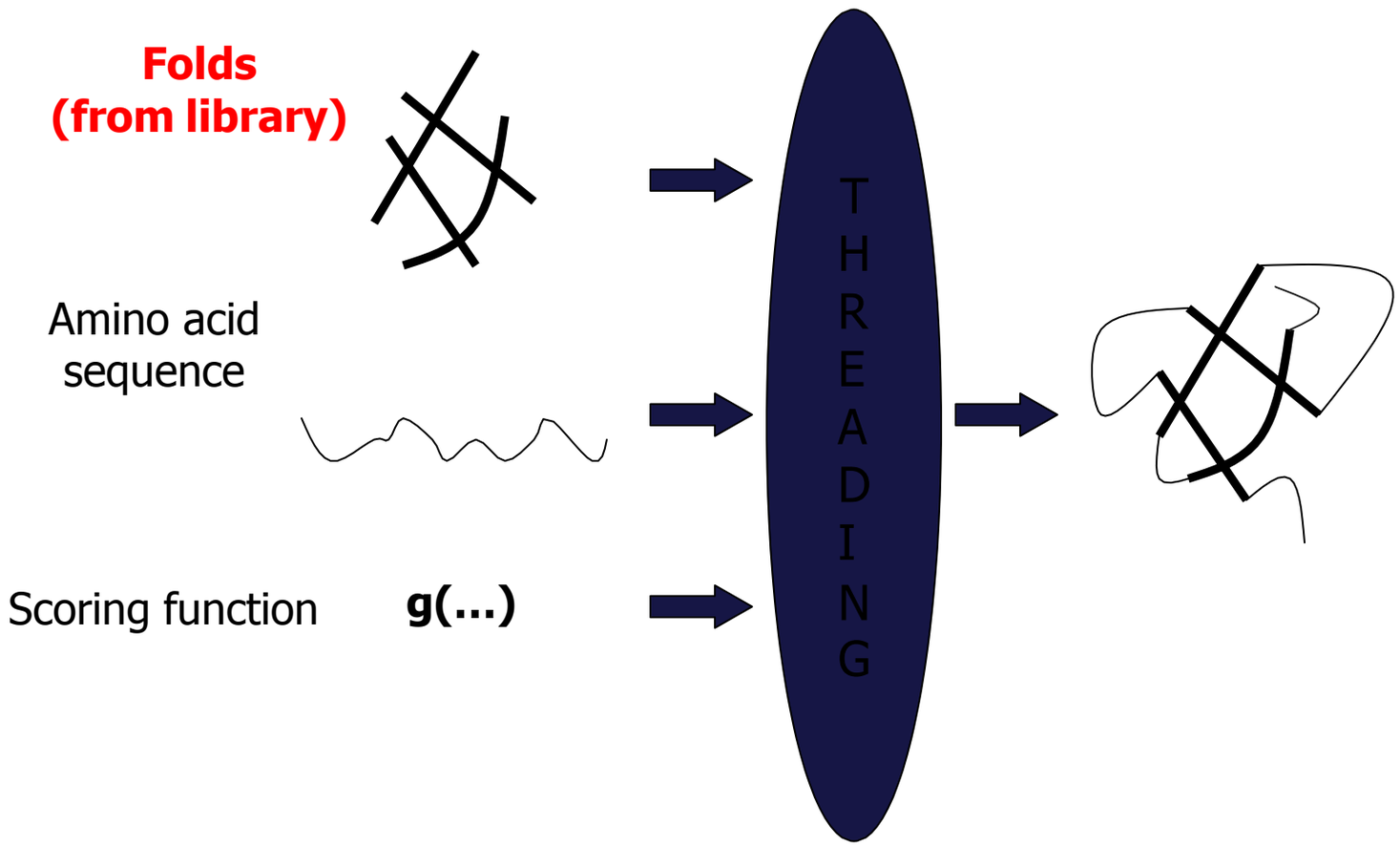
Amina Zee

Summary of Structural Modelling - II



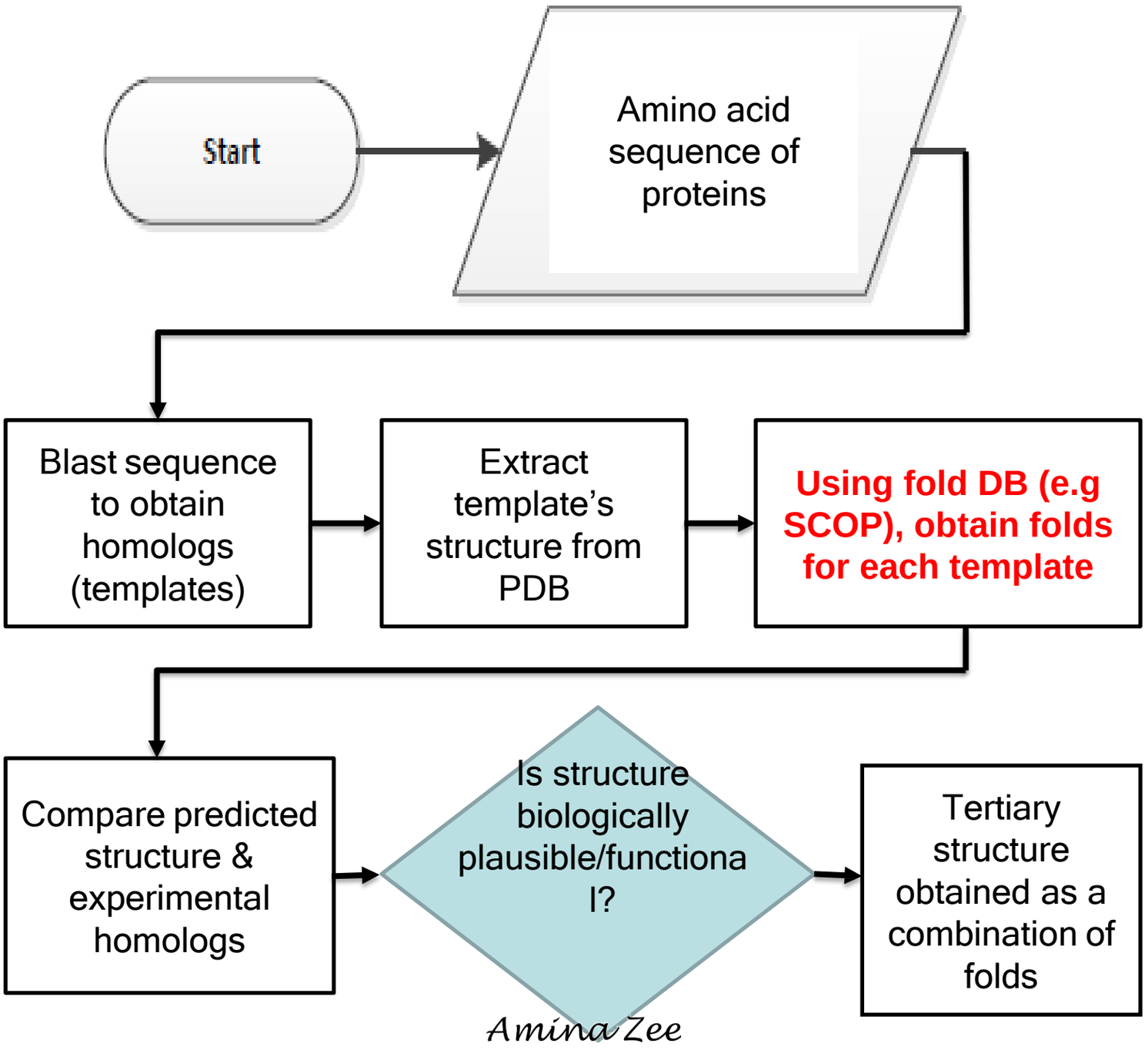
Summary of Structural Modelling - II

Inputs and outputs of threading



Amina Zee

Summary of Structural Modelling - II



Summary of Structural Modelling - II

Conclusion

- For low identity and alignment scores, a “Twilight zone” for structure prediction exists
- Fold recognition / threading is useful in such cases

Amina Zee

Homology Modelling

Lec 141

Summary of Structural Modelling – III

Amina Zee

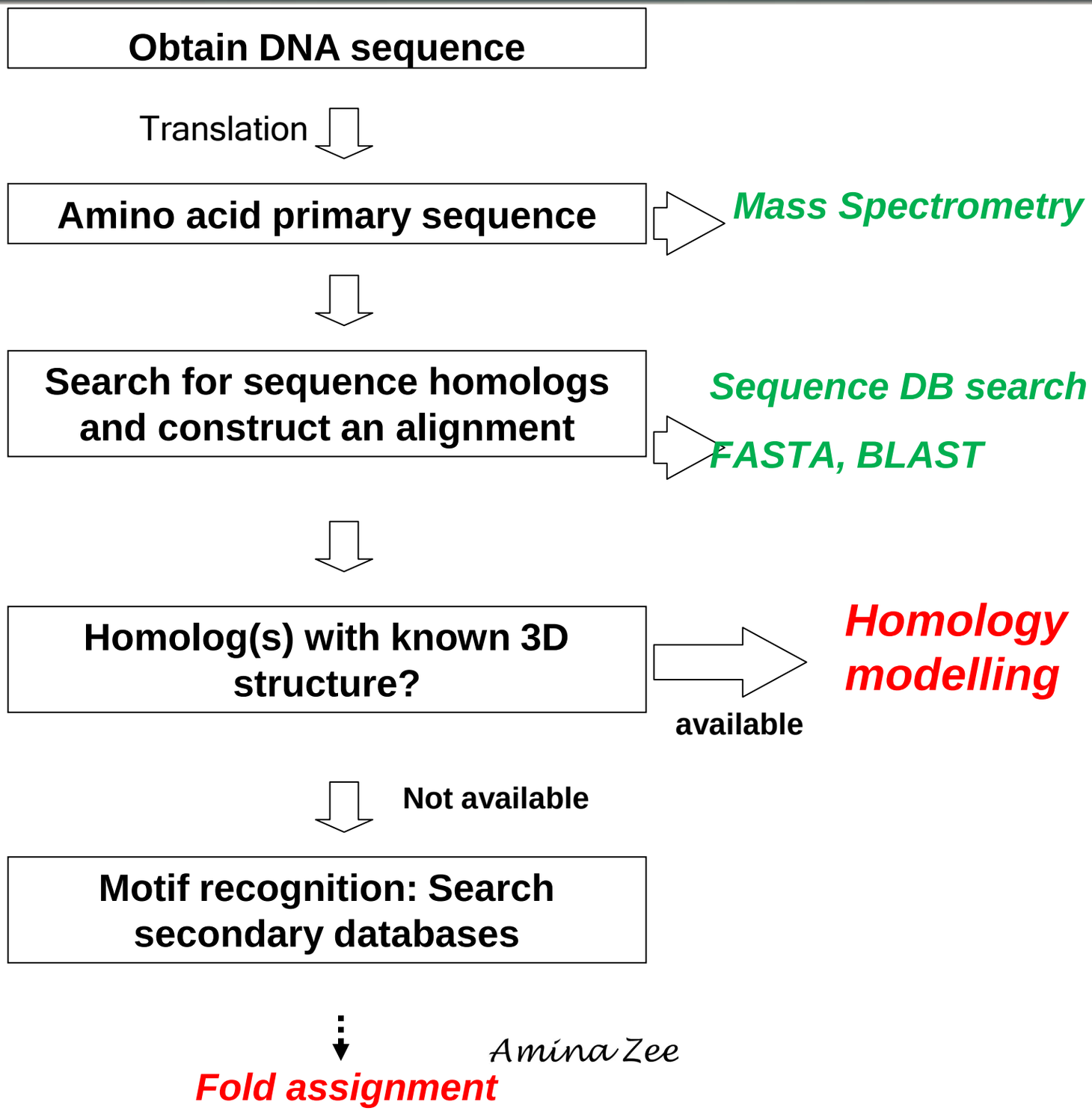
Summary of Structural Modelling - III

Strategies for Structural Modelling

- Homology Modelling
- Fold Recognition
- Ab Initio Modelling

Amina Zee

Summary of Structural Modelling - III



Summary of Structural Modelling - III

Energy Optimization in Ab Initio Modelling

1. **Start with a rough initial model.**

2. **Define an energy function mapping structures to energy values. We have to minimize this later!!**

3. **Solve the computational problem of finding the global minimum.**

Amina Zee

Summary of Structural Modelling - III

Simulation of the Folding Process

1. **Build an accurate initial model (including energy and forces).**

2. **Accurately simulate the dynamics of the protein folding process.**

3. **The native structure will steadily emerge.**

Amina Zee

Summary of Structural Modelling - III

Conclusion

- For cases where even the fold libraries do not give any high scoring matches, Ab Initio strategies can help model the structure
- However, this is a complex and computationally expensive process

Amina Zee

END

Conclusion of the Course

Lec 142

Review of Sequence Analysis

Amina Zee

Review of Sequence Analysis

Important Concepts

How do we sequence:

- Genomes
- Proteomes

Amina Zee

Review of Sequence Analysis

Important Concepts

How do we compare sequences:

- **Pair-wise Sequence Alignment**
- **Multiple Sequence Alignment**

Amina Zee

Review of Sequence Analysis

Important Concepts

Types of Alignments:

Global Alignment (Needle Wunsch)

Local Alignment (Smith Waterman)

Amina Zee

Review of Sequence Analysis

Important Concepts

Advanced Tools:

Fast Alignment (FASTA)

Basic Local Alignment Search Tool (BLAST)

Amina Zee

Review of Sequence Analysis

Important Concepts

Databases:

GenBank

UniProt

Amina Zee

Review of Sequence Analysis

Important Concepts

Online Portals:

Ensemble

Expasy

UniProtKB

Amina Zee

END

Conclusion of the Course

Lec 143

Review of Phylogenetics

Amina Zee

Review of Phylogenetics

Important Concepts

- Molecular Evolution

- Insertions
- Deletions
- Substitutions

Amina Zee

Review of Phylogenetics

Important Concepts

- Phylogenetic Trees

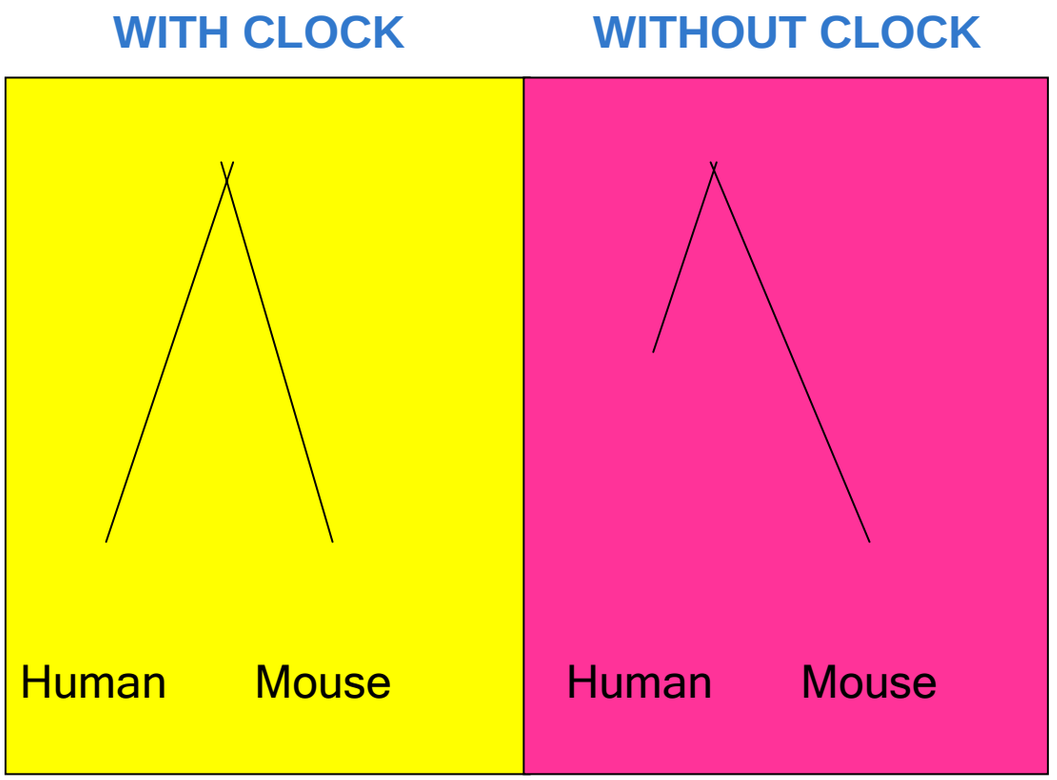
- Scaled Trees

- Unscaled Trees

Amina Zee

Review of Phylogenetics

Edge length with and without a clock!



Since the rate of evolution in species is different, it needs to be considered!

Amina Zee

Review of Phylogenetics

Important Concepts

- Phylogenetic Trees

- Rooted Trees

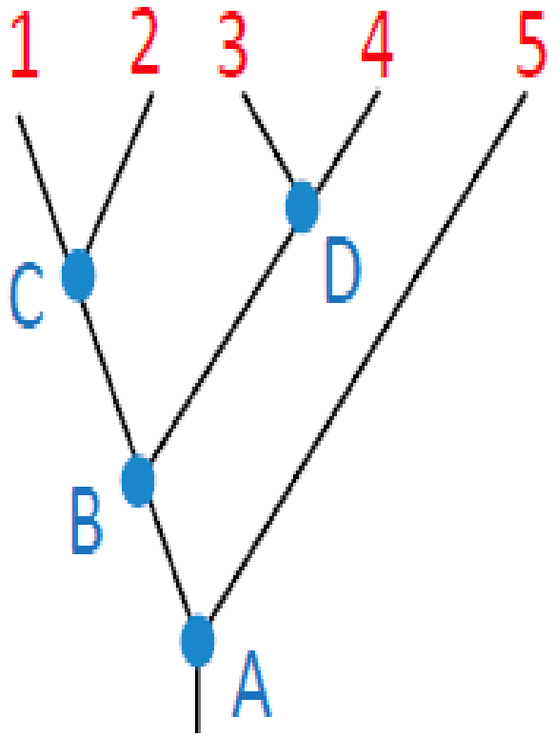
- Unrooted Trees

Amina Zee

Review of Phylogenetics

UPGMA: Unweighted Pair – Group Method using arithmetic Averages

- Two sequences with wit the **shortest evolutionary distance** between them are considered



- These sequences will b **the last to diverge**, and **represented by the most recent internal node.**

Amina Zee

Review of Phylogenetics

Important Concepts

Clustering Vs. Non-clustering Methods:

UPGMA is a clustering method

Maximum Parsimony etc are non-clustering methods (not included in this course).

Amina Zee

END

Conclusion of the Course

Lec 144

Review of Protein Sequencing

Amina Zee

Review of Protein Sequencing

Important Concepts

- Techniques of protein sequencing
- Edman Degradation
- Mass Spectrometry

Amina Zee

Review of Protein Sequencing

Important Concepts

- Protein Ionization
- Mass Analysis
- Protein Fragmentation

Amina Zee

Review of Protein Sequencing

Important Concepts

- MS1
- MS2

Amina Zee

Review of Protein Sequencing

Important Concepts

- Estimating and scoring whole protein mass

Amina Zee

Review of Protein Sequencing

Important Concepts

- Extracting & Scoring Peptide Sequence Tags

Amina Zee

Review of Protein Sequencing

Important Concepts

- Searching Post-translational Modifications

Amina Zee

Review of Protein Sequencing

Important Concepts

Composite Scoring Schemes

Online tools:

- **Mascot**
- **Sequest**
- **Prosight PC**

Amina Zee

END

Conclusion of the Course

Lec 145

Review of Protein Structures

Amina Zee

Review of Protein Structures

Important Concepts

Protein Structures are generally of four types:

- **Primary**
- **Secondary**
- **Tertiary**
- **Quaternary**

Amina Zee

Review of Protein Structures

Important Concepts

Techniques for determining protein structures

- **X-Ray Crystallography**
- **NMR Spectroscopy**

Amina Zee

Review of Protein Structures

Important Concepts

- **Why number of known protein sequences is much larger as compared to known proteins structures?**

Amina Zee

Review of Protein Structures

Important Concepts

Types of Protein Secondary Structures

- Helices
- Beta Sheets
- Coils
- Loops

Amina Zee

Review of Protein Structures

Important Concepts

- Foundation of structure prediction algorithms
- Propensities of certain amino acids to form specific secondary structures

Amina Zee

Review of Protein Structures

Important Concepts

- Algorithm for predicting protein structures
- Chou Fasman Algorithm

Amina Zee

Review of Protein Structures

Important Concepts

- **Protein Structure Database - PDB**
- **Online tools for predicting structures by using proteins sequences**

Amina Zee

END

Conclusion of the Course

Lec 146

Review of Protein Structures

Amina Zee

Review of Protein Structures

Important Concepts

Protein Structures are generally of four types:

- **Primary**
- **Secondary**
- **Tertiary**
- **Quaternary**

Amina Zee

Review of Protein Structures

Important Concepts

Techniques for determining protein structures

- **X-Ray Crystallography**
- **NMR Spectroscopy**

Amina Zee

Review of Protein Structures

Important Concepts

- **Why number of known protein sequences is much larger as compared to known proteins structures?**

Amina Zee

Review of Protein Structures

Important Concepts

Types of Protein Secondary Structures

- Helices
- Beta Sheets
- Coils
- Loops

Amina Zee

Review of Protein Structures

Important Concepts

- Foundation of structure prediction algorithms
- Propensities of certain amino acids to form specific secondary structures

Amina Zee

Review of Protein Structures

Important Concepts

- Algorithm for predicting protein structures
- Chou Fasman Algorithm

Amina Zee

Review of Protein Structures

Important Concepts

- Protein Structure Database - PDB
- Online tools for predicting structures by using proteins sequences

END

Amina Zee

Conclusion of the Course

Lec 147

Review of Homology Modelling

Amina Zee

Review of Homology Modelling

Important Concepts

Four Strata of Protein Structures

- Primary
- Secondary
- Tertiary
- Quaternary

Amina Zee

Review of Homology Modelling

Important Concepts

- Justification for homology modelling
- Number of known protein sequences is much larger as compared to known proteins structures

Amina Zee

Review of Homology Modelling

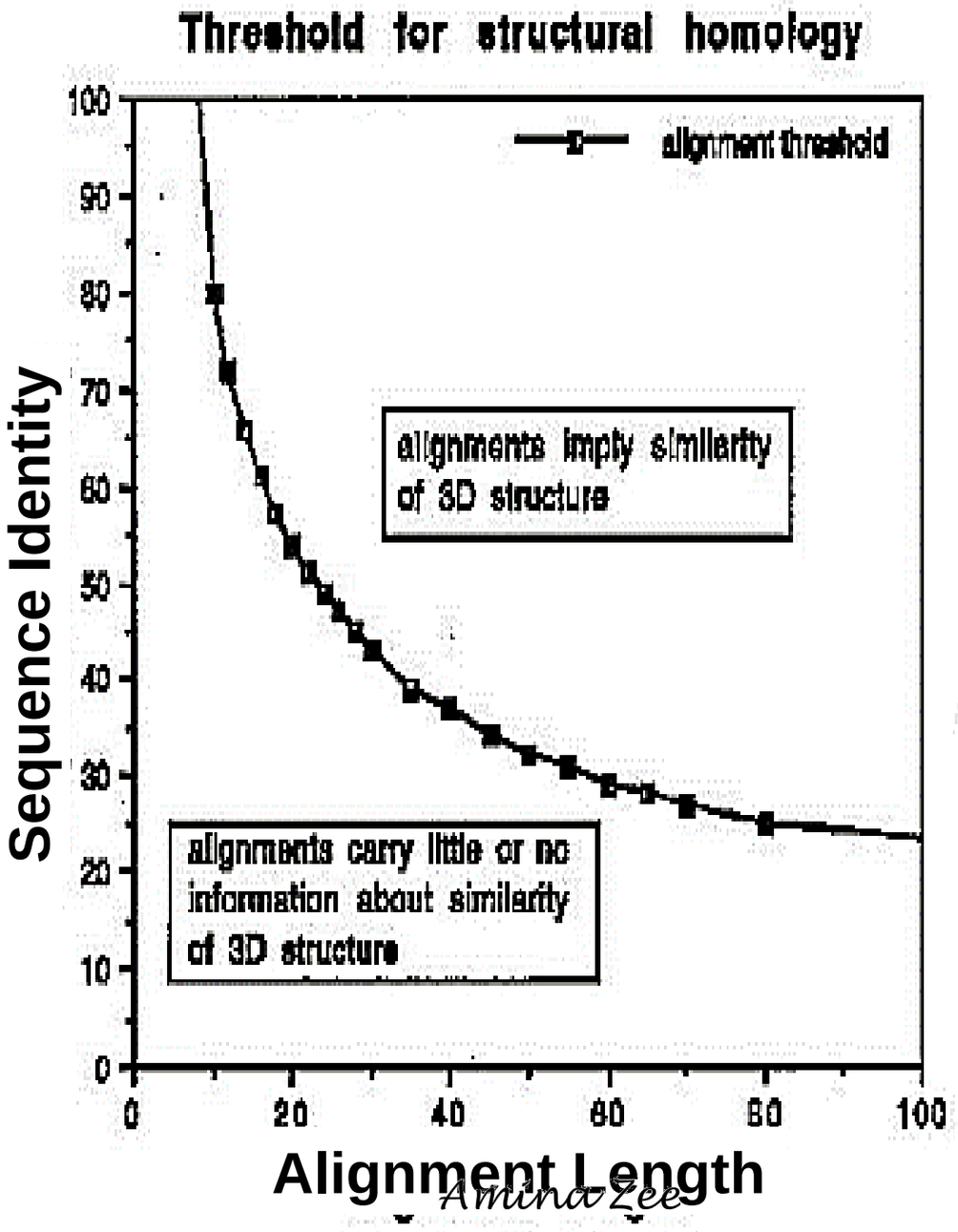
Important Concepts

Three Strategies for Structure Prediction

- Homology Modelling
- Fold Recognition
- Ab Initio Modelling

Amina Zee

Review of Homology Modelling



Amina Zee

http://www.cmbi.ru.nl/gvteach/astra/lectures/homology_modelling.ppt

Review of Protein Structures

Important Concepts

- **Protein Structure Database - PDB**
- **Online tools for predicting structures such as MODELLER and iTASSER**

END

Amina Zee

Lec 148

Conclusions from this Course

Amina Zee

Conclusions from this Course

Important Concepts

- Definition of Bioinformatics
- Need for Bioinformatics
- Areas within Bioinformatics

Amina Zee

Conclusions from this Course

Important Concepts

- Bioinformatics as an interdisciplinary area
- Need to store, process and analyze biological data
- Requirement of newer faster algorithms

Amina Zee

Conclusions from this Course

Important Concepts

Specific areas focused were:

- **Comparing sequences**
- **Comparing structures**
- **Predicting structures**

Amina Zee

Conclusions from this Course

Important Concepts

We looked at:

- Algorithms
- Databases
- Online Tools

for each topic.

Amina Zee

Conclusions from this Course

Important Concepts

- We studied the basic algorithms for each topic
- With evolution and growth of Bioinformatics, newer and better algorithms are now also available!

Amina Zee

END

Conclusion of the Course

Lec 149

Advanced Follow-up Courses

Amina Zee

Advanced Follow-up Courses

Important Concepts

- We looked into the foundations of Bioinformatics
- However, each topic that was studied has a undergone a lot of development

Advanced Follow-up Courses

Important Concepts

- For advanced study in Genomics, you may take "Computational Genomics" course

- **Topics:**
  Genome Assembly, Gene Finding, Annotation, GWAS etc

Amina Zee

Advanced Follow-up Courses

Important Concepts

- For advanced study in Proteomics, you may take "Computational Proteomics" course.

- **Topics:**
  Protein Sequencing, PTM search, Structure Modelling and PPI studies

Amina Zee

Advanced Follow-up Courses

Important Concepts

- For advanced study in Integrative Biology, you may take “Systems Biology” course.

- **Topics: Metabolomics, Transcriptomics, Network Biology etc**

Amina Zee

Advanced Follow-up Courses

Important Concepts

Also, now there are cutting edge courses on:

- **Nano-Bio-IT**
- **Computational Drug Design**
- **Personalized Medicine**

Amina Zee

END

Conclusion of the Course

Lec 150

Careers in Bioinformatics

Amina Zee

Careers in Bioinformatics

Background

- Pakistan as an infrastructure-limited country
- The onset of digital revolution
- Emergence of data as the most precious commodity, globally

Amina Zee

Careers in Bioinformatics

Background

- Specifically, health data as a key commodity of the future
- Health and disease as the primordial challenge of mankind

Amina Zee

Careers in Bioinformatics

Background

- Unique opportunity for us in Pakistan
- Bioinformatics requires two things

1. Smart mind
2. Internet connected computer

Amina Zee

Careers in Bioinformatics

One man company

- You can take public databases and design drugs!
- One man vs. Roche?

Amina Zee

Careers in Bioinformatics

BIGDATA

- You can make a startup company which manages and process health BIGDATA!

- All it needs is basic software development skills coupled with Bioinformatics

Amina Zee

Careers in Bioinformatics

The next disruption

- The next Google, Facebook and Uber is going to emerge from Health and Bioinformatics
- Pharmaceutical companies are investing into bioinformatics human resource development

Amina Zee

Careers in Bioinformatics

Jobs Market

- Pharmaceutical Giants
- Research Centers & Universities
- Hospital & Diagnostic IT departments
- Your own startup company

Amina Zee